

MYERS-BRIGGS TYPE INDICATOR® | GLOBAL STEP I™ AND STEP II™

MBTI® MANUAL for the Global Step I™ and Step II™ Assessments

ISABEL BRIGGS MYERS
MARY H. MCCAULLEY
NAOMI L. QUENK
ALLEN L. HAMMER

Fourth Edition



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NAOMI L. QUENK
ALLEN L. HAMMER

Fourth Edition

With contributions from

Richard C. Thompson
Michael L. Morris
Justin Arneson
Nancy A. Schaubhut
Craig Johnson

With text incorporated from

MBTI® Manual, third edition

Isabel Briggs Myers
Mary H. McCaulley
Naomi L. Quenk
Allen L. Hammer

MBTI® Step II® Manual

Naomi L. Quenk
Allen L. Hammer
Mark S. Majors



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DEDICATION

Following the death of Isabel Briggs Myers in 1980, there could not have been two more stalwart stewards of a body of work than Peter Briggs Myers and Katharine Downing Myers, the son and daughter-in-law of Isabel Myers. During their lifetimes their knowledge, persistence, and commitment to the legacy of the work has served millions of people throughout the world who have benefited from the priceless gift of self-understanding.



Peter Myers and Katharine Myers
Photo courtesy of the Center for Applications of
Psychological Type

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FOREWORD

In creating the *Myers-Briggs Type Indicator*® (MBTI®) assessment, Isabel Myers had two main goals. First, she wanted to provide as many people as possible with the benefits of knowing their own type and how they were similar to, or different from, people of other types. She worked on developing and collecting research data on her own for a number of years. In 1962, Educational Testing Service (ETS) in Princeton, New Jersey, became the first publisher of the *Myers-Briggs Type Indicator*. However, the Indicator was available only as a research instrument so was not yet widely known. It was at ETS that Myers developed and published her first manual for the MBTI work.

When Consulting Psychologists Press (now The Myers-Briggs Company) became the publisher of the instrument in 1975, the MBTI instrument's reach was expanded to qualified professionals, and it gradually moved from being a relatively obscure assessment to one that is widely available to professionals and their clients throughout the world. At this time millions of people have taken the MBTI assessment, either in an English version or in one of the 23 translated forms. As a result, Myers' vision of giving many people access to their own type continues to be fulfilled.

Myers' second goal was to individualize the 16 type descriptions so that each person's uniqueness *within* his or her type could be identified and described. Work on creating an assessment to individualize type reports began in the mid-1980s and reached full fruition with the publication of the MBTI Step II™ instrument and *MBTI® Step II™ Manual* (Quenk, Hammer, & Majors, 2001). Use of the Step II assessment and individualized reports has increased markedly over the years, and Step II information, showing uniqueness within type, has become widely available to practitioners and their clients throughout the world.

At the end of her introduction to the 1962 *Manual*, Myers wrote,

The Indicator aims to ascertain, from self-report of easily reported reactions, people's basic preferences in regard to perception and judgment, so that the effects of the preferences and their combinations may be established by research and put to practical use.

This global manual admirably achieves both of Myers' 1962 goals. Naomi Quenk and Allen Hammer clearly and thoroughly describe the theoretical basis and practical uses of the MBTI Step I and Step II assessments and the relationship between the two instruments. Quenk and Hammer explain the data collection for and construction of the global assessments and summarize original research on their reliability and validity. In addition, they detail the wealth of published research supporting the reliability and validity of the MBTI Step I and Step II assessments as well as the many varied and innovative uses of both instruments. Isabel Myers, Peter Myers, and Katharine Myers would be pleased.

Betsy Styron, Chairman
The Myers & Briggs Foundation

PREFACE

Our preface to the last edition of the MBTI® Step I™ manual (1998), which launched Form M of the MBTI assessment, stated that a “powerful objective of this revision was to reinforce the connection of the MBTI to its roots in the psychological type theory of Carl G. Jung.” Our preface to the MBTI Step II™ manual (2001) described the development of the Step II assessment as the “realization of Myers’ plan to further illuminate the qualities that contribute to each individual’s unique personality.” The goals of these prior manuals are even more relevant if not essential for the current global manual that covers the revisions of both the MBTI Step I and Step II assessments.

By presenting MBTI Step I and MBTI Step II together in the same manual, we were forced to think about and explain just how these two manifestations of type theory were related to each other. MBTI Step I is firmly grounded in Jungian theory, so you will find ample evidence of the ways in which Myers and Briggs carefully adhered to Jung’s theory. Bear in mind that the totality of Jung’s wide-ranging work appears in 20 published volumes, *Psychological Types* being volume 6. As you will see in some of the quotes in chapter 3, “The Theory Behind the MBTI® Assessment,” Jung was a keen observer of people—his analytic patients, his colleagues, historical figures, artists, writers, theorists, and philosophers. He wanted to develop a coherent way to describe the differences he observed. Jung believed he had found what he wanted by identifying people as either extraverts or introverts and much of volume 6, *Psychological Types*, is devoted to examples of this distinction. His further observations convinced him, however, that this classification was inadequate, leading to his describing the opposite ways of perceiving, *Sensation* and *Intuition*, and opposite ways of judging, *Thinking* and *Feeling*. Using this expanded three-category system led Jung to identify eight personality types, each with its own characteristic way of being energized, acquiring information, and coming to conclusions about that information.

Jung’s classification provided Myers and Briggs with the basis for their development of the MBTI Step I assessment. By adding a fourth pair of opposites, the *Judging* versus *Perceiving* orientation to the outside world, Myers and Briggs expanded Jung’s original eight types to encompass 16. The inclusion of the J–P pair of opposites added further complexity to the system as well as much richer individualized descriptions of personality.

Myers’ ultimate goal, however, was to identify and describe individuality within each of the 16 types. For this she needed to move beyond Jung’s original explorations. The MBTI Step II assessment was developed to realize Myers’ goal. This manual is designed to identify and describe the ways that MBTI Step I and MBTI Step II are expressions and extensions of Jung’s original theory.

This manual has two parts, the first part covering theory and interpretation (chapters 1–6) of the MBTI assessment, and the second part covering construction and research (chapters 7–10). Effective use of the assessment can be enhanced by a solid understanding of the theory behind it. Knowing how the MBTI instrument was constructed will help the practitioner respond to questions clients

may ask. The research chapters give evidence for the reliability and validity of the assessment and support appropriate use of the MBTI instrument.

Chapter 1, “Introduction to MBTI® Step I™ and Step II™ Type,” is an overview of the MBTI approach to personality, the current revisions of the assessments, a summary of type theory, the way the MBTI approach differs from other approaches to personality, and individual differences within types.

Chapter 2, “A Closer Look at the Global Step II™ Assessment,” points out the multifaceted nature of the four Step I preferences, which enabled development of the Step II facets. The dynamic nature of the Step II approach to personality theory is explained as well.

Chapter 3, “The Theory Behind the MBTI® Assessment,” details key features of type theory, the fundamentals of type, type as a dynamic system, and the theory of type development.

Chapter 4, “The 16 Types and the 20 Facets,” contains full descriptions of the 16 types, including identification of dynamic elements of each type description, and detailed descriptions of each of the 20 Step II facets.

Chapter 5, “Type Tables and Groupings of MBTI® Preferences,” shows how using arrays of the 16 types into type tables and different groupings of type preferences provide both a research methodology and a way to quickly recognize and make use of type differences.

Chapter 6, “Administering and Interpreting the MBTI® Step I™ and Step II™ Assessments,” provides detailed guidelines for administering the assessments to respondents, interpreting results, and providing feedback to them.

Chapter 7, “Construction of the MBTI® Global Assessments,” describes the lengthy and careful procedures used to obtain global samples of respondents; construct, translate, and administer the research form to international samples; analyze the data obtained, select items, and determine scoring methods for both of the Step I and Step II assessments. The creation of the probability index, new to the global assessments, is covered in chapter 7, and type tables showing the distribution of MBTI types in the global sample are also in that chapter.¹

Chapter 8, “Reliability of the MBTI® Global Step I™ and Step II™ Assessments,” discusses factors that can affect reliability and then presents results from studies of internal consistency reliability and studies of test-retest reliability, that is, if persons take the MBTI assessment more than once will they have the same MBTI type.

Chapter 9, “Validity of the MBTI® Step I™ Assessment,” covers the wealth of research detailing the validity of the Step I assessment.

Chapter 10, “Validity of the MBTI® Step II™ Assessment,” offers a variety of evidence for the validity of the 20 Step II facet scales. Both chapters 9 and 10 contain extensive references to research on the assessments that has been conducted since the publication of the last manual as well as research on the global assessments.

If you are not reading the manual chapters consecutively, you may come across some unfamiliar terminology. Please use the glossary in the back of the book.

Finally, the ever-increasing numbers of users of the two MBTI assessments worldwide is certainly testimony to their usefulness to individuals and groups. At the same time, this popularity raises the likelihood of misunderstanding and misuse of both instruments. As educated and responsible users of the Jung-Myers approach to personality assessment, please pay attention to the accuracy with which type information is portrayed and discussed verbally, in print, and online. Consider yourself an emissary of the accuracy and the appropriate use of the theory and the assessments. Two sayings are relevant in this regard: “A little knowledge is a dangerous thing,” and “People don’t know what they don’t know.” Knowledgeable, fact-based critiques and substantive observations contribute to any field of knowledge and are vital to the credibility, accuracy, and usefulness of the MBTI assessments.

As coauthors we have been privileged to devote much of our lives to studying, exploring, researching, and growing from the depth and breadth of personality type as expressed in the Jung-Myers theory of personality type and in MBTI Step I and Step II. We hope you will be similarly devoted to and enthusiastic about the contents of this manual and your own use of the Step I and Step II assessments.

1. For the distribution of MBTI types within a particular country, see the *MBTI® Manual Global Supplement Series* (Schaubhut, Thompson, Morris, & Arneson, 2018), 23 booklets that can be downloaded from www.themyersbriggs.com/mbti-manual-supplements.

ACKNOWLEDGMENTS

Many people contributed to this combined MBTI® Step I™ and Step II™ manual, most notably our coauthors Isabel Briggs Myers and Mary McCaulley, whose words and ideas appear throughout.

Peter Myers and Katharine Myers provided invaluable guidance on earlier editions of both manuals. We trust they would have approved this combined manual as well. We deeply regret that their continued involvement was not possible.

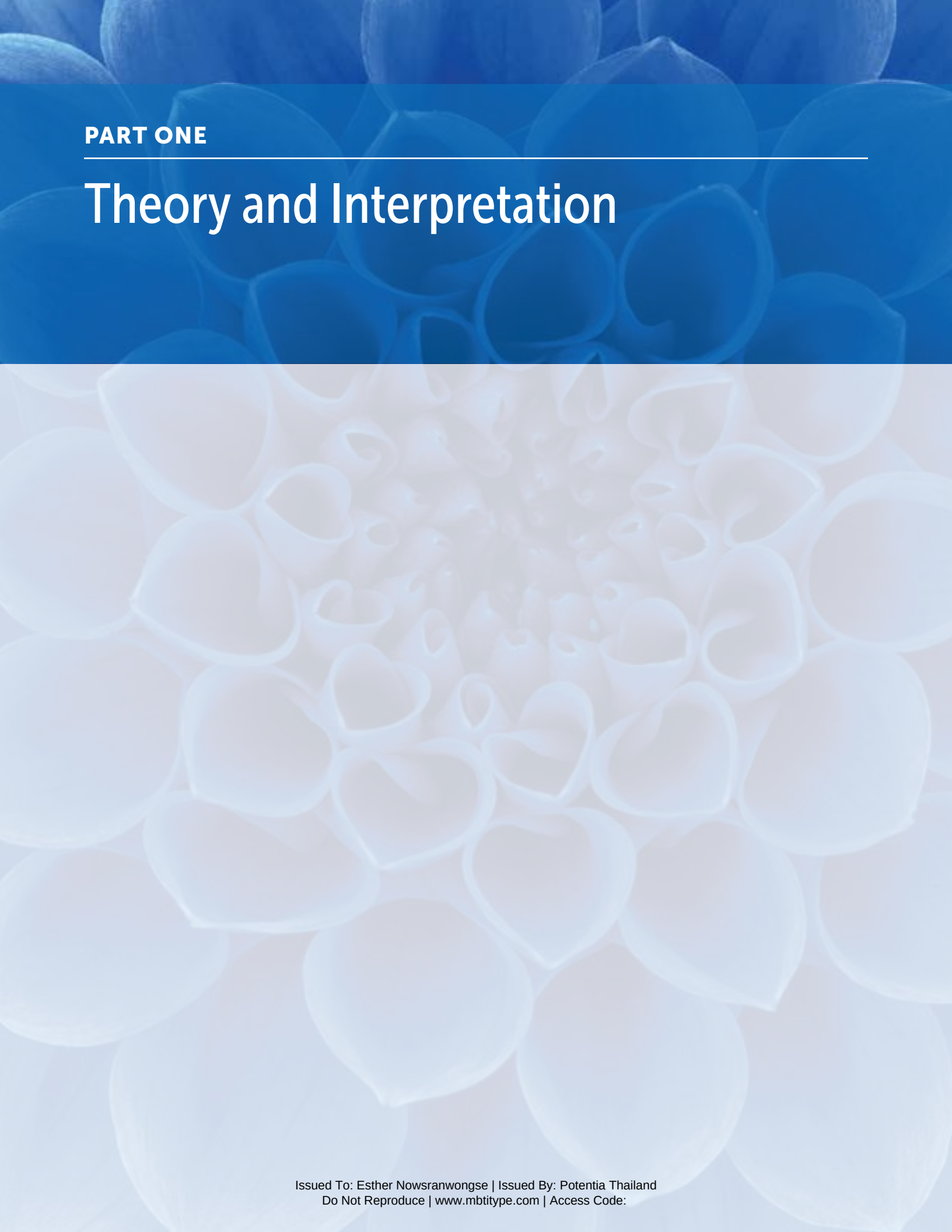
Lee Langhammer-Law, formerly Vice President & Publisher at The Myers-Briggs Company, foresaw the need for the hypothetical planned combined manual long before research to create the global assessment was under way.

We are grateful to Wayne Mitchell, whose deep understanding of the Step II facets and detailed descriptions of them were the basis for the facet descriptions in this manual.

The expertise of the Research division at The Myers-Briggs Company, Rich Thompson, Mike Morris, and Nancy Schaubhut, was invaluable both in developing the revised assessments and in quickly providing whatever analyses were needed.

The research that is the basis of this global manual could not have been done without the cooperation and hard work of the distributors of the MBTI assessment around the world who worked on translations and provided samples. We are grateful to them all as well as to the thousands of unidentified respondents who agreed to take the research form that was the basis for the global instrument.

Without the skilled editing of Jill Anderson-Wilson and Janelle Rohr this project would never have been accomplished. Their expertise and depth of understanding of the type assessments are impressive. We also thank Angela Glenn of The Gasp Company, LLC, for creating a superb design for this publication and Hawkin Chan for his art direction.



PART ONE

Theory and Interpretation

CHAPTER 1

Introduction to MBTI® Step I™ and Step II™ Type

CHAPTER OVERVIEW

Goals of the MBTI® Approach to Personality Description

The Global Revision of the MBTI® Step I™ and Step II™ Assessments

Overview of Personality Type Theory

Differences Between the MBTI® Instruments and Other Personality Instruments

Individual Differences Within Type Preferences

Conclusion

The purpose of the *Myers-Briggs Type Indicator®* (MBTI®) assessment is to make the theory of psychological types described by C. G. Jung (1921/1971) understandable and useful in people's lives. The essence of type theory is that much seemingly random variation in behavior is actually quite orderly and consistent, being due first and foremost to basic differences in the ways individuals use their mind. Jung first identified two opposite ways people gather information (perception) and two opposite ways they come to conclusions (judgment). Perception involves ways of becoming aware of things, people, experiences, and ideas. Judgment involves ways of coming to conclusions about what has been perceived.

In addition to perception and judgment, Jung included in his system a third pair of opposites—two opposite sources of energy. Myers and Briggs' innovative extension and refinement of Jung's theory added a fourth area of mental functioning—two opposite ways people tend to relate to the outside world.

In developing the MBTI assessment, Isabel B. Myers and her mother, Katharine C. Briggs, sought a way to understand and capitalize on these naturally occurring individual differences. They reasoned that if people differ systematically in how and what they perceive, the manner in which they come to conclusions, their source of energy, and how they relate to the outside world, then it is only reasonable for them to also differ in their interests, reactions, values, motivations, and skills. They saw that making practical use of these differences was a way to give individuals, groups, and societies access to the insights of type theory so that people could live their life more effectively and with greater satisfaction.

GOALS OF THE MBTI® APPROACH TO PERSONALITY DESCRIPTION

Briggs and Myers had two related goals in developing the MBTI assessment. First, they wanted to identify which of the opposite ways of using one's mind that are specified or implicit in Jung's theory an individual tends to use most naturally and comfortably. The four pairs of opposites that form the basis of the Jung-Myers theory of personality types are as follows:

Opposite ways of gathering information (perceiving)—
Sensing (S) or Intuition (N)

Opposite ways of coming to conclusions (judging)—
Thinking (T) or Feeling (F)

Opposite sources of energy—Extraverted energy
(E, oriented outward) or Introverted energy
(I, oriented inward)

Opposite ways of relating to the outside world—
a Judging orientation (J) or a Perceiving orientation (P)

The MBTI Step I™ assessment reports an individual's preferences for one of the two opposites in each of these four pairs.

Second, they wanted to show how combinations of preferences operate as dynamic, whole types. An individual's four preferences—the preferred way of perceiving, the preferred way of making judgments, the preferred orientation of energy, and the preferred orientation to the outside world—combine to create one of 16 distinct personality types. The preferences in each of these 16 types combine in specific, dynamic ways that are delineated in the Jung-Myers theory. These "type dynamics" are explained in detail in chapter 3. At this point, it is important to know that the *dynamic character* of the Jung-Myers theory is a central feature of the theory's explanatory power and sets this theory apart from other personality approaches.

The way the preferences in each type combine—their dynamic interaction—is a unique feature of the Jung-Myers personality system.

Achieving the two goals of first identifying the preferences and then showing how combinations of preferences operate as dynamic, whole types enabled Myers and Briggs to describe many of the observable, consistent differences people often notice about each other for which they have no systematic explanation.

Myers also pursued two additional goals during her lifetime. The first was to explore variations in the way people of the same type experience and express their type. The second was to assess an individual's current state of type development and provide guidance for

achieving greater satisfaction and fulfillment within his or her type. The MBTI Step II™ assessment addresses the first of these goals by identifying and measuring facets, or different aspects, of each of the four pairs of opposites. The MBTI Step III™ assessment, which is touched on briefly in this manual, focuses on issues of type development. A separate manual for the Step III assessment (see Myers, McCauley, Quenk, Hammer, & Mitchell, 2009) is available.

The purpose of this initial chapter is to provide readers with the orientation needed to use this manual productively. The chapter describes the differences between the MBTI Global Step I™ and Step II assessments and prior forms of these two assessments. It then discusses basic four-letter (Step I) type and its measurement, as well as the extended MBTI form that expands on Step I information (the Step II assessment). It provides a general overview of the Jung-Myers personality theory, the identification of MBTI type preferences, the interaction of preferences to form the 16 types, differences between the MBTI approach to personality and other approaches, and individual differences within types as revealed by the Global Step II assessment.

THE GLOBAL REVISION OF THE MBTI® STEP I™ AND STEP II™ ASSESSMENTS

The theoretical, structural, philosophical, and ethical features that have long been part of the MBTI Step I and Step II assessments were continued and reinforced in the global revision. These features are described in this chapter and expanded on in subsequent chapters. The global revision rests on the firm foundation of prior forms of the MBTI assessment while taking advantage of innovations in personality measurement and data analysis. In addition, the revision permitted the use of data from a newly compiled global sample. Continuity of features in the Step I and Step II assessments is summarized in table 1.1, along with important changes made since the 1998 Step I revision that produced Form M and the 2001 Step II revision that produced Form Q.

The Global Step I assessment replaces both MBTI Form M and the European Step I assessment, and the Global Step II assessment replaces both MBTI Form Q and the European Step II assessment. This means that there is now *one global* Step I assessment and *one global* Step II assessment, which will be translated for use around the world, rather than separate US and European assessments.

The Global Manual

In the past, separate manuals and manual supplements were published for the Step I and Step II assessments in the United States and in Europe. This global manual combines and integrates into a single volume information that previously was covered in these separate manuals and supplements. Providing information about both the Global Step I and Global Step II assessments in one manual eliminates the duplication that occurred in the previous separate manuals and highlights similarities, differences, and interrelationships among elements of these assessments. Practitioners are thus better equipped to understand and interpret type information and apply it to respondents.

Global Assessments

As stated, the greatest departure in this revision from previous MBTI revisions is the development of *global* forms of the Step I and Step II assessments. Although a number of translations of prior forms have been available for many years, before the current effort there was no uniform set of Step I or Step II items or a standardized scoring system for all translations of either assessment. A number of countries and languages for which commercial translations were not previously available were included in the global sample. For global data collection purposes, the MBTI global assessments were translated into 19 languages in 20 countries within six continents, giving millions of individuals in different cultures access to their personality type. For detailed information on samples, data collection processes, and analyses of data, see chapter 7, the assessment construction chapter of this manual.

Selection and Scoring of Items

Latent class analysis (LCA), which is designed to analyze categorical data such as Step I preferences, was used in selecting and scoring items for the Global Step I assessment. LCA is a relatively new and powerful statistical approach that assumes people belong to one of some number of groups or classes, a view that is consistent with type theory. Most other personality assessments aim to deliver continuous scores on various scales, but the goal of MBTI Step I scoring is to sort individuals into one or the other of two distinct groups (e.g., Thinking or Feeling) that are separated by a midpoint. The prior Step I form, Form M, used item response theory (IRT) for item selection and scoring, as did both the Global Step II assessment and the prior Step II form, Form Q. The prior European Step I form used a prediction ratio method for item selection and scoring. The prior European Step II form used IRT analysis as its basis. For more information about item selection and scoring, see chapter 7.

Table 1.1 | **Continuity and change in MBTI® Step I™ and Step II™ instruments**

Continuity
Firmly grounded in the Jung-Myers theory of personality type
Positive, neutral language describes opposite, healthy type preferences and facets
Preference scales and facet scales named by their two opposite poles
Step I poles separated by a midpoint; Step II poles separated by a midzone for interpretive purposes
Item response theory (IRT) used for Step II item selection and scoring
Based on the philosophy that all preferences are equally valuable
Item selection based on Jung's and Myers' theories of type
Many items with same wording as on prior forms
Same forced-choice item format and use of phrase questions and word pairs as on prior forms
Presentation of extensive validity data
Same 20 facets as on prior Step II forms
Change
Global Step I and Step II assessments developed for many countries and languages
One integrated manual replaces separate manuals for the Step I and Step II assessments
Global Step I assessment replaces MBTI Form M and European Step I assessment; Global Step II assessment replaces MBTI Form Q and European Step II assessment
Probability index replaces Form M preference clarity index (pci) as an estimate of accuracy of Step I reported type
Latent class analysis (LCA) used for Step I item selection and scoring
MBTI reports and support materials aligned to the global instruments
Item weights determined using a large-scale global sample from 20 countries
Data collection and testing of items and scales conducted in 20 countries
Reporting of MBTI research in new areas (e.g., social media)
Improvements in the measurement of some of the Step II facet scales

Readers of this manual who are new to the approach described by the MBTI Step I assessment will want to pay careful attention to sections of this chapter that focus on Jung's theory and its extension by Myers. Readers who are experienced with the Step I assessment but not with the Step II assessment will benefit most from the discussion of the rationale underlying the development of the Step II assessment and the way it describes variations within the 16 personality types.

Both new and seasoned MBTI practitioners will be better able to answer respondents' questions after carefully reviewing the theoretical and psychometric differences between the MBTI instruments and other personality instruments.

Although the emphasis in this manual is on the dynamic, interactive character of the types, information about relationships between individual preferences and a variety of other variables is provided as well. A wealth of such information is also available in previous MBTI manuals (Myers, 1962b; Myers & McCaulley, 1985; Myers, McCaulley, Quenk, & Hammer, 1998) and in the many research and observational studies in the extensive type literature (see www.capt.org/MILO).

OVERVIEW OF PERSONALITY TYPE THEORY

The term *type theory*, as used in this manual, refers to Jung's original theory as described in his writings. The terms *Jung-Myers theory* and *Jung-Myers approach* refer to Jung's theory as interpreted, refined, and expanded by Isabel Myers and Katharine Briggs. The Jung-Myers theory is the basis for all MBTI Step I and MBTI Step II assessments, as well as for the MBTI Step III assessment. Note that although Jung's original volume was titled *Psychological Types*, the term *personality types* also appears in the type literature. These terms are largely interchangeable, but this manual uses *personality types*. This manual also uses a somewhat newer term, *Jung-Myers personality theory*. The latter term reflects the emerging recognition that Myers' writings describe a theory of personality and its development that is broader and more inclusive than the type theory described by Jung. This is especially evident in Myers' writings about type development, reflected in the MBTI Step III assessment and its manual.

The focus of the next section is on each of the pairs of opposites and how preferences interact to form the 16 types.

The Four Preferences That Make Up a Type

The Global Step I assessment identifies four separate pairs of opposites: Extraversion–Introversion, Sensing–Intuition, Thinking–Feeling, and Judging–Perceiving. An individual is assumed to favor one or the other option in each pair of opposites. The four pairs of opposites are described in table 1.2.

The first and fourth pairs of opposites listed (Extraversion–Introversion and Judging–Perceiving) are generally referred to as either *attitudes* or *orientations*, and the middle two pairs (Sensing–Intuition and Thinking–Feeling) are generally referred to as either *functions* or *processes*. Although the terms used most often in this manual are *orientation* rather than *attitude*

and *process* rather than *function*, readers should be aware that the terms *function* and *attitude* appear frequently in existing type literature, including in the Jungian quotations found in this and prior manuals.

EXTRAVERSION OR INTROVERSION (E–I)

The E–I preference pair indicates whether an individual prefers Extraversion or Introversion in the sense intended by Jung (1921/1971). Individuals who prefer Extraversion (termed *Extraverts* by Jung) are oriented primarily toward the outer world; thus they tend to focus their energy on people or objects in their environment and are energized by interacting with people and with the outside world. Individuals who prefer Introversion (termed *Introverts* by Jung) are oriented primarily toward the inner world; thus they tend to focus their energy on concepts, ideas, and internal experiences and sensations and are energized by this inner activity. Jung regarded Extraversion and Introversion as “mutually complementary” attitudes whose differences “generate the tension that both the individual and society need for the maintenance of life” (p. 160). Myers further specified that every individual needs “adequate (but by no means equal) balance

Table 1.2 | The opposites of the MBTI® assessment

Extraversion 	 Introversion
Orientations of energy	
Directing energy mainly toward the outer world of people, objects, and activities	Directing energy mainly toward the inner world of thoughts, impressions, and ideas
Sensing 	 Intuition
Perceiving processes	
Focusing mainly on what can be perceived by the five senses	Focusing mainly on perceiving patterns and interrelationships
Thinking 	 Feeling
Judging processes	
Basing conclusions on logical criteria with a goal of accuracy and objective truth	Basing conclusions on values-centered criteria with a goal of personal and social harmony
Judging 	 Perceiving
Orientations to the outside world	
Preferring the decisiveness and closure that result from dealing with the outside world using one of the judging processes (Thinking or Feeling)	Preferring the flexibility and spontaneity that result from dealing with the outside world using one of the perceiving processes (Sensing or Intuition)

between these two attitudes” (Myers, 1962b, p. 60) in order to function effectively.

It is interesting that one of the dictionary definitions of the word *pair* is “two matching objects that are designed to be used together.” This definition certainly fits for the four pairs of opposites in the Jung-Myers system, as does Jung’s quote above about the tension of opposites, which applies to all opposites in his broad theory. All pairs of opposites in the system complement each other—one does not exist without the other, and both are necessary to conduct one’s life effectively. The “tension” Jung refers to describes the continual shifting back and forth between opposites that allows individuals, groups, and society as a whole to function.

SENSING OR INTUITION (S–N)

The S–N pair is designed to reflect an individual’s preference for one of two opposite ways of perceiving, or gathering information. An individual may primarily rely on the process of Sensing (S), which attends to observable facts or past or current experiences through one or more of the five senses, or on the less obvious process of Intuition (N), which attends to patterns, meanings, relationships, and possibilities that may not be readily apparent to others.

Just as people who prefer Extraversion or Introversion can and do also use the opposite, complementary preference, people who prefer Sensing are free to use Intuition when considering the possibilities that might arise in a situation, while those who prefer Intuition find that using Sensing is necessary to live life on a daily basis.

THINKING OR FEELING (T–F)

The T–F pair is designed to reflect an individual’s preference for one of two opposite ways of deciding or making a judgment. A person may primarily rely on Thinking (T) to decide impersonally on the basis of logical consequences, or on Feeling (F) to decide on the basis of personal or social values. And, as is true for all the opposite preferences, individuals are free to exercise both Thinking *and* Feeling, regardless of their preference for one or the other.

In the Jung-Myers approach, the term *Thinking* does not imply intelligence or competence and the term *Feeling* is not to be confused with being emotional. Intelligence and emotional expression are independent of personality type. Intelligence varies within every type, and people of all types experience emotion.

JUDGING OR PERCEIVING (J–P)

The J–P pair is designed to identify the process an individual tends to use most readily when interacting with the outside world. Individuals who prefer a Judging (J) approach to the outside world have a preference

for using either Thinking or Feeling (whichever of the judging processes they prefer) when dealing with the outside world. Individuals who prefer a Perceiving (P) approach to the outside world have a preference for using either Sensing or Intuition (whichever of the perceiving processes they prefer) when dealing with the outside world. People who prefer a Judging (J) approach to the outside world tend to enjoy deciding what they think or feel about something and taking action as soon as they feel they have sufficient Sensing and/or Intuitive information. Those who prefer a Perceiving approach like to stay open to as much Sensing or Intuitive information as possible before deciding. Myers and Briggs believed that the J–P pair was implicit in Jung’s theory, although it was not described or included in his original typology.

Both the theory and practical observations describe individuals as using each of the eight preferences at least some of the time. “Always” or “never” exercising one or another of the eight preferences is highly unlikely.

Everyone is able to use less preferred processes and orientations regardless of his or her preference for one in each pair.

Identifying MBTI® Preferences

The main objective of the MBTI approach is to identify which pole of each pair of processes and orientations a person prefers. Assessment items present choices between seemingly inconsequential everyday events selected by Myers as stimuli to evoke the more comprehensive underlying area of type functioning. Because the Jung-Myers theory assumes that people use both of the opposites in each pair at least some of the time, determining which pole of a pair an individual prefers requires forced choices on items. Choices offered always reflect opposite preferences within the same pair of opposites. For example, an item never presents a respondent with a choice between a Sensing response and a Thinking response.

To identify what a respondent prefers, MBTI items force the respondent to choose between two neutral or equally valuable opposites, each of which he or she uses at least some of the time.

It is assumed that in everyday life, people tend to respond first, most often, and most comfortably with their preferred processes and orientations. MBTI preferences are thus analogous to right-handedness or left-handedness. Most people can use both their right and left hands, even though they tend to reach first with their preferred hand. The hand individuals prefer can

be determined by observing their behavior or simply by asking them.

To determine a person's MBTI preferences, the Global Step I assessment uses forced-choice items, as described above. Respondents are then sorted into their preferences on the basis of the latent class analysis scoring system described in chapter 7. The purpose of MBTI scoring is twofold. First and foremost, scores sort the individual into one category or the other in each pair of preferences (e.g., Extraversion or Introversion, Thinking or Feeling), as described above. Second, they provide an estimate of how probable or likely it is that the preference indicated is accurate for the respondent and how likely it is that the respondent will choose the same preference again if retested. This estimate for each preference pair is called the *probability index*. Its categories are "very likely" (to be the same on test and retest), "likely," and "somewhat likely." Score ranges and additional information about the probability index can be found in chapter 7.

In contrast to scores obtained on most other personality instruments (see the section "The Meaning of Scores" later in this chapter), MBTI scores do not reflect amounts of whatever is being measured. Nor do they reveal a level of skill with which a respondent exercises a particular preference. An individual who obtains a high numerical score on Intuition (i.e., who, when forced to choose, chose many more responses favoring Intuition than favoring Sensing) does not have "more" or "better" intuition than does an individual who obtains a lower numerical score or a "somewhat likely" probability index on Intuition. In the Jung-Myers approach, it is incorrect to describe such a person as "very intuitive" in the sense of having more or better intuition. The person is appropriately described as being "highly likely to prefer Intuition to Sensing."

Unlike a numerical score on a trait instrument, a numerical MBTI result does not indicate an amount of whatever is being measured or a level of skill in using it.

The 16 Types

Measurement of preference on any one of the four pairs of opposites is designed to be independent of measurement on the other three pairs.¹ Preferences on the four pairs of opposites yield 16 possible combinations called *personality types*. Each type is denoted by four letters identifying which opposite in each preference pair is preferred (e.g., ESTJ, INFP).

According to the Jung-Myers theory, each of the 16 types results from interactions among the preferred opposites in each pair. For ENTP, these are Extraversion (E), Intuition (N), Thinking (T), and Perceiving (P). The theory proposes specific dynamic relationships among the preferences. For each type, one process (Sensing, Intuition, Thinking, or Feeling) is the dominant, or favorite, process and a second serves as an auxiliary, second-favorite, complementary process. Each type has its own pattern of dominant and auxiliary processes and the orientations (Extraversion or Introversion) in which they are typically used.

In the Jung-Myers theory, the auxiliary (second) process complements the dominant (favorite) process by providing access to a reasonably well-developed kind of perception (Sensing or Intuition) or judgment (Thinking or Feeling), whichever is *not* represented by the dominant process. In addition, the auxiliary process works primarily in the *less preferred orientation* of Extraversion or Introversion, whereas the dominant process typically works in the *preferred orientation* of Extraversion or Introversion. The steps used in identifying the specified dynamic relationships within a type are fully explained in chapter 3. The distinctive characteristics of the types, including their expected developmental pathways, follow from the dynamic interplay of all these processes and orientations. These are described in chapter 4.

Understanding dynamic relationships and how they emerge in each of the types is important for MBTI practitioners because such knowledge greatly aids the interpretation and verification process with respondents. More important, when individuals understand the interplay of energy in their and others' personalities, they can discover effective ways to apply these insights in different settings. Knowledge of type dynamics is equally vital for researchers. It suggests appropriate research designs and maximizes the likelihood of obtaining meaningful results and interpreting them appropriately.

Developing Preferences

Myers stated that preferences were "probably innate" (1962b, p. 72). While Jung did not use the term *innate* with regard to the origins of type, he stated that type preferences likely originate at a biological level (1921/1971, pp. 331, 332). Jung and Myers agreed that type was likely to be established very early in a person's life. Therefore, in terms of Jung-Myers theory, people can reasonably be expected to develop greater comfort and facility with the processes to which they are already naturally disposed and that they prefer to use, as well as with the orientations in which they prefer to use those processes.

1. A low to moderate correlation has consistently been found between S–N and J–P. Myers believed, as do many current type experts, that this relationship represents a real and natural association between the two pairs and is not simply attributable to measurement or sample error. Chapter 7 further explicates the rationale for this relationship.

For example, individuals who prefer Extraversion (E) are likely to be more confident and effective in dealing with the outer world of people, objects, and activities than with the inner world of thoughts, impressions, and ideas, which are in the realm of Introversion (I). Those who prefer Sensing perception (S) are likely to be more confident and effective in dealing with facts and realities than in dealing with theories and possibilities, which are in the sphere of Intuition (N). People who prefer the judging process of Thinking (T) are likely to rely more heavily on impartial Thinking judgments than on values-based Feeling (F) judgments. And those who prefer a Judging (J) orientation to a Perceiving (P) one in interacting with the world around them are likely to be more comfortable when they can organize the events of their life rather than experience and adapt to them. On the other hand, if people prefer Introversion, Intuition, Feeling, and Perceiving (INFP), then the converse of the descriptions above is likely to be true. The preferences and the specific dynamics of their interactions are discussed further in chapter 3.

DIFFERENCES BETWEEN THE MBTI® INSTRUMENTS AND OTHER PERSONALITY INSTRUMENTS

The MBTI Step I and Step II instruments differ from most other personality instruments in two ways. First, they are rooted in a personality theory, whereas most other instruments are not tied to a theoretical system. Second, they describe respondents by identifying which one of a pair of opposites (four pairs in all) they prefer to use. The majority of other personality instruments assume that individuals can best be described in terms of universal qualities that vary along a continuum.

Implications of Categorical Measurement

The requirement that the MBTI Step I and Step II instruments force respondents to choose between equally legitimate opposite orientations and processes has been an important factor in the design of items, the use of numerical information to determine type and facet preferences, the identification of general psychometric properties, and the interpretation of results. The following points specify important implications of the MBTI approach.

THE GOAL OF MEASUREMENT

The MBTI Step I instrument is designed to identify a respondent's preferred way of directing and receiving energy (Extraversion or Introversion), perceiving (Sensing or Intuition), judging (Thinking or Feeling), and approaching the outside world (Judging or Perceiving). Opposite preferences are regarded as neutral in relation to emotional health, intellectual functioning,

psychological health, and overall effectiveness. In contrast, approaches to personality that measure variation along a continuum reveal explicit or implicit assumptions about the emotional health, intellectual functioning, psychological health, and overall effectiveness of individuals who differ in their placement on the continuum.

THE NATURE OF CONSTRUCTS

Each of the pairs of opposites included in the MBTI Step I instrument can be experienced and expressed in several different but related ways. Each preference pair has several facets, or different aspects, so the MBTI constructs are referred to as *multifaceted*. In contrast, if only one characteristic of a preference pair were being measured, the construct would be called *unidimensional*. Unidimensional approaches measure a single attribute that is not broken down into identifiable parts. Many other personality instruments use a unidimensional approach.

THE MEANING OF SCORES

The fact that opposite type preferences are viewed as equally valuable and natural leads to quite different meanings of scores compared to those of other instruments. It is important to remember that each preference in each pair of opposites has its own legitimate qualities that are separate and distinct from the qualities of its opposite. For example, according to type theory, Introversion is neither viewed nor described as a lack of Extraversion, nor is Extraversion seen as a deficit of Introversion. An individual with a high probability index favoring Extraversion is best described as likely to show many of the attitudes and behaviors associated with Extraversion and fewer of those associated with Introversion. In contrast, an individual with a high score on the Extraversion scale of the *NEO Personality Inventory*® (NEO-PI®) instrument (McCrae & Costa, 2010), which assigns scores along a continuum, is seen as having more Extraversion than an individual with a low score on that scale, and an individual with a low score on the scale may be viewed as having a deficit of the identified (and implicitly more highly valued) quality, Extraversion.

SORTING, NOT MEASURING

Because the purpose of the MBTI instrument is to sort individuals into opposite categories rather than to measure an amount or degree of a quality, the probability index associated with preferences on the MBTI Step I instrument reflects the likelihood that a respondent has been accurately sorted into a particular type category.

VERIFICATION OF RESULTS BY THE RESPONDENT

One of the unique features of the MBTI instrument is the intent of its developers that it be useful to respondents. Although a respondent’s reported preferences as identified by the MBTI instrument are taken to be trustworthy indicators of underlying complex psychological processes,² verification by the respondent of the accuracy of the indicated type is required, since the individual is viewed as the expert best qualified to judge the accuracy of the type description that results from his or her self-report. (See, however, chapter 6 for factors that may influence the accuracy of self-report and chapter 7 for cautions about using verified type as a research criterion.) In contrast, the results of continuum-based personality instruments are generally taken at face value as estimates of the characteristics being measured. Respondents are not typically asked to verify the accuracy of their results based on their self-knowledge. For these kinds of instruments, the important outcome is the score. For the MBTI instrument, it is the meaning of the results to and for the respondent.

Most personality instruments focus on the score a respondent obtains on a particular scale. The MBTI assessment focuses on the meaning, impact, and importance of results for the respondent.

DYNAMIC RELATIONSHIPS AMONG PREFERENCES

The Jung-Myers theory describes specific, dynamic relationships among the four preferences (Extraversion or Introversion, Sensing or Intuition, Thinking or Feeling, Judging or Perceiving) that lead to the descriptions and characteristics of the 16 personality types. Personality instruments that assume continua do not generally assume theory-driven interactions among measures. Typical personality descriptions reflect a composite of measurements on a specified set of characteristics. Such composites are usually additive in nature, with no specific theoretical underpinning. In other words, for the MBTI instrument, the whole is greater than the sum of the parts, whereas for other instruments the whole is the sum of the parts. Chapters 9 and 10 review research demonstrating the greater explanatory power of dynamic analyses of research results.

THEORY OF PERSONALITY TYPE DEVELOPMENT

MBTI Step I type descriptions reflect a model of development that continues throughout a person’s life span. The types detailed later in this manual and in *Introduction to Myers-Briggs® Type* (Myers, 2015) describe the mature development typical of well-functioning adults. Observation of younger, immature respondents may reveal one-sided, exaggerated, or distorted expression of some of a type’s typical

Table 1.3 | Differences between the MBTI® Step I™ instrument and other personality instruments

MBTI® instrument	Other instruments
Assumes qualitatively distinct categories—individuals prefer one or the other category.	Assume quantitative differences—individuals vary only in the amount of a characteristic they have.
Scores are not assumed to be normally distributed.	Scores are expected to be normally distributed; most scores are in the middle.
Scores are estimates of confidence in the accuracy of the sorting procedure—i.e., placement into the category indicated by assessment results.	Scores are variables that show how much of a characteristic an individual has.
Focus is on individuals scoring near the midpoint, where accuracy of sorting may be in doubt.	Interpretive interest is in individuals at the extremes of the distribution.
Assumes that behavior is an expression of underlying type preferences.	Assume that behavior is caused by the additive effect of the underlying personality characteristics.
Assumes that the four type preferences interact dynamically to form a whole that is more than the sum of its parts.	Assume that the qualities being measured are largely independent of one another.
Both opposites are used in scale names (e.g., Extraversion–Introversion; Sensing–Intuition); each has equal value.	A single word or phrase typically identifies each scale , implying either greater or lesser value for the named quality.
The numerical portion of MBTI results has no diagnostic meaning.	Very high and/or very low scores can be negative or indicative of a flaw or abnormality.

Source: Modified and adapted from Quenk (2009a). Reprinted with permission.

2. Myers described the simple-seeming items in the MBTI instrument as “straws in the wind” that point to underlying complexity (Myers & McCaulley, 1985).

characteristics. Such distortions are expected to diminish with age and life experience. Chapter 6 includes recommendations for using type with young people. In contrast to the type development approach inherent in the Jung-Myers approach, most other personality instruments do not assume that there will be natural and expected changes in expression of the characteristics they are measuring (perhaps with the exception of type expression by adolescents) with increasing age and maturity of respondents.

Table 1.3 summarizes other ways the MBTI Step I instrument differs in approach from typical personality instruments.

INDIVIDUAL DIFFERENCES WITHIN TYPE PREFERENCES

As described earlier, the MBTI preferences are multifaceted; that is, individuals use facets, or different aspects, of each of the four pairs of opposites. For example, the Jung-Myers approach describes several different ways of expressing or exhibiting an extraverted or introverted orientation. These include activity level, comfort expressing thoughts and feelings, kind and amount of stimulation desired, and so on. Identifying such facets of the preferences is the route to individualizing type descriptions, one of Myers' original

goals that she did not fully bring to fruition. The Step II assessment identifies five such facets within each of the four pairs of opposites, for a total of 20 facets. Brief descriptions of these facets appear in the next chapter, and more lengthy descriptions appear in chapter 4. Knowing which pole of each facet an individual prefers, in conjunction with knowing his or her dynamic four-letter type, vastly expands the precision with which that person's individuality can be described. Together, the Global Step I and Global Step II assessments enable understanding of the distinctiveness of personality. An additional advantage of this approach is that it is parsimonious—it uses very few basic elements to explain complex personality processes. All the necessary elements come from the same theoretical system—the Jung-Myers theory.

CONCLUSION

This chapter provided the context for understanding and using the MBTI Global Step I and Step II assessments. It briefly presented Jung's original theory of personality type and its expansion into the Jung-Myers theory. The chapter serves as an introduction to the two chapters that follow, which describe the Step II assessment and greatly expand on the Jung-Myers theory. As an aid to understanding terms that may be unfamiliar to readers, a glossary of terms is provided near the end of this manual.

CHAPTER 2

A Closer Look at the Global Step II™ Assessment

CHAPTER OVERVIEW

The Multifaceted Nature of the MBTI® Preference Pairs

Related Developments of the MBTI Step I™ and Step II™ Assessments

Comparison of the Global Step I™ and Step II™ Assessments

The Dynamic Nature of the Step II™ Assessment

Contribution of the Step I™/Step II™ Approach to the Field of Personality

Conclusion

The Global Step II™ assessment is the result of many years of research and observation that began with Isabel Myers' development of the MBTI® personality inventory in the 1940s (see Myers & McCaulley, 1985; Myers, McCaulley, Quenk, & Hammer, 1998). Various Step I™ forms (e.g., Form M and the earlier Form G) yielded type descriptions that distinguished the 16 types from one another and provided some suggestions about ways inadequate type development might manifest in each of the types (Myers, I. B., with Kirby & Myers, 1998). However, they offered few clues about individual differences among mature, well-functioning members of a type.

Myers always included in the MBTI forms she developed (Forms A through G) a number of items that were not scored for four-letter type. Her intent was to use those items to explore individuality within each of the 16 types at a later time. Myers was in the process of further developing this aspect of her work at the time of her death in 1980. Beginning in the mid-1980s, there were several efforts to individualize type reports. These are discussed in the section "Early Efforts to Individualize Type Reports" later in this chapter.

THE MULTIFACETED NATURE OF THE MBTI® PREFERENCE PAIRS

As noted in chapter 1, each of the four pairs of opposites identified in the MBTI assessment includes different aspects, or facets. Chapter 1 listed several possible facets of the Extraversion–Introversion orientations, such as activity level, expressiveness, and the nature of stimulation desired, to illustrate the multifaceted nature of the MBTI instrument. An assessment that includes different ways an individual uses or expresses parts of a process or orientation provides a way to examine how that individual uses each part, in addition to how he or she uses the overall process or orientation. For example, two individuals may be identified by their MBTI Step I results as preferring Extraversion, but one's extraverted orientation may be primarily focused on expressiveness and activity level, while the other's may be primarily focused on the kind of stimulation desired and gregariousness. In contrast, an instrument with a unidimensional approach to Extraversion–Introversion might be concerned exclusively with only one aspect of this orientation, perhaps sociability. The items on such an instrument would ask questions only about attitudes and behaviors that reflect sociability. An individual's responses to those items would determine his or her status regarding the Extraversion–Introversion orientation of energy. Because all items reflect one aspect exclusively, there are no component parts, so there is no opportunity to individualize respondent results within Extraversion and Introversion.

Based on their observations of people, both Jung and Myers took a multifaceted approach to describing type preferences.¹ Jung was particularly interested in the several ways Extraverts and Introverts live their lives. Much of *Psychological Types* (Jung, 1921/1971) is devoted to examples of Extraverts and Introverts among writers, poets, philosophers, and artists. Jung also noted differences between Sensing² and Intuitive types and between Thinking and Feeling types, although he provided fewer lengthy examples of these differences. Myers and Briggs noticed the same differing expressions of Extraversion and Introversion noted by Jung and made keen observations regarding the additional ways the other three pairs of opposites were expressed. They observed, for example, that Sensing types tended to focus on concrete data, favor practical applications, prefer a tried-and-true approach to tasks, and so on. They found the same kind of diversity of expression in the other pairs. It was therefore necessary for Myers to write items that encompassed the different facets that she and Jung had identified as aspects of each pair of opposites so as to reflect the richness and complexity of each pair.

Myers' goal to represent the diversity of expression of preferences was apparent not only in her careful inclusion of items that reflected the components of each pair of opposites but also in the scales she devised to capture consistent use of specific aspects, or facets, within each pair of opposites. In observing individuals whose type preferences she knew, she noticed that some Introverts seemed to habitually use one or another aspect of Extraversion; that some Feeling types approached things logically some of the time, despite clear indications of their underlying Feeling preference; and that some individuals with a J orientation on the J–P pair, who liked being organized and scheduled, characteristically waited until the last minute to get things done. Myers made similar observations about individuals with the other preferences. She had a hunch that what she had observed was not simply a case of using an opposite preference when circumstances required it. Rather, these individuals seemed to use aspects of an opposite preference consistently. Most important, when Myers examined their answer sheets, she found that they had consistently answered items in a manner congruent with the "out-of-preference" behavior she had observed. Her multifaceted approach to writing items representing the various facets of an orientation or process enabled Myers to capture this kind of diversity in a systematic way in the scales she designed.³ This was the route to individualizing type reports.

In her early efforts, Myers anticipated creating individualized type reports, so she included items and developed scales designed to capture such variations of expression.

RELATED DEVELOPMENTS OF THE MBTI® STEP I™ AND STEP II™ ASSESSMENTS

Myers wrote and tested many items over the years. The items used to indicate four-letter type were included on standard (Step I) forms of the assessment, such as Form F, which was later replaced by Form G. Both Form F and Form G also contained some of Myers' research items—items that either did not meet the criteria for indicating type preference or were included for research purposes, such as for the development of her early version of what is now the Global Step II assessment and, later, for the development of the MBTI Step III™ assessment.

Early Efforts to Individualize Type Reports

Building on Myers' multifaceted approach, David Saunders, a psychologist who had worked closely with

1. Jung did not use the term *preference*. Myers introduced the term to describe the opposites within each pair of orientations/processes.

2. Jung's term was *sensation*. Myers substituted the term *Sensing* to convey its "active" nature.

3. These scales were included in an early version of Myers' type development assessment and are included in the MBTI Step III assessment.

Myers during the MBTI assessment's development, factor-analyzed all 166 Form F items and subsequently all 290 items on Form J, the MBTI research form.⁴ Factor analysis is a technique that statistically groups items together based on the similarity of their content.⁵ Twenty of the 27 factors identified by Saunders represented facets of the four pairs of opposites—5 facets for each pair. (See Saunders, 1987, for a discussion of these 20 factors and the 7 additional factors Saunders identified.) The 131-item Form K was composed of the subset of items on Form J that identified these 20 factors.⁶ The MBTI® Expanded Analysis Report (EAR; Saunders, 1989) provided respondent results for these 20 factors. Interpretation of the MBTI EAR was aided by Kummerow and Quenk's *Interpretive Guide for the MBTI® Expanded Analysis Report* (1992a) and its workbook (1992b). The EAR was later renamed the MBTI® Step II™ Profile (Myers & Briggs, 1996), with a three-page profile replacing the earlier one-page profile. At the same time, a 24-page individualized report of results was made available as the MBTI® Step II™ Expanded Interpretive Report: Form K (Quenk & Kummerow, 1996), the first individualized Step II report. Form Q, the Step II form that replaced Form K, identified the same five facets of each pair of opposites that previous forms had. However, Form Q generated a much more individualized report with greatly improved content and graphics: the MBTI® Step II™ Interpretive Report: Form Q (Quenk & Kummerow, 2001).

Differences Between Form M and Form Q and Previous Forms

Form M, the Step I form that replaced Form G, was the first form to include items that were not written by Myers (see chapter 7 for details). Nearly half the items on Form M were modifications of items that had been written earlier by Myers but never used or that were new to the MBTI assessment. Form M was also the first form that did not contain any research-only items. Form Q, the Step II form that replaced Form K, incorporated all the Form M items, so that four-letter type could be determined. In addition, several Form G items were dropped from the Step II form during the revision to Form Q. Statistical analysis enabled placement of the Form M items on the appropriate facets.

Form M (Step I) was the first MBTI form to use a US national representative sample for scale development. A second US national sample was compiled for the development of Form Q (Step II). Form M and Form Q used item response theory (IRT) for both selection and scoring of responses. Form G, like all previous MBTI

Step I forms, had used Myers' prediction ratio method for selection and scoring of items. Form Q retained most of the names and essential meanings of the Form K Step II facets, although facet items, item weights, and descriptions of facets were modified somewhat based on analyses of the new national sample data. The European Step I and European Step II forms were developed using UK and European samples.

The Global Step II™ Assessment

The Global Step II assessment retains the names and basic descriptions of the 20 facets included in prior forms (EAR, Step II) of the assessment. The facets are briefly described in tables 2.1–2.4.

COMPARISON OF THE GLOBAL STEP I™ AND STEP II™ ASSESSMENTS

There are a number of important differences between these two instruments that especially affect interpretation and verification of results. Being aware of these differences will help practitioners use the global instruments more effectively.

Scale Length and Items Used

The Global Step II instrument contains 143 items, 139 of which are used to score the 20 facets. The additional 4 items are needed to obtain Global Step I scores. The 20 Global Step II scales vary in length from 5 to 11 items. The Global Step I instrument requires 92 items to elicit an individual's preferences and the probable accuracy of those preferences on the four preference scales. The Step I scales vary in length from 21 to 24 items. The Global Step II instrument includes all the Global Step I items. The additional items in the Global Step II instrument, in combination with the Global Step I items, permit scoring of the facet scales.

Scoring Methods

The Global Step I instrument uses latent class analysis (LCA), described in chapter 1 and in greater detail in chapter 7, to score the items that generate results for the four pairs of opposites. The Global Step II instrument uses item response theory (IRT) to score the items that generate results for the 20 facets. See chapter 7 for information about using IRT for selection and scoring of Step II items.

4. Form J included all the items ever written by Myers, as well as 10 to 12 items contributed by Saunders.

5. Myers did not use factor analysis to identify groups of items. She grouped items based on her keen understanding of the items as well as her observation of individuals known to her or for whom she had independent measures of behavior.

6. Saunders' original factors were described in *Type Differentiation Indicator Manual* (1987) and *Manual: MBTI® Expanded Analysis Report* (1989). Relevant details can be found in those sources.

Graphic Array of Results

Step I preferences are traditionally arrayed so that scores to the left of the midpoint are associated with E, S, T, and J preferences. Thus, the graph of Step I results shows E, S, T, and J preferences to the left of the midpoint and I, N, F, and P preferences to the right of the midpoint. In the Global Step I instrument, the probability index replaces the Form M preference clarity index (pci). The probability index is an estimate of the probability or likelihood that the preference reported is accurate for the respondent. A midpoint separates the preferences in each pair of opposites. Figure 2.1 shows a sample Global Step I profile for an individual with INTP preferences.

Step II facets are similarly arrayed, with scores of 1–5 to the left of a 0 center point associated with E, S, T, and J facet poles and scores of 1–5 to the right of 0 associated with I, N, F, and P facet poles, thus yielding an 11-point range of scores. Although a score of 0 separates the two poles of a facet, for interpretive purposes the dividing area is termed a *midzone* that encompasses scores of 1 to the left of 0, 0, and 1 to the right of 0. The rationale for and meaning of this midzone are explained in chapter 6. Figure 2.2 shows the Global Step II profile for the individual whose Global Step I profile appears in figure 2.1.

Goal of Scoring

Step I results identify a respondent as favoring one preference in each pair of opposites: E or I, S or N, T or F, and J or P. The four preferences together permit identification of the individual's whole type and its associated type dynamics. Step II scoring identifies a respondent's distinctive expressions of his or her Step I type by identifying variations within each of the four Step I preference pairs. An individual's Step II facet scores within each preference pair may be on the same side as

the Step I preference (an in-preference result), on the side opposite the Step I preference (an out-of-preference result), or in the midzone (0 or 1).

A respondent whose Step I results indicate a preference for Extraversion could have any of several different facet patterns—for example, scores toward all five E–I facet poles associated with E, three scores on the E side and two scores on the I side, four scores on the E side and one score on the I side, or three scores on the E side and two scores at 0 or 1. Thus, one individual's Step II profile may show scores across the whole range of 11 data points, another individual may have scores that are concentrated on the side that represents his or her Step I preference, and another may have scores concentrated in the midzone.

Combining Step II™ and Step I™ Information

Step II results provide a window into the way individuals express and use each of their Step I preferences. Observing and interpreting a respondent's particular array of Step II scores in conjunction with his or her Step I results enables exploration of the complexity of that individual's personality by first focusing on variations in facet results for each preference pair individually, and then, more broadly, considering the impact of such variations on the individual's whole type. For example, a respondent who prefers I, N, F, and P who has an out-of-preference result on the E–I facet Expressive–Contained (i.e., who scores toward the Expressive pole) would not be expected to manifest this in the same way as would an ISTJ with the same out-of-preference result. The interpretation of Step II results in the context of Step I results thus adds to the richness of individualized information available. Examples of the clarifying effects of using facet information in combination with Step I preference results are presented in chapter 6.

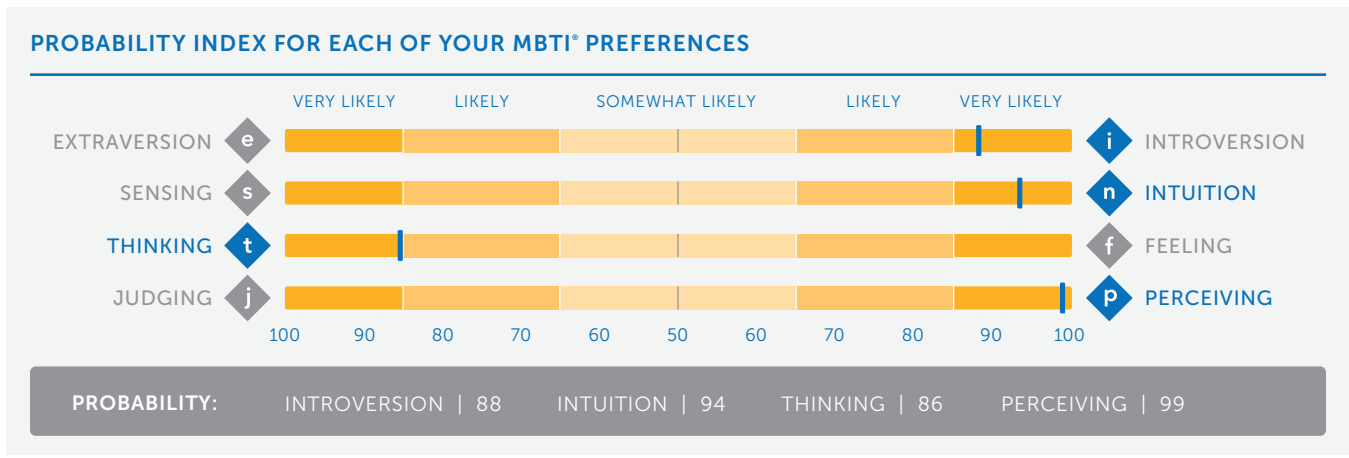


Figure 2.1 | Sample Step I™ profile

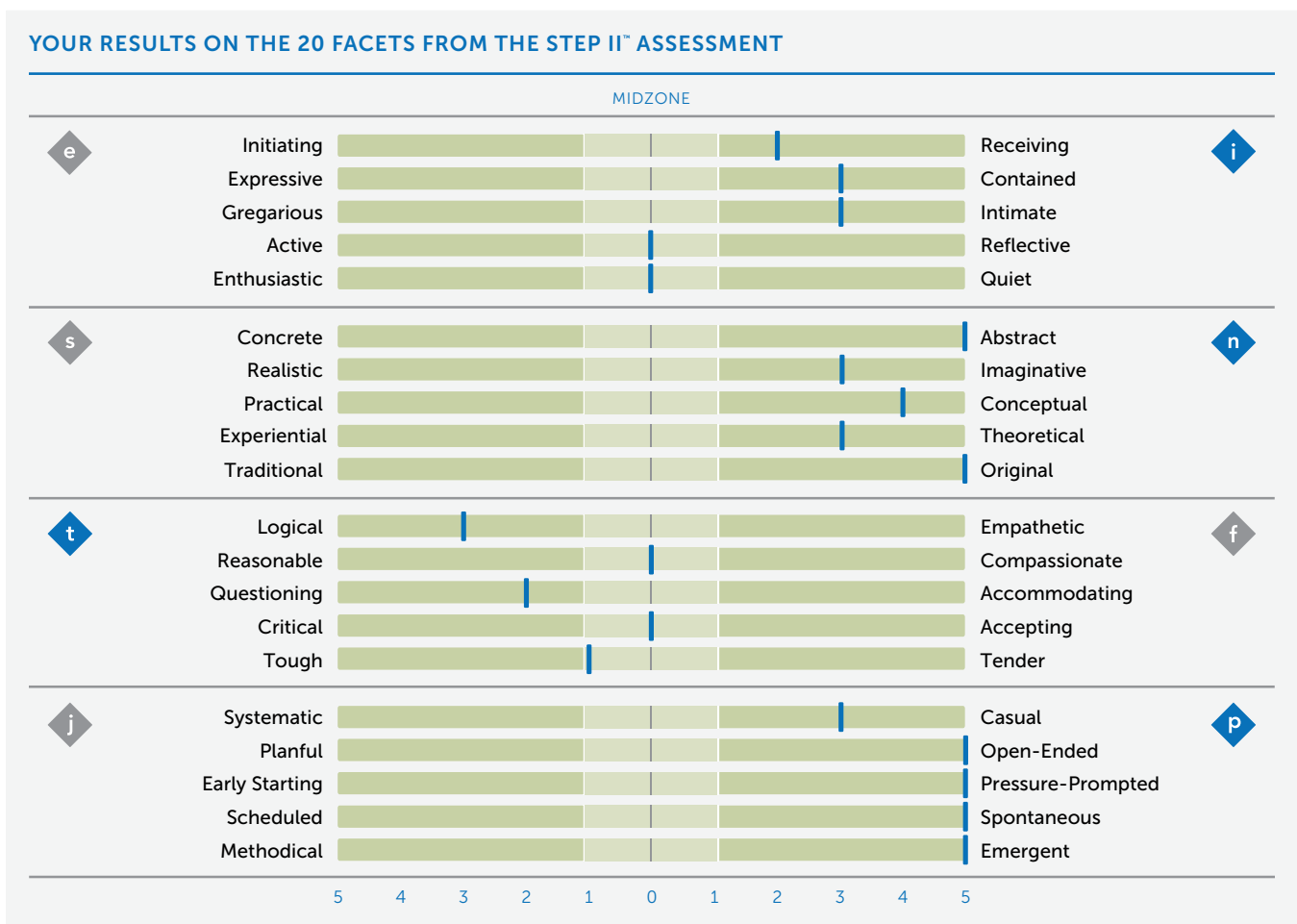


Figure 2.2 | Sample Step II™ profile

Table 2.1 | **Facets of Extraversion and Introversion: opposite orientations of energy**

 Extraversion Directing energy mainly toward the outer world of people, objects, and activities	 Introversion Directing energy mainly toward the inner world of thoughts, impressions, and ideas
Initiating Act as social facilitators; are good at mingling, small talk, meeting new people, conversing at length with almost anyone, networking, and making connections with others.	Receiving Are more at ease being introduced than initiating introductions; in large groups, are more comfortable letting conversations come to them than mixing, mingling, and initiating talk with people they don't know well.
Expressive Readily show their feelings and make their interests obvious; express typically positive feelings; cheerful, warm, and humorous; usually find it easy to confide in others and describe themselves as easy to know; may be puzzled when others do not reciprocate by sharing personal information.	Contained Share thoughts, feelings, interests, and concerns only with people they trust; describe themselves as difficult to know; their primary emotional responses are experienced internally; need time to work through a problem on their own before they are able to discuss it with others; find it difficult to confide in others because the more upset they become, the less they allow it to show.
Gregarious Enjoy maintaining friendships with a diverse group of people with varied interests; are skilled at communicating with different kinds of people in a variety of settings; maintain a wide and diverse circle of friends and acquaintances; enjoy meeting new people.	Intimate Maintain a limited circle of close, trusted friends; tend to have a similarly concentrated range of interests and favorite activities; in group settings, prefer to talk one-on-one with others they know well; may be uncomfortable making chitchat with people they have just met.
Active Favor direct involvement in active environments; learn best by doing, listening, and questioning; generally prefer active rather than passive entertainment; are comfortable meeting strangers.	Reflective Obtain personal meaning through visual, intellectual, or mental engagement rather than through physical or verbal exchange with the environment; learn best in quiet settings permitting extended mental reflection.
Enthusiastic Tend to be talkative, hearty, and lively, enjoying the dynamic flow of energy in conversation for its own sake; like being the center of attention and are adept at making group conversation into an art form; like hearing and telling stories as a way of sharing who they are and learning about others.	Quiet Appear reserved and quiet but often have an abundance of rich and varied internal responses to what is going on around them; can have difficulty finding words to describe this rich interior world; may use understatement as a means of emphasis; may be drained by social interactions.

Sources: Adapted from Myers et al. (1998); Mitchell, Quenk, & Kummerow (1997). Used with permission.

Table 2.2 | Facets of Sensing and Intuition: opposite perceiving processes

s Sensing Focusing mainly on what can be perceived by the five senses	n Intuition Focusing mainly on perceiving patterns and interrelationships
Concrete Prefer to work with and talk about things that are concrete, tangible, literal, and verifiable by the senses—this applies to perceptions, communication, learning styles, view of the world, preferences for entertainment, and values; cautious about going beyond the facts.	Abstract Prefer to focus on concepts and on the abstract meanings of ideas and the relationships among them; are comfortable with nonliteral interpretation.
Realistic Prefer things to have useful and tangible benefits to themselves and others; value common sense because it is practical, realistic, and, being field-tested by experience, has been shown to work; place a high value on personal comfort and family security.	Imaginative Value possibilities over tangibles; find what might be more interesting than what is; are resourceful in dealing with new and unusual experiences; enjoy finding new solutions to any problem.
Practical Prefer working with known materials and objects using familiar and practical methods; are interested in physical comfort that can be experienced in the here and now; prefer building to innovating and are more interested in modest tangible gains than in risky opportunities for much larger gain.	Conceptual Are comfortable with complexity and more intrigued by implied or inferred meanings than by meanings explicitly given, stated, or presented; value ideas and possibilities over the tangible details of what is immediately visible; tend to score well on tests; are more willing to take risks for large potential gain.
Experiential Use experience as the primary criterion for truth and relevance; see no gain in changing methods already proven experientially; for new tasks lacking established procedures, prefer a trial-and-error approach starting with what is already known to be the best approach.	Theoretical Believe that the relevance of something goes beyond its relation to tangible things to its potential relation to other concepts and ideas; are less interested in what is already discovered or well known; prefer to explore the world by seeking new connections among abstract concepts and ideas; are resourceful and insightful in approaching new tasks.
Traditional Prefer traditions and conventions because they provide continuity, security, and validation by the social environment; are uncomfortable with fads and practices running against traditional norms because they lack the validating tests of time, experience, and societal approval.	Original Value originality as a means of self-expression and a way of infusing meaning into day-to-day activities; may be clever, inventive, adventurous, and enterprising; take the initiative to expound their original ideas; find that variety conveys meaning and progress, while sameness detracts from meaning.

Sources: Adapted from Myers et al. (1998); Mitchell, Quenk, & Kummerow (1997). Used with permission.

Table 2.3 | Facets of Thinking and Feeling: opposite judging processes

 Thinking Basing conclusions on logical analysis with a goal of accuracy and impartial truth	 Feeling Basing conclusions on values-centered criteria with a goal of personal and social harmony
Logical Believe that the world and the people in it are best understood and related to through logic and reason; highly value rights, fairness, and reasonableness as standards for relationships; greatly value consistency in all areas of life.	Empathetic Believe relationships are the source of meaning and truth in life; that truth is personal as well as universal and is never detached from people; and that feelings, values, and relationships are more important than rights.
Reasonable Are demonstrably logical in everyday decision making; make effective use of sequential reasoning; are confident and clear thinking.	Compassionate See laws, rules, and procedures as guidelines for relationships rather than their essence; take into consideration the unique and personal individual needs of others; value mercy over fairness.
Questioning Believe that challenging and questioning are justified in order to understand, to solve problems, to find a common ground with other people, or to expose the flaws in an opposing viewpoint.	Accommodating Value harmony highly; believe that challenge and confrontation are less effective than including diverse viewpoints into compromises that will satisfy as many people as possible.
Critical See criticism as an effective and valid way of getting at the truth and assume others will accept it impersonally; believe that criticism is needed to improve ideas, situations, or procedures, and avoid bad outcomes.	Accepting Believe that validating others and their viewpoints is more important than any objective truth; are likely to overlook others' mistakes or deficiencies and hope to be treated the same way; view "win-win" resolutions of problems as possible and worth striving for.
Tough After a considered appraisal of situations, possibilities, and people involved, stand firm in their decisions; believe that to do otherwise would be to reject all of the logic, reason, questioning, and critique that went into the decision that was made; distrust decisions based solely on personal factors.	Tender Use affection and gentleness to achieve their objectives; are aware that there are two mutually contradictory sides to many issues, so purely rational resolutions can't be made.

Sources: Adapted from Myers et al., 1998; Mitchell, Quenk, & Kummerow, 1997. Used with permission.

Table 2.4 | Facets of Judging and Perceiving: opposite orientations to the outside world

 Judging Preferring the decisiveness and closure that result from dealing with the outside world using one of the judging processes (T or F)	 Perceiving Preferring the flexibility and spontaneity that result from dealing with the outside world using one of the perceiving processes (S or N)
Systematic Prefer orderliness in many areas of their life, ranging from the physical environment of work and home, to the planning and scheduling of tasks and leisure activities, to the systematic methods with which they approach all these things.	Casual Have a high tolerance for surprise; are comfortable taking things as they come and adjusting their activities as necessary; are typically described as leisurely; do not necessarily value closure in decisions, preferring to keep their options open for as long as possible.
Planful Prefer to plan and schedule leisure time and activities well in advance; value such planning because it ensures that they will not miss the things that are worth planning for.	Open-Ended Prefer to make decisions on the spur of the moment; value freedom, live in the here and now, and like to “go with the flow”; dislike being tied down by long-range plans; want plans to be flexible.
Early Starting Like to allow plenty of time so that an activity can be carried out efficiently, without fuss and bother; accomplish the most by doing one thing at a time and seek to structure their life in accordance with this rule; may become inefficient under the stress of working until the last minute on a deadline.	Pressure-Prompted Work best, most creatively, and with the most energy under the stress of an imminent deadline; their seeming procrastination serves as a gestational period during which they are generating the ideas and inspiration to do their best work.
Scheduled Are comfortable with routine; regard tried methods and tested routines as the only reliable, efficient way to get things done; find that routine provides an important degree of comfort and security.	Spontaneous Work best and with greatest energy when they have constant variety or the freedom to decide what tasks they will do when; highly value newness and variety; consider the worst fate to be ending up in a rut at work or at home where the routine never changes.
Methodical Strongly prefer to organize the materials, tools, people, and sequence of steps that are required to complete complex projects; are good at following sequenced directions and may be uncomfortable if there is no way to tell what needs to be done when.	Emergent Take an informal, ad hoc approach to solving problems and accomplishing tasks and projects; plunge ahead without detailed plans; may not read directions before starting the assembly of kits; are good at unstructured projects requiring improvisation.

Sources: Adapted from Myers et al. (1998); Mitchell, Quenk, & Kummerow (1997). Used with permission.

THE DYNAMIC NATURE OF THE STEP II™ ASSESSMENT

Recall that the Jung-Myers theory specifies that every person's typological makeup includes the ability to use *each* process and orientation at least some of the time. Particular patterns of facet scores within a pair of opposites *may* reflect the ways less preferred aspects of personality are expressed and can provide insight into an individual's distinctive ways of expressing type. The particular facets involved may be evidence of the expression of less preferred aspects of the Step I preference and will vary with each person, perhaps reflecting individual accommodations to life experiences and environmental pressures. For example, habitual use of one or another aspect of Extraversion–Introversion may be reflected in a particular facet score that occurs at or near the pole opposite the Step I preference (an out-of-preference result) or in the midzone. A variety of score arrays may occur for the other three preference pairs as well.

Hypothetically, the more narrow or context-specific the behavior involved, the more likely it is that out-of-preference and midzone results will be observed. This is especially true if an individual must operate in a context that requires or rewards particular behavior associated with the opposite underlying preference. For example, some individuals who report a preference for Perceiving (P) score as Early Starting (which is on the J side) rather than as Pressure–Prompted (which is on the P side) on the Early Starting–Pressure–Prompted facet. This facet focuses narrowly on a person's approach to meeting deadlines. The demands of the workplace require and reward employees who begin tasks far enough ahead of the deadline that they do not subject themselves and others to last-minute time pressures. So, some people who prefer Perceiving realize that they need to start on projects early, even though that is not their natural inclination. After doing this for many years, starting early may come to feel natural to them. Examples of different arrays of facet scores, how they are experienced by individuals, and the way individuals explain their role in the expression of their type are found in chapter 6.

People can develop a less preferred pole of a facet as a skill when life circumstances or pressures at work or at home require it.

Fluid Nature of the Facets

Just as each of the 16 whole types is more than the sum of its four preferences, each pair of opposite orientations (E–I or J–P) and processes (S–N or T–F) is more than the sum of its five facets. There are two reasons for this. First, there may be additional facets

within each preference pair that have not yet been identified because the number of identifiable facets is limited by the pool of items available. It is possible that a broader or different pool of items might yield additional facets of one or more of the preference pairs. Second, the number of possible interactions among the 20 facets and between the facets and their broader preference pairs is so great and implies such complexity that it is beyond complete understanding. Even if the accuracy of an individual's four-letter type and his or her facet results could be confidently assumed, the totality of that individual's typological makeup would not be fully grasped. Nevertheless, Step II facet information offers considerably increased insight and a better understanding of a person's individuality within his or her type.

CONTRIBUTION OF THE STEP I™/STEP II™ APPROACH TO THE FIELD OF PERSONALITY

Two contrasting approaches to the study of personality have long been an important focus of attention. Although both seek to measure and predict behavior, their foci are quite different. One approach seeks to discover general laws or principles of personality that can be used to predict behaviors based on knowledge of specific variables, typically a set of personality traits. This approach is known as the *nomothetic* (from the Greek *nomos*, meaning law or custom) approach, wherein interest is in laws that apply to everyone, people's common characteristics. The contrasting approach, termed the *idiographic* (from the Greek *idios*, meaning individual or private) approach, focuses on discovering and measuring individual differences, what is unique in people. The goal of this approach is also to predict behavior.

There are advantages and disadvantages to both approaches. The nomothetic approach has been successful in identifying general characteristics of personality that apply to a wide variety of people but has been less helpful in predicting the behavior of a particular individual operating within differing situations. The "Big Five" personality system (McCrae & Costa, 2010), which identifies five traits believed to be universal descriptors, is an example of the nomothetic approach. The idiographic approach provides a rich description of individual differences but does not contribute much to the development of personality measures that may help in comparing different people. Case studies and in-depth interviews are examples of the idiographic approach.

The Jung-Myers theory and the MBTI Step I instrument take an approach to personality that differs from both approaches described above. They use types rather than traits as their units of measurement, so their description of personality begins at a much more specific level than does that of trait instruments. Further, because the

Step I instrument describes types that are defined by particular relationships among its four sets of opposite personality characteristics, it can add details about a person's behavior that are not provided by scores on the four separate pairs of opposites only (scores are generally used in trait descriptions). Thus, with its identification of types at the Step I level of scoring, the MBTI instrument has already taken a step toward providing an idiographic description of personality while remaining anchored to a foundation of generalized, nomothetic characteristics (i.e., the E–I, S–N, T–F, and J–P preference pairs).

When the Step II facets are added to a Step I description, a still more specific level is reached and individual variations within the types can be discerned. The number of variations increases enormously. Although some logically possible Step II patterns rarely turn up, the more common ones allow generalizations about the structure of personality while describing meaningful individual differences. The discussion in chapter 6 of out-of-preference results is of interest in this regard. Thus, as a second-level scoring of the MBTI Step I instrument, Step II information brings personality theorists closer

to reaching the goal of developing individualized (idiographic) descriptions of personality combined with a solid foundation of general (nomothetic) principles.

CONCLUSION

This chapter provided the context for the development of the Step II approach, which permits expansion of the 16 type descriptions. It offered a theoretical, psychometric, and historical rationale for the Step II assessment by focusing on the multifaceted nature of type. The development of the Global Step I and Step II assessments and their similarities and differences were detailed, as was the dynamic nature of the Step II approach to personality description. Finally, the contribution of the combined Step I and Step II assessments to the important distinction between nomothetic and idiographic approaches to personality was explained. This chapter, in conjunction with chapter 1, which focused primarily on personality type in general, has laid the groundwork for the detailed theory chapter that follows.

CHAPTER 3

The Theory Behind the MBTI® Assessment

CHAPTER OVERVIEW

Origins and Key Features of Type Theory

The Fundamentals of Type

The Theory of Type Development

Conclusion

Every individual is unique. Each is a product of heredity and environment and, as a result, is different from everyone else. Yet we often assume that other people's minds work the same way ours does, even when it's clear that the people with whom we interact do not reason as we reason, do not value the things we value, and are not interested in what interests us. While that assumption is probably the easiest and most familiar way for us to understand and try to connect to others, it can promote misunderstanding of their motives and behaviors.

The value of Jung's theory of psychological types (Jung, 1921/1971), which underlies the MBTI® assessment, is that it enables people to expect specific differences in specific individuals and to deal with others and their differences in an effective and constructive way. Briefly, the theory is that much seemingly random, unsystematic variation in human behavior is in fact not due to chance—it is the logical result of a few basic, observable differences in how people use their minds.

Over time, Jung expanded the descriptive elements of his system and their interactions with one another to identify eight personality types. Myers and Briggs clarified the rationale underlying the interactions hypothesized by Jung and expanded the system to encompass the 16 types described in the Jung-Myers approach and the MBTI assessment. Myers also wrote neutral, easily understandable type descriptions, which enabled people to recognize and understand their own and others' types.

This chapter covers a wide range of theoretical and illustrative material. To avoid disrupting the flow of information, some theoretical and historical details are provided in footnotes rather than in the text itself. For comprehensive understanding, readers are encouraged to read these footnotes.

A reminder about terminology: As mentioned in chapter 1, *type theory* as used in this manual refers to C. G. Jung's theory as described in his writings. The *Jung-Myers theory* that is the basis for this manual refers to Jung's theory as interpreted and extended by Isabel Myers and Katharine Briggs.

ORIGINS AND KEY FEATURES OF TYPE THEORY

Jung developed his type theory over a period of many years. He observed variations among people across contexts and cultures that were not explained by gender, personal history, intelligence, culture, psychological health, or any other obvious characteristics. He therefore attributed these variations to universal, natural personality differences.

Opposites in Jungian Psychology

Psychological type is only one aspect of Jung's wide-ranging view of human functioning. The 20 volumes of Jung's work cover areas such as universal archetypes, the psychology of religion, mental illness, and the origins and meaning of symbols, among many others. *Psychological Types* is volume 6 of Jung's *Collected Works* (1921/1971).

All of Jungian psychology rests on the assumption that life processes develop and are maintained through the dynamic flow of energy between opposites: dark and light, inner and outer, conscious and unconscious, male and female, good and evil, and so on. The opposites in Jung's theory are "bipolar," having two opposite poles. In Jung's system, when one member of a pair of opposites is in use, the other cannot also be in use. An individual cannot be both conscious and unconscious at the same moment or focus on specific details (using Sensing) at the same moment that he or she is considering the long-range implications of those details (using Intuition).

Jung conceived of every aspect of personality as bipolar, each member of a pair being qualitatively different in every way from its opposite. It is important to understand that the opposites in any pair share a fixed amount of energy. Some of that energy is under an individual's conscious control, meaning that the individual can direct the energy and use it in a purposeful way. The remainder of the shared energy is not under the individual's conscious control and so cannot be directed and used to accomplish something. This energy is described as unconscious. The Jung-Myers type theory rests firmly within Jung's general theory of opposites and is the part of his wide-ranging theory that is most practical and accessible in daily life.

The amount of energy shared by a pair of opposites that is conscious or unconscious at any given time is constantly shifting. Sometimes one member of the pair is relatively conscious and its opposite is relatively unconscious, and vice versa. It is the shifting of energy between relatively conscious and relatively unconscious opposites that keeps an individual's psychological system active and developing.

We might think that for effective functioning the finite energy available to opposites should be shared equally. However, in order to control and direct energy, one member of the pair has to have considerably more conscious energy than the other. Jung was clear that when individuals devote equal energy to both opposites in a pair, neither opposite has enough energy to rise to a level of consciousness at which it can be controlled and used purposefully. If that occurs, Jung stated, both opposites exist in an unconscious, primitive state because neither one is "differentiated" from the other—that is, they are fused together and not separated into two distinct parts (1921/1971, pp. 514, 515).

Would it not then be best to have one member of a pair totally under conscious control at all times and the other relegated to the unconscious realm? This is no more a recipe for effectively living one's life than is devoting equal energy to two opposites. Neither of these options is recommended in the Jung-Myers system.

Jung's Initial Focus on Extraversion and Introversion

Jung first observed that there are two opposite types of people, Extraverts and Introverts. He described Extraverts as individuals whose energies are primarily directed outward toward people and events in their outer environment and Introverts as those whose energies are primarily directed inward toward thoughts and experiences in their inner environment. Jung's identification of these two types led eventually to the detailed and broad-ranging discussions of Extraversion and Introversion that are found in his book *Psychological Types* (1921/1971). Jung saw Extraversion and Introversion as two different attitudes, or orientations, "a readiness of the psyche to act or react in a certain way, an a priori orientation to a definite thing" (p. 414).¹

About 10 years after his initial studies of Extraversion² and Introversion, Jung recognized that Extraversion and Introversion alone did not provide a complete picture of what he observed. He stated,

What struck me now was the undeniable fact [that] while people may be classed as introverts or extraverts, this does not account for the tremendous differences between individuals in either class. So great, indeed, are these differences that I was forced to doubt whether I had observed correctly in the first place. It took nearly ten years of observation and comparison to clear up this doubt. (1921/1971, p. 535)

1. Myers and Briggs also used both of these terms, *attitude* and *orientation*, as equivalent and interchangeable when referring to Extraversion and Introversion.

2. Note that Jung spelled the term *extraversion*, whereas today the more common spelling is *extroversion*. Jung's spelling preference was continued by Myers and Briggs in their manuals and in other works on the MBTI assessment.

Jung's Addition of the Four Processes

As a result of this realization, Jung subdivided his initial Extravert and Introvert types into eight types by identifying two pairs of opposite mental functions or processes: two opposite perceiving processes, *Sensation* (called *Sensing* by Myers and Briggs) and *Intuition*; and two opposite judging processes, *Thinking* and *Feeling*.³ Jung defined a *function* (process) as “a particular form of psychic activity that remains the same in principle under varying conditions” (1921/1971, p. 436). As explained in chapter 1, *perceiving* refers to the taking in or gathering of information, and *judging* refers to the manner in which individuals come to conclusions about what they have perceived.

Jung saw the two perceiving processes (Sensation and Intuition) as being spontaneous, natural, unplanned, and without a purpose or goal. These processes are *unconstrained* in the sense that they are attuned to the pure flow of events with no requirement that the flow have a purposeful direction or stopping point. Sensing is an unconstrained process in that it entails experiencing a flood of sensations or the accumulation of facts and details while making no attempt to limit or evaluate them. Intuition is an unconstrained process in that it entails being open to a free flow of ideas, connections, and possibilities while making no attempt to choose among or evaluate them.

Jung saw the two judging processes (Thinking and Feeling) as being goal oriented in that they are directed toward the goal of making a decision or coming to a conclusion and thus are constrained by the laws of reason.⁴ These judging processes appraise perceptions by assessing information as either logical or illogical (Thinking) or as important or unimportant (Feeling). Thinking and Feeling judgments thus act as constraints or limits on the free flow of the sensations or intuitions received by the two perceiving processes, which are focused entirely on the general flux of events.

A detailed examination of each of the processes is provided later in this chapter.

The Eight Jungian Types

Jung stated that individuals do not use Extraversion and Introversion in isolation. Rather, they extravert or introvert one or another *mental process* (Sensing, Intuition, Thinking, or Feeling; 1921/1971, p. 520). As a result,

generalizations about “extraverts” and “introverts” are very likely to be inaccurate for many people. Accuracy can be increased greatly if the processes being extraverted or introverted are specified.

Few, if any, generalizations can accurately be applied to all or most “Extraverts” or all or most “Introverts.” We must specify *what* is being extraverted or introverted—Sensing or Intuition or Thinking or Feeling.

Jung further specified that whichever orientation (Extraversion or Introversion) comes most naturally and comfortably to an individual will typically be used in conjunction with whichever of the four mental processes that individual uses most energetically, enthusiastically, and often, and with the greatest comfort and confidence. He called this combination of favorite mental process and favorite orientation the *dominant, primary, or superior* process, in that it dominates, directs, and is most important (primary or superior) for a particular personality.⁵ By combining the two orientations (Extraversion and Introversion) and the four processes (Sensing, Intuition, Thinking, and Feeling), Jung identified eight “dominant types,” as follows:

- Dominant extraverted Sensing types
- Dominant introverted Sensing types
- Dominant extraverted Intuitive types
- Dominant introverted Intuitive types
- Dominant extraverted Thinking types
- Dominant introverted Thinking types
- Dominant extraverted Feeling types
- Dominant introverted Feeling types

An Example of Oppositeness: The Dominant and Inferior Processes

Perhaps the simplest example of how opposites shape and color a personality is provided in what Jung said about the *inferior* function (or process).⁶ He defined the inferior function as the combination of the process that is opposite the dominant process and the orientation that is opposite the preferred orientation (Jung, 1921/1971, p. 519). For example, if the dominant function is extraverted Thinking, the inferior function is introverted Feeling. The more conscious energy individuals devote

3. Myers and Briggs used the term *process* interchangeably with Jung's term *function*. In his writings, Jung occasionally used the term *process* as denoting the same thing as *function*. While the terms are equivalent, this manual uses the term *process* because it better conveys the ongoing, continuous nature of these psychic activities.

4. Jung referred to Sensation and Intuition as *irrational functions* in that they were not directed to any goal so they were free to be experienced in their raw form. He called the judging functions, Thinking and Feeling, *rational* in that they were directed to the goal of making a decision.

5. Jung called the opposite orientation–process combination the *inferior function, or process*, in contrast to the *superior function, or process*.

6. Although the terms used most often in this manual are *process* rather than *function* and *orientation* rather than *attitude*, readers should be aware that the terms *function* and *attitude* appear frequently in existing type literature, especially in the Jungian literature and Jungian quotations, including in this manual. This is especially common and accepted when referring to the *inferior function* and the *dominant function*. These terms are therefore used when citing Jung and in other appropriate instances in this chapter.

to their dominant process, the less conscious energy is available for its opposite, their inferior process. To the extent that the dominant process is conscious, the individual can direct and control it, whereas the opposite, inferior process, which is correspondingly unconscious, cannot be directed and controlled. Such an unconscious process, however, has *unconscious energy*, which may remain hidden and inactive, or at times may erupt and disrupt an individual's conscious desires and intentions. (For more information about the manifestations of the inferior function, see Quenk, 2000, 2002, and 2012.)

Each of Jung's eight personality types, listed above, encompasses a dominant process along with its opposite, inferior process. The characteristics of the dominant and corresponding inferior processes are briefly described in table 3.1.

The Auxiliary and Tertiary Processes

Jung provided a good deal of information about the direction and flow of energy in his system, particularly regarding the dominant and inferior processes. He was somewhat less specific about energy flow for the other two processes. For example, he stated that

apart from the qualities I have mentioned, the undeveloped functions⁷ possess the further peculiarity that, when the conscious attitude (i.e., the dominant function) is introverted, they are extraverted, and vice versa. (1921/1971, p. 521)

In Jung's energy system, the extraverted or introverted orientation attached to the dominant process determines and is balanced by the opposite orientation attached

Table 3.1 | The eight Jungian dominant and inferior processes

Dominant extraverted Sensing Directing most of their conscious energy outward to noticing and focusing on detailed sensory data in the present	Inferior introverted Intuition <i>Directing little energy to exploring internal images, connections, and patterns that might lead to possibilities and insights</i>
Dominant introverted Sensing Directing most of their conscious energy inward to gathering and storing the facts and details of both external reality and internal observations and experiences	Inferior extraverted Intuition <i>Directing little energy to scanning the environment for new ideas, interesting patterns, and future possibilities</i>
Dominant extraverted Intuition Directing most of their conscious energy to scanning the environment for new ideas, interesting patterns, and future possibilities	Inferior introverted Sensing <i>Directing little energy inward to accumulating and storing facts and experiences</i>
Dominant introverted Intuition Directing most of their conscious energy inward to exploring internal images, connections, and patterns that may lead to new possibilities and insights	Inferior extraverted Sensing <i>Directing little energy outward to accumulating detailed, accurate sensory data in the present</i>
Dominant extraverted Thinking Directing most of their conscious energy to seeking and creating logical order in the external environment by applying detached, objective logic and decisive action	Inferior introverted Feeling <i>Directing little energy to seeking meaningful and complex inner harmony through sensitivity to their own and others' values and behavior</i>
Dominant introverted Thinking Directing most of their conscious energy to seeking order, reflecting on their thoughts, and developing logical systems for understanding	Inferior extraverted Feeling <i>Directing little energy to seeking harmony through organizing and structuring the environment to meet their own and others' needs</i>
Dominant extraverted Feeling Directing most of their conscious energy to seeking harmony through organizing and structuring the environment to meet their own and others' needs	Inferior introverted Thinking <i>Directing little energy to seeking accuracy and order in their thoughts through reflecting on and developing a logical system for understanding</i>
Dominant introverted Feeling Directing most of their conscious energy to seeking intensely meaningful and complex inner harmony through sensitivity to their own and others' inner values and outer behavior	Inferior extraverted Thinking <i>Directing little energy to seeking and creating logical order in the external environment by applying detached, objective logic and taking decisive action</i>

7. Jung was referring to the other three functions.

to the three remaining processes:⁸ the *inferior function* (process) described above, as well as a secondary, *auxiliary process*, and a third remaining process (later called the *tertiary* by Myers and Briggs). Note that Jung described the auxiliary process as “in every respect different from the nature of the primary function . . . but not antagonistic to the primary function” (1921/1971, p. 406). “In every respect different” means that the auxiliary process is the opposite “kind” of process (i.e., if the dominant process is a judging process, the auxiliary is a perceiving process, and vice versa) and is used in the orientation opposite that of the dominant process (i.e., if the dominant process is extraverted, the auxiliary process must be introverted, and vice versa). The “antagonistic” process in Jung’s system is the inferior process (function), in that it is opposite the dominant function in every way (e.g., if the dominant process is Sensing that is introverted, the inferior process is Intuition that is extraverted). There is no specific reference in Jung’s work to the third remaining process, the one that is opposite the auxiliary process. Jung included it with his references to “counterbalancing functions,” “unconscious functions,” “undeveloped functions,” and “inferior functions.”⁹

Characteristics of the 16-Type System

Based on their careful reading of Jung’s writings, especially his brief characterization of how the auxiliary process complements the dominant process, Myers and Briggs expanded Jung’s original eight types to account for the moderating effects of an auxiliary process that operates in the attitude opposite that of the dominant process. Jung did not provide any discussion of these effects in his descriptions of his eight types. The efforts of Myers and Briggs resulted in a 16-type system with the following characteristics.

Dominant (favorite) process. For each type, one process leads, or is dominant. The dominant process is also referred to as the favorite, or first, process. Individuals of each type use their dominant process mainly in their favorite orientation. That is, Extraverts use their dominant process mainly in the extraverted world of people, things, and activities, while Introverts use their dominant process mainly in the introverted world of concepts, ideas, and inner experiences.

Auxiliary (second) process. The auxiliary, or second, process complements the dominant process by providing balance to the personality in two ways. First, it provides balance between Extraversion and Introversion. For

Extraverts, the dominant process is extraverted and the auxiliary process is typically used in the inner, introverted world. For Introverts, the dominant process is introverted and the auxiliary process is typically used in the outer, extraverted world. Without the balance provided by the auxiliary process, an individual would have a difficult time functioning effectively because he or she would live exclusively in either the outer world or the inner world.

The auxiliary process enables an individual to develop comfort and facility functioning well in both the outer world and the inner world.

Second, the auxiliary process provides balance between perception and judgment. As Jung specified, if the favorite process is a perceiving one (S or N), the second process is a judging one (T or F). Or, if the favorite process is a judging one (T or F), the second process is a perceiving one (S or N). Without this balancing aspect, an individual using only perception would have no process with which to make decisions and come to conclusions, and an individual using only judgment would have no information on which to base decisions and conclusions.

When using the auxiliary process effectively, an individual gains command of both perception and judgment, that is, of both taking in information and coming to conclusions.

Inferior (fourth) process. The inferior, or fourth, process operates in the less preferred orientation of Extraversion or Introversion. It is the process that is the least appealing to an individual, the least used, and thus the least developed. It has little or no conscious energy and is therefore the process the individual is least able to control and direct. The eruption of the inferior process in times of fatigue and stress is predicted by the Jung-Myers theory and confirmed by everyday experience. (For more information about the inferior process, see Quenk, 2000, 2002, 2012.)

Tertiary (third) process. Opposite the auxiliary process is the tertiary, or third, process. Several views regarding the orientation of the tertiary process have been proposed. All three previous MBTI manuals (Myers, 1962b; Myers & McCaulley, 1985; Myers, McCaulley, Quenk, & Hammer, 1998) took Jung’s position that the dominant process operates in the preferred attitude (orientation) and that the auxiliary, tertiary, and inferior functions (processes) all take the opposite, less preferred attitude. However, many years of clinical, experiential, and anecdotal

8. There are several references in Jung’s writings to the three remaining processes as having an opposite attitudinal (orientational) character. For example, in writing about dominant introverted Thinking types (p. 387), Jung commented that the counterbalancing functions have an extraverted character.

9. Jung seemed to sometimes view the auxiliary process as quite a bit less conscious than the dominant process and thus more similar in lack of “controllability” to the inferior and tertiary processes. At other times, he leaned toward viewing the auxiliary as quite a bit more capable of conscious direction than the tertiary and inferior. This latter view is closer to Myers’ view that the auxiliary is quite a bit more conscious than the tertiary process and certainly the inferior process, and very capable of conscious direction, though not with the same confidence, competence, and energy that characterize the use of the dominant process.

evidence favor the notion proposed by Quenk (1993) that the tertiary process for any individual may operate *in either orientation*, depending on varying situations and individual life experiences and habits.

This does not mean that every individual *habitually* or *consistently* uses his or her tertiary process in one or the other orientation. An individual may use one or the other orientation depending on situational factors, context, training, and myriad other contingencies. In contrast, the dominant and auxiliary processes—processes that are relatively conscious and developed in comparison to the tertiary process—are more habitual, predictable, and stable in their associated orientations. It has been reported (K. D. Myers, personal communication, 2004) that toward the end of her life, Myers herself proposed this hypothesis. Accordingly, this manual does not specify an orientation for the tertiary process.

Given the above assumptions, the first (dominant), second (auxiliary), third (tertiary), and fourth (inferior) processes and, with the exception of the tertiary process, their typical orientations for each of the 16 types can be identified.

Effective Functioning of the 16 Types

The balancing feature of the auxiliary process in the use of Extraversion and Introversion provides an important correction to a misconception commonly held by people who are new to type theory and the MBTI instrument. Extraverts just learning about type may assume that type theory says they never like to be alone, and Introverts may assume that type theory says they never like to be sociable. In truth, however, type theory says everyone lives in both the extraverted world and the introverted world to some extent. Many people attain comfort and competence in both worlds, but for each person one orientation is typically preferred over the other. A well-developed Introvert can deal ably with the extraverted world when necessary by using his or her auxiliary process but will work best, most easily, and with the least fatigue when using his or her dominant process in the inner world, that is, by introverting. Similarly, a well-developed Extravert can use his or her auxiliary process to deal effectively with inner experience but will work with the most interest, satisfaction, and energy when using his or her dominant process externally in action.

Good type development fosters the ability to extravert comfortably and to introvert comfortably but also assumes a natural preference for one orientation over the other.

There are several additional implications of the Jung-Myers model that are important to keep in mind when trying to understand the dynamic nature of personality types:

- Extraverts show their best-developed, favorite process (their dominant process) to the outer world—putting their “best foot forward.”
- Introverts show their second process (their auxiliary process) to the outer world, saving their best-developed process for the inner world of ideas and reflection.
- It follows, then, that Introverts are likely to be underestimated and misunderstood since they use their second-best process rather than their favorite process in their interactions with the outer world. They may therefore appear less confident than Extraverts. Their perceived lower confidence may make them appear less competent as well.
- Extraverts also may be underestimated and misunderstood. They reveal their “best,” extraverted process to the world quickly and confidently. Others, Introverts in particular, who typically think first and then decide whether to express their thoughts out loud, may assume that Extraverts lack depth and that there is not much going on inside them, since they don’t seem to do any thinking before they speak.
- It is important to remember, however, that just as Introverts may feel awkward when extraverting their dominant introverted process, so Extraverts may also feel awkward when extraverting their introverted auxiliary process.

Using All Processes and Orientations

Jung believed that ideally people should be able to use all four functions (processes) because all are vital for living successfully. He stated that

for complete orientation all four functions should contribute equally; thinking should facilitate cognition and judgment, feeling should tell us how and to what extent a thing is important or unimportant for us, sensation should convey concrete reality to us through seeing, hearing, tasting, etc., and intuition should enable us to divine the hidden possibilities in the background, since these too belong to the complete picture of a given situation. (1921/1971, p. 518)¹⁰

10. Myers built on Jung’s notion about using all four processes. She recommended that all four be used consciously and systematically for problem solving: Sensing first to ascertain the facts in a situation, then Intuition to explore the possibilities that could occur, then Thinking to ascertain the logical consequences of one or another course of action, and, finally, Feeling to consider what is important to the people involved (Myers, I. B., 1987). Gordon Lawrence arrayed the four processes in a visual zigzag pattern known as the “problem-solving Z.”

Jung went on to say that

in reality, however, these basic functions are seldom or never universally differentiated and equally at our disposal. As a rule, one or the other function occupies the foreground, while the rest remain undifferentiated in the background. (p. 518)¹¹

Thus all processes and orientations are potentially accessible in all types, although, in practice, the dominant process is most conscious and controllable and the other processes are relatively less so. Jung indicated that the ability and willingness to exercise all functions and attitudes are necessary in order to avoid one-sided, rigid adherence to a single perspective. In Jung's view, one-sided personalities were poorly equipped to function adaptively in the world. Extending Jung's logic, therefore, an individual whose dominant

All four processes—Sensing, Intuition, Thinking, and Feeling—and all four orientations—Extraversion, Introversion, Judging, and Perceiving—are available to every individual, regardless of his or her preferences.

process is extraverted Sensing can (and should) at times access and use introverted Sensing, as well as Intuition, Thinking, and Feeling in one, the other, or both orientations of Extraversion and Introversion.

Being able and willing to use all processes and orientations when appropriate prevents one-sided reliance on only one approach. Jung and Myers believed that one-sided personalities cannot function effectively in the world.

Identifying the Dynamics of Each Type

Jung described his original eight types only in terms of a dominant and its opposite inferior. Jung's extraverted Thinking type, for example, therefore appears to be doing "nothing but" extraverted Thinking unmodified by an auxiliary and a tertiary process but subject to the influence of inferior introverted Feeling. It remained for Myers and Briggs to work out how the interplay of energy Jung hypothesized resulted in the 16 healthy, well-functioning personalities they found to be implicit in Jung's system. This enabled them to expand Jung's model to describe 16 types, with each of Jung's 8 types split into 2 types with the same dominant process but different auxiliary processes. These auxiliary processes are typically used in the orientation opposite that of the dominant process.¹² The list that follows identifies

these 16 types and gives a very brief description of their core features.

Extraverts with dominant Sensing and auxiliary introverted Thinking are most energized when they are directly experiencing the outer world so that they can solve real-world problems using logical analysis.

Extraverts with dominant Sensing and auxiliary introverted Feeling are most energized when they are directly engaging with the outer world so that they can enact their important inner values.

Introverts with dominant Sensing and auxiliary extraverted Thinking are most energized when they are taking in and organizing detailed, factual information that they can use to develop and maintain logical systems in the outer world.

Introverts with dominant Sensing and auxiliary extraverted Feeling are most energized when they are taking in and organizing detailed, factual information that they can use to create and maintain a caring and harmonious outer world.

Extraverts with dominant Intuition and auxiliary introverted Thinking are most energized when they are envisioning future innovations or problem solutions in the outer world that are in accord with logical principles.

Extraverts with dominant Intuition and auxiliary introverted Feeling are most energized when they are envisioning future innovations or problem solutions in the outer world that are in accord with their inner values.

Introverts with dominant Intuition and auxiliary extraverted Thinking are most energized when they are internally creating innovative, complex models and systems that are in accord with logical principles in the outer world.

Introverts with dominant Intuition and auxiliary extraverted Feeling are most energized when they are internally creating complex models and systems that are in accord with their values and standards in the outer world.

Extraverts with dominant Thinking and auxiliary introverted Sensing are most energized when they are organizing the outer world in accordance with logical principles, focusing on efficient use of facts and details.

Extraverts with dominant Thinking and auxiliary introverted Intuition are most energized when they are organizing the outer world in accordance with logical principles, focusing on possible imagined future structures and systems.

11. In Jung's writings, differentiated functions are those that are conscious, while undifferentiated functions are described as unconscious.

12. There are, of course, individuals who use both dominant and auxiliary processes in the same orientation, which negatively affects their functioning in a number of ways. For a discussion of this situation, see Quenk, 1985.

Introverts with dominant Thinking and auxiliary extraverted Sensing are most energized when they are logically analyzing the facts and details in their immediate environment to solve real-world problems.

Introverts with dominant Thinking and auxiliary extraverted Intuition are most energized when they are logically analyzing a broad and often disparate array of information and testing their hypotheses, theories, and models to arrive at the best possible explanations.

Extraverts with dominant Feeling and auxiliary introverted Sensing are most energized when they are creating and maintaining harmonious environments in the outer world by using their internal storehouse of accumulated information about people and relationships.

Extraverts with dominant Feeling and auxiliary introverted Intuition are most energized when they are creating and maintaining harmonious environments in the outer world by using their inner awareness of the possibilities in the complex interrelationships among individuals, groups, and society generally.

Introverts with dominant Feeling and auxiliary extraverted Sensing are most energized when they can live out their deeply held inner values by focusing on external facts and details that support what is most important to them.

Introverts with dominant Feeling and auxiliary extraverted Intuition are most energized when they can find or create outer environments that match their deeply held inner values and important relationships.

As is evident from these brief descriptions of the 16 types, an extraverted dominant process differs from the same process when it is introverted. For example, extraverted Thinking is not the same as introverted Thinking. A dominant process is also modified very differently by its two different auxiliary processes. For example, extraverted Feeling is different when it is complemented by Intuition than when it is balanced by Sensing. Knowing the effects of these differences can help individuals understand and work more effectively with others who share their dominant process but use it in the orientation that is opposite theirs. The same increased understanding occurs when individuals interact with others who share their dominant process and orientation but differ from them in having the opposite auxiliary process.

THE FUNDAMENTALS OF TYPE

This chapter has thus far provided an overall guide to the Jung-Myers theory and its basic history, rationale, and goals, as well as brief descriptions of the ways the parts of the system interact. It has shown that the interactions of particular processes and orientations provide intriguing and useful insights about some of the ways people differ from one another. This section provides a more detailed description of each of the concepts that has been presented.

The Four Processes: Sensing, Intuition, Thinking, and Feeling

The four processes, or functions, as Jung named them, are at the heart of the Jung-Myers theory about how people function typologically. As discussed previously, Jung focused first on differences between Extraverts and Introverts; only much later in his explorations did he describe the four functions. Had Jung limited his observations and descriptions to Extraversion and Introversion, there might not be any type theory, let alone the Jung-Myers theory of personality type.

THE UNCONSTRAINED PERCEIVING PROCESSES: SENSING AND INTUITION

Jung divided all perceiving activities into two categories, *sensation* and *intuition*, which are attuned to the flow of events and operate most broadly when not constrained by rational direction. As described in an earlier section, freely experiencing a flood of sensations either in one's inner world or in the outer world while making no attempt to limit or select among them is an illustration of Sensing as an unconstrained process. During such a state of free receptivity, sensations flow freely and no attempt is made to evaluate them. The technique of brainstorming illustrates one way Intuition is experienced as an unconstrained process. During brainstorming, inspirations are encouraged to flow freely, without the constraint of criticism or appraisal or any need to reach a decision or conclusion.

Sensing (S). Sensing refers to perception of what is observable by the five senses. It establishes what exists. Because the senses can bring to awareness only what is occurring in the present moment, individuals who prefer Sensing tend to focus on the immediate experiences available to their five senses. They often develop characteristics associated with this awareness, such as enjoyment of the present moment, realism, acute powers of observation, the ability to remember the details of both past and present experiences, and practicality.

Intuition (N). Intuition refers to perception of possibilities, meanings, and relationships through insight. Jung characterized intuition as perception by way of the

unconscious. Intuition may come to the surface of consciousness as a hunch or as a sudden discovery of a pattern in seemingly unrelated events. It permits perception beyond what is perceived by the five senses, including possible future events. Individuals who prefer Intuition may develop the characteristics that can follow from that emphasis and become imaginative, theoretical, abstract, future oriented, and original or creative.¹³

Using Sensing and Intuition to perceive an apple. The difference between the two perceiving processes can be illustrated by describing an apple. When individuals who prefer Sensing are asked to describe an apple, they often use descriptors such as “juicy,” “crisp,” “red,” and “white with black seeds.” The focus is on the sensory attributes of the apple. When individuals who prefer Intuition are asked to describe an apple, they may say things like “New York City” (The Big Apple) and “Steve Jobs” (Apple computers). Here the focus is on the associations and possible meanings stimulated by the apple.

THE GOAL-DIRECTED JUDGING PROCESSES: THINKING AND FEELING

Jung used the terms *thinking* and *feeling* in specialized ways to refer to the two processes that can be directed by the person and are in accord with the laws of reason. As mentioned earlier, these judging processes appraise perceptions. Thinking and Feeling judgments act as constraints or limits on the free flow of the sensations or intuitions received by the two unconstrained perceiving processes, which are focused entirely on the general flux of events.

Thinking (T). Thinking is the process by which an individual comes to a decision by linking ideas or facts together through logical connections. It relies on principles of cause and effect and tends to be impartial and impersonal in the application of reason to a decision. The focus of attention when using Thinking is the sensing or intuitive information that provides the raw material or the basis for the particular decision at hand. Thinking relies on objectivity and neutrality with respect to the personal desires and values of both the decision maker and the people who may be affected by the decision. Thinking principles apply to everyone equally. Individuals who prefer Thinking are likely to develop characteristics associated with this way of arriving at conclusions: an

analytical inclination, objectivity, concern with principles of justice and fairness, criticality, an impassive and dispassionate demeanor, and a linear orientation to time—that is, concerned with connections from the past through the present and toward the future.

Feeling (F). Feeling is the process by which an individual comes to a decision by weighing relative values and merits of issues. Feeling relies on an understanding and appreciation of personal and group values; thus it is more subjective than Thinking. Nevertheless, Jung said, “feeling, like thinking, is a rational process,”¹⁴ since values in general are assigned according to the laws of reason” (1921/1971, p. 435).

Because values are subjective and personal, individuals making judgments using the Feeling process are more likely than are those using Thinking to be attuned to the values and feelings of others, as well as to their own values and feelings. They try to understand individual people and to anticipate and take into account the effects of the decision at hand on the people involved and what is important to them. They pay more attention to the human aspects of problems than to the technical aspects and have a desire for affiliation, warmth, and harmony. They are concerned with preserving important human values and carrying them into the future. The focus on subjective values renders the process of making a Feeling judgment less directly observable than the linear, logical process of making a Thinking judgment. As a result, Feeling types may be at a disadvantage when asked to “justify” their judgments from the point of view of logic alone.

Common misconceptions. Although the classical distinction in psychology between being “tough-minded” and “tender-minded” is one aspect of the T–F difference, it does not follow that Thinking individuals do not have values, feelings, or emotions or that Feeling individuals cannot develop or follow a logical argument.¹⁵ Such erroneous views reflect common misconceptions about people who favor one or the other of the two judging processes. Equating *Feeling* as a decision-making process with *feeling* as affect or emotion is another common mistake. In this regard, Jung stated, “Feeling is distinguishable from affect by the fact that it produces no perceptible physical innervations [bodily emotional reactions], i.e., neither more nor less than an ordinary thinking process” (1921/1971, p. 434). *Feeling* as defined

13. Intellectual tradition and various research approaches have associated creativity with a preference for Intuition. When *creativity* is defined as the creation of something new that has not existed before, one can appropriately describe Sensing perception as “creative” as well. The process whereby Sensing develops something new may be the step-by-step, methodical, and detailed procedure characteristic of *introverted* Sensing, or the *extraverted* Sensing process of creating something new by adapting and combining whatever elements are available in the current situation. The Sensing process is in contrast to the flashes of insight that we associate with Intuition. However, the *product* created through either form of Sensing may be a new one that has not existed before. The notion of “thinking outside the box” describes intuitive creativity; the equally valuable innovation that emerges from the Sensing process may very well occur “inside the box.” See the section on creativity in chapter 9, the validity chapter.

14. The meaning of the term *rational* as used by Jung and Myers is different from its everyday, common meaning, which is akin to “logical” or “well-reasoned.”

15. Tough–Tender is one of the five Step II facets in the Thinking–Feeling set. This facet identifies the way a person implements a decision that has been made.

by Jung is a cognitive evaluative process, not an emotional reaction.

Confusing *Feeling* as a decision-making process with *feeling* as affect or emotion is a common mistake.

The use of Thinking and Feeling in a promotion decision. The difference between using Thinking judgment and using Feeling judgment is illustrated by the following example:

In making a decision about which of five employees should be promoted to a supervisory position, a Thinking type's decision process would likely involve developing a set of objective criteria, ranking the criteria relative to their importance to the decision, and rating each of the five employees accordingly. After deciding which candidate or candidates best meet the criteria, a Thinking type might or might not use his or her less preferred Feeling process to consider other factors, such as personal style, acceptability to other employees, and the employee's personal circumstances. Such considerations might aid decision making when a choice must be made between equally logical alternatives. Attending to these more subjective issues might also allow the decision maker to see whether any of these factors might have a significant negative impact on the logical choice.

A Feeling type might focus initially on the candidates' qualifications for the job, as well as on their personal circumstances and their impact on other workers. If two or more candidates were to stand out clearly as a result of this weighing of values, the Feeling type might use his or her less preferred Thinking process to develop a set of objective criteria to aid in final decision making. If only one candidate were to stand out clearly, the Feeling type might apply objective criteria to ensure that nothing important was overlooked, as well as to facilitate a smooth and harmonious transition for all employees affected by the decision.

The Two Orientations of Energy: Extraversion and Introversion

As previously mentioned, a large portion of Jung's *Psychological Types* (1921/1971) is devoted to the historical development and description of the complementary attitudes or orientations of energy. The nature and extent of differences between Extraversion and Introversion translate into profoundly different approaches to life.

EXTRAVERSION

When a person is extraverting, energy and attention flow out, or are drawn out, to the objects and people in the environment. The individual experiences a desire to interact and engage with the environment, to affirm its importance, to welcome its effect, and to increase his or her effect on it. Individuals who habitually take an Extraverted approach may develop some or all of the characteristics associated with Extraversion: awareness of and reliance on the environment for stimulation and guidance; an eagerness to interact with the outer world; an action-oriented, sometimes impulsive way of approaching life; openness to new experiences; ease of communication and sociability; and a desire to "talk things out."

INTROVERSION

When a person is introverting, energy is drawn from the environment toward inner experience and reflection. The individual desires to stay focused on the internal, subjective state, to affirm its value, and to maintain this focus for as long as possible. The main interests of Introverted types are in the world of concepts, ideas, and inner experiences. Individuals who habitually take an Introverted approach may develop some or all of the characteristics associated with Introversion: interest in the clarity of concepts, ideas, and recollected experience; reliance on enduring concepts and experiences more than on transitory external events or fleeting ideas; a thoughtful, contemplative, observant manner; an enjoyment of solitude and privacy; and a desire to "think things out" before talking about them.

To many laypeople, the term *extraverted* means sociable and the term *introverted* means shy.¹⁶ Jung and Myers' conception is different from and much broader than this lay view. Seen as different orientations of energy, Extraversion and Introversion identify the direction in which an individual's energy typically flows, outward or inward. Individuals who prefer Extraversion (Extraverts) are seen as having more energy directed toward the outer world, with correspondingly less energy available for the inner world of reflection. Individuals who prefer Introversion (Introverts) are seen as having more energy directed toward the inner world, with correspondingly less energy available for activities in the outer world. The level of comfort individuals experience with these orientations of energy and the amount of time they devote to them result in the two recognizable variants of normal human personality just described—Extraverts and Introverts. These kinds of individuals have been recognized throughout history and literature, and each has made major contributions to society. Myers'

16. Measures of "shyness" are typically correlated with various measures of introversion, including Introversion as elicited on the MBTI assessment. This is an inevitable consequence of the overlap in content of test items used to identify Introversion and shyness. However, shyness and Introversion are not necessarily related. Some Extraverts describe themselves as shy, and some Introverts describe themselves as socially outgoing or gregarious. Out-of-preference Step II facet scores provide insight into such observations.

descriptions of Extraverts and Introverts reflect the complex, multifaceted nature of Jung's conception.

The Processes in Their Extraverted and Introverted Forms

The four processes were described in an earlier section, but without consideration of the orientation in which they usually operate. Now that a fuller explanation of the nature of Extraversion and Introversion has been provided, this section will explore the way each process is likely to be experienced and expressed when it appears in its extraverted and introverted forms. The focus is on the way *any* individual uses the orientation–process combination being described, regardless of whether the individual prefers that particular combination. As stated earlier, everyone is able to use all four processes in either orientation at least some of the time. Shorthand abbreviations indicating whether a process is extraverted or introverted are commonly used in MBTI literature, as follows:

- Extraverted Sensing: S_e
- Introverted Sensing: S_i
- Extraverted Intuition: N_e
- Introverted Intuition: N_i
- Extraverted Thinking: T_e
- Introverted Thinking: T_i
- Extraverted Feeling: F_e
- Introverted Feeling: F_i

Just as it is easier to “know” and appreciate Extraverts, who present their favorite process to the world, so it is easier to understand and describe the expression of all the processes in their extraverted form. The introverted form of processes is not readily visible to observers, who therefore may be unaware of *any* internal process, be it an individual's preferred form of perception or his or her preferred form of judgment. For example, dominant extraverted Intuitive types with auxiliary introverted Feeling (ENFPs) may mistakenly be viewed as indiscriminately jumping from one idea or project to another. However, their particular enthusiasms are likely to be a reflection of their auxiliary introverted Feeling process, that is, of their inner values and ideals, the depth of which may be hidden from others.

EXTRAVERTED AND INTROVERTED PERCEIVING PROCESSES

As mentioned earlier in this chapter, it is harder to grasp the nature of the unconstrained perceiving processes than it is to understand the goal-directed judging processes. When an individual's perceiving process is introverted (used in the inner world), understanding it becomes even more difficult.

Extraverted Sensing (S_e) focuses on whatever information one's five senses receive from the outside world, without regard for other experiences the individual has had and with no attempt made to categorize, compare, or evaluate the current experience in relation to anything else. Incoming sensory information is unfiltered, so the individual can enjoy the fullest range of what is out there. Individuals who use extraverted Sensing naturally and habitually typically strive to experience as much of the outside world as possible and to enjoy the present moment to the fullest, savoring each experience and giving little attention to past or future. When they are engaged in using extraverted Sensing they are likely to be seen as energetic, outwardly focused, fun loving, willing to take physical risks, and at times irresponsible.

Introverted Sensing (S_i), like extraverted Sensing, involves using one's five senses to observe and take in information in the environment. However, when individuals are introverting their Sensing, that environment also includes information that they acquired in the past and categorized and stored internally. They may therefore sort new incoming information into pre-existing categories. If there is no relevant category that quite fits the new information, they may establish a subcategory or add a new category. Individuals who engage in introverted Sensing naturally and habitually use their acute powers of observation, memory for details, and storehouse of accumulated knowledge to maintain efficient systems and procedures and provide factual, accurate, relevant evidence when needed. When they are engaged in using introverted Sensing, they are likely to be seen as conscientious, devoted, accurate, consistent, and at times inflexible.

Extraverted Intuition (N_e) refers to perception of possibilities, meanings, interrelationships, and patterns in the outside world. The process of extraverting Intuition is similar to the process of extraverting Sensing in the sense that both involve an openness to the unrestricted flow of incoming information. Recall, however, that Jung characterized intuition as perception by way of the unconscious. In the case of extraverted Intuition, incoming perceptions quickly combine with one another, with previous perceptions, and with all manner of conscious and unconscious information. This results in ideas, patterns, speculations, and theories that may move far beyond what is seen, heard, felt, tasted, or smelled in the moment when information is perceived. When individuals are extraverting their Intuition they become aware of possibilities, hypotheses, and potentialities in the future. This process is so rapid that they may be unable to trace the sequence that led to a hunch, a new idea, or a vision of the future. Individuals who perceive in this way naturally and habitually may have a seemingly uncanny ability to forecast future events, anticipate new models and systems, and propose innovative explanations of

complex events in the world. They are typically seen as imaginative, theoretical, abstract, future oriented, and at times unreliable.

Introverted Intuition (Ni) can be quite mystifying when one is engaged in using it, as the process draws on an individual's internal storehouse of connections, associations, images, and input from the environment, which swiftly combine with the individual's internal connections. The vastness of the inner perceptions that may be accessed often defies description. Unlike with extraverted Intuition, which is likely to be verbalized, individuals who are introverting their Intuition may not be able or willing to tell others what is going on inside them. They may express what they are processing internally in some metaphorical or symbolic way. Individuals who perceive in this way naturally and habitually seem to have a sixth sense with which they can pick up the thoughts, feelings, and motives of others long before anyone else is aware of them. Others tend to see them as innovative, independent, complex, and at times unfathomable.

EXTRAVERTED AND INTROVERTED JUDGING PROCESSES

The two judging processes are somewhat easier to grasp because people can see their result: a conclusion or judgment. In the case of extraverted judgment, the result of the Thinking or Feeling process is likely to be readily observed and communicated. Although the process of introverting Thinking or Feeling may not be observable, the conclusion or judgment involved is likely to be known in the actions or inactions of both introverted judging types.

Extraverted Thinking (Te) is perhaps the easiest of the orientation–process combinations to observe and understand because it involves coming to logical conclusions about people, events, and systems in the outside world. Using linear logic and stated or assumed premises that can be analyzed in a systematic way, individuals who extravert their Thinking tend to arrive at a judgment that appears as a statement of truth about the subject at hand. When using this process they systematically apply logic to the available facts or possibilities and come to an inevitable logical conclusion. Individuals who use this process naturally and habitually are likely to be seen as decisive, forceful, organized, confident, and at times domineering.

Introverted Thinking (Ti) is less observable than is extraverted Thinking because the logical judgments are being made internally. When individuals use introverted Thinking to make judgments they apply linear logic and stated or assumed premises to arrive at a coherent internal approximation of truth and validity. People who use this process naturally and habitually often come to conclusions that take the form of complex models, theories, or unique ways of looking at

something in the physical, mechanical, psychological, or sociological world. When they are engaged in using introverted Thinking they are likely to be seen as calm, complex, dispassionate, unemotional, fair-minded, and at times stubborn.

Extraverted Feeling (Fe), like extraverted Thinking, is quite visible to others because when individuals are extraverting their Feeling they are directly informing others of the important values that underlie their decision or judgment. Their goal in using extraverted Feeling judgment is to arrive at a decision that is in accord with their core values. This may involve supporting their own or others' well-being or enacting some strongly held belief about what is right or wrong. People who use this process naturally and habitually are likely to enthusiastically and energetically work to correct what they believe to be undesirable for themselves, others, and society and to implement what they see as desirable. When they are engaged in using extraverted Feeling they are likely to be seen as caring, responsible, enthusiastic, and at times overbearing.

Introverted Feeling (Fi), like introverted Thinking, is harder to comprehend because the strongly held values involved are unlikely to be shared directly with others. When individuals are introverting their Feeling, they may find it difficult to talk about their values, even though their values are just as intensely felt and held as are those of individuals who extravert their Feeling. Instead, they may pursue enactment of what is important to them by quietly and unobtrusively encouraging others to take up their "cause." People who use this process naturally and habitually are likely to be particularly persuasive in their writing and to use images and stories with which others can readily connect. When they are engaged in using introverted Feeling they are likely to be seen as creative, serious, caring, and at times hypersensitive.

The Two Orientations to the Outside World: Judging and Perceiving

No explicit mention is made by Jung of a judging versus perceiving attitude or orientation toward the outer world. Katharine Briggs described such a type classification in her unpublished work prior to the publication of Jung's *Psychological Types* in 1921. When Briggs read *Psychological Types* in 1923 she discontinued her own work in favor of Jung's better-developed approach. However, one part of her work was retained and became a vital part of the MBTI assessment. Briggs had observed that some individuals habitually used *judgment* in interacting with the outside world and were likely to come to conclusions and achieve closure quickly. Myers and Briggs would later identify these individuals as having a *Judging* orientation. Briggs described other individuals as habitually interacting with the outside world using *perceiving* and liking to continue taking in

information for as long as possible before comfortably coming to closure. These individuals would later be identified as having a *Perceiving* orientation. Briggs' work in conjunction with Jung's incomplete discussion of this area formed the basis for the Judging–Perceiving preference pair used in the MBTI assessment.

The Judging and Perceiving orientations have two uses. First, together with the Extraversion and Introversion orientations, they are used to identify which of a type's two preferred processes is the dominant process and which is the auxiliary process. Second, they describe identifiable orientations to the outside world. The recognition and development of the character of the Judging and Perceiving orientations were a major contribution of Briggs and Myers to the theory of personality types.

The concept of the Judging and Perceiving orientations was developed over a period of several decades, before any MBTI items were written. Myers' observation of different behavioral patterns formed the basis for the J–P distinction. Behaviors characteristic of individuals who used either Thinking or Feeling in their outer life (i.e., who extraverted one of the judging processes) formed the basis of the Judging orientation. Behaviors characteristic of individuals who used either Sensing or Intuition in their outer life (i.e., who extraverted one of the perceiving processes) formed the basis of the Perceiving orientation. Others readily notice how individuals behave in the outside world and, as a result, often recognize someone as having a Judging or a Perceiving orientation relatively early on in their acquaintance.

One of the most overlooked and misunderstood characteristics of the J–P pair of opposites is that it describes the orientation to the outer, or extraverted, world for every type, that is, for both Extraverts and Introverts.

In any new activity, it is appropriate to use a perceiving process (S or N) *first* in order to observe or take in the situation and then to use a judging process (T or F) to decide on the appropriate action. Perceiving types typically remain longer in the observing mode because that is more comfortable and natural for them. Judging types tend to move more quickly through perception in order to reach conclusions and achieve closure, as they are most comfortable and satisfied when a plan has been developed or a decision has been made.

Perceiving types enjoy taking in as much information as possible so that they can keep observing as long as possible. Judging types move quickly through perception to judgment so that they can enjoy the closure of coming to a conclusion.

THE JUDGING ORIENTATION (J)

In the Judging orientation, individuals are concerned with making decisions, seeking closure, planning current and future activities, or organizing their environment. For Thinking Judging (TJ) types, the decisions and plans are more likely to be based on logical analysis. For Feeling Judging (FJ) types, the decisions and plans are more likely to be based on weighing and assessing values. But for both TJ and FJ types, who characteristically use a Judging (J) approach in their outer life, perception tends to be shut off as soon as they have observed enough to make a decision. Individuals who prefer Judging often seem in their outer behavior to be organized, purposeful, and decisive, typically assuring themselves or others that they indeed have enough information to make the decision needed.

THE PERCEIVING ORIENTATION (P)

In the Perceiving orientation, an individual is attuned to incoming information. For Sensing Perceiving (SP) types, the information is more likely to be the immediate realities in the environment, what is happening and observable. For Intuitive Perceiving (NP) types, the information is more likely to be new ideas, interesting patterns, and future possibilities. But for both SP and NP types, the Perceiving approach is open, curious, and interested. They will often suspend judgment in order to take another look, reporting, "I don't yet know enough to make a decision." Individuals who prefer the Perceiving orientation seem in their outer behavior to be spontaneous, curious, adaptable, and open to what is new and changeable. Their aim is to continue to receive information as long as possible in an effort to miss nothing that might be important.

THE NEED FOR BOTH PERCEPTION AND JUDGMENT

An individual must use *both* perception and judgment in order to function effectively. For adequate functioning, judgment should be preceded by perception. However, remaining too long in the perceiving mode before coming to a conclusion becomes unhelpful procrastination. Similarly, bypassing or spending too little time in the perceiving mode and forming a judgment or reaching a conclusion before acquiring sufficient information can result in obstinacy.

Individuals new to the MBTI assessment sometimes erroneously assume that Judging types are necessarily *judgmental*, confusing their readiness to make judgments with the *nature* of their judgments. It is also important to distinguish between a preference for using a perceiving process in the outer world (P) and *perceptiveness*, in the sense of being especially insightful. The most accurate statement that can be made is that individuals who favor a Judging orientation enjoy making decisions as soon as

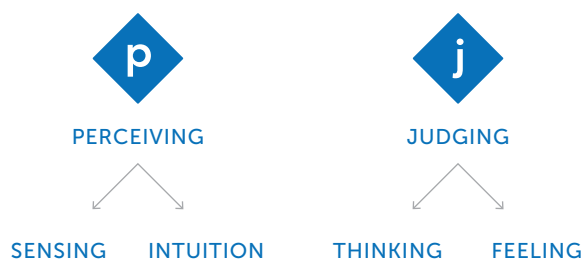


Figure 3.1 | The J–P preference pair

Source: From *Introduction to Myers-Briggs® Type* (7th ed., p. 2) by I. B. Myers, 1998. Sunnyvale, CA: The Myers-Briggs Company.

possible and those with a Perceiving orientation enjoy collecting information as long as possible.

It is also important to recognize that the Judging orientation encompasses the two judging processes, Thinking and Feeling, and the Perceiving orientation encompasses the two perceiving processes, Sensing and Intuition. This fact is often overlooked by new MBTI users and can be a source of confusion, especially when they are trying to understand and determine the dynamics of each type. Figure 3.1 shows this relationship between the Judging and Perceiving orientations and the four processes.

The Dynamic Interaction of the Preferences

This chapter has introduced all the parts of the Jung-Myers theory: the favorite, or first (dominant), second (auxiliary), third (tertiary), and fourth (inferior) processes; the processes in their extraverted and introverted forms (e.g., extraverted Thinking, introverted Sensing); the four orientations (Extraversion and Introversion; Judging and Perceiving); and how the orientations and processes interact with each other. This section provides the details needed to understand the dynamic characteristics of each of the 16 types identified by the MBTI assessment.

When individuals respond to the items on the MBTI assessment, they are not only casting votes for Extraversion (E) or Introversion (I), Sensing (S) or Intuition (N), Thinking (T) or Feeling (F), and Judging (J) or Perceiving (P); they are also providing the information needed to form a hypothesis about the flow of energy typical of their type, which defines their type's *dynamics*. Each four-letter type stands for a complex set of dynamic relationships among the mental processes (S, N, T, and F), the orientation of energy (E or I), and the orientation to the outside world (J or P). The letters in any type appear in this order: E or I, which shows which of the two orientations of energy an individual prefers; S or N, which identifies the kind of perception the person prefers; T or F, which shows which of the two judging processes the person prefers; and J or P, which shows whether the person prefers to use his or her preferred judging process (T or F) or preferred perceiving process (S or N) when

dealing with the outside world, regardless of his or her preference for Extraversion or Introversion. All 16 possible combinations of letters occur—ESTJ, ISFP, INFJ, ENTP, and so on.

Understanding what each letter in the four-letter type signifies and the dynamic relationships specified in the Jung-Myers theory provides the information necessary to determine the favorite, second, third, and fourth processes for each type. This is demonstrated in the following example.

IDENTIFYING THE HIERARCHY OF THE PROCESSES

The four letters of a type represent the following, as indicated in table 3.2.

First letter: preference for orientation of energy, Extraversion (E) or Introversion (I)

Second letter: preference for kind of perception used, Sensing (S) or Intuition (N)

Third letter: preference for Thinking (T) or Feeling (F) judgment

Fourth letter: preference for a Judging (J) or Perceiving (P) orientation to the outside world

It is important to remember that for any type, the middle two letters identify the favorite and second processes. The following steps present a straightforward way to determine the dynamics of type for any four-letter type.

Step 1: The first letter of an individual's type indicates which orientation of energy is preferred and thus the orientation in which the person's favorite process is used. For ESTJ, the first letter is E, indicating that ESTJs prefer Extraversion and that they extravert their favorite process. The next step is to determine which process they extravert in order to identify their favorite process.

Step 2: The last letter of a type, J or P, indicates whether individuals of that type extravert a judging process (T or F) or a perceiving process (S or N). For ESTJ, the J shows that a judging process is extraverted. The T in ESTJ indicates that in this case the judging process is Thinking. So the J in the formula points to Thinking as the extraverted process for ESTJ. Thus the favorite process for ESTJs is extraverted Thinking.

Table 3.2 | Positions of the preferences

e	i	s	n	t	f	j	p
Orientations		Perceiving processes		Judging processes		Orientations	
Extraversion or Introversion		Sensing or Intuition		Thinking or Feeling		Judging or Perceiving	

Source: Modified and adapted from table 2.2, Myers et al. (1998). Used with permission.

Step 3: Now that the favorite process is known, the next step is to determine the second process. The second process is identified by default—it is the other middle letter in the four-letter type. For ESTJ, the favorite process is extraverted Thinking. The other middle letter for ESTJ is S for Sensing; thus the second process for ESTJ is Sensing. And since for an Extraverted type the favorite process is extraverted, the second process (in this case, Sensing) must be introverted in order to provide balance.

Using subscripted letters to indicate the orientation in which the processes are used, the type can now be shown as follows: ES_iT_eJ. So T_e indicates that Thinking is extraverted, or used in the outer world, and S_i shows that Sensing is introverted, or used in the inner world.

Step 4: To finish identifying the dynamics of ESTJ, the two processes that do not appear in the type, N and F, should be looked at next. One of these is the third process and the other is the fourth process. To determine which is which, it is important to remember two simple rules. (1) The third process is the opposite of the second process. Since Sensing (S) is the second process for ESTJ, Intuition (N) is the third process. In accordance with current MBTI convention, an orientation (E or I) is not specified for the third process. (2) The fourth process is the opposite of the favorite process. Since it has been determined that Thinking is the favorite process for ESTJ, Feeling (F) is the fourth process. Furthermore, according to type theory, the fourth process takes the orientation opposite that of the dominant process. Since for ESTJ Thinking is dominant and extraverted, the fourth process is introverted Feeling.

The same four steps are used for all types. Following is an illustration for an introverted type, INFP.

Step 1: The first letter of an individual's type indicates which orientation is preferred and thus the orientation in which the person's favorite process is used. For INFP, the first letter is I, indicating that INFPs prefer Introversion and that they introvert their favorite process. The next step is to determine which process they introvert in order to identify their favorite process.

Step 2: The last letter of a type, J or P, indicates which process is extraverted, either a judging process (T or F) or a perceiving process (S or N). Because this is an introverted type, the process that is introverted must be determined next, as this will indicate the favorite process. So for INFP, the P shows that a perceiving process (either S or N) is extraverted. In accordance with the principle of balance, the introverted process must therefore be a judging process (either T or F). For INFP, the judging process is indicated by F for Feeling. So the favorite process is introverted Feeling.

Step 3: Now that the favorite process is known, the next step is to determine the second process. The second

process is identified by default—it is the other middle letter in the four-letter type. For INFP, the favorite process is introverted Feeling. The other middle letter for INFP is N for Intuition; thus the second process for INFP is Intuition. And since for an introverted type the favorite process is introverted, the second process (in this case, Intuition) must be extraverted in order to provide balance.

Using subscripted letters to indicate the orientation in which the processes are used, the type can now be shown as follows: IN_eF_iP. So F_i indicates that Feeling is introverted, or used in the inner world, and N_e shows that Intuition is extraverted, or used in the outer world.

Step 4: To finish identifying the dynamics of INFP, the two processes that do not appear in the type, S and T, should be looked at next. One of these is the third process and the other is the fourth process. To determine which is which, it is important to remember two simple rules. (1) The third process is the opposite of the second process. Since Intuition is the second process for INFP, Sensing is the third process. In accordance with current MBTI convention, an orientation (E or I) is not specified for the third process. (2) The fourth process is the opposite of the favorite process. Since it has been determined that Feeling is the favorite process for INFP, Thinking (T) is the fourth process. Furthermore, according to type theory, the fourth process takes the orientation opposite that of the favorite process. Since for INFP introverted Feeling is the favorite process, the fourth process is extraverted Thinking.

Table 3.3 shows the hierarchy of the processes for all 16 types.

THE THEORY OF TYPE DEVELOPMENT

In Jung's view, every type moves along predictable pathways that enable personality to develop over time. The goal Jung envisioned for personality development was for individuals to complete themselves, to become the most inclusive version of their individual, unique self that is possible for them. He called this kind of development *individuation*. From a typological perspective, this means having some, but by no means equal, ability to access all parts of one's typological makeup.

Type Development During Childhood

Type theory assumes that children are born with a predisposition to prefer some processes to others. Children are most interested in using their preferred processes in their preferred orientation of Extraversion or Introversion. They are motivated to exercise their favorite process and, to a somewhat lesser extent, their second process, becoming more skillful, adept, and discriminating in the way they exercise them over

Table 3.3 | Hierarchy of the processes for the 16 types

	ISTJ	ISFJ	INFJ	INTJ
1. Dominant	S _i	S _i	N _i	N _i
2. Auxiliary	T _e	F _e	F _e	T _e
3. Tertiary	F	T	T	F
4. Inferior	N _e	N _e	S _e	S _e

	ISTP	ISFP	INFP	INTP
1. Dominant	T _i	F _i	F _i	T _i
2. Auxiliary	S _e	S _e	N _e	N _e
3. Tertiary	N	N	S	S
4. Inferior	F _e	T _e	T _e	F _e

	ESTP	ESFP	ENFP	ENTP
1. Dominant	S _e	S _e	N _e	N _e
2. Auxiliary	T _i	F _i	F _i	T _i
3. Tertiary	F	T	T	F
4. Inferior	N _i	N _i	S _i	S _i

	ESTJ	ESFJ	ENFJ	ENTJ
1. Dominant	T _e	F _e	F _e	T _e
2. Auxiliary	S _i	S _i	N _i	N _i
3. Tertiary	N	N	S	S
4. Inferior	F _i	T _i	T _i	F _i

Source: Modified and adapted from table 3.3, Myers et al. (1998). Used with permission.

time. A sense of competence comes from exercising a process well, and with the reinforcement of constant practice these processes become more controlled and trustworthy. The pleasure of using their processes generalizes to other activities requiring use of them and leads to the surface characteristics, behaviors, and skills associated with these comfortable habits.

While this development of preferred processes is occurring, there is relative neglect of the opposite processes. In this sequence of events, for example, a child who prefers Sensing perception and a child who prefers Intuitive perception will develop along divergent lines. Each will become relatively experienced in an area in which the other remains relatively inexperienced. Both will channel their interests and energies into activities that give them a chance to use their mind in the ways they prefer, and each will acquire a set of surface characteristics that grow out of their basic preferences.

Type Development During Adolescence and Adulthood

As a child becomes an adolescent and then an adult, continued exercise of the favorite and second processes promotes further enjoyment, development, and differentiation of those processes as compared to their opposite fourth and third processes. For example, an individual who develops Sensing in a highly differentiated way is likely to become an astute observer of the immediate environment. Along the way, the person will acquire some of the characteristics assumed to follow from this acute awareness: realism, practicality, and common sense. While paying attention to the specifics of the environment, the person will spend less time and energy using Intuition, which is the perceiving process opposite Sensing. Focusing on the present (S) gives individuals who prefer the Sensing process less time for focusing on the future (N); focusing on the concrete (S) gives them less energy for focusing on the abstract (N); focusing on practical applications (S) gives them less interest in theoretical issues (N); and focusing on reality (S) gives them less time for focusing on the imaginary or fanciful (N).

Recognizing Types

As has been shown, individuals create or “grow into” their type by exercising their preferences with regard to their use of perception and judgment. Individuals who share the same preferences tend to have in common whatever qualities result from exercising those preferences. The interests, values, needs, and habits of mind that naturally result tend to produce a recognizable kind of person. People who don’t know anything about type theory often recognize and describe similarities and differences among people they know. That such observations often pick up on type characteristics is evident in the “aha!” reactions many people have when they are first introduced to type preferences and type descriptions. Suddenly their unsystematic observations seem to fall neatly into place. Of course, with further experience, ideally they will moderate a natural tendency to attribute all differences and similarities to type. Just as our type itself develops over time, so does our awareness of what we can legitimately attribute to type and what is better explained by myriad other factors.

Type Development Over One’s Life Span

Type development is seen as a lifelong process of gaining greater command of the processes or powers of perception and judgment. For each type, two of the four processes—the favorite (dominant) and second (auxiliary)—are assumed to be more interesting and thus are more likely to be consciously developed and used. The two less preferred processes—the third (tertiary) and fourth (inferior)—are assumed to be less interesting

and thus are more likely to be relatively neglected. Development comes from striving for excellence in those processes that hold the greatest interest and from becoming at least passable in using the other, less interesting but essential processes.

In youth and adulthood, the task is to find a comfortable and effective identity and place in the world. To do that, individuals need to develop their favorite and second favorite processes. Both are needed because, as discussed earlier, they complement and balance each other: one is extraverted and the other is introverted, and one is a judging process and the other is a perceiving process. Type theory assumes that exercising these natural processes is the best way for an individual to move through adolescence and young adulthood. In general, “specializing” by devoting a great deal of energy to one’s favorite and second processes continues to be useful and appropriate during the first half of life.

During midlife, individuals appear to be naturally motivated toward completing their typological development by gradually adding the previously neglected third and fourth processes to the sphere of conscious operation. Therefore, in the second half of life, it is appropriate to be a “generalist” rather than a specialist. Development of this kind allows individuals to adopt new perspectives and have experiences that previously were not fulfilling or rewarding. Giving greater energy and attention to the third and fourth processes involves giving correspondingly less energy and attention to the favorite and second processes. However, this does not involve a change in an individual’s type. The assumption is that type does not change over one’s life span. Rather, the experience or expression of type may vary in accordance with different stages of life and changing life circumstances.

Type theory assumes that individuals’ type remains the same over their life span, but the way they experience and express their type may be different at different life stages.

A very few exceptional individuals may reach a stage of development at which they can consciously use each process relatively easily as the situation requires. For most people, however, striving for a comfortable and effective expression of the four mental processes is an interesting lifelong challenge, and they have no expectation that they will arrive at a predetermined level of development. Jung saw the lifelong striving for individuation as more important than a hypothetical, typically unattainable, end point.

Influence of the Environment on Type Development

In the Jung-Myers type model, environment becomes extremely important because it can foster development of an individual’s natural preferences, or it can discourage it by reinforcing activities that are less satisfying, less comfortable, and less motivating. When the environment does not support an individual’s natural preferences, or when it actively discourages them, that individual may have difficulty acquiring confidence and skill in using his or her most natural and fulfilling personality characteristics.

In fact, environmental interference with type development can distort or suppress an individual’s natural type. Individuals who experience this may become skillful in using an initially less preferred process but may also be less content, feel less competent, or be out of touch with their best qualities. In contrast, other individuals may recognize and affirm their type but, for one reason or another, develop one or more aspects or facets of a less preferred process or orientation. These developments may become habitual, comfortable, and adaptive parts of their personality. Out-of-preference facet results on the MBTI Step II™ assessment provide evidence of this kind of type development. Facet results can also help individuals who have doubts about their natural preferences differentiate between their natural Step I™ preferences and accommodations they may have made in response to environmental pressures. Chapter 6, which focuses on the administration and interpretation of the MBTI Step I and Step II assessments, discusses strategies for helping individuals better identify their true preferences by understanding their Step II facet results.

The MBTI assessment does not necessarily reveal the extent to which natural development has been fostered or thwarted or what effect this may have had on a particular individual. Hypotheses and insights into this area can be found in the *MBTI® Step III™ Manual* (Myers et al., 2009), which provides an explanation of and an abundance of material supporting Isabel Myers’ theory of personality development.

Rationale for Type Development Theory

As described earlier in this chapter, when a person tries to attend to two opposites at the same time, the opposites “jam” each other and no clear signal comes through. For example, an individual who is “pulled” equally toward Sensing and Intuition might focus on concrete facts in the present, move quickly to future possibilities, switch to an entirely different set of facts, and so on. Or such an individual might just be frozen, not knowing what to focus on or what kind of information he or she needs to make a decision. Persistent vacillation between perceiving processes would likely produce confusion and frustration for the individual. At the point at which

a decision needed to be made based on available information, Sensing facts and Intuitive possibilities might be indistinguishable from one another. Or, such an individual might have a clear preference for Sensing or Intuition but vacillate between Thinking and Feeling decision making. The simultaneous pull of logic and values might cancel each other out, again causing confusion and frustration. So if either one of a pair of processes is to develop, the other must be suppressed most of the time in order to give the individual a chance to gain experience and confidence in his or her natural, preferred process. Once a preferred process is secure and habitual, which might take years or decades, however, the individual can (and probably should) consider the less preferred process in order to enrich his or her perception and judgment.

In Jungian typology, therefore, optimum use of the four processes is to be obtained not through equality but rather through selective development of each process in proportion both to its relative importance to the individual and to its useful relationship to the other processes. The general goal of development is as follows:

- **Become differentiated with one's preferred processes**—that is, adequately develop a favorite, most conscious, process used primarily in the preferred orientation and a complementary and balancing, but somewhat less conscious, second process that is used primarily in the less preferred orientation.
- **Eventually admit the least developed processes to conscious, purposeful use** in the service of the favorite process, or for selected tasks, or for limited periods of time. This may require the individual to temporarily “turn off” the favorite and second processes so that the third or fourth process can become more accessible. For example, when an INFP writer reviews and edits a draft of her manuscript she uses her favorite and second processes (Feeling and Intuition) to be sure that the text is communicating what she intended in the way she intended. She then turns off Feeling and Intuition so that she can notice problems in the logical flow of the work, inconsistencies in headings, grammatical errors, and other issues or errors of logic and fact. The writer barely notices these Thinking and Sensing concerns while the work is being developed and, if she does detect them, is unable to make those corrections while at the same time paying attention to the combination of her Feeling judgment and Intuitive focus that naturally inform her writing. She must purposely select a time to do the Thinking and Sensing tasks required.
- **Use each of the processes for the tasks for which it is best fitted.** For example, an individual whose favorite process is Intuition may find that using his fourth process, Sensing, is the most effective way

of handling important financial records, and an individual whose favorite process is Thinking may find employees to be less resistant to organizational change when she uses her Feeling process to provide them with an opportunity to voice their personal concerns and also to really listen to and value those concerns. As shown in the next section, MBTI Step II results both reveal and encourage the use of less preferred aspects of an individual's type.

Contribution of MBTI® Step II™ Facets to Type Development

As components of each of the four pairs of opposites, facets give individuals insight into their distinctive ways of expressing their type. The particular patterns of facet scores may show the specific ways they habitually use the less preferred parts of their type. To the extent that such use moderates or prevents the one-sidedness that Jung warned about, one might, with some caution, hypothesize that this is evidence of type development. However, because MBTI Step II results are provided for each of the four pairs of opposites *separately*, there is no direct information about an individual's favorite orientation–process combination. Consider the following example:

- Step I results show John's favorite process to be introverted Feeling (his fourth process is extraverted Thinking).
- John is out-of-preference on the T–F Reasonable–Compassionate facet (i.e., he favors the Reasonable pole of the facet).
- John is also out-of-preference on the E–I Expressive–Contained facet (i.e., he is Expressive).

Can one assume that John habitually introverts the four T–F facets on which he scores on the Feeling side and extraverts his out-of-preference Reasonable result (attaching it to his out-of-preference Expressive facet)? Though it is tempting to make that assumption, there is no evidence to support taking such a leap. And the complexity of unknown and unmeasured Step I and Step II variables would make empirical verification problematic.

Exploring such variations within Step I type can enrich our hypotheses about individual type development and differentiation. This may be especially important if an individual must operate in a context that requires or rewards behavior associated with facet characteristics that are opposite his or her basic preference. Chapter 2 provided an example of individuals who prefer a perceiving orientation to the outside world (P) nevertheless scoring as Early Starting (the J side of the facet) rather than Pressure–Prompted (the P side) on the Early Starting–Pressure–Prompted facet of the J–P preference pair. The proposed hypothesis for this out-of-preference result is that demands of the workplace

reward Early Starting behavior and discourage employees from habitually completing tasks at the last minute. Perceiving employees whose natural inclination is to start working on a task only when the pressure of an impending deadline is upon them may over time learn to start earlier (and thus may respond to the Step II items toward Early Starting) in response to rebukes from their manager and from co-workers who need them to finish their assignment so they can finish theirs. This tactic may not be natural or comfortable for Pressure-Prompted individuals, but it may be a necessary adaptation to the demands of the workplace.

Another possible Step II indication of type development can be seen in Thinking types who are out-of-preference on the Critical–Accepting facet. Accepting Thinking types typically explain that they learned through experience that taking the time to solicit and take into account others' views instead of automatically offering their objective criticism actually improved their decision making by broadening their perspective and increased the chance of buy-in from others. Repeated success with this approach eventually made it a comfortable part of their personality. Anecdotal evidence indicates that Accepting Thinking types are highly regarded and sought after as teachers and supervisors.

Universality of Type

The growing evidence regarding the biological basis of personality type, which is described in chapter 9, gives credence to Jung's assumption that type is universal, that is, that the personalities of all human beings are structured in accordance with typological opposites. Further support for universality is reflected in the MBTI Global Step I and Step II assessments and in additional translations of other forms of the MBTI Step I and Step II assessments that are in use or in development. International users in widely divergent cultures report excellent results in applying personality type concepts. All four pairs of opposites and all 16 types have been recognized and validated, as have the Step II facets, in translated form. However, the way the opposite preferences are expressed and the variations within the Step II facets may vary somewhat across cultures. A discussion of this and related issues regarding the universality of type can be found in *Type and Culture* (Kirby, Kendall, & Barger, 2007). Some interesting observations have been reported by users in various countries. For example, on a management team in China, managers with a preference for Critical did not usually express their criticism directly or overtly. They would often ask employees many questions, hoping that their questions would lead the employees to realize their mistake on their own.

The Dynamics of the Type Descriptions

Once an individual has grasped the principles and mechanical steps of type dynamics, he or she can figure out the dynamics for all 16 types, as detailed in table 3.3. When the hierarchy of processes has been determined, the dynamic relationships for a type can be understood. These predictable dynamic relationships are the basis for and are evident in the type descriptions in *Introduction to Myers-Briggs® Type* (Myers, 2015) and in chapter 4, where the dynamic that is described in each paragraph of the 16 type descriptions is shown. As you will see, Myers' type descriptions are not random collections of adjectives or observations but are carefully constructed content intended to illustrate the dynamics of each type. The following example describes the way the four processes come into play and interact for the type ENTP. Note that, as in table 3.3, the numbers 1 through 4 are used to designate the hierarchy of the four processes. Number designations (or *first*, *second*, *third*, *fourth*) are often used in publications such as *Introduction to Myers-Briggs® Type*.

ENTP extraverts Intuition (N) as the favorite process and introverts Thinking (T) as the second process. In theory, therefore, ENTPs trust, use, and develop their extraverted Intuition (#1) most and shape their life to give themselves maximum freedom to pursue the goals of their extraverted Intuition (#1). They use their introverted Thinking (#2) in pursuit of something challenging to their Intuition but may not let their introverted Thinking (#2) veto anything their extraverted Intuition (#1) seriously desires. Being oriented to the challenge of new possibilities in the outer world (#1), they may often find themselves committed to too many projects. Intuitive types, especially those who extravert their Intuition, typically underestimate the time it takes to bring activities to completion. When an ENTP decides that there are too many activities for the time available, the introverted Thinking (#2) process is needed. The ENTP withdraws into the inner, introverted world, uses introverted Thinking to analyze the costs and consequences of continuing or dropping each activity, and then makes a logical judgment (#2) about which activities to drop. With more mature type development, the ENTP may learn to take time to pull back from the exciting challenges for action (#1, extraverted Intuition) and consider (#2, introverted Thinking) *before* taking on new commitments and decide whether the costs permit adding new activities.

The fourth, or inferior, process of ENTP is introverted Sensing (#4). ENTPs need to train themselves not to neglect inner realities (#4) in their enthusiasm for possibilities in the outside world (#1). Type theory says that for an ENTP introverted Sensing (#4) and Feeling (in either attitude; #3) develop more slowly than do the favorite and second processes. One way for an ENTP to develop introverted Sensing (#4) and Feeling (#3)

is by working on a project that appeals to his or her extraverted Intuition (#1). To accomplish the project, the ENTP will need to know many facts and be able to deal with many practical details; thus introverted Sensing (#4) comes into play to help extraverted Intuition (#1). Further, the project may require working as part of a team or persuading others; thus Feeling (#3) also comes in to help extraverted Intuition (#1). With good type development, individuals tend to use their less preferred processes *in the service of* their preferred processes rather than out of attraction to the less familiar processes themselves.

Individuals gain experience with their less preferred processes and develop their type by using their third (#3) and fourth (#4) processes *in the service of* their favorite (#1) and second (#2) processes.

The preceding description of the way an ENTP uses his or her type dynamics should not be taken literally as a hard-and-fast, methodical procedure that all ENTPs follow. As the MBTI Step II instrument shows, there is variation within Step I type. Some ENTPs who are out-of-preference on particular facets may engage the world somewhat differently. The point here is that the dynamics of any particular type tend to flow naturally and seamlessly from process to process, usually without the conscious attention of the individual.

The descriptions of the 16 types in *Introduction to Myers-Briggs® Type* are built on the same kinds of dynamic principles that were illustrated in the description of ENTP, above. The booklets in the *Introduction to Myers-Briggs® Type* series are valuable guides for describing the MBTI types. It is important to understand that the type descriptions are firmly based on the dynamics of type theory, supplemented by years of observation of the types and empirical results from research. Also note that these descriptions do not take into account the possibility of being out-of-preference on a facet.

CONCLUSION

This chapter placed the MBTI assessment in the context of Jung's development of the theory of psychological

types, beginning with his attempt to classify individuals into the two categories of extraverts and introverts, and his later addition of the two pairs of opposite processes (sensation versus intuition and thinking versus feeling). Myers and Briggs extended Jung's three-category system of pairs of opposites to include a fourth pair, the Judging and Perceiving orientations. The J–P pair describes identifiable behavior that is used by an individual in dealing with the outside world, in addition to enabling the determination of type dynamics.

The chapter also looked at definitions, descriptions, and examples of the opposite processes and orientations; the contrast between characteristics of the favorite and fourth processes; the four processes as expressed in their extraverted and introverted forms; and the dynamic interactions among the processes and orientations as illustrations of both the complexity and the practical applications implicit in using the MBTI assessment.

Jung's notion of type development and the goals of that development over one's life span were explained and illustrated, as was the contribution of the MBTI Step II facets to type development. The rationale and method for determining the dynamics of any four-letter type were shown, and a detailed example of how type dynamics emerge in one type was provided.

In summary, the MBTI assessment helps individuals identify their personality preferences on four separate pairs of opposites: E–I, S–N, T–F, and J–P. Two of these, S–N and T–F, describe opposite perceiving and judging processes. The other two, E–I and J–P, reflect opposite orientations. Together, these processes and orientations influence how an individual perceives a situation and decides on a course of action. Insight into variations within each type are described and put to practical use through the MBTI Step II facets.

The way individuals experience and express their type is akin to finding forks in the road of human development, offering different paths that lead toward different kinds of excellence. How far different individuals will go, how much excellence they will actually achieve, depends in part on their energy and aspirations. The kind of excellence toward which they are headed is determined, according to type theory, by their natural preferences that direct them at each fork in the road and by the way their environment may support or suppress their natural choices.

CHAPTER 4

The 16 Types and the 20 Facets

CHAPTER OVERVIEW

A Brief History of Myers' Type Descriptions

The 16 Step I™ Types

The 20 Step II™ Facets

Conclusion

This chapter provides descriptions of the 16 Step I™ types and the 20 Step II™ facets. The preceding chapter provided a solid grounding in the Jung-Myers theory, its logical rationale, and examples of how its orientation–process combinations result in specific and observable differences among the types. The first part of this chapter puts all that information together to show how each of the 16 whole types is more than the sum of its parts. The second part focuses on the 20 Step II facets and how they reveal individual variations among people of the same type.

A BRIEF HISTORY OF MYERS' TYPE DESCRIPTIONS

The first edition of the *Introduction to Type®* (ITT) booklet was self-published by Isabel Myers (1962a). The type descriptions therein followed essentially the same presentation as those in Myers' 1962 *MBTI® Manual* (1962b). Both publications devoted several paragraphs to the dominant function (favorite process), followed by two brief paragraphs describing the two auxiliary processes that complement that dominant process. Later editions (after 1980) of the ITT booklet devoted one full page to each of the 16 types, arranged so that the two types that share the same dominant process but have opposite auxiliary processes appeared on facing pages. The seventh edition of ITT (Myers, 2015) benefited from years of research and practitioner experience and contains type descriptions that are substantially more detailed than Myers' original descriptions. The ITT booklet is a standard vehicle used by practitioners to provide feedback to respondents and help them verify and understand their type and how it differs from the other 15 types.

Practitioners are encouraged to have the newest edition of the ITT booklet on hand when presenting type to respondents (see additional interpretation recommendations in chapter 6). The type descriptions presented in this chapter are from the seventh edition of ITT (Myers, 2015).

THE 16 STEP I™ TYPES

The unique contribution the MBTI® assessment makes to people's understanding of personality is its focus on the 16 types as broad, complex structures whose parts interact dynamically and develop over a person's life span. This section includes the following:

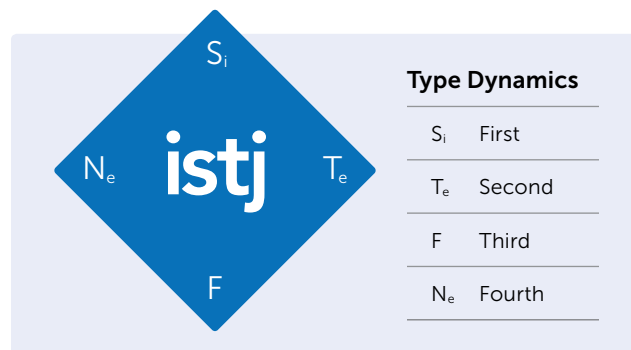
- A summary of the dynamics of each of the 16 types
- A type description for each type, including indications of which dynamic element is described in each paragraph

Table 4.1 provides a helpful introduction to and overview of the 16 types.

Descriptions of the 16 Types

The order of presentation of the 16 types follows that in the type table (see table 4.1), reading across each row, starting with ISTJ, which is the first type in the first row of the type table, and ending with ENTJ, the last type in the last row of the table. Note also that to reinforce readers' learning from chapter 3, the terms *dominant*, *auxiliary*, *tertiary*, and *inferior* are included in addition to the terms *favorite* (or *first*), *second*, *third*, and *fourth* to designate dynamic elements.

Each paragraph (or paragraphs) of each type description is preceded by a heading specifying the dynamic element that is being described—the type as a whole; the favorite (dominant) process; the second (auxiliary) process; the third (tertiary) process; the fourth (inferior) process—so that practitioners can better understand and explain to respondents the dynamic character of the types.



INTROVERSION SENSING THINKING JUDGING

At Their Best

WHOLE TYPE

ISTJs have a strong sense of responsibility and great loyalty to their organizations, families, and relationships. They work steadily to fulfill their commitments as promised and on time. They will go to almost any effort to complete something they see as necessary but will resist doing anything that doesn't make sense to them.

ISTJs generally prefer to work alone and be accountable for the results. However, they are comfortable working as part of a team when that is necessary to do the job right, when roles are clearly defined, and when everyone fulfills assigned responsibilities. Competence and responsibility are extremely important to ISTJs, who expect others to be as dutiful and trustworthy as they require themselves to be.

DOMINANT INTROVERTED SENSING

Characteristics of ISTJs ISTJs have a profound respect for facts. They use their Sensing primarily internally, where they have a storehouse of information upon which they draw to help them understand the present. Thus, they are likely to be

- Practical, sensible, and realistic
- Systematic

Source: Myers (2015), pp. 10–41. Copyright 1998, 2015 by Peter B. Myers and Katharine D. Myers. Used with permission.

Table 4.1 | Characteristics frequently associated with each type

ISTJ	ISFJ	INFJ	INTJ
Quiet, serious, succeed by being thorough and dependable. Practical, matter-of-fact, realistic, and responsible. Decide logically what should be done and work toward it steadily, regardless of distractions. Take pleasure in making everything orderly and organized—their work, their home, their life. Value traditions and loyalty.	Quiet, friendly, responsible, and conscientious. Committed and steady in meeting their obligations. Thorough, painstaking, and accurate. Loyal, considerate, notice and remember specifics about people who are important to them, concerned with how others feel. Strive to create an orderly and harmonious environment at work and at home.	Seek meaning and connection in ideas, relationships, and material possessions. Want to understand what motivates people and are insightful about others. Conscientious and committed to their firm values. Develop a clear vision about how best to serve the common good. Organized and decisive in implementing their vision.	Have original minds and great drive for implementing their ideas and achieving their goals. Quickly see patterns in external events and develop long-range explanatory perspectives. When committed, organize a job and carry it through. Skeptical and independent, have high standards of competence and performance—for themselves and others.
ISTP	ISFP	INFP	INTP
Tolerant and flexible, quiet observers until a problem appears, then act quickly to find workable solutions. Analyze what makes things work and readily get through large amounts of data to isolate the core of practical problems. Interested in cause and effect, organize facts using logical principles, value efficiency.	Quiet, friendly, sensitive, and kind. Enjoy the present moment, what's going on around them. Like to have their own space and to work within their own time frame. Loyal and committed to their values and to people who are important to them. Dislike disagreements and conflicts, don't force their opinions or values on others.	Idealistic, loyal to their values and to people who are important to them. Want to live a life that is congruent with their values. Curious, quick to see possibilities, can be catalysts for implementing ideas. Seek to understand people and to help them fulfill their potential. Adaptable, flexible, and accepting unless a value is threatened.	Seek to develop logical explanations for everything that interests them. Theoretical and abstract, interested more in ideas than in social interaction. Quiet, contained, flexible, and adaptable. Have unusual ability to focus in depth to solve problems in their area of interest. Skeptical, sometimes critical, always analytical.
ESTP	ESFP	ENFP	ENTP
Flexible and tolerant, take a pragmatic approach focused on immediate results. Bored by theories and conceptual explanations; want to act energetically to solve the problem. Focus on the here-and-now, spontaneous, enjoy each moment that they can be active with others. Enjoy material comforts and style. Learn best through doing.	Outgoing, friendly, and accepting. Exuberant lovers of life, people, and material comforts. Enjoy working with others to make things happen. Bring common sense and a realistic approach to their work, and make work fun. Flexible and spontaneous, adapt readily to new people and environments. Learn best by trying a new skill with other people.	Warmly enthusiastic and imaginative. See life as full of possibilities. Make connections between events and information very quickly, and confidently proceed based on the patterns they see. Want a lot of affirmation from others, and readily give appreciation and support. Spontaneous and flexible, often rely on their ability to improvise and their verbal fluency.	Quick, ingenious, stimulating, alert, and outspoken. Resourceful in solving new and challenging problems. Adept at generating conceptual possibilities and then analyzing them strategically. Good at reading other people. Bored by routine, will seldom do the same thing the same way, apt to turn to one new interest after another.
ESTJ	ESFJ	ENFJ	ENTJ
Practical, realistic, matter-of-fact. Decisive, quickly move to implement decisions. Organize projects and people to get things done, focus on getting results in the most efficient way possible. Take care of routine details. Have a clear set of logical standards, systematically follow them and want others to also. Forceful in implementing their plans.	Warmhearted, conscientious, and cooperative. Want harmony in their environment, work with determination to establish it. Like to work with others to complete tasks accurately and on time. Loyal, follow through even in small matters. Notice what others need in their day-to-day lives and try to provide it. Want to be appreciated for who they are and what they contribute.	Warm, empathetic, responsive, and responsible. Highly attuned to the emotions, needs, and motivations of others. Find potential in everyone, want to help others fulfill their potential. May act as catalysts for individual and group growth. Loyal, responsive to praise and criticism. Sociable, facilitate others in a group, and provide inspiring leadership.	Frank, decisive, assume leadership readily. Quickly see illogical and inefficient procedures and policies, develop and implement comprehensive systems to solve organizational problems. Enjoy long-term planning and goal setting. Usually well informed, well read, enjoy expanding their knowledge and passing it on to others. Forceful in presenting their ideas.

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AUXILIARY EXTRAVERTED THINKING

ISTJs use Thinking in decision making, taking an objective, logical, and tough-minded approach. Their focus is on the task or system as a whole rather than on individuals. Thus, ISTJs tend to be

- Logical and analytical
- Detached and reasonable

WHOLE TYPE

ISTJs are clear and steadfast in their opinions because they have arrived at them by carefully and thoroughly applying logical criteria based on their experience and knowledge. They believe standard procedures exist because they work. They will support change only when facts demonstrate that it will bring about better results.

How Others May See Them

ISTJs are sociable when they are comfortable in the roles they are playing. However, they generally don't share their wealth of rich observations and memories except with close friends. Others see their standards and opinions, their desire for structure and schedules, but they may not see their individual, sometimes humorous, private reactions.

Although it can be hard for ISTJs to see the sense in needs that are very different from theirs, once they are convinced that something matters to a person they care about, that need becomes a fact. Then they will go to great lengths to meet the need, even while continuing to think it doesn't make sense.

Others usually see ISTJs as

- Calm, reserved, and serious
- Consistent and orderly
- Valuing traditions

Potential Areas for Growth

UNDEVELOPED INTROVERTED SENSING AND EXTRAVERTED THINKING

Sometimes life circumstances have not supported ISTJs in the development and expression of their favorite processes, Sensing and Thinking.

- If they have not developed their Sensing, ISTJs may rush into premature decisions and actions without considering new information.
- If they have not developed their Thinking, they may not have reliable ways of dealing with the outside world and instead may focus solely on their memories and internal data.

EXAGGERATED ISTJ

If ISTJs don't find a place where they can use their gifts and be appreciated for their contributions, they usually feel frustrated and may

- Become rigid about time, schedules, and procedures
- Be critical and judgmental
- Find it difficult to delegate—to trust anyone else to do the job right

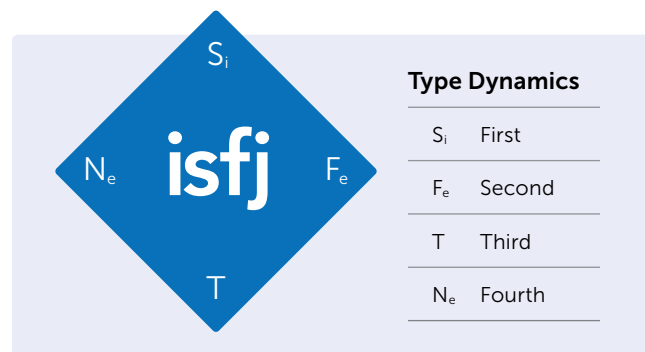
NEGLECT OF INFERIOR INTUITION AND TERTIARY FEELING

It is natural for ISTJs to give less attention to their less favored processes, Intuition and Feeling. If they neglect them too much, however, they may

- Not see the wider ramifications of current, expedient decisions
- Concentrate on logic so much that they don't consider impacts on people
- Fail to respond appropriately to others' needs for connection and intimacy

INFERIOR FUNCTION FORM

When greatly stressed, ISTJs may be unable to use their customary calm, reasonable judgment and get caught up in imagining negative possibilities for themselves and others in both the immediate and distant future.



INTROVERSION SENSING FEELING JUDGING

At Their Best

WHOLE TYPE

ISFJs are dependable and considerate, committed to the people and groups they are associated with, and faithful in carrying out their responsibilities. They work steadily to complete jobs fully and on time. They will go to great lengths to do something they see as necessary but dislike doing anything that doesn't make sense to them.

ISFJs focus on what people need and want, and they establish orderly procedures to ensure that those needs and wants are met. They take roles and responsibilities seriously and want others to do the same. Family is extremely important to ISFJs, who fulfill their roles conscientiously and expect other family members to do likewise.

DOMINANT INTROVERTED SENSING

Characteristics of ISFJs ISFJs have a realistic and practical respect for facts. They use their Sensing primarily internally, where they have a large storehouse of information. They can clearly remember details about things that have personal meaning for them, such as tones of voice and facial expressions. Thus, ISFJs are likely to be

- Practical and realistic
- Concrete and specific

AUXILIARY EXTRAVERTED FEELING

ISFJs use Feeling to make decisions based on their personal values and concern for others. They value harmony and cooperation and work to create them. Thus, they are likely to be

- Cooperative and thoughtful
- Kind and sensitive

WHOLE TYPE

Their opinions are firm because their decisions are based on careful application of their clear values and their wealth of stored information. ISFJs respect established procedures and authority, believing that these have persisted because they work well. Therefore, they will support change only when new data show it will be of practical benefit to people.

How Others May See Them

ISFJs are unassuming and quiet in their interactions, often putting the needs of others—especially family members—ahead of their own. They are uncomfortable with confrontation and will try hard to accommodate others, though their respect for traditions and people's feelings can lead them to challenge actions they perceive as hurtful or insensitive. People see their values, their desire for structure and closure, their kindness. What others may not see is their wealth of rich, accurate internal Sensing impressions and memories.

Others usually see ISFJs as

- Quiet, serious, and conscientious
- Considerate, good caretakers
- Honoring commitments, preserving traditions

Potential Areas for Growth

UNDEVELOPED INTROVERTED SENSING AND EXTRAVERTED FEELING

Sometimes life circumstances have not supported ISFJs in the development and expression of their favorite processes, Sensing and Feeling.

- If they have not developed their Sensing, ISFJs may rush into value judgments or taking care of others without considering the practical realities.
- If they have not developed their Feeling, they may not have reliable ways of dealing with the outside world and instead may focus solely on their impressions and memories.

EXAGGERATED ISFJ

If ISFJs don't find a place where they can use their gifts and be appreciated for their contributions, they usually feel frustrated and may

- Become rigid in supporting hierarchy, authority, and procedures
- Feel unappreciated and resentful—complain a lot
- Be overly focused on immediate impacts of decisions

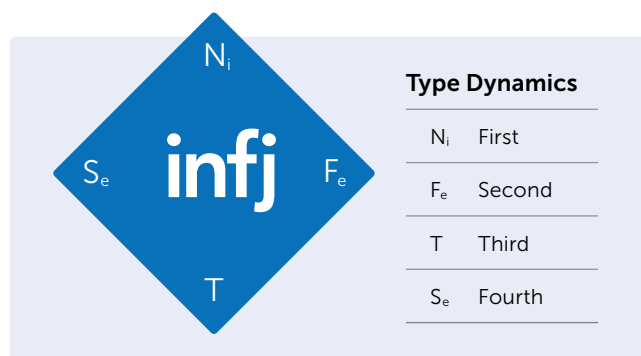
NEGLECT OF INFERIOR INTUITION AND TERTIARY THINKING

It is natural for ISFJs to give less attention to their less favored processes, Intuition and Thinking. If they neglect them too much, however, they may

- Not see the wider ramifications of current decisions or procedures
- Find it difficult to assert their needs
- Be uncomfortable applying impersonal criteria to decisions, even when necessary

INFERIOR FUNCTION FORM

When greatly stressed, ISFJs can get caught up in imagining negative possibilities. They may then express these without their usual consideration for the impact on people around them.



INTROVERSION INTUITION FEELING JUDGING

At Their Best

WHOLE TYPE

INFJs have a gift for intuitively understanding complex meanings and human relationships. They have faith in their insights and find that they often empathically understand people's feelings and motivations before the people themselves are aware of them. They combine this empathic understanding with the drive and organization to implement global plans for enhancing people's lives.

INFJs have a visionary grasp of human relationships and possibilities, which, when shared, can elevate and inspire others.

DOMINANT INTROVERTED INTUITION

Characteristics of INFJs INFJs seek meaning and connection in their lives. They have little use for details unless they fit with their inner vision. They use their Intuition primarily internally, where they develop sweeping, complex pictures and understandings. INFJs are likely to be

- Insightful, creative, and visionary
- Conceptual, symbolic, and metaphorical
- Idealistic, complex, and deep

AUXILIARY EXTRAVERTED FEELING

INFJs make decisions by applying their personal values and empathizing with others. They are loyal to people and institutions that exemplify their values but have little interest in those that don't. INFJs prefer to lead persuasively by sharing their vision. They are likely to be

- Sensitive, compassionate, and empathetic
- Deeply committed to their values

WHOLE TYPE

INFJs want meaning and purpose in their work, their relationships, even their possessions. They are invested in growth and development for themselves

and their significant others and are willing to consider unconventional paths to achieve these. They value the depth and complexity of their insights and creative gifts as well as those of others. They want to see these insights come to fruition.

How Others May See Them

INFJs readily show compassion and caring for others, but they share their internal intuitions only with people they trust. Because they keep this most valued, important part of themselves private, others may find them difficult to know. When they try to communicate their internal sense of "knowing," they often express it metaphorically and with complexity. They especially value authenticity and commitment in relationships.

Though INFJs are usually reserved, they don't hesitate to assert themselves when their values are violated. Then they can be persistent and insistent.

Others usually see INFJs as

- Private, even mysterious
- Intense and individualistic

Potential Areas for Growth

UNDEVELOPED INTROVERTED INTUITION AND EXTRAVERTED FEELING

Sometimes life circumstances have not supported INFJs in the development and expression of their favorite processes, Intuition and Feeling.

- If they have not developed their Intuition, INFJs may take in insufficient information or only information that fits with their internal pictures. Then they will make poor decisions based on distorted or limited information.
- If they have not developed their Feeling, they may not have reliable ways of making decisions and accomplishing their goals. Then, their valuable insights and creativity stay locked inside.

EXAGGERATED INFJ

If INFJs don't find a place where they can use their gifts and be appreciated for their contributions, they usually feel frustrated and may

- Not share with others the information they used to arrive at a decision, and thus seem arbitrary
- Base their conclusions on little data, on a sense of "knowing" that has little basis in reality
- Withdraw their energy and insight
- Become resentful and critical

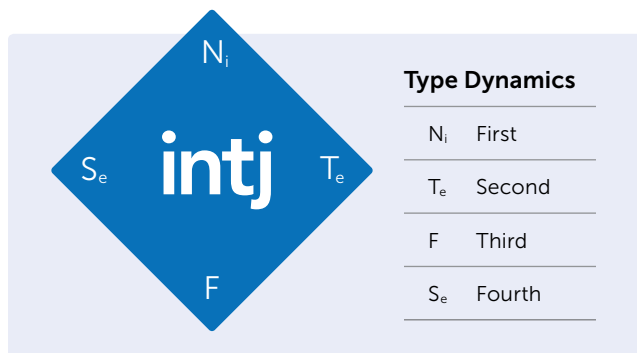
NEGLECT OF INFERIOR SENSING AND TERTIARY THINKING

It is natural for INFJs to give less attention to their less favored processes, Sensing and Thinking. If they neglect them too much, however, they may

- Be unable to verbalize their insights in a way others can understand
- Fail to check their insights against reason and practicality and end up pursuing a vision that has little possibility of being realized
- Become single-minded in pursuit of a vision

INFERIOR FUNCTION FORM

When greatly stressed, INFJs may become obsessed with information they normally would consider irrelevant or overindulge in Sensing activities, such as watching too much TV, overeating, or buying things that have little meaning for them.



INTROVERSION INTUITION THINKING JUDGING

At Their Best

WHOLE TYPE

INTJs have a clear vision of future possibilities coupled with the drive and organization to implement their ideas. They love complex challenges and readily synthesize complicated theoretical and abstract matters. Once they have created their general structure, they devise strategies to achieve their goals. Their global thinking leads them to develop visionary goals and an overarching plan for achieving these within large organizational structures.

INTJs value knowledge and expect competence from themselves and others. They hate confusion, mess, and inefficiency.

DOMINANT INTROVERTED INTUITION

Characteristics of INTJs INTJs see things from a global perspective and quickly relate new information to overall patterns. They trust their insightful connections regardless of established authority or popular opinion. Dull routine smothers their creativity. INTJs use their Intuition primarily internally, where they develop complex mental images of the future. They are likely to be

- Insightful, creative synthesizers
- Conceptual, long-range thinkers

AUXILIARY EXTRAVERTED THINKING

INTJs use their Thinking to make logical decisions. They assess everything with a critical eye, quickly identify problems to solve, and are tough and decisive when the situation calls for it. INTJs tend to be

- Clear and concise
- Rational, detached, and objectively critical

WHOLE TYPE

INTJs are excellent long-range planners and often rise to positions of leadership in groups and organizations. They are independent, trust their own perceptions and opinions more than those of others, and apply their high standards of knowledge and competence most rigorously to themselves.

How Others May See Them

INTJs present a calm, decisive, and assured face to the world, though they may find it difficult to engage in social conversation. They usually don't directly express their most valued and valuable part, their creative insights. Instead, they translate their insights into logical decisions, opinions, and plans, which they often express clearly. Because of this, others sometimes experience INTJs as intractable, much to the surprise of the INTJs, who are very willing to change an opinion when new evidence emerges.

Others usually see INTJs as

- Private, reserved, hard to know
- Conceptual, original, and independent

Potential Areas for Growth

UNDEVELOPED INTROVERTED INTUITION AND EXTRAVERTED THINKING

Sometimes life circumstances have not supported INTJs in the development and expression of their favorite processes, Intuition and Thinking.

- If they have not developed their Intuition, INTJs may take in insufficient information or only information that fits their insights. Then they may make poor decisions based on limited information.
- If they have not developed their Thinking, they may not have reliable ways to translate their valuable insights into achievable goals.

EXAGGERATED INTJ

If INTJs don't find a place where they can use their gifts and be appreciated for their contributions, they usually feel frustrated and may

- Appear aloof and abrupt, not giving enough information about their internal processing
- Be critical of those who don't see their vision right away
- Become single-minded and unyielding in pursuing their vision

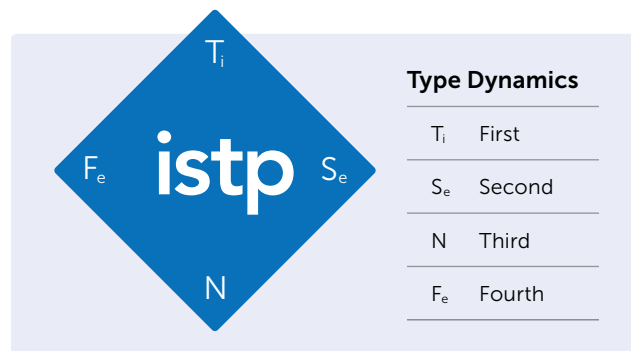
NEGLECT OF INFERIOR SENSING AND TERTIARY FEELING

It is natural for INTJs to give less attention to their less favored processes, Sensing and Feeling. If they neglect them too much, however, they may

- Overlook details or facts that don't fit into their Intuitive patterns
- Engage in "intellectual games," raising frivolous objections to abstract issues and terms that have little meaning or relevance to others
- Not fully appreciate the impact of their decisions on individuals
- Fail to give as much praise or intimate connection as others desire

INFERIOR FUNCTION FORM

When greatly stressed, INTJs may overindulge in Sensing activities—for example, watching too much TV, playing video games for long periods, or overeating—or become overly focused on specific details in their environment that they normally don't notice or usually see as unimportant.



INTROVERSION SENSING THINKING PERCEIVING

At Their Best

WHOLE TYPE

ISTPs carefully observe what is going on around them. Then, when the need arises, they move quickly to get to the core of a problem and solve it with the greatest efficiency and the least effort. They are interested in how and why things work but find abstract theories uninteresting unless they can quickly apply them. They often act as troubleshooters.

ISTPs resist regimentation and rules, thrive on variety and novelty, and enjoy the challenge of solving a new, concrete, extensive problem.

DOMINANT INTROVERTED THINKING

Characteristics of ISTPs ISTPs use their Thinking primarily internally to look for the essential structure underlying the facts. Their minds seem to work almost like computers, organizing data, reasoning impersonally and objectively. They make rational decisions based on a great deal of concrete data. ISTPs are likely to be

- Detached and objective critics
- Analytical and logical problem solvers

AUXILIARY EXTRAVERTED SENSING

ISTPs are realists, focusing on what is and what can be done with it rather than on theoretical possibilities. They are often creative at dealing with immediate problems and good at accomplishing hands-on tasks. ISTPs are likely to be

- Practical and realistic
- Factual and pragmatic

WHOLE TYPE

ISTPs are expedient and believe in economy of effort—doing only what is necessary with the least possible discussion and hassle. Their focus is on getting the desired results.

How Others May See Them

ISTPs want everyone treated equally and generally will tolerate a wide range of behaviors—until their ruling logical principles are attacked. At that point, they can surprise others by expressing their firm and clear opinions. ISTPs listen and seem to agree because they are not disagreeing; later, others may find that the ISTP was analyzing and making internal judgments.

With their constant scanning for information and focus on results, ISTPs will change course readily if they see another, more efficient way. Because of this, others sometimes have trouble “reading” them. They tend to be quiet and reserved, though they can be quite talkative in areas in which they have a lot of knowledge.

Others usually see ISTPs as

- Adaptable, action-oriented risk takers
- Confident, independent, and self-determined

Potential Areas for Growth

UNDEVELOPED INTROVERTED THINKING AND EXTRAVERTED SENSING

Sometimes life circumstances have not supported ISTPs in the development and expression of their favorite processes, Thinking and Sensing.

- If they have not developed their Thinking, ISTPs may get caught up in what is happening around them and not take time to do the internal logical processing they need to make good decisions. Then their actions may be haphazard responses to immediate needs.
- If they have not developed their Sensing, they may have no reliable way of getting accurate data about the external world or of translating their thoughts into action.

EXAGGERATED ISTP

If ISTPs don't find a place where they can use their gifts and be appreciated for their contributions, they usually feel frustrated and may

- Become cynical and negative critics
- Withdraw their attention and energy
- Postpone decisions

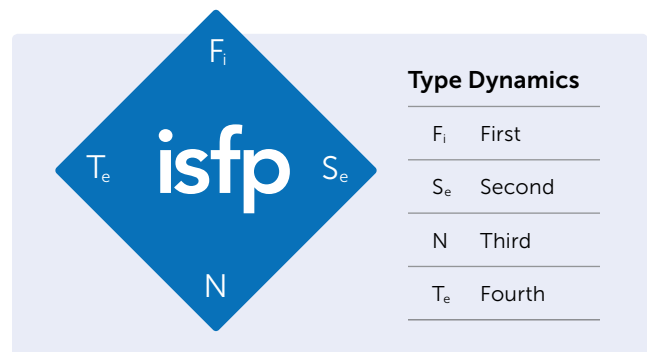
NEGLECT OF INFERIOR FEELING AND TERTIARY INTUITION

It is natural for ISTPs to give less attention to their less favored processes, Feeling and Intuition. If they neglect them too much, however, they may

- Overlook others' emotional needs and values
- Not fully appreciate the impact of their decisions on others
- Focus so intently on immediate results that they lose track of the long-term ramifications of their decisions and actions

INFERIOR FUNCTION FORM

When greatly stressed, ISTPs may erupt outwardly in inappropriate displays of emotion. The resulting explosive anger or hurt tearfulness is quite unnerving to others and embarrassing to the usually calm and controlled ISTP.



INTROVERSION SENSING FEELING PERCEIVING

At Their Best

WHOLE TYPE

ISFPs live in the present with a quiet sense of joyfulness. They want time to experience each moment. They prize the freedom to follow their own course, have their own space, and set their own schedule, and they extend the same freedom and tolerance to others. They are faithful in fulfilling obligations to people and things that are important to them.

ISFPs develop intimacy with others slowly, but once they do, they consider those relationships very important. They express their devotion to others spontaneously in many quiet ways.

DOMINANT INTROVERTED FEELING

Characteristics of ISFPs ISFPs are guided by their strong inner values and want their outer life to demonstrate those values. They want their work to be more than just

a job; they want to contribute to people's well-being or happiness. They don't enjoy routine but will work with energy and dedication when doing something they believe in. ISFPs are likely to be

- Trusting, kind, and considerate
- Sensitive and gentle

AUXILIARY EXTRAVERTED SENSING

ISFPs are acutely aware of the people and the world around them. They learn by doing more than by reading or hearing and get involved in day-to-day activities that take care of necessary tasks. ISFPs are likely to be

- Observant
- Realistic, practical, concrete, and factual

WHOLE TYPE

ISFPs are attuned to the feelings and needs of others and flexible in responding to them. They often have an affinity for nature and for beauty in all living things—people, plants, and animals. They prize most those who take time to understand their values and goals and who support them in achieving those goals in their own way.

How Others May See Them

ISFPs are adaptable and flexible unless something that matters strongly to them is endangered. Then they stop adapting. They care deeply about people but may show it through doing things for others more than through words.

ISFPs tend to be quiet and unassuming, and their warmth, enthusiasm, and playful humor may not be apparent to people who don't know them well. They prefer to observe and support rather than organize situations. They don't like to dominate.

ISFPs may be underestimated by others and may also underrate themselves. They often take for granted what they do well and overemphasize the contrast between their inner standards and their actual behavior and accomplishments.

Others usually see ISFPs as

- Quiet, reserved, and private—hard to know well
- Spontaneous and tolerant

Potential Areas for Growth

UNDEVELOPED INTROVERTED FEELING AND EXTRAVERTED SENSING

Sometimes life circumstances have not supported ISFPs in the development and expression of their favorite processes, Feeling and Sensing.

- If they have not developed their Feeling, ISFPs may become overly caught up in the details of everyday life and not take time for the internal valuing process by which they make their best decisions. They may avoid making decisions, allowing others or circumstances to decide for them.
- If they have not developed their Sensing, they may have no reliable way of getting accurate data about the external world or of actualizing their values. Their decisions will be based on little information and be overly personal.

EXAGGERATED ISFP

If ISFPs don't find a place where they can use their gifts and be appreciated for their contributions, they usually feel frustrated and may

- Withdraw from people and situations
- Be excessively self-critical
- Passively resist structures and rules
- Feel unappreciated and undervalued

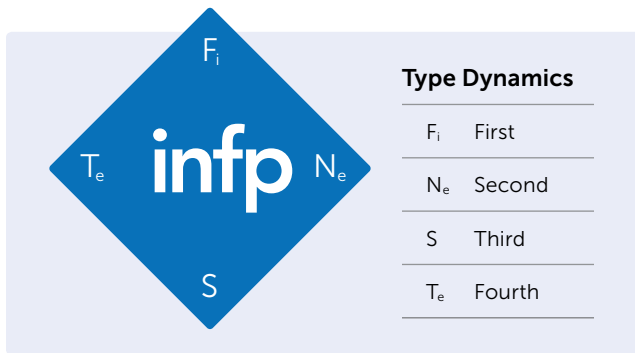
NEGLECT OF INFERIOR THINKING AND TERTIARY INTUITION

It is natural for ISFPs to give less attention to their less favored processes, Thinking and Intuition. If they neglect them too much, however, they may

- Reject or not take seriously logical systems
- Feel ill-equipped to deal with complexity
- Not always see the wider ramifications of their specific, immediate decisions

INFERIOR FUNCTION FORM

When greatly stressed, ISFPs can become very critical of themselves and others, verbalizing harsh and negative judgments.



INTROVERSION INTUITION FEELING PERCEIVING

At Their Best

WHOLE TYPE

INFPs have an inner core of values that guides their interactions and decisions. They want to be involved in work that contributes to their own and others' growth and inner development—to have a purpose beyond their paycheck. They make it a priority to clarify their values and live in congruence with them.

INFPs recognize and honor the emotional and psychological needs of others, even when those others may not have recognized or expressed their own needs.

DOMINANT INTROVERTED FEELING

Characteristics of INFPs INFPs use their Feeling primarily internally, where they make decisions based on their values of self-understanding, individuality, and growth. They live by a moral commitment to what they believe in and are likely to be

- Sensitive, concerned, and caring
- Idealistic and loyal to their ideas

AUXILIARY EXTRAVERTED INTUITION

INFPs enjoy reading, discussing, and reflecting on possibilities for positive change in the future. They are curious about ideas and quick to see connections and meanings. INFPs are likely to

- Be curious and creative
- Have long-range vision

WHOLE TYPE

INFPs are usually fascinated by opportunities to explore the complexities of human personality. They tend to work in bursts of energy and are capable of great concentration and output when fully engaged in a project. They are generally faithful in fulfilling obligations related to people, work, or ideas to which they are

committed, but they can have difficulty performing routine work that has little meaning for them.

How Others May See Them

INFPs find structures and rules confining and prefer to work autonomously. They are adaptable and flexible until something violates their inner values. Then they stop adapting. The resulting expression of value judgments can emerge with an intensity that is surprising to others.

INFPs tend to be reserved and selective about sharing their most deeply held values and feelings. They value relationships based on depth, authenticity, true connection, and mutual growth. INFPs prize most those who take time to understand their values and goals.

Others usually see INFPs as

- Sensitive, introspective, and complex
- Original and individual
- Sometimes difficult to understand

Potential Areas for Growth

UNDEVELOPED INTROVERTED FEELING AND EXTRAVERTED INTUITION

Sometimes life circumstances have not supported INFPs in the development and expression of their favorite processes, Feeling and Intuition.

- If they have not developed their Feeling, INFPs may not take time for the inner valuing process by which they make their best decisions, instead going from one exciting possibility to another and achieving little.
- If they have not developed their Intuition, they may not have reliable ways to take in information and may fail to notice the realities of situations. Then they may make decisions based solely on personal values and find it difficult to translate their values into action.

EXAGGERATED INFP

If INFPs don't find a place where they can use their gifts and be appreciated for their contributions, they usually feel frustrated and may

- Have uncharacteristic difficulty expressing themselves verbally
- Withdraw from people and situations
- Not give enough information to others, especially about important values

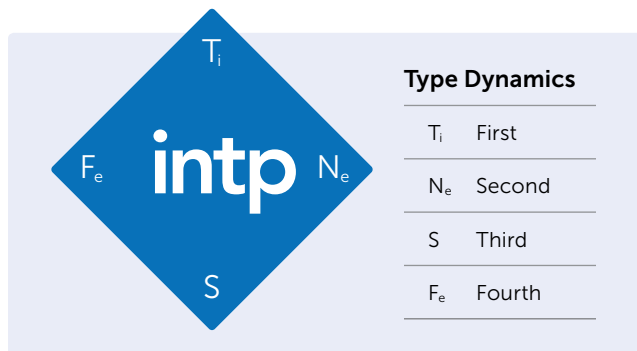
NEGLECT OF INFERIOR THINKING AND TERTIARY SENSING

It is natural for INFPs to give less attention to their less favored processes, Thinking and Sensing. If they neglect them too much, however, they may

- Become easily discouraged about the contrast between their ideals and their accomplishments
- Reject logical reasoning even in situations that require it, asserting the supremacy of their internal viewpoint
- Be impractical and have difficulty estimating the resources required to reach a desired goal

INFERIOR FUNCTION FORM

When greatly stressed, INFPs may doubt their own and others' competence, becoming overly critical and judgmental.



INTROVERSION INTUITION THINKING PERCEIVING

At Their Best

WHOLE TYPE

INTPs are independent problem solvers who excel at providing a detached, concise analysis of an idea or situation. They ask the difficult questions, challenging others and themselves to find new logical approaches.

INTPs' best work may emerge when they are allowed to work independently on a problem whose solution requires an approach that runs counter to prevailing wisdom or knowledge. Though they typically work best alone, their incisive critiques and summaries can assist a group in getting to the core of complex problems.

DOMINANT INTROVERTED THINKING

Characteristics of INTPs INTPs use their Thinking primarily internally to find or develop underlying principles and logical structures for understanding and explaining the world. They approach almost everything with skepticism, form their own opinions and standards, and apply these standards rigorously to themselves. They highly value intelligence and competence. INTPs are likely to be

- Logical, analytical, and objectively critical
- Detached and contemplative

AUXILIARY EXTRAVERTED INTUITION

INTPs see possibilities and connections beyond the present and obvious. They are curious and seek knowledge for its own sake. They love to theorize and discuss abstractions. INTPs are usually

- Mentally quick, insightful, and ingenious
- Intensely curious about ideas, theories, and what makes things work

WHOLE TYPE

INTPs quickly see inconsistencies and illogical thinking and enjoy taking ideas apart and reworking them. They naturally build complex theoretical systems to explain the realities they see. They find it difficult to work on routine tasks but bring great energy, intensity, and focus to researching or analyzing a complex problem that arouses their curiosity.

How Others May See Them

INTPs are usually quiet and reserved, though they can be talkative in areas in which they are especially knowledgeable. Unless their work requires action, they are more interested in the challenge of finding solutions than in putting solutions to practical use. They prefer not to organize people or situations.

INTPs are tolerant of a wide range of behavior, arguing and raising issues only when they believe it is reasonable to do so. This flexibility disappears, however, when their ruling principles are challenged. Then they stop adapting.

INTPs prize precision in communication and dislike redundancy and stating the obvious. They want to express the exact truth, but they may make it so complex that others have difficulty understanding.

Others usually see INTPs as

- Quiet, contained, calm, and detached observers
- Independent, valuing autonomy

Potential Areas for Growth

UNDEVELOPED INTROVERTED THINKING AND EXTRAVERTED INTUITION

Sometimes life circumstances have not supported INTPs in the development and expression of their favorite processes, Thinking and Intuition.

- If they have not developed their Thinking, INTPs may go from insight to insight, never analyzing them critically or integrating them into a whole.

- If they have not developed their Intuition, they may have no reliable way of taking in information and so may become immersed in their internal logical systems. Then they find it difficult to actualize or even communicate their ideas.

EXAGGERATED INTP

If INTPs don't find a place where they can use their gifts and be appreciated for their contributions, they usually feel frustrated and may

- Become cynical and negative critics
- Be sarcastic and destructively critical
- Isolate themselves and put off action
- Engage in verbal sparring and arguments

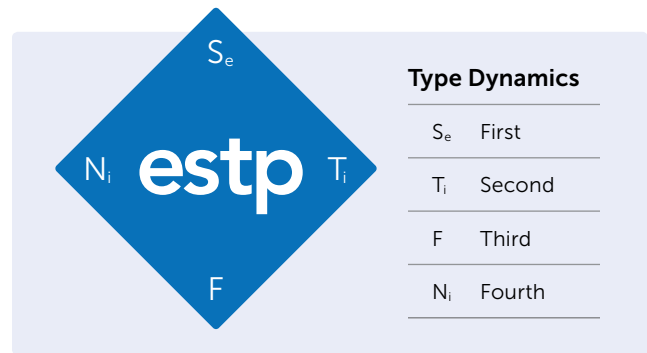
NEGLECT OF INFERIOR FEELING AND TERTIARY SENSING

It is natural for INTPs to give less attention to their less favored processes, Feeling and Sensing. If they neglect them too much, however, they may

- Be insensitive to others' needs for information and emotional connection
- Decide that something they or others value is not important because it is "not logical"
- Fail to consider the impact on others of their ideas or style of expression
- Be impractical, forgetting details such as appropriate dress, unpaid bills, physical needs

INFERIOR FUNCTION FORM

When greatly stressed, INTPs may erupt outwardly in inappropriate displays of emotion. The resulting explosive anger or hurt tearfulness is quite unnerving to others and embarrassing to the usually calm and controlled INTP.



EXTRAVERSION SENSING THINKING PERCEIVING

At Their Best

WHOLE TYPE

ESTPs are energetic, active problem solvers who respond creatively to challenging situations in their environment. They seldom let rules or standard procedures interfere, finding new ways to use existing systems. ESTPs develop easy ways to do difficult things and make their work fun. They are flexible, adaptable, inventive, and resourceful, can pull conflicting factions together, and are good team members.

They are popular companions for activities (parties, sports, or work) because of their zest for life and their enjoyment of the moment.

DOMINANT EXTRAVERTED SENSING

Characteristics of ESTPs ESTPs are interested in everything going on around them—activities, food, clothes, people, the outdoors, and anything that offers new experiences. Because they learn more from doing than from studying or reading, they tend to plunge into things, learning as they go, trusting their ability to respond resourcefully. ESTPs are likely to be

- Observant
- Practical and realistic
- Active, involved in immediate experiences

AUXILIARY INTROVERTED THINKING

ESTPs make decisions using logical analysis and reasoning and can be tough when the situation calls for it. They usually are

- Analytical, rational problem solvers
- Straightforward and assertive

WHOLE TYPE

ESTPs are expert at seeing the needs of the moment and reacting quickly to meet them. They generally prefer to deal flexibly with what is happening at the moment and not dwell on potential future possibilities. They good-naturedly take things as they are and seek satisfying solutions rather than impose a “should” or “must” of their own.

How Others May See Them

ESTPs love life and immerse themselves in it. Others respond to their enthusiasm and good humor. ESTPs are people of action. They usually dislike and avoid theory and written directions. They may have difficulty with traditional schooling, although they do well when they see the relevance of what they are learning and are allowed to experiment.

Others usually see ESTPs as

- Gregarious, fun loving, and spontaneous
- Adventurous risk takers
- Pragmatic troubleshooters

Potential Areas for Growth

UNDEVELOPED EXTRAVERTED SENSING AND INTROVERTED THINKING

Sometimes life circumstances have not supported ESTPs in the development and expression of their favorite processes, Sensing and Thinking.

- If they have not developed their Sensing, ESTPs may focus only on what’s immediately before them. Their decisions may then be limited to gratifying their sensual desires, particularly those involving physical challenge and risk.
- If they have not developed their Thinking, they will not have a useful way of selecting among the barrage of incoming sensory data. They may then have difficulty setting priorities or make poor decisions.

EXAGGERATED ESTP

If ESTPs don’t find a place where they can use their gifts and be appreciated for their contributions, they usually feel frustrated and may

- Have trouble accepting structure and meeting deadlines
- Focus entirely on excitement and activity, getting caught up in the moment
- Put enjoying life ahead of fulfilling important obligations

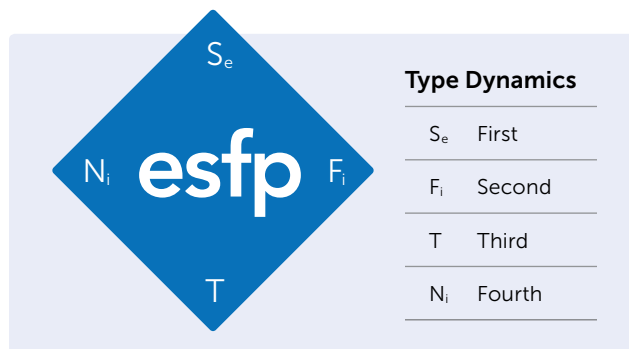
NEGLECT OF INFERIOR INTUITION AND TERTIARY FEELING

It is natural for ESTPs to give less attention to their less favored processes, Intuition and Feeling. If they neglect them too much, however, they may

- Not see the wider ramifications of their actions and decisions
- Forget dates and events that have special meaning to others
- Be unaware of the impact of their actions on others
- Be impatient with discussion or exploration of relationships

INFERIOR FUNCTION FORM

When greatly stressed, ESTPs may make negative inferences based on random observations. They may imagine that others don’t really care about them, then distort the facts to provide themselves with “evidence” of this neglect.



EXTRAVERSION SENSING FEELING PERCEIVING

At Their Best

WHOLE TYPE

ESFPs are exuberant lovers of life. They live in the moment and find enjoyment in people, food, clothes, animals, the natural world, and activities. They seldom let rules interfere with their lives, focusing on meeting human needs in creative ways.

ESFPs are excellent team players, oriented to accomplishing tasks with maximum fun and minimum hassle.

DOMINANT EXTRAVERTED SENSING

Characteristics of ESFPs ESFPs are interested in people and new experiences. Because they learn more from doing than from studying or reading, they tend to plunge into things, learning as they go. They appreciate and enjoy their possessions. ESFPs are likely to be

- Observant
- Practical, realistic, and specific
- Active, involved in immediate experiences

AUXILIARY INTROVERTED FEELING

ESFPs use their Feeling internally to make decisions based on their personal values by identifying and empathizing with others. They are good at interpersonal interactions and often play the role of peacemaker. Thus, ESFPs are likely to be

- Generous, optimistic, and persuasive
- Warm, sympathetic, and tactful

WHOLE TYPE

ESFPs are keen observers of people's behavior. They seem to sense what is happening with others and respond quickly to their practical needs. They are especially good at mobilizing people to deal with a crisis.

How Others May See Them

ESFPs have a lot of fun and are fun to be with. Their exuberance and enthusiasm draw others to them. They are flexible, adaptable, congenial, and easygoing. They seldom plan ahead, trusting their ability to respond in the moment and deal effectively with whatever presents itself. They hate structure and routine and will generally find ways to get around them.

ESFPs tend to learn by doing, by interacting with their environment. They usually dislike theory and written explanations. They may have difficulty with traditional schooling, although they do well when they see the relevance of what they are learning and are allowed to interact with people or the topics being studied.

Others usually see ESFPs as

- Resourceful and supportive
- Gregarious, fun loving, playful, and spontaneous

Potential Areas for Growth

UNDEVELOPED EXTRAVERTED SENSING AND INTROVERTED FEELING

Sometimes life circumstances have not supported ESFPs in the development and expression of their favorite processes, Sensing and Feeling.

- If they have not developed their Sensing, ESFPs may focus only on what's immediately before them. Their decisions may then be limited to gratifying their sensual desires, particularly those involving interactions with other people.

- If they have not developed their Feeling, they may get caught up in what's happening in the moment, with no mechanism for weighing or evaluating the situation, or for anchoring themselves.

EXAGGERATED ESFP

If ESFPs don't find a place where they can use their gifts and be appreciated for their contributions, they usually feel frustrated and may

- Become distracted and impulsive
- Have trouble accepting and meeting deadlines
- Personalize others' actions and decisions

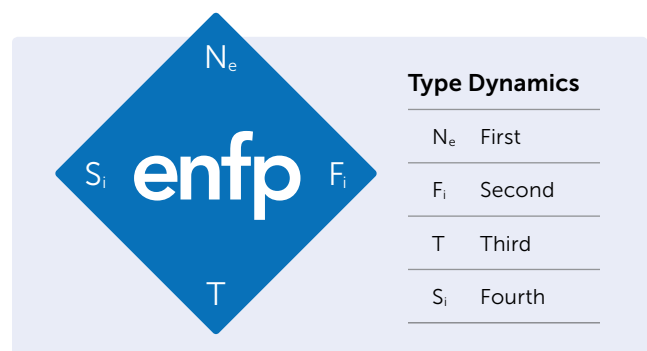
NEGLECT OF INFERIOR INTUITION AND TERTIARY THINKING

It is natural for ESFPs to give less attention to their less favored processes, Intuition and Thinking. If they neglect them too much, however, they may

- Fail to look at long-term consequences, acting on their own and others' immediate needs
- Avoid what they view as complex or ambiguous situations and people
- Put enjoyment ahead of obligations

INFERIOR FUNCTION FORM

When greatly stressed, ESFPs may feel overwhelmed by negative and baseless possibilities. They then put energy into justifying their interpretation of what is happening.



EXTRAVERSION INTUITION FEELING PERCEIVING

At Their Best

WHOLE TYPE

ENFPs see life as a creative adventure full of exciting possibilities. Unusually perceptive about people and the world, they are insightful about the present and future. ENFPs experience a wide range of feelings and intense

emotions. They need affirmation from others and readily give appreciation and support.

ENFPs are good at understanding how people and groups work and are persuasive and compelling in pursuing what is important to them. They are adaptable, making the best of any situation. Their energy and enthusiasm encourage others to make the best of things, too.

DOMINANT EXTRAVERTEDED INTUITION

Characteristics of ENFPs ENFPs are innovators, initiating projects and putting a lot of energy into getting them going. Using Intuition primarily externally, they are stimulated by new people, ideas, and experiences. They readily find meaning and significance in things and see connections that others don't. They are likely to be

- Curious, creative, and imaginative
- Energetic, enthusiastic, and spontaneous

AUXILIARY INTROVERTED FEELING

ENFPs value harmony and goodwill. They like to please people and will adapt to their needs and wishes when possible. ENFPs primarily use Feeling internally, making decisions by applying their personal values while keeping others' needs in mind. They are likely to be

- Warm, friendly, and caring
- Cooperative and supportive

WHOLE TYPE

ENFPs have exceptional insight into possibilities for others and the energy and motivation to help bring them to fruition. They feel confident moving ahead based on their insights, and their enthusiasm tends to bring others along with them.

How Others May See Them

ENFPs are usually lively, gregarious, and sociable, with a large circle of friends. They are interested in almost everything and have a zest for life that draws others to them. At the same time, they value depth and authenticity in their close relationships and will go to great lengths to create and support open and honest communication.

ENFPs hate routine, schedules, and structure and usually manage to avoid them. They are normally verbally fluent, even in extemporaneous situations; however, when talking about their deepest values, they may suddenly become awkward and express their opinions with uncharacteristic intensity.

Others usually see ENFPs as

- Personable, perceptive, and persuasive
- Enthusiastic, spontaneous, and versatile
- Giving and seeking affirmation

Potential Areas for Growth

UNDEVELOPED EXTRAVERTEDED INTUITION AND INTROVERTED FEELING

Sometimes life circumstances have not supported ENFPs in the development and expression of their favorite processes, Intuition and Feeling.

- If they have not developed their Intuition, ENFPs may fail to take in enough information, mistrust their own insights, be uncertain, and accept others' perceptions too quickly.
- If they have not developed their Feeling, they may go from enthusiasm to enthusiasm, never committing the energy needed to actualize their insights, or they may make overly personal decisions.

EXAGGERATED ENFP

If ENFPs don't find a place where they can use their gifts and be appreciated for their contributions, they usually feel frustrated and may

- Become scattered, have trouble focusing, be easily distracted
- Fail to follow through on decisions
- Become rebellious, excessively nonconforming
- Ignore deadlines and procedures

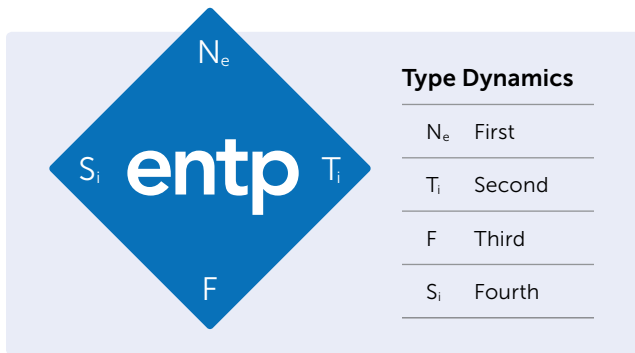
NEGLECT OF INFERIOR SENSING AND TERTIARY THINKING

It is natural for ENFPs to pay less attention to their less favored processes, Sensing and Thinking. If they neglect them too much, however, they may

- Not take care of the details and routine required for implementing their inspirations
- Overextend themselves—have trouble saying no to interesting possibilities and people
- Fail to apply reason and logic to assess their inspirations and decisions

INFERIOR FUNCTION FORM

When greatly stressed, ENFPs may get overwhelmed by detail and lose their normal sense of perspective and options. Then they tend to focus on an unimportant or distorted detail, fixating on it to the point that it becomes the central fact of their universe.



EXTRAVERSION INTUITION THINKING PERCEIVING

At Their Best

WHOLE TYPE

ENTPs constantly scan their environment for opportunities and possibilities. They see patterns and connections not obvious to others and at times seem able to see into the future. They are adept at generating conceptual possibilities and then analyzing them strategically.

ENTPs are good at understanding how systems work and are enterprising and resourceful in maneuvering within them to achieve their goals.

DOMINANT EXTRAVERTED INTUITION

Characteristics of ENTPs ENTPs are enthusiastic innovators. Their world is full of possibilities, interesting concepts, and exciting challenges. They are stimulated by difficulties, quickly devising creative responses and plunging into activity, trusting their ability to improvise. They use their Intuition primarily externally and enjoy exercising their ingenuity in the world. ENTPs are likely to be

- Creative, imaginative, and clever
- Theoretical, conceptual, and curious

AUXILIARY INTROVERTED THINKING

ENTPs use their Thinking primarily internally to analyze situations and their own ideas and to plan. They admire competence, intelligence, precision, and efficiency. ENTPs are usually

- Analytical, logical, rational, and objective
- Assertive and questioning

WHOLE TYPE

ENTPs are enterprising, resourceful, active, and energetic. They respond to challenging problems by creating complex and global solutions. They are usually adept at

“reading” other people, seeing how to motivate them, and assuming leadership. They can do almost anything that captures their interest.

How Others May See Them

ENTPs are spontaneous and adaptable. They find schedules and standard operating procedures confining and work around them whenever possible. They are remarkably insightful about the attitudes of others, and their enthusiasm and energy can mobilize people to support their vision.

Their conversational style is seen by many as challenging and stimulating because they love to debate ideas. They are fluent conversationalists, are mentally quick, and enjoy verbal sparring. When they express their underlying Thinking principles, however, they may speak with an intensity and abruptness that seem to challenge others.

Others usually see ENTPs as

- Independent, autonomous, and creative
- Lively, enthusiastic, and energetic
- Assertive and outspoken

Potential Areas for Growth

UNDEVELOPED EXTRAVERTED INTUITION AND INTROVERTED THINKING

Sometimes life circumstances have not supported ENTPs in the development and expression of their favorite processes, Intuition and Thinking.

- If they have not developed their Intuition, ENTPs may not take in enough relevant information, resulting in “insights” unrelated to current reality.
- If they have not developed their Thinking, they may not have reliable ways to evaluate their insights and make plans to carry them through. Then they will go from enthusiasm to enthusiasm with little actually accomplished.

EXAGGERATED ENTP

If ENTPs don’t find a place where they can use their gifts and be appreciated for their contributions, they usually feel frustrated and may

- Behave in a brash, rude, and abrasive manner
- Criticize others, especially those they deem inefficient or incompetent
- Become rebellious and combative
- Become scattered—unable to focus

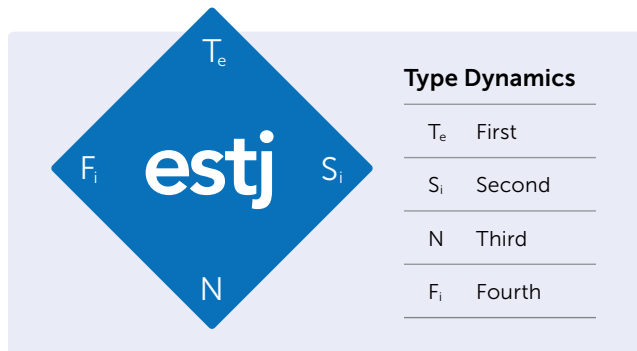
NEGLECT OF INFERIOR SENSING AND TERTIARY FEELING

It is natural for ENTPs to give less attention to their less favored processes, Sensing and Feeling. If they neglect them too much, however, they may

- Not take care of the details and routine required to implement their insights
- Not fully appreciate the impact on others of their ideas and plans
- Be excessively and inappropriately “challenging and stimulating”

INFERIOR FUNCTION FORM

When greatly stressed, ENTPs can get overwhelmed by details and lose their ability to generate possibilities. Then they will focus on a minor or distorted detail, treating it as extremely important.



EXTRAVERSION SENSING THINKING JUDGING

At Their Best

WHOLE TYPE

ESTJs like to organize projects, procedures, and people and then act to get things done. They live by a set of clear standards and beliefs, make a systematic effort to follow these, and expect the same of others. They value competence, efficiency, and results and display them in their work and play.

ESTJs enjoy interacting and working with others, as long as the others are responsible about meeting deadlines and completing assigned tasks. They work best in situations where clear, known problems can be solved with proven techniques.

DOMINANT EXTRAVERTED THINKING

Characteristics of ESTJs ESTJs take an objective approach to problem solving and are tough when the situation requires it. They use their Thinking primarily externally to organize their life and work, and they have

little patience with confusion, inefficiency, or lack of follow-through. ESTJs are likely to be

- Logical, analytical, and objectively critical
- Decisive, clear, and assertive

AUXILIARY INTROVERTED SENSING

ESTJs focus on the present—what is real and actual. They apply and adapt relevant past experience to deal with problems, and they prefer jobs where results are immediate, visible, and tangible. ESTJs are likely to be

- Practical, realistic, and matter-of-fact
- Systematic and pragmatic

WHOLE TYPE

ESTJs are usually excellent administrators because they understand systems and logistics. They can project the steps needed to accomplish a task, foresee potential problems, assign responsibilities, and marshal resources. They deal with all aspects of a situation, leaving no unresolved issues or unfinished business, and get things done on time. When they see that something is not working, they will plan and act to correct it. Otherwise, they prefer proven procedures and systems. Their orientation is toward tasks, action, and the bottom line.

How Others May See Them

Because ESTJs naturally devise systems, procedures, and schedules, others rely on them to take charge and get results. Others may find ESTJs overpowering at times because they are so certain about how things should be. Because they are clear and straightforward in their communication, people seldom have to wonder where they stand.

ESTJs can be quite gregarious and generally enjoy interacting with people, especially around tasks, games, traditions, and family activities. They take relationship roles seriously and fulfill them responsibly.

Others usually see ESTJs as

- Conscientious and dependable
- Decisive, outspoken, and self-confident

Potential Areas for Growth

UNDEVELOPED EXTRAVERTED THINKING AND INTROVERTED SENSING

Sometimes life circumstances have not supported ESTJs in the development and expression of their favorite processes, Thinking and Sensing.

- If they have not developed their Thinking, ESTJs may not have a reliable way of evaluating information and thus may make inconsistent or overly harsh decisions.

- If they have not developed their Sensing, they may make decisions too quickly, before they have taken in enough information. Then their decisions will reflect their previously formed judgments or biases.

EXAGGERATED ESTJ

If ESTJs don't find a place where they can use their gifts and be appreciated for their contributions, they usually feel frustrated and may

- Become rigid and dogmatic
- Be intrusive, "know-it-all" experts, overpowering others and refusing to listen
- Get picky about details and be impatient with those who don't follow procedures exactly

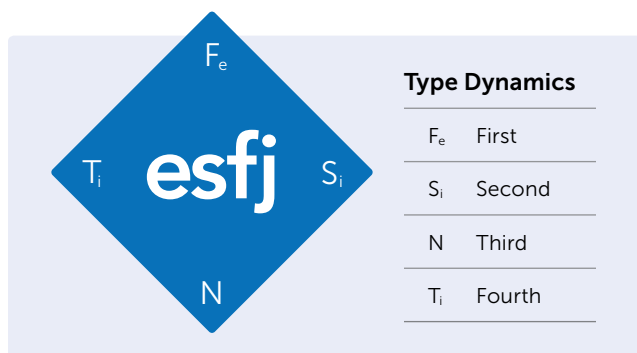
NEGLECT OF INFERIOR FEELING AND TERTIARY INTUITION

It is natural for ESTJs to give less attention to their less favored processes, Feeling and Intuition. If they neglect them too much, however, they may

- Apply logic even when emotions and impacts on people need primary consideration
- Fail to respond to others' needs for intimate connection and processing of feelings
- Not always see the wider ramifications of a seemingly simple, direct action

INFERIOR FUNCTION FORM

When greatly stressed, ESTJs may feel alone and unappreciated and be unable to communicate their feelings of distress and despair.



EXTRAVERSION SENSING FEELING JUDGING

At Their Best

WHOLE TYPE

ESFJs like to organize people and situations and then work with others to complete tasks accurately and on time. They are conscientious and loyal, following through

even in small matters, and they want others to be the same. They value security and stability.

Sociable and outgoing, ESFJs enjoy celebrations and traditions and bring personal caring to their workplace and home. They want to be appreciated for who they are and for what they give to others.

DOMINANT EXTRAVERTED FEELING

Characteristics of ESFJs ESFJs use their Feeling primarily externally and radiate warmth and energy. They are encouraged by approval and hurt by indifference or unkindness. Conflict-filled or tense situations make them uncomfortable, and they work to ensure that these don't occur. ESFJs are likely to be

- Warm, sympathetic, and helpful
- Personable, cooperative, and tactful

AUXILIARY INTROVERTED SENSING

ESFJs focus on the present and base their decisions on experience and facts. Though they enjoy variety, they adapt well to routine and don't like work that demands mastery of abstract ideas or impersonal analysis. They enjoy their possessions and take good care of them. ESFJs are likely to be

- Practical, realistic, and down-to-earth
- Decisive, thorough, and consistent

WHOLE TYPE

ESFJs are sensitive to the needs of each individual in their environment and good at providing practical caring. Much of their pleasure and satisfaction comes from the comfort and pleasure of others.

How Others May See Them

ESFJs are energized by interaction with others and genuinely interested in others' lives and concerns. They feel most comfortable in structured situations and enjoy creating order, structure, and schedules. They prefer to do things the traditional and accepted way.

For the sake of harmony, ESFJs will agree with others when they can. However, they also have strong values, which they express clearly and confidently when they think it is appropriate.

ESFJs value family and social ties. They enjoy belonging and are good at celebrations and traditions.

Others usually see ESFJs as

- Sociable, outgoing, enthusiastic, and energetic
- Organized and orderly
- Committed to preserving traditions

Potential Areas for Growth

UNDEVELOPED EXTRAVERTED FEELING AND INTROVERTED SENSING

Sometimes life circumstances have not supported ESFJs in the development and expression of their favorite processes, Feeling and Sensing.

- If they have not developed their Feeling, ESFJs may be tentative and uncertain, accepting the thoughts and opinions of others too quickly.
- If they have not developed their Sensing, they may not take in much information before making decisions and thus may jump to conclusions before fully understanding a situation. They may then impose their decisions on everyone around them.

EXAGGERATED ESFJ

If ESFJs don't find a place where they can use their gifts and be appreciated for their contributions, they usually feel frustrated and may

- Doubt themselves and focus their attention entirely on satisfying the needs of others
- Worry and feel guilty
- Become controlling in their push for harmony—"we will all get along"
- Become overly sensitive, imagining slights where none was intended

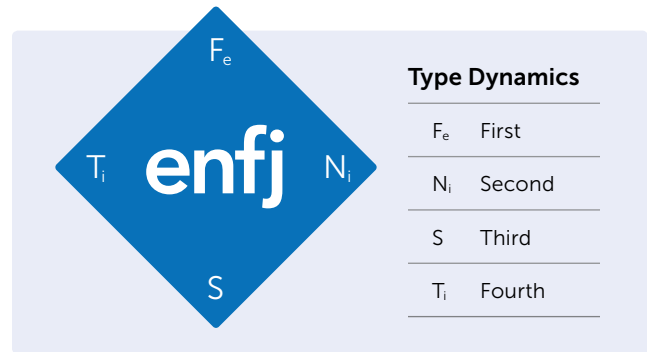
NEGLECT OF INFERIOR THINKING AND TERTIARY INTUITION

It is natural for ESFJs to give less attention to their less favored processes, Thinking and Intuition. If they neglect them too much, however, they may

- Find it difficult to acknowledge and deal with negative realities and problems that can't be ignored, especially when people or things they care about are involved
- Support those in charge or the standard procedures too uncritically
- Fail to see wider possibilities or alternative ways of doing things

INFERIOR FUNCTION FORM

When greatly stressed, ESFJs may find themselves uncharacteristically critical of themselves and others. Their negative thoughts and opinions then trouble them greatly.



EXTRAVERSION INTUITION FEELING JUDGING

At Their Best

WHOLE TYPE

ENFJs are highly attuned to others, using empathy to quickly understand emotional needs, motivations, and concerns. Their focus is on supporting others and encouraging their growth.

ENFJs are friendly persuaders who can build consensus among people whose interests and motives are quite diverse. They often act as catalysts, including everyone and drawing out the best in others. They can be inspiring leaders as well as loyal followers.

DOMINANT EXTRAVERTED FEELING

Characteristics of ENFJs ENFJs base their decisions on personal values. They use their Feeling primarily externally, radiating warmth and energy. They look for and find the best in others and prize harmony and cooperation.

ENFJs are warmed by approval, responding with energy and devotion, and are especially sensitive to criticism and tension. They are likely to be

- Warm, compassionate, and supportive
- Loyal and trustworthy

AUXILIARY INTROVERTED INTUITION

ENFJs see meanings and connections and can be very insightful about people. They are curious about new ideas and stimulated by possibilities for contributing to the good of humanity. ENFJs are likely to

- Be imaginative and creative
- Like variety and new challenges

WHOLE TYPE

ENFJs naturally see the potential for growth in others and devote energy to helping them achieve it. They are sensitive facilitators. ENFJs take responsibility for organizing interactions of colleagues, friends, and family

so that all are involved, harmony prevails, and people have fun.

How Others May See Them

ENFJs are energetic, enthusiastic, and very aware of others. Their genuine interest can usually draw out and involve even the most reserved person. They listen to and support others but also have very definite values and opinions of their own, which they will express clearly. ENFJs are energized by people and are socially adept. However, they also have a strong need for authentic, intimate relationships. They bring great enthusiasm and intensity to creating and maintaining these.

ENFJs like their lives to be organized and will work to bring closure to ambiguous relationships or situations. However, if people's needs conflict with schedules and rules, they will put people first.

Others usually see ENFJs as

- Sociable, personable, congenial, and gracious
- Expressive, responsive, and persuasive

Potential Areas for Growth

UNDEVELOPED EXTRAVERTED FEELING AND INTROVERTED INTUITION

Sometimes life circumstances have not supported ENFJs in the development and expression of their favorite processes, Feeling and Intuition.

- If they have not developed their Feeling, their decisions may be inconsistent and poorly formulated. They may then accept the thoughts and opinions of others too readily.
- If they have not developed their Intuition, ENFJs may not see possibilities, making decisions too quickly without taking in enough information or considering factors beyond their own personal values.

EXAGGERATED ENFJ

If ENFJs don't find a place where they can use their gifts and be appreciated for their contributions, they usually feel frustrated and may

- Worry, feel guilty, and doubt themselves
- Become insistent and controlling in their desire for harmony
- Be overly sensitive to criticism, real or imagined

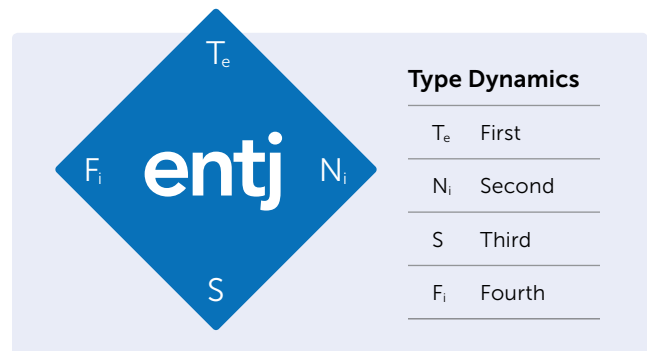
NEGLECT OF INFERIOR THINKING AND TERTIARY SENSING

It is natural for ENFJs to give less attention to their less favored processes, Thinking and Sensing. If they neglect them too much, however, they may

- Make decisions based solely on personal values when logic is needed also
- Find it difficult to admit to problems or disagreements with people they care about
- Overlook details required to realize their ideals

INFERIOR FUNCTION FORM

When greatly stressed, ENFJs may find themselves suddenly and uncharacteristically critical and fault-finding with others. They generally keep these negative opinions to themselves, but they find such thoughts troubling and upsetting.



EXTRAVERSION INTUITION THINKING JUDGING

At Their Best

WHOLE TYPE

ENTJs are natural leaders and organization builders. They conceptualize and theorize readily and translate possibilities into plans to achieve short-term and long-term objectives. They are good at seeing illogical and inefficient procedures and feel a strong urge to correct them—to organize people and situations to get them moving in the right direction.

ENTJs are strategic visionaries, adept at planning for the future needs of the people and organizations for which they are responsible.

DOMINANT EXTRAVERTED THINKING

Characteristics of ENTJs ENTJs use their Thinking primarily externally and are thus natural critics. They set their own standards and are forceful in applying them to others, to organizations, and to themselves. They value intelligence and competence and hate inefficiency and ignorance. They can be tough when the situation calls for it and are likely to be

- Analytical, logical, and objectively critical
- Decisive, clear, and assertive

AUXILIARY INTROVERTED INTUITION

ENTJs are intellectually curious, seek new ideas, and like complex problems. They use their Intuition primarily internally to conceive possibilities and create the insights they use in making decisions and plans. ENTJs are likely to be

- Conceptual and global thinkers
- Innovative theorizers and planners

WHOLE TYPE

ENTJs are usually excellent solvers of organizational problems. They are keenly aware of the intricate connections within organizations and are action oriented and strategic—they think ahead, anticipate problems, devise broad plans and systems, and marshal human and material resources to achieve goals. They are generally not interested in routine maintenance activities, preferring new challenges.

How Others May See Them

ENTJs love, and are energized by, stimulating interactions with people. They often challenge people's statements and behaviors, expecting that others will defend them and that, as a result, mutual learning will take place. ENTJs admire and seek out people who are knowledgeable and who stand up to them, say what they think, and argue persuasively.

ENTJs prefer that things be settled and clear, but their love of ideas can pull them into wide-ranging exploration and discussions. Their verbal fluency, decisiveness, self-confidence, and urge to organize others can overpower people at times.

Others usually see ENTJs as

- Direct, challenging, and decisive
- Objective, fair, and stimulating

Potential Areas for Growth

UNDEVELOPED EXTRAVERTED THINKING AND INTROVERTED INTUITION

Sometimes life circumstances have not supported ENTJs in the development and expression of their favorite processes, Thinking and Intuition.

- If they have not developed their Thinking, ENTJs may not have a reliable way to evaluate their insights and make plans. Then their decision making will be inconsistent and changeable.

- If they have not developed their Intuition, they may make decisions too quickly without considering alternatives or exploring possibilities. In this case, their decisiveness can become dictatorial.

EXAGGERATED ENTJ

If ENTJs don't find a place where they can use their gifts and be appreciated for their contributions, they usually feel frustrated and may

- Become overly impersonal and critical
- Be intrusive and directive—giving orders without listening
- Become abrasive and verbally aggressive

NEGLECT OF INFERIOR FEELING AND TERTIARY SENSING

It is natural for ENTJs to give less attention to their less favored processes, Feeling and Sensing. If they neglect them too much, however, they may

- Fail to notice or value another's need for personal connection, appreciation, or praise
- Fail to factor into their plans people's needs for support and processing time
- Overlook specifics and realistic factors that are necessary to carry their plans to completion

INFERIOR FUNCTION FORM

When greatly stressed, ENTJs can be overwhelmed by self-doubt, feel alone and unappreciated, and be unable to express their distress to others.

THE 20 STEP II™ FACETS

This section presents detailed, in-depth descriptions of the 20 facets that are the essence of the Global Step II assessment. In reading these descriptions it is important to recognize that, consistent with the way the four pairs of Step I preferences (E–I, S–N, T–F, J–P) are described, the opposites that make up the 20 facets are described as qualitatively different, with each opposite considered legitimate and valuable in its own right. Readers will no doubt like or feel more drawn to one or the other pole of each facet, as is often the case when people read descriptions of the Step I opposites. However, neither Step I nor Step II descriptions favor one side or the other of a preference pair or facet. Both ways of being are seen as inherently equal, valued, and acceptable.¹

1. Clearly, some facet pole preferences are more highly valued than others in certain contexts. For example, Early Starting is more highly valued than is Pressure-Prompted in many organizational settings. There is a good deal of both research and observational evidence indicating that qualities associated with J facet poles (and a Judging orientation in general) are favored in both educational and work contexts in Western, as well as in many other, cultures.

Context for Viewing Facet Descriptions

In looking at individual Step II facets within any of the Step I preference pairs, we are focusing a microscope on certain aspects of each four-letter type. This magnified view can be valuable because it enables examination of narrower parts of the types in detail. But it is important to remember that in reality the facets described here are exhibited by people who are whole types and not just Introverts or Thinking types or other fragments of whole types.

Each facet must be considered in the context of one of the 16 whole types, not merely as an aspect of the preference pair within which it is found.

The facet descriptions in this chapter may be used as benchmarks when comparing individuals whose Step I type preferences are known. In general, respondents' type preferences and whole type should be used as a context within which to more fully interpret their facet results. This is because the descriptions of the Step II facets do not take into account interactions among Step I preferences. Consider the following:

- The description of the Intimate pole of the Gregarious–Intimate facet is generalized across all Introverted types. It does not consider how the expression of Intimate by an Introvert might vary depending on whether his or her other preferences are Sensing or Intuition, Thinking or Feeling, or Judging or Perceiving.
- The description of the Gregarious facet pole does not consider how the expression of Gregarious by an Introvert with an out-of-preference score might differ from that of an Extravert whose score is in-preference—that is, consistent with his or her preference for Extraversion.² See chapters 2 and 6 for a discussion of in-preference and out-of-preference facet scores.

The factors that affect a particular individual's expression of the facets may explain how two people who prefer Thinking, for example, would use their logic differently, or the different ways two Sensing types might display their concreteness. These facet-level differences may not appear as great as those between Thinking and Feeling types, or between people who prefer opposite poles on any of the other preference pairs. However, although facet differences are usually less noticeable than preference pair differences, sometimes they are more predictive of specific behaviors than are overall preference differences. For examples of this effect, see chapter 9.

In addition to differences that are directly related to type, variations in age, gender, developmental history, and life experiences can also influence how individuals express their preferences on the facets measured by the Step II assessment. As noted in chapter 2, type-specific variations in facet scores are described by the Step II assessment. Age and gender differences are discussed in the Step II sections of chapter 6 (assessment administration and interpretation) and chapter 7 (assessment construction). Bear in mind also the following important features of Step II results:

- Facet descriptions in this chapter refer to the poles of the facets, so they are thrown into sharp relief compared to descriptions of individuals scoring in the midzone of the facets.
- The behavior of individuals scoring in the midzone of a facet may look somewhat like a mixture or a muted expression of both poles of that facet, or may differ from the behavior of people scoring at either pole.
- Remember that since types are more than the sum of their parts, they have features that go beyond their four component Step I preferences. Certain facets that have a narrow focus and more normal-looking distributions may be tapping behaviors and qualities that result from exercising an underlying type preference rather than showing intrinsic aspects of that preference.

Step II facets vary in their source. Some are closely tied to their underlying preference; other, more narrowly defined, facets likely result from habitual exercise of an underlying type preference.

Order of the Facets

The order in which the facets appear in this manual was established in the first Step II guide (Mitchell, Quenk, & Kummerow, 1997). The manuals for the earlier Expanded Analysis Report (Saunders, 1989) and its predecessor, the *Type Differentiation Indicator Manual* (Saunders, 1987), followed a different order. Traditionally, the facets within each preference pair have been ordered based on one or both of the following criteria: (1) their conceptual breadth and/or generality; (2) the natural sequence in which they are likely to be used in relation to one another. For E–I, S–N, and T–F, both criteria apply. The facet that is listed first fulfills the criterion of breadth and generality from a conceptual point of view and is the most highly correlated of the five facets with the underlying preference scale (E–I, S–N, or T–F; see table 10.2). In addition, the order of presentation of the

2. The influence of in-preference and out-of-preference scores on facet descriptions is, however, reflected in the MBTI® Step II™ Interpretive Report.

facets within these preference pairs approximates the natural sequence in which the facets would be used.

Within the E–I (source of energy) facets, Initiating–Receiving reflects a fundamental difference in energy flow. In addition to correlating more highly than the other four E–I facets with the underlying Step I E–I scale, it shows the highest percentage of in-preference scores of the five E–I facets and the lowest percentage of out-of-preference scores. The remaining E–I facets focus on more specific ways that energy flow occurs.

Within the S–N (perceiving processes) facets, Concrete–Abstract is listed first because it captures the broad focus of the overall Sensing–Intuition preference pair. Research evidence shows that the Concrete–Abstract facet has the highest percentage of in-preference scores, followed by Realistic–Imaginative. In addition, data for the Concrete–Abstract facet indicate that it has the lowest percentage of out-of-preference scores of all 20 facets. The next three S–N facets, Realistic–Imaginative, Practical–Conceptual, and Experiential–Theoretical, have to do with how individuals begin to make meaning of their initial Concrete or Abstract perceptions. Finally, the Traditional–Original facet describes how individuals assemble their initial perceptions and the meanings they have assigned to them to influence what they see and look for in their social environment.

Within the T–F (judging processes) facets, Logical–Empathetic is listed first because when using Thinking and Feeling, the first step is to establish whether to apply a Logical or an Empathetic criterion to make a decision or come to a conclusion. The individual can then enact the criterion chosen by using either a Reasonable or a Compassionate approach. These two closely related facets show the first- and second-highest percentages of in-preference scores and are tied for the lowest percentage of out-of-preference scores. The next facet, Questioning–Accommodating, describes what people tend to do when they have differences of opinion or viewpoint: they employ either a Questioning or an Accommodating approach. They then follow up by taking a Critical or Accepting approach to the issue being discussed. Finally, once a decision has been made, they will implement or defend it using either a Tough or a Tender approach.

Within the J–P (approach to the outside world) facets, Systematic–Casual describes a fundamental orientation toward dealing with the external environment. However, it is not the most highly correlated of the five J–P facet scales with the Step I J–P scale, and its percentages of in-preference, midzone, and out-of-preference scores do

not provide statistical evidence for its position of primacy among the five facet scales. (The same was true for the ordering of the J–P facets in Form Q, the preceding Step II assessment.) Nevertheless, the Systematic–Casual facet meets the criterion of breadth and generality.

From a conceptual point of view, the four J–P facets listed after Systematic–Casual describe instances in which a general Systematic or Casual approach plays out in more specific ways. Unlike with the E–I, S–N, and T–F facets, the placement of these four facets within the set of J–P facets does not necessarily reflect the order in which individuals would likely use them when relating to the outside world. Rather, it is likely based on the particular demands of situations in which they find themselves.

For example, whether an individual takes a Methodical or an Emergent approach to a task may depend on the nature of the task involved and/or the situation in which the task must be performed. This facet has the second-highest percentage of out-of-preference scores of all 20 facets, as well as the lowest test-retest reliability of all the facets. Practitioner experience based on participant feedback suggests that the behaviors associated with the J–P facets can be learned in response to environmental demands, particularly in the workplace. In addition, one or more of the other Step I preferences may be at play. For example, analysis of data from the MBTI global sample confirmed a hypothesis that Perceiving (P) types with a preference for Thinking (T) are significantly more likely than are Perceiving types with a preference for Feeling (F) to score toward the Systematic pole rather than the Casual pole on the Systematic–Casual facet. See chapter 6 for in-preference, midzone, and out-of-preference comparisons.

Descriptions of the Facets

The facet descriptions below are based solely on the item content of the facets.³ We might think of these descriptions as mini-worldviews held by people who choose all or most of the item responses associated with one or the other pole of the individual facets.⁴

Undoubtedly, additional characteristics associated with these facets exist that have not yet been directly measured. Such characteristics may be discovered as a result of ongoing research with the Step II assessment or practitioners' work with individuals who provide additional examples and elaboration of behaviors measured by the facets. Research evidence that shows the relationship of the facet scales to the scales of other instruments is reported in chapter 9.

3. Facet descriptions in this chapter are modified and updated versions of the facet descriptions developed by Wayne D. Mitchell, whose research with earlier Step II scales formed the basis for the facet descriptions in the *MBTI® Step II® Manual* (Quenk, Hammer, & Majors, 2001).

4. Wayne Mitchell, who developed the original facet descriptions, related that in order to do justice to the opposite poles of the facets, he used a mind-set of looking at the world through the lens of someone who was fully committed to a given pole. After writing a facet pole description, he waited a day or so to clear his mind before adopting a mind-set for the opposite pole. An interesting way to read these descriptions, therefore, is to imagine how you yourself would deal with the world if you had that particular perspective. The S–N facets are particularly illuminating for this exercise.

READING THE DESCRIPTIONS

Some of the facet descriptions may seem quite similar to one another on first reading. However, it is important to keep in mind that each facet, along with four others, “belongs to” one basic orientation (E–I or J–P) or process (S–N or T–F). All the Step II facet scales are at least moderately correlated with their underlying Step I preference scale (E–I, S–N, T–F, or J–P), depending on the facet block under consideration. For example, the E–I facet scales correlate from .74 (Gregarious–Intimate) to .87 (Initiating–Receiving) with the E–I scale, and the S–N facet scales correlate from .63 (Experiential–Theoretical) to .87 (Concrete–Abstract) with the S–N scale. All the facets within a preference pair also correlate with one another. Table 10.3 presents the correlations of

the Global Step II facet scales with one another. These correlations illustrate the overlap among facets within each preference pair.

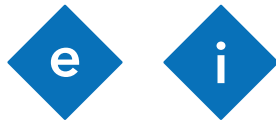
Based on these data, some overlap in the item content of some facet scales is to be expected. In addition, unlike the facets on prior Step II forms, the Global Step II facets include some item overlap across Step I preference pairs. Such overlap also contributes to similarities in facet descriptions. Because of such overlaps, facet descriptions must be read carefully.

Table 4.2 provides a succinct overview of facet content and may be used as a guide to the differences and similarities among the facets in each preference pair.

Table 4.2 | **Summary of facet content**

<i>E–I facets</i>	
Initiating–Receiving	Describes people’s basic orientation to communicating with and connecting to others
Expressive–Contained	Focuses on how people communicate their thoughts, feelings, interests, and experiences
Gregarious–Intimate	Describes the breadth and depth of people’s connections to others
Active–Reflective	Describes how people engage with the outer environment, including other people, for entertainment, socializing, and acquiring information
Enthusiastic–Quiet	Focuses on the level and kind of energy people bring to exchanges with others rather than on the content of what is exchanged
<i>S–N facets</i>	
Concrete–Abstract	Focuses broadly on how people generally perceive the world and the kinds of things to which they direct their attention
Realistic–Imaginative	Describes how people develop something new through dealing with the tasks and problems of daily life and work
Practical–Conceptual	Deals with the products or outcomes of people’s perceptions rather than the process of perception itself
Experiential–Theoretical	Emphasizes the process by which people make something either tangible or symbolic from their perceptions
Traditional–Original	Focuses on the social context that gives meaning to people’s perceptions
<i>T–F facets</i>	
Logical–Empathetic	Emphasizes the criteria people use to reach a judgment
Reasonable–Compassionate	Emphasizes the standards people use to maintain relationships when making Thinking or Feeling judgments
Questioning–Accommodating	Focuses on how people deal with differences of opinion
Critical–Accepting	Describes what people do after making their initial judgments
Tough–Tender	Focuses on the impact of people’s judgments and how they carry out a decision once a judgment has been made
<i>J–P facets</i>	
Systematic–Casual	Focuses on how people organize their physical environment, including the flow of events, activities, tasks, and projects
Planful–Open-Ended	Emphasizes people’s approach to current and future events and goals, including leisure time and social activities
Early Starting–Pressure–Prompted	Focuses narrowly on how people manage time with regard to deadlines
Scheduled–Spontaneous	Centers on the degree of structure in people’s daily activities
Methodical–Emergent	Focuses narrowly on how people approach smaller tasks required to complete larger projects

Another way to get a better picture of the content of each facet is to try the following procedure. First, focus on one Step I preference pair by reading all the descriptions for its facets, one after the other. This will give you a general idea of the scope of the material. Next, read through the descriptions again, this time keeping in mind what they have in common while noting the subtle differences. For example, the J–P facets are all concerned with people’s approach to the outside world, either by organizing the outer environment to meet their needs or by responding or adapting to the world as it is. However, one facet focuses on how people organize their leisure time, while another focuses on the smaller tasks needed to complete a larger project and a third emphasizes the way people approach deadlines. Similar differences are found among facets within the other preference pairs as well.



THE EXTRAVERSION–INTROVERSION FACETS

The five facets within the Extraversion–Introversion preference pair are

- Initiating–Receiving
- Expressive–Contained
- Gregarious–Intimate
- Active–Reflective
- Enthusiastic–Quiet

INITIATING–RECEIVING

Describes people’s basic orientation to communicating with and connecting to others

Initiating

- Get pleasure from mingling with others in large or small gatherings
- Enjoy connecting with others and helping people connect with each other
- Are adept at and enjoy light conversations typical of receptions, meeting breaks, parties
- Are comfortable introducing themselves to people they know about but have never met, finding common interests to discuss with those people
- Are constantly on the lookout for new people with whom to make contact
- Are able to keep a conversation going with almost anyone using socially relevant information
- Enjoy talking to many people in informal settings, both those they know and those they don’t; feel stifled if unable to circulate

- Have the ability to start and maintain a conversation while scanning the group for others with whom to make contact
- Enjoy getting things going and facilitating interactions in social situations using their large store of information to make connections with others

Receiving

- Are much more comfortable letting conversations come to them than initiating contact, especially in a group of people they don’t know
- Are much more likely to be introduced to new people than to introduce themselves or connect people they know to each other
- Find that it takes a good deal of effort to introduce themselves to people who share their interests but whom they have never met
- Are often introduced by an Initiating friend to someone who shares the Receiving person’s interests, skills, etc.
- Feel much more comfortable talking to people they know than making new connections
- Find it difficult to keep a conversation going with someone with whom they have nothing in common because they are unlikely to have a broad range of interests to call on
- Are able to talk at great length with a new acquaintance who shares an important interest
- May feel awkward and adrift in informal group settings, so may corner a person they know and spend much time talking to him or her

EXPRESSIVE–CONTAINED

Focuses on how people communicate their thoughts, feelings, interests, and experiences

Expressive

- Are ready and willing to share feelings, thoughts, personal history, interests, and opinions with others
- Are easy to get to know since they readily express themselves
- Feel that not sharing personal information is dishonest and may damage their relationships
- Seek two-way, open conversations about how they and others feel about an issue or about other people and find relationships that lack such conversations to be incomplete or not genuine
- Are likely to tell the other person involved when their relationship lacks mutual expressiveness
- Are likely to bring into the open unacknowledged feelings in a group, allowing the group to deal with what is not being said

- Find that others' expression of their thoughts and feelings helps them discover and clarify their own thoughts and feelings
- May unintentionally reveal confidential information by expressing things that others want to be kept private
- Assume that others want to know, and will benefit from hearing, what they have to say

Contained

- Are selective about what information they share about themselves, including their thoughts, feelings, personal history, interests, and opinions, and with whom they share it
- Require a certain level of comfort with and trust in someone before they will share information about personal matters, and will do so only when they feel it is relevant to the conversation at hand
- Are likely to reveal personal interests only when they come up in conversation
- Tend to use few words and little body language, making it difficult for others to get a read on their thoughts and feelings
- Find it difficult to talk about it to others when they are distressed, believing that no one else can understand
- Require sufficient time to assess a problem or situation internally before they are able or willing to discuss it with others because such a discussion may interfere with or confuse their own assessment
- Assume that others do not necessarily want to hear about their inner life
- May reveal information only if specifically asked

GREGARIOUS–INTIMATE

Describes the **breadth and depth of people's connections to others**

Gregarious

- Maintain a satisfying social life by enjoying friendships and associations with many and varied people, typically in group settings
- Enjoy group interactions much more than one-on-one conversations and develop a personal identity by being popular and known by a wide circle of people
- Value the freshness and adventure of making new friends and connecting with others who have large social networks
- Make the most of their contacts by expanding the network of people with whom they can connect
- Find being cut off from their network stressful because they can't get the latest news about everyone, impairing their sense of self

- Are able to respond to varied expectations of others without compromising their own identity
- Can juggle several people-oriented tasks simultaneously, doing all of them well
- May be unaware of or not sensitive to the opinions and emotions of more reserved group members, especially in larger groups where there is lively social mingling
- In meetings, may be able to suggest compromises to achieve consensus due to their familiarity with individual group members' views and concerns
- May strive to be the center of attention in small and large groups
- In large-group settings, are aware of the flow of the group's mood or sentiment
- Can respond rapidly in social contexts that involve a series of quick exchanges with others

Intimate

- Are most at ease in social exchanges with others they know well
- Prefer lengthy one-on-one conversations in which they can go deeply into a topic
- Find social mingling and chitchat unappealing and avoid them if possible, especially in large-group situations where they don't know anyone
- In a large group, are likely to spend the entire time talking to one or a few friends with whom they feel connected
- Very much dislike it when people change the subject or insert other topics into a conversation
- Feel that their in-depth approach does not lend itself to responding to rapid remarks by different people in large groups, so their possibly valuable remarks may remain unspoken
- Prefer a smaller number of friendships because friendships require time to develop and maintain; share personal parts of their life with very few people
- Value the intimacy involved in sharing hidden parts of themselves with a trusted friend much more than being popular
- Enjoy concentrating their energy on specific interests they can explore in depth over long periods of time
- Losing or not having at least some deep friendships makes them feel incomplete
- Seek out environments and experiences that permit intimate interactions
- Value and engage in networking only when an important goal is involved
- Are better at identifying nuances of meaning among intimate friends than recognizing nuances of group dynamics

ACTIVE–REFLECTIVE

Describes **how people engage with the outer environment**, including other people, for entertainment, socializing, and acquiring information

Active

- Actively engage with the environment, especially through energetic face-to-face interactions with others
- Prefer active involvement over passive viewing and find attending parties more entertaining than watching a performance
- Choose leisure activities that involve direct contact with the outside world
- Prefer to communicate to others and contribute to projects by speaking rather than through writing
- Learn better by doing, listening, and questioning than through independent study, reflection, or reading
- Are comfortable meeting strangers and taking the lead in social interactions
- Like to deal with problems or misunderstandings by speaking to others face-to-face
- Feel that active participation allows them to better know themselves, the world, and how they and the world are dynamically connected

Reflective

- Seek out and enjoy experiences that evoke internal responses—visual, intellectual, or emotional
- Enjoy the way creative, artistic, intellectual, or cultural works stimulate their thoughts
- Learn best by reading on their own, which permits them to freely associate other known information with the material they are reading
- Discover the meaning of things through active mental engagement, which makes actually interacting with the outside world unnecessary
- Find that group processing is not effective for learning or using leisure time because it interrupts their internal dialogue
- Communicate and contribute to projects best through writing and reading
- Find that meeting and interacting with strangers is uncomfortable, so prefer to let others keep conversations going in social situations

ENTHUSIASTIC–QUIET

Focuses on the **level and kind of energy people bring to exchanges with others** rather than on the *content* of what is exchanged

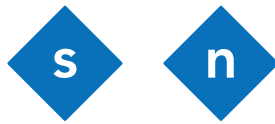
Enthusiastic

- Are talkative, hearty, and lively; enjoy conversation for its own sake

- Give and receive energy by talking to others
- Especially enjoy discussions in group settings where lively input from several people creates a spontaneous “event”
- Bring their enthusiasm for connecting directly and immediately with others to one-on-one conversations as well
- Are typically among the first to know what is going on in their network of friends and acquaintances
- Easily pick up and remember information about people they know as well as people they haven’t actually met
- Seek out group settings that have high, upbeat collective energy
- Are attracted to places and events with lots of action and people who like a high level of energy
- Enjoy being the center of attention and in their enthusiasm may embellish accounts of events
- Believe that the goal of conversation is to entertain and have fun, not to transmit facts, so telling stories is their primary way of interacting with others

Quiet

- Have a calm, reserved, quiet bearing, even in groups with enthusiastic people
- Expend little energy when responding to group and social exchanges
- Tend to be drained by interacting with others rather than stimulated or energized by it
- Are interested in others and interesting themselves but operate with a subdued level of social input and response
- May find that their reserve causes the subtleties and meanings of their responses to be overlooked or misunderstood
- Are likely to have highly active internal responses to social stimuli since in general they are more highly aroused by external stimuli (e.g., sights and sounds) than are Enthusiastic people
- Are at their best in quiet, calm environments due to their higher level of arousal
- Find it hard to put their rich inner experience into words, so they may understate things, be more succinct, and reveal less than they intended
- Have a highly active internal response to social stimuli, although it might not look that way to an observer
- May find that those who don’t know them are especially likely to miss or dismiss their contributions
- May be the last on the grapevine to hear what is going on, but they do have ways to get relevant information



THE SENSING–INTUITION FACETS

The five facets within the Sensing–Intuition preference pair are

- Concrete–Abstract
- Realistic–Imaginative
- Practical–Conceptual
- Experiential–Theoretical
- Traditional–Original

CONCRETE–ABSTRACT

Focuses broadly on **how people generally perceive the world** and the kinds of things to which they direct their attention

Concrete

- Are grounded in physical, material aspects of the world, so they favor tangible and concrete things over intangible and abstract ideas
- Like factual information and trust words and descriptions that point to what is real and tangible; communicate, learn, and form their view of the world on this basis
- Use literal and specific words that convey the details of sensory impressions; understand best when others communicate to them in a similar fashion
- Believe that the world consists of things perceived by the senses and that, to be credible, those things must be based on some physical, sensory process that is directly experienced by themselves or others
- Tend to dismiss abstract concepts as less believable, less important, and largely irrelevant compared to sensory information
- See what is abstract and what is imaginary as equivalent
- Much prefer having specific instructions for building things, making repairs, and so on, rather than proceeding primarily by knowing the underlying principles involved
- Judge the value of something according to how grounded it is in the tangible world
- Base their decisions on hard data rather than abstract principles

Abstract

- Believe that the world of symbols and ideas is more important than the tangible world
- Create meaning by forming associations among tangible things, which can generate more ideas and associations

- Believe that physical reality has no meaning in itself and can be misleading, distracting, and/or annoying
- Like to consider the number and variety of possibilities that emerge from abstracting the meaning of things
- Find meaning in relationships among combinations of facts rather than in isolated facts themselves
- See the tangible world as just a stimulus for associations that provide significant meanings
- Find that ideas with many interconnections are valuable, interesting, and exciting
- Feel that ideas need not be abandoned when they contradict evidence, because those ideas may later stimulate useful modifications
- Communicate primarily using words, expressions, and descriptions that evoke a variety of associations
- Use language as a way to imply or suggest rather than specify something concrete

REALISTIC–IMAGINATIVE

Describes **how people develop something new** through dealing with the tasks and problems of daily life and work

Realistic

- Focus on practical, useful things that make a difference
- Emphasize objects, activities, methods, and the kind of knowledge needed to accomplish the goal at hand
- Find relevance in tasks ranging from accomplishing daily routines to acquiring the skills and knowledge needed to accomplish a wide variety of jobs in life, from building a toy chest to planning a wedding
- Value efficiency in the use of time, energy, and money, especially when a useful or tangible purpose is involved
- Give more attention, credence, and time to practical things than to intangibles
- Believe that if something has no tangible impact on doing a task, it has little value
- Value things for themselves as well as for the uses to which they can be put
- Use literal descriptions of things and situations in what they find funny, in contrast to the inferential humor and plays on words used by Imaginative people
- Find that they benefit from acquiring factual knowledge and experience outside their own area of expertise
- Believe that their hands-on knowledge of the material world is a commonsense way of living

- Find that their common sense adds stability and efficiency to daily life and helps them feel grounded in the world
- Feel that their life experiences confirm that complications inevitably arise when common sense is lacking
- Devote much of their time and energy to finding sensible, matter-of-fact solutions to problems; tend to dismiss anything that is merely fascinating or imaginative
- Want solutions that can be implemented within current constraints, such as time or budget

Imaginative

- Feel that the possibilities that tangible things suggest are more important than the tangible things themselves
- See facts as valuable primarily for the associations and images they evoke
- Believe that images have a reality of their own; the physical origin of images is secondary, so once something tangible triggers a chain of associations and images, the initial stimulus may be forgotten
- Enjoy and value the process of stringing together creative images as much as the ideas generated by it
- Tend to make mental leaps rather than using practical, step-by-step procedures for getting something done
- Value creativity for its own sake; find the very newness of an idea appealing
- Tend to question “tried-and-true” procedures and implicit assumptions about things, operations, objectives, people
- Are able to come up with an original way to solve a practical problem by ignoring the limits imposed by the details of the problem
- Find that a solution they suggest may not be workable in its original form but may be reworked to yield an ingenious solution
- Are attracted to strategic planning, envisioning new markets, services, products, and an organization’s future
- Are able to recognize a problem or need and envision a new program to meet it

PRACTICAL–CONCEPTUAL

Deals with the **products or outcomes of people’s perceptions** rather than the process of perception itself

Practical

- Are attracted to others who display practicality and common sense
- Value and find ideas useful only when they are applied to practical, down-to-earth problems

- Enjoy putting things together from known objects and materials, using familiar and practiced methods
- Derive their creativity from experience that is grounded in understanding the materials used and how they fit together
- Believe that new solutions or products evolve from step-by-step changes that are tested at each stage before going on to the next stage
- Are likely to be builders rather than innovators
- Prefer what is solid to what is hazy or imprecise, certainty to vagueness, things that have substance to things that are purely symbolic
- Are oriented to the day-to-day details of living and working rather than to trends or possible future events
- Are likely to accept new ideas if they are immediately applicable or represent a tangible improvement over current practices

Conceptual

- Look for meaning in what is around them, believing that tangible things imply a greater reality that lies beyond what is perceived by the five senses
- Focus on inferences drawn at the conceptual level rather than on what is immediately present, so they enjoy making connections and finding relationships among a variety of ideas
- Enjoy tracing the development of an idea or concept from its origins
- Value abstract intellectual discourse and are stimulated by people with a quick and insightful mind with whom they can exchange ideas
- Are attracted to the exceptional rather than the routine
- Value intellectual and scholarly pursuits over activity and practicality and are more interested in the concept of a new thing than in the thing itself
- Especially enjoy creating things that originate in or are stimulated by their mental images, often communicating to others the meaning the creation has for them
- Feel that step-by-step processes are not as satisfying as the symbolic meaning that arises when they have completed a product or creation
- Tend to be oriented to the future because they value most the insights and images that lie beyond the immediately tangible
- Believe that ideas and concepts stand alone and have meaning independent of any practical application

EXPERIENTIAL–THEORETICAL

Emphasizes the **process by which people make something either tangible or symbolic** from their perceptions

Experiential

- Give serious attention only to things that are validated by experience, as experience is their primary criterion for truth and relevance
- Value and trust the certainty of their own active participation in the world
- Find it hard to discuss truth apart from the trial and error of experience
- Like instructions with specific action steps based on practical experience
- Get pleasure from expertly applying their experience and enjoy being actively involved in whatever is going on
- Find gratification and pleasure when immersed in active, tangible projects and activities
- Feel that producing something using what they already know and have experienced is much more satisfying than trying out an unknown approach with new materials
- See no need to change methods that have worked well repeatedly
- Where there are no established procedures, adeptly use trial and error to discover what works, beginning with what is known from similar situations

Theoretical

- Tend to work at a level or two removed from the immediately tangible, so they find immediate and tangible relationships far less interesting
- Believe that understanding and knowledge exist in an abstract series of principles, explanations, and theories
- Seek patterns and see meanings in those patterns by exploring and seeking new connections among concepts of interest
- Find that facts that are not connected to a theory or put in a larger context have little meaning to them and are quickly forgotten
- Enjoy weaving concepts together to form a coherent set of principles or explanations
- Find that new associations create fresh meanings that promote further exploration
- Find that work that offers no new ideas or approaches to try quickly becomes boring
- Learn and understand better when given theories, concepts, and contexts in addition to factual information

- Recognize beauty in their physical surroundings but find the beauty in concepts and theories even more awe inspiring

TRADITIONAL–ORIGINAL

Focuses on the **social context that gives meaning** to people's perceptions

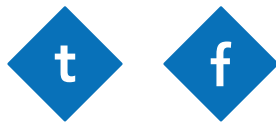
Traditional

- Like doing things in established ways that are shared by most other people
- Enjoy the security that comes from fitting in with the community or group; like feeling that they belong to or are part of something larger than themselves
- Feel that engaging in conventional behavior and having a lifestyle similar to that of others provides validation that their way of being is valued and correct
- Adopt a style of living, including style of dress, housing, and recreational activities, that is informed by tradition
- Find that traditions give meaning to their own and family members' lives and are associated with what is good
- Are uncomfortable going against custom, culture, and traditional norms, so they are reluctant to adopt new products, methods, and activities until they have been validated by time and experience
- Consider change for its own sake pointless and are reluctant to make a change unless it proceeds gradually and is grounded in what was done before
- Want to pass on traditions to the next generation and are puzzled when this is not appreciated

Original

- Find that repetition and sameness are unstimulating but can often serve as a trigger for innovation
- Feel that coming up with new ways to do established things is a way to express themselves and contribute to their family or social group; change is the major motivation—not improvement—because it's boring to do things the same way all the time
- Are bored by work that doesn't allow opportunities to invent different ways of doing things and to put their best efforts into work that rewards originality
- Feel that performing familiar activities in new ways allows for the unexpected and gives new energy to an experience
- May want to retain a basic traditional approach to a holiday or other family event but with variations, believing that the variations enhance the meaning of the activity

- Respect tradition but find that doing things exactly the same way every time diminishes the meaning of an event
- Do not feel the need to be original in all areas of their life, only in especially meaningful ones
- Admire people who are willing to be different, finding virtue in the unconventional and eccentric



THE THINKING–FEELING FACETS

The five facets within the Thinking–Feeling preference pair are

- Logical–Empathetic
- Reasonable–Compassionate
- Questioning–Accommodating
- Critical–Accepting
- Tough–Tender

LOGICAL–EMPATHETIC

Emphasizes the **criteria people use to reach a judgment**

Logical

- Understand the things that happen in the world best when they make logical sense
- Analyze objects, events, and statements using reason
- Believe that analyses that are internally consistent and logical are true and when that is not the case the issue in question is untrue or not understandable
- See illogical statements as a waste of time and attention, so they are persuaded only by arguments that make logical sense, where facts support conclusions
- Typically start with a set of assumptions or facts and use specific rules to make deductions
- Assume that there are universal rules that permit logical deductions
- Search for the premises and assumptions involved when they don't understand something
- Are convinced that reason leads to accuracy and that others will understand and accept conclusions based on reason because logic speaks for itself and should be obvious to everyone
- Consider facts about emotional states "objective" if they fit into a consistent analytic framework
- Respect clear and consistent thinking; disrespect illogic, inconsistency, and contradiction
- May strongly disagree with others while still respecting them, but will cast doubt on their

credibility or viewpoint when their argument is inconsistent

- Tend to apply general, impersonal principles to a broad range of relationships
- Value rights, fairness, and reasonableness as standards for relationships and expect the same from others

Empathetic

- Believe that the world operates within a rational framework of relationships among people
- See relationships, life experiences, and personal meanings as more important than detached logic
- Are convinced that truth is not separate from people and their lives, so applying logically consistent principles detached from specific people makes no sense to them
- See truth as both personal and universal
- Believe that understanding others comes from knowing about their experiences or from a mutual sharing of experiences
- Value others for their basic humanness and uniqueness
- Feel that logic and reason are included in their framework of experience and understanding as one element among many others necessary to grasp life's meaning
- In intimate relationships, handle differences by accepting them as part of the other person's uniqueness—as long as key values are not at issue
- Pay more attention to people's feelings than to their rights and believe that the use of power in the name of rights is heartless self-interest, especially when someone is hurt by it
- Feel that the impacts of judgments on people and their feelings are of far higher value than any rights or standards of fairness that do not consider the well-being of the individuals involved

REASONABLE–COMPASSIONATE

Emphasizes the **standards people use to maintain relationships** when making Thinking or Feeling judgments

Reasonable

- See relationships as necessary for accomplishing tasks, so they factor people's needs into their problem-solving logic
- Demonstrate caring by analyzing and solving problems and by fixing or changing things
- Believe that sympathizing is not helpful to problem solving or decision making, seeing mercy as equal to letting sympathy for one person overrule principles of fairness

- See the need for consistency in dealing with people because exceptions made for one person must also be made for others with similar problems
- Believe that making exceptions based on individual needs is the same as playing favorites
- Find it difficult to work for or with someone who is inconsistent
- Try to resolve disagreements by balancing the positive and negative consequences for the people involved
- Believe that giving and receiving fair treatment is much more important than attending to one's own or others' feelings or emotional states

Compassionate

- See the world as personal and interconnected rather than impersonal and detached
- Believe that laws, rules, and procedures define the boundaries of relationships but are not their essence
- Pay attention to the unique needs others bring to situations
- See mercy as the basis on which judgments should be made, as it takes into account people's unique and possibly mitigating circumstances and characteristics
- Value the individual person over applying abstract principles to everyone
- Believe that the failure to act compassionately hurts those involved
- Respond best in relationships when they are treated sympathetically and compassionately
- Highly value being recognized as an individual connected to others in a network of friendships
- Are likely to be devoted and loyal when treated with sympathy, understanding, and compassion but not when merely treated with fairness and consistency

QUESTIONING–ACCOMMODATING

Focuses on **how people deal with differences of opinion**

Questioning

- Require reasons that make logical sense and need questions, statements, and events to fit together logically
- Ask direct questions to clarify others' point of view and want what others say to be logically consistent with what the questioner already knows
- Ask questions to arrive at common ground from which they and the other person can proceed
- See questioning as a way to solve problems and arrive at objective truth

- Approach all statements with some degree of skepticism and distrust people who seek agreement before all questions have been dealt with
- Are likely to ask questions or object when asked to accept or conform to things that don't make logical sense
- See being asked to accept a decision not thoroughly examined as an insult to their intelligence
- See truth as independent of people, so their questions are not meant to disparage, disregard, or personally attack anyone
- Believe that if something doesn't work, the assumptions about how its parts go together are wrong; questioning those assumptions may lead to a solution to the problem
- Question doing something simply because it has always been done that way
- Usually trust their assessment of a situation over others' assessments, especially when the legitimacy of a leader is in doubt

Accommodating

- Define reality in terms of interpersonal and social relationships
- Are more concerned with how truth is understood, valued, and used than with an assumed objective truth
- Believe that the most important truths are validated by group consensus or personal relationships, even if an objective assessment may show otherwise
- Primarily use criteria that focus on their own and others' experiences and understanding of the world
- Will gently correct, but never disparage, others when their understanding is incorrect
- Believe that if directly questioning someone's statement can be taken as a personal attack and/or might threaten a relationship or group membership, it is better not to question the statement or the other person's motives
- May value harmony so highly that they are unwilling to question another's viewpoint
- Feel that agreement and harmony are not merely signs of a good relationship; they are the mortar that holds it together
- Will aim for a decision that satisfies everyone when there are differences of opinion
- Believe that groups cannot function well if different perspectives are not accommodated; work to ensure harmony in family and other personal situations by avoiding disagreements and confrontations
- May follow another person's lead in adhering to established ways of doing things, even if they see an appropriate alternative

- Avoid hurting people's feelings and expect the same from others

CRITICAL-ACCEPTING

Describes **what people do after making their initial judgments**

Critical

- Seek the truth in order to improve things, situations, or procedures, so focus on correcting what is incorrect
- Tend to point out what is wrong before noticing or commenting on what is right, believing that mentioning what is right wastes time since it will not lead to improvement
- Believe that critiquing is necessary to determine an issue's relative merits; not evaluating things critically is irresponsible
- Are convinced that truth is more important than people's feelings; may not use tact in communicating their judgments
- See being honest as kinder than being untruthful, since people get over hurt feelings but the consequences of wrong judgments tend to hurt them more in the long run
- Believe that the future consequences of a decision are more important than any good or bad feelings in the present
- Find it dishonest and distasteful when others smooth over disagreements or only mildly disagree
- Believe that anything done merely to create harmony compromises their beliefs and values
- Are able to tolerate others' shortcomings and accept lower standards when it is absolutely necessary

Accepting

- Believe that truth that affirms the value and worth of others' ideas and viewpoints is far more important than "objective" truth
- Feel that facts, nonpersonal issues, and possible long-term negative consequences are secondary to valuing and accepting others
- First notice and comment on what is right and well done, then offer corrections in a mild, supportive way
- Tend to overlook others' lapses and deficiencies
- Feel that accepting others enhances relationships and helps people achieve their full potential
- Actively affirm others despite possible risks to the affirming person (e.g., suffering financial loss or setting aside their own needs)
- Take a collaborative approach, at times putting aside their own desires and ambitions in favor of those of others

- See the essence of the world as primarily human and social, so passing critical judgment on another person's ideas or contributions may harm that person and destroy the relationship
- Believe that there are many good ideas or opinions in the world and that better solutions come from accepting that there are a variety of viewpoints; are willing to put up with others' shortcomings and tolerate less-than-optimal standards, performance, or outcomes

TOUGH-TENDER

Focuses on the **impact of people's judgments and how they carry out a decision once a judgment has been made**

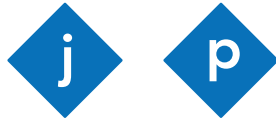
Tough

- Stand firm on judgments because they believe that their thorough and careful evaluation inevitably led to the best possible decision
- Believe that once a judgment is made, no negative effect can justify changing it, even if it has a negative impact on the decision maker
- Believe that one's personal opinions and those of others are irrelevant to the outcome of a well-made decision
- Distrust considerations based solely on feelings, personal attachments, or a desire for warmth and harmony
- Feel that compromising on a decision is inappropriate, even if it would avoid outcomes that would hurt some people
- Believe that giving in to sentiment is irresponsible, as it will likely cause future harm

Tender

- Believe that the effects one's decision may have on others far outweigh any logical process of reaching a decision
- Feel that kindness, caring, and consideration of others are the primary concerns in both making and implementing a decision
- Feel that bringing warmth and concern for others to decision making is important because the well-being of people is integral to the decision-making process
- Express tenderness, warmth, and concern when communicating their decisions to others, especially when a decision will hurt some people or be unpopular
- Insist that others be treated with kindness and consideration; respond best when they are treated with the same gentleness and warmth they give to others

- Are much more concerned with anticipated impacts of a decision on the well-being of the people involved than with any possible logical outcomes
- Are likely to be just as convinced of the rightness of their conclusions as are Tough people



THE JUDGING–PERCEIVING FACETS

The five facets within the Judging–Perceiving preference pair are

- Systematic–Casual
- Planful–Open-Ended
- Early Starting–Pressure–Prompted
- Scheduled–Spontaneous
- Methodical–Emergent

SYSTEMATIC–CASUAL

Focuses on **how people organize their physical environment**, including the flow of events, activities, tasks, and projects

Systematic

- Employ a variety of ways to achieve orderliness in both small, brief tasks and large, long-term ones
- Like structure, methods, and deliberate, systematic approaches to efficient time use, including meeting deadlines and estimating how long a task will take
- Feel a need to schedule tasks and activities efficiently, so they value being punctual and starting and ending things on time
- Find that their approach gives them freedom to enjoy other things; manage their leisure time and activities systematically so that they can make the most of their time and enjoy it more
- Believe that not being systematic sacrifices time, effort, and resources, allowing inefficiency and waste
- Organize tasks and activities every day and at the outset of a project to avoid wasting time looking for items that have been misplaced
- Find that unexpected negative consequences and disorder ensue when they don't control their time, physical environment, and approach to work
- Find that chaotic work or home environments interfere with getting things done
- Are willing to yield control to another trusted systematic person
- Need closure on decisions and believe that delaying decisions creates tension and depletes energy that could be better used on something else

Casual

- Prefer a spontaneous approach to accomplishing their tasks, believing that systems and order are impediments to working effectively
- Feel that too much orderliness inhibits the excitement, energy, and variety allowed by spontaneity
- Are most comfortable with an easygoing approach to schedules, deadlines, decision making, the physical environment, using time, and performing tasks
- Are at their best in work settings where procedures are not well established and projects allow for rapid changes in the way they approach and accomplish goals
- Feel that being casual allows openness to seeing and experiencing things in new, fresh ways
- Believe that a loose, unstructured approach allows them to see, experience, or respond to opportunities they might otherwise miss
- Actively seek variety and newness, avoid constraining structures; are at their best when acting on impulse, working on whatever they wish, and responding to whatever tasks arise
- Are aware that others likely see their preferred work environment as chaotic but find that the stimulation promotes positive outcomes
- Like surprises and have great energy and enthusiasm when given unexpected tasks, so they don't mind being interrupted
- Are able to work systematically when necessary but find it less satisfying
- Are comfortable postponing decisions until all angles have been explored
- Are able to make decisions before they feel ready but may change a decision with new information
- Are reluctant to change a thoroughly considered decision because they believe the time they invested will have been wasted

PLANFUL–OPEN-ENDED

Emphasizes **people's approach to current and future events and goals**, including leisure time and social activities

Planful

- Like to schedule leisure time, including exactly when they will do something, to ensure that everything they plan will happen
- Like to plan in advance how they will spend each day, including vacations and recreation
- Prefer knowing the dates and times of social activities in advance so that they can avoid scheduling two things at the same time

- Put time and energy into advance planning for educational and financial objectives because they are uncomfortable leaving them to chance in the future

Open-Ended

- Prefer leaving leisure time unscheduled so that they can take advantage of unexpected opportunities
- Feel that planning in advance guarantees that something more attractive will turn up and they will not have the freedom to choose from what is available at the time
- Find that variety and being able to improvise are as meaningful as the activities themselves
- May not regret missing something they failed to plan for because they value having the freedom to choose something appealing in the moment
- Enjoy matching themselves to the flow of life events and activities, which is not possible if they plan everything in advance
- Tend to avoid devoting time and energy to advance planning for educational and financial goals unless pushed to do so by others

EARLY STARTING—PRESSURE-PROMPTED

Focuses narrowly on how people manage time with regard to deadlines

Early Starting

- Deal with deadlines by starting on a task early enough to allow plenty of time to finish
- Prefer knowing assignments well in advance so that they can start working on them early; do their best work when they can avoid a last-minute rush
- Find it stressful having to work until the last minute on projects with deadlines; the stress is likely to impact their relationships with co-workers and family
- Feel that time pressure makes it hard for them to concentrate on tasks and leads to inefficiency
- Feel like they have failed when they miss a deadline, were too rushed to do their best work, or had no time to check their work before submitting it
- Find it stressful being given unexpected tasks even without a deadline because it interferes with time they had reserved for scheduled tasks

Pressure-Prompted

- Do their best work when there is sufficient time pressure to challenge their ability to meet a deadline
- Very much enjoy the adrenaline rush that occurs when meeting a tight schedule
- Find it hard to work without the pressure of a deadline, leading others to question whether they will get the work done

- May put off working until the deadline is close enough to require significant effort to complete the task, which others may see as procrastination
- May appear to be doing “nothing,” but in reality they are working on the task in their head; this serves as a “gestation” period that stimulates their energy
- Find that working on several projects at the same time can stimulate solutions or methods for one incomplete task that are relevant to a different incomplete task
- Feel that they do their best work when under stress from severe time constraints, as the energy stimulated by an approaching deadline yields more ideas, efficiency, and clarity of thought
- May not know how long a task will take, but the time pressure enables them to structure their time so that they can better decide what to include or discard
- If a task is finished early, they may feel that they began too soon, wasting time they could have used for something else—such as getting started on another project with a deadline

SCHEDULED—SPONTANEOUS

Centers on the degree of structure in people's daily activities

Scheduled

- Are comfortable with routine because it enables efficient functioning by ensuring that time and energy are not wasted
- Find that routine promotes the continuity of work so that everything flows together as smoothly and flawlessly as possible
- Especially enjoy routine that allows blending their own abilities and energy with those of others
- Like to structure daily activities in a known sequence so that shifting gears will not be necessary
- Feel that scheduling their time allows them to look forward to all parts of their day and that routine is restful and comfortable because it minimizes surprises
- Favor routines that cover long spans of time, including weekly, seasonal, and yearly patterns of work, leisure, and family time
- Find that repetition can be meaningful because it has a stabilizing or “anchoring” effect on friendships and family relations
- Are able to function without a fixed pattern, but may be uncomfortable without an organizing structure that covers days and weeks

Spontaneous

- Are energized by variety in their daily work, so they work best when doing tasks that involve constant variety, which they strive to introduce into their work
- Like having the freedom to decide what tasks to do and when; may tackle a favorite task first, regardless of any deadlines
- Find variety and freedom to respond on the spur of the moment essential to their mental health, so the worst thing for them is being stuck in a rut with no escape
- Feel cramped and confined by the very thought of following the same routine day after day
- Break the monotony of having too much routine in their days by exploring new activities
- Are able to work within a routine when necessary but find it boring or even painful
- May be turned off by the thought of longer-term routines, such as family vacations, holiday and birthday celebrations, or other recurring events, so they will seek ways to add variety

METHODICAL–EMERGENT

Focuses narrowly on **how people approach smaller tasks required to complete larger projects**

Methodical

- Tackle large projects by organizing themselves and whatever materials, tools, and people they will need in advance of starting the project
- Make lists of needed items, necessary steps to be taken, and the order of taking them
- Make sure to have necessary materials available or to acquire them so that they will have what they need when they need it
- Enjoy organizing tools and materials so that they will be free to focus on getting the job done and prepared to do it well
- Will contact others in advance if they need their help and make sure to remind them as the date they will be needed approaches
- Like to read the complete instructions in advance of starting work so that they will have the big picture and know what to expect
- Believe that a methodical approach saves time, minimizes mistakes, and reduces costs—in addition to producing better work and greater personal satisfaction
- Tend to set subgoals or targets to help assess whether they are on track
- Are likely to find it frustrating and inefficient when they have to work on a task with an unmethodical person

- As teachers, make orderly lesson plans, with explicit learning objectives that build sequentially on students' knowledge
- As trainers and presenters, prepare and follow detailed agendas so that new material can be understood and remembered by their trainees or audience members

Emergent

- Tend to treat projects as explorations or discoveries, delighting in finding out what to do and how to do it as they go along
- Usually don't start a large task with step 1, and may not complete any steps in a particular order unless absolutely necessary
- May make a very general outline or list of required tasks, but find that taking time to organize tools, materials, and people before beginning a task is not satisfying, interesting, or efficient
- Are eager to first tackle the parts of a task they find interesting; then they will proceed to other, less appealing requirements in the same unsystematic way
- May discover by trial and error when a strict sequence of tasks is required
- Read directions in advance rarely, if at all, since that would remove the exploratory spirit they enjoy
- Look for tools only when they are needed—though they may have trouble finding them
- Are likely to discover whether a part is missing or mislabeled only after they have already assembled the piece
- Tend to tackle the core piece first, spending most of their time and energy on what is most important; completing one task leads to another, so by the end an overall picture of how everything fits together will emerge
- Find that their approach is especially suitable for new situations or projects with many unknowns or where improvisation is needed
- As teachers, take an adaptable approach to lesson plans; may diverge from the plan in response to student interest and opportunities for new learning, while still covering major learning objectives
- As trainers, prepare a sketchy agenda with few details, finding that the sequencing of presentation arises dynamically from their unique interaction with a particular group on a particular day

CONCLUSION

This chapter presented type descriptions for the 16 MBTI Step I types and the 20 Step II facets. It provided a brief history of Isabel Myers' type descriptions as they developed over time. The chapter also provided a helpful context for viewing the facet descriptions, as well as a summary of the rationale used in ordering the facets within each block of five facets. Suggestions were made about how to approach reading the facet descriptions, which summarize the descriptions that were written in 1997 (Mitchell et al.) and revised for inclusion in the Step II manual for Form Q (Quenk, et al., 2001).

Practitioners who are new or relatively new to the MBTI Step II assessment may feel somewhat overwhelmed by the subtleties of meaning reflected in the five facets within each facet block. As is true when using any assessment that acknowledges the complexity of human personality, the practitioner's comfort with the Step II assessment will increase with experience in using it with respondents. By providing feedback to respondents and paying close attention to what they say about their facet results, practitioners will increase their understanding and skill with surprising speed. Keeping this chapter available when working with respondents is highly recommended.

CHAPTER 5

Type Tables and Groupings of MBTI® Preferences

CHAPTER OVERVIEW

Using Type Tables to Promote Understanding

Illustrative Type Tables

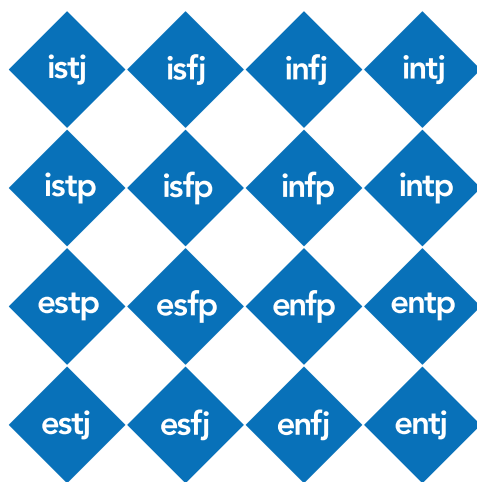
Groupings of the Preferences

Conclusion

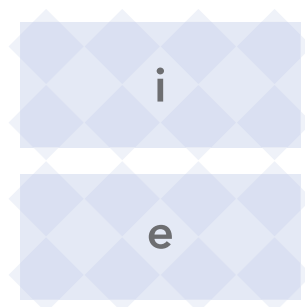
The preceding chapters provided an understanding of the theory underlying the MBTI® Global Step I™ and Step II™ assessments. This chapter focuses on characteristics that the Step I types have in common by exploring Isabel Myers' graphic array of the types in the form of a type table. Type tables can be used to show the frequencies of types in groups of people who have some characteristic in common—for example, the same occupation, college major, avocation, management philosophy, or communication style. A wide variety of variables of interest to researchers and practitioners can be explored using type tables, allowing easy visual identification of types who share some common variable. This feature encourages the exploration of hypotheses so that information can be used to counsel and advise respondents. Using a type table, one can tell at a glance which types tend to be found more or less frequently in the group. However, further statistical analysis of the type table is required to determine whether these frequencies are significant. Research showing the validity of type-related hypotheses in occupational and organizational choice is discussed in chapter 9.

USING TYPE TABLES TO PROMOTE UNDERSTANDING

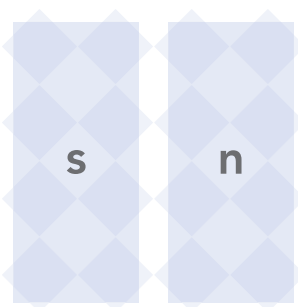
The type table was designed by Isabel Myers as a way to highlight similarities and differences among the types. Familiarity with the placement of types within the type table enables quick recognition of which types predominate and which appear with less frequency in different samples and populations. Figure 5.1 depicts the format of the type table and illustrates the placement within the table of each Step I type. Note that each type has three letters in common with any adjacent type. The placement of each of the preferences and the rationale given by Myers for these placements are described below.



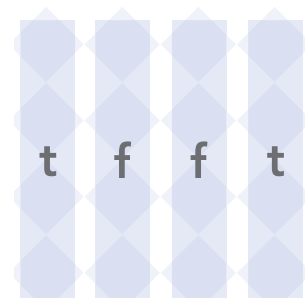
Extraversion–Introversion



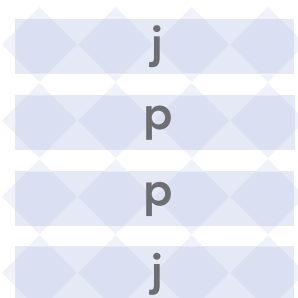
Sensing–Intuition



Thinking–Feeling



Judging–Perceiving



Extraversion (E) and Introversion (I). Introverted types appear in rows 1 and 2 of the type table. Extraverted types appear in rows 3 and 4. Isabel Myers' mnemonic for E–I placement is that Introverts are more likely to have their head *up* in the clouds and Extraverts are more likely to have their feet *down* on the ground.

Sensing (S) and Intuition (N). Sensing types and Intuitive types are positioned on the type table in the same order as in the name of the S–N preference pair, with Sensing in the two left-hand columns and Intuition in the two right-hand columns.

Thinking (T) and Feeling (F). Thinking types appear in the two outer columns of the type table. Feeling types appear in the two inner columns. Myers chose this placement so that it would be easy to remember that the Feeling types, with their greater need for affiliation, are in the inner columns, surrounded by other types. The more objective Thinking types are in the outer columns, with open space on their outer side.

Judging (J) and Perceiving (P). The decisive Judging types are found in the top and bottom rows, providing boundaries for the flexible and adaptable Perceiving types, who are in the two middle rows.

ILLUSTRATIVE TYPE TABLES

To show the kind of useful information that can be derived from type tables, a type table with data for 8,707 accountants is shown in table 5.1 and another for 2,054 training and development specialists is shown in table 5.2.

For each type in tables 5.1 and 5.2, the following information is shown:

- *n* (the number of people of that type in the sample)
- % (the percentage of people of that type in the sample)
- SSR (self-selection ratio, explained below)

A self-selection ratio is a number showing whether the frequency of a particular type in the group for which a type table was created is the same as, greater than, or less than the frequency that would be expected in an appropriate base comparison group. For example, the base group for accountants would be a large sample of college graduates representing a wide range of majors since self-selecting to be an accountant requires a person to have a college degree. A base comparison group for training and development specialists might be a large sample of individuals who hold an associate or bachelor's degree. An SSR of 1 means the frequency of a type in the group shown in the type table is the same as the frequency that would be expected in the base group; an SSR greater than 1 means that the type is overrepresented in the group shown in the type table.

Figure 5.1 | Format of type table

compared to the base group; an SSR less than 1 means that the type is underrepresented in the group shown in the type table compared to the base group. SSRs are also discussed in chapter 9.

Type theory leads to the hypothesis that the frequency of types in a type table is the result of choices individuals make that are directly or indirectly influenced by their type preferences. It is not surprising, therefore, that in table 5.1 (accountants in the US sample) all but one of the types in the two outer columns (the eight types who prefer Thinking) are overrepresented and all but one

of the types in the two inner columns (the eight types who prefer Feeling) are underrepresented. In table 5.2 (training and development specialists in the US sample) all the types in the two right-hand columns (the eight types who prefer Intuition) are overrepresented and all but two of the types in the two left-hand columns (the eight types who prefer Sensing) are underrepresented. These kinds of frequencies in type tables for occupations provide evidence for the validity of type theory because the theory predicts that individuals will be attracted to occupations that permit and encourage them to use their type preferences. Such data are also useful for

Table 5.1 | Type table of accountants in the US sample

Sensing				Intuition										
Thinking		Feeling		Thinking										
ISTJ <i>n</i> = 1,787 20.5% SSR = 1.77		ISFJ <i>n</i> = 804 9.2% SSR = 0.67		INFJ <i>n</i> = 186 2.1% SSR = 1.40		INTJ <i>n</i> = 233 2.7% SSR = 1.29		Judging	Introversion					
ISTP <i>n</i> = 592 6.8% SSR = 1.26		ISFP <i>n</i> = 356 4.1% SSR = 0.47		INFP <i>n</i> = 244 2.8% SSR = 0.64		INTP <i>n</i> = 264 3.0% SSR = 0.91				Perceiving				
ESTP <i>n</i> = 547 6.3% SSR = 1.47		ESFP <i>n</i> = 378 4.3% SSR = 0.51		ENFP <i>n</i> = 355 4.1% SSR = 0.51		ENTP <i>n</i> = 334 3.8% SSR = 1.19		Judging	Extraversion					
ESTJ <i>n</i> = 1,383 15.9% SSR = 1.83		ESFJ <i>n</i> = 755 8.7% SSR = 0.71		ENFJ <i>n</i> = 196 2.3% SSR = 0.92		ENTJ <i>n</i> = 293 3.4% SSR = 1.89								
Preferences			Orientation pairs		Process pairs		Orientation of energy and perceiving pairs		Judging and external orientation pairs					
	<i>n</i>	%		%		%		%		%				
E	4,241	48.7	EJ	2,627	30.2	ST	4,309	49.5	ES	3,063	35.2	TJ	3,696	42.4
I	4,466	51.3	EP	1,614	18.5	SF	2,293	26.3	EN	1,178	13.5	TP	1,737	19.9
S	6,602	75.8	IJ	3,010	34.6	NF	981	11.3	IS	3,539	40.6	FJ	1,941	22.3
N	2,105	24.2	IP	1,456	16.7	NT	1,124	12.9	IN	927	10.6	FP	1,333	15.3
T	5,433	62.4												
F	3,274	37.6												
J	5,637	64.7												
P	3,070	35.3												

Note: *N* = 8,707 (3,125 men; 5,475 women; 107 not reporting gender). Highest percentage and highest SSR are in bold type.

Average age: 37 years; education: 66% associate/bachelor's degree, 25% graduate degree; satisfied with work: 92%; average time in occupation: 11 years; most common organizational level: 44% nonsupervisory.

Source: Data from The Myers-Briggs Company archive database, Schaubhut & Thompson (2012).

Table 5.2 | Type table of training and development specialists in the US sample

Sensing				Intuition										
Thinking			Feeling			Thinking								
ISTJ <i>n</i> = 206 10.0% SSR = 0.86			ISFJ <i>n</i> = 123 6.0% SSR = 0.43			INFJ <i>n</i> = 61 3.0% SSR = 1.98			INTJ <i>n</i> = 80 3.9% SSR = 1.85			Judging	Introversi	
ISTP <i>n</i> = 68 3.3% SSR = 0.61			ISFP <i>n</i> = 38 1.9% SSR = 0.21			INFP <i>n</i> = 96 4.7% SSR = 1.06			INTP <i>n</i> = 93 4.5% SSR = 1.37					
ESTP <i>n</i> = 93 4.5% SSR = 1.05			ESFP <i>n</i> = 105 5.1% SSR = 0.60			ENFP <i>n</i> = 234 11.4% SSR = 1.41			ENTP <i>n</i> = 156 7.6% SSR = 2.37			Perceiving	Extroversion	
ESTJ <i>n</i> = 220 10.7% SSR = 1.23			ESFJ <i>n</i> = 186 9.1% SSR = 0.74			ENFJ <i>n</i> = 155 7.6% SSR = 3.02			ENTJ <i>n</i> = 140 6.8% SSR = 3.79					
Preferences			Orientation pairs			Process pairs			Orientation of energy and perceiving pairs			Judging and external orientation pairs		
	<i>n</i>	%		<i>n</i>	%		<i>n</i>	%		<i>n</i>	%		<i>n</i>	%
E	1,289	62.8	EJ	701	34.1	ST	587	28.6	ES	604	29.4	TJ	646	31.5
I	765	37.2	EP	588	28.6	SF	452	22.0	EN	685	33.3	TP	410	20.0
S	1,039	50.6	IJ	470	22.9	NF	546	26.6	IS	435	21.2	FJ	525	25.6
N	1,015	49.4	IP	295	14.4	NT	469	22.8	IN	330	16.1	FP	473	23.0
T	1,056	51.4												
F	998	48.6												
J	1,171	57.0												
P	883	43.0												

Note: *N* = 2,054 (647 men; 1,383 women; 24 not reporting gender). Highest percentage and highest SSR are in bold type.

Average age: 39 years; education: 49% associate/bachelor's degree, 33% graduate degree; satisfied with work: 95%; average time in occupation: 11 years; most common organizational level: 64% nonsupervisory.

Source: Data from The Myers-Briggs Company archive database, Schaubhut & Thompson (2012).

career counselors. If type preferences had no effect on occupational choice, the frequencies in the type tables would be expected to be about equal or to be randomly distributed.

GROUPINGS OF THE PREFERENCES

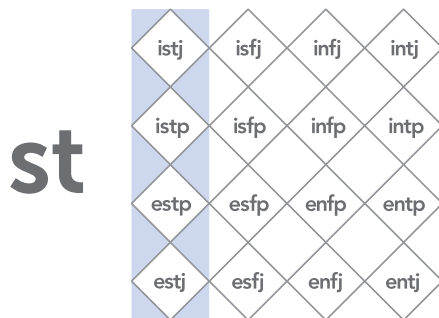
Dividing the type table into groups of types who have in common particular combinations of preferences allows hypotheses to be made, based on type theory, about what characteristics members of a particular grouping are likely to have in common. Therefore, being familiar with the type table and with the characteristics of people who have specific preference combinations aids in understanding the individual differences that are identified through psychological type.

The discussion of each type grouping begins with the distinctive dynamic feature (e.g., the ST process pair or the EJ orientation pair) that the types in that group share. The descriptive phrases that are associated with each grouping were created by Isabel Myers to identify and help people remember that group's central distinctive feature.

The Columns of the Type Table: The Four Mental Process Pairs (ST, SF, NF, NT)

Isabel Myers considered the columns of the type table to be the most important groupings of the types, particularly when it comes to career choices. This grouping focuses on the combination of a perceiving process (S or N) with a judging process (T or F). The characteristics of the types in this group are assumed to stem from the preferred use of these mental processes. The process pairs have received a great deal of research attention over the years. They are often identified as *learning styles* or *cognitive styles*. This is an area of interest to educators, career counselors, and organizational development specialists.

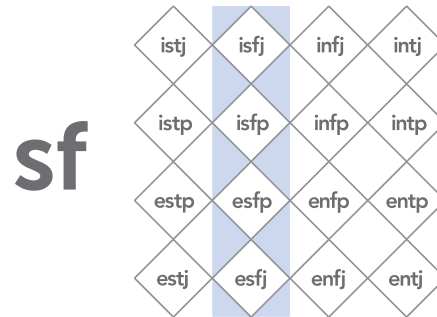
STs: THE PRACTICAL AND MATTER-OF-FACT TYPES



For the four Sensing Thinking (ST) types, one of the two processes Sensing or Thinking is the dominant process and the other is the auxiliary process.

ST types rely primarily on Sensing for taking in information and on Thinking for making decisions and drawing conclusions about what has been perceived. Their main interests focus on noticing and remembering facts, because facts can be collected and verified directly by the senses—by seeing, hearing, touching, tasting, or smelling. The ST types typically approach their decisions about facts using objective analysis—because what they trust is Thinking, with its linear and logical process of reasoning from cause to effect, from premise to conclusion. Consequently, STs tend to be practical and matter-of-fact. Type theory predicts that STs will be attracted to career fields that demand nonpersonal analysis of concrete facts, such as economics, law, surgery, business, accounting, production, and the handling of machines and materials. The frequency with which ST types are found in such fields bears this out.

SFs: THE SYMPATHETIC AND FRIENDLY TYPES

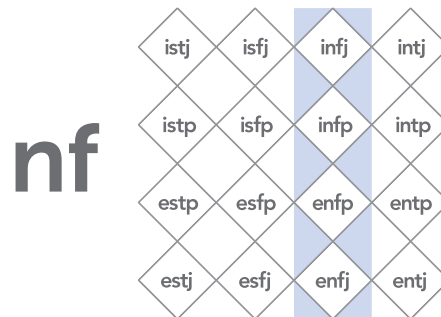


For Sensing Feeling (SF) types, one of the two processes Sensing or Feeling is the dominant process and the other is the auxiliary process.

SF types, like STs, rely primarily on Sensing for taking in information but rely on Feeling for making decisions and drawing conclusions about what has been perceived. They too are mainly interested in facts they can gather directly through the senses, but they approach their decisions with a subjectivity that is based on their personal value system. This subjectivity and the warmth they convey come from their trust of Feeling, with its power to weigh how much things matter to them and others. In contrast to Sensing Thinking types, they are more interested in facts about people than in facts about things.

SFs tend to be sympathetic and friendly. In theory, we would expect them to be attracted to fields in which their personal warmth can be effectively applied to concrete situations. The combination of Sensing and Feeling can be valuable in service jobs, selling tangibles, teaching (especially in the early grades and applied fields), nursing, pediatrics, and other health fields involving direct patient care. The sympathetic and friendly SFs are frequently found at all levels of direct patient care.

NFs: THE ENTHUSIASTIC AND INSIGHTFUL TYPES



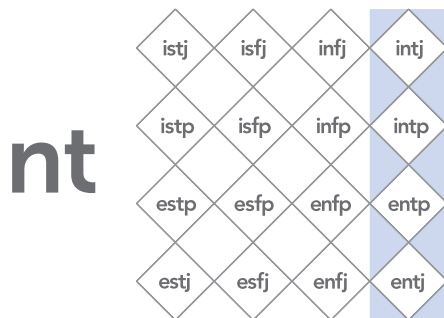
For Intuitive Feeling (NF) types, one of the two processes Intuition or Feeling is the dominant process and the other is the auxiliary process.

NF types rely primarily on Intuition for taking in information and on Feeling for making decisions and drawing conclusions about what has been perceived.

They typically possess the same personal warmth as SF types. However, since they prefer Intuition to Sensing, they focus their interest on possibilities rather than concrete realities. They are likely to be attracted to new projects, things that have never happened but might be made to happen, or the exploration or discovery of new truths. NF types are typically interested in the complexities of communication, and their Intuition provides them with an interest in symbolic meanings, theoretical relationships, and patterns that underlie immediate facts. Feeling provides the interest in using these intuitive insights to benefit human relationships.

The personal warmth and commitment with which NF types seek and follow up on possibilities tend to make them enthusiastic as well as insightful. They often have a marked gift for the spoken or written word and can communicate both the possibilities they see and the values they attach to those possibilities. They are likely to be attracted to work that involves the unfolding of possibilities, especially possibilities for people, such as in teaching (particularly in the upper grades and college), counseling, writing, research, and selling intangibles, such as raising funds for charitable organizations or asking people to volunteer for worthy causes.

NTs: THE LOGICAL AND INGENIOUS TYPES



For Intuitive Thinking (NT) types, one of the two processes Intuition or Thinking is the dominant process and the other is the auxiliary process.

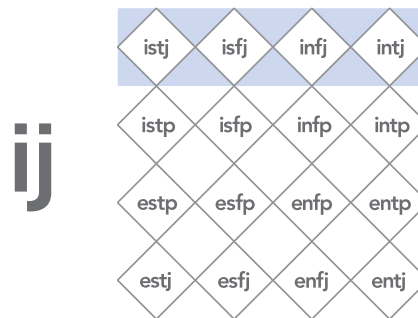
Like NFs, NT types rely primarily on Intuition for purposes of perception, but they prefer the objectivity of Thinking for purposes of judgment. They too focus on possibilities, theoretical relationships, and abstract patterns, but they judge these from a nonpersonal, cause-and-effect perspective. They often pursue possibilities that are technical, scientific, theoretical, or executive, where attention to the human issues may be secondary.

NTs tend to be logical and ingenious. They are best at solving problems within their field of special interest, whether scientific research, mathematics, the more intricate aspects of finance, or any sort of pioneering or innovative endeavor in a wide variety of technical or administrative areas.

The Rows of the Type Table: The Four Orientation Pairs (IJ, IP, EP, EJ)

The rows of the type table group the introverted and extraverted perceiving types (IJs, whose dominant process is an *introverted* perceiving process, and EPs, whose dominant process is an *extraverted* perceiving process) and the introverted and extraverted judging types (IPs, whose dominant process is an *introverted* judging process, and EJJs, whose dominant process is an *extraverted* judging process).

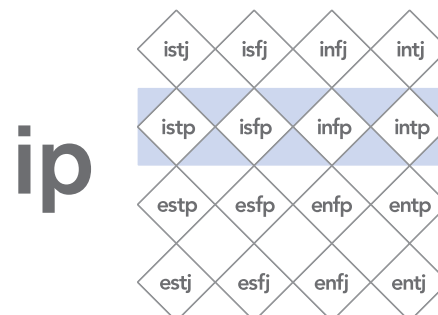
IJs: THE DECISIVE INTROVERTS



All four Introverted Judging (IJ) types have a dominant introverted perceiving process and an auxiliary extraverted judging process.

IJs are introspective, persevering, and hard to convince or change, unless compelling data are provided that override a decision or foregone conclusion. They can appear resistant to change because they first extravert their preferred auxiliary judging process (Thinking or Feeling). Thus they state their conclusions first, before providing the data for their conclusions, which come from their dominant introverted perceiving process, Intuition or Sensing. They can therefore appear to others to be adamant and inflexible. However, since their perceiving process is the dominant, most crucial one, IJs will readily relinquish a conclusion if they are given new information that contradicts their earlier conclusion, even when it seems to other people that their decision is firm and intractable.

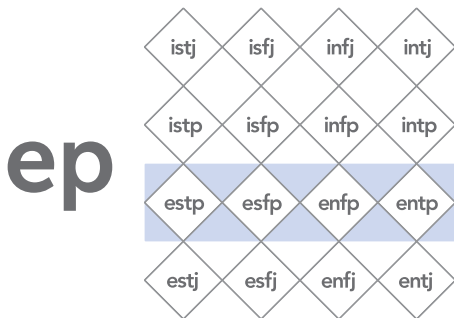
IPs: THE ADAPTABLE INTROVERTS



All four Introverted Perceiving (IP) types have a dominant introverted judging process and an auxiliary extraverted perceiving process.

IPs are introspective, adaptable in little things, and firm on important issues because their dominant judging process is central to their personality. Others may assume that the outer flexibility and adaptability of IPs reflects their internal state as well. Such an assumption is usually unjustified, especially when important inner judgments are at issue. In such circumstances, when IPs are satisfied that they have considered the relevant data and used their dominant Thinking or Feeling judgment to reach a conclusion, they are unlikely to be open to changing that judgment. When others erroneously assume that IPs are as flexible in their judgments as they are in their perceptions, they may be surprised and confused. They may conclude that IPs are by nature intractable and resistant to change. For IPs, however, in order for new information to be effective in promoting a willingness to change, it must address the logic or values of their dominant judging process or it will be rejected as irrelevant.

EPs: THE ADAPTABLE EXTRAVERTS

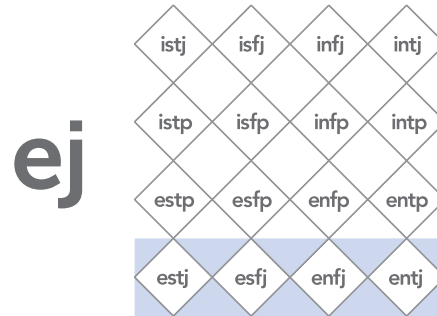


All four Extraverted Perceiving (EP) types have a dominant perceiving process (S or N) that is extraverted and a judging process (T or F) that is the introverted auxiliary process.

EPs are active, energetic, and sociable and often seek new experiences. They adapt even more easily to new outer conditions than do the IPs (for whom the extraverted perceiving process has only an auxiliary status). EPs' dominant extraverted perceiving process leads them to trust and eagerly engage the outside world. They are optimistic about what that outside world has to offer and about their ability to interact with it in a satisfying way. Their optimism may be perceived as excessive risk taking by types who approach the world more cautiously (often the types who introvert their perceiving process—for example, dominant introverted Sensing [ISJ] and dominant introverted Intuitive [INJ] types). EPs embrace new opportunities with optimism and see obstacles as merely temporary setbacks or challenges to be met. At times, the pull to new

experiences or possibilities in the world can override their auxiliary introverted judgment, and they may in fact fail to achieve their optimistic intent.

EJs: THE DECISIVE EXTRAVERTS

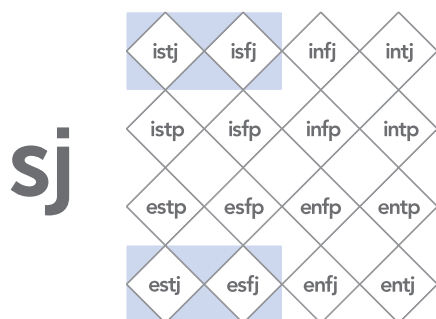


All four Extraverted Judging (EJ) types have a dominant extraverted judging process (T or F) and an auxiliary introverted perceiving process (S or N).

EJs are fast moving and decisive, appear confident, and enjoy making things happen. It can be difficult for them to let go of a decision or conclusion because their dominant judging process (Thinking or Feeling) tends to be more persuasive than their auxiliary perceiving process (Sensing or Intuition). They are therefore harder to convince than the IJs, for whom judgment is auxiliary and has less authority in their personality. For EJs, new data are less compelling than the consequences (Thinking or Feeling) of persisting in a possibly faulty decision. They are therefore more likely to change their mind if a likely negative effect of a course of action is pointed out. EJs are seen by others as natural leaders, and they relish that role, as it permits them to exercise their dominant judging process in the outside world to reach decisions and get things done.

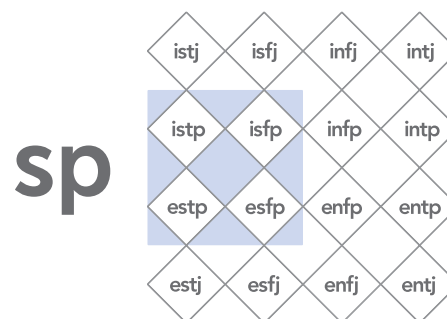
Combinations of Perception and Orientation to the Outside World: SJ, SP, NP, and NJ

The SJ, SP, NP, and NJ grouping of types identifies types who use the same perceiving process in the same orientation of Extraversion or Introversion, regardless of whether their perceiving process is dominant or auxiliary in their type. Focusing on this grouping enables understanding of the different dynamics involved in *introverting Sensing* (characteristic of the SJ types) as compared to *extraverting Sensing* (characteristic of the SP types); and *extraverting Intuition* (characteristic of the NP types) as compared to *introverting Intuition* (characteristic of the NJ types). The descriptions below are adapted from *Introduction to Type® Dynamics and Development* (Myers & Kirby, 1994).

SJs: THE REALISTIC DECISION MAKERS

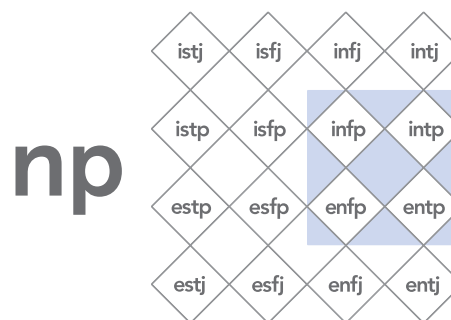
All four Sensing Judging (SJ) types introvert their Sensing process, regardless of whether it is dominant (ISTJ, ISFJ) or auxiliary (ESTJ, ESFJ), and extravert their preferred judging process, Thinking or Feeling. SJs find it difficult to describe what is happening when they are introverting their Sensing because their energy and attention are being used inside to remember both past and present experiences. They are also focusing on their thoughts, feelings, bodily sensations, and memories. They take in new information and integrate it into an internal storehouse for retrieval when needed. When they are introverting Sensing, their goal is to form a rich, solid, substantial, and accurate description of the world and their place in it.

The effect of introverting Sensing can be seen in the way SJ types are typically described. They are seen as seeking order in their environment, which leads them to be organized, dependable, and conservative. They tend to solve problems by relying on past experiences (retrieving data from their internal databank), and they dislike ambiguity. People are often unaware of the wealth of factual and experiential data SJs bring to bear in making decisions because the data tend to remain inside their head. When they are called upon to give evidence and provide the detailed bases for their decisions, other types may feel overwhelmed by the information and see it as unnecessary minutiae. The SJs, however, see it as relevant and necessary. At times, SJs, especially those for whom introverted Sensing is dominant (ISTJ and ISFJ), can run the risk of being distracted from their goals by their focus on collecting the facts that are relevant to the goal at hand.

SPs: THE ADAPTABLE REALISTS

All four Sensing Perceiving (SP) types extravert Sensing, regardless of whether it is dominant (ESTP, ESFP) or auxiliary (ISTP, ISFP), and introvert their preferred judging process, Thinking or Feeling. The process of extraverting Sensing involves directing energy and attention outward, noticing sights, sounds, smells, touch, human reactions—in short, everything present—in great detail and accurately. SPs' focus is on intensely experiencing what is happening in the present and on seeking stimulation through the variety and vividness of their sensations. When extraverting Sensing, SPs' goal is to experience as much of the world as possible by having an unending variety of Sensing experiences.

The effects of extraverting Sensing can be seen in the way SP types are described. They tend to seek new experiences in the present moment and are curious about the world around them. They adapt to situations as they arise and are good observers of the immediate situation. SPs are attracted to work that focuses on attending to facts and details in the immediate environment. They are relatively uninterested in the long-range aspects or implications of what they do. They constantly seek stimulus and flourish when they can experience a lot of it. If the external environment is not stimulating enough, they will create some stimulation themselves.

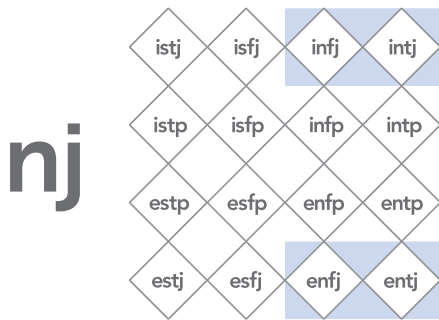
NPs: THE ADAPTABLE INNOVATORS

All four Intuitive Perceiving (NP) types extravert Intuition, regardless of whether it is their dominant (ENFP, ENTP) or auxiliary (INFP, INTP) process, and introvert their preferred judging process, Thinking or Feeling. The process of extraverting Intuition involves scanning the

outside world for whatever is new—new ideas, new people, or new possibilities—in order to understand whatever patterns and relationships they observe. NPs' focus in extraverting perception is on exploring and experiencing the world in all its potentialities. When extraverting Intuition, they seek to find and explore possibilities through new and exciting challenges.

The process of extraverting Intuition is reflected in the way NP types are described. They constantly seek the challenge of the unknown and adapt to new possibilities as they arise. They are unconventional, independent spirits who hate to be fenced in. They enjoy seeking new solutions for both old and new problems and like the challenge of the apparently impossible or the very difficult, especially in the realm of possibilities in the world.

NJs: THE VISIONARY DECISION MAKERS



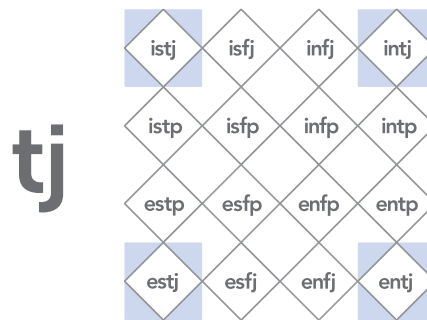
All four Intuitive Judging (NJ) types introvert Intuition, regardless of whether it is their dominant (INFJ, INTJ) or auxiliary (ENFJ, ENTJ) process, and extravert their preferred judging process, Thinking or Feeling. The process of introverting Intuition involves attending to internal images and connections that may lead to new patterns and new ways of seeing things. NJs' focus is on changing and reshaping the environment to align it with their inner vision and insight by finding patterns and meaning in the world. They try to make the outside world conform to their inner vision. When introverting Intuition, they develop inner intuitive patterns for understanding the world.

The process of introverting Intuition is reflected in the way NJ types are typically described. They strive to accomplish the goals that result from their inner vision and thus are driving, persistent, and determined. They can be charismatic leaders who attract dedicated followers. They have the ability to present their inner vision in a compelling, convincing way by framing it in terms of either long-range logical consequences or long-range impact on important values.

Combinations of Judgment and Orientation to the Outside World: TJ, TP, FP, and FJ

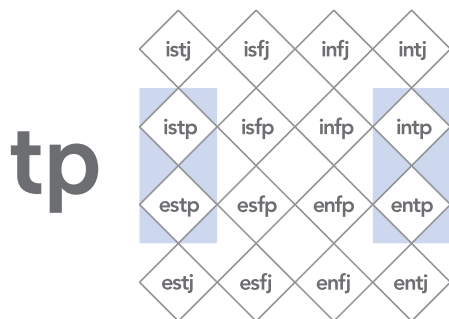
The grouping of TJ, TP, FP, and FJ identifies types who use the same judging process in the same orientation of Extraversion or Introversion, regardless of whether their judging process is dominant or auxiliary in their type. Focusing on this grouping enables an understanding of the different dynamics involved in the processes of *extraverting Thinking* (characteristic of TJ types) as compared to *introverting Thinking* (characteristic of TP types); and *introverting Feeling* (characteristic of FP types) as compared to *extraverting Feeling* (characteristic of FJ types).

TJs: THE LOGICAL DECISION MAKERS



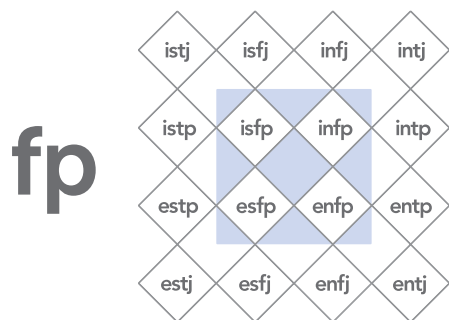
The corners of the type table contain the four Thinking Judging (TJ) types. All TJs extravert Thinking, regardless of whether it is their dominant (ESTJ, ENTJ) or auxiliary (ISTJ, INTJ) process, and introvert their preferred perceiving process, Sensing or Intuition. The process of extraverting Thinking involves seeking to bring order to the external environment, which is accomplished through expressing thoughts and judgments with directness and clarity. TJs' focus is on critiquing systems, procedures, and ideas so that consequences can be anticipated and acted upon quickly. When they are extraverting Thinking, their goal is to create logical order in the outside world by making the outer environment rational.

The process of extraverting Thinking is apparent in the way TJ types are described. These occupants of the four corners of the type table are tough-minded, executive, analytical, and instrumental leaders. They readily take on leadership roles in whatever setting they are in, and they communicate their confidence in the viability and primacy of focusing on logical outcomes. TJs are seen as confident, in control, and able to implement decisions quickly and effectively. They can easily impose a logical organizational structure in a situation with the goal of effective problem solving. TJs may be seen by others as hypercritical, too quick to judge and act, and tactless in their style of communication, which tends to be direct and to the point. They generally use categorical statements that represent logical conclusions and avoid modifying terms such as *may*, *tends to*, or *generally*.

TPs: THE ADAPTABLE THINKERS

All four Thinking Perceiving (TP) types introvert Thinking, regardless of whether it is their dominant (ISTP, INTP) or auxiliary (ESTP, ENTP) process, and extravert their preferred perceiving process, Sensing or Intuition. The process of introverting Thinking involves mentally developing a logical model of the world. TP's focus is on using reflective observation to critique reality and identify the inconsistencies and irrelevancies often overlooked by others. When they are introverting Thinking, their goal is to create logical order by developing rational principles for understanding the world.

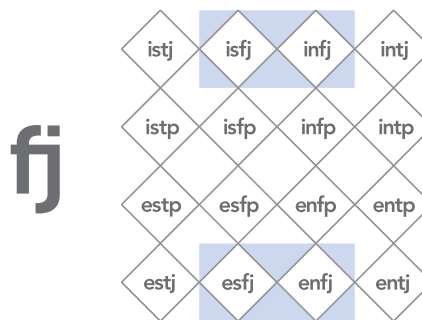
The process of introverting Thinking can be observed in the way TP types are described. They are objective, skeptical, observant, and curious, especially about materials, events, or possibilities that have or can be made to fit into consistent and orderly frameworks. They can be effective troubleshooters with the ability to home in on the essence of problems and find logical solutions for them. They are able to consider a broad range of facts, data, or ideas in order to come to the most accurate conclusion and are unlikely to rule out anything arbitrarily, regardless of how unorthodox or unlikely it may seem to others. Because TP's Thinking is introverted, others may have difficulty following the logic in their shifts of focus and changes in direction when they are deeply involved in a project. The internal logic may seem so obvious to TP's that they will assume that everyone else sees it too.

FPs: THE GENTLE TYPES

All four Feeling Perceiving (FP) types introvert Feeling, regardless of whether it is their dominant (ISFP, INFP) or auxiliary (ESFP, ENFP) process. The process of introverting

Feeling involves seeking an intense, meaningful, and complex inner life through being attuned to possible contradictions between inner values and outer life. FP's focus is on clarifying and maintaining the consistency of their values and actions, although the intensity of their values may seldom be expressed to others. When they are introverting Feeling, their goal is to identify an inner core of values and establish an external life that is congruent with them.

The process of introverting Feeling is evident in the way FP types are described. They are adaptable harmony seekers who are concerned with the human aspects of problems. They often have a knack for enlisting support from others, communicating a sincerity that makes them likable and promotes trust. They welcome others' ideas and genuinely appreciate the contributions people make. Because FP's Feeling process is introverted, however, others may not be aware of the central values that motivate them until one of those values is attacked or disregarded. At such times, the typical flexibility and acceptance of FP's can change to stubborn insistence on doing things the "right" way.

FJs: THE BENEVOLENT ADMINISTRATORS

All four Feeling Judging (FJ) types extravert Feeling, regardless of whether it is their dominant (ESFJ, ENFJ) or auxiliary (ISFJ, INFJ) process. The process of extraverting Feeling involves seeking smooth and harmonious relationships with everyone in their environment through being highly attuned to people's desires and expectations. FJ's focus is on organizing and structuring the environment to meet people's needs and to facilitate the achievement of individual and group goals. When they are extraverting Feeling, their goal is to create harmony and cooperation in the external environment and to facilitate others' getting what they need and want. The process of extraverting Feeling is often misjudged as excessive emotionality. When FJ's are extraverting Feeling, their important values are expressed with the goal of ensuring that those values become actualized in the world. Strong affect often accompanies this expression but is secondary to the actualization of personal values.

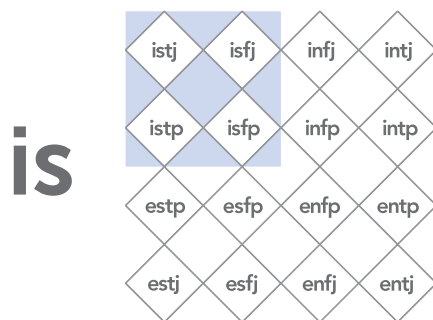
The process of extraverting Feeling can be seen in the way FJ's are described. They are observant about people

and their needs and are often expressive leaders. They expend energy trying to make people happy and bringing harmony into relationships. They may be judged as overly accommodating and insufficiently attentive to their own needs and desires since the natural inclination of their extraverted Feeling involves bringing harmony into the outer world. In fact, at times they may risk overcommitment to others' welfare, which may prove stressful when they attempt to honor and fulfill all their important obligations. FJs are generally quite loyal to others who share important values and goals and may judge those who deviate from those personal values in negative, sometimes harsh ways. Including others and being included is an important value that, when not met, may lead FJs to experience a sense of failure and hurt feelings.

Combinations of Orientation of Energy and Perception: IS, IN, ES, and EN

The grouping of IS, IN, ES, and EN is referred to as the "quadrants" because these types make up the four quadrants of the type table. The quadrants combine the processes of perception (Sensing and Intuition) with the Extraversion or Introversion orientation. Note that the types in these quadrants are defined as having Extraversion or Introversion with Sensing or Intuition. For example, the four types in the IN quadrant are described as having Introversion with Intuition because the grouping does not refer to a dynamic commonality, since only one process is involved. The same style of designation is used for the other three quadrants.

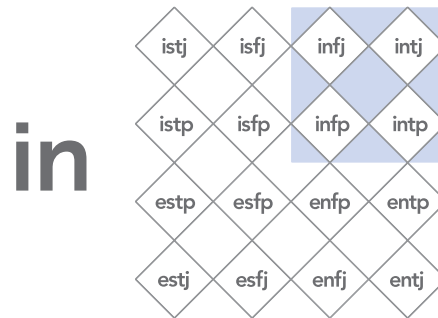
ISs: THE THOUGHTFUL REALISTS



Introverts with Sensing (ISs) like to test ideas to see whether they are supported by facts. They like to deal with what is real and factual in a careful, unhurried way and can become frustrated and irritable when they feel rushed and unable to devote adequate attention to the task at hand. They typically do not put themselves forward and tend to stay in the background. Their quiet and persistent work style may draw attention away from their often significant contributions at home and at work. Regardless of whether their Sensing is dominant and introverted (ISTJ, ISFJ) or auxiliary and extraverted

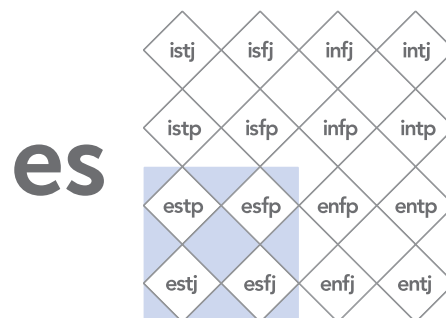
(ISTP, ISFP), they may be reluctant to take on leadership roles if those roles do not make use of their skills and abilities. If unwanted leadership positions are pushed on them, they may experience considerable discomfort. The stress of leading may come to overshadow and diminish their satisfaction with other, enjoyable aspects of their environment.

INs: THE THOUGHTFUL INNOVATORS



Introverts with Intuition (INs) are introspective and scholarly. They are interested in knowledge for its own sake, as well as in ideas, theory, and depth of understanding. They are the least practical of the types, preferring the complexity of theory to the pragmatism of accomplishments in the world. Regardless of whether their Intuition is dominant and introverted (INFJ, INTJ) or auxiliary and extraverted (INFP, INTP), these types enjoy discussing ideas with each other. Such discussions help them feel affirmed in their intellectual or philosophical interests and in their disinterest in and sometimes disdain for the practical world that seems to be so attractive to other types. They may be judged by others as too serious and as missing out on many of the activities and experiences available in the outside world.

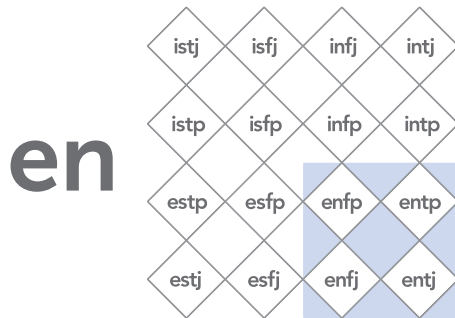
ESs: THE ACTION-ORIENTED REALISTS



Extraverts with Sensing (ESs) are active, realistic doers and are the most practical of the types. They learn best when they can immediately see a practical application for a new skill or idea. They enjoy the material world and devote attention to the appearance of things, both aesthetically and from a practical, utilitarian perspective. The pleasure and accomplishment involved in experiencing as much of the world as is possible can be

a persistent goal of ESs, and they may go to great lengths to seek out particularly interesting places, activities, modes of travel, and experiences. Regardless of whether their Sensing is extraverted and dominant (ESTP, ESFP) or introverted and auxiliary (ESTJ, ESFJ), the types in this quadrant typically appear confident of their relationship to the world and all it contains.

ENs: THE ACTION-ORIENTED INNOVATORS

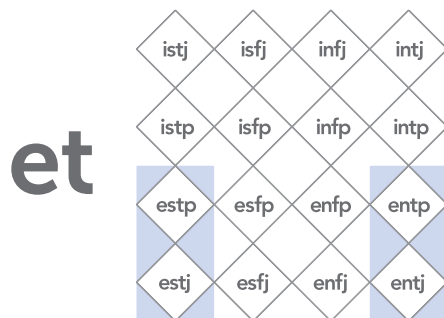


Extraverts with Intuition (ENs) are change agents; they see possibilities as challenges to make something happen. They have wide-ranging interests and like to discover new patterns and relationships. Like the types in the ES quadrant, ENs are comfortable with and excited by their interactions with the outer world, but they are more future oriented in their pursuits and goals. Regardless of whether their Intuition is dominant and extraverted (ENFP, ENTP) or auxiliary and introverted (ENFJ, ENTJ), these four types share a vision of future potentialities in the world, whether for people, structures, institutions, or the general future of human activities.

Combinations of Orientation of Energy and Judgment: ET, EF, IF, and IT

This grouping, like that of E and I with the perceiving processes, does not identify dynamic entities but rather identifies the combination of an orientation with a judging process, which may be either the dominant or the auxiliary process.

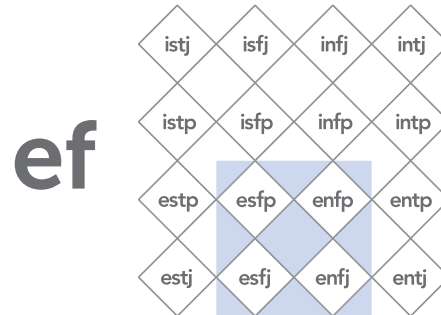
ETs: THE ACTION-ORIENTED THINKERS



Extraverts with Thinking (ETs) are active and energetic. They are objective and like to make things happen in

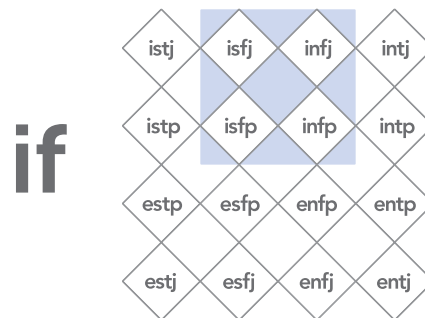
reasoned, analytical, and logical ways. They expect the same approach from others. Regardless of whether their Thinking is dominant and extraverted (ESTJ, ENTJ) or auxiliary and introverted (ESTP, ENTP), these four types share an expectation that they and those around them should be confident, competent, and effective. ETs can be impatient and annoyed with others' self-doubt, hesitation, and low self-confidence. Consequently, Feeling types in particular may experience ETs as harsh, unsympathetic, and intolerant of others' needs.

EFs: THE ACTION-ORIENTED COOPERATORS



Extraverts with Feeling (EFs) are sociable, friendly, and sympathetic. They like to make things happen for the pleasure or welfare of others. They focus on liking others and being liked and on connecting people with each other. All are sensitive to the nuances of emotion that they pick up from those around them, and they find it hard to function at their best when they are in an environment characterized by frequent conflict and divisiveness. Regardless of whether their Feeling is dominant and extraverted (ESFJ, ENFJ) or auxiliary and introverted (ESFP, ENFP), EF types try to defuse contentious situations or mediate among people who are at odds with each other. The restoration of harmony is not only a strong value for them; it may also be necessary if they are to remain in any situation.

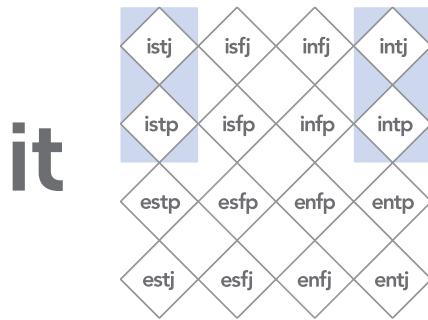
IFs: THE REFLECTIVE HARMONIZERS



Introverts with Feeling (IFs) are quiet and caring. They have concern for deep and enduring values, as well as for people and the way people feel. Regardless of whether their Feeling is dominant and introverted (ISFP, INFP) or auxiliary and extraverted (ISFJ, INFJ), these

types feel things quite intensely and in a manner that can sometimes be puzzling to others. The depth of feeling about individual values can make them appear overly serious and excessively sensitive to their own and others' emotional state. Unlike the EFs, who strive to eliminate disharmony when they find it, IFs are likely to withdraw from such situations rather than deal directly with the distressing feelings that exist in the situation.

ITs: THE REFLECTIVE REASONERS



Introverts with Thinking (ITs) are quiet and contemplative. They have concern for basic principles that explain the causes and consequences of events or the way things work. ITs are the least comfortable with and least interested in daily social intercourse and the slowest to develop social skills of all the types. They tend to be indifferent to or dislike small talk and may find many

social interactions unwelcome and stressful. They can be seen as socially awkward or may come across as overly formal and even unfriendly. Regardless of whether their Thinking is introverted and dominant (ISTP, INTP) or auxiliary and extraverted (ISTJ, INTJ), these types are natural and enthusiastic critics of whatever is being presented. The introverted approach they take while evaluating and critiquing contributes to their shared tendency to maintain an impassive facial expression or body language that seems to convey disapproval or disregard of others. As a result, they may be seen, often erroneously, as overly critical, disdainful, or, at best, uninterested in what is going on in the outer world.

CONCLUSION

This chapter showed readers the rationale underlying the type table, a standard way to array data that shows the frequency of the 16 types in a sample of interest, such as occupational groups, college majors, specialization within an occupation such as medicine or psychology, and any other group under investigation. Using type table data as a way to validate type theory and to provide useful information for career advisement was also pointed out. The remainder of the chapter focused on combinations of grouping of four preferences, what each grouping has in common, and distinguishing characteristics of each grouping.

CHAPTER 6

Administering and Interpreting the MBTI® Step I™ and Step II™ Assessments

CHAPTER OVERVIEW

Administering the Assessments

The Process of Interpreting and Verifying
Step I™ and Step II™ Results

Interpreting and Verifying Step I™ Results

Interpreting and Verifying Step II™ Results

Conclusion

The preceding five chapters provided the foundation needed to understand and use the MBTI® Global Step I™ and Step II™ assessments. This chapter covers guidelines for administration and interpretation that are applicable in all languages and countries included in the global revision of the two MBTI assessments and for all or most respondents. To be able to purchase and administer the assessments, practitioners worldwide are required to complete MBTI certification training.¹

As a practitioner, you² will likely get the most from this chapter by being aware of the underlying assumptions that guide administration and interpretation of the MBTI assessments. Keeping the following assumptions in mind as you read will help anchor the detailed information in the chapter:

- Personality type differences are real and readily observable.
- Many people implicitly “know” and recognize their type regardless of whether they are aware of type theory or the MBTI assessment.
- Type preferences guide people’s interests, habits, reactions to life experiences, and general outlook over their life span.
- The MBTI assessments are reliable and valid tools for assessing and identifying a person’s type.
- A number of factors can influence people’s willingness and ability to give candid responses to MBTI items.

ADMINISTERING THE ASSESSMENTS

It is extremely important to ensure that when individuals respond to the MBTI items they adopt a frame of mind that reflects the way they function most naturally and effortlessly—not what they *can do* when they have to, but what they *prefer* and enjoy doing when free from external pressures. Ensuring this frame of mind is vital because most people take the MBTI assessments in their work environment, where their mind-set may push them to respond to items as their “work self.” Administration conditions are critical when administering the MBTI Step I and Step II assessments because the goal is to ensure respondents’ willingness to reveal their personality *preferences* rather than their skills, abilities, or attitudes. Each item asks respondents to choose between opposite approaches or behaviors, each of which they use at least some of the time. Anticipating the kinds of questions or concerns respondents may have and providing them with accurate information is important to prevent respondents from automatically answering as their “work self.” They have little or no opportunity to ask questions or seek clarification while they are sitting at their computer or using a mobile device.

Isabel Myers recommended that respondents answer the MBTI items using a frame of mind that reflects their “shoes-off self,” the way they function most naturally, smoothly, and effortlessly when they are not “going against the grain.”

Encouraging Accuracy of Responses

The specific goal in administering the MBTI assessments is to enable individuals to respond to the items in a manner that will reflect their personality type and/or facet preferences. A straightforward way to describe the MBTI assessments is to say that they provide information about naturally varying ways that people use their mind. In line with the first three assumptions listed above, you can further say that respondents’ results will provide information they may already know about themselves, but it will be organized in a way that will help them use it to better understand themselves and others. For individuals who have previously taken the Step I assessment and received an interpretation of their results, and who are responding to the Step II items for the first time, you can make one or more of the following points, depending on their level of inquiry:

- The Step II assessment provides information about people’s individuality within their Step I type.
- Step II results show how people of the same Step I type express five different aspects of each of their Step I preferences.
- Step II information can help resolve doubts for people who are unsure of one or more of their Step I preferences.

Regardless of which form of the MBTI assessment you will be administering or the setting in which you will be administering it, you can encourage individuals to respond to the items candidly by anticipating questions they may have. Whether you give them information in person or via email, phone, or videoconference prior to their taking the Step I or Step II assessment online, state the purpose for administering the assessment, as well as the following:

- Taking the Step I or Step II assessment is voluntary.³
- Respondents’ results are confidential and will not be released to anyone without their permission.
- Their results will not be used as a basis for hiring or advancement in their organization.
- Unlike with many other assessments, MBTI results belong to the respondent, who may use them for his or her own purposes, such as career exploration, devising learning strategies, improving interpersonal relationships, team building, management development, and so on.
- The Step I and Step II assessments do not assess mental or emotional health, intelligence, competence, or maturity, because the items were not designed for those purposes.
- Respondents are the judge of the accuracy of their results. This follows from the assumptions above and is the basis for the standard review, interpretation, and verification process required for all MBTI assessments. When respondents know that they will be invited to confirm or correct the report of their type or facet results, their responses are more likely to be genuine.
- When the MBTI assessment is administered solely for research purposes, researchers may wish to offer to review results with respondents so that they can have an opportunity to verify their results. Such a practice can increase cooperation.

1. For information about becoming certified, contact the publisher, The Myers-Briggs Company, or the authorized distributor in your country. In addition, familiarize yourself with your state or country guidelines on using assessment instruments and, in the United States, with the guidelines of the Americans with Disabilities Act, the Civil Rights Act, Title VII, and other legislation dealing with privacy issues. In other countries, attend to requirements of any comparable entities.

2. In this chapter, the authors will be addressing *you*, the practitioner, directly.

3. Be aware that respondents who are *required* to take the assessment or who are subject to any form of coercion, such as employer or peer-group pressure to participate in the assessment process, may not respond candidly and therefore may not receive accurate results.

Appropriate Respondent Populations

The MBTI assessments vary in the extent to which they are suitable for different kinds of respondents. This section discusses ages appropriate for administration, reading levels required, and global use.

APPROPRIATE AGE GROUPS

Most research on the MBTI Step I and Step II assessments is based on an adult population, ages 18 to 70, so that is the optimal age group. Both assessments can be used with individuals as young as 12 or 13 (seventh- to eighth-grade reading level); however, generally lower reliabilities have typically been found for such younger respondents. Young people may have little or no experience with the content of and language used in some items. Therefore, interpret results for this younger age group with caution.

You can most effectively elicit type for students ages 7 through 13 using the *Murphy-Meisgeier Type Indicator for Children®* (MMTIC®) assessment (Murphy & Meisgeier, 2008) in countries where it is available. The MMTIC assessment is administered online and has a separate manual and interpretation materials (Murphy & Meisgeier, 2008).⁴ It can also be used successfully with high school-age students. A study by McPeck et al. (2013) showed that results on the MBTI Step I and MMTIC assessments are equivalent when both assessments are administered to students (mean age: 16.54 years).

Step II results can also be considered reasonably accurate for students and adults who read at the recommended level, but the Step II assessment has not been used with enough younger respondents to confirm that the facet descriptions are accurate for them. The facet scales were constructed using an adult population ages 18 and older, and that is the age group for which the Step II assessment is recommended. In addition, Step II facets deal with behaviors that are more narrowly defined and more specific than those described by the four Step I preference pairs. Without extensive validation with younger age groups, it cannot be known how well these facets describe them. For example, many young people have not had enough experience handling projects to know whether their preference is for Methodical or Emergent. The same may be true for other narrow areas of preference associated with other facets. Therefore, anyone administering the Step II assessment to and interpreting results for young people should proceed with caution.

READING LEVEL REQUIRED

The reading level for a set of items can be approximated using a variety of approaches (DeVellis, 1991). However, conventional methods for assessing reading difficulty

(e.g., using formulas based on sentence and word length) were not designed for test items, particularly in an instrument that mixes forced-choice phrase items and word pairs. Syntactic and semantic factors related to items—for example, simple versus complex sentence structure, the fact that each phrase item actually forms two sentences, and the context of the words for the reader—will affect reading levels. Thus, you should use common sense when determining the appropriateness of MBTI instruments for various populations.

The phrase items for the prior Step I assessment, Form M, were analyzed using the Fry readability formula, which uses the average number of sentences and syllables in 100-word samples to assign approximate grade equivalents to written passages (Fry, 1977). This method yielded a reading level for the Form M phrase items of about seventh grade (individuals approximately 12 to 13 years old) by treating each item as a single sentence. Additional items that appeared in the prior Step II assessment, Form Q, and are included in the Global Step I and Step II assessments are not substantially different from the Form M and Form Q items tested using the Fry method. Remember, however, that reading level is only an approximation and is not a substitute for knowing your respondents well enough to assess the suitability of administering the MBTI assessments to them.

ADMINISTERING MBTI® ASSESSMENTS ACROSS LANGUAGES AND CULTURES

Jung's theory attempts to describe basic human mental processes that cut across cultural differences. It is therefore reasonable to expect to find evidence that type and type differences are recognizable and measurable in a wide range of cultures and countries. The successful development of the Global Step I and Step II assessments can be viewed as convincing evidence in support of Jung's assertion regarding the universal existence of personality type.

Developing a single set of items that successfully assess MBTI Step I type and Step II facet pole preferences in many diverse countries and languages is a lengthy, arduous, and costly task due to cultural and language differences that can affect item content and meaning. For information on the sample compilation, item selection, and translation processes that went into the development of the two global forms, see chapter 7, which discusses the construction of the global assessments. Also see chapters 8, 9, and 10 for information on the reliability and validity of the Step I and Step II assessments in different countries and languages. A number of country/language supplements to this manual are available at www.themyersbriggs.com/mbti-manual-supplements.

4. The MMTIC assessment is published by the Center for Applications of Psychological Type (CAPT). For further information about administering the MMTIC assessment, visit www.capt.org.

Selection of Available Step I™ and Step II™ Forms

All forms of the MBTI assessment provide basic four-letter type information.⁵ They differ in the amount and kind of additional information provided. In selecting a form, consideration should be given to the purpose for using the assessment, the time available for testing, and the testing situation.

The Global Step I assessment replaces Form M in the United States and several other English-speaking countries, as well as all other forms that assess solely four-letter MBTI type, including the European Step I assessment, in other countries composing the global sample.⁶ The Global Step II assessment replaces Form Q and the European Step II assessment. Both the Global Step I and Global Step II assessments contain items selected from a large item pool (see chapter 7 for a description of the pool).

The Global Step I assessment contains 92 items, all of which are scored for four-letter type. The Global Step II assessment contains all 92 Global Step I items plus an additional 51 items needed to score the Step II facets, for a total of 143 items. Table 6.1 summarizes information about the forms of the MBTI assessment that are available in the US and other selected countries.

The reports generated by the different forms of the MBTI assessment reflect the many applications for which it has proved useful. A full list and descriptions of available MBTI reports can be found at

www.themyersbriggs.com or the website of the authorized distributor in your region.

Administration Guidelines

Regardless of whether you are administering the MBTI assessments in person or online, the following information should be covered with individuals before they respond to the assessment items:

- The MBTI assessments are *indicators* of natural and healthy personality differences. They are not “tests” that have right and wrong answers. However, it is common for people to think of and refer to the MBTI assessments as tests: they contain items that ask respondents to choose one or the other option and yield numerical information in addition to narrative descriptions of personality types.
- The MBTI Step I assessment is also referred to as the Myers-Briggs® personality inventory. The Step II assessment is an extended version of the MBTI assessment that identifies differences among people of the same Step I type.
- Respondents do not have a time limit for taking an MBTI assessment, but it is best that they not think too long about any one item, instead working rapidly and making spontaneous choices. When individuals cannot decide on a response to an item, they should skip it but return to it later and try to respond to as many items as possible in order to ensure accurate results.

Table 6.1 | Administration options for the MBTI® Global Step I™ and Step II™ assessments

Option	Number of items	Administration time (approx.)	Scoring method	Uses
MBTI Global Step I	92	15–25 minutes	LCA online scoring	For all settings with Internet access
MBTI Global Step II	143	25–30 minutes	IRT scoring via secure website	For providing Step II facet results as well as Step I type
MBTI®Complete (practitioner chooses to use Step I or Step II)	92 (Step I); 143 (Step II)	60–90 minutes (includes an interactive learning session about the Step I preferences)	LCA scoring via secure website (Step I); IRT scoring via secure website (Step II)	For convenience, where type information is needed prior to respondent’s contact with practitioner or for use in application workshop session
MBTI®Online	92	60–90 minutes (includes an interactive learning session about the Step I preferences)	LCA scoring via secure website	When type information is needed prior to respondent’s contact with practitioner or for use in application workshop sessions; unlike with MBTI®Complete, practitioner has no access to results

5. In addition to the Global Step I and Step II forms, there is the MBTI Step III™ instrument, briefly described in chapter 1.

6. Countries composing the global sample include Australia, Canada, South Africa, and the United States (North American English); Ireland and the United Kingdom (European English); Brazil (Brazilian Portuguese); Canada (Canadian French); China (Simplified Chinese and Traditional Chinese); Denmark (Danish); Finland (Finnish); France (European French); Germany (German); Greece (Greek); Italy (Italian); Mexico (Latin American Spanish); the Netherlands (Dutch); Norway (Norwegian); Portugal (European Portuguese); South Africa (Afrikaans); Spain (European Spanish), and Sweden (Swedish).

- A majority of respondents take the MBTI assessments in a work or school setting. Before they begin, encourage them to adopt a frame of mind that reflects the way they function most naturally and effortlessly—when they are being themselves rather than behaving in accordance with the expectations of others. This step is important in optimizing accuracy in responding.

If you can anticipate the likely type distribution in your group, be aware of these preference-related issues:

- Sensing types will want to know that the information they will receive after taking an MBTI assessment will have some practical value.
- Intuitive types will want to see possible future benefits of taking the assessment. They may also find having to choose between alternatives frustrating; they like the widest range of possibilities.
- Thinking types can be expected to be skeptical, since skepticism is an important aspect of their judging process. It will help if you acknowledge their skepticism and suggest that they “wait and see.”
- Introverted types may be especially concerned with privacy issues. Assure them that the MBTI assessments will not reveal anything hidden about them, nor will results be given to anyone without their permission.

MBTI assessments are virtually self-administering; all necessary instructions are provided for respondents. As an administrator, you must be aware of the following related issues:

- Verification and interpretation processes should not be conducted through the mail or solely by providing printed or online results without also ensuring that respondents will receive an interpretation by a certified practitioner.
- The online administration and interpretation option, MBTI®Complete, follows the recommended procedure of presenting the items on screen, having respondents enter their responses, and then, before they see their reported results, guiding them through an interactive learning session (ILS) and verification process so that they can decide whether the reported type or a different type best fits them, thus arriving at best-fit (verified) type.
- A Step I self-scorable form (Form M) is available. It is intended for use primarily with groups when administration time is an issue and circumstances do not permit online administration and scoring. Respondents should not be permitted to self-score the instrument before the interpreter provides them with information about type. Respondents should be encouraged to double-check their scoring to reduce the chance of error.

RESPONSE OMISSIONS

Respondents can omit (not respond to) up to four items on the MBTI Step I and Step II assessments without jeopardizing the validity of their results. Myers permitted such omissions if respondents were unable to choose between alternatives, on the assumption that an item cannot reliably contribute useful information about a respondent's type unless choices are understood and the item lies within the respondent's experience. Forcing respondents to respond to every item is tantamount to asking them to guess and reduces the validity of the response record. However, too many omissions can negatively affect validity of the administration. A Step II administration will not be scored if more than four omissions occur over the 143 items, because a few facets are composed of only 5 or 6 items. Scoring a profile with more than four omissions would likely affect the accuracy of the facet scale results.

In practice, only about 2% of respondents omit more than two items. However, where all or most omissions occur on one preference or facet scale, concern arises about the interpretability of those results. Respondents who exceed four omissions on either assessment will be prompted by the online system to complete more items to enable scoring.

Scoring Global Step I™ and Step II™ Administrations

The Global Step I assessment uses latent class analysis (LCA) to score respondents' results and derive their four-letter type. The Global Step II assessment uses item response theory (IRT) to score respondents' facet results. The LCA procedure (for Step I scoring) and the IRT-based procedure (for Step II scoring) yield precise classification of respondents' type and facet preferences. For more information on LCA and IRT scoring, see chapter 7.

THE PROCESS OF INTERPRETING AND VERIFYING STEP I™ AND STEP II™ RESULTS

Briggs and Myers developed an assessment approach that relies on respondents' being able and willing to communicate what they know about themselves. Given the assumptions listed at the outset of this chapter, the best way to find a satisfactory fit between respondents' reported type and their self-knowledge is to ask them. Carefully following the administration guidelines presented earlier should do a great deal to discourage respondents from believing that the “test” knows them better than they know themselves or assuming that the results will reveal something negative or unacceptable about them. However, it is natural for respondents to apply their previous test-taking experiences to the MBTI assessments. MBTI assessments are unique in the field

of personality testing in that (1) the results are given directly to respondents and (2) respondents' judgment concerning the accuracy or "fit" of those results is honored and accepted.

Purposes of Interpretation

The requirement that each respondent participate in an interactive review, discussion, and verification of MBTI results—i.e., an interpretation, or feedback, session—was initiated by Isabel Myers. The purposes of this session are as follows:

- To ensure that respondents understand the terms used, their neutral or positive character, and the equal value accorded both opposites in all four preference pairs, all 16 types, and both poles of all 20 Step II facets
- To communicate to respondents that any set of assessment items, no matter how carefully developed and tested, is subject to error
- To clearly convey respect for respondents' self-knowledge as the key criterion in assessing their type and the individual expression of their type through the Step II facets
- To make sure that respondents experience the interpretation and verification process as a collaborative effort between practitioner and respondent
- To at least tentatively arrive at respondents' verified, or best-fit, Step I type, defined as the type they verify after they have participated in a type interpretation and verification process with a certified practitioner, and to assist respondents in arriving at judgments concerning the accuracy of their Step II facet results⁷

Some respondents may wish to continue their exploration of their best-fit type after the interpretation and verification session. No one should be pressured to identify a best-fit type, especially in a group feedback session.

The Review Process

As with any self-report instrument, the accuracy of MBTI results depends in part on how willing or able respondents are to self-report their preferences. If people taking an MBTI assessment feel that they have nothing to gain from taking the assessment, they may respond carelessly or even at random. If they fear that they have something to lose, they may respond as they assume they should. Following the administration recommendations presented earlier in this chapter

should help maximize the accuracy of self-reporting. However, if you have reason to believe that a respondent has not answered candidly, you might explore possible reasons for that.

When you are reviewing with respondents their type and facet results, use statements such as, "This report shows how you responded to the items. Does it describe you? Does it fit you?" Or for Step II interpretation, "Does the description of how people with this pattern tend to react or behave seem accurate for you?" An interpreter should never say, "You are [this or that] type" or "You must be a Questioning Feeling type," as such statements bypass respondents' judgment and give more credence to assessment results than to respondents' self-knowledge.

Essential Points to Make to Respondents

Each item in the MBTI Step I and Step II assessments requires respondents to choose between two opposite alternatives. Making these choices may seem difficult because both options describe activities that are necessary and valuable. The following are points you should make to respondents:

- *The terms and letters used in Step I results and the names of the 20 Step II facets may be unfamiliar, but you are likely to recognize the behaviors they describe.*
- *The MBTI assessments are "indicators," not tests in the sense of having right or wrong answers. There are no good or bad, acceptable or unacceptable, types or facet results. All types and both poles of each facet are valuable, though the contexts in which they are more or less effective may vary.*
- *Whatever your personality type or facet preferences, you likely use both members of each pair of opposites, though not with equal ease or liking. A helpful analogy used for Step I interpretation is that preferences are like "handedness." For the most part, people are either right-handed or left-handed. However, people do use both hands, usually reaching first with their preferred hand because it is more comfortable and natural to do so. The same is true for the poles of the Step II facets. People who report a preference for one pole over the other can use the less preferred pole when necessary. Other people habitually use one or the other pole of a particular facet depending on circumstances, which is represented by a score in the midzone. This may be because the five facets describe more narrowly defined behaviors than does their underlying Step I preference, so it may be easier to habitually choose whichever pole fits a particular situation.*

7. The terms *best-fit type* and *verified type* are used interchangeably when interpreting Step I results. There are no comparable terms for a Step II interpretation.

- *People generally find it easier to use the Step I processes and orientations they like than their opposites. But they may not necessarily use those preferences skillfully. You should therefore not assume that having a preference ensures that you will use it effectively. Given the specificity of the Step II facet scales, it may be that you or someone else will be skilled and effective in using some facets of a preference and not others.*
- *Type and facet preferences are important for understanding how people assimilate information and make decisions, the settings in which they interact most comfortably, and how they like their outside world to be organized. You probably recognize and are familiar with the characteristics and behaviors in your and others' type descriptions because they are natural, everyday qualities people show each other. Knowing about the equally valuable opposites in the Step I preference pairs and their variations in the Step II facets will help you use the information you already know about yourself and others in organized, thoughtful, and practical ways.*

INTERPRETING AND VERIFYING STEP I™ RESULTS

The recommended process provided here constitutes basic instructions for conducting an initial Step I interpretation session either with an individual or in a group setting.⁸ Following it will ensure that a cooperative and collegial tone is maintained throughout your interpretation and verification sessions. The important points to cover listed below should be followed by an explanation of each preference pair, one at a time, and an opportunity for respondents to self-assess on each pair. You then should give respondents their reported result for the pair. Some practitioners have respondents self-assess on each pair and wait until all have been covered before they provide results on all four pairs. Either approach increases understanding, conveys respect for respondents' self-knowledge, and discourages respondents from agreeing too readily with their reported type.

Important Points to Cover

The first step in interpreting and verifying type with respondents is to make sure they understand that no assessment items, however carefully they were devised, can elicit answers that explain all human complexity. MBTI Step I results give respondents and interpreters a hypothesis about an individual's type that can be confirmed or modified through an interpretation session.

In addition, if a reported process, orientation, or whole type does not seem right when MBTI results are provided, respondents can explore and decide for themselves, with help from you, which preference or which type more accurately describes them.

Describing the Step I™ Preference Pairs

The next step is to describe the nature of the two opposite alternatives that compose each preference pair. The following is a recommended procedure for presenting each preference pair:

1. Define the preference pair by focusing on what it describes. For example: "Extraversion and Introversion deal with how we are energized and how we prefer to use our energy."
2. Define both poles of the preference pair in a neutral and unbiased manner and give examples relevant to respondents' interests and life circumstances and their purpose in taking the assessment. Using language such as the following helps emphasize the important fact that everyone uses both members of a pair of opposites at least some of the time: "When we are extraverting, we are focusing on the outside world of people, the environment, and action. When we are introverting, our attention is on our internal world of ideas, inner experiences, and reflection."
3. For each preference pair, point out the ways people who prefer one member of the pair differ from those who prefer its opposite. For example, "People who prefer Extraversion are likely to be most energized by being around people and activities. They therefore typically seek out situations that permit them to experience and interact with the outside world. People who prefer Introversion are likely to be most energized when they can spend time alone in reflection. They therefore typically seek out situations that permit them to do more introverting than extraverting." When conducted carefully, this procedure provides an illustration of type differences as well as a first check on the accuracy of reported type.

It is helpful to modify the descriptions of the opposite preferences in each preference pair to accommodate respondents' type and the time available. Using concrete behavioral examples can help explain the theoretical abstractions in everyday terms, especially when you are explaining type concepts to Sensing types. Visual aids and illustrative anecdotes that highlight differences are helpful for all types.

8. The process of interpreting and verifying MBTI Step II results differs substantially from the Step I process. For example, it is very important that respondents self-assess their Step I preferences prior to being provided with their MBTI reported results. However, such a procedure is not recommended for Step II facet results. For this reason and others, the interpretation and verification process for Step I results is presented first, followed by the process for Step II results.

Verifying Step I™ Type

Any type identification method, including responding to an MBTI assessment, is subject to the errors in understanding and interpretation that are associated with all self-assessment methods. In light of the reliability and validity evidence reported in chapters 8, 9, and 10, reported type as elicited through the MBTI Step I assessment appears to be the best available personality type estimate for individuals. Nevertheless, respondent verification, as described in this section, is recommended in order to ensure that reported type is confirmed by the respondent as his or her best-fit type or to enable identification of a different, better-fitting type. After each preference pair has been described, respondents should be given an opportunity to self-assess what they believe their preference is based on their responses to the Step I assessment. This can be accomplished simply by asking them how they think they responded to the items relevant to each preference pair. An alternative feedback procedure is described in *Verifying Your Type Preferences* (Kummerow, 1986).

Some respondents may prefer to have their results on the four preference pairs given to them all at once after they have heard about all four pairs. You can ask them which procedure they prefer. If respondents' self-assessment agrees with their reported results, the correctness of the reported preferences is tentatively confirmed. When a preference is not confirmed, it should be considered to be in doubt and evaluated further.

At this point in the feedback process, be careful to avoid implying that respondents are "wrong" in their assessment or that their reported type carries greater weight than their self-knowledge. Language such as "Your responses indicated the opposite preference on that preference pair, so let's explore that further" presents the required neutral stance for providing feedback.

VERIFICATION AIDS

You can choose from among a number of verification aids, depending on your level of experience with MBTI interpretation and the needs of your respondents. You can also choose MBTI®Complete, the online assessment administration and interpretation/verification option.⁹

One way of checking reported type is by using the table "Characteristics Frequently Associated with Each Type" in the *Introduction to Myers-Briggs® Type* booklet (Myers, 2015), which also appears as table 4.1 in this manual. This table contains an abbreviated type description for each of the 16 types. Respondents often find it easier to recognize themselves among these miniature portraits of whole types than to recognize their preferences one at a time. When using the type snapshots for

verification, respondents should read the description that corresponds to their reported type. If they identify one or another part of the description that doesn't fit, you can recommend one or more alternatives for consideration.

Ideally, respondents should also read one or more of the full type descriptions in *Introduction to Myers-Briggs® Type* to help them identify their best-fit type. Respondents tend to get the most out of interpretation when they have access to all 16 type descriptions both during and after verification of their type. By reading descriptions of other types in addition to their own, they gain understanding of and insight into people who are different from them. The type booklet also contains additional information about mutual usefulness of opposite preferences, career choice, learning styles, relationships, and more. Type descriptions found in other *Introduction to Myers-Briggs® Type* series booklets that are designed for specific applications can also be helpful. (See www.themyersbriggs.com for a complete listing.)

EXPLAINING TYPE DYNAMICS AND DEVELOPMENT

An explanation of the dynamic nature of MBTI type is typically reserved for a later discussion, depending on the time available and the respondent's purpose in taking the assessment. However, if you will not have a later opportunity to explain type dynamics and development to an individual or group, it is essential that you include a general explanation of this area during the interpretation session. When such an explanation is not provided, respondents can easily dismiss the MBTI types as a group of rigid categories or "boxes," thus limiting their ability to make practical and effective use of their results and the type information provided. At a minimum, therefore, when you have only a single opportunity to explain MBTI type, you should cover the following aspects of dynamics and development, at least briefly:

- A four-letter type reflects a dynamic and developmental system for describing personality. Individuals of each type typically use their favorite perceiving (Sensing or Intuition) and judging (Thinking or Feeling) processes in one or the other orientation of energy (Extraversion or Introversion), but they are free to use their less preferred processes and orientation as well. People's type is not seen as changing over time; however, individuals may express their type in somewhat different ways at different times and at different ages and stages of life. One consequence of this is that some individuals may respond to the MBTI items differently at different life stages.
- Each type has a *dominant*, or leading, favorite process and tends to use it in its preferred orientation of Extraversion or Introversion. Each type also has an

9. For any other online administration option for the Myers-Briggs assessments, the practitioner must use standard type verification and feedback procedures for Step I type either in person or in a comparable interaction.

auxiliary, balancing, second process that tends to be used in the less preferred orientation. For example, ESTJs' dominant, or favorite, process is extraverted Thinking, and their auxiliary, balancing process is introverted Sensing. For INTPs, introverted Thinking is the dominant process and extraverted Intuition is the auxiliary process.

- The process that is the exact opposite of the dominant, or favorite, process, called the *inferior*, or fourth process, tends to be the least developed, least used or experienced, and least comfortable part of people's type. For example, the inferior process for ESTJs, whose dominant process is extraverted Thinking, is introverted Feeling, the exact opposite of extraverted Thinking.
- The opposite of the auxiliary process is called the *tertiary*, or third, process. It tends to be somewhat more accessible and comfortable than the inferior process (fourth process).
- The dynamics of type, as identified by the dominant, auxiliary, tertiary, and inferior processes, describe the individuality and richness of personality in ways that people can easily recognize and understand. These dynamics are embodied in the type descriptions in the *Introduction to Myers-Briggs® Type* booklet.

If time permits, you might have the respondent or group look at their full type description in *Introduction to Myers-Briggs® Type* and point out paragraphs or sentences that discuss aspects of type dynamics. For example, the first two paragraphs of each type description, under the heading "At Their Best," describe the whole type; the next paragraph (characteristics of ESTJs, ISFPs, etc.) discusses the dominant process; and the fourth paragraph shows the modifying effects of the auxiliary process. Recall that the descriptions of the 16 types in chapter 4 of this manual include designations of type dynamics elements.

In addition to covering these basic points, you can direct respondents to the booklet *Introduction to Type® Dynamics and Development* (Myers & Kirby, 1994), which presents dynamics information in a practical and accessible way, and to *In the Grip* (Quenk, 2000), which provides details about the inferior function (fourth process).

ACCURACY OF SELF-REPORTING

Each stage of the interpretive process gives information that respondents can consider and use in evaluating the accuracy of their type. In some cases, they instantly recognize the correctness of their results. With these respondents, you will spend most of your interpretation time presenting type examples and suggesting practical applications for type information. Other individuals will have doubts about their reported type. With these respondents, you can use the suggestions provided in the

section "Verifying Step I™ Type," above, to bring out new information that may help resolve questions about which type is likely to be their best-fit type.

Throughout the explanation process, keep in mind possible reasons why respondents' MBTI results may not be an accurate estimate of their type. The working assumption of type theory is that each person is born with a tendency to develop certain type preferences rather than others. However, life circumstances may not support those natural typological tendencies. For example,

- There may be a lack of differentiation of type. Such a lack occurs more often in young people, whose development into a type may still be in the exploration and experimentation stage. For instance, during adolescence great value is placed on many of the qualities associated with Extraversion. Adolescents who are trying to "fit in" might therefore respond to the assessment as an Extravert would, when in fact Introversion is their natural and comfortable preference.
- Respondents may have difficulty choosing between the expectations of their parents and their own preferences. For example, Sensing individuals raised in a family in which their parents and siblings prefer Intuition may learn to value Intuitive qualities over Sensing qualities, attempt to develop those qualities, and therefore respond as Intuitive types to the MBTI assessment.
- Respondents may feel torn between the demands of work and their own preferences. This source of bias can be a major issue since most people take the assessment in a work setting. MBTI instructions describe the appropriate mind-set, so it is a good idea to draw respondents' attention to them.
- Respondents may be experiencing a life crisis and thus not using their typical mode of coping. Administering the MBTI assessment when respondents are experiencing unusual distress is generally not recommended.
- The MBTI assessment may have been administered by an authority figure (e.g., a teacher or supervisor). In such situations, individuals sometimes respond to items in accordance with their perception of the authority figure's preferences instead of their own.
- The words used to explain the MBTI assessment may have been misunderstood, and respondents may have rejected them because of an assumed negative connotation, such as when Introversion is interpreted to mean neurotic or shy, when Judging is interpreted to mean judgmental, or when Feeling is interpreted to mean overly emotional. When explaining the MBTI assessment, be sure to include a discussion of the meanings of these words in the context of MBTI type theory.

- Respondents may have been influenced by perceived social pressures. Such an effect is most likely to occur on the T–F preference pair for those respondents who may equate Thinking with masculinity and Feeling with femininity.
- Respondents may have believed that a type description needed to fit them perfectly to be accepted. Carefully explaining, before providing MBTI results, that there are individual differences within each type should minimize this effect. If you will be reviewing both Step I and Step II results, you might mention that such individual differences will be discussed later.
- Respondents may be in a growth period during which previously unused or unappreciated processes are being developed. During this period, they may become uncertain about previously trusted processes as their less developed processes become differentiated. For example, during midlife a dominant Feeling type may become fascinated with logically analyzing life problems and correspondingly be less interested in solving problems by maintaining harmony.

All self-report instruments are subject to the kinds of effects described above and to many other effects not intended by those who administer them. Since the MBTI assessments present pairs of opposites, and since the Jung-Myers theory postulates natural, probably biologically based preferences, you should be particularly careful during each stage of interpretation to help individuals discover for themselves which of each pair of opposites best fits them.

It is not necessary for respondents to discover a best-fit type during the interpretation session, but it is necessary for the interpreter to set the stage for the search. For respondents who are unsure about one or more preferences, you might suggest that they spend a period of time observing what they tend to do “automatically,” similar to which hand they reach with first to pick something up. For example, if a respondent is unsure whether she prefers Thinking or Feeling, the next time she needs to make a decision she might attend to what pops into her head first—what seems logical and impartial, or how the decision will affect her or others’ well-being. Any doubts respondents have should be taken seriously. With experience, an interpreter can judge which of the above steps are necessary for a particular individual and which may be omitted; however, the issues and strategies discussed above should be kept in mind during any interpretation session.

POTENTIAL BIASES IN VERIFICATION OF TYPE

There are several factors that can influence type interpretation and verification sessions despite the best efforts of respondents and interpreters. Keeping these factors in mind during your sessions will help minimize their influence.

- Respondents may have made incorrect assumptions about the MBTI assessment of which you are unaware. Asking them at the outset of your interpretation session what they know or have heard can provide an opportunity to correct misinformation.
- Your natural communication style may be more effective with some types than with others. Use your prior knowledge of respondents’ reported type, if available, to adopt a style that may appeal to their type.
- You may spend more time, have better examples, and be more enthusiastic when describing your own type preferences than those opposite yours. Try to develop and adhere to a presentation that gives equal time to opposite preferences and obtain some good descriptors and examples from colleagues whose type differs from yours.¹⁰

Interpreting the Step I™ Probability Index

The numerical scores associated with MBTI Step I preferences are known as probability indexes. The probability indexes reported for respondents’ categorical preferences indicate how likely it is that those preferences are correct for them.¹¹ Answers to respondent questions about the probability index should focus on this meaning and can be used for further exploration of the preference at issue.

None of the accuracy or clarity estimates (i.e., probability index, preference clarity index, preference score) used at various times in reports of MBTI results are interpretable as “quantities” of a preference. MBTI data, unlike those for most personality instruments, do not involve a continuum of scores and thus do not reflect “amounts” of the characteristics being measured.

In addition to the numerical probability index, a probability category (“somewhat likely,” “likely,” or “very likely”) is shown on the graph of respondents’ Step I results. The meanings of these categories are discussed next, but first some cautions are in order.

The meaning of high indexes. A common error in regard to probability indexes is assuming that a higher number implies excellence or superior development.

10. *Essentials of Myers-Briggs Type Indicator® Assessment* (Quenk, 2009a) suggests ways to anticipate and deal with practitioner–respondent differences.

11. The probability index replaces the preference clarity index (pci) associated with Form M, which in turn replaced preference scores reported for Form G.

For example, it is incorrect to assume that a person with a probability index of 90 associated with Intuition has a better command of Intuition than does a person with a probability index of 60. A larger number simply means that the respondent, when forced to choose, frequently chose N over S; a person with a lower probability index likely chose N over S less frequently. It often happens (and it is reasonable to expect) that individuals who report a high probability index exercise their preference more often than its opposite and thus (1) are more likely to have developed the behaviors associated with that preference and (2) are more likely to develop the characteristics and habits associated with exercising those behaviors.

However, this development sequence may be interrupted in any given individual. For example, a respondent's Step I results may indicate a high likelihood that he or she prefers Thinking as a way of making decisions, but for unknown reasons he or she may frequently make poor Thinking decisions. Another respondent with the same very high index for Thinking may use this preferred process to make generally consistent and satisfying decisions. Thus, the probability index for these individuals by itself does not indicate their inadequacy or excellence in using the Thinking process or how developed or differentiated they are, how well they will perform a task, or how likely it is that they will succeed or fail in an occupational setting.¹²

The numerical value of a probability index does not indicate how strong a preference is, how well it is used, how developed it is, or any other inference typically associated with scores on other personality measures.

Potential effects of Step II information on Step I results.

Respondents' Step II facet scores on each preference pair can help shed light on the meaning of their Step I results. Bear in mind that the facets associated with each MBTI preference pair do not necessarily identify all the possible components of the underlying type construct, nor are such known and potential components equivalent to the complexity and breadth of the Jung-Myers constructs. Rather, each preference pair not only is hypothesized to be greater than the sum of its identified parts but also is assumed to be influenced by each of the other preference pairs and the dynamic way they are combined.

However, observed patterns of Step II facet scores within the Step I preference pairs provide useful information about some lower probability indexes. It is

fairly common for respondents to have one or two facet scores on the side opposite their underlying preference (i.e., out-of-preference scores) and/or to have one or more scores in the midzone. For example, they may have an underlying preference for Sensing, yet their Step II pattern of facet scores may show them to habitually use one or more components of Intuition. Such individuals may respond to enough Step II items in the direction of Intuition that their Step I probability category is only "somewhat likely." If you have both Step I and Step II results available, you will have ready access to the clarifying effects of Step II information. Respondents find that their out-of-preference and midzone Step II results often help them resolve doubt or confusion about their Step I preferences.

GUIDELINES FOR INTERPRETING THE PROBABILITY INDEX AND PROBABILITY CATEGORIES

The guidelines provided here offer information about the confidence you can have that respondents actually have the preference they reported on each preference pair.

"Very likely." For respondents who obtain a probability index for a particular preference in the "very likely" category (85–100), the probability is reasonably high that the reported preference is accurate and that the respondent has and uses the reported preference and many of the characteristics and skills that accompany it. It can be useful to inquire during the interpretation session how the person uses the opposite preference in each pair when the situation demands it. For example, you might discuss how a respondent whose probability index is "very likely" for Perceiving meets required deadlines. Do not be surprised to find some respondents who are always on time or early for appointments despite their clear preference for Perceiving.

"Likely." With a probability index in the "likely" category (65–84), there is a reasonable probability that the reported preference is accurate and that the respondent will agree with the reported preference. However, you should inquire whether the preference fits and be alert for questions about the preference during the interpretation session. It is quite likely that the respondent makes habitual use of one or more aspects of the opposite preference in the pair and may spontaneously describe such use. If the respondent's Step II information is available, you can check this out.

"Somewhat likely." When a probability index is in the "somewhat likely" category (50–64) the respondent does not seem to be consistently drawn toward either preference in the pair. You should ask the respondent

12. It is not uncommon for critics of the MBTI assessment to conclude that it is invalid because preferences and associated scores do not predict job success. No such claim has ever been made for MBTI measures. Rather, evidence shows that MBTI types are more likely to be *attracted* to college majors and careers that allow them to use their preferences. There is no claim that people of some types *should* be more or less successful than other types in certain occupations, although employees who are able to exercise their type preferences in their job are likely to report greater work satisfaction.

whether the reported preference is a good fit based on his or her self-knowledge. The interpretation and verification methods described earlier in the chapter will prove helpful here. For some individuals, probability indexes in the “less likely” category reflect discomfort and dissatisfaction in using either pole of a preference pair. For example, a respondent with a “somewhat likely” result on S–N may report feeling simultaneously pulled toward attending to details and needing to step back and see the big picture. For other individuals, such a result may be affected by the consistent and comfortable exercise of particular Step II facets associated with the S–N preference pair. For example, respondents who prefer Sensing overall may like using their imagination when dealing with problems, which may reflect using the Imaginative (N) pole of the Realistic–Imaginative facet. Similarly, some Feeling types use a Questioning approach in decision making (a component of Thinking) rather than an Accommodating approach (a component of Feeling). Nevertheless, the Imaginative person can still verify a preference for Sensing, and the Questioning person a preference for Feeling.

It is interesting to think about possible factors that influence the observed differences among respondents’ probability indexes. The next section considers some possibilities.

INFLUENCE OF SOCIETAL BIASES ON MBTI® RESULTS

Social expectations are most likely to affect results on the Thinking–Feeling preference pair, a bias that seems to be pervasive in many cultures. There may be regional biases within some countries as well. For example, in the states of the southern United States, girls and women seem to be socialized toward Extraversion and Feeling. Men in general tend to be encouraged toward Thinking, but in some cultures that value the group over the individual, this may look like Feeling. In today’s multicultural world, practitioners are advised to keep the cultural backgrounds of their respondents in mind when verifying type results. Practitioners who use the MBTI Step I and Step II assessments within different cultures will most certainly be sensitive to such cultural influences.

Gender biases. In the global sample, 67.5% of men and 43.1% of women prefer Thinking (T) and 32.5% of men and 56.9% of women prefer Feeling (F). (See tables 7.10 and 7.11.) An examination by gender of the percentage of sample participants in the three probability categories can be clarifying. It shows that 40% of men who prefer T fall into the “very likely” category compared to 21% of women; 33% of women who prefer F appear in the “very likely” category compared to 13% of men; and 18% of men and 13% of women who prefer T are in the “likely” category. The percentages of men who prefer F and women who prefer T that appear in the “somewhat likely” category are virtually the same. It appears that, in general, men who prefer Thinking and women who

prefer Feeling have a higher probability of preferring Thinking and Feeling, respectively, than do men and women whose preferences are not in accordance with societal expectations.

Gender biases modified by Step II facets. It can be useful to ask individuals whose T–F preference is in doubt whether they responded to the Step I items in a socially expected manner. If so, their results may indicate a conflict between gender role expectations and their natural type preferences. If their Step II results are available, they may shed light on the issue. For example, a man who prefers Thinking may have adopted one or more facets of Feeling as a way to better interact with colleagues and family members, and a man who prefers Feeling may have developed some facets of Thinking to fulfill societal gender expectations. Similar accommodations may be made by women who are natural Thinking or Feeling types, given pressures at work and at home. Evidence of such Step II variations and adaptations can be particularly enlightening for respondents who are in doubt about their T–F preference. You might keep these tendencies in mind when reviewing both Step I and Step II results with respondents.

Preference frequency biases. As shown in table 6.2, the percentage of participants in the global sample with probability indexes in the “somewhat likely” category ranges from 5.8% for those who prefer Sensing to 9.1% for those who prefer Thinking and Judging. The percentage in the “likely” category ranges from 12.1% for those who prefer Intuition to 15.7% for those who prefer Perceiving. The percentage in the “very likely” category ranges from 11.7% for those who prefer Intuition to 49.2% for those who prefer Sensing. This last comparison, which shows a 4.5:1 ratio of Sensing types over Intuitive types in the “very likely” category, is striking. The global sample contains 67.5% Sensing types compared to 32.5% Intuitive types. So not only are participants who prefer Intuition in the minority (a result consistent with data from prior national samples), but they appear to respond to a fair number of Step I items in the direction of Sensing. This may be because surviving and thriving in the real world of details and facts demands some Sensing skills. People who are by nature equipped with

Table 6.2 | Percentages for each preference with a probability index of “very likely,” “likely,” or “somewhat likely” in the global sample

Very likely (%)		Likely (%)		Somewhat likely (%)	
E 22.4	I 36.6	E 13.1	I 12.7	E 7.7	I 7.5
S 49.2	N 11.7	S 12.5	N 12.1	S 5.8	N 8.8
T 29.9	F 23.5	T 15.4	F 13.7	T 9.1	F 8.4
J 23.1	P 27.6	J 15.6	P 15.7	J 9.1	P 8.9

Note: N = 16,773.

Sensing skills typically are not similarly pressured to adopt the skills that come naturally to Intuitive types, especially since the majority of people around them are also Sensing types. There is corroboration of this inference in table 10.6, which shows that there are fewer adjectives that show a large difference between Sensing and Intuitive types than might be expected from the Intuitive participants in the global sample because so many Intuitive types endorse the Sensing adjectives.

You might keep these tendencies in mind when reviewing both Step I and Step II results with respondents. Intuitive types' sensitivity to pressures to "go against their grain" at work and in other arenas of their life could be a fruitful and enlightening area to explore. Work environments that require employees to pay a lot of attention to detail but provide little opportunity or reward for innovation and big-picture thinking may be very unsatisfying for Intuitive types. Sensing types may find pressure to adopt Intuitive skills equally unsettling. Work environments that encourage or require frequent change, reward innovation over steady accomplishment, and place a high value on thinking outside the box can be quite stressful and unfulfilling for Sensing types.

INTERPRETING AND VERIFYING STEP II™ RESULTS

Because the Step II assessment is interpreted in the context of Step I knowledge, respondents must receive a Step I interpretation prior to receiving their Step II facet information. If respondents already know their Step I type from an earlier administration of an MBTI assessment, make sure that they have had an opportunity to verify their type and that their basic type knowledge is adequate. Providing a review of the preferences and confirming respondents' best-fit Step I type, as well as consolidating their understanding of type as a dynamic system, will ensure maximum benefit from a Step II interpretation. Using the brief review of type dynamics provided earlier in this chapter may be helpful.

Interpretation Materials Available

You can use either a brief profile report or a longer interpretive report when providing Step II feedback to respondents. The information in the brief Profile report is included in the Interpretive Report. Regardless of which report you use, you will find that reviewing respondents' reports prior to each interpretation session is an excellent way to gain knowledge of and experience with report contents, in addition to getting prepared for each respondent.

MBTI® Global Step II™ Profile. This report provides a brief description of the respondent's Step I and Step II results; graphs showing (1) probability indexes for the four Step I preference pairs and (2) results on the five facets of each preference pair; and an interpreter's summary, which summarizes all results. For a sample Global Step II Profile report, go to www.themyersbriggs.com.

MBTI® Global Step II™ Interpretive Report. This report includes the following:

- A brief introduction to Myers-Briggs personality type and the four Step I preference pairs
- The respondent's reported Step I type with a corresponding type description and a graph showing the respondent's probability indexes for the four preference pairs
- An introduction to the Step II facets and instructions for reading Step II results
- Detailed information on the respondent's facet results, including a graph of results on the five facets in each preference pair (four graphs in all) and several brief statements about each facet result, whether it is at one of the facet poles or in the midzone¹³
- Individualized information in five application areas
- Graphic display and explanation of type dynamics and development
- The respondent's Step II individualized type, or "Step II name" (respondent's four-letter type modified by his or her out-of-preference facet results)
- An overview of results
- An interpreter's summary

Figure 6.1 shows a Step II Interpretive Report page displaying the E–I facet results for a respondent who verified Extraversion as her Step I preference. Go to www.themyersbriggs.com for a sample Step II Interpretive Report.

When choosing between the Step II Profile and the Step II Interpretive Report, bear in mind that the latter contains a great deal of individualized information that is unavailable in the brief Profile. To aid your decision, compare the sample Profile and Interpretive Report at www.themyersbriggs.com. For detailed explanations and examples relevant to both, see the *MBTI® Step II® User's Guide* (Quenk & Kummerow, 2019).

Facet Scoring Categories

As mentioned in chapter 2, facet scores are arrayed as 11 data points, with scores of 1–5 to the left and 1–5 to the right of a 0 center point. Scores of 0, 1 to the left of 0, and 1 to the right of 0 are in the midzone. Such scores

13. These brief descriptions combine consistent self-reports obtained from a broad range of respondents and workshop participants, the item-based descriptions in chapter 4, and empirical research data.

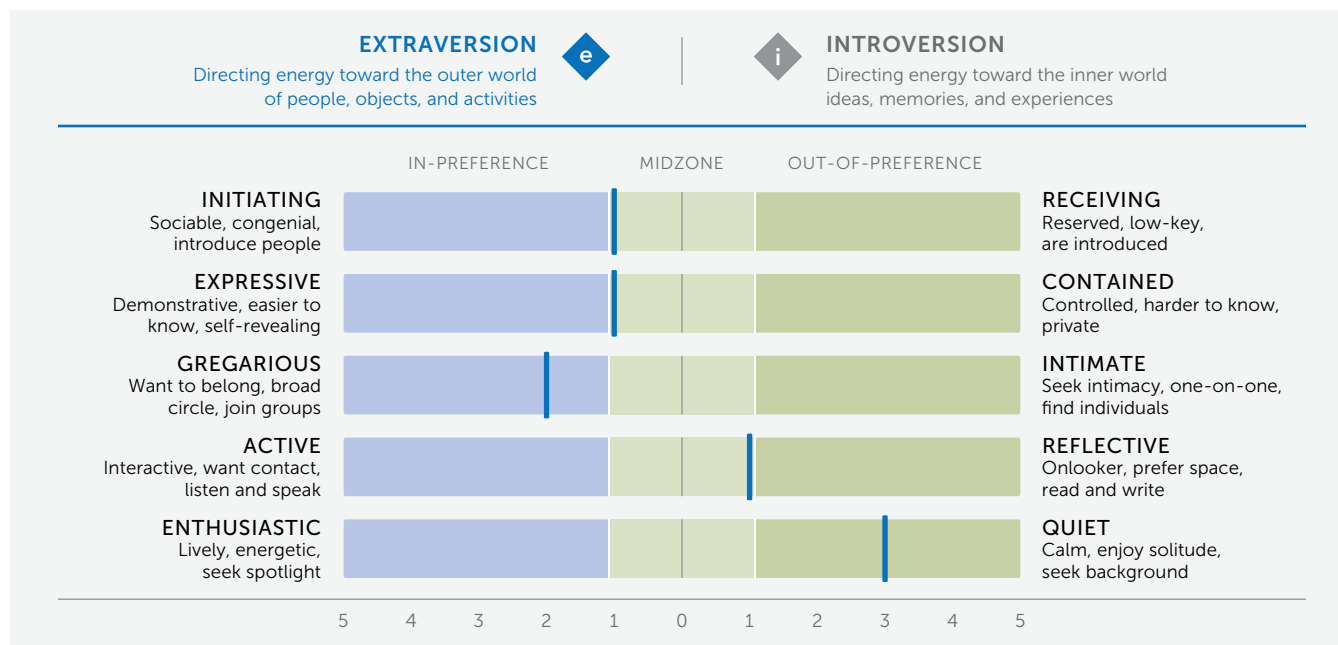


Figure 6.1 | E–I facet results for a respondent who verified Extraversion as her Step I™ preference

do not favor either pole of a facet. Scores to the left of the midzone indicate facet poles associated with E, S, T, or J, while scores to the right of the midzone show facet poles associated with I, N, F, or P. Within any of the four preference pairs, an individual's facet scores can fall into one of three categories:

In-preference scores. These are scores on the same side (to the left or right of the midzone) as the underlying preference. They conform to the respondent's underlying preference. For an individual with a Sensing preference, a score of 5 to the left of the midzone on the Concrete–Abstract facet is in-preference. For an individual with a Thinking preference, a score of 3 to the left of the midzone on the Logical–Empathetic facet is in-preference. And for an individual with an underlying preference for Perceiving, a score of 2 to the right of the midzone on the Early Starting–Pressure–Prompted facet is also in-preference.

Out-of-preference scores. These are scores on the side opposite an individual's Step I preference. Such scores do not conform to the assumption that respondents' Step II facet scores will most likely be on the same side as their underlying preference. For an individual with a preference for Introversion, a score of 3 to the left of 0 (toward the Enthusiastic pole) on the Enthusiastic–Quiet facet is out-of-preference. For an individual whose Step I preference is Judging, a score of 4 to the right of 0 (toward the Casual pole) on the Systematic–Casual facet is also out-of-preference.

Midzone scores. These are scores of 0, 1 to the left of 0, and 1 to the right of 0. Such scores do not favor either pole of a facet. The respondent cannot be adequately

described by the qualities associated with either of the facet poles.

THE BASIS FOR OUT-OF-PREFERENCE AND MIDZONE SCORES

The Jung-Myers theory predicts that individuals commonly use their less preferred processes and orientations some of the time or in a few habitual ways. In Step II terms, some of their facet results are out-of-preference. The theory does not specify which facets of a preference pair will be out-of-preference for a particular individual. Without such foreknowledge, the best prediction regarding any facet result is that it will align with the underlying preference—in other words, that it will be on the same side as the respondent's underlying orientation or process.

Midzone scores are quite common in Step II profiles and are associated with some consistent behavioral characteristics, depending on the particular facet or individual involved. Whereas scores at a pole reflect endorsement of all or most of the items describing that pole, scores in the midzone indicate endorsement of items reflecting *both* poles to some extent and/or some of the time.

There are three likely reasons respondents may score in the midzone, as described below. Respondents may come up with one or more of these reasons on their own. Sometimes, however, they recognize that the description of one or the other facet pole fits them better than does the midzone description. In discussing midzone results with respondents, it is generally more productive to allow them to describe their experience than to suggest the three possibilities. Practitioners

report that their skill in helping respondents arrive at satisfying facet category placements increases markedly with experience. Ongoing research indicates that there are meaningful relationships between midzone scores on particular facets and observable behaviors (see chapter 10).

Interpreters should encourage respondents to provide their own description and explanation of why they scored in the midzone on a particular facet rather than immediately suggesting the three possibilities offered here.

SOURCES OF MIDZONE SCORES

As described above, scores of 1 to the left of 0, 0, and 1 to the right of 0 are interpreted as midzone scores and may be attributed to one or more of the following respondent behaviors.

Habitual consideration of both poles of a facet.

Respondents may report being comfortable using both poles of a facet, feel equally satisfied using either pole, and be able to choose when and where to use each pole. For example, an ENFJ may explain that she finds it necessary to use both poles of several S–N facets because the behaviors described by both poles are valuable and necessary in her daily life. Thus she may hypothetically envision (Abstract) a family activity and then plan the details (Concrete) of how to make it happen given the practical constraints of her immediate situation (Concrete–Abstract facet). She may also explain that although the activity is a new one, it incorporates some of the familiar routines her family enjoys (Traditional–Original facet).

Habitual uncertainty or ambivalence when using a facet. Respondents may find that neither of the opposite poles of a facet is consistently appealing or satisfying as an approach. For example, an ISTP may want to remain receptive to possibilities (Spontaneous) when solving problems at work, yet he may feel pulled toward sticking to a schedule (Scheduled) so that he can coordinate his work with that of others (Scheduled–Spontaneous facet). Moving back and forth between the two opposites may drain his energy and be unsatisfying. Ideally, this individual will have developed a way of accommodating the type dynamics of ISTP, combining the extraverted Sensing energy of his auxiliary process, which wants to stay open to experience, with the introverted Thinking energy of his dominant process, which exerts pressure toward reaching an efficient, timely, and effective solution.

Habitual situational use of a facet. Respondents who score in the midzone on a facet sometimes describe their use of one or the other pole of that facet as being dependent on circumstances. For example, an ENTP may describe his tenacious approach (Tough) to

colleagues at work and his lenient way (Tender) of relating to his children (Tough–Tender facet), or an INFJ may explain her midzone score on Methodical–Emergent as a function of whether she is working and managing her household or on vacation. Sometimes the choices may not be so clear-cut, so the respondent may take some extra time to determine which approach is appropriate in a given situation.

Evidence of Type Development

An out-of-preference result that is verified by the respondent may provide evidence of that individual's type dynamics or type development. Similarly, verified midzone results may provide evidence of dynamically appropriate use of the tertiary or inferior process. Many respondents are aware of nuances of their personality functioning and enjoy giving examples of their motivations and behaviors and of how they have played out in their life. You might ask one or more of the following questions:

- Have you always functioned in accordance with your out-of-preference or midzone result?
- Can you identify life experiences or circumstances that may have stimulated development of this approach?
- How and when do you use the opposite (in-preference) pole of the facet (or both poles, for a midzone result) comfortably?

Both out-of-preference and midzone results are worth discussing with respondents as evidence of the distinctiveness of their type. For example, an ENTJ may adaptively express her inferior introverted Feeling process by taking an Accommodating approach rather than a Questioning one. She may explain that she learned to use this approach as a way of accomplishing important goals at work. She now uses her Accommodating approach habitually in many situations, and she finds it comfortable and satisfying. Or an INTP, whose auxiliary process is extraverted Intuition, may have a score of 3 on the Practical side of the S–N facet Practical–Conceptual. He expresses a deep appreciation of underlying concepts but also a desire to see his ideas result in practical improvements in his areas of interest. This could be a way of expressing his tertiary Sensing process. An ENTP may explain his midzone score on the Methodical–Emergent facet as behavior he consciously learned for his job as a programmer. Although he has made good use of a Methodical approach over the years, he is aware of the effort it requires and knows it is not his natural one.

When interpreting midzone scores to respondents, inquire about their particular experience and understanding of the facets involved using the verification and interpretation procedures described later in this chapter. As suggested above, respondents may feel better described by one or the other pole of a facet,

or they may describe a unique way of using the facet due to individual experiences and circumstances. Respondent verification of a midzone expression of a facet provides an opportunity to explore its meaning by considering one or more of the possibilities covered above. As an orientation to midzone interpretation, table 6.3 presents some of the general ways of interpreting midzone scores on the 20 facets. When using the information in the table with respondents, bear in mind that the descriptions provided there do not take into account the underlying preference, as do the midzone descriptions in the Step II Interpretive Report. It is therefore best to elicit respondents' spontaneous comments about their experience of a facet as a way of helping them understand their individuality.

Interpretation Value of Facet Scores

Determining whether a particular score is in-preference, out-of-preference, or midzone provides the basis for recognizing and understanding individual differences within people of the same type. When providing

feedback, the most important verification issue is the accuracy with which the scores reflect respondents' placement at either pole or in the midzone. If individuals disagree with their result on a facet, interpretation will be affected. For example, respondents may decide that a facet result that was reported as midzone is actually out-of-preference or in-preference. In such instances, statements about that facet in the report will be incorrect, so you will need to provide appropriate interpretive information that corresponds to the respondents' verified facet preference.

Table 6.4 shows the percentages and rank order of in-preference, midzone, and out-of-preference scores on the 20 Step II facets for individuals in the global sample. You may find this table useful because it shows which facets are more or less likely to yield scores in these three categories. There are wide variations in the frequency with which facet results are likely to be out-of-preference. In the global sample, the J–P facet Early Starting–Pressure–Prompted leads at 21%, followed by Methodical–Emergent (also J–P) at 17% and Experiential–Theoretical (S–N) at 13%. The J–P facet

Table 6.3 | General midzone descriptions for the 20 Global Step II™ facets

Global Step II™ facets	Midzone descriptions
E–I facets	
Initiating–Receiving	Seem socially at ease with people they know well; initiate conversations in familiar surroundings; willing to make introductions if they are not being made by others.
Expressive–Contained	Show interest in others but are selective in what they reveal about themselves; share quite a bit of personal information with friends and little with strangers; speak readily about topics they know well.
Gregarious–Intimate	Are moderately comfortable relating to strangers, depending on circumstances; comfort with being outgoing is situational; prefer to have deep, intimate relationships with very few people.
Active–Reflective	Communicate in person about personal issues, in writing about technical issues; like learning new information face-to-face; prefer to update information through reading.
Enthusiastic–Quiet	Show enthusiasm with topics and people they know well, otherwise stay in the background; desire for action or calm depends on how active or calm their day has been; people who typically see only one side of them may have very different perceptions of them.
S–N facets	
Concrete–Abstract	Want selected facts as well as the meaning behind those facts; can see the bigger picture without all the details but believe the big picture must be grounded in facts; under pressure, may over-rely on one fact and miss the meaning.
Realistic–Imaginative	Move quickly to apply a new idea and understand its limitations; stay in touch with commonsense aspects of situations; dislike “far-fetched” ideas and those that take too long to implement.
Practical–Conceptual	Blend pragmatism and curiosity; enjoy juggling ideas and their applications; want their best ideas to be used, not merely considered.
Experiential–Theoretical	Are interested in theories that explain what is important to them; are unlikely to pursue theories intensively or in great depth; are more interested in applying the patterns of a theory than in the patterns themselves.
Traditional–Original	Like following established methods that work, but readily change those that don’t; value selected traditions and consistency in certain family activities; enjoy novelty that doesn’t conflict with selected traditional values.

CONTINUED

Scheduled–Spontaneous appears least likely to elicit out-of-preference responses (1%), followed by the T–F facets Logical–Empathetic and Reasonable–Compassionate (both 3%). The three facets most likely to have midzone scores are Questioning–Accommodating (43%), Traditional–Original (40%), and Scheduled–Spontaneous (38%).

Table 6.4 provides general information about the frequency with which global sample participants' results were in-preference, midzone, and out-of-preference on the 20 facets, while table 6.5 shows whether the types preferring one pole of a preference pair in the sample received out-of-preference results on each facet more or less frequently than did those preferring the opposite preference and which of the 40 ways of being out-of-preference are most prevalent. Looking at tables such as these during your interpretation sessions can help respondents gain insight into their individuality.

For example, respondents may find it useful to know whether their out-of-preference results on particular facets are relatively common or unusual for people with

their preferences. Table 6.5 provides this information. As the table shows, for the E–I facets, Intimate Extraverted type is the most frequently occurring out-of-preference pattern among global sample participants, and Receiving Extraverted type is the least common. For the S–N facets, Experiential Intuitive type is by far the most frequently occurring out-of-preference pattern, followed by Conceptual and Theoretical Sensing types; the least common pattern is Concrete Intuitive type. For the T–F facets, Accepting Thinking type is the most common out-of-preference pattern, followed by Accommodating Thinking type, while the least common pattern is Logical Feeling type. For the J–P facets, Early Starting Perceiving type is the most common out-of-preference pattern. It is also the most common of all 40 out-of-preference patterns. Methodical Perceiving type is the second most frequently occurring J–P pattern, and Spontaneous Judging type is the least common. We might hypothesize that for this large global sample there is pressure to adopt the qualities associated with a Judging orientation, which is likely to be an uphill battle for many natural Perceiving types.

Table 6.3 | **General midzone descriptions for the 20 Global Step II™ facets** *CONTINUED*

Global Step II™ facets	Midzone descriptions
T–F facets	
Logical–Empathetic	Believe an ideal decision should consider both logical consequences and feeling values and appear neither overly logical nor overly concerned about others' feelings; may experience some tension in deciding between an analytical and a personal approach.
Reasonable–Compassionate	Make decisions based on concern for others as well as the logic of the situation; may vacillate in decision making if they feel that using objective criteria will hurt someone important to them; others may sometimes see them as ambivalent.
Questioning–Accommodating	Ask questions only when necessary and then in a mild and tactful way; are neither noticeably confrontational nor overly conciliatory; may become confrontational and direct when an important value or principle is threatened.
Critical–Accepting	Critique ideas and actions readily, taking little for granted; see flaws quickly and clarify what is wrong in situations; may critique silently or aloud, depending on the circumstances.
Tough–Tender	Prefer being conciliatory at first but will be tough when needed; are comfortable pushing others to necessary action; are devoted and loyal to those close to them.
J–P facets	
Systematic–Casual	Like general plans with some contingencies; inhibited by too much detail; don't mind interruptions if no plan is in place but dislike them once a project is under way.
Planful–Open-Ended	Exhibit varying degrees of planning in everyday life, depending on the activity; often plan for big social events but not little ones; may feel uneasy without at least some long-range plans.
Early Starting–Pressure-Prompted	Have difficulty beginning a task too far in advance of the deadline; need some pressure to get started; work best when the deadline is close enough to create some pressure.
Scheduled–Spontaneous	Welcome a moderate amount of routine, which provides them with both predictability and freedom to respond to the unexpected; see schedules as helpful at work but confining at home, or vice versa.
Methodical–Emergent	Are comfortable without a plan in familiar situations; don't need to have all the steps in place before starting a task; are seen as generally flexible.

Note: The descriptions provided in this table are based on behavioral reports and observations sampled from the more detailed midzone descriptions that appear in the MBTI® Global Step II™ Interpretive Report.

Table 6.4 | In-preference, midzone, and out-of-preference percentages and rankings for the Global Step II™ facets in the global sample

Global Step II™ facet	In-preference		Midzone		Out-of-preference	
	%	Rank	%	Rank	%	Rank
E–I facets						
Initiating–Receiving	63	4	33	10	4	16
Expressive–Contained	56	17	37	5	8	9
Gregarious–Intimate	59	13	30	17	10	5
Active–Reflective	59	14	35	6	6	13
Enthusiastic–Quiet	62	6	30	16	8	8
S–N facets						
Concrete–Abstract	60	11	37	4	4	17
Realistic–Imaginative	64	1	31	14	5	15
Practical–Conceptual	59	12	32	13	9	7
Experiential–Theoretical	57	16	30	18	13	3
Traditional–Original	53	19	40	2	7	10
T–F facets						
Logical–Empathetic	64	3	34	8	3	19
Reasonable–Compassionate	62	5	35	7	3	18
Questioning–Accommodating	47	20	43	1	10	6
Critical–Accepting	56	18	32	12	12	4
Tough–Tender	60	8	33	11	7	12
J–P facets						
Systematic–Casual	60	10	33	9	7	11
Planful–Open-Ended	64	2	31	15	6	14
Early Starting–Pressure-Prompted	58	15	21	20	21	1
Scheduled–Spontaneous	61	7	38	3	1	20
Methodical–Emergent	60	9	23	19	17	2

Note: N = 16,773.

A natural preference for Perceiving rather than Judging appears to be globally disfavored and discouraged, beginning in childhood, continued throughout the educational system, and further reinforced in the workplace. That Perceiving types manage to adapt, thrive, and excel in spite of this may be seen as a very positive triumph of nature over nurture.

The Polarity Index

Chapter 8 presents information on the reliability of the Step II assessment as reflected in the consistency with which individuals respond to the set of items (internal consistency reliability) and the stability of its results over time (test-retest reliability). The interpreter's summary included in the Global Step II Profile and the Global Step II Interpretive Report provides an additional measure of consistency in the form of the *polarity index*, which is a way of looking at the reliability of a single individual's scores. The polarity index was designed by Saunders (1987) to describe how consistently a respondent has scored *toward the poles of the facets*. It thus provides

information on an individual level that traditional reliability statistics cannot provide.

The polarity index ranges from 0 to 100 (see chapter 7 for details). One of Saunders' initial goals in including such a measure was to provide a way of estimating whether a profile is valid—the result of candid responses to the items. The operating assumption derived from type theory is that respondents are more likely to score toward one or the other pole of a facet than in the midzone. Individual variations in the polarity index enable the formation of some hypotheses that can be useful in individual interpretations. Important basic information about the polarity index is summarized as follows:

- A polarity index of 100 occurs when scores on all facets are at one or the other pole, that is, when all are at 5 to the left of 0 or 5 to the right of 0, or a combination of both.
- A polarity index of 0 occurs when all facet scores are 0.
- Most adults (77% in the global sample) obtain a polarity index of 50 or above.

- Responding to the items randomly yields an average polarity index of about 30.
- Low polarity indexes (<45) may indicate an invalid profile.
- Polarity indexes above 80 are not unusual.

The meaning of a high polarity index. In most cases, respondents with a high polarity index (70 or above) very clearly prefer *one or the other pole* of each facet, regardless of whether a particular score is in-preference or out-of-preference. In other words, the numerical value of the index does not indicate whether they have clear preferences for the facet poles that correspond to their four underlying Step I preferences. For example, one INTJ with a polarity index of 75 may clearly favor all I, N, T, and J poles of the facets, while another INTJ with the same polarity index of 75 may have some facet scores of 4 or 5 that correspond to her INTJ preferences and others that correspond to the opposite poles (the E, S, F, and P poles). The polarity index indicates that these two INTJs are both clear in their facet preferences, but it does not describe the particular array of those preferences.

A high or low polarity index tells you nothing at all about how consistently an individual's facet preferences correspond to his or her type preferences. It reflects only how consistently that individual scored toward the poles.

High polarity indexes with all or most scores

in-preference. In addition to verifying respondents' preferences and Step I type, a standard review of the Step I Profile provides an opportunity to explore unique ways they use their type that are not revealed in the Step II Profile for the following reasons:

- The Jung-Myers theory specifies that people use all the processes and orientations at least some of the time and that this is necessary for adaptation. Given this general working assumption, inquiry into how and when a respondent calls on the less preferred personality attributes is important and often enlightening for both respondent and interpreter.
- The Step II assessment does not necessarily encompass all the possible facets of each preference pair. Respondents whose Step II results are mostly or entirely consistent with their

Table 6.5 | The 40 out-of-preference facet patterns by percentage of individuals in the global sample

E-I out-of-preference pattern	%	S-N out-of-preference pattern	%
Initiating Introverted type	5.6	Concrete Intuitive type	2.5
Receiving Extraverted type	1.9	Abstract Sensing type	4.4
Expressive Introverted type	7.8	Realistic Intuitive type	4.2
Contained Extraverted type	7.1	Imaginative Sensing type	5.7
Gregarious Introverted type	8.8	Practical Intuitive type	6.2
Intimate Extraverted type	12.5	Conceptual Sensing type	10.5
Active Introverted type	7.1	Experiential Intuitive type	19.4
Reflective Extraverted type	4.0	Theoretical Sensing type	10.1
Enthusiastic Introverted type	8.9	Traditional Intuitive type	4.1
Quiet Extraverted type	7.3	Original Sensing type	8.3
<i>Average % for Introverted types</i>	<i>7.6</i>	<i>Average % for Intuitive types</i>	<i>8.2</i>
<i>Average % for Extraverted types</i>	<i>6.6</i>	<i>Average % for Sensing types</i>	<i>7.8</i>
T-F out-of-preference pattern	%	J-P out-of-preference pattern	%
Logical Feeling type	2.5	Systematic Perceiving type	9.8
Empathetic Thinking type	2.7	Casual Judging type	3.7
Reasonable Feeling type	3.5	Planful Perceiving type	7.0
Compassionate Thinking type	2.7	Open-Ended Judging type	3.9
Questioning Feeling type	5.9	Early Starting Perceiving type	23.0
Accommodating Thinking type	12.8	Pressure-Prompted Judging type	10.6
Critical Feeling type	6.3	Scheduled Perceiving type	1.9
Accepting Thinking type	17.4	Spontaneous Judging type	0.3
Tough Feeling type	4.7	Methodical Perceiving type	18.0
Tender Thinking type	8.5	Emergent Judging type	14.9
<i>Average % for Feeling types</i>	<i>4.6</i>	<i>Average % for Perceiving types</i>	<i>11.9</i>
<i>Average % for Thinking types</i>	<i>8.8</i>	<i>Average % for Judging types</i>	<i>6.7</i>

Note: N = 16,773.

underlying preferences may be using opposite Step I preference poles in ways that are not detectable by the Step II assessment.

Sometimes people may be so firmly committed to their type that they become one-sided—unable or unwilling to use opposite approaches when necessary. However, neither the Step I probability index nor the Step II facet scores by themselves allow such a conclusion. Facet scores that are very consistent with a respondent's Step I type (i.e., that are all or mostly in-preference) are one variation on the theme of individuality and worthy of further exploration. For example, as part of a Step I interpretation session, a respondent who consistently chooses Sensing over Intuition might be asked to give an example of when and how she uses Intuition. This general interpretation technique reminds her that she uses a less preferred personality characteristic at times and that a preference for Sensing does not proscribe the use of Intuition. In fact, such inquiry might encourage the respondent to describe ways of using Intuition that are not tapped by the Step I assessment. A similar approach can be even more fruitful in interpreting Step II results. Such a discussion may also reveal distinctive features of the respondent's type dynamics and development, which can provide important insights for both respondent and practitioner.

It may be easier for a respondent to come up with examples of using the opposite poles of the five facets of a preference pair than examples of using the opposite preferences in that pair because the facets focus on more specific features than does the general underlying preference pair.

High polarity indexes with both in-preference and out-of-preference results. Having a profile with both very clear in-preference facet results and very clear out-of-preference results is not unusual. When verified by the respondent, such a profile often elicits many insights regarding individual type dynamics and development. The technique of asking respondents to come up with examples of using both the preferred facet poles and their opposite, nonpreferred poles can be especially interesting when the preferred pole is out-of-preference and the nonpreferred pole is in-preference. For example, an Enthusiastic (facet score of 5 in the direction of Extraversion) INFJ musician may remark that when it comes to his passion for music, he is lively and energetic and likes to be at the center of things, whereas in most other circumstances he is content to stay quietly in the background and reserve his energy for his music. He appears to have developed a habitual way of using the Enthusiastic facet pole of his nonpreferred Extraversion orientation as a way of fulfilling his goals as a musician.

The meaning of a low polarity index. Possible explanations for a polarity index below 45 may emerge

during a standard interpretation session. For example, respondents may not have formed clear preferences for the facet components and will have difficulty verifying a best-fit type, or they may habitually function in a manner consistent with many midzone scores yet readily verify a best-fit type. Other respondents may be unwilling to reveal themselves, perhaps due to mistrust of the testing situation. (Note that providing adequate information to respondents before they take the assessment, as described in the administration part of this chapter, will reduce such occurrences.) In some cases, individuals respond to the items falsely or arbitrarily, or respond inaccurately due to language, reading, or vocabulary issues.

Interpreters should be cautious in assessing the possible meaning of low polarity indexes. As with all aspects of type interpretation, respondents' judgment regarding the accuracy and personal meaning of their results must be respected. Practitioners have reported a fair number of instances in which respondents verified their low polarity index as reflecting their consistent behavior on a number of facets. However, occasionally respondents with low indexes are unable to verify any type and may not find the typological approach particularly meaningful.

Research Information on the Polarity Index

Research evidence based on data from the global sample reveals some interesting information about differences in polarity indexes among sample participants. Such differences may be useful to interpreters of the MBTI assessments.

For example, table 6.6 lists mean polarity indexes for the 16 types in rank order from highest to lowest and the corresponding standard deviation (SD) for each type. It is interesting that ISTJ has the highest mean polarity index and ISTP, which differs by only one letter (P vs. J) has the second-lowest mean index. Both types have similarly high standard deviations, indicating extensive variability among respondent scores. Perhaps the different dynamics of these two types contribute to their polarity index results: ISTJs are dominant introverted Sensing types with auxiliary extraverted Thinking, tertiary Feeling, and inferior Extraverted Intuition; ISTPs, in contrast, are dominant introverted Thinking types with auxiliary extraverted Sensing, tertiary Intuition, and inferior extraverted Feeling. Thus both the energy devoted to each of their preferences and the orientations associated with them are quite different. (See chapter 3 to refresh your memory about type dynamics.) It might be hypothesized that on average the ISTJs in this large global sample behave more consistently with regard to the facet poles they prefer, while ISTPs on average behave less consistently. Readers might find it interesting to review the type descriptions of these two types in chapter 4 to speculate on a possible rationale for this

Table 6.6 | Rank order of types in the global sample by mean polarity index

MBTI® type	<i>n</i>	<i>M</i>	<i>SD</i>
ISTJ	2,671	65.17	12.95
ENFP	1,370	63.57	11.60
ISFJ	1,404	61.51	11.52
INFP	1,065	61.13	11.76
ENFJ	361	61.02	10.64
INFJ	387	60.61	11.27
ESTJ	1,509	60.58	11.86
INTJ	429	60.31	11.49
ENTJ	309	60.01	10.47
ESFJ	951	59.94	10.52
ENTP	722	57.62	11.95
INTP	809	57.32	12.95
ESFP	1,001	53.61	10.44
ISFP	1,106	53.25	11.05
ISTP	1,651	52.70	13.16
ESTP	1,028	51.09	11.53

Note: *N* = 16,773.

difference. For more evidence of the effects of type dynamics on research results, see chapter 9.

Influence of age and gender on polarity indexes. Age and gender differences also affect average polarity indexes. Though we can't apply group data to a particular respondent, it may be useful to bear in mind that for the eight age groups shown in table 6.7, ranging from 15–18 years to 75–88 years in the global sample, the mean polarity index generally rose with increasing age except in the oldest age groups, where sample sizes were considerably smaller and thus subject to greater variation. So it appears that with increasing age, people in general become more sure or clear about the facet poles they prefer.

Even more interesting are the gender differences shown in table 6.8. In the first four age groups (ages 18–44), women have higher indexes. Men start to “catch up” at ages 45–54, but not until age 75+ (where the sample size is very small for both genders) do men's indexes noticeably exceed those of women. It may be that men as a group take longer to solidify their facet pole preferences than do women, perhaps due to societal expectations for men that are not an issue for women.

Type-Specific Score Ranges

Differences in Step II results among the 16 types—such as variations in mean polarity index and percentages of in-preference and out-of-preference scores—can provide information about type dynamics and offer interpretational clues for individual respondents. Differences among members of the same type can also be explored by looking at the average range of scores on

each facet that is typical for that type. Known as norms in quantitatively varying instruments, these averages specify the mean, or average, score for a type on each facet and show the range of scores that are plus or minus one standard deviation above and below that mean (about 68% of the sample, assuming a normal distribution).

Type-specific ranges are shown on the interpreter's summary. For an example, see figure 6.2, which compares an ISFJ respondent's scores on the J–P facets to the average range of scores for all ISFJs in the global sample. The length of the bar shown on the profile for each facet indicates the range of scores that fall between 1 standard deviation above and 1 standard deviation below the mean score. The mean itself is shown as a vertical line. For any facet, about 68% of the people of that type can be expected to obtain scores in the range shown. Participants who obtain a score above or below the average range are reporting either a greater or a lesser preference for that facet pole than others of their type. Facet scores that fall outside the standard deviation for a type are often, but not always, out-of-preference scores. Bear in mind that these type-specific

Table 6.7 | Mean polarity indexes by age group in the global sample

Age group	<i>n</i>	<i>M</i>	<i>SD</i>
15–18 years	101	53.70	11.74
19–24 years	2,526	55.82	13.12
25–34 years	3,665	57.32	13.30
35–44 years	3,622	59.98	12.62
45–54 years	3,283	60.56	11.85
55–64 years	2,092	61.74	11.68
65–74 years	1,263	60.99	11.76
75–88 years	221	55.28	15.14
<i>Combined</i>	<i>16,773</i>	<i>59.08</i>	<i>12.73</i>

Note: *N* = 16,773.

Table 6.8 | Mean polarity indexes by age group and gender in the global sample

Men		Women	
Age group	Mean polarity index	Age group	Mean polarity index
<18 years	51.49	<18 years	55.63
18–24 years	52.86	18–24 years	57.85
25–34 years	54.63	25–34 years	59.42
35–44 years	58.74	35–44 years	61.00
45–54 years	60.25	45–54 years	60.80
55–64 years	61.13	55–64 years	62.37
65–74 years	61.12	65–74 years	60.75
75+ years	56.01	75+ years	53.72

Note: *N* = 16,773.

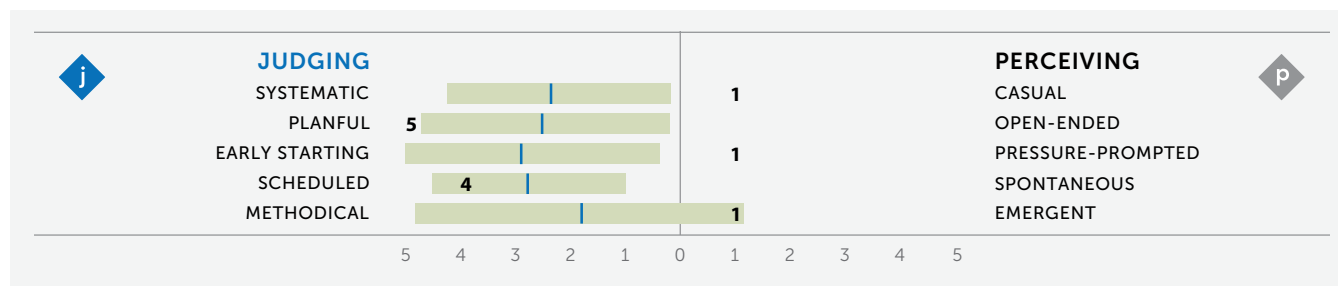


Figure 6.2 | Scores for an ISFJ compared to average range of scores for ISFJs in the global sample

ranges include respondents of both genders and across all ages sampled. They are thus of limited interpretive value for an individual respondent. However, when a respondent's basic four-letter type is in question and other clarification methods have not helped, comparing the score ranges for two or more types can be informative. For example, if a respondent finds that both ESTJ and ISTJ descriptions are equally appropriate, comparing his scores to the average ranges for both these types might suggest reconsideration of the type that more closely approximates the respondent's scores. However, you should employ such a tactic judiciously so as not to "convince" respondents to verify their type based on statistical evidence alone. As always, a variety of verification approaches should be used. Appendix A provides the means, standard deviations, and average score ranges on the 20 facets for each of the 16 types.

Interpreting Step II™ Results

As mentioned earlier in this chapter, the recommended procedure for interpreting the Step II assessment differs markedly from that for interpreting the Step I assessment. Unlike with Step I verification, Step II respondents are not asked to self-assess their results on each facet before viewing their Step II Profile or Interpretive Report. The interrelationship of the facet scales with one another and with their underlying preference pairs makes this task too difficult for respondents and impractical given the number of facets involved. The recommended approach is to describe the essence of each facet, give some information or relevant examples, look at the results with the respondent, and then ask the respondent to assess whether the score placement and the description provided fit.

Although this procedure may risk facile agreement from some respondents, practitioners do not report this as a problem. Studies of the Global Step II assessment in this area are not yet available. However, a study that asked Form Q workshop participants to assess the accuracy of their placement on the 20 facet scales showed a very high level of agreement. Disagreements primarily involved moving from the midzone to an in-preference or out-of-preference placement, or from an in-preference placement to the midzone. There were no reported

instances of moving from one pole to the opposite pole (Quenk, 2009b). Results from an even earlier survey of six practitioners who had regularly administered Form K of the Step II assessment over a five-year period similarly showed that it was rare for respondents to judge that they were more accurately described by the pole opposite a reported result (Quenk, Hammer, & Majors, 2001).

The preceding sections of this chapter laid the groundwork for conducting a Step II interpretation. This section covers logistics as well as suggested methods of eliciting information from respondents. It is followed by a summary of Step II information for each of the 16 types that may prove useful for individual interpretation.

TIME AND TIMING OF STEP II™ INTERPRETATION

Allow at least twice as much time for an individual Step II interpretation as would be customary for a Step I interpretation. When interpreting to groups, a half day is desirable, and longer if an application of the Step II assessment is included in the feedback process. Ideally, a follow-up session should be planned for individuals and for groups. Step II results present a great deal of information, and respondents usually need time to consider its relevance to their life.

Many respondents benefit from having a period of time during which they can absorb and explore the meaning of their Step I results. Interpreting both Step I and Step II results in the same session may therefore not be useful. However, with some respondents, introducing Step II results early in the process of verifying Step I type can be effective in helping them arrive at a best-fit type. You should decide what procedure to use on the basis of individual circumstances and respondent needs.

STEPS IN PROVIDING STEP II™ FEEDBACK

The following seven steps are recommended for the Step II feedback process.

Step 1: Conduct a standard Step I verification/interpretation session for respondents who have not already had one. Give those who are new to MBTI personality type and the theory underlying it time to absorb this information. If respondents have had an

interpretation session previously, review and confirm their understanding of critical characteristics of their type.

Step 2: Explain the basis and purpose of the Step II assessment. Ask respondents about their particular interests and goals for the interpretation session.

Step 3: Explain and show how Step II information is arrayed using a sample profile. This is preferable to showing the respondents' profile, which may cause distraction by presenting personal results before understanding has been achieved. However, make sure respondents receive a copy of their Step II Profile or Interpretive Report to retain.

Step 4: Before reviewing respondents' results, explain the verification procedure you will use—that you will describe the content of each facet, give some examples, and then together look at the results to see whether the result on each facet fits respondents' self-knowledge.

- For facet scores that are in-preference (or for selected in-preference scores that are of particular interest or relevance to respondents), ask respondents to give examples of using their preferred and nonpreferred poles, as described earlier in the discussion on interpreting polarity indexes.
- For out-of-preference scores, ask respondents to give examples of using facets in an out-of-preference manner as well as in a manner consistent with the underlying preference.
- For midzone scores, have respondents give examples of their typical behavior in regard to the facets in question. You might also ask them to describe their thoughts or actions when forced to choose between the behaviors described by the poles of those facets. This may shed light on whether their scores result from an inability to decide which behavior to use and perhaps a feeling of being pulled in opposing directions; comfort with using both regardless of circumstances; or situational and personal influences, such as fatigue level and time of day.

Step 5: After verifying each facet, review and reinforce the relevance of respondents' whole type, even if they already have a firm grasp on this concept. Explore issues that pertain to type dynamics and development in the context of their distinctive expression of their type. If there were doubts, clarity may have been achieved during the verification process. If not, explore possible type dynamics evidence (see Myers & Kirby, 1994), including the typical expression of the respondents' inferior function (see Quenk, 2000).

Step 6: If the Step II Interpretive Report has been generated for the respondents, they should not try to read all of it during the interpretation session. Instead,

you should point out particular areas that are relevant to their reasons for taking the Step II assessment and ask them to read the report in private after the session is completed. Suggest an opportunity for comment and follow-up.

Step 7: Schedule a follow-up session if possible, either as a single meeting or as part of a continuing consultative process. If you conducted a group interpretation session, provide an opportunity for individuals to follow up with you at a later time.

RECOMMENDED SUPPORT MATERIALS

It is helpful to have facet descriptions available that are briefer than the descriptions presented in chapter 4, although sharing all or part of the longer descriptions with respondents can be useful in an interpretation session. Even experienced interpreters of Step II results often find that they need some brief facet descriptions available in their sessions. Practitioners often report having the most difficulty distinguishing for respondents the differences between the S–N facets.¹⁴ When reviewing respondents' Interpretive Report, have them read the bulleted statements for each of their facet results. If you are using the Step II Profile rather than the Interpretive Report, you may also want to provide information from table 4.2 (content summaries of the facets), the brief descriptions of facet poles that appear in tables 2.1–2.4, and the general descriptions of midzones that appear in this chapter (table 6.3). Respondents often provide pertinent examples of the meanings of the facets that will enrich interpreters' subsequent interpretation sessions.

Several resources provide guidance for group and individual interpretation, including *Understanding Your MBTI® Step II™ Results* (Kummerow & Quenk, 2018) and the *MBTI® Step II™ User's Guide* (Quenk & Kummerow, 2019). Visit www.themyersbriggs.com for descriptions of the most current resources available.

ADDITIONAL CONSIDERATIONS

Individual respondent needs and the nature of respondent–practitioner interaction will inevitably present distinctive interpretation issues. In addition, the following areas can influence results and therefore should be taken into consideration.

Influence of gender on Step II results. Jung did not address gender differences in his theory, but he did make some observations based on his experience as a clinician. Myers discovered gender differences in the way men and women responded to items on the T–F preference pair. These differences occurred on all MBTI forms prior to Form M. Items that elicited different responses from men

14. This may be a consequence of the perceiving processes being “unconstrained” as discussed in chapter 3, that is, there is no goal or endpoint involved.

and women were removed from Form M and were not used in constructing the MBTI Global Step I and Global Step II assessments.

It is not known whether the differences Myers found were the result of cultural or biological factors or a combination of both. Regardless of their source, real gender differences in the expression of type can be readily observed and are likely a natural outgrowth of the different socialization experiences and expectations of women and men. For example, women who verify INTP as their type tend to appear more “sociable” than do male INTPs, probably a result of societal expectations for girls and women that are not typically present for boys and men.

Support for the existence of differing gender-based expressions of type can be seen in the different patterns

of correlations of Step II facet scales with the scales of other assessments. For example, table 6.9 shows Step II facet scales that highly correlated with a selection of *Adjective Check List* (ACL) scales for male and female participants in the global sample (see appendix B for brief definitions of ACL scales). For ease of understanding, facet poles that are unique to each gender appear in bold type in the table. The table shows, for example, that for the ACL Achievement scale (striving to be outstanding in one’s pursuits), men’s scores are uniquely associated with the Step II facet poles Original, Tough, and Questioning, while women’s scores are uniquely associated with the facet poles Expressive and Enthusiastic. It is tempting to infer that the qualities associated with men striving to achieve are not also associated with women pursuing that same goal.

Table 6.9 | **Global Step II™ facet scales in the global sample that are highly correlated with *Adjective Check List* (ACL) scales by gender**

Men Women		Men Women		Men Women	
Achievement		Dominance		Endurance	
Original	Expressive	Expressive ^a	Questioning	Methodical	Systematic
Tough	Enthusiastic	Questioning ^a	Gregarious^a		Methodical
Active	Active	Enthusiastic ^a	Enthusiastic ^a		Early Starting
Initiating	Initiating ^a	Active ^b	Active ^b		Sheduled
Questioning		Initiating ^b	Initiating ^b		
Affiliation		Nurturance		Exhibition	
Original	Accepting	Accepting ^a	Accepting ^b	Original	Original
Gregarious	Enthusiastic	Tender ^a	Tender ^a	Gregarious ^a	Casual
Active	Active	Compassionate	Compassionate	Expressive ^b	Open-Ended
Enthusiastic	Initiating	Empathetic	Empathetic	Enthusiastic ^b	Gregarious ^a
Initiating	Expressive	Accommodating	Accommodating	Active ^c	Expressive ^b
Expressive		Contained	Enthusiastic	Initiating ^c	Enthusiastic ^b
			Active		Active ^b
			Initiating		Initiating ^b
			Expressive		
Autonomy		Aggression		Change	
Original ^a	Original ^a	Expressive	Enthusiastic ^a	Original ^b	Original ^a
Conceptual	Open-Ended	Questioning	Active ^a	Imaginative ^a	Casual ^a
Critical	Spontaneous	Enthusiastic ^a	Initiating ^a	Casual ^a	Open-Ended
Initiating	Casual	Initiating ^a	Expressive	Abstract ^a	Abstract
Active	Conceptual	Active ^a	Questioning	Conceptual ^a	Imaginative
Tough	Questioning		Gregarious	Expressive	Spontaneous
Questioning ^a			Casual	Gregarious^a	Conceptual
				Enthusiastic ^a	Initiating
				Active ^a	Expressive
					Enthusiastic
					Active

CONTINUED

Another example concerns the ACL scale Autonomy, which involves acting independently of others or of social values and expectations. For men, high scores on Autonomy are correlated with facet scores in the direction of Critical, Initiating, Active, and Tough; for women, high scores on Autonomy are correlated with facet scores in the direction of Open-Ended, Spontaneous, and Casual. It is hard to imagine reversing those results for men and women—women being critical, initiating, active, and tough and men being open-ended, spontaneous, and casual.

Similar contrasts in the table may also suggest societal influences for men and women. Bear in mind, however, that these are correlations that account for a relatively small amount of the variation between genders, in spite of all results being statistically significant. Nevertheless, such comparisons are notable, particularly in light of the

size, language, and cultural heterogeneity of the global sample. A similar table of correlations appeared in the *MBTI® Step II™ Manual* (Quenk, Hammer, & Majors, 2001) based on the Form Q US national sample ($N = 1,378$), which was used in constructing the Form Q assessment. For that much smaller and more homogeneous sample, many more unique facet poles were found to differentiate men and women. Those found for the global sample appear to occur globally rather than nationally or regionally.

Awareness of the influence of these kinds of gender differences should sensitize interpreters to possible distinctions in the ways respondents experience themselves typologically. Insights resulting from such considerations can be particularly useful in the workplace as a way of differentiating between personality differences that are natural, those that may have been

Table 6.9 | **Global Step II™ facet scales in the global sample that are highly correlated with Adjective Check List (ACL) scales by gender** *CONTINUED*

Men Women		Men Women		Men Women	
Deference		Self-Control		Self-Confidence	
Accommodating ^a	Accommodating ^a	Reflective ^a	Reflective ^a	Original^a	Gregarious
Tender ^a	Tender	Receiving ^a	Quiet ^a	Conceptual	Expressive ^a
Reflective	Accepting	Quiet	Receiving	Gregarious ^a	Enthusiastic ^a
Accepting	Receiving	Accommodating	Contained	Expressive ^a	Active ^b
Receiving	Traditional	Contained	Early Starting	Enthusiastic ^a	Initiating ^b
Traditional		Concrete	Traditional	Active ^a	
		Realistic	Scheduled	Initiating ^b	
		Systematic	Planful		
		Traditional	Systematic ^a		
Personal Adjustment		Ideal Self		Creative Personality	
Expressive	Accepting	Original	Conceptual	Original ^b	Original ^a
Initiating	Enthusiastic	Conceptual	Gregarious	Conceptual ^a	Conceptual ^a
	Active	Expressive	Expressive	Abstract ^a	Imaginative
	Initiating	Active	Enthusiastic	Casual ^a	Abstract
	Expressive	Gregarious	Active	Enthusiastic	Casual
		Initiating ^a	Initiating ^a	Questioning	Enthusiastic
				Active	Active
				Initiating	Initiating
Order					
Logical	Planful				
Reasonable	Early Starting				
Scheduled	Methodical				
Systematic	Systematic				
Methodical	Scheduled ^a				

Note: $N = 16,773$. Names of scales unique to that gender are in bold type. All scales listed are significant at a minimum level of $p < .01$. Scales with no superscript have a correlation coefficient of .20–.29.

^a Correlation coefficient = .30–.39.

^b Correlation coefficient = .40–.49.

^c Correlation coefficient = .50–.59.

^d Correlation coefficient = .60+.

developed as coping strategies, and those that result from social expectations.

Influence of age on Step II results. Jung did not regard types as static. He believed that a developmental process takes place over an individual's life span that unfolds in distinctive ways for each type. While the details of this part of Jungian theory are beyond the scope of this manual, the main point is that Jung believed that there were natural developmental changes that occur with age for each type. He did not attribute such changes to chronological age itself, nor did he see such changes as inevitable. People of the same type do not necessarily develop in exactly the same way or at the same rate. Nevertheless, over their lifetime individuals of each type face common developmental issues that differ from those for individuals of other types. The "Developmental Personality Descriptions" section included in MBTI Step III™ reports (see Myers, McCaulley, et al., 2009) describes developmental advantages and challenges for each of the 16 types.

Age and developmental stage differences in the expression of type are to be expected. Because using one's type fully and effectively is not inevitable and does not occur according to any fixed schedule, there are no age brackets that can be used to identify an individual's

stage of type development. However, practitioners who work with respondents of different ages have observed maturational differences in the expression of type. In general, expression by younger members of a type tend toward the extremes of the type's qualities, while expression by older members appears more nuanced. One could say that older people wear their type more comfortably, assuming adequate development.

Interpreting Step II™ Results in the Context of Step I™ Information

The two major goals of Step II interpretation are to (1) give respondents insight into their unique expression of type as revealed in their Step II results and (2) help them clarify their four-letter type and understand the dynamics that contribute to the distinctiveness of each type.

The Step II assessment provides an opportunity to explore type dynamics and development issues by considering facet information for each of the 16 types. This can include looking for trends that might reflect generalized expressions of the less preferred poles as a function of each type's unique dynamics. For individual interpretation, it can also be helpful to summarize Step II information for each type (see tables 6.11–6.26 later in

Table 6.10 | **Step II™ facet poles within each preference pair with highest mean score for each type in the global sample**

	ISTJ	ISFJ	INFJ	INTJ
E–I	Quiet	Receiving	Receiving	Contained
S–N	Concrete	Traditional	Abstract	Theoretical
T–F	Logical	Accommodating	Compassionate	Logical
J–P	Systematic	Scheduled	Scheduled	Scheduled
	ISTP	ISFP	INFP	INTP
E–I	Quiet	Receiving	Receiving	Contained
S–N	Realistic	Practical	Abstract	Original
T–F	Logical	Empathetic	Empathetic	Logical
J–P	Spontaneous	Open-Ended	Casual	Spontaneous
	ESTP	ESFP	ENFP	ENTP
E–I	Active	Initiating	Enthusiastic	Initiating
S–N	Concrete	Practical	Abstract	Original
T–F	Reasonable	Empathetic	Empathetic	Reasonable
J–P	Spontaneous	Open-Ended	Casual	Spontaneous
	ESTJ	ESFJ	ENFJ	ENTJ
E–I	Initiating	Expressive	Initiating	Initiating
S–N	Concrete	Traditional	Imaginative	Conceptual
T–F	Reasonable	Empathetic	Empathetic	Logical
J–P	Systematic	Scheduled	Planful	Scheduled

Note: N = 16,773.

this chapter). This information will personalize and enrich your interpretation by enabling you to present Step II information in the context of respondents' whole type.

For example, table 6.10 shows the facet poles within each preference pair that have the highest mean scores for each type. The table illustrates some similarities and differences in emphasis between dynamically similar types. The observations you make from these and other arrays of Step II results will add to your understanding and expand your sensitivity to interpretation issues. Although no single way of looking at the data can be considered definitive, evidence gleaned from a variety of analyses will enhance your interpretation efforts.

The tables ranking Step II variables that appeared earlier in this chapter (tables 6.4, 6.6, 6.10) enable comparison of one type to another across the 16 types. In contrast, tables 6.11–6.26 summarize data from the global sample for each type so that you will have easy access to information about the probable type (or alternative types) of your respondent. Each table shows the sample size for the type, the mean polarity index, the average number of out-of-preference responses overall for the type, and the percentage of out-of-preference results on each facet.¹⁵ For ease of comparison, the tables are ordered by dominant process, starting with the two dominant extraverted Thinking types (ESTJ and ENTJ). The brief comments below are limited to differences between the two types that have the same dominant process. However, you will find many other intriguing comparisons relevant to type-specific data in tables 6.11–6.26. Note that in the tables the facet with the highest percentage of out-of-preference results within each preference pair is in bold type.

DOMINANT EXTRAVERTED THINKING TYPES (ESTJ AND ENTJ)

TABLES 6.11 AND 6.12

These two types have similar and moderately high mean polarity indexes. ENTJs have slightly more out-of-preference scores than ESTJs overall.

Among the E–I facets, ESTJs have their highest percentage of out-of-preference scores on Quiet (Enthusiastic–Quiet facet), while ENTJs have their highest percentage on Intimate (Gregarious–Intimate facet). Among the S–N facets, ESTJs' highest percentage is on Conceptual (Practical–Conceptual facet), while ENTJs' highest percentage (a very high 22%) is on Experiential (Experiential–Theoretical facet). Among the T–F facets, the highest percentage for both ESTJs and ENTJs is on Accepting (Critical–Accepting facet). And among the J–P facets, ESTJs' highest percentage is on Emergent (Methodical–Emergent facet), while ENTJs'

is on Pressure–Prompted (Early Starting–Pressure–Prompted facet).

DOMINANT INTROVERTED THINKING TYPES (ISTP AND INTP)

TABLES 6.13 AND 6.14

The mean polarity index for INTPs is somewhat higher than that for ISTPs, while ISTPs show a somewhat higher average number of out-of-preference scores than do INTPs.

Among the E–I facets, ISTPs have their highest percentage of out-of-preference scores on Gregarious (Gregarious–Intimate facet), while INTPs have their highest percentage on Enthusiastic (Enthusiastic–Quiet facet). Among the S–N facets, ISTPs' highest percentage is on Theoretical (Experiential–Theoretical facet), while INTPs' is on the opposite pole of that facet, Experiential. Among the T–F facets, ISTPs' highest percentage is on Accommodating (Questioning–Accommodating facet), while INTPs' is on Accepting (Critical–Accepting facet). And among the J–P facets, the highest percentage for both ISTPs and INTPs is on Early Starting (Early Starting–Pressure–Prompted facet).

DOMINANT EXTRAVERTED FEELING TYPES (ESFJ AND ENFJ)

TABLES 6.15 AND 6.16

The mean polarity index for ESFJs is slightly lower than that for ENFJs, as is the average number of out-of-preference scores.

Among the E–I facets, both ESFJs and ENFJs have their highest percentage of out-of-preference scores on Intimate (Gregarious–Intimate facet). Among the S–N facets, ESFJs' highest percentage is on Conceptual (Practical–Conceptual facet), with Theoretical (Experiential–Theoretical facet) a close second, while ENFJs' highest percentage by far is on Experiential (Experiential–Theoretical facet). In fact, Experiential is the facet pole most likely to be out-of-preference for ENFJs. Among the T–F facets, ESFJs' highest out-of-preference percentage is on Questioning (Questioning–Accommodating facet), with Reasonable (Reasonable–Compassionate facet) and Critical (Critical–Accepting facet) only slightly lower; ENFJs' highest percentage is also on Questioning (Questioning–Accommodating facet). Among the J–P facets, ESFJs' highest percentage by far is on Emergent (Methodical–Emergent facet), while ENFJs' is on Casual (Systematic–Casual facet).

15. The tables also provide interesting insights into the dynamic qualities of each type. For example, compare ISFPs (table 6.17; dominant introverted Feeling types with *auxiliary extraverted Sensing*) to INFPs (table 6.18; dominant introverted Feeling types with *auxiliary extraverted Intuition*). Note the different out-of-preference percentages for these two types, particularly on the J–P facets.

Table 6.11 | Percentage of out-of-preference facet scores for ESTJs in the global sample

ESTJ							
E-I	%	S-N	%	T-F	%	J-P	%
Receiving	1.6	Abstract	2.4	Empathetic	2.3	Casual	0.9
Contained	7.2	Imaginative	4.4	Compassionate	1.7	Open-Ended	2.7
Intimate	10.5	Conceptual	14.0	Accommodating	11.7	Pressure-Prompted	10.2
Reflective	4.4	Theoretical	8.6	Accepting	20.1	Spontaneous	0.4
Quiet	11.1	Original	8.9	Tender	6.5	Emergent	14.3

Note: ESTJ portion of the global sample, $n = 1,509$. Facet with highest percentage of out-of-preference scores in each preference pair is in bold type. Average number of out-of-preference facet scores = 1.44. Mean polarity index = 60.58.

Table 6.12 | Percentage of out-of-preference facet scores for ENTJs in the global sample

ENTJ							
E-I	%	S-N	%	T-F	%	J-P	%
Receiving	0.6	Concrete	2.9	Empathetic	3.6	Casual	5.8
Contained	12.0	Realistic	6.8	Compassionate	3.2	Open-Ended	3.6
Intimate	13.3	Practical	4.9	Accommodating	7.4	Pressure-Prompted	17.8
Reflective	7.8	Experiential	22.0	Accepting	19.7	Spontaneous	0.3
Quiet	4.2	Traditional	3.9	Tender	11.0	Emergent	13.3

Note: ENTJ portion of the global sample, $n = 309$. Facet with highest percentage of out-of-preference scores in each preference pair is in bold type. Average number of out-of-preference facet scores = 1.64. Mean polarity index = 60.01.

Table 6.13 | Percentage of out-of-preference facet scores for ISTPs in the global sample

ISTP							
E-I	%	S-N	%	T-F	%	J-P	%
Initiating	5.6	Abstract	4.2	Empathetic	3.6	Systematic	21.3
Expressive	7.9	Imaginative	4.5	Compassionate	4.1	Planful	7.1
Gregarious	13.3	Conceptual	11.9	Accommodating	15.4	Early Starting	27.9
Active	11.1	Theoretical	13.0	Accepting	14.2	Scheduled	2.7
Enthusiastic	7.6	Original	12.4	Tender	8.6	Methodical	15.8

Note: ISTP portion of the global sample, $n = 1,651$. Facet with highest percentage of out-of-preference scores in each preference pair is in bold type. Average number of out-of-preference facet scores = 2.12. Mean polarity index = 52.70.

Table 6.14 | Percentage of out-of-preference facet scores for INTPs in the global sample

INTP							
E-I	%	S-N	%	T-F	%	J-P	%
Initiating	8.5	Concrete	2.8	Empathetic	3.5	Systematic	10.0
Expressive	4.9	Realistic	7.4	Compassionate	3.8	Planful	8.0
Gregarious	10.4	Practical	6.7	Accommodating	11.5	Early Starting	26.3
Active	8.9	Experiential	14.3	Accepting	19.8	Scheduled	1.1
Enthusiastic	14.6	Traditional	3.0	Tender	9.5	Methodical	22.1

Note: INTP portion of the global sample, $n = 809$. Facet with highest percentage of out-of-preference scores in each preference pair is in bold type. Average number of out-of-preference facet scores = 1.97. Mean polarity index = 57.32.

Table 6.15 | Percentage of out-of-preference facet scores for ESFJs in the global sample

ESFJ							
E-I	%	S-N	%	T-F	%	J-P	%
Receiving	2.0	Abstract	4.3	Logical	2.8	Casual	5.4
Contained	3.9	Imaginative	8.7	Reasonable	5.3	Open-Ended	4.1
Intimate	15.6	Conceptual	9.4	Questioning	5.5	Pressure-Prompted	7.3
Reflective	2.8	Theoretical	9.1	Critical	5.2	Spontaneous	0.3
Quiet	9.5	Original	3.9	Tough	4.9	Emergent	13.7

Note: ESFJ portion of the global sample, $n = 951$. Facet with highest percentage of out-of-preference scores in each preference pair is in bold type. Average number of out-of-preference facet scores = 1.24. Mean polarity index = 59.94.

Table 6.16 | Percentage of out-of-preference facet scores for ENFJs in the global sample

ENFJ							
E-I	%	S-N	%	T-F	%	J-P	%
Receiving	1.9	Concrete	2.5	Logical	2.5	Casual	17.5
Contained	5.3	Realistic	3.6	Reasonable	4.4	Open-Ended	3.9
Intimate	15.8	Practical	6.6	Questioning	8.9	Pressure-Prompted	11.1
Reflective	3.9	Experiential	22.7	Critical	4.2	Spontaneous	0.6
Quiet	6.1	Traditional	10.5	Tough	3.9	Emergent	9.4

Note: ENFJ portion of the global sample, $n = 309$. Facet with highest percentage of out-of-preference scores in each preference pair is in bold type. Average number of out-of-preference facet scores = 1.45. Mean polarity index = 61.02.

Table 6.17 | Percentage of out-of-preference facet scores for ISFPs in the global sample

ISFP							
E-I	%	S-N	%	T-F	%	J-P	%
Initiating	5.2	Abstract	8.3	Logical	2.9	Systematic	10.0
Expressive	11.7	Imaginative	9.0	Reasonable	5.2	Planful	6.7
Gregarious	11.7	Conceptual	9.0	Questioning	4.5	Early Starting	35.9
Active	7.6	Theoretical	11.7	Critical	10.3	Scheduled	3.3
Enthusiastic	10.5	Original	8.4	Tough	5.7	Methodical	17.5

Note: ISFP portion of the global sample, $n = 1,106$. Facet with highest percentage of out-of-preference scores in each preference pair is in bold type. Average number of out-of-preference facet scores = 1.95. Mean polarity index = 53.25.

Table 6.18 | Percentage of out-of-preference facet scores for INFPs in the global sample

INFP							
E-I	%	S-N	%	T-F	%	J-P	%
Initiating	4.8	Concrete	1.5	Logical	2.1	Systematic	2.3
Expressive	10.4	Realistic	2.9	Reasonable	2.3	Planful	7.3
Gregarious	9.3	Practical	5.9	Questioning	7.8	Early Starting	30.6
Active	5.2	Experiential	18.1	Critical	7.0	Scheduled	1.9
Enthusiastic	16.2	Traditional	3.9	Tough	5.1	Methodical	20.4

Note: INFP portion of the global sample, $n = 1,065$. Facet with highest percentage of out-of-preference scores in each preference pair is in bold type. Average number of out-of-preference facet scores = 1.65. Mean polarity index = 61.13.

Table 6.19 | Percentage of out-of-preference facet scores for ESTPs in the global sample

ESTP							
E-I	%	S-N	%	T-F	%	J-P	%
Receiving	2.7	Abstract	5.0	Empathetic	3.6	Systematic	15.0
Contained	9.5	Imaginative	8.2	Compassionate	2.1	Planful	7.5
Intimate	12.8	Conceptual	14.5	Accommodating	13.5	Early Starting	26.3
Reflective	3.0	Theoretical	13.7	Accepting	17.9	Scheduled	1.8
Quiet	8.7	Original	17.1	Tender	7.6	Methodical	16.3

Note: ESTP portion of the global sample, $n = 1,028$. Facet with highest percentage of out-of-preference scores in each preference pair is in bold type. Average number of out-of-preference facet scores = 2.07. Mean polarity index = 51.09.

Table 6.20 | Percentage of out-of-preference facet scores for ESFPs in the global sample

ESFP							
E-I	%	S-N	%	T-F	%	J-P	%
Receiving	1.8	Abstract	7.2	Logical	1.7	Systematic	7.0
Contained	5.4	Imaginative	9.4	Reasonable	3.4	Planful	7.8
Intimate	11.3	Conceptual	8.7	Questioning	5.1	Early Starting	34.6
Reflective	3.8	Theoretical	8.6	Critical	6.1	Scheduled	1.5
Quiet	7.1	Original	10.4	Tough	5.7	Methodical	18.5

Note: ESFP portion of the global sample, $n = 1,001$. Facet with highest percentage of out-of-preference scores in each preference pair is in bold type. Average number of out-of-preference facet scores = 1.65. Mean polarity index = 53.61.

Table 6.21 | Percentage of out-of-preference facet scores for ISTJs in the global sample

ISTJ							
E-I	%	S-N	%	T-F	%	J-P	%
Initiating	5.7	Abstract	2.5	Empathetic	1.6	Casual	1.3
Expressive	5.7	Imaginative	2.5	Compassionate	1.9	Open-Ended	4.5
Gregarious	5.9	Conceptual	9.5	Accommodating	13.9	Pressure-Prompted	11.3
Active	6.0	Theoretical	8.8	Accepting	14.6	Spontaneous	0.3
Enthusiastic	5.1	Original	5.7	Tender	9.1	Emergent	15.7

Note: ISTJ portion of the global sample, $n = 2,671$. Facet with highest percentage of out-of-preference scores in each preference pair is in bold type. Average number of out-of-preference facet scores = 1.32. Mean polarity index = 65.17.

Table 6.22 | Percentage of out-of-preference facet scores for ISFJs in the global sample

ISFJ							
E-I	%	S-N	%	T-F	%	J-P	%
Initiating	4.8	Abstract	4.9	Logical	4.1	Casual	3.3
Expressive	8.7	Imaginative	5.5	Reasonable	3.3	Open-Ended	3.7
Gregarious	6.7	Conceptual	7.5	Questioning	4.1	Pressure-Prompted	8.9
Active	4.7	Theoretical	9.0	Critical	6.6	Spontaneous	0.2
Enthusiastic	7.4	Original	2.6	Tough	3.4	Emergent	18.3

Note: ISFJ portion of the global sample, $n = 1,404$. Facet with highest percentage of out-of-preference scores in each preference pair is in bold type. Average number of out-of-preference facet scores = 1.18. Mean polarity index = 61.51.

Table 6.23 | Percentage of out-of-preference facet scores for ENTPs in the global sample

ENTP							
E-I	%	S-N	%	T-F	%	J-P	%
Receiving	1.4	Concrete	3.6	Empathetic	4.3	Systematic	6.0
Contained	9.4	Realistic	4.7	Compassionate	3.9	Planful	6.8
Intimate	12.5	Practical	6.1	Accommodating	9.3	Early Starting	25.2
Reflective	5.7	Experiential	22.9	Accepting	23.4	Scheduled	1.4
Quiet	4.7	Traditional	2.9	Tender	8.9	Methodical	18.3

Note: ENTP portion of the global sample, $n = 722$. Facet with highest percentage of out-of-preference scores in each preference pair is in bold type. Average number of out-of-preference facet scores = 1.81. Mean polarity index = 57.62.

Table 6.24 | Percentage of out-of-preference facet scores for ENFPs in the global sample

ENFP							
E-I	%	S-N	%	T-F	%	J-P	%
Receiving	2.0	Concrete	1.8	Logical	1.2	Systematic	1.7
Contained	6.7	Realistic	2.2	Reasonable	2.4	Planful	5.4
Intimate	12.3	Practical	6.4	Questioning	7.6	Early Starting	28.9
Reflective	3.6	Experiential	21.2	Critical	3.6	Scheduled	0.9
Quiet	3.0	Traditional	2.8	Tough	4.7	Methodical	17.6

Note: ENFP portion of the global sample, $n = 1,370$. Facet with highest percentage of out-of-preference scores in each preference pair is in bold type. Average number of out-of-preference facet scores = 1.36. Mean polarity index = 63.57.

Table 6.25 | Percentage of out-of-preference facet scores for INTJs in the global sample

INTJ							
E-I	%	S-N	%	T-F	%	J-P	%
Initiating	6.3	Concrete	5.8	Empathetic	1.6	Casual	3.5
Expressive	4.4	Realistic	6.1	Compassionate	2.1	Open-Ended	4.7
Gregarious	7.5	Practical	6.5	Accommodating	9.6	Pressure-Prompted	14.2
Active	5.6	Experiential	15.9	Accepting	20.7	Spontaneous	0.0
Enthusiastic	8.2	Traditional	3.7	Tender	9.3	Emergent	11.7

Note: INTJ portion of the global sample, $n = 429$. Facet with highest percentage of out-of-preference scores in each preference pair is in bold type. Average number of out-of-preference facet scores = 1.47. Mean polarity index = 60.31.

Table 6.26 | Percentage of out-of-preference facet scores for INFJs in the global sample

INFJ							
E-I	%	S-N	%	T-F	%	J-P	%
Initiating	3.9	Concrete	1.3	Logical	3.6	Casual	13.2
Expressive	9.6	Realistic	4.4	Reasonable	1.8	Open-Ended	3.4
Gregarious	6.7	Practical	5.2	Questioning	5.4	Pressure-Prompted	11.4
Active	8.8	Experiential	18.9	Critical	5.9	Spontaneous	0.5
Enthusiastic	10.9	Traditional	7.5	Tough	2.6	Emergent	11.4

Note: INFJ portion of the global sample, $n = 387$. Facet with highest percentage of out-of-preference scores in each preference pair is in bold type. Average number of out-of-preference facet scores = 1.36. Mean polarity index = 60.62.

DOMINANT INTROVERTED FEELING TYPES (ISFP AND INFP)

TABLES 6.17 AND 6.18

The mean polarity index for ISFPs is considerably lower than that for INFPs, while the average number of out-of-preference scores is higher.

Among the E–I facets, Expressive (Expressive–Contained facet) and Gregarious (Gregarious–Intimate facet) are tied for highest percentage of out-of-preference scores for ISFPs, while INFPs’ highest percentage is on Enthusiastic (Enthusiastic–Quiet facet). Among the S–N facets, ISFPs’ highest percentage is on Theoretical (Experiential–Theoretical facet), while INFPs’ is on the opposite pole of the same facet, Experiential. Among the T–F facets, ISFPs’ highest percentage is on Critical (Critical–Accepting facet), while INFPs’ is on Questioning (Questioning–Accommodating facet). Among the J–P facets, Early Starting (Early Starting–Pressure–Prompted facet) has by far the highest percentage of out-of-preference scores for both ISFPs and INFPs.

DOMINANT EXTRAVERTED SENSING TYPES (ESTP AND ESFP)

TABLES 6.19 AND 6.20

The mean polarity index is slightly higher for ESFPs than for ESTPs, while the average number of out-of-preference scores is higher for ESTPs.

Among the E–I facets, both ESTPs and ESFPs have their highest percentage of out-of-preference scores on Intimate (Gregarious–Intimate facet), with ESTPs slightly higher. Among the S–N facets, both types have their highest percentage of out-of-preference scores on Original (Traditional–Original facet), with ESTPs nearly seven points higher on average. Among the T–F facets, ESTPs’ highest percentage is on Accepting (Critical–Accepting facet), while ESFPs have a somewhat low percentage of out-of-preference scores on all the Thinking poles, Critical being the highest at just 6.1%. Among the J–P facets, both types have their highest percentage on Early Starting (Early Starting–Pressure–Prompted facet), with ESFPs about eight points higher than ESTPs.

DOMINANT INTROVERTED SENSING TYPES (ISTJ AND ISFJ)

TABLES 6.21 AND 6.22

The mean polarity index for ISTJs is somewhat higher than that for ISFJs, as is the average number of out-of-preference scores.

Among the E–I facets, ISTJs have their highest percentage of out-of-preference scores on Active (Active–Reflective facet), followed closely by Gregarious (Gregarious–Intimate facet), while ISFJs have their highest percentage on Expressive (Expressive–Contained

facet). Among the S–N facets, ISTJs’ highest percentage is on Conceptual (Practical–Conceptual facet), while ISFJs’ is on Theoretical (Experiential–Theoretical facet). Among the T–F facets, ISTJs’ highest percentage is on Accepting (Critical–Accepting facet), while ISFJs’ is on the opposite pole of that facet, Critical. Among the J–P facets, Emergent (Methodical–Emergent facet) is highest for both types, though somewhat higher for ISFJs.

DOMINANT EXTRAVERTED INTUITIVE TYPES (ENTP AND ENFP)

TABLES 6.23 AND 6.24

The mean polarity index for ENFPs is higher than for ENTPs, but the average number of out-of-preference scores is higher for ENTPs.

Among the E–I facets, Intimate (Gregarious–Intimate facet) has the highest out-of-preference scores for both ENTPs and ENFPs. Among the S–N facets, Experiential (Experiential–Theoretical facet) is highest for both types. Among the T–F facets, ENTPs’ highest percentage is on Accepting (Critical–Accepting facet), while ENFPs’ is on Questioning (Questioning–Accommodating facet). Among the J–P facets, both types have their highest percentage of out-of-preference scores on Early Starting (Early Starting–Pressure–Prompted facet).

DOMINANT INTROVERTED INTUITIVE TYPES (INTJ AND INFJ)

TABLES 6.25 AND 6.26

The polarity indexes for INTJs and INFJs are about the same, while the average number of out-of-preference scores is slightly higher for INTJs.

Among the E–I facets, both INTJs and INFJs have their highest percentage of out-of-preference scores on Enthusiastic (Enthusiastic–Quiet facet), with INFJs nearly three points higher. Among the S–N facets, both types have their highest percentage on Experiential (Experiential–Theoretical facet), again with INFJs three points higher. Among the T–F facets, INTJs’ highest percentage (20.7%) is on Accepting (Critical–Accepting facet), while INFJs’ is on the opposite pole, Critical, though it is much lower (5.9%). The percentages of out-of-preference scores for these two types show the greatest difference on the J–P facets; INTJs’ highest percentage is on Pressure–Prompted (Early Starting–Pressure–Prompted facet), while INFJs have about the same percentage of out-of-preference scores on Casual (Systematic–Casual facet).

CONCLUSION

The administration and interpretation of the MBTI Step I and Step II assessments are linked enterprises. By following administration guidelines, practitioners increase the likelihood that respondents will have a positive and insightful interpretation experience. Deviating from administration guidelines makes the task of interpretation more difficult and less fulfilling for both respondent and practitioner. Adhering to those guidelines will pay off in more candid responses, which in turn will

minimize the time necessary for “correcting” type or facet results that do not fit. If possible, providing a way for respondents to have their questions answered prior to taking the assessment is desirable as well. In conjunction with the individual practitioner’s interpretation experience, the information in this chapter offers respondents a range of insights into their individuality. This will help them become aware of their similarities to and differences from individuals of their own type as well as those of other types.



PART TWO

Construction and Research

CHAPTER 7

Construction of the MBTI® Global Assessments

CHAPTER OVERVIEW

Priorities of the Step I™ and Step II™ Assessment Revisions

Phase 1: Construct and Translate a Global Research Version of the Assessment

Phase 2: Administer the Global Research Version of the Assessment to Individual Samples and Collect Data

Phase 3: Select Global Step I™ Items and Determine Scoring Method

Phase 4: Select Global Step II™ Items and Determine Scoring Method

Global Sample Type Summaries

Conclusion

This chapter provides an overview of the construction of the MBTI® Global Step I™ and Global Step II™ assessments. First, the project goals and priorities for the global assessments are explained. Second, the phases of the research are delineated. The phases include the construction and translation of a global research version of the MBTI assessment, the administration of the global research version to individual samples and the collection of data, and the selection of items and scoring methods for the Global Step I and Step II assessments. It is important to note at the outset that it was planned to revise both assessments from a single research initiative using a common research form and sample. As such, while the Global Step I and Step II assessments were developed using different statistical approaches and criteria, much in the development process was shared. The first two phases, construction/translation and administration/data collection, apply to both assessments. The selection of items for each assessment, on the other hand, was conducted using the same sample and item pools but followed different paths. This chapter also includes discussion of how best-fit type research was used to finalize the scoring methods. Additionally, type tables of the MBTI global sample are presented.

PRIORITIES OF THE STEP I™ AND STEP II™ ASSESSMENT REVISIONS

As steward of the MBTI assessment, CPP, Inc. (now The Myers-Briggs Company) had two overarching goals in undertaking the revision of the assessment to create global forms: (1) preserve the integrity of the Step I and Step II forms of the assessment and (2) improve the reliability and validity of the assessment overall. More specifically, the company sought to update existing representative samples and compile new representative samples in additional countries based on translations (or *adaptations*)¹ of the assessment into additional languages, use a statistical model consistent with type theory, and, if supported by data analysis, use the same scoring method globally, so that scores could be compared across all those countries and languages.

Broadening and compiling new representative samples was a high priority. The prior revision of the MBTI assessment culminated in 1998 in the publication of MBTI Form M, which replaced the earlier Form G. Form Q was subsequently published in 2001 and replaced Form K. In the United Kingdom, the European Step I assessment was published in 1997. The European Step II assessment was published in 2003 based on pan-European samples compiled by OPP Ltd. Although all these assessments have served their audiences well, no additional representative samples in the United States or the United Kingdom had been compiled subsequent to their publication. It was therefore important to update those samples, as well as to compile a number of new representative samples in additional countries based on translations of the assessment into additional languages to reflect the increasingly global reach of the MBTI assessment. From the outset, larger representative samples were planned for the United States, the United Kingdom, Canada, and Australia, and smaller representative samples for other countries or regions. Convenience samples, compiled by distributors, did not have targets to ensure representation. The goal for those samples was to have them be as large as possible.

(Note: All individual samples making up the MBTI global sample were created based on participants' country of residence or origin, as well as the language in which the assessment was administered. Each sample in this manual will hereafter be referred to as *country/language* sample, as reflected in its name. However, a particular sample may be referred to by either country or language for convenience in a particular context. Please refer as needed to the sample names listed in table 7.1.)

Another priority was to investigate the use of latent class analysis (LCA) for building and scoring the revised Step I assessment. This investigation is detailed in the section "Latent Class Analysis Overview" later in this chapter. Latent class analysis offers a statistical approach that is both powerful and consistent with type theory, in that it presumes that latent categories explain the underlying associations between items.

Achieving greater consistency across translated MBTI assessment versions and scoring was also a high priority. Prior to the release of the Global Step I assessment, the United States, Japan, and Europe had different scoring methods and item weights for the assessments they used. To simplify matters, one of the priorities was to create MBTI translations and revised scoring that were effective across many languages, had as many items in common as possible, had as much common scoring as possible, and produced output that was comparable across languages and cultures. Such an approach offers many advantages to practitioners, as well as to individuals learning about their personality type, who gain the ability to directly compare their results with those of anyone they know or work with around the world.

The global revision of the MBTI assessment offered the opportunity to revise, add, and delete items from the assessments in use. To accomplish the major priorities of the revision (update existing representative samples and collect new representative samples, create consistency across translated versions, and explore the use of latent class analysis as the primary scoring approach for the Global Step I assessment), a systematic process was employed to test and select items that would compose the Global Step I and Step II assessments.

The major phases in the process were as follows:

1. Construct and translate a global research version of the assessment
2. Administer the global research version of the assessment to individual samples and collect data
3. Select Global Step I items and determine scoring method
4. Select Global Step II items and determine scoring method

1. The terms *translation* and *adaptation* are often used interchangeably in the testing and measurement literature. Historically, *translation* has been used to describe the process by which an assessment is converted to a language other than the one in which it was originally constructed. However, the term *adaptation* is increasingly being used to reflect the fact that an effective conversion of assessment items from one language to another often requires not a word-for-word translation but rather a modification intended to maintain the general sense or purpose of those items in a particular language. Nevertheless, as the more readily understood term, *translation* is used here, but it should be read as including adaptation.

PHASE 1: CONSTRUCT AND TRANSLATE A GLOBAL RESEARCH VERSION OF THE ASSESSMENT

One of the priorities of the revision was to select item sets that would enable all respondents around the world to respond to the same items and obtain their MBTI Step I type preferences and whole type, as well as their results on the MBTI Step II facets. To do this, it was necessary to use a global research version that contained enough MBTI assessment items to enable an adequate number of acceptable final items that would be effective across many countries and languages around the world for both the Step I and Step II assessments.

Prior international MBTI research had used a 230-item research version of the MBTI assessment to develop the European Step I and Step II assessments. That research version was used for this project as well. However, existing translations were reviewed and updated as needed, and the items were also translated into additional languages. The global research version contained all the items from the commercially available European Step I and Step II assessments and the commercially available US Form M and Form Q assessments, as well as additional items not used in any of those forms.

Translation Method

In order to ensure high-quality translations of the different forms of the MBTI assessment, CPP used industry-standard assessment translation methods (for a review, see International Test Commission, 2005). In brief, the process involves four steps: (1) search for previous translations, (2) create an initial draft translation, (3) conduct an in-country review, and (4) reconcile all reviewers' comments to arrive at a mutually agreeable translation.

The first step in creating a new translation is to look for prior translations that have been developed. As described above, the 230-item MBTI global research version was used to develop the European Step II assessment, but in other parts of the world, translations of prior forms (Form F, Form G, etc.) were also available. To begin the translation process, these prior translations, which typically used only a subset of the 230-item set, were provided to professional linguists, along with background information on the MBTI assessment and a glossary of key terms.

The second step is to complete an initial draft translation through a modified "double-forward" process. In this process, the initial translation is created by an accredited professional linguist who is a native-language speaker and a similarly accredited native-language speaker who acts as an editor and reviews the draft translation and source text. The linguist and editor then work through the initial translation and come to agreement regarding the translated content. The result of this stage is a first draft of the translated instructions, items, and response options.

The third step is an in-country expert review by one or more fluent native-language speakers who have subject matter expertise with respect to the MBTI framework and type theory. Whenever possible, multiple in-country expert reviewers were used for each language. Each in-country expert reviewed the materials independently and provided his or her review to the linguist and editor.

The fourth step is reconciliation, where the goal is to arrive at an acceptable, mutually agreeable translation. For reconciliation, a serial process was used whereby the in-country reviewers' comments were provided to the linguist and editor, and they, in turn, responded to suggestions and edits. Typically, they accepted some of those but rejected others, and made some minor modifications. The updated version with comments then went back to the in-country reviewers, who made additional recommendations and comments. For each round of edits and responses, all parties were asked to justify their recommendations. This process was repeated until all parties accepted the translation. In some cases, a second approach to reconciliation was used, in which the linguist, editor, and reviewers assembled in the same room or on the telephone to work out differences with the help of a facilitator.

PHASE 2: ADMINISTER THE GLOBAL RESEARCH VERSION OF THE ASSESSMENT TO INDIVIDUAL SAMPLES AND COLLECT DATA

To address the need for updated and new representative samples, a large, targeted data collection effort was undertaken to gather samples. Twenty-three individual samples were compiled and included in the global sample used to develop the Global Step I and Global Step II assessments.² Within the 23 samples, data were collected in 19 languages and in 20 countries.³ Specific sampling targets were set for 20 of the 23 samples to

2. The initial data collection effort also included samples from India (North American English) and Saudi Arabia (Arabic). However, these samples were eliminated from consideration in the global sample due to concerns about the quality of the data provided. The Saudi Arabia sample was from a single company and was overwhelmingly ESTJ, and therefore was not representative or sufficiently diverse. The India sample had a very large percentage of cases whose response patterns indicated inattentive responding to the 230-item global research version, such as all A or all B responses.

3. Data using the global research version of the assessment were also collected in Indonesia (Indonesian), Israel (Hebrew), Japan (Japanese), Poland (Polish), Russia (Russian), South Korea (Korean), Thailand (Thai), and Turkey (Turkish). For a variety of reasons, these data were not included in the global sample. They are, however, addressed in technical briefs for each of the samples (www.themyersbriggs.com).

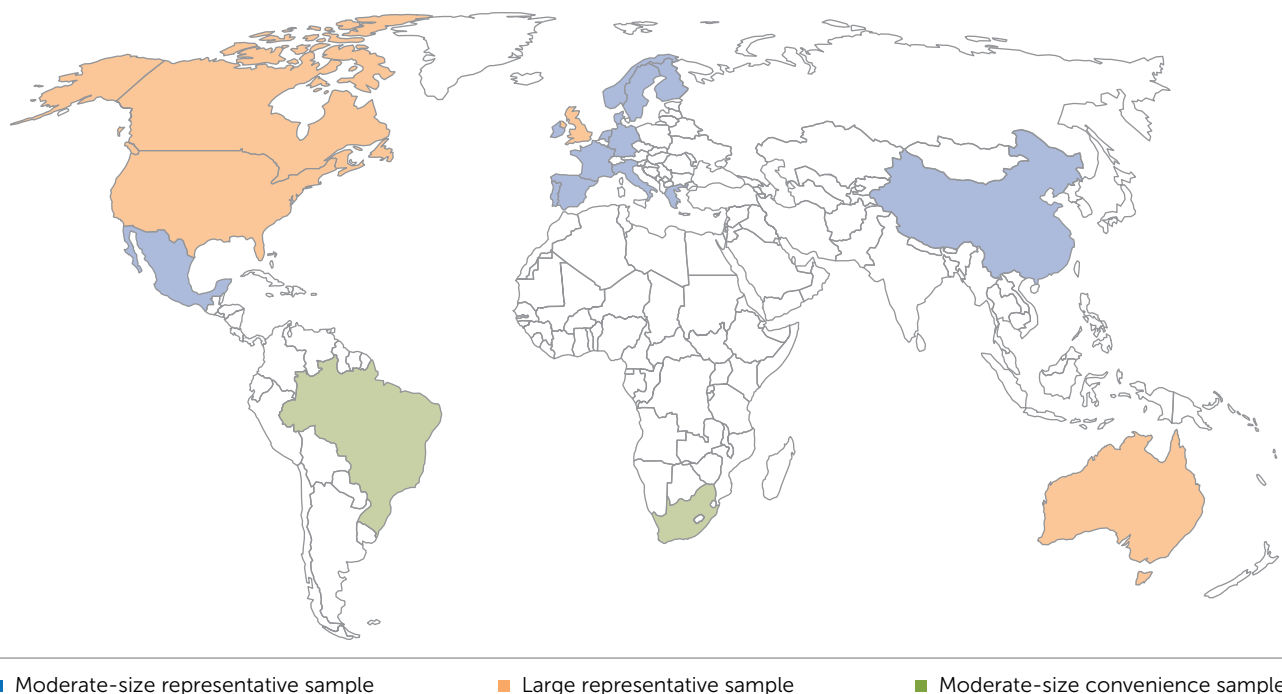


Figure 7.1 | Country/language samples in the MBTI® global sample

create representative samples, while the remaining 3 were convenience samples. In the United States, for instance, CPP sought to represent the US adult population broadly, so targets were set for gender, age, employment status, and ethnicity, among other demographics. These targets were selected based on statistics obtained from a variety of official sources, such as the US Census and Bureau of Labor Statistics. For other samples, MBTI distributors helped select the targets for samples in their respective countries or regions by choosing appropriate official sources of statistics. In general, sampling targets were designed to mirror the working-age population. The *MBTI® Manual Global Supplement Series* (Schaubhut, Thompson, Morris, & Arneson, 2018) details the targets that were set and the actual numbers obtained (www.themyersbriggs.com/mbti-manual-supplements). Figure 7.1 shows the geographic location of each sample.

Data were collected exclusively online through two websites. The first site was for participants who primarily read and understand a language other than English. The site was multilingual, and participants were directed to the appropriate language based on the results of a prescreening survey conducted by a third-party provider. Participants could change the language of the site, but only at the outset of data collection. The site hosted the demographic items and the 230-item MBTI global research version (GRV), along with any other translated validity assessments that were used. The second website was a special modification of MBTI®Complete created for this research initiative, and it was accessed by

participants who read and understand English. It used the 230-item MBTI GRV, followed by MBTI®Complete's online interpretation session (known as the interactive learning session), yielding respondents' best-fit type results. The primary differences between the two sites were the inclusion of the interpretation session and the gathering of best-fit type data from participants who read and understand English.

MBTI®Complete was modified in the following way: US, Australian, Canadian, and South African respondents who read and understand North American English were administered the 230-item MBTI GRV (presented in North American English) rather than the 93-item Form M. When respondents completed the assessment, the 93 items of Form M were scored and the interpretation and verification process proceeded normally (additional manipulations of the scoring were used in researching the "prompting effect"; see the section "Best-Fit Type Analysis" later in this chapter). This procedure allowed the collection of demographic data, GRV results, and reported and best-fit type data. For European respondents who read and understand European English, a similar procedure was followed, except that the MBTI GRV was presented in European English. For administration of additional assessments for validity purposes, respondents who completed the MBTI GRV on the MBTI®Complete site were then directed to the first site.

Once the websites were prepared and the sampling targets were set, data collection began. The majority of sample participants were provided incentives by an

external market research company. Such companies maintain panels of participants who have expressed willingness to be contacted for research purposes. A separate panel was used for each representative country/language sample obtained. As a result, the new panel members could be solicited in the language spoken by established members of that panel, and it was possible to identify the country/language background of each panel member. These participants were paid to complete some combination of demographic surveys, the MBTI GRV, retests, and other assessments used for validity studies.

The process typically involved inviting panel members to complete a short prescreening survey, in which they responded to basic demographic items, such as gender and age. If a panel member was not part of any excluded group (e.g., younger than 18 years old in the United States) or groups for whom data collection had already been completed (e.g., the target of females with full-time employment in the United States had been reached), he or she was then asked to complete the assessment's battery of items on one of the two websites being used for data collection. Additional evaluations of the samples against the targets were applied later.⁴ For the sample compiled in Brazil (Brazilian Portuguese) and the two samples compiled in South Africa (one administered the assessment in North American English and the other in Afrikaans), locally based MBTI distributors led the data collection effort by directing respondents to the appropriate site, resulting in convenience samples. These efforts are described in each sample's supplement found in the *MBTI® Manual Global Supplement Series* (www.themyersbriggs.com/mbti-manual-supplements).

Data Integrity

Once data were collected, all cases were thoroughly examined and invalid cases were removed. This cleaning step, while reducing final sample sizes, was required to ensure that only the highest-quality data remained for analysis. Cases were removed for several reasons, with criteria including the following:

Duplicates. Duplicate cases were identified first by checking for identical first and last names. Other demographics were then compared to determine whether cases were truly duplicates. If duplicate cases were identified, those already flagged for exclusion for other reasons were removed. If identified cases were not flagged for exclusion for other reasons, the case completed last was kept.

Age or gender nonresponse. Responses to these key demographic items were considered a requirement for all participants.

Language. The external market research company solicited participants from panels it had established in the countries identified for inclusion in the research. Therefore, there was an objective indicator of the country from which the sample was drawn. However, respondents were asked to self-report their country of origin and residence. For individual samples, cases were excluded if (1) the country of origin and the country of residence indicated were mismatched given the linguistic and cultural history of the sample, based on a series of necessarily subjective judgments; (2) if the country of residence was missing and the country of origin did not align with the assessment language; or (3) if the country of residence did not align with the assessment language (regardless of country of origin). For example, all US sample cases resided in the United States, while the Germany sample included participants who resided in Germany and Switzerland. Details about which countries were represented in each sample are provided in the sample's corresponding supplement found in the *MBTI® Manual Global Supplement Series* (www.themyersbriggs.com/mbti-manual-supplements).

Nonidentical responses to identical items on prescreening survey and CPP demographic items.

Selected basic items (e.g., gender) were identical on the prescreening survey and in the CPP website's demographic section. If responses to these items were provided in both places, they were required to be identical (e.g., "male" on the prescreening survey and "male" in the CPP website's demographic section). If they were not identical, the case was excluded. If one of the matching items was skipped on either the prescreening survey or the CPP website, the case was not excluded. This check was done for gender, age, and, for some samples, employment status. For age, a one-year difference was tolerated to allow for the possibility of a slight participant error or time elapsing between completion of the prescreening survey and the CPP demographic items.

Best-fit type missing. All respondents who completed the GRV in North American English or European English did so through the modified MBTI®Complete website described previously. Cases were excluded in which the respondent did not complete the interpretation session and thus did not select a best-fit type.

Suspicious response patterns. Cases were excluded if more than 12 items from the 230-item MBTI GRV were omitted (i.e., not responded to). This allowed for approximately 5% of MBTI items to be skipped. Additionally, cases were excluded if response option A, or response option B, was selected more than 200 times on the 230-item MBTI GRV.

4. More than 8,000 cases collected were later excluded from the global sample for a variety of reasons, primarily related to adjustments to the sampling targets.

Global Sample Composition

Once data quality and integrity had been established, data from all country/language samples were combined into the global sample of 16,773 participants. Table 7.1 summarizes the country/language composition of the global sample. The global sample includes 9,002 women and 7,771 men. At the time of data collection, the average age of the sample was 41 ($SD = 15$ years, with ages ranging from 15 to 88 years). Over half the members of the sample (54%) were employed in a full-time job, and 15% reported being employed part-time. Full-time students made up 8% of the sample, with the remainder retired (10%), not working for income (6%), or not responding to the item (7%). Further details for each of the samples in table 7.1 can be found in the respective sample supplements in the *MBTI® Manual Global Supplement Series* (www.themyersbriggs.com/mbti-manual-supplements). The global forms of the assessment were developed using this global sample.

Different sample sizes were desired at the outset of the data collection effort. “Large” samples were targeted to

include 1,000 or more participants, to exceed the sample size of existing representative samples (specifically, in the United States and the United Kingdom), and to reflect the size of the market for the MBTI assessment. As shown in table 7.1, large samples were compiled in Australia (North American English), Canada (North American English), the United Kingdom (European English), and the United States (North American English). “Moderate-size” samples were compiled in areas where use of the MBTI assessment was not as prevalent as in the aforementioned four countries and where no then-current representative sample had been collected. All but 3 of the 19 moderate-size samples had targets to ensure the representativeness of the sample obtained.

To provide validity and test-retest data, subsets of participants from most samples completed one or more additional assessments. Specifically, for validity data the CPI 260® assessment or the *Adjective Check List* (ACL) was administered if available in the local language. For test-retest evaluation, the global research version of the MBTI assessment was administered a second time after a target interval of 1 to 6 weeks (some respondents took up to 15 weeks). Respondents were allowed to complete more than one follow-up if they wished. A total of 69 subsamples, summarized in table 7.2, were collected across all languages represented in the global sample. This highly consistent data collection effort yielded samples that shared responses to a common MBTI global research form, demographic items, and other validation assessments.

Table 7.1 | Country/language samples in the MBTI® global sample

Country/language sample	N
Large samples	
Australia (North American English)	776
Canada (North American English)	939
United Kingdom (European English)	2,831
United States (North American English)	3,578
Moderate-size samples	
Brazil (Brazilian Portuguese)*	839
Canada (Canadian French)	176
China (Simplified Chinese)	521
China (Traditional Chinese)	477
Denmark (Danish)	468
Finland (Finnish)	524
France (European French)	472
Germany (German)†	440
Greece (Greek)	277
Ireland (European English)	383
Italy (Italian)	458
Mexico (Latin American Spanish)	359
Netherlands (Dutch)	506
Norway (Norwegian)	493
Portugal (European Portuguese)	503
South Africa (Afrikaans)*	505
South Africa (North American English)*	189
Spain (European Spanish)	564
Sweden (Swedish)	495

Note: Global sample, $N = 16,773$.

*Data collection for this sample was distributor led; it is not a representative sample.

†Germany sample includes one individual residing in Switzerland.

PHASE 3: SELECT GLOBAL STEP I™ ITEMS AND DETERMINE SCORING METHOD

Item selection for the Global Step I and Step II assessments had much in common but also differed in some respects. The most notable difference is that latent class analysis (LCA) was used as the primary statistical approach for the Global Step I assessment, while item response theory (IRT) was used as the primary statistical approach for the Global Step II assessment. This section details the item analysis and selection process for the Global Step I form. It was necessary to develop the Global Step I assessment first so that (1) those items could be considered for use in developing the Global Step II facet measures; (2) the preferences could be used in identifying the item content of the Global Step II facet scales; and (3) in-preference and out-of-preference percentages could be examined (calculation of such rates cannot be done without the preferences).

Latent Class Analysis Overview

Methods of scoring the MBTI assessment have varied over the decades and, more recently, in different regions of the world. For example, MBTI Form M and Form Q,

Table 7.2 | Overview of the data collection process

Country/language sample	Demographic survey and MBTI® GRV data	MBTI® GRV retest data	ACL data	CPI 260® data
Large samples				
Australia (North American English)	•	•	•	•
Canada (North American English)	•	•	•	•
United Kingdom (European English)	•	•	•	•
United States (North American English)	•	•	•	•
Moderate-size samples				
Brazil (Brazilian Portuguese)	•			
Canada (Canadian French)	•		•	
China (Simplified Chinese)	•	•	•	•
China (Traditional Chinese)	•	•	•	
Denmark (Danish)	•	•	•	•
Finland (Finnish)	•	•		
France (European French)	•	•	•	•
Germany (German)	•	•	•	•
Greece (Greek)	•	•		
Ireland (European English)	•		•	•
Italy (Italian)	•	•		•
Mexico (Latin American Spanish)	•		•	•
Netherlands (Dutch)	•	•	•	•
Norway (Norwegian)	•	•		•
Portugal (European Portuguese)	•	•		
South Africa (Afrikaans)	•			
South Africa (North American English)	•			
Spain (European Spanish)	•	•	•	•
Sweden (Swedish)	•	•	•	•

Note: Global sample, $N = 16,773$.

which are used in the United States and Japan, are scored using IRT-based methods, while the European Step I and Step II assessments use prediction ratios and IRT to select items but use integer weights for scoring. Since the publication of Form M, there has been some discussion about latent class analysis and its potential usefulness for scoring the MBTI assessment (Morris, 2012), especially after LCA was used successfully on another personality type indicator, the *Murphy-Meisgeier Type Indicator for Children®* (MMTIC®) assessment, a Jungian type indicator for use with children (Murphy & Meisgeier, 2008). Because LCA may be unfamiliar to many readers, some of its advantages and key terminology are described next. For a general introduction to LCA, see Magidson & Vermunt (2004). The use of latent class analysis for the development of the Global Step I assessment is described in detail following this overview of LCA.

The key advantage of latent class analysis is that it assumes that people belong to one of some number of groups, or classes, a view that is consistent with MBTI type theory (Magidson, 1997). Unlike many current models of personality that assume a trait that is distributed along a continuum, the MBTI framework,

based on the Jung-Myers theory, assumes four categorical (dichotomous) latent factors, one associated with each of the preference pairs (E–I, S–N, T–F, and J–P). The purpose of the MBTI assessment is “to sort people into equally valuable groups to which . . . they already belong” (Myers et al., 1998), for example, by identifying individuals who prefer Extraversion and those who prefer Introversion (E–I). Latent class analysis is the natural means of accomplishing this goal, as it is designed to classify individuals based on their item responses into that particular latent category into which they best fit. For example, the E–I factor consists of two latent categories, called latent classes: one associated with people who prefer Extraversion and the other associated with people who prefer Introversion. Based on their pattern of responses on the E–I items, individuals are classified into that latent class (E class or I class) for which the *posterior membership probability* (PMP) is highest. Posterior membership probabilities are computed using Bayes’ theorem (Magidson & Vermunt, 2004).

The key advantage of latent class analysis is that it assumes that people belong to one of some number of groups, or classes, a view that is consistent with MBTI type theory.

Although IRT was used successfully with previous MBTI and other personality assessments, it is designed to measure an underlying continuous latent trait. LCA, in contrast to IRT, postulates a categorical latent variable to explain the associations between item responses. Latent class analysis offered the promise of a statistical approach that is both consistent with MBTI theory and powerful enough to provide accurate classifications. As Magidson (1997, p. 40) stated,

If, on the one hand, the correlations among the items are due to one or more underlying latent *continuous* variables, then the appropriate tools should be selected from item response theory and a latent trait model should be employed. If, on the other hand, the correlations among the items in the MBTI are due to the Jungian typology, which postulates latent dichotomies, then the appropriate scoring tools should come from latent class analysis.

The shift to LCA requires nothing new of MBTI respondents. Practitioners and researchers, however, are advised to learn some of the basic changes in terminology associated with the Global Step I assessment. Particularly important are the continuous score based on log odds called *logits*, which replaces the theta from IRT, and the *probability index*, which replaces the preference clarity index (pci) used to report scores to respondents for Form M. These are explained below.

While the latent class model is a multiplicative model, posterior membership probabilities can be transformed to odds, and then the logarithm of the odds (the logit) can be computed to achieve a continuous score expressed as a simple weighted sum of the item responses where the item response scores are multiplied by the associated item weights. For example, suppose a person's probability of having a preference for E turns out to be .8. This means that his or her probability of having a preference for I is .2 ($1.0 - .8 = .2$), and the odds of having a preference for E are thus $.8/.2 = 4$. That is, this person is four times more likely to have a preference for E than for I.

RELATIONSHIP BETWEEN PROBABILITIES AND LOGITS

Latent class models can be used to classify individuals into the most likely category (e.g., E or I) based on an equation that is very similar to a linear regression. When scale scores are computed using latent class analysis, simple item weights are applied to the raw item responses and summed along with an intercept term. The

continuous score that results is the logit and is analogous to a theta score in IRT. A logit for, say, Extraversion can be transformed through a simple formula into a probability of having a preference for E. If researchers or others are interested in using continuous scores in their research on the MBTI preference scales, it is the logits or corresponding probabilities that they should use.⁵

The logits on the MBTI Global Step I assessment are set such that 0 is the midpoint, or *cut point*, for each preference. (A logit of 0 corresponds to a probability of .5.) Logit scores below 0 are assigned to one preference (e.g., E), while scores above 0, corresponding to probabilities greater than .5, are assigned to the other preference (e.g., I). Logits can also be transformed into posterior membership probabilities through a simple formula, which then provides an intuitive estimate of the likelihood of a correct classification. For practitioners, this is an important change, because these probabilities are used to replace the previously used preference clarity index. Instead of explaining that some people have differing levels of clarity on a 1- to 30-point scale, respondents will have probability estimates of the likelihood of a correct classification for each preference. Posterior membership probabilities were adjusted further after the final items were selected, a process that is described below.

Another advantage of LCA is that an extended LCA model is available to ensure that the latent preference pairs measured are uniform across all countries (Kankaras, Moors, & Vermunt, 2010). These extended models allow for differential levels of clarity of preference for different respondents.

Evaluation of the Latent Class Analysis Approach to Item Selection and Scoring

Once the data were collected and evaluated and the global sample was created, the development of the MBTI Global Step I form was undertaken. As an initial step, analyses were conducted using a subsample from the larger global sample. Specifically, the samples from the United States, the United Kingdom, Canada (North American English), and Australia were combined and evaluated. This intermediate step was chosen to make the analysis more manageable, and because if cross-sample consistency could not be found in these more culturally similar samples, it was unlikely that it would be found when analyzing the vastly more diverse global sample.

Because prior MBTI assessments had been developed using both traditional approaches (e.g., prediction ratios) and IRT, and because latent class analysis was to be evaluated for this revision, a three-pronged analysis approach was undertaken in order to determine whether

5. To correlate the continuous score with another variable, using the logits is recommended. When preparing histograms to examine the distribution, using probability index scores is recommended.

latent class analysis would work for the MBTI assessment, and if it would lead to conclusions similar to those that resulted from using prior approaches. Specifically, the large representative sample data were analyzed by three independent researchers, with each using a different statistical approach. All three analysts independently used the same data and initial item pools to develop MBTI Global Step I measures under very broad direction. Each analyst was instructed to develop a measure of the four preference pairs with no more than 100 total items in his or her recommended assessment. One used a traditional scale construction approach, including analyses associated with classical test theory as well as some analyses (e.g., prediction ratios) that have been widely used in previous MBTI research. The second used item response theory (IRT), which underlies the MBTI Form M, Form Q, and European Step II assessments. The third used latent class analysis (LCA). The items selected using the traditional, IRT, and LCA approaches overlapped substantially. Almost without exception, items that were judged to be high quality under the LCA perspective were also judged to be high quality under the traditional and IRT perspectives (Thompson, 2014).

Based on the initial analyses and the consistency of the identified experimental measures of the MBTI preference pairs across the three statistical approaches, the next step was to complete the same analyses with the entire global sample. Again, it was found that similar items were identified for each of the preferences across the three approaches. Given this consistency, it was decided that further analyses would focus on latent class, for the reasons mentioned previously, but the final item selection would still draw on information from the traditional and IRT approaches when appropriate (e.g., considering traditional measures such as Cronbach's alpha and IRT information such as item characteristics).

Extension of the Measurement Model

Having determined that latent class analysis was the appropriate approach for the development of the Global Step I assessment, the next step was to examine whether an extension of the standard LCA model was necessary to achieve measurement equivalence for the MBTI assessment across countries and languages. These investigations were undertaken because statistical analysis of the response patterns for each of the four preference pair measures demonstrated two clear sources of difference or heterogeneity. First, differences existed between individuals whose scores were on one side of the preference pair or the other. This is already reflected in the one-factor model with two latent classes for each preference scale. However, additional heterogeneity was found to exist for each response

pattern associated with the level of consistency exhibited by individuals on either side of the preference pair.

To account for both sources of heterogeneity, the fit of the standard LCA model was compared to that of an extended two-factor model. This two-factor model consisted of the dichotomous factor representing the desired theoretical construct (E–I, S–N, T–F, J–P) along with a second categorical latent variable identifying respondents exhibiting more or less consistency in their responses. In LCA terms, these are called *latent scale classes*. Note that the latent categories believed to underlie these scale classes are people with different degrees of clarity about their preferences. Those who are very clear are believed to respond more consistently and have a greater likelihood of having accurate results. The results of this analysis suggested that in addition to the dichotomous factor representing the preferences, five scale classes were needed to achieve measurement equivalence across the global samples for the Global Step I assessment. For example, scale classes for the T–F preference pair were labeled as follows:

1. **Low consistency**—respondents selected a similar number of responses for T and F
2. **Moderate consistency**—respondents selected a preponderance of responses in one direction
3. **Very high consistency T**—respondents selected nearly all T responses
4. **Very high consistency F**—respondents selected nearly all F responses
5. **High consistency**—respondents selected a very large number of responses for *either* T or F

Further analysis showed that the respondents exhibiting a moderate amount of consistency made up the largest group (50%–78% of respondents in each country/ language sample). A relatively small percentage of people (0%–15% for most samples) had responses that revealed little consistency in choosing between the two preference categories. The remaining respondents comprised those who exhibited high or very high consistency by responding to all or almost all items in the direction of one preference category. Since these very high consistency classes differed only in the few particular out-of-preference item responses chosen, they were combined, resulting in the three consistency classes summarized in table 7.3.⁶

Formally, each of the five scale (clarity) classes or categories is associated with an estimated scale factor: s_1 , s_2 , s_3 , s_4 , or s_5 . The weight for the j^{th} preference item is scaled accordingly for respondents in a particular scale class k :

6. In the analyses, the order of the scale classes differs from the order presented here. Briefly, the less clear class was class 2, and the moderate clarity class was class 1. The order was changed for simplicity in presentation.

Table 7.3 | Frequency of Thinking–Feeling scale classifications for country/language samples

Country/language sample	% showing low consistency (scale class 1)	% showing moderate consistency (scale class 2)	% showing very high consistency (scale classes 3, 4, 5)
Large samples			
Australia (North American English)	1	62	37
Canada (North American English)	3	58	39
United Kingdom (European English)	7	57	36
United States (North American English)	5	56	38
Moderate-size samples			
Brazil (Brazilian Portuguese)	0	52	48
Canada (Canadian French)	12	67	21
China (Simplified Chinese)	3	76	20
China (Traditional Chinese)	15	65	20
Denmark (Danish)	2	62	37
Finland (Finnish)	3	66	31
France (European French)	7	68	24
Germany (German)	5	58	37
Greece (Greek)	5	62	33
Ireland (European English)	2	55	43
Italy (Italian)	10	65	26
Mexico (Latin American Spanish)	6	50	44
Netherlands (Dutch)	0	65	35
Norway (Norwegian)	0	72	28
Portugal (European Portuguese)	2	78	20
South Africa (Afrikaans)	0	64	36
South Africa (North American English)	0	64	36
Spain (European Spanish)	14	62	24
Sweden (Swedish)	0	75	25

Note: Global sample, $N = 16,773$.

$$\beta_j^{(k)} = s_k \beta_j \quad k = 1, 2, 3, 4, 5$$

Thus, the item weights are largest in magnitude for individuals with very clear preferences.

As shown in table 7.3, the percentage of low clarity, moderate clarity, and very high clarity respondents varies by sample. Allowing variation in scale class size across samples enabled more accurate item weight estimates to be obtained. However, use of scale classes in the latent class models was for estimation and calibration purposes only. In order to achieve measurement equivalence, for scoring purposes the item weights associated with the moderate clarity scale class are used for all respondents in the scoring of the final Global Step I assessment.

Once these scale classes were established, analyses were conducted to determine the items to include in

the final Global Step I assessment, beginning with all available items for each preference scale for the entire global sample. Each model tested provided item weights adjusted for the scale classes associated with the levels of clarity, as discussed above; significance tests for those weights (testing whether the item weight is statistically significantly different from zero); and bivariate residuals (BVRs). BVRs essentially measure the relationship between any pair of items. A high BVR value suggests that the two items being evaluated are highly related and possibly redundant. Based on these results, certain items were eliminated and the analysis was rerun, iteratively, until satisfactory item and scale characteristics were obtained. Weak, nonsignificant item weights, indicating a lack of predictive power, and high BVRs, indicating a violation of the assumption of local independence,⁷ were the primary reasons that items were dropped from

7. Local independence is an assumption made by many statistical analysis approaches, such as factor analysis, IRT, and structural equation modeling (SEM). Local independence means that the associations between the variables are due solely to the underlying latent construct and not some other variable. In particular, this implies that among respondents with the same latent variable score (or in the same latent class), any two items are independent of each other. When examining responses on an assessment, this means that a response to a prior item does not affect or influence a response on another item. Bivariate residuals, available with latent class models, can provide an indication that the local independence assumption is violated. A violation of local independence would occur, for example, if two items contained very similar wording, in which case there would be a large bivariate residual (indicative of a large residual correlation) remaining after accounting for the measured latent construct.

analysis. This process ensured that each item retained made a strong, independent contribution to the measurement of the preference pairs.

Final Item Choices

For final item selection, a group of experts convened to make decisions in light of the vast array of information assembled. The following evidence and factors were considered:

Content of the items. Each item was examined to make certain of its theoretical link to the preferences. Moreover, looking ahead to the Step II revision, counts of the number of items associated with each facet scale within each preference pair were also examined. Although the preferences are not the sum of the facets, and representation is in no way necessary theoretically, some representation from each facet was seen as desirable in order to maintain the breadth and richness of the preference scales.

Overlap with prior forms. Each item that had been in use on one of the existing commercial forms (i.e., Form M, European Step I) received special consideration under the assumption that it had been selected previously for good reason. Inclusion of items from existing Step I forms also boosted the associations between the prior commercial forms and the Global Step I assessment.

Factor loadings. Each item should contribute meaningful unique variance to the latent preference.

Bivariate residuals. Creating a model in which items are locally independent ensures that each item contributes uniquely to the latent construct. Violations can take the form of repeated or highly similar wording of items, which can artificially inflate measures such as Cronbach's alpha. For the MBTI global research initiative, careful attention was paid to identifying item pairs with high BVRs and reducing redundancy among the items by keeping only one item from these pairs.

Item and scale characteristic and information curves and parameters. As detailed in the Form M manual, the MBTI Form M preference scales were built so as to maximize precision around the midpoint separating one preference from its opposite. Items that tended to offer information around the midpoint were favored here as well. As mentioned earlier, the midpoint selected was associated with the scale class that reflected moderate clarity in order to achieve measurement equivalence.

Internal consistency and test-retest reliability. High internal consistency and test-retest estimates were desired at the scale level. However, it was recognized that the new emphasis on eliminating items with high BVRs might have the effect of lowering internal consistency estimates compared to prior forms.

Prediction ratios. As detailed in prior MBTI manuals, researchers generally sought prediction ratios above .60.

Correlations of the continuous measures with each other. It was expected that S–N and J–P would be moderately correlated, as they were on prior MBTI forms, but otherwise the preference scales were expected to remain relatively independent. (See chapter 9 for these correlations.)

Correlations with other measures. The CPI 260 and ACL assessments were incorporated into the data collection effort in many country/language samples. In addition, several existing sets of MBTI data available from the publisher were rescored. MBTI Form Q had enough items to make rescoring archival Form Q data into Global Step I data possible, although doing so in some cases meant that the maximum number of omissions (items not responded to) for the Global Step I assessment was exceeded. Some of these analyses are included in chapter 9 (e.g., correlations between the MBTI Global Step I assessment and the TKI® and FIRO-B® assessments).

Demographic differences across forms. Demographic differences consistent with those found on prior forms were expected. For example, women were expected to disproportionately prefer Feeling, while men were expected to disproportionately prefer Thinking, and individuals who prefer Extraversion were expected to be slightly more satisfied with their job than were those who prefer Introversion.

Performance of items across samples. Items and scales were evaluated separately for the country/language samples detailed in table 7.1. Items that performed well across many samples were selected over those that performed well in only a small number of samples. Although most items worked well in most languages, inevitably there were items that did not perform well in one or just a few. Further examination of possible adjustments for these differences is discussed below.

In summary, the traditional, IRT, and latent class approaches all tended to point to the same set of optimal items. For example, an item that had a large and significant item weight in a latent class analysis also tended to have a large *a* parameter in an IRT analysis and a large item–total correlation in the traditional analysis.

This process resulted in 92 items being selected to create a Global Step I assessment measuring the four preference pairs: 24 E–I items, 24 S–N items, 23 T–F items, and 21 J–P items. The Global Step I 92-item set overlaps substantially with the item sets of prior commercial forms. MBTI Form M shares 15 items on E–I, 20 on S–N, 20 on T–F, and 16 on J–P with the Global Step I assessment, while the European Step I assessment shares 11 items on E–I, 11 on S–N, 8 on T–F, and 9 on J–P with the Global Step I assessment.

Scoring Decisions

Important steps in finalizing the scoring of the Global Step I assessment were as follows: (1) setting the cut point for each preference scale; (2) evaluating differences between the country/language samples listed in table 7.1, making adjustments for items that the respondent did not answer; and (3) evaluating the need for and establishing a tiebreaker rule.

CUT POINTS

The cut point for each preference scale was set by adjusting the intercept term in the LCA scoring equations. Cut points were set to maximize agreement with best-fit type using all cases from the global sample who finished MBTI®Complete (all such cases were from the United States, the United Kingdom, Canada, Australia, or Ireland). Use of best-fit type data to aid in setting cut points follows the pattern of previous revisions. Logits were then centered again so that cut points were set to 0.

SCORING DIFFERENCES ACROSS SAMPLES

After finalizing the measurement model, country/language-level differences in the model were investigated. The purpose was to determine whether further adjustments were needed for the different country/language samples within the global sample in order to improve the estimation of type preferences. In these analyses, item weights adjusted for scale heterogeneity (i.e., differences in the clarity of responses as reflected in the scale classes) were estimated separately for each of the individual samples within the global sample. Two primary analyses were conducted to evaluate differences among the item weights in the samples.

First, for each of the four measured preference pairs, the estimated item weights identified for each sample were compared. Sample by sample, each estimated item weight was evaluated to determine whether it had the correct sign and differed significantly from zero, which meant that it was contributing to the estimation of the respective preference within that sample. The analysis showed that nearly all the items contributed significantly within each sample. Specifically, for the 552 E–I comparisons (24 E–I items in 23 samples), only 4 were nonsignificant. For the 552 S–N comparisons, only 25 were nonsignificant. Of the 529 T–F comparisons, only 9 were nonsignificant. Finally, of the 483 J–P comparisons, only 6 were nonsignificant.

Second, correlations between the global sample item weights and the individual sample item weights were computed. These are summarized in table 7.4. The table shows that for nearly all the individual samples, the item weights are similar to the item weights for the global sample, with an average correlation across the preference scales of .94. In the moderate-size

samples, where correlations were expected to be lower, correlations range from a low of .76 for T–F in a sample from China (Traditional Chinese) to a high of .98 in six of these samples. In brief, these results show that the same items that are most important on each scale in the global sample are also found to be most important in each of the individual samples.

A final adjustment was considered, in which items were evaluated to determine whether there was a benefit of not scoring, or “zeroing out,” certain items (effectively applying a weight of 0 to those items) when they were found to be nonsignificant in a particular country/language sample. Investigations concentrated on the three individual samples with the weakest measurement properties (Simplified Chinese, Traditional Chinese, and Mexico). The results indicated that zeroing out even a handful of items did not improve the resulting categorizations or the overall fit of the model in a meaningful way for that sample. Because of the benefit of using the same scoring formula across samples and the minimal benefits (if any) of making item weight adjustments by individual sample, no items were zeroed out. Thus, the same item weights are used for all the samples in table 7.1 and will likely be used for any new translation (or adaptation) created. As a result, the Global Step I scoring and results are consistent and comparable worldwide.

Global Step I scoring and results are consistent and comparable worldwide.

MISSING ITEM RESPONSES

Upon concluding that a consistent set of items and weights could be used for all the individual samples within the global sample, the issue of missing items—that is, items that lack responses—was dealt with next. Again, different strategies were examined. One approach was to use the missing-at-random (MAR) assumption (Vermunt & Magidson, 2013) to set a score to 0 for items without a valid response and re-estimate the intercept for all item configurations for which missing responses were possible. Since this approach was impractical due to the large number of potential item configurations, an equivalent strategy was used that replaced any missing responses with values determined by LCA. Specifically, these are values that are very close to the mean values for the missing items but estimated from the latent class model to minimize bias. This means that for any item that is not responded to on the Global Step I assessment, a value is substituted into the scoring equation for that missing value that differs only slightly from the mean for that item in the global sample.

Table 7.4 | Global MBTI® sample item weight correlations for country/language samples

Country/language sample	E–I	S–N	T–F	J–P
Large samples				
Australia (North American English)	.97	.99	.99	.96
Canada (North American English)	.97	.99	.98	.98
United Kingdom (European English)	.97	.99	.99	.99
United States (North American English)	.98	.99	.98	.99
Moderate-size samples				
Brazil (Brazilian Portuguese)	.97	.93	.93	.97
Canada (Canadian French)	.93	.88	.93	.93
China (Simplified Chinese)	.95	.95	.91	.87
China (Traditional Chinese)	.89	.96	.76	.94
Denmark (Danish)	.98	.94	.98	.97
Finland (Finnish)	.96	.98	.90	.89
France (European French)	.93	.95	.96	.90
Germany (German)	.93	.94	.93	.95
Greece (Greek)	.87	.92	.91	.88
Ireland (European English)	.87	.98	.92	.98
Italy (Italian)	.93	.91	.93	.93
Mexico (Latin American Spanish)	.91	.92	.98	.91
Netherlands (Dutch)	.88	.97	.98	.94
Norway (Norwegian)	.97	.91	.97	.88
Portugal (European Portuguese)	.95	.92	.92	.96
South Africa (Afrikaans)	.98	.96	.95	.96
South Africa (North American English)	.97	.96	.96	.96
Spain (European Spanish)	.97	.93	.97	.96
Sweden (Swedish)	.92	.88	.95	.88
<i>Mean</i>	.94	.94	.94	.94

Note: Global sample, $N = 16,773$.

To further minimize the impact of missing values for each preference pair, only complete response sets and response sets containing at most a small number of missing items were allowed. For the Global Step I assessment, a maximum of four omissions, or items not responded to, are allowed in the online version.

TIEBREAKER RULE

Historically, ties in scoring were broken in the direction of I, N, F, and P. The rationale for this methodology (Myers et al., 1998) was that in US culture and perhaps in that of many other Western countries, social desirability might affect a person's responses in the direction of E, S, T, or J. Thus, any ties might be the result of responding in a socially desirable manner. Therefore, a tie should be broken toward the *less* socially desirable preference.

While alternative rules were considered, the traditional tiebreaking rule is preserved for this revision. However, due to the complexity of latent class scoring, ties are highly unlikely for the Global Step I assessment, and the rule is unlikely ever to be used.

Probability Index Development

A replacement had to be found for the preference clarity index used on Form M, since the pci was derived from IRT scoring and the Global Step I assessment uses latent class scoring. As described previously, latent class yields a *logit*, which is computed for each preference scale for each respondent. Logits are used for analyses requiring continuous preference scores. These logits can be transformed into the more easily interpretable estimates, called *posterior membership probabilities*, that are bounded between 0 and 1 and are expected to be useful to practitioners and respondents in understanding the results. The MBTI items are designed to elicit a respondent's bipolar preferences (e.g., E vs. I) on each of the four preference pairs, E–I, S–N, T–F, and J–P. Latent class analysis, used in the global revision of the MBTI assessment, provides, for each respondent, posterior membership probabilities of being in each of these four latent categories. The probability index shown on the MBTI® Global Step I® Profile and other reports displaying scored Global Step I results is derived from these posterior membership probabilities. The process by

which these probabilities were evaluated and adjusted to create a useful probability index is explained next.

UNADJUSTED POSTERIOR MEMBERSHIP PROBABILITIES

Plots of these probabilities reveal that the distributions for all four theoretical dimensions are clearly bimodal (see Magidson, 2014, for a more detailed exposition). For example, figure 7.2 shows the unadjusted posterior membership probabilities (UPMPs) for the global sample for the T–F preference pair. Note that the UPMP ranges from 0 to 1; however, for the purposes of displaying the figure, a range from $-.99$ to $+.99$ was used, with the midpoint being $.50$. The bimodal nature of this distribution is evident, as the right tail displays the large number of respondents with a high probability of reporting a preference for Feeling, while the left tail shows respondents with a high probability of reporting a preference for Thinking (1 minus the probability of reporting a preference for Feeling).

The figure shows a very clear bimodal distribution in terms of the percentage of respondents scoring along the UPMP continuum for each of the scale classes. It can be seen in the figure that, of the respondents in the scale class representing those with very high clarity, nearly half have a UPMP of $.99$ (on either end of the continuum). Similarly, individuals in the moderate clarity scale class are clearly distributed at the ends of the continuum.

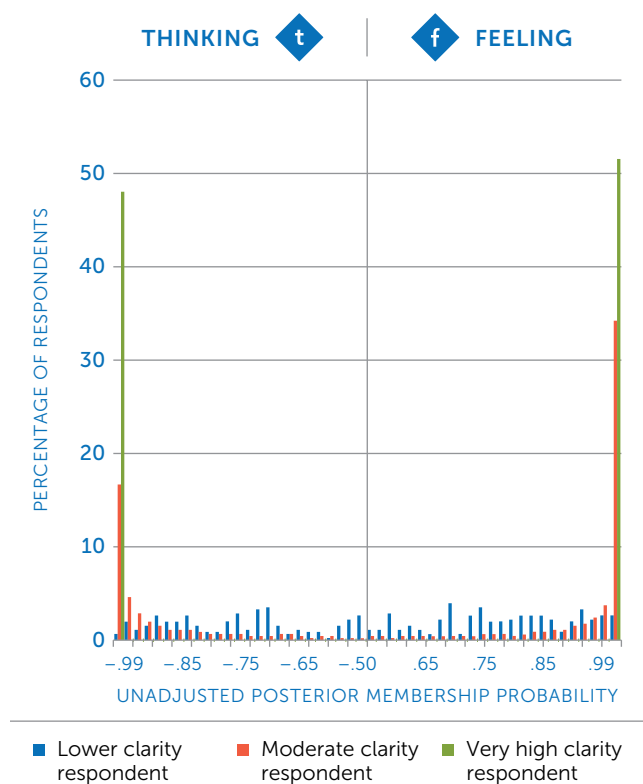


Figure 7.2 | Clarity class unadjusted posterior membership probabilities for the T–F preference pair

Only those in the lower clarity scale class are distributed along the entire range of the continuum, although still clustering away from the center. Based on a statistically appropriate analysis technique, figure 7.2 provides evidence of bimodality of the T–F preference pair for all the scale classes, with increasingly obvious bimodality as the response pattern more clearly favors one side or the other of the T–F preference pair.

ADJUSTED POSTERIOR MEMBERSHIP PROBABILITIES

The unadjusted posterior membership probabilities summarized in figure 7.2 show that for most people who take the MBTI assessment, the probability of correct classification is quite high. Indeed, nearly all respondents in the global sample have an unadjusted posterior membership probability of greater than $.80$ (in the appropriate direction). While accurate, there was a concern that this would be confusing to practitioners and respondents, possibly result in too strong a likelihood of endorsing the reported type, and unduly influence the interpretation process. Statistician George Box (1979) is often quoted as saying, “All models are wrong, but some are useful.” All models attempt to model the real world, but none does so perfectly. As is the case with all latent class and IRT models based on many variables, local independence is not fully achieved. While the data show clear signs of bimodality, it is expected that the extent of bimodality suggested by figure 7.2 is exaggerated

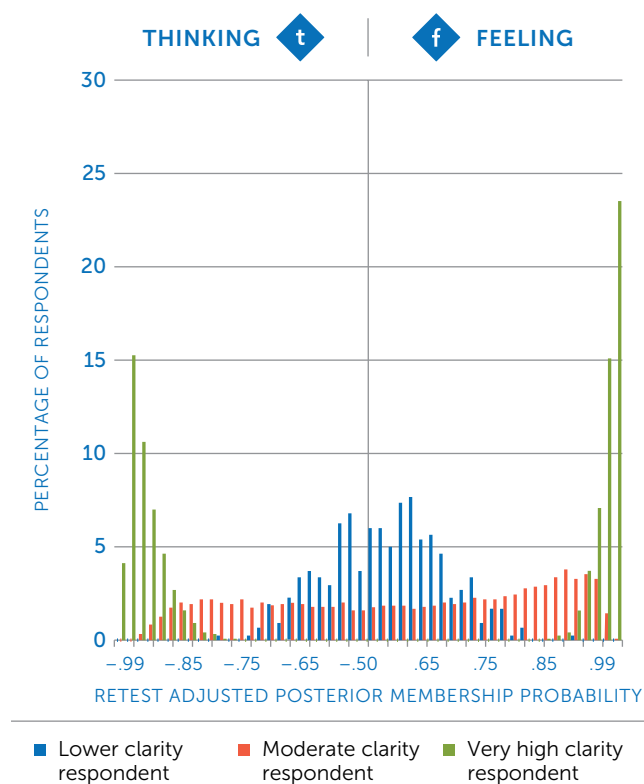


Figure 7.3 | Clarity class adjusted posterior membership probabilities for the T–F preference pair

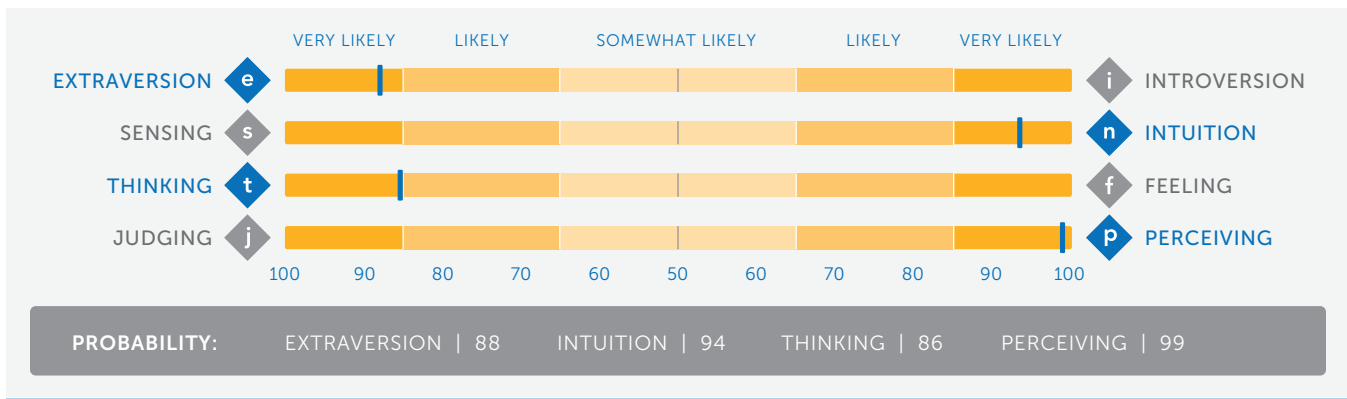


Figure 7.4 | Sample of probability indexes

somewhat due to the local dependence that remains in the data. As a result, an adjustment was made to the posterior membership probabilities to make them more useful to practitioners and respondents.

In order to make these probabilities more useful to practitioners, test-retest data were used to estimate the amount of shrinkage of these probabilities required to predict the likelihood of being classified into the same category (e.g., T or F) following two different MBTI administrations, thus obtaining a more meaningful probability index. Formally, a logit model was used to predict agreement between an initial test and a retest. The prediction was found to be very strong. The following logit model was used:

$$\text{Logit}(\text{Agree}_{T1,T2}) = \alpha + \beta_j(Y)$$

$\text{Logit}(\text{Agree}_{T1,T2})$ represents the logit in favor of test-retest classification agreement, Y denotes the current unadjusted logit score, α and β are the estimated parameters, and $j = 1, 2$ denotes individuals classified into category 1 (e.g., T) and those classified into category 2 (e.g., F) of the latent preference pair.

It was found that $\alpha = 0$ and $\beta_2 = -\beta_1$ for each of the four latent preference pairs. The β estimate corresponds to the amount of shrinkage, and $\beta_2 = -\beta_1$ means that the amount of shrinkage required to predict test-retest agreement is identical for both preference categories, resulting in a natural symmetry. The estimates for β serve to shrink the posterior membership probabilities, and the amount of shrinkage was found to be about the same for all four preference pairs ($E-I = -.30$, $S-N = -.38$, $T-F = -.28$, $J-P = -.33$). The negative signs in the adjustments indicate that, for each preference, the unadjusted posterior membership probability is reduced toward the cut point, set at .5, to identify each preference category. These predicted test-retest agreement logits were then transformed to test-retest agreement probabilities, resulting in the new probability index, using the following formula:

$$\text{Prob}(\text{Agree}_{T1,T2}) = \frac{\exp[\text{Logit}(\text{Agree}_{T1,T2})]}{1 + \exp[\text{Logit}(\text{Agree}_{T1,T2})]}$$

Figure 7.3 shows the change in the posterior membership probabilities following this adjustment. Specifically, comparing figure 7.2 and figure 7.3 shows that the bimodality is retained for respondents in the very high clarity scale class and that the distribution remains bimodal for those in the lower clarity scale class, although it is more closely centered around the cut point. Similarly, the distribution within the scale class representing moderate clarity is also bimodal, but flatter compared to those of the other two scale classes. In addition, the figure shows that for most respondents, the *probability index*, as the adjusted posterior membership probability is called, reflects a smaller value and can be interpreted as the probability of getting the same preference categorization on a second administration of the MBTI assessment. For example, if a respondent gets a result of E with a posterior probability of .95, she has an approximately 95% chance of reporting a preference for Extraversion after taking the assessment again.

DESCRIPTION OF THE PROBABILITY INDEX

As a result of these adjustments, a probability index is reported for each of the four MBTI preference pairs (see figure 7.4). The probability index is a numerical value ranging from 50 to 100 that applies to the side of the preference pair that the respondent favors. It indicates the likelihood of a respondent being sorted into the same preference on a subsequent administration of the MBTI assessment. In addition, the probability index is categorized into one of three categories: "somewhat likely," "likely," or "very likely."

A probability index is reported for each of the four MBTI preference pairs that indicates the likelihood of a respondent being sorted into the same preference on a subsequent administration of the MBTI assessment.

Table 7.5 | **Unadjusted and adjusted posterior membership probabilities and the probability index categories for the MBTI® global sample**

Probability index category	Unadjusted posterior membership probability (%)	Retest adjusted posterior membership probability (%)
E–I preference pair		
Somewhat likely (50–64)	4.7	15.8
Likely (65–84)	8.4	26.0
Very likely (85–100)	86.9	58.3
S–N preference pair		
Somewhat likely (50–64)	5.3	15.1
Likely (65–84)	9.7	25.0
Very likely (85–100)	85.0	59.9
T–F preference pair		
Somewhat likely (50–64)	5.1	18.0
Likely (65–84)	9.2	29.5
Very likely (85–100)	85.7	52.5
J–P preference pair		
Somewhat likely (50–64)	6.4	18.6
Likely (65–84)	10.9	31.7
Very likely (85–100)	82.8	49.7

Note: Global sample, $N = 16,773$. Probabilities may not total 100 due to rounding of decimals.

Table 7.5 shows these categories, along with the percentage of respondents who fall into each one for both the unadjusted and adjusted posterior membership probability values discussed previously. The table further demonstrates the impact of the adjustment made to the posterior membership probabilities in order to arrive at the probability index. It shows that if the unadjusted PMP had been used, only about 5% of respondents would ever be categorized as somewhat likely to have the E or I preference, whereas after the adjustment, the percentage of respondents assigned to the “somewhat likely” category is approximately three times larger. This adjusted probability, based on the retest data collected, is empirically more accurate. It is also likely to be more appealing to practitioners and respondents, because those who choose the opposite preference as their best fit during interpretation will not be confronted with an overly large probability index.

Best-Fit Type Analysis

Participants in the US, Canada, and Australia (North American English) samples and the UK and Ireland (European English) samples who completed the 230-item global research version of the assessment were also administered a modified version of MBTI®Complete so that best-fit type data could be collected along with

item responses. The best-fit type data were believed to be important for three reasons: (1) when the MBTI assessment was initially developed, Myers used a criterion group made up of individuals for whom she could confidently identify preferences, and she used this group to identify items and scoring weights that accurately measured those preferences (Myers & McCaulley, 1985, p. 142); (2) because of Myers’ work, agreement between best-fit type and reported type has been used as a criterion in a number of validity studies over the years (see Myers et al., 1998, and chapter 9 in this manual); and (3) best-fit type was used as an aid in determining the midpoint for Form M preference scales constructed using IRT (Myers et al., 1998). The intent, therefore, was to draw on best-fit type as a criterion when developing the Global Step I assessment either by including it as one of the factors or classes in the extended LCA model or by using it to adjust posterior membership probabilities. However, in the process of examining the potential usefulness of best-fit type in constructing the Global Step I assessment, some challenges arose and unexpected results emerged.

The challenges were evident after a review of the literature on best-fit type. As detailed in chapter 9, many uncontrolled factors influence the choice of a best-fit type by a respondent. This is partially demonstrated by the range of agreement between best-fit type and reported type (i.e., the type that results from scoring responses to the assessment). Agreement between the two requires matching all four preferences; by chance, such agreement would occur in 6.25% of cases. The *Form M MBTI® Manual* (Myers et al., 1998) summarizes several studies of best-fit type–reported type agreement. One study comparing different methodological approaches found an overall 64% agreement rate based on IRT scoring (pp. 144–145), and a second study using IRT scoring found 72% agreement for unadjusted midpoints, which was raised to 78% after adjusting the midpoints for best-fit type (p. 149). The latter result is among the higher agreement rates reported. The European Step I form of the MBTI assessment has a similar rate of best-fit type–reported type agreement, with OPP (2011) reporting an overall agreement rate (unweighted by sample size) of 63% for a set of European samples (Danish practitioners, Danish media, Dutch qualifying course delegates, Dutch practitioners, French qualifying course delegates, French business students, German practitioners, a Polish convenience sample, and a Russian convenience sample). The range reported by OPP was lowest in a Danish sample (56%) and highest in two Dutch samples (72%). A study in the United States using MBTI®Complete found an overall agreement rate of 76% (Krause & Thompson, 2008). From the studies conducted to date, it appears that the best-fit type–reported type agreement rate for the MBTI assessment on a variety of forms shows considerable range but is much greater than chance.

It was hoped that some of the methodological variability used to determine best-fit type in past studies could be controlled by using a standardized process, that is, the interpretation and verification process embedded in MBTI®Complete. Therefore, as part of the global research initiative, best-fit type was collected from samples of participants who regularly used North American English or European English, which included representative samples from the United States, the United Kingdom, Canada, Australia, and Ireland. All five of these samples completed the interpretive session and verification process embedded in MBTI®Complete. As part of this process, participants completed the 230-item global research version (GRV) of the MBTI assessment in either North American English or European English, but in all cases the Form M scoring method was used to provide reported type for the verification process. However, because the 230-item GRV included items from both Form M and the European Step I form, scores for both prior forms could be computed and the resulting reported types could be compared to best-fit type. Note that the advantage of pairing MBTI®Complete with the 230-item GRV is that three reported types can be scored from this form: one based on Global Step I scoring, one based on Form M scoring, and one based on European Step I scoring.

When proceeding through the interpretation portion of MBTI®Complete, respondents are first provided with a definition and description of a preference pair, along with behavioral examples, and then asked to estimate their preference on the pair (referred to as the self-estimate). They are then presented with their reported result for the preference pair, determined by Form M scoring. This is repeated for all four preference pairs. Finally, if the self-estimated type and the reported type differ, respondents are asked to read brief type descriptions of both and to select the one that reflects their best-fit type. Respondents also have the opportunity to reject both of these and to read other type descriptions and choose a different best-fit type. Note that reported type based on scored responses is always presented before respondents select a best-fit type, so it is possible for reported type to have an influence on best-fit type selection.

Best-fit type was compared against reported type for Form M, European Step I, and Global Step I scoring options. The median best-fit type–reported type agreement rate across the five samples using Form M scoring was 67.4%. Using European Step I scoring it was 46.3%, and using Global Step I scoring it was 58.4%. These results demonstrate that Form M scoring, which was the reported type scored and presented to the respondents during the MBTI®Complete interpretation session, had the highest best-fit type–reported type agreement rate of the three scoring approaches. At first glance, this might suggest that US-based Form M scoring is the best approach for identifying type. However, past

research on European Step I scoring (Deakin, 2006; Hackston, 2005; OPP, 2011) shows a much higher best-fit type–reported type agreement rate when the European Step I assessment was the basis of the reported type. This raised the question of whether the best-fit type–reported type agreement rate is in any way influenced by which scoring approach is used to obtain the reported type used during the verification process. In order to more fully investigate this possibility, several additional studies of best-fit type were completed.

THE “PROMPTING EFFECT”

The hypothesis was that respondents may be inclined to agree with the assessment results that are presented to them during the verification process (i.e., the reported type). If so, the best-fit type–reported type agreement rate may to some extent be affected by the scored results of whichever assessment was administered. This possibility was referred to as the “prompting effect” because respondents are literally prompted with their reported type results before they determine their best-fit type. This is similar to what in behavioral economics research is referred to as “priming” or “anchoring” (Kahneman, 2011).

To investigate the impact of reported type on the selection of best-fit type, the publishers undertook two studies. In each study, the 230-item GRV of the MBTI assessment was used, which contained all the items necessary to score and compare results of the European Step I, Form M, and Global Step I assessments. What varied across studies was the scoring method used to obtain the reported type presented during interpretation. Both studies used samples of participants who had no prior experience with MBTI type. The participants also used a modified MBTI®Complete verification process, the modification being which reported type (based on European Step I, Form M, or Global Step I scoring) was presented to them.

The first study (study 1) examined agreement between best-fit type and European Step I reported type in a sample of 128 participants (52% women, 47% men, 1% did not report gender; average age = 45 years) who were recruited by a third-party vendor. Upon completing the assessment, participants received a follow-up interpretation session with an MBTI expert by phone. Afterward, they selected a best-fit type. The results of this study showed that when participants were prompted with a reported type based on European Step I scoring, the agreement rate between best-fit type and reported type was 72.7%, while the agreement rate between best-fit type and both Global Step I reported type and Form M reported type was 44.5%.

To further evaluate the impact of the prompt, a second study (study 2) was conducted to examine the agreement rate between the respondents’ self-estimated type and

reported type. The respondents obtained their self-estimated type after they completed all the assessment items but before they were presented with their reported type. A total of 435 people completed the assessment, of which 205 resided in the United Kingdom and 230 resided in the United States. In this study, after obtaining their self-estimated type, respondents were prompted with a reported type based on Global Step I scoring. For comparison purposes, reported type was also calculated using Form M and European Step I scoring procedures. The results showed that before scoring results were introduced in the interpretation session, the agreement rate between self-estimated type and reported type was nearly identical across the three scoring approaches. This provides further confirmation of the prompting effect.

GLOBAL STEP I™ BEST-FIT TYPE RESULTS

A key concern in this chapter is the role of best-fit type in the construction of the Global Step I assessment. Since study 2 was the only one to use a reported type based on Global Step I items and scoring as part of the verification process, the study 2 results provide the best estimate of best-fit type–reported type agreement rates for the Global Step I form of the assessment. The agreement rates between best-fit type and reported type can be seen in table 7.6. The Global Step I reported type used as a prompt led to the highest agreement rates across both samples. The table shows that for the two samples combined, there was about a 63% whole-type agreement rate for Global Step I scoring, which is slightly lower than, but generally consistent with, what was found in past studies.

Given these results, the combined sample from this study was used to determine each possible level of agreement between best-fit type and reported type, as was often reported in the past, both by country of

residence and for the sample combined. The combined agreement rate for people matching on three letters and on four letters is also reported in table 9.43 and is in line with what was found for prior versions of the MBTI assessment. So, while the single best estimate of agreement between best-fit type and reported type comes from the study using the Global Step I prompt, further studies using samples more consistent with the commercial application of the MBTI assessment need to be completed in order to determine a better estimate of the actual best-fit type–reported type agreement rate.

In summary, prior research demonstrated that the specifics of the verification process can have an impact on the best-fit type selected by the respondent, and the research reported here also shows that the scoring method used to generate the reported type can affect the best-fit type that is selected. It is important to emphasize that the requirement that all respondents have an opportunity to discuss and determine for themselves which type best describes them is entirely separate and distinct from the discussion and conclusions reported above. As detailed in chapter 6, it is critical that practitioners verify type with each MBTI respondent. Use of a verification process by a trained professional, especially one who is aware of the various influences on the determination of best-fit type outlined here and in chapter 9, is an integral part of the MBTI experience. Likewise, the use of best-fit type as a criterion in validity studies is also a separate issue. Best-fit type can still be used as just one piece of evidence for validity if the limitations of the method are acknowledged and taken into account (see chapter 9).

Table 7.6 | Best-fit and reported type agreement rates among participants when prompted with Global Step I™, Form M, and European Step I™ scoring

		Percentage with matching reported and best-fit type				
		E–I	S–N	T–F	J–P	Whole type
Combined sample (N = 435)	Global Step I scoring	84.1	78.3	85.7	86.2	63.0
	Form M scoring	77.7	73.6	83.9	83.0	49.9
	European Step I scoring	76.1	72.9	74.5	72.2	37.2
UK participants (n = 205)	Global Step I scoring	81.5	75.6	84.4	86.3	60.0
	Form M scoring	76.6	70.7	84.4	84.9	51.2
	European Step I scoring	77.1	70.7	74.1	73.2	38.0
US participants (n = 230)	Global Step I scoring	86.5	80.9	87.0	86.1	65.7
	Form M scoring	78.7	76.1	83.5	81.3	48.7
	European Step I scoring	75.2	74.8	74.8	71.3	36.5

Note: N = 435 participants.
Source: Unpublished best-fit type study conducted by CPP, Inc., as part of the MBTI® global research initiative.

PHASE 4: SELECT GLOBAL STEP II™ ITEMS AND DETERMINE SCORING METHOD

The same global sample and 230-item set described above were also used in developing the Global Step II assessment. For the Global Step II assessment, all 230 items were examined for the purpose of evaluating and, if possible, improving the measurement of the 20 facets included in the Form Q and European Step II assessments, while maintaining the underlying meaning of the facets. (See chapter 4 and Quenk & Kummerow, 2019, for further descriptions of facet meaning.) Nonetheless, the assessment developers were open to adding, removing, and interchanging (i.e., moving from one facet scale to another) items on the basis of their complete set of analyses.

MBTI Global Step II item selection decisions were made with consideration of both global sample and country/language sample data, with primary focus given to the global sample. The focus on the global data was thought to be most effective, given that it was a large, globally representative data set. Country/language sample data were also examined to ensure that, to the extent possible, the selected items were applicable across country/language samples and translations.

Note that while latent class analysis was used to drive Global Step I item selection in an effort to align theoretical and statistical considerations, an IRT-based approach continued to drive Global Step II item selection. There were several reasons for the use of IRT for the development of the Global Step II assessment. First, the Step II measures have historically included a midzone. Experience with interpretation of Step II results leads to the conclusion that respondents find value and accuracy in the midzone descriptions, which differ from the in-preference or out-of-preference descriptions. For some facets, these midzones may constitute a separate class; for other facets, people scoring in the midzone may respond more like those scoring at one of the opposite preferences (see chapter 10 for examples). This makes it difficult to estimate latent classes across the 20 facets. Second, from a measurement perspective, some of the facet scales are more dichotomous, but others seem to be more like a continuum. Forcing a categorical solution, as required by LCA, onto a somewhat continuous measure is inappropriate. Third, the facet scales are more narrowly defined than are the preference scales, and some facet preferences result from using a type preference in a particular situation or in response to specific environmental pressures (see chapter 6 for examples). Again, this makes estimating a consistent set of latent classes inappropriate. Thus, the approaches used to evaluate the Global Step II facets focused on both traditional analyses and IRT analyses, with more emphasis given to the IRT results.

An iterative process was used to identify the final item sets to compose each of the Global Step II facet scales. Two experts performed analyses independently, and their results were considered together to arrive at initial measures for each of the facets. Specifically, analyses included

- Exploratory factor analysis
- Item–facet correlations
- Item response theory (IRT)
- Expert judgment of item content relevance, similar to the decision-making process elaborated previously for Global Step I development

Exploratory Factor Analysis

As an initial step, the two experts conducted a series of exploratory factor analyses using the global data. A principal components analysis (PCA) was used to determine whether the factor structure of the data differed significantly from that of data used in prior analyses and whether new items were appropriately related to their hypothesized facet scales. Results indicated that, for the most part, items grouped into factors that corresponded to the existing 20 facet scales. Given that the goal was not to reformulate the Global Step II assessment as a whole, the exploratory analysis was viewed as a guide to determine whether factors were more or less aligned, which they were.

As a further examination of the overall appropriateness of the 20-factor solution as a representation of the facets, the principal components analyses were repeated, this time specifying 15-, 20-, and 25-facet solutions. An examination of the grouping of the factor loadings indicated that the 20-facet specification model provided the most interpretable representation of the data. Taken together, these results indicate that the diverse, global data set is similar to previous, more homogeneous samples with respect to the factor structure underlying Step II items. That is, items tend to load most strongly on the facet scales on which they have historically loaded. Certainly, there is overlap in loadings given the similarity of content of facets within preference pairs. This is an issue that will be addressed subsequently.

Correlations Among the Items and the Facets

Although the initial factor analyses indicated that the general factor structure was as expected, the researchers still evaluated not only whether different research items added meaningfully to a facet scale (i.e., whether the items improved scale reliability while reflecting the meaning of the scale) but also whether items previously assigned to other facet scales might fit better elsewhere. This evaluation was accomplished in a number of ways. First, a comprehensive matrix correlating items with each of the Form Q facet scales was created. Similar to the

factor analyses, this allowed the researchers to determine not only whether existing items correlated best with their current or hypothesized facets but also whether there was opportunity to remove items from facets or shift items among facets. Since this was an iterative process, suspect items, or items with potential to be relocated, were flagged and revisited in subsequent stages of evaluation.

Item Response Theory

Before discussing the IRT analyses for the Global Step II analysis, a brief overview of IRT is provided. In IRT, the functional relationship between a chosen response to an item and the underlying construct the associated item is intended to measure can be estimated using what is called an *item characteristic curve* (ICC). The horizontal axis of an ICC for a particular item shows the varying levels of the psychological characteristic the item is designed to measure. The vertical axis shows the expected rate of item endorsement, or the probability that a respondent will endorse the item in the keyed direction. The function describing different probabilities of item endorsement at different levels of the scale is the ICC.

The scores measuring the psychological characteristic on the horizontal axis are called *thetas* in IRT terminology and are denoted as θ . Theta corresponds roughly to a nonlinear rescaling of the “true score” of classical test theory (Drasgow & Hulin, 1990). In IRT, the scale of measurement is arbitrary in nature; by convention, it is typically expressed with 0 defining the theta midpoint and scores below the midpoint being negative and those above being positive. However, in the specific case of MBTI Step II items, thetas are further converted to an 11-point integer scale with no negatives, thus avoiding the pejorative connotation when reporting scores. The direction of the keyed response—that is, which pole of the facet scale is defined as the keyed direction—is also entirely arbitrary for the purposes of IRT. However, for the sake of consistency with previous MBTI conventions, the I, N, F, and P poles of the facet scales are defined as the keyed direction.

Figure 7.5 provides a theoretical ICC for three different items. For each of the three theoretical items, the shape and location of the ICC are determined by the *a* parameter, or the degree of slope in the ICC, along with the *b* parameter, or location parameter. The resulting ICC shows differing probabilities of item endorsement at different values of theta (the horizontal axis). The first item, illustrating good location and slope, shows that for this theoretical item, the dividing point for the item is very near the midpoint of theta, or 0 on the theta continuum. In addition, it has a large *a* parameter, indicating that the shift in probability from a response in the keyed direction to a response in the non-keyed direction occurs sharply

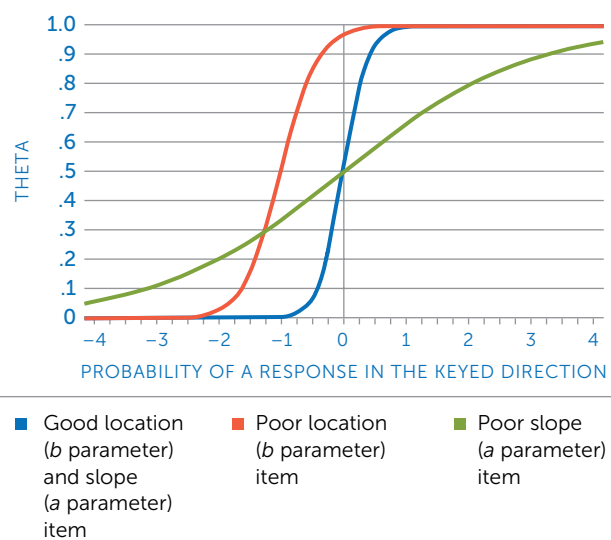


Figure 7.5 | Theoretical item characteristic curves for different-quality items

within a very narrow range of theta. The second item, illustrating poor location (*b* parameter), has a good slope, or *a* parameter, but is found in a suboptimal location for the end goal of MBTI categorization. That is, even individuals with a moderately clear preference in the non-keyed direction (e.g., $\theta = -1$) have a fairly high probability of endorsing the item in the keyed direction. The third item, with a poor slope (*a* parameter) and a good *b* value, has an equal likelihood of being endorsed in the keyed or non-keyed direction exactly at the theta midpoint, but it has a poor slope, or *a* parameter, indicating that the probability of being in the keyed or non-keyed direction can occur across a broad range of values for theta. So, the probability of being assigned to the keyed or non-keyed direction can be nontrivial and similar along the entire theta continuum. Both the *a* and *b* parameters provide important information to use in determining a good item versus a poor one.

The factor analyses and correlation analyses, along with the prior Form Q and European Step II item sets, provided a set of candidate items for use in exploring the Global Step II facet measures using IRT. However, prior to starting the IRT analyses, several additional decisions needed to be made regarding the model to be used. These decisions included the number of parameters to estimate—a one-, two-, or three-parameter logistic (1PL, 2PL, or 3PL) model.

IRT analyses have historically been most closely associated with measures of ability, in which a three-parameter logistic model is typically most appropriate (a 3PL model was used with MBTI Form M and Form Q). However, a review of the literature indicates that in the personality assessment domain, a two-parameter logistic model is often appropriate (e.g., Reise & Waller, 1990), and exploratory analyses on the global sample

indicated that a two-parameter IRT model fit the MBTI data as well as (or better than) the three-parameter model.⁸ Additionally, the traditional third IRT parameter (called *pseudo-guessing*) was deemed theoretically unnecessary. The third parameter would give every respondent a nonzero probability of endorsing items in the keyed direction regardless of underlying trait location. As a result, analyses proceeded with a two-parameter logistic model, estimating slope and location parameters, which are described above.

In addition to IRT model decisions, the number of quadrature points needed to be determined. Quadrature points, while complex and beyond the scope of this manual, can be defined for our purposes as the number of points along the theta continuum to be estimated in arriving at an ICC. The higher the number of quadrature points, the more accurate the ICC. However, the increase in accuracy comes at the expense of a more time-consuming analysis. For the Global Step II form of the assessment, analyses were conducted with different numbers of quadrature points, with the retention of an odd number of theta points considered important (Bess & Harvey, 2002). Ultimately, 31 (which allowed for a “zero” point and 15 theta values on either side of zero) was identified as a reasonable compromise between accuracy and analysis efficiency, and thus the decision was to use 31 quadrature points.

Final Item Selection and Expert Judgment

Once the model characteristics were determined (2PL with 31 quadrature points), IRT analyses proceeded with the goal of identifying the best item sets for each of the 20 facet scales in the Global Step II assessment. These item sets were compared to Form Q item parameters, also computed in the global sample. The European Step II assessment was not compared due to its very high correlations with Form Q for the 20 facets in those measures. For the global sample and for each of the country/language samples (equated to the US sample metric), measures were developed and the *a* and *b* parameters were examined, along with ICCs. This information was then used in conjunction with the results from traditional analyses performed by a panel of experts to identify a final item set for the Global Step II assessment. The procedure largely duplicated what was done for the Global Step I analyses, with the exception of latent class analysis.

To arrive at the final pool of items and their facet scale assignments, a group of experts convened to discuss each candidate item individually, based on all data that could be brought to bear on that item (e.g., factor loading, IRT-based item characteristic curves, content congruence, incremental contribution to the reliability of the scale), and to consider inclusion versus exclusion. Their primary approach in modifying and developing the facet measures was to preserve the meaning of the original scale, with the goal of improving facet scale reliability whenever possible. Nonetheless, given overlap in item content, the researchers recognized the potential to incorporate new items and create even stronger measures of the facets. The IRT analyses became a critical part of this process. Those items exhibiting strong ICCs (items with steep slope centered near 0 on the *x* axis, or theta) were given preference over those with less distinct curves, due to their utility in differentiating among respondents near the midpoint of the scale. Ultimately, the group reached a decision about whether each candidate and existing item should be included in the respective facet scale.

Given the volume of information available to the researchers, combinations of items contributing to a facet were iteratively evaluated based on their content and contribution to overall facet scale reliability (e.g., two items might have had strong psychometric properties but replicated content, so one would need to be discarded). In some cases, there was clear indication that a previously used item was contributing less to scale reliability than a candidate item. In those cases, the researchers discussed the content of the item, factored in its psychometric results, and made a decision to retain or discard it. During this comprehensive review process, they made a decision to allow certain items to be scored for more than one facet. This decision was made for several reasons. First, facets are intended to measure different aspects of the same construct (one of the preference pairs). As such, they are not expected to be orthogonal, or fully independent of one another. In general, the researchers were able to select items that contributed most uniquely to a specific facet but found that certain items contributed greatly to more than one facet with respect to item–facet correlations and increments in scale reliability. Second, allowing certain items to cross load (be included on more than one facet scale) provides some consistency in the direction of the facets within the overall preference for those items that are also measuring one of the four preference pairs.

8. Fit of the data to the IRT models was evaluated by splitting Australian (*N* = 776), UK (*N* = 2,831), and US (*N* = 3,578) respondent groups into calibration and validation samples and investigating model/empirical fit plots as well as chi-square estimates of predicted versus actual item endorsement. The sampling rate was based on a desired validation sample of 3,000 respondents (facilitating comparison of fit coefficient “adequacy” across different samples and measures; e.g., Drasgow, Levine, Tsien, Williams, & Mead, 1995). Fit indices were inspected for both 2PL and 3PL estimates using the 4,185 (calibration) and 3,000 (validation) respondents. Visual inspection of fit-plots indicated good fit for most items using either the 2PL or 3PL specification. Chi-square singles, doubles, and triples (Drasgow et al., 1995) indicated better 2PL fit to the data (than 3PL) for the T–F, S–N, and J–P scales. Although the 3PL fit the E–I data moderately better than the 2PL, both the 2PL and 3PL models had comparably acceptable absolute chi-square fit indices. Generally, the results of the fit analyses therefore support the 2PL as both good fitting for the MBTI data and better fitting than the 3PL. Because of the conceptual mismatch (with the MBTI assessment), generally low-magnitude empirical *c* parameter estimates, and comparable-to-slightly-favorable fit statistics, the 2PL was retained for all substantive IRT analyses.

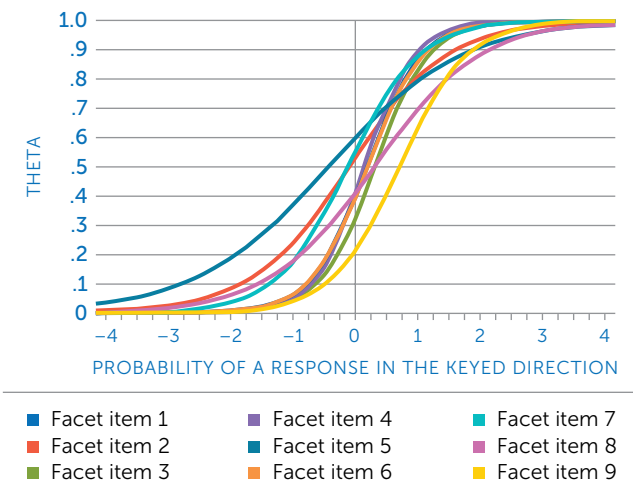


Figure 7.6 | Item characteristic curves for the Initiating–Receiving items

When responded to in the same direction as the overall preference, such items will reduce out-of-preference patterns. (“Out-of-preference” means a facet score is in the direction opposite that of the overall preference and not in the midzone.) When responded to in a direction inconsistent with the overall preference, these items will contribute to potential out-of-preference scores on those facets. In addition, allowing cross loadings increased the number of items available to measure the facets without simultaneously increasing the overall length of the assessment.

The effect on the internal consistency of the scale of adding or subtracting an item was one of two final factors influencing the ultimate selection of an item, the

second being how well the item content matched the intent of the scale (a decision based on expert judgment). An example of a final measure’s facet-level ICCs is shown in figure 7.6. Specifically, figure 7.6 shows the ICCs for the items that remained at the completion of the item finalization process for the Initiating–Receiving facet of the E–I preference pair. The figure most prominently shows that in the global sample the item characteristics of the scale are very good in that they show strong discrimination and desirable midpoint locations.

In addition to examining how facet item sets behaved in the global sample, the researchers also evaluated each facet item set separately within each of the samples detailed in table 7.1. Here, the item sets were equated using the US parameter estimates as an anchor. Empirical estimates of differential item functioning (DIF) were not conducted across samples due to the sheer volume of comparisons (25 countries and 206 candidate items produce 61,800 pairwise comparisons). Instead, visual inspection of ICCs was performed as a proxy DIF investigation. DIF essentially compares the item parameters and provides an indication of whether the item parameters for two samples differ, and, if they do, by how much. With a large, diverse set of samples as used here, some amount of DIF is expected. However, an effort was made to minimize the amount of DIF across samples in the facet item sets. Examples of the sample-level ICCs for a good item and a poorer item can be found in figures 7.7a and 7.7b.

The examination of Global Step II item and scale properties resulted in the selection of 139 items to be administered to measure the Global Step II facets. Because the Global Step II assessment also scores the

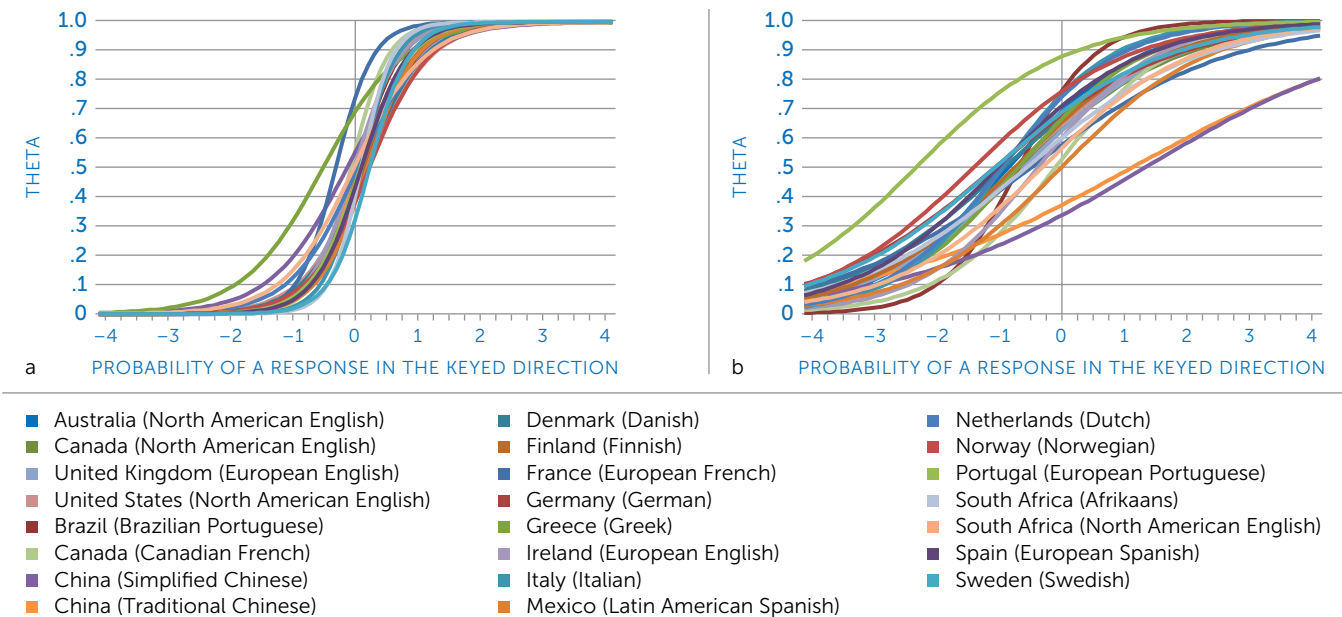


Figure 7.7 | Examples of sample-level ICCs: (a) good item with similar locations and slopes across samples; (b) poorer item with different locations and slopes across samples

Table 7.7 | Composition of Global Step II™ facets and overlap with the Global Step I™ assessment and prior Step II™ forms

Global Step II™ facet	Total number of items on facet	Number of Global Step I™ items on facet	Number of Form Q items on facet	Number of European Step II™ items on facet
E–I facets				
Initiating–Receiving	9	5	8	9
Expressive–Contained	8	6	8	7
Gregarious–Intimate	7	3	6	7
Active–Reflective	7	3	4	7
Enthusiastic–Quiet	7	5	7	7
S–N facets				
Concrete–Abstract	10	7	8	9
Realistic–Imaginative	7	5	7	7
Practical–Conceptual	5	4	5	5
Experiential–Theoretical	7	4	6	7
Traditional–Original	11	6	9	9
T–F facets				
Logical–Empathetic	9	8	8	8
Reasonable–Compassionate	9	8	9	7
Questioning–Accommodating	10	3	9	10
Critical–Accepting	7	2	6	7
Tough–Tender	7	5	7	6
J–P facets				
Systematic–Casual	9	7	7	8
Planful–Open-Ended	7	5	7	7
Early Starting–Pressure-Prompted	6	4	6	6
Scheduled–Spontaneous	11	9	11	8
Methodical–Emergent	5	3	5	5

Note: One item on the Systematic–Casual facet scale is also on the Global Step I S–N scale.

Global Step I assessment, it has 143 items, with four items used to measure Global Step I preferences but not used to score any of the facets. Due to the allowance of cross loadings, however, the facets are scored based on a total of 158 item assignments. Specifically, 17 items are assigned to two facets, and one item is used to score three facets. All but one of these is cross-loaded onto a facet within the same preference pair. One item cross loads between facets on S–N and J–P, which is not surprising, given the correlations between S–N and J–P preference scales. The Global Step II form has one item fewer than MBTI Form Q and 23 items fewer than the European Step II assessment. A summary of the distribution of items across the facets is presented in table 7.7 which also shows the overlap with the Global Step I assessment and with prior Step II forms.

The reliability and validity of the Global Step II facet scales are discussed in chapters 8 and 10, respectively, and those of each country/language sample are reported in that sample's corresponding manual supplement found in the *MBTI® Manual Global Supplement Series* (www.themyersbriggs.com/mbti-manual-supplements).

As appropriate and when data are available, each of these supplements reports on the translation process, the sample demographic characteristics, performance against established sampling targets, and, in addition, reliability and validity information for the Global Step II facets.

Each of the facets on the Global Step II assessment is scored using the global sample IRT parameter estimates. Theta scores are computed using the Bayes estimated a priori (Bayes EAP) method, which was also used with Form Q. Theta scores are then divided into 11 categories to reflect the clarity of the facet preferences, as was done on Form Q. In order to ensure that Global Step I results can also be scored from the Global Step II assessment, a limit on the number of omissions was set to four, the same number as for the Global Step I assessment. As on Form Q, missing item responses do not contribute to the estimate of theta for a facet. For example, on a six-item facet measure, if one item were omitted, the facet would be computed using the five items for which there was a valid response. While highly unlikely, it is possible for a respondent to skip four items within a single facet. As a

Table 7.8 | MBTI® type distribution in the global sample

Sensing		Intuition		Judging	Introversion
Thinking	Feeling	Thinking	Feeling		
ISTJ <i>n</i> = 2,671 15.9%	ISFJ <i>n</i> = 1,404 8.4%	INFJ <i>n</i> = 387 2.3%	INTJ <i>n</i> = 429 2.6%		
ISTP <i>n</i> = 1,651 9.8%	ISFP <i>n</i> = 1,106 6.6%	INFP <i>n</i> = 1,065 6.3%	INTP <i>n</i> = 809 4.8%		
ESTP <i>n</i> = 1,028 6.1%	ESFP <i>n</i> = 1,001 6.0%	ENFP <i>n</i> = 1,370 8.2%	ENTP <i>n</i> = 722 4.3%	Judging	Extraversion
ESTJ <i>n</i> = 1,509 9.0%	ESFJ <i>n</i> = 951 5.7%	ENFJ <i>n</i> = 361 2.2%	ENTJ <i>n</i> = 309 1.8%		

Note: *N* = 16,773.

Table 7.9 | MBTI® type preference and preference combination distributions in the global sample

Preferences		Orientation pairs		Process pairs		Orientation of energy and perceiving pairs		Judging and external orientation pairs	
<i>n</i>	%	<i>n</i>	%	<i>n</i>	%	<i>n</i>	%	<i>n</i>	%
E	7,251 43.2	EJ	3,130 18.7	ST	6,859 40.9	ES	4,489 26.8	TJ	4,918 29.3
I	9,522 56.8	EP	4,121 24.6	SF	4,462 26.6	EN	2,762 16.5	TP	4,210 25.1
S	11,321 67.5	IJ	4,891 29.2	NF	3,183 19.0	IS	6,832 40.7	FJ	3,103 18.5
N	5,452 32.5	IP	4,631 27.6	NT	2,269 13.5	IN	2,690 16.0	FP	4,542 27.1
T	9,128 54.4								
F	7,645 45.6								
J	8,021 47.8								
P	8,752 52.2								

Note: *N* = 16,773.

result, practitioners should encourage respondents to complete all items.

GLOBAL SAMPLE TYPE SUMMARIES

Type distributions based on the Global Step I assessment are summarized in six tables. Tables 7.8 and 7.9 provide the type summary and preference information for the entire global sample. Tables 7.10 and 7.11 provide the type and preference information for men, and tables 7.12 and 7.13 for women. (See chapter 5 for background on the preference combinations in the type summary tables.) It is important to keep in mind that while the

information in these tables provides the best estimate of type distributions to date based on a global sample, it is not a definitive summary of type distributions for the more than 7 billion people in the world. Differences in samples will always occur. Therefore, direct numerical comparisons with prior representative type samples are not provided here. However, readers will note some differences from prior type distributions. Specifically, in the global sample, more than 50% of both women and men reported a preference for Introversion, whereas in the Form M US representative sample, only women exceeded 50%. (Note, too, that in the UK representative sample both men and women reported a preference for

Table 7.10 | MBTI® type distribution for men in the global sample

Sensing		Intuition		Judging	Introversion
Thinking	Feeling	Thinking	Feeling		
ISTJ <i>n</i> = 1,470 18.9%	ISFJ <i>n</i> = 383 4.9%	INFJ <i>n</i> = 110 1.4%	INTJ <i>n</i> = 232 3.0%		
ISTP <i>n</i> = 1,035 13.3%	ISFP <i>n</i> = 428 5.5%	INFP <i>n</i> = 382 4.9%	INTP <i>n</i> = 450 5.8%		
ESTP <i>n</i> = 604 7.8%	ESFP <i>n</i> = 394 5.1%	ENFP <i>n</i> = 449 5.8%	ENTP <i>n</i> = 400 5.1%	Judging	Extraversion
ESTJ <i>n</i> = 878 11.3%	ESFJ <i>n</i> = 272 3.5%	ENFJ <i>n</i> = 107 1.4%	ENTJ <i>n</i> = 177 2.3%		

Note: Men in global sample, *n* = 7,771.

Table 7.11 | MBTI® type preference and preference combination distributions for men in the global sample

Preferences		Orientation pairs		Process pairs		Orientation of energy and perceiving pairs		Judging and external orientation pairs	
<i>n</i>	%	<i>n</i>	%	<i>n</i>	%	<i>n</i>	%	<i>n</i>	%
E	3,281 42.2	EJ	1,434 18.5	ST	3,987 51.3	ES	2,148 27.6	TJ	2,757 35.5
I	4,490 57.8	EP	1,847 23.8	SF	1,477 19.0	EN	1,133 14.6	TP	2,489 32.0
S	5,464 70.3	IJ	2,195 28.2	NF	1,048 13.5	IS	3,316 42.7	FJ	872 11.2
N	2,307 29.7	IP	2,295 29.5	NT	1,259 16.2	IN	1,174 15.1	FP	1,653 21.3
T	5,246 67.5								
F	2,525 32.5								
J	3,629 46.7								
P	4,142 53.3								

Note: Men in global sample, *n* = 7,771.

Introversion.) This difference across samples does not necessarily mean that the global distribution of people who prefer Extraversion and Introversion has shifted from E to I. It is possible that changing methodologies (telephone vs. internet) used to collect the data are responsible for some of these differences. Another hypothesis is that a sample is an estimate of a population, and as samples grow larger and more diverse, both of which are true of the global sample compared to the Form M sample, it is more likely that the sample will be a better estimate of the true population distribution.

For the other three preference pairs, the distributions in the global sample are more similar to those in the

Form M sample. The gender difference on Thinking and Feeling remains, with more women reporting a preference for Feeling and more men reporting a preference for Thinking, although the difference is smaller than in the Form M representative sample. The Sensing and Intuition distribution is similar to that in the Form M US national representative sample, with about two-thirds of individuals reporting a Sensing preference. Finally, the Judging and Perceiving distribution is close to 50% reporting a Judging preference and 50% reporting a Perceiving preference.

Table 7.12 | MBTI® type distribution for women in the global sample

Sensing		Intuition		Judging	Introversion
Thinking	Feeling	Thinking	Feeling		
ISTJ <i>n</i> = 1,201 13.3%	ISFJ <i>n</i> = 1,021 11.3%	INFJ <i>n</i> = 277 3.1%	INTJ <i>n</i> = 197 2.2%		
ISTP <i>n</i> = 616 6.8%	ISFP <i>n</i> = 678 7.5%	INFP <i>n</i> = 683 7.6%	INTP <i>n</i> = 359 4.0%		
ESTP <i>n</i> = 424 4.7%	ESFP <i>n</i> = 607 6.7%	ENFP <i>n</i> = 921 10.2%	ENTP <i>n</i> = 322 3.6%	Judging	Extraversion
ESTJ <i>n</i> = 631 7.0%	ESFJ <i>n</i> = 679 7.5%	ENFJ <i>n</i> = 254 2.8%	ENTJ <i>n</i> = 132 1.5%		

Note: Women in global sample, *n* = 9,002.

Table 7.13 | MBTI® type preference and preference combination distributions for women in the global sample

Preferences		Orientation pairs		Process pairs		Orientation of energy and perceiving pairs		Judging and external orientation pairs	
<i>n</i>	%	<i>n</i>	%	<i>n</i>	%	<i>n</i>	%	<i>n</i>	%
E	3,970 44.1	EJ	1,696 18.8	ST	2,872 31.9	ES	2,341 26.0	TJ	2,161 24.0
I	5,032 55.9	EP	2,274 25.3	SF	2,985 33.2	EN	1,629 18.1	TP	1,721 19.1
S	5,857 65.1	IJ	2,696 29.9	NF	2,135 23.7	IS	3,516 39.1	FJ	2,231 24.8
N	3,145 34.9	IP	2,336 25.9	NT	1,010 11.2	IN	1,516 16.8	FP	2,889 32.1
T	3,882 43.1								
F	5,120 56.9								
J	4,392 48.8								
P	4,610 51.2								

Note: Women in global sample, *n* = 9,002.

CONCLUSION

At the outset of the MBTI global research initiative, it was unclear whether the available forms of the MBTI assessment could be made globally consistent or if the measures could be improved over those of the prior forms, which already had very strong measurement properties. At the conclusion of this effort, it became clear that creating consistent Global Step I and Global Step II assessments was viable. Moreover, while the effort resulted in developing item sets and scoring methods for a significantly more diverse sample in terms of languages

and cultures, there were some small tradeoffs against measurement properties of the different forms of the assessment. (These differences are explored more fully in chapters 8, 9, and 10.)

Despite those small tradeoffs, the goals of the research initiative were largely met. Larger and more current representative samples were compiled in the United States and the United Kingdom. Sixteen additional representative samples from other countries were compiled and integrated into a large global sample. In addition, measures of the Step I preferences and Step II facets were developed, which allowed for use

on a worldwide scale. Further, insight was gained regarding the use of latent class analysis, demonstrating that it is appropriate for the Step I form of the MBTI assessment. Latent class analysis is more appropriate theoretically and offers the opportunity to introduce MBTI results as probabilities, which should be more intuitive for respondents.

The MBTI global research initiative resulted in a 92-item Global Step I assessment that works well. The research initiative also resulted in a 143-item Step II assessment and in a better understanding of the nature of the Step II facets.

CHAPTER 8

Reliability of the MBTI® Global Step I™ and Step II™ Assessments

CHAPTER OVERVIEW

Factors That Affect Reliability

Considerations in Evaluating MBTI® Reliability

Internal Consistency Reliability Estimates

Test-Retest Reliability Estimates

Polarity Index

Conclusion

This chapter addresses the reliability of the scales of the MBTI® Global Step I™ and Step II™ assessments in a variety of samples. The reliability of a scale is an estimate of how consistently it yields the same results across its set of items or over time. The terms *reliability* and *consistency* are synonymous. A measure is said to be reliable when it produces a consistent, though not necessarily identical, result with a particular group of people. Reliability is an issue because when items are used to measure some aspect of human nature, it cannot be assumed that they will yield entirely consistent responses from everyone or perfectly consistent responses over time.

Two kinds of reliability are commonly considered in measuring a construct:

- Internal consistency reliability: How consistently does a set of items measure the construct?
- Test-retest reliability: How consistent are the measure's scores over time?

A *reliability coefficient* is an index of the consistency of responses made by a particular group at a particular time, or over different times, to a specific set of items designed to measure a construct (Capraro & Capraro, 2002; Dawis, 1987).

FACTORS THAT AFFECT RELIABILITY

A reliability coefficient is limited by the amount of error in the measurement. In this chapter, the error is referred to as *noise*, which is defined as “irrelevant or meaningless data or output occurring along with desired information” (www.merriam-webster.com/dictionary/noise) or “random fluctuations that obscure or do not contain meaningful data or other information” (<https://en.oxforddictionaries.com/definition/noise>). The less noise there is in the measurement, the more consistency there will be in measuring the desired construct. Noise in reliability coefficients primarily derives from two sources. One is the items themselves. How items are worded, and how consistently they, as a set, tap the construct of interest, will affect reliability. The other source of noise is the differing characteristics of the individuals who are responding to the items. These include the characteristics of the individuals at a particular point in time and characteristics that may change over time.

In terms of the first potential source of noise, the items, words often have slightly different meanings for different people. It is uncertain whether two questions, both designed to measure the same construct, will be answered in the same way, even by the same person. An item may be worded in a manner so confusing or vague that it elicits a different response from the same person under different circumstances, depending on mood, time of day, physical environment, or preceding activities. Also, an item may be responded to differently over time by the same person.

Since the response to any given item is affected in part by the wording of the item, the strategy applied in building a scale is to use more than one item to tap the construct. Each item may attempt to approach the construct from a slightly different angle or with slightly different wording. Or various items may attempt to capture how the construct manifests in different contexts. It follows that the richer and more complex the construct is, the more varied the items must be. And the more items there are, the better chance there is to achieve consistency in responses because there will be less chance that one “different” response will point the results in another direction. If consistency is the goal, all the items should attempt to measure the same underlying construct and therefore elicit the same response. In terms of the MBTI Step I items, then, all the items on one preference scale would address some aspect of a person’s preference for one of two opposites.

The second potential source of noise that affects reliability is the characteristics of the people in the sample. The reliability of a scale can only be as good

as the consistency or stability of the quality being measured in a particular group of people. The characteristics of these people essentially set a limit for reliability. Factors such as gender, age, ethnicity, and developmental level of the respondents, to name a few, can affect how consistently people respond to a certain set of items. For example, research with prior forms of the MBTI assessment has demonstrated that reliability coefficients are affected by the developmental level of the respondents. Underachieving students show lower consistency in responses than do “overachieving” or high-achieving students (Myers, McCaulley, Quenk, & Hammer, 1998). Lower reliabilities in samples presumed to be less developed are to be expected.¹ For some populations, it is even possible that the reliability of a certain scale might be too high. Unexpectedly high reliabilities for some samples might suggest that the scale consists of many items with very slight variations in wording; that is, the scale is tapping a very narrow construct or is focused on a very narrow part of a broad construct. While such a scale may be useful in some contexts, its validity is also limited to the narrow construct.

Note: Several characteristics of the sample under consideration dictate that when interpreting the conventional estimates of reliability discussed in this chapter, it is important to bear in mind that these are *group-level* statistics. For this reason, it is somewhat misleading to designate the reliability of an assessment or scale as if the estimate were some internal property of the assessment itself. Instead, users should ask about the reliability of *scores* derived from the assessment or scale when used with a particular population. Nevertheless, reliability of an assessment (or scale) is typically used in the literature to simplify expression of the concept and therefore is used in this chapter.

These same potential sources of noise—wording and differing sample characteristics—will also affect the reliability of the measure over time. The consistency or replicability of item responses over time is referred to as *stability*, or *test-retest reliability*. The question here is: If the items are administered to the same person at two different times, will the person receive the same results both times?

In addition to the factors discussed above, there are two other important factors to consider when evaluating stability coefficients. One is whether the characteristics of the people in the sample would be expected to change over time. Just as exceedingly high internal consistency reliabilities for some groups might indicate a problem with the measurement, so too might high test-retest reliabilities raise concerns about the scale if

1. Some reviewers have ignored this fact and have mistakenly claimed that a few relatively lower reliabilities of selected MBTI preference scales in developmentally challenged groups is evidence of the lack of reliability of the MBTI assessment as a whole, whereas such results are actually predictable, appropriate, and meaningful.

they mistakenly indicate stability in a construct that is not expected to be stable in a particular population over a specified time period.

The second factor is the interval between administrations. If the retest interval is too short, respondents' memory of the items and their previous answers may influence their subsequent responses, artificially inflating the correlations between the two administrations. On the other hand, if the retest interval is too long, and if the qualities being measured vary over time, then real changes in the measured quality may be confused with estimates of an instrument's reliability.

CONSIDERATIONS IN EVALUATING MBTI® RELIABILITY

Given this discussion of reliability, a number of factors must be taken into account when the reliability of the MBTI preference scales in particular populations is being evaluated.

One important factor is that the MBTI Step I preference scales attempt to measure multifaceted constructs. The existence of the five Step II facet scales within each of the four Step I preference scales is *prima facie* evidence of this. While many respondents' preferences point in the same direction on both the overall preference scale and all of the facet scales within that preference scale, there are also people who have some facet results that point in the direction opposite their preference (see the discussion on out-of-preference facet results in chapter 2). To the extent that this occurs, it will reduce the reliability of the overall preference scales. In other words, a feature (the possibility of facet results being out-of-preference) that provides valuable information in terms of understanding an individual may be considered noise from the perspective of the reliability of scores in a group.

Another factor concerns how the psychological constructs underlying the MBTI assessment manifest in different cultures, which will also have an effect on reliability. As Kirby, Kendall, and Barger (2007) point out, the behaviors exhibited by people of different types can be influenced by culture. Thompson, Anderson, Johnson, et al. (2013) asked 35 type experts from 22 different countries to rate standard descriptions of the eight preferences on how similar they were to the way the preferences were described in their culture. For the E, I, S, N, T, and J preferences, 89% of the respondents said the descriptions were similar or very similar. For the F preference the agreement was 83%, and for the P preference it was 91%. So, while the preference descriptions appear to be quite similar across cultures, the slight differences could manifest as differential reliabilities in different samples. Although the MBTI items

performed quite well across cultures, which is why global forms of the Step I and Step II assessments are possible (see the discussion in chapter 7), there can still be cross-cultural differences in the response to some items that could introduce noise, and thus inconsistency, into the measurement.

A related but distinct issue is that many of the respondents in the country/language samples that compose the global sample used to develop the Global Step I and Step II assessments were responding to a translation of the items in their own language. While the translation process (described in chapter 7) was rigorous and carefully controlled, any translation has the potential to introduce noise into a measurement.

Another factor affecting reliability is that the way type preferences are exhibited can be affected by interactions with specific environments. Type does not exist in a vacuum. Its expression can be influenced by psychosocial processes, such as opportunities to express preferences, acceptance of one's preferences, pressure to behave in ways that are not consistent with one's preferences, or rewards or incentives for behaving in certain ways. Environmental considerations can affect both internal consistency and test-retest reliability estimates. For example, Salter, Evans, and Forney (1997) and Hammer and Treeger (1984) suggested that test-retest results on the T–F scale were influenced by the participants' exposure to training programs in which there were strong environmental pressures toward behaving in accordance with a Feeling preference. Similar pressures on managers can be observed in organizations, but in those contexts the pressure usually favors Thinking and Judging preferences.

As stated earlier, the developmental level of the respondent can also affect reliability. While, theoretically, an individual's underlying type preferences are believed to be stable over time, their manifestation in behavior (including the behavior of responding to an item) is influenced by the person's developmental level. This has implications for both internal consistency estimates and test-retest estimates. In the former, respondents with less effective command of perception and judgment would be expected to show less consistency in their responses. In the latter, if respondents develop better command of their preferences as they grow older and mature, test-retest estimates would be expected to become more consistent.

The final factor to consider, which concerns the Step II facet scales, is the length of the scale (i.e., the number of items on the scale). As discussed above, longer scales will have higher reliabilities, all else being equal. This has implications for the facet scales, which have many fewer items than the preference scales.

Reliability estimates of the Global Step I preference scales can be affected by cross-cultural factors, translation factors, environmental pressures on individuals, differing developmental levels of respondents, length of scales, and the effect of out-of-preference Step II results on Step I preferences.

Finally, it is important to consider reliability in context. Reliability is just one of many criteria for determining the usefulness of a scale. It is possible to develop a scale using internal consistency reliability as the sole criterion for selecting items and for evaluating the overall quality of the scale. Long scales with very slight changes in wording across items are likely to yield high internal consistency coefficients when used with people who consistently manifest the psychological construct in question. But such scales would likely be measuring a very narrow construct and one that would have limited applicability in other samples or across a variety of settings. So, while it is possible to build redundancy into a scale and thus increase its consistency, such a scale might not prove useful to practitioners or respondents. When building a scale, applying a variety of criteria, such as those discussed in the validity chapters (chapters 9 and 10) and the construction chapter (chapter 7) of this manual, is likely to yield a scale that is richer, more meaningful, and more useful.

INTERNAL CONSISTENCY RELIABILITY ESTIMATES

This section reports continuous-score reliability estimates for the four preference scales of the MBTI Global Step I assessment and the 20 facet scales of the MBTI Global Step II assessment. To estimate internal consistency reliability, the response to each item on a scale is correlated with the response to every other item on that scale. A high correlation coefficient indicates that people in the sample tended to respond to each item on the scale in the same way. The average of all of these item-to-item correlations is Cronbach's alpha, more commonly called *coefficient alpha* (Cronbach, 1951).

Internal Consistency of Global Step I™ Preference Scales

The internal consistency of the four Global Step I preference scales was estimated using coefficient alpha. For each preference pair, table 8.1 shows alpha coefficients in the global sample and in 23 country/language samples. In these samples, the E–I scale reliabilities range from .85 in the Mexico (Latin American Spanish) sample to .92 in the Brazil (Brazilian Portuguese) sample, with a median of .89. The reliabilities for the remaining three preference scales are lowest in the sample compiled in mainland China of participants taking

the assessment using simplified Chinese characters (hereafter, referred to as the Simplified Chinese sample). For the S–N scale, the range is from .74 for the Simplified Chinese sample to .91 for the South Africa (Afrikaans) sample, with a median of .86. For the T–F scale, the range is from .79 for the Simplified Chinese sample to .91 for the US (North American English) sample, the Brazil sample, and the South Africa (North American English) sample, with .89 as the median. And for the J–P scale, the range is from .77 for the Simplified Chinese sample to .92 for the South Africa (Afrikaans) sample, with a median of .88. The sample with the largest difference in reliabilities across the four scales is the Simplified Chinese sample (from .74 for S–N to .88 for E–I), followed by the Mexico sample (from .79 for S–N to .88 for T–F). The reliabilities reported here for the two Chinese samples—the Simplified Chinese sample and a sample compiled in Hong Kong and mainland China of participants taking the assessment using traditional Chinese characters (hereafter, the Traditional Chinese sample)—are somewhat lower than those reported by Beuke, Freeman, and Wang (2006) for the Form M Step I scales in two much smaller samples.

Comparing the internal consistency reliability estimates of the four Global Step I preference scales to those of other personality assessments is subject to two considerations. First, as discussed in chapter 7, latent class analysis eliminates items with high bivariate residuals. This may have the effect of lowering internal consistency reliability estimates compared to those of prior MBTI forms and those of instruments that use a more classical approach to scale development that allows for more redundancy in item content and wording. Second, because almost all reliability estimates available for other instruments are based on US samples, the most appropriate comparison for evaluating the reliabilities is the US sample.

Few if any other assessments provide reliability estimates on such a large and diverse group of languages as that found in the MBTI global sample. As shown in the table, the mean alpha coefficient for the four preference scales in the US sample was .89. The mean alpha for the four scales of the NEO instrument (excluding the Neuroticism scale) was .89 for the NEO Five-Factor Inventory (NEO-FFI), .90 for the NEO Personality Inventory (PI)-3 Form S, and .88 for the NEO PI-R Form S (McCrae & Costa, 2010). For the DiSC® instrument, the mean alpha was .79 for the adaptive scoring of the four scales and .77 for the natural scoring (Inscape Publishing, 2005). The seven major scales of the *Birkman Method*® had a mean alpha coefficient of .70 (Birkman, Elizondo, Lee, Wadlington, & Zamzow, 2008). Despite the strictures imposed by the use of bivariate residuals in scale development, the internal consistency reliabilities of the MBTI Global Step I scales compare quite favorably with those of the scales of other well-known personality instruments.

Table 8.1 | Internal consistency reliabilities of the MBTI® Global Step I™ preference scales in the global and country/language samples

Sample	N	Cronbach's alpha			
		E–I	S–N	T–F	J–P
Global sample	16,773	.89	.87	.89	.88
Large country/language samples					
Australia (North American English)	776	.90	.90	.90	.90
Canada (North American English)	939	.89	.89	.90	.88
United Kingdom (European English)	2,831	.89	.88	.89	.88
United States (North American English)	3,578	.90	.88	.91	.88
Moderate-size country/language samples					
Brazil (Brazilian Portuguese)*	839	.92	.88	.91	.90
Canada (Canadian French)	176	.86	.84	.81	.84
China (Simplified Chinese)	521	.88	.74	.79	.77
China (Traditional Chinese)	477	.86	.79	.82	.79
Denmark (Danish)	468	.91	.89	.90	.88
Finland (Finnish)	524	.91	.86	.89	.86
France (European French)	472	.86	.83	.85	.84
Germany (German)†	440	.86	.81	.89	.82
Greece (Greek)	277	.89	.83	.88	.91
Ireland (European English)	383	.88	.89	.90	.89
Italy (Italian)	458	.89	.82	.86	.83
Mexico (Latin American Spanish)	359	.85	.79	.88	.85
Netherlands (Dutch)	506	.89	.86	.89	.88
Norway (Norwegian)	493	.89	.87	.87	.86
Portugal (European Portuguese)	503	.87	.85	.86	.85
South Africa (Afrikaans)*	505	.91	.91	.90	.92
South Africa (North American English)*	189	.91	.88	.91	.90
Spain (European Spanish)	564	.87	.83	.86	.82
Sweden (Swedish)	495	.88	.84	.84	.89
Median	—	.89	.86	.89	.88

*Data collection for this sample was distributor led and nonrepresentative.

†Germany sample includes one individual residing in Switzerland.

The internal consistency reliabilities of the MBTI Global Step I preference scales are comparable to or better than those of the scales of other well-known personality instruments.

Table 8.2 shows alpha coefficients for the four Step I preference scales in the global sample by gender. Women had slightly higher reliability coefficients, although the differences are not statistically meaningful.

Table 8.3 shows alpha coefficients for the four Step I preference scales in the global sample for nine age groups. Isabel Myers believed that people with a better command of perception and judgment, who are better able to understand the items and more likely to accurately self-report their preferences, tend to respond more consistently to the items. If this assumption is correct, samples of older individuals should have higher reliability estimates than samples of younger individuals.

As predicted, the alpha coefficients in this table show an increase in consistency with older-person samples, with the exception of the over-70 group.

These findings are mirrored by those in table 8.4, which shows alpha coefficients for different levels of employment status. Again, moving from student

Table 8.2 | Internal consistency reliability of the MBTI® Global Step I™ preference scales by gender in the global sample

Gender	n	Cronbach's alpha			
		E–I	S–N	T–F	J–P
Men	7,771	.88	.86	.87	.87
Women	9,002	.90	.87	.89	.88

Note: N = 16,773.

samples to samples of people who are retired, there was a slight increase in consistency. The largest increase in reliabilities across those age groups and by employment status was on the S–N scale. As predicted, the internal consistency reliabilities increased with presumed increase in the command of perception and judgment, although these changes were relatively minor, with the exception of those on the S–N scale.

An analysis of a very large commercial database (almost 2 million people who took Form M) examined reliabilities separately for individuals at each age from 11 to 80 (not shown for space reasons). Similar to what is seen in the data from the global sample, the S–N scale showed the greatest increase in internal consistency reliabilities across age, starting with the lowest alpha coefficient (.78) for 11-year-olds and ending with the largest alpha coefficient (.93) for 80-year-olds. A difference was that the reliabilities on the J–P scale were consistently high in the larger database and did not exhibit the increase with age seen in the data from the global sample. In the large commercial database,

Table 8.3 | Internal consistency reliability of the MBTI® Global Step I™ preference scales by age group in the global sample

Age group	n	Cronbach's alpha			
		E–I	S–N	T–F	J–P
<18 years	101	.86	.80	.82	.81
18–21 years	1,230	.87	.80	.85	.85
22–25 years	1,669	.87	.83	.87	.85
26–30 years	1,844	.88	.83	.87	.86
31–40 years	3,689	.90	.86	.89	.88
41–50 years	3,439	.90	.87	.90	.88
51–60 years	2,652	.90	.90	.91	.89
61–70 years	1,722	.90	.90	.90	.89
>70 years	427	.88	.87	.88	.84

Note: N = 16,773.

Table 8.4 | Internal consistency reliability of the MBTI® Global Step I™ preference scales by employment status in the global sample

Employment status	n	Cronbach's alpha			
		E–I	S–N	T–F	J–P
Working full-time	9,107	.89	.87	.89	.88
Working part-time	2,423	.89	.87	.89	.88
Not working for income	1,032	.89	.88	.89	.88
Retired	1,664	.90	.89	.90	.88
Full-time student	1,384	.88	.83	.86	.86
None of the above	1,116	.89	.86	.89	.87
No response	47	—	—	—	—

Note: N = 16,773.

the young people were as internally consistent in their responses as the older people on the E–I and J–P scales and less consistent on the S–N and T–F scales. In this data set, regardless of the changes across age in either sample, the absolute level of the alpha coefficients even in the youngest groups (median = .85; $n = 41,266$, ages 11 to 17) was quite good, indicating that the MBTI Global Step I assessment is appropriate for use with young populations, as well as with more mature populations.

Table 8.5 shows alpha coefficients for the four Step I preference scales for ethnic groups in three different countries: the United States, the United Kingdom, and Canada. The lowest alpha coefficient (.79) is on the S–N scale for the “Other” group, a composite, in the Canada (North American English) sample. The two next-lowest alpha coefficients (both .82) are on the S–N scale for blacks in the United States and the “Other” group in the United Kingdom. All of these alpha coefficients are well within the acceptable range for personality assessments.

Table 8.5 | Internal consistency reliability of the MBTI® Global Step I™ preference scales by ethnicity in three country/language samples

Ethnicity	n	Cronbach's alpha			
		E-I	S-N	T-F	J-P
United States (North American English)					
White	2,388	.91	.90	.92	.89
Black	445	.87	.82	.86	.85
Hispanic/Latino	311	.88	.85	.86	.86
Asian or Pacific Islander	89	.87	.84	.88	.85
Mixed ethnicity	153	.86	.85	.88	.84
Other	103	.89	.85	.90	.88
No response	89	—	—	—	—
United Kingdom (European English)					
White (British)	2,155	.89	.88	.89	.89
White (other European)	144	.88	.87	.86	.88
Other	195	.85	.82	.84	.84
No response	337	—	—	—	—
Canada (North American English)					
White	689	.90	.90	.91	.88
Other	112	.88	.79	.87	.84
No response	138	—	—	—	—

Note: US (North American English) portion of the global sample $N = 16,773$. Due to small sample sizes, Asian Indians, Middle Easterners, and American Indians are combined with the group that selected “Other.”

UK (European English) portion of the global sample $N = 2,831$. Due to small sample sizes, Bangladeshi; African, Caribbean, and other blacks; Asian Indians; Chinese; other Asians; and individuals of mixed ethnicity are combined with the group that selected “Other.”

Canada (North American English) portion of the global sample $N = 939$. Due to small sample sizes, aboriginals, blacks, Chinese, Filipinos, Latin Americans, South Asians, and Southeast Asians are combined with the group that selected “Other.”

Table 8.6 | Internal consistency reliability of the MBTI® Global Step II™ facet scales in the global and country/language samples

Sample	Initiating–Receiving	Expressive–Contained	Gregarious–Intimate	Active–Reflective	Enthusiastic–Quiet	Concrete–Abstract	Realistic–Imaginative	Practical–Conceptual	Experiential–Theoretical	Traditional–Original
Global sample	.82	.73	.62	.64	.69	.74	.72	.66	.68	.72
Large country/language samples										
Australia (North American English)	.85	.76	.65	.65	.71	.79	.76	.67	.72	.75
Canada (North American English)	.85	.76	.60	.65	.71	.79	.75	.67	.75	.74
United Kingdom (European English)	.83	.74	.63	.63	.71	.75	.73	.63	.68	.72
United States (North American English)	.84	.74	.63	.64	.72	.77	.75	.64	.75	.71
Moderate-size country/language samples										
Brazil (Brazilian Portuguese)	.85	.81	.73	.72	.74	.78	.75	.72	.81	.76
Canada (Canadian French)	.76	.69	.49	.60	.55	.66	.74	.73	.43	.76
China (Simplified Chinese)	.71	.66	.66	.69	.66	.45	.57	.69	.51	.65
China (Traditional Chinese)	.74	.60	.56	.64	.65	.51	.65	.65	.50	.62
Denmark (Danish)	.83	.79	.68	.64	.74	.80	.76	.75	.67	.75
Finland (Finnish)	.86	.78	.59	.69	.73	.76	.78	.72	.45	.74
France (European French)	.78	.65	.56	.58	.57	.59	.67	.55	.61	.72
Germany (German)	.76	.63	.51	.62	.63	.67	.68	.64	.45	.71
Greece (Greek)	.82	.74	.64	.61	.67	.65	.73	.65	.64	.73
Ireland (European English)	.82	.75	.61	.63	.69	.79	.75	.67	.70	.73
Italy (Italian)	.78	.74	.60	.57	.66	.69	.65	.66	.57	.70
Mexico (Latin American Spanish)	.75	.66	.67	.52	.51	.62	.66	.58	.45	.66
Netherlands (Dutch)	.83	.77	.58	.64	.67	.70	.71	.71	.52	.75
Norway (Norwegian)	.81	.75	.63	.66	.70	.73	.76	.73	.55	.76
Portugal (European Portuguese)	.82	.68	.60	.59	.69	.74	.64	.64	.72	.73
South Africa (Afrikaans)	.86	.79	.74	.70	.74	.84	.77	.71	.75	.79
South Africa (North American English)	.84	.69	.74	.71	.74	.77	.70	.65	.72	.72
Spain (European Spanish)	.74	.70	.61	.62	.63	.64	.74	.65	.48	.69
Sweden (Swedish)	.83	.77	.57	.62	.64	.67	.74	.64	.47	.70
Median	.82	.74	.61	.64	.69	.73	.74	.66	.61	.73

CONTINUED

Internal Consistency of Global Step II™ Facet Scales

The alpha coefficients of the Step II continuous scores are shown in table 8.6 for the global sample and for the 23 country/language samples that compose it. In the global sample, the alpha coefficients range from .59 for the Critical–Accepting scale to .82 for the Initiating–Receiving scale, with an overall median alpha coefficient of .72. As shown in the table, the Step II facet scales with the highest median alpha coefficients across all the country/language samples are Initiating–Receiving (.82), Logical–Empathetic and Scheduled–Spontaneous (both .80), and Planful–Open–Ended (.78). The facet scales with the lowest median alpha coefficients across all the samples are Critical–Accepting (.54) and

Gregarious–Intimate, Experiential–Theoretical, and Questioning–Accommodating (all .61). The samples with the highest median alpha coefficients across all 20 Step II facet scales are South Africa (Afrikaans) (.77), Australia and Brazil (both .76), and US, Canada (North American English), and Denmark (all .75). The country/language samples with the lowest median alpha coefficients are Traditional Chinese (.63), Simplified Chinese (.64), France (European French) (.64), and Canada (Canadian French) (.65).

Table 8.7 shows the internal consistency reliability of the Global Step II facet scales by gender. As with the Step I preference scales, women evidence just slightly higher alpha coefficients (median = .73) on all but three

Table 8.6 | Internal consistency reliability of the MBTI® global Step II™ facet scales in the global and country/language samples *CONTINUED*

Sample	Logical– Empathetic	Reasonable– Compassionate	Questioning– Accommodating	Critical– Accepting	Tough– Tender	Systematic– Casual	Planful– Open-Ended	Early Starting– Pressure-Prompted	Scheduled– Spontaneous	Methodical– Emergent	Median
Global sample	.80	.76	.62	.59	.73	.76	.79	.65	.80	.64	.72
Large country/language samples											
Australia (North American English)	.82	.78	.62	.65	.78	.81	.84	.66	.85	.68	.76
Canada (North American English)	.83	.78	.67	.65	.76	.79	.81	.64	.83	.67	.75
United Kingdom (European English)	.79	.75	.63	.62	.76	.78	.81	.64	.83	.66	.73
United States (North American English)	.82	.80	.64	.64	.77	.76	.80	.63	.82	.66	.75
Moderate-size country/language samples											
Brazil (Brazilian Portuguese)	.87	.76	.68	.55	.66	.80	.82	.78	.80	.70	.76
Canada (Canadian French)	.68	.57	.51	.53	.64	.74	.66	.49	.76	.56	.65
China (Simplified Chinese)	.67	.62	.48	.39	.51	.67	.55	.49	.67	.54	.64
China (Traditional Chinese)	.74	.62	.43	.32	.63	.65	.63	.49	.66	.46	.63
Denmark (Danish)	.80	.75	.65	.67	.81	.80	.80	.67	.82	.67	.75
Finland (Finnish)	.83	.73	.61	.55	.78	.81	.73	.64	.78	.58	.73
France (European French)	.73	.64	.52	.45	.70	.75	.71	.64	.75	.54	.64
Germany (German)	.79	.72	.61	.52	.75	.72	.74	.59	.70	.55	.66
Greece (Greek)	.85	.79	.39	.47	.65	.83	.78	.71	.84	.66	.69
Ireland (European English)	.81	.78	.64	.65	.77	.79	.83	.64	.85	.65	.74
Italy (Italian)	.80	.67	.52	.46	.68	.68	.73	.60	.72	.50	.67
Mexico (Latin American Spanish)	.82	.76	.45	.42	.67	.71	.75	.60	.71	.60	.66
Netherlands (Dutch)	.80	.75	.66	.66	.74	.82	.78	.77	.80	.53	.73
Norway (Norwegian)	.77	.71	.55	.47	.68	.75	.78	.70	.76	.64	.72
Portugal (European Portuguese)	.81	.64	.57	.39	.70	.75	.77	.69	.72	.60	.69
South Africa (Afrikaans)	.84	.76	.67	.61	.77	.81	.86	.81	.87	.71	.77
South Africa (North American English)	.82	.81	.64	.61	.80	.80	.84	.72	.85	.71	.73
Spain (European Spanish)	.79	.68	.44	.43	.66	.68	.79	.55	.70	.60	.66
Sweden (Swedish)	.74	.71	.53	.54	.67	.77	.81	.61	.80	.67	.67
Median	.80	.75	.61	.54	.70	.77	.78	.64	.80	.64	

Note: Global sample, $N = 16,773$.

of the facets than do men (median = .71). On the three exceptions, the alpha coefficients are the same.

Table 8.8 shows the internal consistency reliability of the Global Step II facet scales by age group. As with the Step I preference scales, the youngest group in the global sample evidenced the lowest reliability, and the alpha coefficients generally increase slightly with age but then drop off considerably in the 75-and-over group. Two of the changes in reliability by age stand out. First, the largest increase from one age group to another occurs on the T–F facets when moving from the under-18 age group to the 18–24 group: the median alpha coefficient (not shown in the table) for the former is .55, and for the latter it is .69. This finding conforms to the prediction that younger groups would not yet have well-developed

judgment. Second, the largest decrease is on the J–P facet scales when moving from the 65–74 age group, with a median alpha of .79, to the 75-and-over group, with a median alpha of .65. All the J–P facets show a drop in internal consistency reliability across these two age groups, with the largest decrease on Early Starting–Pressure-Prompted (from .69 to .52). Additional research to explore the cause of these changes is needed.

Type theory predicts less consistent responding and therefore lower reliabilities for younger people than for older ones, and reliability studies confirm this expectation.

Comparison of the MBTI Step II facet scales with the scales or subscales of other instruments is based on the North American English sample compiled in the United States. The mean alpha coefficient in this US sample across the 20 facet scales was .75. For the subscales of the 16PF, the mean alpha was .74 (Conn & Rieke, 1994); for the subscales of the NEO-PI-3 Form S and PI-R (excluding the Neuroticism subscales), the mean alpha coefficients were .75 and .71, respectively (McCrae & Costa, 2010). The internal consistencies of the MBTI Global Step II facet scales are similar to those of the subscales of other well-known personality assessments.

The internal consistency reliabilities of the MBTI Global Step II facet scales are comparable to those of the subscales of other well-known personality assessments.

Table 8.7 | Internal consistency reliability of the MBTI® Global Step II™ facet scales by gender in the global sample

Global Step II™ facet scale	Cronbach's alpha	
	Men (n = 7,771)	Women (n = 9,002)
E–I facet scales		
Initiating–Receiving	.80	.84
Expressive–Contained	.70	.76
Gregarious–Intimate	.62	.62
Active–Reflective	.62	.66
Enthusiastic–Quiet	.68	.71
S–N facet scales		
Concrete–Abstract	.72	.75
Realistic–Imaginative	.71	.74
Practical–Conceptual	.65	.68
Experiential–Theoretical	.69	.69
Traditional–Original	.71	.73
T–F facet scales		
Logical–Empathetic	.77	.80
Reasonable–Compassionate	.73	.76
Questioning–Accommodating	.58	.62
Critical–Accepting	.54	.59
Tough–Tender	.72	.72
J–P facet scales		
Systematic–Casual	.75	.77
Planful–Open-Ended	.77	.81
Early Starting–Pressure-Prompted	.64	.67
Scheduled–Spontaneous	.79	.81
Methodical–Emergent	.62	.66
<i>Median</i>	.71	.73

Note: N = 16,773.

As mentioned earlier, the size of an internal consistency reliability estimate is partly a function of the length of the scale—longer scales will generally have higher reliabilities than shorter scales, all else being equal. Fortunately, it is possible to determine the effect of scale length by applying the Spearman-Brown prophecy formula. Using the average length of the four Global Step I preference scales (23 items), the number of items on each Global Step II facet scale, and the internal consistency reliabilities of the Step II facet scales in the global sample, what the reliability of the facet scales would be if they were all the same length as the overall preference scales can be estimated. The average reliability of facet scales would be .88, which is exactly the same as the average reliability of the preference scales in the global sample.

The alpha coefficients of the MBTI Step I and Step II scales in the global sample are comparable to or greater than those of the scales or subscales of other mainstream personality instruments. As predicted, there was variation in reliability estimates across country/language samples, but, again, the coefficients compare favorably with those of other assessments. The exceptions are the two Chinese samples, which generally exhibit the lowest internal consistency and stability. As predicted, the coefficients of age groups believed to have more command of perception and judgment are higher than those of age groups expected to have less command, that is, young adults and students.

TEST-RETEST RELIABILITY ESTIMATES

Test-retest reliability is an estimate of the stability of the measures of a particular construct in a particular group over time. To estimate the test-retest reliability of the MBTI continuous preference or facet scores, the scores from two different administrations are correlated. To accomplish this, the global research version of the MBTI assessment was administered twice to respondents who agreed to participate; the second time was after a target interval of 1 to 6 weeks (although some respondents took up to 15 weeks).

A high test-retest coefficient indicates that the group's responses to items are consistent over the time period between the two administrations. In addition, for the four preference pairs, reliabilities that reflect their stability are also reported, since MBTI Step I scale scores are assumed to reflect underlying preferences and the major interest of most MBTI respondents is the consistency of their reported type. Similarly, in regard to their Global Step II results, respondents are interested in knowing whether they will score toward the same pole (or in the midzone) of the facet scales over time. Reliabilities in this section include the following:

- Correlations of continuous scores for Step I and Step II scales

Table 8.8 | Internal consistency reliability of the MBTI® Global Step II™ facet scales by age group in the global sample

Global Step II™ facet scale	Cronbach's alpha							
	<18 years (n = 101)	18–24 years (n = 2,526)	25–34 years (n = 3,665)	35–44 years (n = 3,622)	45–54 years (n = 3,283)	55–64 years (n = 2,092)	65–74 years (n = 1,263)	≥75 years (n = 221)
E–I facet scales								
Initiating–Receiving	.77	.79	.80	.83	.84	.84	.82	.77
Expressive–Contained	.71	.68	.73	.74	.75	.77	.75	.65
Gregarious–Intimate	.57	.59	.63	.64	.63	.62	.60	.47
Active–Reflective	.57	.62	.65	.67	.65	.65	.60	.43
Enthusiastic–Quiet	.68	.65	.67	.72	.71	.72	.71	.64
S–N facet scales								
Concrete–Abstract	.68	.68	.70	.74	.76	.80	.76	.72
Realistic–Imaginative	.57	.64	.68	.72	.74	.79	.77	.71
Practical–Conceptual	.55	.59	.63	.68	.68	.72	.69	.51
Experiential–Theoretical	.58	.65	.66	.69	.69	.74	.72	.70
Traditional–Original	.57	.62	.67	.73	.76	.79	.77	.67
T–F facet scales								
Logical–Empathetic	.76	.77	.78	.81	.83	.82	.80	.79
Reasonable–Compassionate	.55	.71	.73	.76	.79	.79	.78	.72
Questioning–Accommodating	.53	.56	.62	.62	.64	.64	.62	.52
Critical–Accepting	.41	.52	.55	.60	.61	.64	.62	.59
Tough–Tender	.66	.69	.70	.75	.76	.77	.76	.66
J–P facet scales								
Systematic–Casual	.68	.73	.75	.77	.77	.79	.79	.65
Planful–Open-Ended	.71	.74	.77	.81	.81	.81	.80	.66
Early Starting–Pressure-Prompted	.51	.61	.62	.67	.66	.68	.69	.52
Scheduled–Spontaneous	.73	.77	.79	.81	.81	.82	.81	.73
Methodical–Emergent	.58	.61	.61	.67	.66	.66	.66	.55
<i>Median</i>	.58	.65	.68	.73	.75	.77	.76	.66

Note: N = 16,773.

- The percentage of cases assigned the same letter (direction of preference) on retest
- The percentage of cases reporting the same four preferences on retest (i.e., the same type), three preferences, two preferences, one preference, or no preferences
- The percentage of cases occurring in the same interpretive category on retest for the Step II facets

Test-Retest Reliability of Global Step I™ Continuous Scores

Table 8.9 shows test-retest correlations in two subsets of participants in the global sample who were administered the global research version of the MBTI assessment at two different times, as well as the composite of these two samples. Table 8.10 shows test-retest correlations for 17 country/language samples over the entire test interval.

For the E–I scale, the correlations range from .67 for the Traditional Chinese sample to .91 for the Norway sample, with a median of .87. For the S–N scale, the range is from .59 for the Simplified Chinese sample to .87 for the US, Norway, and Finland samples, with a median of .83. For the T–F scale, the range is from .51 for the Simplified Chinese sample to .88 for the Germany and Denmark samples, with a median of .81. For the J–P scale, the range is from .59 for the Traditional Chinese sample to .87 for the Netherlands sample, with a median of .81.

When test-retest coefficients of the MBTI Global Step I assessment are being compared with those of other personality assessments, some caution should be employed due to differences in the kinds of samples, the sample sizes, and the time intervals between administrations of the various assessments. For this purpose, the coefficients from the US sample provide the best comparison, because most of the instruments

Table 8.9 | Test-retest correlations of the MBTI® Global Step I™ preference scales in the global sample

Sample (interval)	<i>n</i>	E-I	S-N	T-F	J-P
Global (≤6 weeks)	588	.86	.82	.81	.81
Global (7–15 weeks)	1,133	.86	.83	.83	.81
Combined global (≤15 weeks)	1,721	.86	.83	.82	.81

Note: Retest portion of global sample, *n* = 1,721.

being compared are also based on US samples and North American English. The mean test-retest coefficient over the four preference scales in this sample is .87 over an interval of up to 15 weeks. For the NEO-FFI scales (excluding the Neuroticism subscales) over a 12-week period, the mean test-retest coefficient is .79; for the NEO-PI-R, it is .83 in a small sample (*N* = 132) over a very short interval of 1 week (McCrae & Costa, 2010). For the *Birkman Method* scales, the mean test-retest coefficient is .84 over a 2-week period (Birkman et al., 2008). The MBTI Global Step I assessment has test-retest coefficients higher than those of many other personality instruments over a longer time interval.

Test-retest reliabilities for the MBTI Global Step I assessment exceed those of many other personality instruments—and over a longer time interval between assessment administrations.

A number of other studies of the internal consistency of the Step I preference scales are worth mentioning here, even though they were conducted on prior Step I forms of the MBTI assessment. Capraro and Capraro (2002) conducted a meta-analytic reliability study for Form M. The alpha coefficients in the reviewed studies range from .55 to .97. The lowest alpha (.55) is found on the T–F scale in a sample of Italian women who are heterogeneous with respect to educational background and job experience. The highest alpha (.97) is for a sample of 343 managers from six continents (the preference scale on which this alpha was achieved was not reported). The average alpha over the four preference scales is .82. Capraro and Capraro concluded that, overall, the MBTI assessment tends to yield acceptable reliabilities for the preferences.

Salter, Forney, and Evans (2005) used a generalizability theory (GT) approach to arrive at reliability estimates. GT is useful because it can control for multiple sources of error and makes fewer assumptions about the nature of the data than does the classical approach. The procedure yields the generalizability coefficient. The

Table 8.10 | Test-retest correlations of the MBTI® Global Step I™ preference scales by country/language sample

Sample	<i>n</i>	E-I	S-N	T-F	J-P
Large country/language samples					
Australia (North American English)	92	.87	.82	.87	.81
Canada (North American English)	68	.87	.79	.80	.72
United Kingdom (European English)	88	.88	.81	.85	.85
United States (North American English)	236	.88	.87	.87	.85
Moderate-size country/language samples					
China (Simplified Chinese)	96	.78	.59	.51	.62
China (Traditional Chinese)	89	.67	.63	.58	.59
Denmark (Danish)	81	.89	.86	.88	.83
Finland (Finnish)	133	.90	.87	.87	.85
France (European French)	80	.86	.83	.77	.70
Germany (German)	82	.87	.73	.88	.80
Greece (Greek)	64	.85	.86	.78	.77
Italy (Italian)	81	.85	.83	.81	.77
Netherlands (Dutch)	150	.88	.86	.85	.87
Norway (Norwegian)	82	.91	.87	.79	.86
Portugal (European Portuguese)	87	.86	.85	.79	.81
Spain (European Spanish)	115	.77	.79	.75	.77
Sweden (Swedish)	97	.90	.81	.81	.83
Median	—	.87	.83	.81	.81

Note: Retest portion of global sample, *n* = 1,721. Coefficients are all for test-retest period ≤15 weeks. Retest data were not collected for Brazil (Brazilian Portuguese), Canada (Canadian French), Ireland (European English), Mexico (Latin American Spanish), South Africa (Afrikaans), and South Africa (North American English). See table 7.2 for an overview of the data collection process.

simplest GT design (individuals by items on a scale) produces a reliability estimate that is equal to Cronbach’s alpha. However, in a test-retest study such as this, the design also includes in the analysis the measurement occasion or time. For continuous scores, the estimates for occasions (time) are from .71 for T–F to .80 for J–P. Based on their results, the authors concluded, “If the goal of the MBTI is to help individuals to become aware of their ‘true type’ dispositions, which should remain relatively stable over time, then our results seem consistent with that objective” (p. 217).

Test-Retest Reliability of Global Step I™ Preference Scales

Table 8.11 shows the test-retest percentages of agreement for the four preference scales in the two test-retest samples and the composite from the global sample. It is interesting that the agreement rates are somewhat higher for the longer time period. The T–F and J–P scales show slightly lower stability coefficients.

Table 8.12 shows the percentages of individuals reporting four, three, two, one, or no preferences the same over the two time intervals and in the composite. If the Global Step I assessment assigned people to whole types randomly, the probability of obtaining the same whole type (i.e., coming out the same type) at test and retest by chance would be 1 in 16, or 6.25%. The actual test-retest agreement rates are considerably higher than what would be expected by chance. Most of the changes that occurred over time were a change in only one preference (i.e., three of the four preferences composing the whole type remained the same over the time interval). In the composite sample, 84% of individuals changed either no preferences or one preference over the period.

While significant improvements over chance have been observed, the test-retest agreement will never reach 100%. One of the factors limiting test-retest agreement over time for whole types (i.e., all four preferences the same at test and retest) is that whole types are arrived at by measuring the four preference scales separately. Calculation of test-retest agreement on combinations of scales, such as is done with MBTI scales, is exceedingly rare in personality psychology and is a very stringent test. Scales of other instruments are almost always evaluated separately. As discussed above, responses to the items on each scale are subject to measurement error and to normal fluctuations in the behavior and characteristics of individuals that may affect how those individuals self-report on certain items at certain times. When these factors are aggregated across the four preference scales, it creates a ceiling on the upper limit of agreement over time. For example, if the categorical test-retest reliabilities of the preference scales for the samples shown in table 8.11 are considered as independent measures and multiplied together to arrive at an expectation of what

Table 8.11 | Test-retest percentage agreement of MBTI® Global Step I™ preference pairs in the global sample

Sample (interval)	n	Agreement of preference pairs at retest (%)			
		E–I	S–N	T–F	J–P
Global (≤6 weeks)	588	83	84	77	77
Global (7–15 weeks)	1,133	85	86	80	80
Combined global (≤15 weeks)	1,721	84	85	79	79

Note: Retest portion of global sample, n = 1,721.

Table 8.12 | Percentage of individuals with preferences the same at retest for the MBTI® Global Step I™ assessment in the global sample

Sample (interval)	n	Number of preferences the same at retest (%)				
		4	3	2	1	0
Global (≤6 weeks)	588	45	36	14	4	1
Global (7–15 weeks)	1,133	49	36	11	3	1
Combined global (≤15 weeks)	1,721	48	36	12	3	1

Note: Retest portion of global sample, n = 1,721.

the agreement for whole types should be over time, the results for the three samples are 41%, 47%, and 45%, respectively. These are very close to the actual figures of 45%, 49%, and 48% for the same samples, as shown in table 8.12. To achieve test-retest agreement on whole types over time in, say, the mid-80% range, the test-retest reliabilities of the individual scales would all have to be over 96%. Expecting such a result over any time period is unrealistic for scales that measure such complex and multifaceted constructs as the MBTI preferences, just as it would be for any scale measuring a meaningful multifaceted psychological construct.

Most studies in personality testing estimate the reliability of scales separately, whereas MBTI studies examine combinations of the four MBTI preference scales, that is, the stability of whole types over time. For the global sample, 84% changed either no preferences or only one preference at retest.

Table 8.13 extends the analysis of the percentage of individuals with preferences the same at retest to 17 country/language samples using the time interval of less than or equal to 15 weeks. The country/language samples with the highest percentage of individuals yielding the same whole type (i.e., all four preferences the same at test and retest) are Australia (60%), Denmark and

Finland (both 57%), Norway (54%), and the Netherlands (53%). The samples with the lowest whole-type test-retest agreement are the Simplified Chinese (25%) and Traditional Chinese (33%) samples. Considering individuals who changed either no preferences or only one preference over the time interval, the range is from 92% for the Denmark and Finland samples to 66% for the Simplified Chinese sample. The mean is 83%.

A number of other test-retest studies are worth mentioning here because they illustrate a particular method, sample, or retest interval of interest. Salter, Forney, and Evans (2005) conducted a study on the stability of MBTI Step I scores using longitudinal configural frequency analysis (CFA) across three administrations of the MBTI assessment. The subjects were 231 graduate students who took Form G three times. Students received their MBTI results and an interpretation in the first semester. Using CFA, Salter, Forney, and Evans examined the eight possible categorical results for each of the four preference scales over the three administrations. For example, for the Thinking–Feeling scale, one possible (inconsistent) outcome would be to report a preference for Feeling on the first administration, a preference for Thinking on the second administration, and then a preference for Feeling again on the third administration, a pattern that can be designated as F–T–F. The results of this study

indicate that all eight perfectly consistent preference patterns (e.g., E–E–E) show highly statistically significant stability. Moreover, inconsistent patterns for three of the preferences—Sensing (e.g., S–S–N), Thinking (e.g., T–F–T), and Perceiving (e.g., P–P–J)—are statistically underrepresented. Across all three administrations, 31% of profiles did not change at all and 42% changed only one preference, so 74% changed no preferences or only one preference over a three-year period.

In another study, Salter, Evans, and Forney (1997) used log-linear analysis to examine the stability of the dominant function (favorite process) as derived from MBTI Step I results. The subjects were 99 graduate students who were administered Form G 20 months apart. These students were participating in counseling training that the authors believed to be “biased” toward Intuition, Feeling, and Perceiving approaches and possibly subject to confounding effects of environmental influences (also see Hammer & Treeger, 1984). The subjects received their Step I results and an interpretation of them between the first and second administrations. Test-retest correlations for preference scores range from .69 (T–F) to .77 (E–I and J–P). Log-linear analysis was used to examine the categorical results. Lambda (λ) estimates range from .59 (J–P) to .86 (S–N). Examination of standardized residuals shows that Extraversion, Feeling, and Judging are relatively less stable than their

Table 8.13 | **Percentage of individuals with number of preferences the same at retest for the MBTI® Global Step I™ assessment by country/language sample**

Sample	n	Number of preferences the same at retest (%)				
		4	3	2	1	0
Large country/language samples						
Australia (North American English)	92	60	29	5	5	0
Canada (North American English)	68	50	32	10	4	3
United Kingdom (European English)	88	47	33	17	2	1
United States (North American English)	236	51	39	9	2	0
Moderate-size country/language samples						
China (Simplified Chinese)	96	25	41	26	7	1
China (Traditional Chinese)	89	33	37	22	6	2
Denmark (Danish)	81	57	35	6	2	0
Finland (Finnish)	133	57	35	3	3	2
France (European French)	80	49	38	10	4	0
Germany (German)	82	50	34	15	1	0
Greece (Greek)	64	47	33	16	3	2
Italy (Italian)	81	41	42	14	4	0
Netherlands (Dutch)	150	53	37	8	2	0
Norway (Norwegian)	82	54	32	13	1	0
Portugal (European Portuguese)	87	43	36	16	5	1
Spain (European Spanish)	115	43	37	15	4	1
Sweden (Swedish)	97	49	36	12	2	0

Note: Retest portion of global sample, $n = 1,721$. Time between test-retest: ≤ 15 weeks. Retest data were not collected for Brazil (Brazilian Portuguese), Canada (Canadian French), Ireland (European English), Mexico (Latin American Spanish), South Africa (Afrikaans), and South Africa (North American English). See table 7.2 for an overview of the data collection process.

opposites. In terms of dominant functions, the subjects whose dominant is Intuition or Feeling have the highest stability, those whose dominant is Sensing have somewhat less, and those whose dominant is Thinking have the lowest stability. The authors hypothesize that the lower stability of Thinking is partly a reflection of Myers' hypothesis that T–F judgment is the last to develop and therefore the least reliable in test-retest studies with young people. They also attribute the lower Thinking stability partly to environmental influence; the participants were graduate students in a training program that was believed to influence them toward Feeling responses.

Another study deserves specific mention here, even though it also used Form G, because it helps answer a question often asked by participants in workshops. This was a study of the effect of mood on MBTI reliabilities conducted by Howes and Carskadon (1979). The authors administered Form G of the MBTI assessment to introductory psychology student volunteers. Mood changes were then effected by asking three groups of students to complete one of three 50-item self-report questionnaires. The mood-elevating form contained positive, optimistic self-statements that most people could honestly agree with; the mood-depressing form contained negatively worded statements; a neutral form contained bland, nonpersonal attitude statements. A post-treatment measure of mood demonstrated that completing these forms had the intended effect. The results of a second MBTI administration showed that mood changes did not significantly affect test-retest MBTI correlations or percentage of agreement on the preference scales. There is no reason to believe that results would be any different for the Global Step I assessment, although that remains to be tested.

A Form G study found no effect of mood on test-retest reliability.

Capraro and Capraro (2002) reviewed test-retest coefficients in their meta-analytic reliability study of Form M. The overall mean test-retest coefficient across the four preference scales is .81. The coefficients for the specific preference scales are .84 for E–I and S–N, .76 for T–F, and .82 for J–P. The studies included in the meta-analysis used different time intervals.

A test-retest study by Johnson (1992a) also deserves mention. Although these data are from a small sample ($n = 74$) and from a prior form of the MBTI assessment, the results are informative because of the long period (30 months) between test and retest. The test-retest correlations of the continuous scores over a two-and-a-half-year period range from .62 for T–F to .83 for S–N, with a mean of .77.

Finally, another test-retest study is of interest for both historical and theoretical reasons. Isabel Myers administered one of the first forms of the MBTI assessment to the 87 members of the Swarthmore High School class of 1943. At the 50th reunion of this class, Katharine Myers (personal communication) administered Form G to 39 of these same people. Over this 50-year period, 8 people (21%) had the same type, 13 (33%) had changed one preference, 16 (41%) had changed two preferences, 2 (5%) had changed three preferences, and no one in the group had changed all four preferences. Over a 50-year interval, then, 54% changed either no preferences or just one preference. One interpretation of these data is that personality preferences are fairly stable over long periods of time but there are some changes. Note, however, that changes over time observable in these data were compounded by changes in both the items and the scoring weights on the MBTI forms that were used 50 years apart, so it is likely that these test-retest percentages overestimate the changes in the people themselves.

In conclusion, test-retest reliabilities of the MBTI Step I continuous scores, categorical preferences, and, in one study, dominant functions, show good consistency over weeks and months and acceptable consistency over years. When subjects report a change in type, it is most likely to occur in only one preference. Over very long periods of time, as the Swarthmore sample demonstrates, there is evidence of both stability and change, as would be expected.

Test-Retest Reliability of Global Step II™ Continuous Scores

Test-retest reliability estimates are also available for the Global Step II assessment. Table 8.14 shows test-retest reliability estimates for the global sample and for 17 country/language samples. The interval between administrations varied from 1 week to 15 weeks.

As shown in the table, the median stability coefficients in the global sample range from .58 for Experiential–Theoretical to .83 for Initiating–Receiving. The median coefficient in the global sample was .75. The Step II facet scales with the highest median coefficients across all the country/language samples are Initiating–Receiving (.83), Expressive–Contained and Traditional–Original (both .77), and Enthusiastic–Quiet, Realistic–Imaginative, Logical–Empathetic, and Systematic–Casual (all .76). The facet scales with the lowest median coefficients across all the country/language samples are Experiential–Theoretical (.58) and Methodical–Emergent (.59). The country/language samples with the highest median stability coefficients across all 20 Step II facet scales are the US sample (.79) and the Denmark and Finland samples (both .78). The samples with the lowest medians

across the 20 facet scales are the Simplified Chinese sample (.52) and the Traditional Chinese sample (.53).

Test-retest results for the facet scales in the global sample were examined by age group. These data are shown in table 8.15. This analysis allows a comparison of the stability of the Step II facet scales in younger versus more mature samples. It is expected, for the reasons stated above regarding the internal consistency reliability analysis, that the stability of the scores in the samples of younger participants will be lower than those in samples of older participants. As can be seen in the table, the prediction holds true, as the stability coefficients for every facet scale increased with age. The median stability coefficient for the youngest group is .62, and for the oldest group it is .80. The largest differences in the coefficients (>.20) from the youngest to the oldest are for Active–Reflective, Realistic–Imaginative, Traditional–Original, Early Starting–Pressure–Prompted, and Methodical–Emergent. The largest change in the coefficients between contiguous age groups occurs between the 25–34 (median = .68) and 35–44

(median = .76) groups. It seems that as people move into middle age, their facet scores become more stable. This suggests that Myers' developmental theory, which states that by middle age most people will have both a well-developed dominant function and a balancing auxiliary function, also manifests in the facets.

When comparing the stability coefficients of the Global Step II facet scales with those of the scales or subscales of other instruments, the US sample again seems the most appropriate. The mean test-retest coefficient across the 20 facet scales in the US sample is .78 (for up to a 15-week interval). The mean for the 16PF scales over an 8-week period is .70 (Conn & Rieke, 1994). For the NEO (excluding the Neuroticism subscales) in a small sample ($N = 132$) of students over a 1-week period, it was .83 (McCrae & Costa, 2010).

Test-retest stability coefficients of the Global Step II facet scales compare favorably to those of the scales or subscales of other instruments and over longer time intervals.

Table 8.14 | Test-retest reliability of MBTI® Global Step II™ continuous scores for the global and country/language samples

Sample	Initiating–Receiving	Expressive–Contained	Gregarious–Intimate	Active–Reflective	Enthusiastic–Quiet	Concrete–Abstract	Realistic–Imaginative	Practical–Conceptual	Experiential–Theoretical	Traditional–Original
Global sample	.82	.78	.73	.75	.76	.76	.76	.71	.62	.75
Large country/language samples										
Australia (North American English)	.84	.74	.73	.71	.74	.79	.76	.65	.55	.71
Canada (North American English)	.85	.80	.76	.82	.76	.72	.72	.63	.58	.68
United Kingdom (European English)	.78	.85	.74	.66	.83	.71	.77	.67	.64	.77
United States (North American English)	.84	.83	.77	.78	.77	.81	.81	.76	.74	.79
Moderate-size country/language samples										
China (Simplified Chinese)	.65	.63	.64	.66	.60	.47	.58	.57	.41	.51
China (Traditional Chinese)	.62	.52	.61	.62	.52	.53	.62	.52	.54	.51
Denmark (Danish)	.83	.83	.70	.80	.77	.83	.82	.78	.69	.80
Finland (Finnish)	.89	.77	.77	.82	.73	.79	.81	.75	.67	.81
France (European French)	.85	.73	.54	.72	.64	.65	.57	.49	.49	.77
Germany (German)	.82	.65	.71	.69	.77	.66	.64	.73	.46	.72
Greece (Greek)	.82	.70	.68	.61	.76	.71	.83	.69	.76	.67
Italy (Italian)	.84	.78	.76	.68	.69	.72	.62	.74	.67	.82
Netherlands (Dutch)	.83	.86	.73	.74	.72	.78	.79	.72	.62	.82
Norway (Norwegian)	.81	.87	.74	.86	.78	.75	.81	.79	.56	.79
Portugal (European Portuguese)	.81	.73	.67	.77	.82	.77	.76	.72	.67	.80
Spain (European Spanish)	.76	.66	.73	.69	.77	.79	.71	.66	.39	.63
Sweden (Swedish)	.86	.88	.68	.80	.82	.81	.74	.68	.52	.75
Median	.83	.77	.73	.72	.76	.75	.76	.69	.58	.77

CONTINUED

Test-Retest Reliability of Global Step II™ Interpretive Categories

Although the test-retest coefficients of Step II continuous scores are of interest, the primary units of *interpretation* for Step II facet scales are the categories (see chapter 4). The most important piece of information for understanding an individual's result on a given facet scale is whether he or she scored toward one of the poles or in the midzone. It follows, therefore, that test-retest statistics designed to estimate the likelihood that an individual will remain in the same category are important.

Thus, three events are of interest in determining the stability of the Step II facet scales: (1) an individual's score remains in the same category (left pole, right pole, or midzone) at retest; (2) the individual's score moves from one of the poles toward the midzone or from the midzone toward one of the poles at retest; (3) the individual's score crosses from one pole to the opposite

pole at retest. Obviously, the frequency of the first event (remaining in the same category) should be high and the frequency of the third event (crossing over from one pole to its opposite) should be low. The frequency of movement into or out of the midzone should fall between these two.

Table 8.16 shows the frequency of these three events in a subset of the global sample over intervals that ranged from 1 week to 15 weeks. Adults age 25 and older in the sample whose scores remain in the same category at retest display frequencies that range from 61% for Methodical–Emergent to 71% for Initiating–Receiving, with a median of 66%. The two facet scales showing the greatest degree of movement from one pole to its opposite are from the J–P preference pair: Methodical–Emergent (7%) and Early Starting–Pressure–Prompted (6%). The median for those who crossed from one pole to the other is 3%.

Table 8.14 | Test-retest reliability of MBTI® Global Step II™ continuous scores for the global and country/language samples *CONTINUED*

Sample	Logical– Empathetic	Reasonable– Compassionate	Questioning– Accommodating	Critical– Accepting	Tough– Tender	Systematic– Casual	Planful– Open-Ended	Early Starting– Pressure–Prompted	Scheduled– Spontaneous	Methodical– Emergent	Median
Global sample	.78	.74	.73	.70	.73	.77	.74	.70	.76	.61	.75
Large country/language samples											
Australia (North American English)	.79	.75	.85	.75	.82	.80	.75	.76	.79	.59	.75
Canada (North American English)	.70	.72	.78	.68	.73	.70	.62	.63	.68	.71	.72
United Kingdom (European English)	.75	.76	.71	.77	.79	.73	.79	.77	.74	.70	.76
United States (North American English)	.86	.80	.76	.73	.79	.81	.79	.65	.80	.67	.79
Moderate-size country/language samples											
China (Simplified Chinese)	.53	.46	.33	.43	.29	.53	.46	.48	.57	.41	.52
China (Traditional Chinese)	.60	.43	.52	.36	.34	.60	.53	.55	.57	.46	.53
Denmark (Danish)	.85	.81	.74	.78	.78	.76	.75	.78	.84	.59	.78
Finland (Finnish)	.82	.82	.73	.66	.74	.84	.73	.80	.81	.66	.78
France (European French)	.77	.67	.67	.73	.69	.77	.58	.61	.68	.50	.67
Germany (German)	.83	.84	.79	.70	.76	.70	.75	.75	.76	.57	.73
Greece (Greek)	.75	.67	.73	.68	.69	.80	.70	.69	.74	.48	.70
Italy (Italian)	.74	.68	.70	.65	.68	.72	.70	.62	.70	.45	.70
Netherlands (Dutch)	.78	.73	.75	.74	.80	.88	.73	.77	.79	.60	.76
Norway (Norwegian)	.76	.70	.73	.65	.61	.76	.80	.81	.81	.70	.77
Portugal (European Portuguese)	.82	.71	.65	.77	.71	.77	.72	.74	.72	.58	.74
Spain (European Spanish)	.70	.66	.66	.60	.67	.66	.78	.58	.72	.52	.67
Sweden (Swedish)	.76	.76	.72	.63	.66	.77	.80	.63	.79	.63	.76
Median	.76	.72	.73	.68	.71	.76	.73	.69	.74	.59	

Note: Retest portion of global sample $n = 1,721$. Retest data were not collected for Brazil (Brazilian Portuguese), Canada (Canadian French), Ireland (European English), Mexico (Latin American Spanish), South Africa (Afrikaans), and South Africa (North American English). See table 7.2 for an overview of the data collection process.

Table 8.15 | Test-retest of the MBTI® Global Step II™ facet scales by age group in the global sample

Global Step II™ facet scale	Age group					
	18–24 years (n = 240)	25–34 years (n = 329)	35–44 years (n = 357)	45–54 years (n = 357)	55–64 years (n = 293)	65–74 years (n = 151)
E–I facet scales						
Initiating–Receiving	.77	.74	.84	.84	.83	.88
Expressive–Contained	.70	.70	.79	.78	.82	.84
Gregarious–Intimate	.62	.68	.77	.75	.73	.80
Active–Reflective	.62	.70	.73	.82	.76	.84
Enthusiastic–Quiet	.68	.68	.79	.79	.78	.83
S–N facet scales						
Concrete–Abstract	.70	.68	.76	.75	.82	.80
Realistic–Imaginative	.61	.71	.73	.79	.83	.83
Practical–Conceptual	.59	.64	.71	.75	.77	.73
Experiential–Theoretical	.57	.55	.61	.62	.68	.66
Traditional–Original	.58	.61	.78	.80	.82	.80
T–F facet scales						
Logical–Empathetic	.72	.70	.78	.83	.80	.80
Reasonable–Compassionate	.68	.64	.75	.80	.77	.74
Questioning–Accommodating	.60	.64	.80	.77	.73	.80
Critical–Accepting	.62	.59	.74	.76	.72	.75
Tough–Tender	.68	.64	.74	.77	.73	.80
J–P facet scales						
Systematic–Casual	.68	.70	.77	.81	.78	.85
Planful–Open-Ended	.60	.70	.73	.78	.78	.78
Early Starting–Pressure-Prompted	.57	.64	.73	.70	.70	.80
Scheduled–Spontaneous	.66	.69	.76	.79	.79	.82
Methodical–Emergent	.46	.51	.66	.69	.63	.70
<i>Median</i>	.62	.68	.76	.78	.78	.80

Note: Retest portion of global sample, $n = 1,721$. The <18 and ≥ 75 age groups had too few retest cases to be included.

Test-retest frequencies by interpretive category for the younger respondents (ages 18–24) are also shown in table 8.16. While the stability of the scales is quite high over this time period for the younger adults, the frequencies, as expected, are lower than those of the older adults. The frequencies of those remaining in the same category for young adults range from 51% for Methodical–Emergent to 70% for Initiating–Receiving, with a median of 58%. The two facet scales showing the greatest degree of movement from one pole to the other were the same two scales from the J–P preference pair as those in the older adult groups: Methodical–Emergent (15%) and Early Starting–Pressure-Prompted (13%). The median for those who crossed from one pole to the other is 4%. The percentage of crossovers for both of these J–P facet scales is more than double that of the older adults. These results could be interpreted as attempts by younger adults to determine how much order and structure they want or need. These two facets represent specific behaviors in the J–P domain for which

many young people may not yet have determined a preference, perhaps because their environments have not required them to adopt the behaviors associated with the J poles of these facets. The lower internal consistency and stability of the Methodical–Emergent scale may be related to a suggestion discussed in chapter 4 that the use of either a methodical or an emergent approach may be more task or situation dependent than the behaviors associated with some of the other facets.

To further explore these patterns, the facet scales were ranked from highest to lowest crossover percentage and the 10 highest crossovers were examined. It is interesting to note that, among the 10 facet scales with the greatest crossover in the younger sample, 4 are within the J–P preference scale. Newly independent young adults may need to learn unfamiliar judging behaviors while at the same time taking advantage of expanded opportunities for a more spontaneous lifestyle.

Table 8.16 | Test-retest of the MBTI® Global Step II™ facet scales interpretive categories by age group in the global sample

Global Step II™ facet scale	Young adults (18–24 years) (n = 164)			Adults (≥25 years) (n = 1,144)		
	Remaining in same category (%)	Moving to/from midzone (%)	Moving to opposite pole (%)	Remaining in same category (%)	Moving to/from midzone (%)	Moving to opposite pole (%)
E–I facet scales						
Initiating–Receiving	70	30	1	71	28	1
Expressive–Contained	67	29	4	69	29	2
Gregarious–Intimate	63	31	6	64	32	4
Active–Reflective	57	38	5	70	28	2
Enthusiastic–Quiet	64	32	4	68	29	3
S–N facet scales						
Concrete–Abstract	59	38	4	69	30	1
Realistic–Imaginative	57	38	5	68	29	3
Practical–Conceptual	55	37	7	66	30	4
Experiential–Theoretical	59	38	4	64	32	4
Traditional–Original	58	39	3	69	30	1
T–F facet scales						
Logical–Empathetic	63	34	3	68	29	3
Reasonable–Compassionate	57	40	4	66	32	3
Questioning–Accommodating	61	36	3	63	35	1
Critical–Accepting	52	45	4	63	33	3
Tough–Tender	57	37	5	64	32	4
J–P facet scales						
Systematic–Casual	59	40	1	67	31	2
Planful–Open-Ended	55	41	4	64	34	3
Early Starting–Pressure-Prompted	54	33	13	66	27	6
Scheduled–Spontaneous	62	34	4	66	32	1
Methodical–Emergent	51	34	15	61	32	7
<i>Median</i>	58	37	4	66	30	3

Note: Retest portion of global sample, n = 1,721. Percentages within a sample across a row may total more than 100% due to rounding of decimals.

For younger respondents, the facet scales with the most movement from one pole to its opposite on retest were Methodical–Emergent and Early Starting–Pressure-Prompted, and 4 of the 10 scales with the greatest crossover for this sample were from the J–P preference pair.

In the sample of adults age 25 and older, 4 of the 10 facet scales with the highest crossover percentage are within the T–F preference scale and 3 are within the J–P scale. People in business settings often move from Thinking to Feeling, or vice versa, at retest. In the first instance, managers may have received feedback that their Thinking preference manifests in a critical style that becomes a disadvantage in working with colleagues and teams. They then sought to become more oriented toward behaviors associated with the Feeling poles of the facet scales and consequently responded to more items

in the Feeling direction. The crossover from Feeling to Thinking facet poles may be the result of some Feeling managers making a conscious effort to be tougher and more analytical so as to fit better into what are primarily TJ environments. In all these instances, environmental pressures can influence how a person reports his or her preferences on the facets, which would affect test-retest coefficients.

For older adults, 3 of the 10 facet scales with the greatest crossover at retest were from the J–P preference pair, but 4 facet scales were from the T–F preference pair, perhaps due to pressures in the workplace.

POLARITY INDEX

Using group-level statistics such as those described above to estimate the reliability of the Step II facet scales provides no information about individual respondents. The statistics apply only to the mythical average or typical respondent from a specific group of participants of a particular age, ethnic group, gender, and—probably most important for scales that measure type preferences and facets—a certain developmental level. When working with individuals in any setting, however, it is important to know whether the scores on that person's Step II profile are consistent.

For example, it would be helpful to know whether a respondent consistently scored toward the poles of the facet scales. If the individual's responses to the items on a given facet scale consistently point to one of the poles, then the score will fall close to the 5 on either side of the scale. If the individual's responses instead favor one pole about half the time and the opposite pole the other half, then the score will show this by being close to the 0 point. Overall, then, we can tell something about the consistency of a person's preferences for the poles of the facet scales by noting whether the person's scores are mostly close to the 0 point on the scales or instead are distributed near 5 on either side. A profile that has many scores near the 0 point and few near the poles implies less consistency on the facet scales. A profile that has many scores near the poles and few scores near the 0 point implies greater consistency. It is important to note, however, that there may be other reasons besides lack of clarity or consistency that explain midzone scores. See chapter 6 for a discussion of this topic.

Saunders (1989) developed the polarity index to make assessment of individual consistency more rigorous. The polarity index was designed to describe how consistently an individual has scored toward the poles of the facet scales for the profile as a whole. By providing information about an individual profile, the polarity index gives us data that traditional reliability statistics are unable to supply: the consistency of one individual's responses on a single administration of the Step II assessment.

Computation and Meaning of the Polarity Index

The polarity index is computed using the following formula:

$$\text{Polarity index} = 20 \cdot \sqrt{\frac{\sum x_i^2}{20}}$$

where x is the score on each facet scale.

A low polarity index indicates that a respondent has many facet scores near 0, and a high polarity index indicates that a respondent has many facet scores near the poles. Based on the formula above, a person whose score on all

20 Step II facet scales is 0 will have a polarity index of 0, while another whose score on all 20 facet scales is at one of the poles (all 5s) will have an index of 100. An index of 100 is obtained when a person's facet scale scores are all 5 in one direction or a mixture of 5s in *either* direction, with no score falling in between these extremes. A person who scores 0 on half the facet scales and 1 on the other half will have a polarity index of 14. Step II scores for these two people are illustrated in figure 8.1. A person who scores 4 toward either pole on all the facet scales will have a polarity index of 80. Random responses to the MBTI items can produce a polarity index as high as 30.

When examining a polarity index, it is important to understand that despite the fact that the polarity index is summed over all the scales, it does not measure the consistency with which a respondent's scale scores all point toward the *same* overall preference in each separate MBTI preference pair. For example, an individual might have a score of 5 on each of the S–N facet scales, but those 5s might point toward opposite poles on succeeding facet scales—a 5 toward Concrete, Imaginative, Practical, Theoretical, and Traditional. That individual might also have a similar pattern of 5s on all the facet scales within the other three preference pairs. He or she would thus have a polarity index of 100 because all the responses to the items on each scale were at the poles. However, within each overall preference pair, the facet scores would obviously be switching back and forth from one preference to the other. The polarity index does not tell us anything about such alternations between facet poles within the four Step I preference pairs. Alternations of this nature would instead be identified by out-of-preference scores, which are described in chapter 4.

The polarity index does *not* show consistency in choosing facet poles toward one preference or the other within a preference scale—just consistency in choosing scores at *either* pole over all 20 facets.

Polarity Index and the Global Sample

Table 6.6 displays information about the polarity index for each type in the global sample, rank ordered from high to low. The mean polarity index in the global sample was 59.08 ($SD = 12.73$), which indicates that 95% of the participants have a polarity index between 34 and 84. The mean polarity index by type ranges from 51.09 for ESTPs to 65.17 for ISTJs. It is interesting to note that, with just a couple of exceptions (ENFPs and INFPs), individuals with a preference for Judging have a higher polarity index ($M = 61.14$) than do those with a preference for Perceiving ($M = 56.29$).

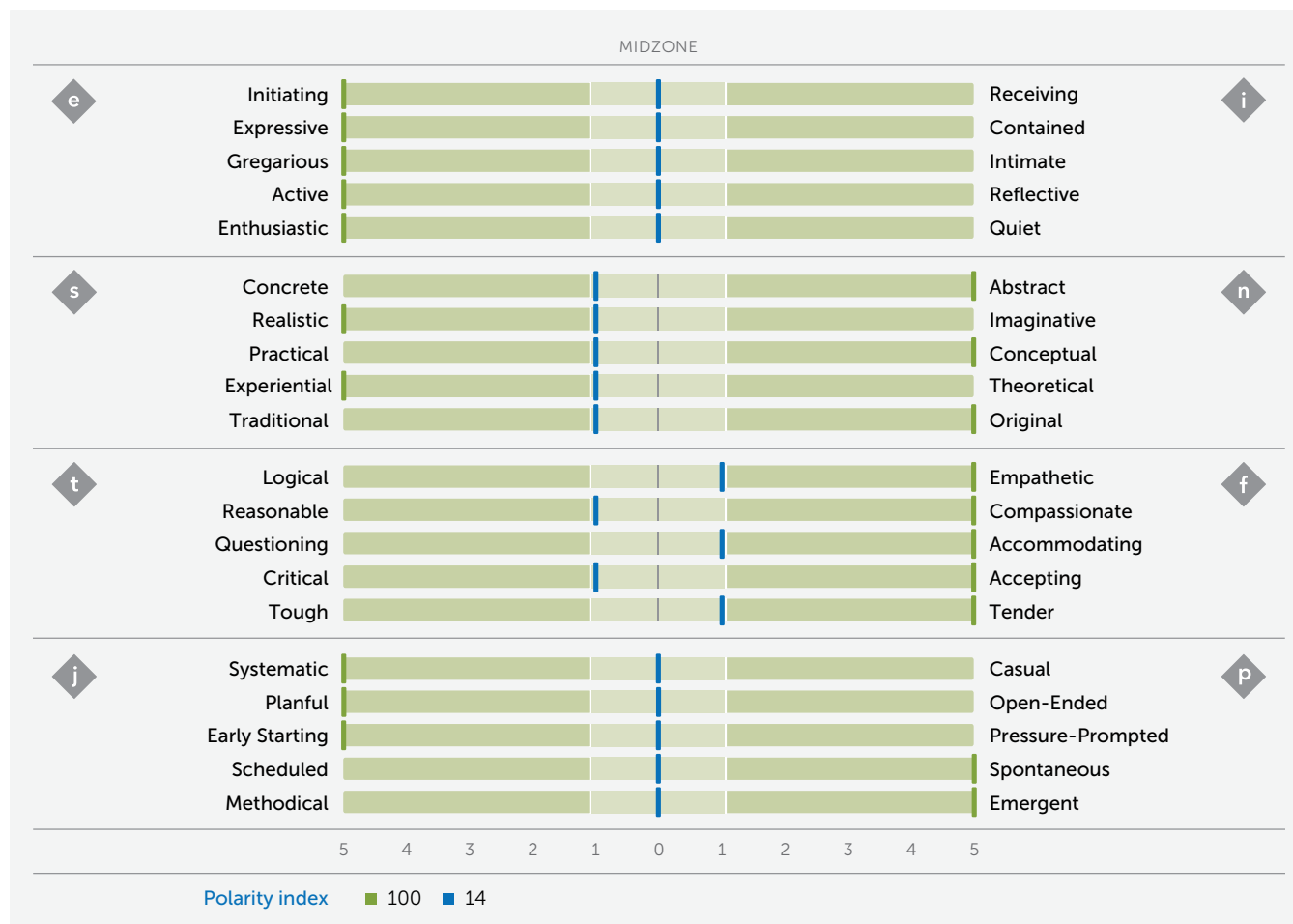


Figure 8.1 | Polarity index for two people

The mean polarity index was also computed for the 23 country/language samples that compose the global sample. These polarity indexes are shown in table 8.17. As indicated in the table, there is little difference in mean polarity index across the four large samples. There is more of a range among the moderate-size samples. In the latter subset, the samples with the highest means are Brazil (63.62) and South Africa (Afrikaans) (63.44). The two samples with the lowest means are Traditional Chinese (52.80) and Simplified Chinese (53.38). Some of these differences may be due to sampling factors, given the smaller size and makeup of these samples.

Gender. The overall difference on the polarity index between men ($M = 57.87$) and women ($M = 60.13$) was examined and found to be negligible. However, among participants who prefer Feeling, women have a higher polarity index ($M = 60.67$) than do men ($M = 56.62$; $d = 0.34$). Women who prefer Feeling, which may be considered a sanctioned preference in many cultures, responded more consistently toward the poles of the Step II facet scales than did men.

Age group. The relationship of the polarity index to age is shown in table 6.7. The groups with the lowest

mean polarity index are the two at the ends of the age distribution, that is, in the 15–18 and 75–88 categories; these are also the groups with the smallest sample sizes. An interesting pattern can be observed in this table. As predicted by type theory, the youngest age group has a relatively lower polarity index, since preferences on constructs measured by at least some of the Step II facet scales are still being formed. There is a slight increase at every age level, perhaps indicating that some preferences have crystallized, until the 65–74 age group is reached, at which point there is a slight decrease. The biggest difference between successive age groups is between the 65–74 group and the 75–88 group ($d = 0.45$). Is the lower polarity index in the oldest group due to transcending opposites, or is this another period of challenge, comparable to adolescence and midlife, during which reevaluation takes place? Much larger samples of older people will be needed to answer this question.

Educational level. As shown in table 8.18, the polarity index increases with higher levels of education. Perhaps education offers people more opportunities to learn about themselves and experience challenges and new environments that promote clarity of preference on the Step II facet scales. The biggest difference in successive

levels of education is between individuals with some high school experience and those who have received a high school diploma. The next-largest difference is between individuals with some college experience and those with an associate degree. Although these differences are small, they raise the interesting question of whether perseverance and goal attainment are related to clarity or consistency of self-knowledge, and what is cause and what effect.

CPI 260® vector 3 scale. The polarity index was also examined using the vector 3 (orientation toward self) scale of the CPI 260 assessment (Gough & Bradley, 2005). This scale is designed to measure seven “levels of realization, or competence, or ego integration,” with Level 7 representing the highest level of actualization.

Table 8.17 | Means and standard deviations of polarity index in the country/language samples

Sample	N	M	SD
Large country/language samples			
Australia (North American English)	776	61.76	12.66
Canada (North American English)	939	60.97	12.85
United Kingdom (European English)	2,831	59.18	13.72
United States (North American English)	3,578	60.53	13.31
Moderate-size country/language samples			
Brazil (Brazilian Portuguese)	839	63.62	11.63
Canada (Canadian French)	176	54.51	13.05
China (Simplified Chinese)	521	53.38	13.05
China (Traditional Chinese)	477	52.80	12.28
Denmark (Danish)	468	60.91	11.15
Finland (Finnish)	524	59.21	10.94
France (European French)	472	55.10	10.77
Germany (German)	440	55.21	11.05
Greece (Greek)	277	58.92	11.09
Ireland (European English)	383	60.36	12.69
Italy (Italian)	458	56.02	12.50
Mexico (Latin American Spanish)	359	56.07	11.09
Netherlands (Dutch)	506	59.31	10.76
Norway (Norwegian)	493	58.69	10.06
Portugal (European Portuguese)	503	57.16	10.64
South Africa (Afrikaans)	505	63.44	11.70
South Africa (North American English)	189	61.40	13.27
Spain (European Spanish)	564	55.64	12.50
Sweden (Swedish)	495	56.50	10.55

Note: Global sample, N = 16,773.

The mean polarity index by level is shown in table 8.19. Overall, the median polarity index is higher for individuals at the four highest levels (60.53) compared to those in the three lowest levels (57.25). Individuals who manifest higher levels of efficacy and realization tend to respond more consistently toward the poles of the Step II facet scales.

Gender has no effect on the polarity index, but age, educational level, and, to some extent, a measure of “self-actualization” all have an influence.

Table 8.18 | Means and standard deviations of polarity index by educational level in the global sample

Educational level	n	M	SD
Some high school	162	55.57	13.76
High school diploma	1,145	59.54	12.37
Trade or technical training	226	58.58	12.76
Some college (no degree)	916	59.51	13.21
Associate degree	282	62.18	13.37
Bachelor's degree	892	61.52	13.92
Master's degree	353	62.21	13.99
Professional degree (e.g., DDS, JD, MD)	70	62.64	14.12
Doctorate (e.g., PhD, EdD)	48	64.40	12.21

Note: Portion of global sample, n = 4,094.

Table 8.19 | Means and standard deviations of polarity index by CPI 260® level of self-realization in the global sample

CPI 260®	n	M	SD
Level 1	198	55.90	13.44
Level 2	226	57.25	11.31
Level 3	293	58.99	12.48
Level 4	462	60.70	12.07
Level 5	320	61.43	11.14
Level 6	146	60.36	11.06
Level 7	83	59.16	11.35

Note: Portion of global sample that completed the CPI 260 assessment, n = 1,728. Level 7 represents the highest level of actualization.

CONCLUSION

Given all the factors described above that could potentially introduce noise into the measurement of preferences in a global sample (e.g., cultural differences, translations) and also in more narrow samples (e.g., developmental differences, environmental influences), it is encouraging that the internal consistency and test-retest reliabilities of the preference scales and facet scales are comparable to, and in some cases greater than, those of the scales of other personality instruments, whose reliability estimates are based on much smaller and more narrow samples.

The internal consistency of the continuous scores of the four MBTI preference scales is quite high in almost all the country/language samples making up the global sample, and it is comparable to that of the scales of other personality assessments. Test-retest reliabilities of the MBTI scales show consistency over time, with levels of

agreement much greater than those expected by chance. When subjects report a change in type, it almost always occurs on only one preference pair, and on scales in which the original probability index was low. The test-retest coefficients of the facet scale continuous scores are also respectable and comparable to those of the subscales of other personality instruments. Of special interest are the categorical test-retest results for the facet scales. The median percentage of older and younger adults who moved from one pole to the opposite pole over an interval of up to 15 weeks was only 3% and 4%, respectively, demonstrating the high stability of the Step II category placements over this time interval.

Although additional reliability studies, particularly longitudinal studies, would be helpful, the evidence indicates that the MBTI assessment, especially in regard to the four preference scales, is very reliable in a wide range of groups and compares favorably to other major personality assessments.

CHAPTER 9

Validity of the MBTI® Step I™ Assessment

CHAPTER OVERVIEW

Intercorrelations Among Global Step I™ Scales

Factor Analyses and Construct Validity

Type Differences Observed on Scales from Other Assessments

Within-Preference Differences

Heritability and Brain Activity Patterns

Occupational Selection and Careers

Organizations

Education

Health, Stress, and Coping

Validity Shown in Other Studies of Type Differences

Validity of Whole Types and Type Dynamics

Outcome Studies

Conclusion

The MBTI® Global Step I™ assessment is designed to implement the Jung-Myers theory of personality types. Therefore its validity is determined by its ability to demonstrate relationships and outcomes predicted by the theory. The theory suggests that individuals are, or become, different types. Consequently, the assessment attempts to classify individuals into whole types based on their preferences. The item content, item format, and scoring method are consistent with this goal.

The theory postulates that the basic preferences for perception and judgment lead to observable differences in point of view, motivation, values, and behaviors, all of which are considered surface indicators of the effects of the basic preferences and attitudes defined by the theory and measured by the assessment. In other words, if the Jung-Myers theory describes preferences that do exist, and if the MBTI assessment adequately indicates those preferences, then surface behaviors should be in the directions predicted by the theory.

This chapter presents evidence for the validity of the MBTI Step I assessment. The next chapter covers validity evidence for the MBTI Step II™ assessment. The discussion of Step I evidence will include results from the Global Step I™ assessment, as well as some data collected using prior Step I forms, particularly Form M, the assessment that immediately preceded the global form. Given the high correlations of the preference scales in the Global Step I assessment with those in prior forms (Form M and the European Step I assessment), shown in table 9.1, and in turn the correlations between Form M and still earlier forms of the assessment (see Myers, McCauley, Quenk, & Hammer, 1998), research conducted with other forms of the MBTI assessment is useful in establishing validity.

Table 9.1 | **Correlations of continuous scores between MBTI® Global Step I™ and prior assessments**

Global Step I™ preference scale	Form M preference scale	European Step I™ preference scale
E–I	.96	.92
S–N	.96	.90
T–F	.98	.88
J–P	.96	.88

Note: Global sample, $N = 16,773$.

For interested readers, additional evidence for Form G and Form M can be found in the 1998 *MBTI® Manual* (Myers et al., 1998) and in *MBTI® Applications: A Decade of Research on the Myers-Briggs Type Indicator®* (Hammer, 1996b), which summarizes 10 years of research. In addition, the Center for Applications of Psychological Type (CAPT) maintains the Isabel Briggs Myers Memorial Library (www.capt.org/MILO), a comprehensive, continually updated, searchable database of research on the MBTI assessment, and CAPT can provide access to research articles.

Two broad categories of evidence can be brought to bear on the issue of the validity of MBTI Step I results. The first category includes evidence for the validity of the four separate preference scales that compose the assessment. This is the level at which measurement takes place with the Step I assessment. The separate preferences are the component parts needed to achieve the goal of helping individuals identify a whole type, so it is important to establish the validity of the separate preference scales. The second category of evidence concerns the validity of whole types or of particular combinations of preferences, particularly those combinations that have theoretical importance. Since the primary goal in using the Step I assessment is to help an individual arrive at a complete four-letter type, it is important to show that each of the 16 types has unique characteristics and that these characteristics are not entirely predictable from knowledge of the separate preferences.

The validity evidence presented here is subject to a variety of influences, among which are measurement error, respondents' developmental level, and environmental pressures. All measurement is subject to error; there is no perfect or pure measurement. Measurement error creates "noise" that should be taken into account when interpreting the scores derived from the assessment, as well as when interpreting validity evidence (see chapter 8 for a discussion of factors that create noise in measurement). The stage of development of the individuals being tested also must be taken into account when interpreting scores. The relationship of development to MBTI results is covered in depth in the *MBTI® Step III® Manual* (Myers, McCaulley, Quenk,

Hammer, & Mitchell, 2009). However, development is relevant in this chapter because samples composed of individuals with less developed or less consistent preferences would be expected to have lower validity coefficients. Finally, overriding environmental pressures can interfere with expression of type preferences, which would also affect validity coefficients.

The validity evidence reported in this chapter is organized around several topics that represent the major areas of both research and application for the Step I assessment. As such, they demonstrate the validity of the assessment across a wide variety of samples, contexts, and variables. They also demonstrate the usefulness of the Step I assessment in these areas, a usefulness that derives from the assessment's validity. This link between validity and application is demonstrated in a survey of users of six common assessments who were asked to rate important attributes they considered when selecting an assessment (Thompson, Walling, Mackey, Anderson, Schaubhut, Weber, & Kulapaditharom, 2013). Of MBTI users, 90% were concerned that "results are scientifically backed and have proven reliability."

The major topic areas used to organize this chapter are as follows:

- Intercorrelations among the four preference scales*
- Factor analyses as evidence of construct validity*
- Type differences observed on scales of other assessments*
- Within-preference differences
- Heritability and brain mapping
- Occupational selection and careers
- Organizations*
- Education
- Health, stress, and coping*
- Other specific areas of research*
- Research on whole type and type dynamics

For readers who are particularly interested in the validity of the Global Step I assessment, the asterisks above indicate those topics that contain data from the MBTI global sample or portions thereof. However, all the topics on the list incorporate new research on the validity of the MBTI Step I assessment, although not necessarily based on data from the global sample.

INTERCORRELATIONS AMONG GLOBAL STEP I™ SCALES

On the basis of both theory and previous research, it was expected that the four Global Step I preference scales would be mostly independent. Specifically, the interscale correlations were expected to be small, except for that between S–N and J–P, which previous research

Table 9.2 | **Intercorrelations of MBTI® Global Step I™ continuous scores**

Global Step I™ preference scale	Global Step I™ preference scale			
	E–I	S–N	T–F	J–P
E–I	—	—	—	—
S–N	–.20	—	—	—
T–F	–.15	.27	—	—
J–P	–.15	.48	.23	—

Note: Global sample, $N = 16,773$.

had demonstrated to be of medium magnitude (the correlation was .47 in the Form M US national sample, $N = 3,036$). Table 9.2 displays the correlations among the four Global Step I preference scale logits—the continuous scores underlying the four preference pairs. Similar to what was found with prior Step I forms, the largest correlation was between S–N and J–P, with small correlations between the other three preference scales. This confirms the expected pattern, showing that the four preference scales are relatively independent, with the possible exception of S–N and J–P.

The correlation between S–N and J–P indicates that Sensing types are more likely to prefer Judging and Intuitive types are more likely to prefer Perceiving. This probably reflects a fact about the types themselves. Sensing types typically prefer to rely on past experience and dislike unexpected events that require a rapid assessment of new possibilities. A preference for Judging typically leads to a life that is planned and consistent, decreasing the number of such unexpected events. Intuitive types, on the other hand, are attracted to future possibilities and the identification of new relationships. The Perceiving attitude keeps the door open to an incoming stream of these new possibilities.

FACTOR ANALYSES AND CONSTRUCT VALIDITY

The factor structure of MBTI item pools provides evidence of the construct validity of the MBTI Step I assessment, that is, the degree to which the assessment measures the construct it was designed to measure. When conducting factor analytic studies of the Step I assessment it is important to select an appropriate item pool. If the research question involves the construct validity of the four preference scales, then the items of interest are only those items that are used to score the preference scales: the 92 items on the Global Step I assessment or the 93 items used in Form M scoring.¹

Once the correct set of items has been selected, the appropriate factor analytic tool must also be used. Two basic kinds of factor analytic studies can be performed: exploratory and confirmatory. Exploratory factor analysis is the most widely used, although it provides only limited information about a hypothesized factor structure of a set of items. Although very useful for the purpose of developing hypotheses with respect to the factor structure of an assessment, exploratory factor analysis is not particularly useful in testing the degree to which an already hypothesized structure fits the observed data. The MBTI Step I assessment has a clearly hypothesized structure based on theory—the four-factor model that represents the four preference scales—and thus confirmatory studies will yield more information about the construct validity of the assessment. The discussion of these two kinds of factor analysis below relies on the review by Harvey (1996).

Exploratory Factor Analyses

Several exploratory studies of various forms of the MBTI Step I assessment have produced results that are virtually identical to the hypothesized four-factor model—for example, Harvey, Murry, and Stamoulis (1995); Thompson and Borrello (1986); Tzeng, Outcalt, Boyer, Ware, and Landis (1984). Tischler (1994), analyzing a large sample, found “unusually strong evidence that MBTI items are correlated with (measure) their intended scales; the scales are almost factorially pure” (p. 30). Factor analytic studies have also been conducted with a large sample in South Africa ($N = 10,705$; Van Zyl & Taylor, 2012) and with a Chinese translation of a prior form of the Step I assessment (Chen, Tian, & Miao, 2009), with results that supported the four-factor structure.

Two instances have also been reported in the literature where the predicted factor structure was not obtained: Comrey (1983) and Sipps, Alexander, and Friedt (1985). Both of these studies can be questioned regarding one or more of the decisions that guided their analyses: retaining factors based on questionable criteria; forcing an orthogonal rotation on the data; using samples with very low ratios of sample participants to items; using the principal components model when it is obvious that MBTI items contain considerable levels of unique variance. The factor structures that emerged from both of these studies were tested against the predicted four-factor structure using the more rigorous confirmatory approach, which is discussed in the following section.

1. Previous forms of the MBTI assessment (i.e., Forms G, J, K, and Q) contained research items or items used to score for facets or both. Unfortunately, although these additional items were never intended to be used as measures of the four preference pairs, they were included in some factor analyses designed to test the four-factor structure of the MBTI assessment. Not surprisingly, results were other than predicted, as more than four factors emerged.

Confirmatory Factor Analyses

This section summarizes the results of a confirmatory factor analysis (CFA) with dichotomous latent factors conducted to test the construct validity of the four latent preferences based on 92 dichotomous items used in the Global Step I assessment. Confirmatory factor analysis (e.g., James, Mulaik, & Brett, 1982; Jöreskog & Sörbom, 1981) is able to provide a much more definitive test of the plausibility of a hypothesized factor structure than is the exploratory approach. In a confirmatory factor analysis, a specific structural model is fitted to the data. The goal of the MBTI Step I assessment is to make explicit the personality types, which are based on four latent dichotomies specified in the theory of Isabel Myers. Thus, the form of CFA that is appropriate for dichotomous factors was used.²

Of the total 92 MBTI items, 24 are used to measure the E–I dichotomy, 24 for S–N, 23 for T–F, and the remaining 21 for J–P. The goal of CFA as conducted here is to test whether the items used to measure each dichotomy load only on that dichotomy. Therefore, the CFA model estimates loadings for each item only on that factor that it is assumed to measure, with all other loadings being restricted to 0. CFA also allows correlation parameters to be estimated between those pairs of factors assumed to be correlated based on the theory being tested. Consistent with previous analyses of the MBTI scales and the results presented above, only the S–N and J–P latent preferences were assumed to be correlated.

MBTI theory allows for respondents with the same preference to differ with respect to their degree of clarity in choosing that preference. Therefore, what is called “scale heterogeneity” in latent class analysis (LCA) was taken into account using an extended latent class model (Magidson, 2014) containing both preference classes (the four preference scales) and “scale classes,” where each scale class represented cases having a different level of clarity of preference (e.g., low clarity, moderate clarity, high clarity classes, etc.).

Table 9.3 shows the results of four tested models: one exploratory factor analysis (EFA) model and three different CFA models—one that does not take into account the scale classes (4-DFactor CFA) and two additional models, one with two scale classes and one with three. When the standard confirmatory factor model was extended to take into account scale heterogeneity via the scale classes, the model fit improved, resulting in an extended CFA model that fit much better than the EFA model, thus supporting the hypothesis that items designed to measure a particular dichotomous factor loaded only on that latent preference factor. Table 9.3 summarizes these analyses. The table shows the Bayesian

Table 9.3 | Comparison of confirmatory factor analysis model fit for EFA and other CFA models

Model	Log-likelihood (LL)	Number of parameters estimated (Npar)	Bayesian information criterion (BIC)
EFA with 4 DFactors	–795072	464	1594604
4-DFactor CFA	–811046	189	1623909
4-DFactor CFA + 2 scale classes per preference factor	–790367	197	1582627
4-DFactor CFA + 3 scale classes per preference factor	–788590	205	1579150

Note: Global sample, N = 16,773.

information criterion (BIC), the log-likelihood (LL), and the complexity of the model estimates (number of parameters estimated; Npar) for each of the models that was specified and tested. The model having the lowest BIC is the preferred model. This is the CFA model that takes into account the clarity of the preference (specifically, the model with three classes of preference clarity), which had the best overall fit, although it required more parameters to be estimated.

Thompson and Borrello (1989) evaluated the fit of the hypothesized four-factor model, concluding that it fit well in their sample. However, they did not test any competing models against which the reported levels of fit could be compared. Harvey, Murry, and Stamoulis (1995) tested three competing views of the latent structure of the Step I assessment. The three models were the predicted four-factor model and the two five-factor models developed by Sipps et al. (1985) and Comrey (1983) in their exploratory factor analyses. The results of the Harvey et al. study provided strong support for the validity of the predicted four-factor model. In contrast, the Sipps et al. and Comrey five-factor models both exhibited severe flaws that suggested that they were fundamentally misspecified.

A confirmatory factor analysis was also conducted on Form M, using data from the Form M US national representative sample (N = 3,009). A diagonal weighted least squares procedure was used to estimate the model (Jöreskog & Sörbom, 1981). Also, because the S–N and J–P scales are known to correlate, the latent variables were allowed to correlate. The adjusted goodness of fit was .95 (values above .90 indicate an acceptable model

2. The particular kind of factor analysis appropriate for latent dichotomies is known as latent class factor analysis when the items are dichotomous or categorical, as is the case with the MBTI items. For details of the latent class factor (also known as discrete factor or DFactor) approach as implemented in the Latent GOLD program (Vermunt & Magidson, 2013), see Magidson and Vermunt (2001, 2004) and Vermunt and Magidson (2003).

fit; Hu & Bentler, 1999), and the non-normed fit index was .97 (values above .95 indicate a good model fit). The median of the fitted residuals was $-.008$ (values less than .08 suggest an acceptable model). These results indicate an excellent fit to the four-factor model.

Bess, Harvey, and Swartz (2003) conducted a hierarchical confirmatory factor analysis of the 94 items used on Form G of the MBTI assessment. Their results clearly supported the view that the 94-item pool for Form G was dominated by the predicted four-factor structure. In discussing their results, they wrote,

The fact that this *a priori* factor structure was yet again recovered via exploratory means and found to fit reasonably well using confirmatory analysis, in a sample of “real world” managerial employees that is arguably quite different from the college student based samples that have been seen in earlier confirmatory and exploratory factor analyses, speaks directly to the robustness and generalizability of past claims of support for the predicted 4-factor MBTI latent structure. . . . The criticisms [of the MBTI assessment] that have been offered by its vocal detractors (e.g., Pittenger, 1993) have led some psychologists to view it as being of lower psychometric quality in comparison to more recent tests based on the FFM (e.g., McCrae & Costa, 1987). In contrast, [our findings]—especially when viewed in the context of previous confirmatory factor analytic research on the MBTI, and meta-analytic reviews of MBTI reliability and validity studies (Harvey, 1996)—provide a very firm empirical foundation that can be used to justify the use of the MBTI as a personality assessment device in applied organizational settings. (p. 4)

In sum, the exploratory and confirmatory factor analytic results reported here provide strong support for the construct validity of the MBTI Step I assessment. Several large-sample, carefully conducted exploratory studies produced “textbook” four-factor structures that almost exactly matched the hypothesized pattern of loadings (Harvey, 1996). Of even greater importance, several confirmatory studies of Form G, Form M, and the Global Step I form have unanimously supported the validity of the hypothesized factor structure and in one case conclusively rejected the competing models proposed by Sipps et al. (1985) and Comrey (1983). It is especially noteworthy that the CFA on the Global Step I form used a method that was consistent with both the underlying type theory, which assumes latent classes, and the latent class scoring procedure introduced with the Global Step I form. Although additional confirmatory studies need to be conducted to demonstrate the generalizability of the above findings, there is no question that the results of the factor analytic studies have been very supportive of the validity of the four-scale structure of the MBTI Step I assessment.

Confirmatory factor studies of prior MBTI Step I forms strongly support the validity of the four-factor structure of the Step I assessment. These results also hold for the Global Step I form, which is the first form to be subjected to latent class factor analysis, congruent with the assumptions of type theory.

TYPE DIFFERENCES OBSERVED ON SCALES FROM OTHER ASSESSMENTS

This section examines differences between individuals with opposite preferences on a wide variety of scales from other assessments. As part of the development of the MBTI Global Step I assessment, participants in various country/language samples were administered different research items and assessments at the discretion of the entity doing the research in each country. (See chapter 7 for a discussion of the data collection process.)

In prior editions of this manual, tables of correlations showing the relationship of the continuous scores for the four MBTI preference scales with continuous scores for scales from other assessments were presented. Many of these tables also contained levels of statistical significance. This manual focuses primarily on categorical analyses using mean differences between individuals with opposite preferences, an analytical approach that is consistent with the theory that underlies the MBTI Global Step I assessment, the item format, and the scoring procedure, which employs latent class analysis. (Relevant to our emphasis on categorical analyses, Healy [2000], Schneider, Smith, Taylor, and Fleenor [1998], and Leary, Reilly, and Brown [2009] compared continuous and categorical approaches to analyzing MBTI results and noted that the two produced almost identical outcomes.) Furthermore, researchers are increasingly moving away from null hypothesis significance testing and focusing more on effect sizes (Cohen, 1994; and see below). This is appropriate for the analyses reported throughout this chapter and the next because they are based on very large samples, for which most tests would be statistically significant. Effect size is more helpful to use as a guide for understanding these relationships and is especially useful given that our purpose is not to test specific hypotheses but to marshal data from a wide variety of sources for the purpose of identifying patterns and relationships that provide evidence for the validity of the MBTI Step I assessment. Although the emphasis in the tables included in this chapter is on categorical analyses and effect size, some of the tables are drawn from other research studies that used correlational analysis and standard significance tests.

Most of the tables presented here show the mean and standard deviation of individuals with each pole preference on the four MBTI preference scales. In addition, Cohen’s *d*, an estimate of effect size, is

presented as a summary statistic (Cohen, 1988). Cohen's d is the difference between the means of the two groups divided by the pooled or common standard deviation. In other words, it expresses the difference between the means of the two groups in standard deviation units. In type terms, the larger the absolute value of d , the more separation or difference there is between individuals at the two poles of the preference scale on the dependent variable. Positive values of d indicate higher means by those with I, N, F, or P preferences, while negative values show that those with E, S, T, or J preferences had higher means. (Note that the sign of d used here is in the same direction as the sign of correlation coefficients used in other tables in this manual, in prior MBTI manuals, and in most research articles on the MBTI assessment.) For example, the difference between Extraverts and Introverts on the CPI 260® Sociability scale (see table 9.4) is $d = -1.34$, indicating that the scores of Extraverts and Introverts on this particular dependent variable are separated by one and one-third standard deviation units and that the Extraverts had the higher mean score on Sociability.

To aid in interpreting d , Cohen proposed "conventions or operational definitions" for ranges of values. Small effects are considered to be around $d = .20$. Regarding the interpretation of small effects, Cohen stated, "When phenomena are studied which cannot be brought into the laboratory, the influence of uncontrollable extraneous variables ('noise') makes the size of the effect small relative to these (makes the 'signal' difficult to detect)" (p. 25). Putting small effect sizes into perspective, he pointed out that small effect sizes are quite common and meaningful and that "many effects sought in personality, social and clinical-psychological research are likely to be small effects" (p. 13). Moving to greater values of d , effect sizes in the medium range, around .5, are considered large enough to be visible to a careful observer. For example, $d = .5$ is the magnitude of the difference in height between 14- and 18-year-old girls. A large effect size is designated as around $d = .8$ or larger. The example Cohen provides of a large effect size is the degree of separation found in IQ between college freshmen and individuals holding a PhD degree.

In addition to the d value, the right column in table 9.4 designates the letter for the preference or preferences for which d is at least .20. The .20 cutoff is used simply as a convention, so that by scanning the right column in the tables the reader can quickly identify differences associated with each preference and can judge whether these characteristics make sense in light of the expectations of the theory. Readers should note, however, that the use of this convention means that there may be important differences shown in the main body of the table that are not flagged in the right column because d was less than .20. While small, such differences may be theoretically as well as statistically significant, and readers

are urged to seek these out. The sample sizes for some of these analyses are so large that many of the differences below $d = .20$ would be statistically significant. In fact, in a sample of $N = 1,000$, a value of $d = .10$ would be all that would be necessary for a t-test to be significant at $p = .01$. For example, given the number of Extraverts ($n = 654$) and Introverts ($n = 896$) in one analysis, an effect size of $d = .20$ would yield $t = 3.89$, which would be significant at $p < .0005$. For theoretical purposes, the direction and trend of the differences for a given preference scale across a variety of scales and samples is more important than the significance or absolute level of one difference in one sample for one scale.

Another point to consider when examining these tables is that for the purposes of validating the theory, it is just as important that certain differences are not significant as that other differences are in the expected directions. For example, Extraverts should score higher than Introverts on measures of sociability, whereas Thinking types should not score higher than Feeling types. In other words, the flagged d may not be the most important one for a given hypothesis, as the lack of a difference may be as or more meaningful. What are important are overall patterns, across many scales, in both size and direction.

The direction and trend of differences for a preference scale across various scales and samples is more important than the significance levels or differences in any one sample or scale.

Note also that the effect sizes reported in this chapter will be affected by both the reliability of the MBTI scale and the reliability of the scale from the other assessment. Most of the reliability coefficients for the preference scales on the Global Step I assessment are in the .86 to .91 range (see chapter 8). However, many of the reliability coefficients for the other measures that are displayed in the tables or discussed in this chapter are unknown. Given that some of these scales were created only for research purposes or have not been subjected to extensive development, it is safe to say that their reliabilities stand a good chance of being lower. Unreliability shrinks observed effect sizes, so the d values reported here may be underestimating the true effect of being at opposite poles on a preference scale.

It should also be noted that differences between individuals with opposite preferences on one of the preference scales, while important, have their limitations as evidence for construct validity of type theory. These differences are analyzed for the four preference scales one at a time and do not show the 16 types as whole or dynamic entities. Differences between whole types and interactions among the preferences are discussed throughout this chapter, as well as in a separate section titled "Validity of Whole Types and Type Dynamics." Differences between individuals with opposite

preferences do not show how differences in probabilities of being classified as one of the opposites affect results on dependent variables. These differences are discussed in the section “Within-Preference Differences,” later in this chapter.

CPI 260® Assessment

Table 9.4 shows differences in scores on the scales of the CPI 260 assessment for MBTI global sample participants with opposite preferences. (See appendix C for a description of the CPI 260 scales.) As shown in the table, Extraverts tend to score higher on almost all these scales, with some large effect sizes. For example, as predicted, Extraverts scored higher on Sociability ($d = -.134$). Introverts scored higher on the CPI 260 scales Self-control and Sensitivity, as well as on vector 1, on which high scores indicate Introversion as defined by the CPI 260 assessment. The table also shows that Intuitive types tend to score higher than Sensing types on CPI 260 scales, although the effect sizes are lower than those for Extraversion versus Introversion. For example, as predicted by theory, Intuitive types scored higher than Sensing types on Creative Temperament, Conceptual Fluency, and Flexibility. Sensing types scored higher on Self-control and Good Impression, as well as on vector 1 (toward internality), vector 2 (norm-favoring), and Law Enforcement Orientation (high scores on the latter two scales indicate an inclination to follow rules). For the T–F scale, differences on fewer CPI 260 scales met the criterion. Feeling types scored higher on Sensitivity. Thinking types scored higher on scales indicating social dominance (Dominance, Leadership, Capacity for Status), as well as on scales indicating high self-acceptance and a preference for operating independently. See more on the relationship between Thinking and Feeling types and leadership in the section “Organizational Leadership,” below. For the J–P scale, the table shows that Perceiving types scored higher on Creative Temperament and Flexibility than did Judging types. Perceiving types also scored higher on Empathy, perhaps because they tend to be more open to taking in information. Judging types scored higher on scales that indicated a desire to follow common conventions and rules, such as vector 2, Responsibility, Social Conformity, and Achievement via Conformance.

A similar analysis of differences on CPI 260 scales was available from a large data set ($n = 18,345$) compiled by the publisher. These were Form Q data and thus did not contain the same exact pool of MBTI items, but the item responses were rescored with the latent class analysis scoring procedure used with the Global Step I form. Although not shown here for space reasons, the results closely matched those of the MBTI global sample shown in table 9.4. On a few CPI 260 scales, the mean differences for the two large samples were in opposite directions; for example, the T–F scale showed opposite

results on eight CPI 260 scales. However, on none of these CPI 260 scales were the d values greater than .10, suggesting that the few differences between the two data sets were not meaningful and were possibly due to chance.

Another way to look at the relationship between MBTI Step I results and CPI 260 scales is to examine whole types that score high or low on each scale of the CPI 260 assessment. Table 9.5 shows the types whose mean score was at least .5 standard deviation higher or lower than the overall sample mean on each CPI 260 scale. The results in the table generally conform to predictions from type theory. For example, two of the three types scoring highest on Creative Temperament are the dominant extraverted Intuitive types, and the third type (INTP) extraverts its auxiliary Intuition. For another example, the three types scoring highest on Sensitivity (indicating a solicitude and sensitivity toward others) are Introverted Feeling types, while the four types scoring lowest on this scale are Extraverted Thinking types. It is interesting that the seven scales tapping self-management (Responsibility, Social Conformity, Self-control, Good Impression, Communality, Well-being, and Tolerance) resulted in the smallest number of types that met the criterion. This may be because these CPI 260 scales tend to tap an awareness of and conformance to societal rules, and since the MBTI global sample is composed of individuals from a variety of cultures, the difference in societal rules across these cultures may be washing out the type differences.

Another way to investigate the effect of whole types is to compare the results in table 9.5 with those in table 9.4. For example, based on table 9.4, one would expect that those scoring highest on Dominance would be individuals with Extraversion, Intuition, and Thinking preferences, as these were the preferences with the highest means on this CPI 260 scale. Indeed, the two types with these three preferences (ENTJ, ENTP) have the highest scores on this scale and the two types with opposite preferences on these three scales (ISFP, ISFJ) have the lowest means. However, one of the other Sensing types (ESTJ) and two Feeling types (ENFJ, ENFP) also have high scores on Dominance. This suggests that there are differences in the way an individual's whole type affects the behavior encompassed by a particular CPI 260 scale that is not captured by differences in the independent preferences.

Table 9.4 | MBTI® global sample opposite-preference differences on CPI 260® scales

CPI 260® scale	Extraversion (n = 715)			Introversion (n = 1,013)			Sensing (n = 1,218)			Intuition (n = 510)		
	M	SD		M	SD		M	SD		M	SD	
Dominance (Do)	21.59	5.71		14.76	6.23	-1.14	17.06	6.81		18.84	6.94	0.26
Capacity for Status (Cs)	14.74	3.86		10.74	3.95	-1.02	11.64	4.19		14.21	4.28	0.61
Sociability (Sy)	16.05	3.41		10.80	4.22	-1.34	12.51	4.62		14.08	4.66	0.34
Social Presence (Sp)	18.44	3.76		14.66	3.96	-0.98	15.56	4.09		17.81	4.39	0.54
Self-acceptance (Sa)	14.65	3.20		11.04	3.93	-0.99	12.10	3.98		13.58	4.04	0.37
Independence (In)	14.44	3.80		11.62	4.20	-0.70	12.39	4.20		13.73	4.31	0.31
Empathy (Em)	14.41	3.24		11.39	3.23	-0.93	11.99	3.38		14.20	3.50	0.65
Responsibility (Re)	15.24	3.61		14.65	3.52	-0.17	14.72	3.60		15.32	3.48	0.17
Social Conformity (So)	19.62	4.42		18.61	4.39	-0.23	19.18	4.39		18.67	4.49	-0.11
Self-control (Sc)	15.02	4.95		16.37	4.57	0.29	16.23	4.77		14.81	4.65	-0.30
Good Impression (Gi)	13.98	4.53		13.30	4.20	-0.16	13.93	4.33		12.76	4.31	-0.27
Communality (Cm)	18.17	2.67		18.02	2.71	-0.05	18.06	2.79		18.13	2.44	0.03
Well-being (Wb)	14.45	3.79		12.49	4.10	-0.49	13.21	4.10		13.53	4.07	0.08
Tolerance (To)	11.49	4.03		10.28	3.88	-0.31	10.37	3.96		11.75	3.89	0.35
Achievement via Conformance (Ac)	19.67	4.39		18.31	4.14	-0.32	18.92	4.29		18.76	4.32	-0.04
Achievement via Independence (Ai)	14.25	3.95		13.16	3.95	-0.28	13.00	3.78		15.09	4.08	0.54
Conceptual Fluency (Cf)	19.62	4.45		17.15	4.50	-0.55	17.61	4.56		19.52	4.56	0.42
Insightfulness (Is)	12.43	2.99		11.52	3.09	-0.30	11.60	3.02		12.59	3.12	0.32
Flexibility (Fx)	8.62	3.87		8.48	3.84	-0.04	7.79	3.59		10.33	3.86	0.69
Sensitivity (Sn)	12.59	3.29		14.55	3.58	0.57	13.63	3.53		13.99	3.74	0.10
Managerial Potential (Mp)	14.88	4.19		12.07	4.11	-0.68	12.95	4.29		13.90	4.49	0.22
Work Orientation (Wo)	15.48	3.84		14.24	3.96	-0.32	14.64	3.98		15.00	3.90	0.09
Creative Temperament (Ct)	15.17	4.11		12.73	4.03	-0.60	12.71	3.80		16.21	4.21	0.89
Leadership (Lp)	25.62	6.11		19.40	6.47	-0.98	21.44	6.95		23.25	7.03	0.26
Amicability (Ami)	16.99	4.89		15.85	4.78	-0.24	16.31	4.83		16.36	4.92	0.01
Law Enforcement Orientation (Leo)	17.47	3.16		16.30	3.19	-0.37	17.05	3.20		16.15	3.22	-0.28
Vector 1 (externality/internality)	8.73	4.43		12.57	4.35	0.88	11.48	4.73		9.81	4.69	-0.35
Vector 2 (norm-favoring/norm-doubting)	13.27	3.27		12.52	3.33	-0.23	13.18	3.29		12.02	3.26	-0.35
Vector 3 (self-actualization)	15.68	5.90		14.08	5.52	-0.28	14.20	5.68		16.05	5.65	0.33

CONTINUED

Table 9.4 | MBTI® global sample opposite-preference differences on CPI 260® scales *CONTINUED*

CPI 260® scale	Thinking (n = 934)			Feeling (n = 794)			Cohen's d	Summary	Judging (n = 871)			Perceiving (n = 857)			Cohen's d	Summary
	M	SD		M	SD				M	SD		M	SD			
Dominance (Do)	18.57	6.81		16.42	6.81		-0.32	T	17.31	7.19		17.86	6.57		0.08	
Capacity for Status (Cs)	12.84	4.19		11.87	4.53		-0.22	T	11.99	4.44		12.81	4.28		0.19	
Sociability (Sy)	13.13	4.64		12.79	4.73		-0.07		12.72	4.94		13.23	4.40		0.11	
Social Presence (Sp)	16.51	4.16		15.90	4.45		-0.14		15.59	4.26		16.87	4.25		0.30	P
Self-acceptance (Sa)	12.99	3.90		12.01	4.16		-0.24	T	12.11	4.11		12.97	3.94		0.21	P
Independence (In)	13.57	4.06		11.86	4.33		-0.41	T	12.51	4.33		13.07	4.20		0.13	
Empathy (Em)	12.75	3.59		12.51	3.52		-0.07		12.19	3.63		13.09	3.43		0.26	P
Responsibility (Re)	14.98	3.61		14.80	3.53		-0.05		15.38	3.52		14.41	3.56		-0.27	J
Social Conformity (So)	19.19	4.45		18.84	4.39		-0.08		19.89	4.32		18.15	4.37		-0.40	J
Self-control (Sc)	15.62	4.66		16.03	4.91		0.09		16.79	4.50		14.81	4.84		-0.42	J
Good Impression (Gi)	13.68	4.33		13.46	4.38		-0.05		14.49	4.23		12.66	4.28		-0.43	J
Communality (Cm)	17.97	2.87		18.21	2.46		0.09		18.47	2.51		17.69	2.81		-0.29	J
Well-being (Wb)	13.68	4.00		12.86	4.15		-0.20	T	13.58	4.05		13.02	4.12		-0.14	
Tolerance (To)	10.91	3.89		10.63	4.09		-0.07		10.93	3.93		10.63	4.04		-0.08	
Achievement via Conformance (Ac)	19.23	4.35		18.46	4.20		-0.18		20.02	4.05		17.71	4.23		-0.56	J
Achievement via Independence (Ai)	13.88	3.85		13.30	4.11		-0.15		13.60	3.88		13.62	4.08		0.01	
Conceptual Fluency (Cf)	18.66	4.54		17.60	4.70		-0.23	T	18.40	4.72		17.94	4.55		-0.10	
Insightfulness (Is)	12.30	3.04		11.42	3.06		-0.29	T	12.01	3.08		11.77	3.08		-0.08	
Flexibility (Fx)	8.20	3.77		8.94	3.90		0.19		7.51	3.55		9.58	3.86		0.56	P
Sensitivity (Sn)	12.82	3.50		14.82	3.40		0.58	F	13.94	3.66		13.53	3.52		-0.12	
Managerial Potential (Mp)	13.85	4.23		12.50	4.42		-0.31	T	13.45	4.49		13.00	4.23		-0.10	
Work Orientation (Wo)	14.99	3.90		14.46	4.00		-0.13		15.18	3.89		14.32	3.98		-0.22	J
Creative Temperament (Ct)	13.82	4.15		13.65	4.34		-0.04		12.63	3.83		14.87	4.33		0.55	P
Leadership (Lp)	22.84	6.97		20.95	6.95		-0.27	T	22.14	7.36		21.80	6.67		-0.05	
Amicability (Ami)	16.46	4.81		16.17	4.91		-0.06		16.86	4.81		15.78	4.85		-0.23	J
Law Enforcement Orientation (Leo)	17.14	3.24		16.37	3.17		-0.24	T	17.34	3.17		16.21	3.20		-0.36	J
Vector 1 (externality/internality)	10.35	4.77		11.72	4.68		0.29	T	11.46	4.75		10.50	4.76		-0.20	J
Vector 2 (norm-favoring/norm-doubting)	13.17	3.26		12.44	3.35		-0.22		13.86	3.15		11.79	3.17		-0.66	J
Vector 3 (self-actualization)	14.80	5.64		14.68	5.85		-0.02		14.68	5.59		14.82	5.88		0.02	

Note: Portion of the global sample that completed the CPI 260 assessment, n = 1,728. Summary column shows the preference for which d ≥ .20 in either direction.

Table 9.5 | Highest- and lowest-scoring MBTI® types on the CPI 260® scales

CPI 260® scale	Highest-scoring types	Lowest-scoring types
Dominance (Do)	ENTP, ENTJ, ESTJ, ENFJ, ENFP	INFP, ISFP, INFJ, ISFJ
Capacity for Status (Cs)	ENTJ, ENTP, ENFP, ENFJ, ESTJ	ISTP, INFJ, ISTJ, ISFP, ISFJ
Sociability (Sy)	ENTJ, ENTP, ENFJ, ESTJ, ENFP	INFP, ISTP, ISTJ, ISFP, INFJ, ISFJ
Social Presence (Sp)	ENTP, ENFP, ENTJ, ENFJ	ISFP, ISTJ, INFJ, ISFJ
Self-acceptance (Sa)	ENTP, ENTJ, ENFP, ENFJ, ESTJ	ISFP, INFJ, ISFJ
Independence (In)	ENTP, ENTJ, ESTJ	ISFP, INFJ, ISFJ
Empathy (Em)	ENTJ, ENTP, ENFJ, ENFP	ISTP, ISFP, INFJ, ISTJ, ISFJ
Responsibility (Re)	ENTJ	[ESTP]
Social Conformity (So)	ESTJ	[INFP]
Self-control (Sc)	[ISFJ]	ESTP
Good Impression (Gi)	ESTJ	[INTP]
Communality (Cm)	[ENFJ]	[ISTP]
Well-being (Wb)	ENTJ	INFJ
Tolerance (To)	ENTJ, ENFJ	[ISFJ]
Achievement via Conformance (Ac)	ESTJ, ENTJ, ENFJ	ISTP
Achievement via Independence (Ai)	INTJ, ENFJ	ISFJ
Conceptual Fluency (Cf)	ENTJ, ENTP, ENFJ	ISFP, ISFJ
Insightfulness (Is)	ENTJ, INTJ	ISFJ
Flexibility (Fx)	INTP, ENFP	ISTJ, ESTJ
Sensitivity (Sn)	INFJ, INFP, ISFJ	ENTJ, ESTP, ESTJ, ENTP
Managerial Potential (Mp)	ENTJ, ESTJ, ENTP	INFP, ISFP, INFJ, ISFJ
Work Orientation (Wo)	ENTJ	[ISFJ]
Creative Temperament (Ct)	ENTP, ENFP, INTP	ISFP, ISTJ, ISFJ
Leadership (Lp)	ENTJ, ENTP, ESTJ, ENFJ	INFP, ISFP, INFJ, ISFJ
Amicability (Ami)	[ENTJ]	[INTP]
Law Enforcement Orientation (Leo)	[ESTJ]	INTP, INFP
Vector 1 (externality/internality)	ISFJ, ISFP, INFJ	ESTP, ENTJ, ENTP
Vector 2 (norm-favoring/norm-doubting)	ESTJ	ENFP, INFP, INTP
Vector 3 (self-actualization)	ENTJ	[ISFJ]

Note: Portion of the global sample that completed the CPI 260 assessment, $n = 1,728$. Types listed in this table are those for which the mean for that CPI 260 scale is at least 1 standard deviation higher or lower than the overall sample mean. Types in brackets are the highest or lowest for that CPI scale, but their difference from the overall sample mean is less than .5 standard deviation.

Adjective Check List

Table 9.6 shows differences in scores on selected scales from the *Adjective Check List* (ACL) for MBTI global sample participants with opposite preferences. (See appendix B for a description of these ACL scales.) Extraverts tend to describe themselves more favorably overall than do Introverts; the same is true for Intuitive types versus Sensing types. Extraverts scored higher on 14 of the 24 ACL scales. (This is consistent with the pattern of Extraverts tending to rate themselves higher on surveys of management behaviors discussed in the section “Behaviors, Competencies, and Performance,” below.) Introverts scored higher on Succorance (solicit sympathy or support from others), Deference, and Self-Control. Intuitive types scored higher on 15 of the 24 scales. Sensing types scored higher on Deference and Self-Control. Thinking types scored higher on eight scales. Feeling types scored higher on Nurturance, Affiliation, Succorance, and Deference. There were fewer differences on the J–P scale. Judging types scored

higher on Endurance, Order, Deference, Self-Control, and Adult (objectivity, industriousness). These results are consistent with type predictions.

FIRO-B® Assessment

The relationship between the MBTI Step I assessment and the FIRO-B assessment, which evaluates interpersonal needs, has been examined in a number of studies. (See appendix D for a description of FIRO-B scales.) The FIRO-B Expressed Control scale measures a need to interact with others by setting goals and directing others. The Wanted Control scale taps a need to have others set the direction. The Expressed Inclusion scale measures a need to include others in one’s activities and to keep them in the loop. The Wanted Inclusion scale indicates a need to be included by others. Expressed Affection measures a need to be open and friendly with others, and the Wanted Affection scale measures a need to have others behave toward oneself in this manner. Table 9.7 shows differences in scores on the FIRO-B scales for

Table 9.6 | MBTI® global sample opposite-preference differences on Adjective Check List scales

ACL scale	Extraversion (n = 654)		Introversion (n = 896)		Summary	Sensing (n = 1,049)		Intuition (n = 501)		Cohen's d	Summary
	M	SD	M	SD		M	SD	M	SD		
Sum of number checked	71.90	37.24	65.86	34.92		64.11	33.80	77.40	38.83	0.37	N
Sum of favorable checked	35.04	16.19	29.50	15.67	E	29.40	15.63	36.93	15.96	0.48	N
Sum of unfavorable checked	5.31	6.86	6.43	7.14		5.83	6.73	6.24	7.65	0.06	
Communality	7.67	4.08	7.62	4.54		7.21	4.31	8.56	4.31	0.31	N
Achievement	8.33	5.47	5.92	5.18	E	6.52	5.39	7.81	5.43	0.24	N
Dominance	4.84	4.33	1.27	4.76	E	2.50	4.85	3.37	5.00	0.18	
Endurance	6.38	5.19	5.48	5.32		5.82	5.28	5.93	5.29	0.02	
Order	5.22	4.67	5.23	4.71		5.29	4.69	5.08	4.70	-0.04	
Intrception	9.16	5.19	8.46	5.20		8.02	5.08	10.30	5.12	0.45	N
Nurturance	9.77	6.06	7.68	6.45	E	7.88	6.35	10.00	6.17	0.34	N
Affiliation	15.19	7.31	11.60	7.02	E	12.16	7.13	15.11	7.44	0.41	N
Exhibition	2.82	3.73	-0.91	3.99	E	0.14	4.04	1.75	4.61	0.38	N
Autonomy	2.52	3.59	1.39	3.61	E	1.48	3.44	2.68	3.92	0.33	N
Aggression	0.18	3.61	-2.30	4.22	E	-1.55	4.06	-0.65	4.31	0.22	N
Change	3.26	3.04	1.68	3.18	E	1.66	2.88	3.79	3.41	0.69	N
Succorance	-0.17	2.69	0.64	3.11	I	0.40	2.92	0.09	3.07	-0.10	
Deference	0.15	3.85	1.52	3.97	I	1.25	3.84	0.31	4.18	-0.24	S
Self-Control	-0.97	3.19	0.85	3.31	I	0.48	3.23	-0.75	3.53	-0.37	S
Self-Confidence	7.30	4.74	3.45	4.58	E	4.47	4.89	6.34	5.05	0.38	N
Personal Adjustment	6.31	3.85	4.70	3.98	E	5.08	4.00	5.99	3.94	0.23	N
Ideal Self	5.41	5.44	2.77	5.29	E	3.10	5.40	5.53	5.37	0.45	N
Creative Personality	3.35	3.39	1.96	3.44	E	1.75	3.13	4.21	3.58	0.75	
Military Leadership	6.61	4.75	5.53	4.72	E	5.62	4.77	6.75	4.67	0.24	N
Adult	5.53	5.14	4.82	5.13		4.95	5.01	5.46	5.39	0.10	

CONTINUED

Table 9.6 | MBTI® global sample opposite-preference differences on Adjective Check List scales CONTINUED

ACL scale	Thinking (n = 858)		Feeling (n = 692)		Cohen's d	Summary	Judging (n = 777)		Perceiving (n = 773)		Cohen's d	Summary
	M	SD	M	SD			M	SD	M	SD		
Sum of number checked	67.61	35.57	69.39	36.60	0.05		65.86	35.07	70.96	36.82	0.14	
Sum of favorable checked	31.04	16.13	32.83	16.06	0.11		31.26	16.01	32.41	16.23	0.07	
Sum of unfavorable checked	6.23	7.11	5.62	6.95	-0.09		5.34	6.82	6.58	7.21	0.18	
Communality	7.43	4.32	7.90	4.38	0.11		7.61	4.39	7.67	4.31	0.01	
Achievement	7.46	5.58	6.29	5.18	-0.22	T	7.16	5.38	6.72	5.48	-0.08	
Dominance	3.28	4.89	2.16	4.88	-0.23	T	2.75	4.86	2.80	4.97	0.01	
Endurance	6.33	5.38	5.27	5.09	-0.20	T	6.68	5.20	5.02	5.23	-0.32	J
Order	5.96	4.75	4.32	4.46	-0.36	T	6.15	4.72	4.29	4.48	-0.41	J
Intracception	8.49	5.21	9.08	5.19	0.11		8.70	5.13	8.81	5.29	0.02	
Nurturance	7.04	6.31	10.46	5.93	0.56	F	8.34	6.41	8.78	6.33	0.07	
Affiliation	12.43	7.39	13.97	7.23	0.21	F	12.73	7.25	13.51	7.45	0.11	
Exhibition	0.72	4.26	0.59	4.34	-0.03		0.09	4.05	1.23	4.46	0.27	P
Autonomy	2.31	3.64	1.33	3.58	-0.27	T	1.31	3.50	2.43	3.71	0.31	P
Aggression	-1.08	4.22	-1.47	4.07	-0.09		-1.73	4.05	-0.78	4.21	0.23	P
Change	2.09	3.11	2.67	3.33	0.18		1.56	2.97	3.14	3.27	0.51	P
Succorance	-0.18	2.82	0.90	3.04	0.37	F	0.29	2.96	0.31	2.98	0.01	
Deference	0.23	3.88	1.84	3.91	0.41	F	1.46	3.90	0.43	3.99	-0.26	J
Self-Control	0.05	3.27	0.12	3.51	0.02		0.77	3.22	-0.61	3.39	-0.42	J
Self-Confidence	5.26	4.92	4.83	5.14	-0.09		4.84	4.90	5.30	5.13	0.09	
Personal Adjustment	5.24	4.10	5.55	3.88	0.08		5.45	4.05	5.31	3.95	-0.03	
Ideal Self	4.43	5.61	3.21	5.30	-0.22	T	4.04	5.60	3.73	5.41	-0.06	
Creative Personality	2.73	3.43	2.32	3.54	-0.12	T	2.04	3.32	3.06	3.57	0.29	P
Military Leadership	6.10	4.86	5.85	4.63	-0.05		6.23	4.83	5.74	4.68	-0.10	
Adult	5.71	5.25	4.38	4.91	-0.26	T	5.75	5.08	4.48	5.13	-0.25	J

Note: Portion of the global sample that completed the Adjective Check List, n = 1,550. Summary column shows the preference for which d ≥ .20 in either direction.

participants with opposite preferences in a very large database compiled by the publisher ($n = 91,794$). Again, these were data from prior forms of the MBTI Step I assessment that were rescored using LCA. As would be expected given the emphasis on interpersonal needs, Extraverts scored significantly higher than Introverts on all but one of the FIRO-B scales, resulting in very large effect sizes. The exception is Wanted Control, on which there was no difference. The largest effect sizes were on Total Expressed needs and Expressed Inclusion. Participants with a preference for Intuition scored higher than Sensing types on many of the scales, although the effect sizes were in the small range. On the T–F scale, Feeling types scored higher than Thinking types on the majority of the FIRO-B scales, with small to medium effect sizes. The exception is Expressed Control, on which Thinking types scored higher. This is consistent with the higher scores by Thinking types on the Dominance scales of the CPI 260 assessment

and the ACL. There were no significant differences in interpersonal needs on the J–P scale, indicating that interpersonal needs are driven primarily by other type preferences.

This sample was also analyzed for whole types. ENFJs or ENFPs ranked either first or second for Expressed Inclusion, Wanted Inclusion, Expressed Affection, Wanted Affection, Total Inclusion, Total Affection, Total Wanted, and Total Needs. ENTJs and ENTPs ranked first and second, respectively, on Expressed Control and Total Control, the latter driven almost entirely by their very high scores on the expressed dimension of this need. The types scoring highest on Wanted Control were the ISFPs and ESFPs.

The relationship between the MBTI Step I and FIRO-B assessments was also examined by Schaubhut, Herk, and Thompson (2009) using a sample of 1,900 individuals (50% women, 50% men) who completed

Table 9.7 | MBTI® opposite-preference differences on FIRO-B® scales

FIRO-B® scale	Extraversion ($n = 53,461$)		Introversion ($n = 38,333$)		Cohen's d	Summary	Sensing ($n = 52,463$)		Intuition ($n = 39,331$)		Cohen's d	Summary
	M	SD	M	SD			M	SD	M	SD		
Expressed Inclusion	5.26	1.85	3.11	1.95	–1.14	E	4.18	2.20	4.60	2.10	0.19	
Wanted Inclusion	4.17	3.40	2.11	2.96	–0.64	E	3.02	3.33	3.70	3.41	0.20	N
Expressed Control	4.37	2.66	3.25	2.59	–0.42	E	3.75	2.70	4.10	2.67	0.13	
Wanted Control	3.44	2.08	3.42	2.12	–0.01		3.42	2.12	3.44	2.06	0.01	
Expressed Affection	4.78	2.30	2.82	1.86	–0.92	E	3.74	2.30	4.27	2.36	0.23	N
Wanted Affection	5.59	2.23	4.36	2.30	–0.54	E	4.87	2.35	5.35	2.29	0.21	N
Total Inclusion	9.43	4.63	5.22	4.19	–0.95	E	7.20	4.90	8.30	4.87	0.22	N
Total Control	7.81	3.34	6.67	3.29	–0.34	E	7.18	3.40	7.55	3.31	0.11	
Total Affection	10.37	4.02	7.18	3.60	–0.83	E	8.60	4.13	9.62	4.12	0.25	N
Total Expressed	14.41	4.69	9.18	4.26	–1.16	E	11.67	5.20	12.97	5.10	0.25	N
Total Wanted	13.20	5.65	9.88	5.37	–0.60	E	11.31	5.75	12.49	5.73	0.21	N
Total Needs	27.62	9.02	19.07	8.11	–0.99	E	22.98	9.61	25.46	9.46	0.26	N

FIRO-B® scale	Thinking ($n = 65,864$)		Feeling ($n = 25,930$)		Cohen's d	Summary	Judging ($n = 55,724$)		Perceiving ($n = 36,070$)		Cohen's d	Summary
	M	SD	M	SD			M	SD	M	SD		
Expressed Inclusion	4.24	2.17	4.68	2.13	0.20	F	4.34	2.17	4.40	2.17	0.03	
Wanted Inclusion	3.13	3.34	3.77	3.44	0.19		3.17	3.36	3.53	3.41	0.11	
Expressed Control	4.17	2.70	3.22	2.54	–0.36	T	3.96	2.71	3.81	2.66	–0.06	
Wanted Control	3.28	2.03	3.81	2.21	0.25	F	3.40	2.09	3.48	2.11	0.04	
Expressed Affection	3.67	2.25	4.70	2.40	0.45	F	3.89	2.33	4.07	2.36	0.08	
Wanted Affection	4.81	2.34	5.75	2.19	0.41	F	4.99	2.34	5.20	2.34	0.09	
Total Inclusion	7.37	4.87	8.44	4.94	0.22	F	7.51	4.89	7.93	4.94	0.09	
Total Control	7.46	3.34	7.03	3.41	–0.13		7.37	3.39	7.29	3.33	–0.02	
Total Affection	8.48	4.06	10.45	4.07	0.48	F	8.89	4.15	9.27	4.16	0.09	
Total Expressed	12.08	5.17	12.60	5.24	0.10		12.20	5.20	12.28	5.19	0.02	
Total Wanted	11.22	5.66	13.33	5.79	0.37	F	11.56	5.74	12.21	5.81	0.11	
Total Needs	23.31	9.49	25.92	9.70	0.27	F	23.76	9.62	24.48	9.62	0.08	

Note: FIRO-B commercial sample, $N = 367,176$. Summary column shows the preference for which $d \geq .20$ in either direction.

both assessments. The correlations between the MBTI preference scales and the FIRO-B scales were very similar to those shown in table 9.7. They were also consistent with correlations reported in the *FIRO-B® Technical Guide* (Hammer & Schnell, 2000), the *FIRO Business® Technical Guide* (Herk, Thompson, Morris, & Schaubhut, 2009), and the 1998 *MBTI® Manual* (Myers et al., 1998). Furnham and Crump (2007) examined the relationship between the MBTI Step I and FIRO-B assessments in another large sample ($N = 4,143$) of managers in Britain and also found correlations between FIRO-B scales and MBTI preference scales in the expected directions.

People with opposite MBTI preferences scored in the predicted direction on scales of the CPI 260, ACL, and FIRO-B assessments. Sixty percent of the comparisons yielded effect sizes $\geq .20$.

Other Measures

The relationship of the MBTI preference scales to scales from additional personality assessments are presented here for which only correlational data are available, not mean differences between preferences. Table 9.8 shows correlations of MBTI Global Step I continuous scores with a measure of the Big Five scales (Extraversion, Agreeableness, Conscientiousness, Openness, and Neuroticism) derived from the ACL and correlations with the Big Five scales as measured by the NEO assessment. John (1989, 1990) used the ACL to create measures of the Big Five factors and found, as expected, that the MBTI E–I scale correlated with the Big Five Extraversion scale in the E direction; T–F correlated with Agreeableness in the F direction; J–P correlated somewhat with Conscientiousness in the J direction; and S–N correlated with Openness in the N direction. There were no correlations $\geq .20$ with Neuroticism. The bottom half of table 9.8 shows correlations of MBTI Form G preference scale continuous scores with the NEO scales based on a meta-analysis of seven samples (Arneson & Landowski, 2015). The correlations were weighted by sample size and corrected for reliability in both the NEO and the MBTI scales. The correlations show a pattern similar to those based on the ACL. The difference in magnitude of some of the correlations compared to the ACL study are likely due to the fact that the correlations from the meta-analysis were corrected for reliability and used different MBTI and different Big Five measures.

The *Murphy-Meisgeier Type Indicator for Children®* (MMTIC®) assessment (Murphy & Meisgeier, 2008) was designed to assess personality type preferences in children. Although the MMTIC was designed for a different population (children), some age ranges overlap with the accepted age range for administering the MBTI items, making it possible to examine the relationship

between the two assessments. McPeck and Breiner (2013) correlated the continuous scores of MBTI Form M with the MMTIC in a sample of teenagers ($N = 123$) at two different times. The correlations of the continuous scores at the first administration (time 1) were .81 for E–I, .78 for S–N, .65 for T–F, and .74 for J–P. All the correlations at the time of the second administration (time 2) were slightly higher than at time 1. At time 1, 76.4% of the sample had either the same or three preferences alike; at time 2, the percentage was 78.2%. No sample participants had all four preferences different across the two assessments. Participants with higher preference clarity indexes on the MBTI scales and with higher percentage scores on the MMTIC were more likely to yield the same type on the two assessments.

The *Birkman Method®* personality assessment measures occupational preferences (Interests), effective behaviors (Usual behaviors), interpersonal and environmental preferences (Needs), and ineffective behaviors (Stress behaviors) (Birkman, Elizondo, Lee, Wadlington, & Zamzow, 2008). The *Birkman Method* scales were correlated with the MBTI Step I preference scales using a sample of 216 adults who had completed both assessments. Social Orientation (Usual), which measures the degree of social interaction sought by an individual, was related to Extraversion. Communication (Usual), a sensitivity construct that includes things such as shyness, concerns about being embarrassed, correcting others, and being corrected, was related to Introversion.

Table 9.8 | Correlations of MBTI® Step I™ continuous scores with Big Five scales scored from the ACL and the NEO™ assessment

Big Five scale	Global Step I™ continuous scores			
	E–I	S–N	T–F	J–P
ACL				
Neuroticism	.07	.03	.14	.06
Extraversion	–.47	.14	–.01	.08
Openness	–.15	.30	–.07	.08
Agreeableness	–.13	.18	.29	.06
Conscientiousness	–.01	.00	–.06	–.16
Big Five scale	Form G Step I™ continuous scores			
	E–I	S–N	T–F	J–P
NEO™ assessment				
Neuroticism	.22	.05	.12	.13
Extraversion	–.69	.03	.06	.06
Openness	–.09	.53	.05	.18
Agreeableness	–.04	–.03	.31	–.02
Conscientiousness	–.07	–.18	–.22	–.40

Note: Top half of table from a portion of the global sample that took the ACL, $n = 1,550$. Bottom half of table from a meta-analysis of NEO and MBTI Form G studies (Arneson & Landowski, 2015), $N = 1,662$.

Interaction (Usual), which measures an individual's desire to be talkative, spend time in groups, and enjoy parties, as well as comfort level in talking to strangers, was related to Extraversion. Personal Autonomy (Need), which measures the degree to which an individual's responses are conventional or unconventional, was related to Intuition. Activity (Need), which measures how quickly one prefers to take action, think quickly, and express energy physically, was related to Thinking. Empathy (Need), a measure of how comfortable one is with expressing feelings and emotions, was related to Feeling. Finally, Process Orientation (Need), which measures how much an individual wants to give or receive clear directions, follow instructions carefully, use systematic methods, and complete tasks, was related to Judging. These relationships are consistent with the intended meaning of the MBTI preference scales.

The *DiSC Validation Research Report* (Inscape Publishing, 2005) provides correlations between the MBTI Step I assessment and the DiSC instrument. The following relationships were reported: A preference for Extraversion and Feeling was related to the DiSC Influence scale (optimistic, outgoing, energizing, "people people"), a preference for Introversion was related to the Conscientiousness scale (concerned, correct, accurate), a preference for Feeling was related to the Steadiness scale (sympathetic, cooperative, helpful, consistent, good listeners), and a preference for Thinking was related to the Dominance scale. With the possible exception of the relationship of Introversion to Conscientiousness, these findings are consistent with type theory. It should also be noted that individuals who scored high on the DiSC Conscientiousness scale were described as quiet and private (Inscape Publishing, 2005).

Mitchell (1981) reported on data from 475 bank employees at all levels of the organization in a study concerned with the social climate of the organization. Included were data from the FIRO-B® assessment (Schutz, 1978), the *Strong-Campbell Interest Inventory* (SCII; Campbell & Hansen, 1981), the *Edwards Personality Preference Schedule* (EPPS; Edwards, 1954), and *Super's Work Values Inventory* (WVI; Super, 1970), as well as data about job level, type of job, and performance. The results of this study were consistent with theoretical predictions: sociability and concern for others were associated with Extraversion and Feeling; focus on achievement with Judging; and independence, autonomy, and intellectual achievement with Intuition and Perceiving. The data were also analyzed by whole type: ENTPs scored highest on variety and challenge, while ISFJs and ISFPs scored lowest. The four SF types scored highest on harmonious interpersonal relationships, while ENTPs scored lowest. ESTJs scored highest on achievement; ESTPs, INTPs, and ENTPs scored highest on autonomy, and ISTJs and ISFJs scored lowest. ESFJs were highest and ISTPs lowest on outgoing affiliation. ENFJs and INFPs scored highest

on nurturing affiliation, and ENTJs scored lowest. These results are clearly congruent with predictions of type theory for both the preferences and whole types.

Another point to consider when examining the tables presented above is that for the purposes of the Jung-Myers theory it is just as important that certain differences do not show a meaningful effect size, or that correlations are not significant. For example, Extraverts would be predicted to score higher than Introverts on CPI 260® Sociability, which was indeed the case ($d = -1.34$), but Thinking types would not be expected to score higher than Feeling types on this scale, which was also the case ($d = -.07$). The flagged differences or correlations, which illustrate convergent validity, may not be the most important ones for a given hypothesis. For some investigations, overall patterns (both size and direction) may be important, and a lack of a difference or a low correlation may be meaningful.

WITHIN-PREFERENCE DIFFERENCES

The Jung-Myers personality theory posits the existence of dichotomies. A promising technique for showing evidence of dichotomies is to plot the means of a dependent variable on both sides of a given preference scale. A dichotomy should show a division point with a change in slope or level, or some other disparity, between the two sides of the scale. If possible, the dependent variable should be one in which individuals with opposite preferences behave quite differently. It is not easy to identify variables relevant to a given scale that are also relatively free of variance from the other three scales. When a disparity can be found, it supports the hypothesis that two distinct categories of individuals, with opposite preferences, have been separated by MBTI scores.

When plotting scores of a relevant variable on MBTI scores, a crucial question is whether the observed disparities between the two halves of the distribution are consistent with the assumption of a continuum, which is the case with most personality scales, or are better explained by the hypothesis of two different populations. If the plots were found to show unusual characteristics that can best be explained as reflecting a dichotomy, then both the theory and the validity of the MBTI Step I assessment would be supported. The basic trend for each half of the distribution should be clearly evident, and each half of the distribution should be allowed to start at zero as though the other did not exist. The trend of each half can then be determined independently. No a priori assumption of a continuum should be imposed on the data by connecting the two halves. The clearest kind of disparity in slope is a reversal of direction, for example, a V-shaped or U-shaped line. According to the Jung-Myers theory, each distribution should divide the

population into two categories. The slope of the line within each category should reflect the effect of differing clarities of preference or probabilities.

In the 1998 *MBTI® Manual* (Myers et al.), plots of the four MBTI preference scales on IQ and vocabulary are shown. The E–I, T–F, and J–P scales show a trend indicating that clearer preferences toward each pole (a U-shaped curve) are associated with higher scores on these variables. On the J–P scale, the Perceiving side of the scale is higher than the Judging side of the scale, which is typical for J–P plots on aptitude/achievement data. The plots for the S–N scale show that this scale does not operate the same way on intelligence and vocabulary as do the other scales. Rather than clearer preferences for *either* pole leading to higher IQ and vocabulary scores, only clearer preferences for Intuition lead to higher scores. This also demonstrates why it is important to choose a variable for each scale that is theoretically appropriate.

Data demonstrating disparities are also available for scores on the CPI assessment and ratings on the *Benchmarks®* assessment from managers who participated in programs at the Center for Creative Leadership® (CCL®). One scale that shows a clear disparity between Judging and Perceiving types is shown in figure 9.1. Sample participants were rated by their bosses on the *Benchmarks* scale Leading Employees. The plot clearly does not suggest a continuum; the slope of the two poles was reversed. The difference between individuals with a slight Judging preference and those with a slight Perceiving preference is significant ($t = 2.56$, $p < .01$). Another Judging–Perceiving difference is shown in figure 9.2. The variable plotted here is the sample participants' self-ratings on the *Benchmarks* scale Decisiveness. Again, the slopes reverse and the two sides show different patterns of self-ratings. Yet another J–P difference is shown in figures 9.3 and 9.4, which show sample participants' scores on the CPI scales Dominance and Responsibility. The plot of Dominance shows more of a U-shaped pattern. The difference on Dominance between individuals with a slight Judging preference and those with a slight Perceiving preference is significant ($t = 1.81$, $p < .05$). The Responsibility scale shows a different pattern, with an increasing score associated with greater clarity of preference for Judging types and a constant and lower score across the scale for Perceiving types.

In another sample (the Form M US national sample, $N = 3,036$), some illustrative plots are available that show endorsement rates for adjectives from the *Adjective Check List*. Two variables that show interesting patterns for the Thinking–Feeling scale are the adjectives *aggressive* and *argumentative*, which are shown in figures 9.5 and 9.6, respectively. Thinking types are more likely to say that they are aggressive; as shown in figure 9.5, as the clarity of the preference for Thinking increases, the endorsement of this adjective also

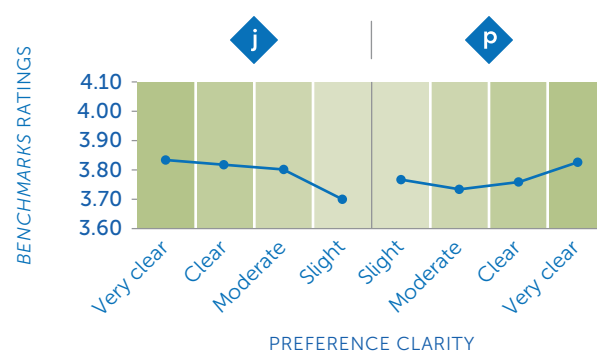


Figure 9.1 | Bosses' ratings on *Benchmarks®* Leading Employees scale for J and P managers by MBTI® Form M preference clarity

Note: Respondents rated on a 5-point scale, from "not at all" to "a very great extent," the extent to which they engage in the behavior.

Source: Center for Creative Leadership. Used with permission.

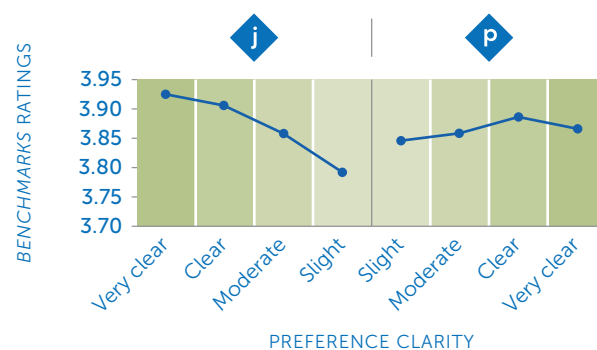
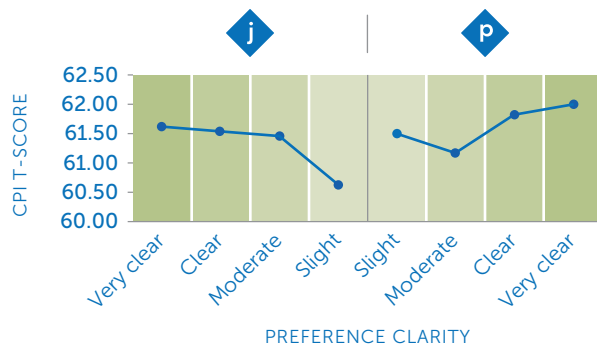


Figure 9.2 | Self-rating on *Benchmarks®* Decisiveness scale for J and P managers by MBTI® Form M preference clarity

Note: Respondents rated on a 5-point scale, from "not at all" to "a very great extent," the extent to which they engage in the behavior.

Source: Center for Creative Leadership. Used with permission.

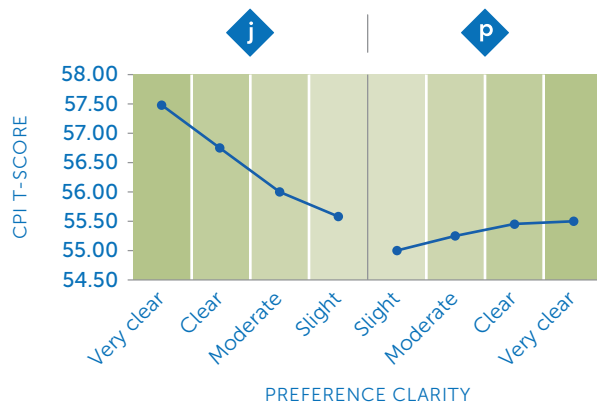
increases (from 26% for those with a slight preference for Thinking to 53% for those with a very clear preference). Feeling types' average endorsement of this adjective is 28% and does not change much with increasing clarity of preference. As shown in figure 9.6, the adjective *argumentative* shows a reverse U-shape; those with a very clear preference for Thinking or Feeling are less likely to endorse this adjective than are those with a slight preference. Another adjective that shows a U-shaped curve for the T–F scale is *efficient*. This is shown in figure 9.7. The endorsement rates range from about



Number of respondents	Very clear	683	Very clear	101
	Clear	908	Clear	330
	Moderate	997	Moderate	894
	Slight	552	Slight	657

Figure 9.3 | Scores on CPI™ Dominance scale for J and P managers by MBTI® Form M preference clarity

Source: Center for Creative Leadership. Used with permission.



Number of respondents	Very clear	683	Very clear	101
	Clear	908	Clear	330
	Moderate	997	Moderate	894
	Slight	552	Slight	657

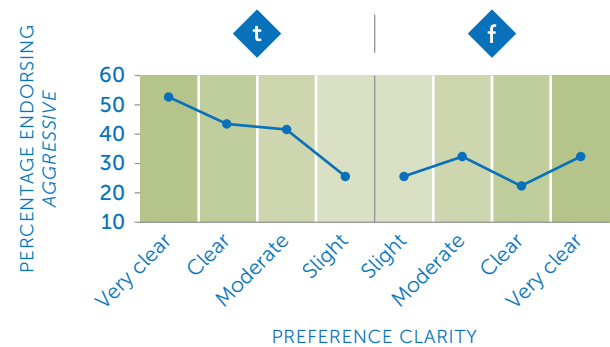
Figure 9.4 | Scores on CPI™ Responsibility scale for J and P managers by MBTI® Form M preference clarity

Source: Center for Creative Leadership. Used with permission.

55% for those with a slight preference to about 71% for those with a very clear preference for either Thinking or Feeling.

Studies of relevant variables that show discontinuities in the distribution of their scores across MBTI preference scales suggest two distinct populations, which supports both type theory and the validity of the MBTI Step I assessment.

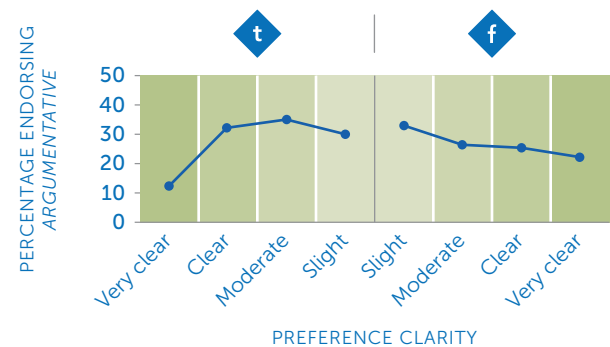
A comparison of the number of books reported being read per year by individuals with different preferences provides an example of a large difference in level between opposite poles of a preference scale. Hicks



Number of respondents	Very clear	17	Very clear	27
	Clear	60	Clear	110
	Moderate	101	Moderate	193
	Slight	80	Slight	109

Figure 9.5 | Endorsement rates for ACL adjective aggressive for T and F types by MBTI® Form M preference clarity

Note: Portion of Form M US national sample, $n = 697$.



Number of respondents	Very clear	17	Very clear	27
	Clear	60	Clear	110
	Moderate	101	Moderate	193
	Slight	80	Slight	109

Figure 9.6 | Endorsement rates for ACL adjective argumentative for T and F types by MBTI® Form M preference clarity

Note: Portion of Form M US national sample, $n = 697$.

(1984) described the number of books reported being read by Sensing types and Intuitive types in the past year. Figure 9.8 shows the mean number of books reported being read by individuals with different levels of preference for Sensing or Intuition (Hicks, 1984). Individuals with different clarity of preference at the opposite poles clearly behaved differently with respect to number of books read. Those with a Sensing preference reported reading about five books per year on average, regardless of the clarity of their preference, while Intuitive types read about six times as many books and the amount increased with clarity of preference.

The plots shown in these figures demonstrate that for selected variables, including rating by others, scores

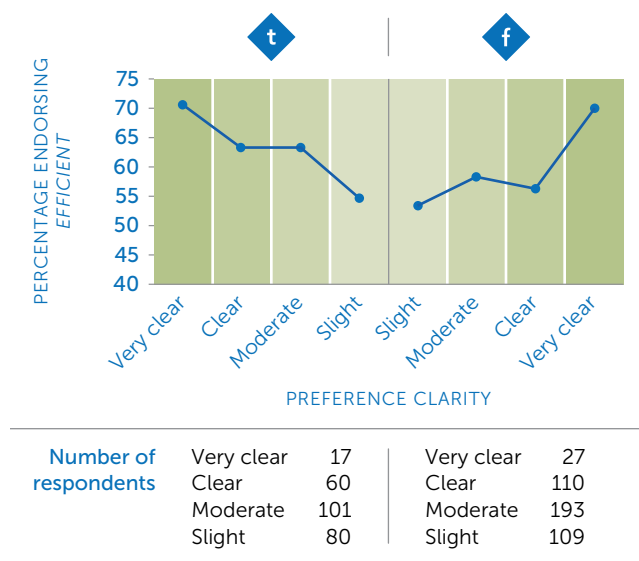


Figure 9.7 | Endorsement rates for ACL adjective *efficient* for T and F types by MBTI® Form M preference clarity

Note: Portion of Form M US national sample, $n = 697$.

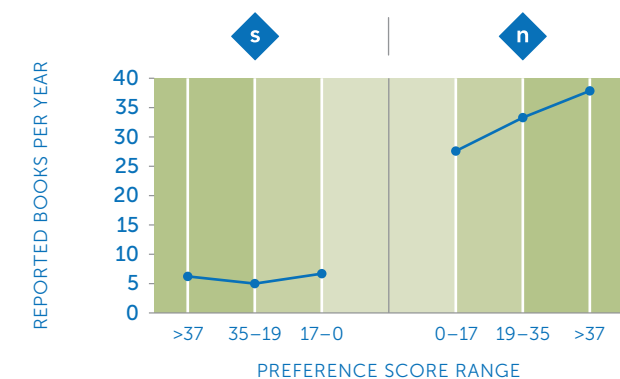


Figure 9.8 | Books read per year as reported by S and N types

Source: Hicks (1984).

on other assessments, endorsement of adjectives, and reading behavior, there are observable differences between individuals at the opposite poles of the MBTI preference scales, depending on the clarity of the preference on each side. These differences support the type hypothesis.

HERITABILITY AND BRAIN ACTIVITY PATTERNS

Jung, Briggs, and Myers viewed type preferences as inborn dispositions to develop along specific lines. Type may thus be viewed as a basic template that directs and organizes life experiences and contributes to differential responses to similar life events. Jung saw personality as being biologically based and concluded that type must have some kind of biological foundation as well

(1921/1971). Thorne, Korobov, and Morgan (2007), in their research on identity, referred to the performative approach to development as explicated by Caspi (1998). This model of developmental elaboration views individuals as actively selecting, construing, and evoking situations that are compatible with their predispositions. So, for example, a preference for Extraversion, while genetically based, becomes developmentally elaborated through psychosocial processes. This view is entirely consistent with a type theory approach to development.

Bouchard and Hur (1998) provided support for the inborn nature of type preferences in their study of sets of monozygotic and dizygotic twins and triplets reared apart. MBTI Form F continuous scores were used after correcting for gender, age, and age x gender interactions. The model-fitting heritability estimate was .57 for E-I, .46 for S-N, .60 for T-F, and .39 for J-P, with an average of .50. Overall, these estimates of genetic influence on the MBTI scales are very much like those found for most other personality measures (Bouchard & McGue, 2003).

Methods such as topographic brain mapping and EEG recordings offer a means of examining biological dispositions as manifested in neurological activity. Differences between Extraverts and Introverts have been studied by a number of researchers. Wilson and Languis (1989) discussed an approach to construct validation that involved an attempt to “establish a link between neurocognitive psychophysiological data and psychological type data through the investigation of differences in brain electrical activity patterns” (p. 14). Significant and consistent differences were found across a number of tests between Extraverts and Introverts, suggesting that Extraverts demonstrate less cortical arousal than do Introverts. The authors hypothesized that the relatively greater cortical arousal of Introverts leads them to seek a reduction in stimulation from the environment.

In other research, Stenberg’s (1990) studies of blood flow showed lower activity in anterior temporal lobes for Extraverts, which may be related to Extraverts being geared to respond to and seek excitement in order to avoid underarousal. In contrast, Introverts, who are geared to introspect, may be threatened by overarousal and seek peaceful pursuits in solitude and tranquility. Evidence from longitudinal studies of temperament (Kagan & Snidman, 1991) focusing on measures of selected limbic sites provide similar support for a physiological basis for Extraversion and Introversion.

Some evidence is also available regarding physiological differences in Sensing and Intuition. Newman (1985), using EEG alpha symmetry ratios, found that Intuitive types as a group showed greater levels of activity in the left hemisphere of the brain and Sensing types greater levels of activity in the right hemisphere. Wilson and Languis (1989) reported a similar finding for adults

with clear preferences for Extraverted Intuition and Extraverted Sensing; and in preliminary exploratory research, Wilson, Laposky, and Languis (1991) found a pattern of generalized posterior arousal among middle school students who preferred Intuition.

Gram, Dunn, and Ellis (2005) studied the relationship of EEG patterns to all four MBTI preference pairs. The sample consisted of 35 students who gazed at a blank screen while baseline EEGs were recorded from 19 brain sites according to a standard international system. The MBTI Step I assessment was administered after the EEG recordings were made. Similar to what occurred in other studies, Extraverts produced significantly less internal arousal than did Introverts. However, the largest differences were between individuals who preferred Sensing and those who preferred Intuition. Sensing types produced greater theta amplitudes than did Intuitive types, suggesting that with little environmental stimulation the Sensing types had little to process. They were apparently able to “quiet their minds.” In contrast, those who preferred Intuition did considerably more internal processing in the “sparse” environment. The authors also found some differences on T–F and J–P, but these were less significant and more mixed across sites. They concluded that their data provided “additional strong evidence for Eysenck’s (1967) basic notions concerning Extraversion and Introversion and suggest that the Extraversion–Introversion scale of the MBTI assessment may be a more reliable measure of this dichotomy than [Eysenck’s EPI] assessment” (p. 44). Their findings for Sensing–Intuition also support type theory.

The growing evidence regarding the biological basis of psychological type gives credence to Jung’s assumption that type is universal, that is, that the personalities of all human beings are structured in accordance with type dichotomies.

Estimates of genetic influence on the MBTI scales are very much like those found for most other personality measures.

OCCUPATIONAL SELECTION AND CAREERS

The purpose of this section is to provide evidence for the validity of the MBTI Step I assessment by demonstrating the relationship between the eight preferences and occupational choice. An ancillary goal is to demonstrate the practical usefulness of the MBTI assessment in career selection. These two goals are directly related in that the usefulness of the assessment in career choice is a consequence of its demonstrated validity. The use of the MBTI assessment in career counseling to help people find meaningful and productive work was one of Isabel Myers’ original motivations in developing the assessment

(Saunders, 1991). Career counseling applications were one of the first areas of applied research on type, and the field continues to generate a large number of research studies and the accrual of very large data sets.

Hammer and Macdavid (1992a) stated the underlying premise as follows:

Psychological type theory postulates that the distribution of individuals into various occupations is not random, but rather represents an effort by the individual to find

- Opportunities to express their personality preferences
- Career satisfaction
- Rewards for doing what they like to do and what they are good at doing
- Tasks and problems that they will find interesting and challenging

To the extent that individuals succeed in this effort, there will be an observable relationship between type and occupational choice. (pp. 3–4)

To state the converse of the type hypothesis is to suggest that if type does not influence career choice or occupational attraction in any way—in other words, if the movement of types into occupations is indeed random—then the frequency of a given type in a given occupation should be the same as the frequency of that type in the general population (or in the appropriate base population). Therefore the questions that need to be answered are these:

- To what kinds of occupations or work environments would we expect individuals with different preferences to be attracted?
- What kind of evidence is needed to test the type hypothesis?
- How do we know whether individuals with different preferences or of different types are distributed in occupations by chance?

Occupational Hypotheses

Myers and McCaulley (1985, p. 78) noted the kind of work typically preferred by people with the eight preferences:

- **Extraverted types:** Work that allows interaction with a succession of people or that has activities outside the office or away from the desk
- **Introverted types:** Work that permits some solitude and time for concentration
- **Sensing types:** Work that requires attention to detail and careful observation, such as in applied occupations
- **Intuitive types:** Work that provides a succession of new problems to be solved
- **Thinking types:** Work that requires logical order, especially with ideas, numbers, or physical objects,

such as in occupations with technical/scientific components

- **Feeling types:** Work that provides service to people and a harmonious and appreciative work environment, such as in occupations emphasizing communications and interpersonal interaction
- **Judging types:** Work that imposes a need for system and order
- **Perceiving types:** Work that requires adapting to changing situations or where understanding situations is more important than managing them

It is important to keep in mind, however, that people of all 16 types can and do work in a variety of occupations. Furthermore, those whose type is different from the modal type in the occupation or organization can often bring new and needed perspectives.

One method used to test these hypotheses is to examine data presented in the form of type tables that show samples from various occupations. While type tables are the basic method for presenting MBTI group data, they are sometimes seen as providing merely descriptive information. On the contrary, type tables can be used to provide evidence for validity. When the type table for a given occupation demonstrates that significantly more of the types predicted by theory are indeed more attracted to that particular occupation, analysis of the type table contributes to construct validity.

Type tables are important as descriptive career information for practitioners and respondents and also offer evidence of the construct validity of the MBTI Step I assessment because the differential attraction of types to specific occupations is consistent with predictions from type theory.

Interests

As stated above, type theory predicts that people will be attracted to occupations or jobs where there are tasks and problems that they will find interesting and challenging. This suggests a parallel between type preferences and interests. Holland's typology of persons and work environments is a major force in career psychology (Spokane, 1996); it is measured by assessing the interests of individuals. The theory states, "Individuals seek and remain in congruent environments; and environments recruit, retain, and reward congruent people. Congruence is assessed according to the degree of match between the vocational personality of an individual and the environmental type of an occupation or position" (Gottfredson & Holland, 1996, p. 6). Thus liking the activities associated with one's job and with the people doing the job leads to job satisfaction and tenure. These hypotheses mirror those of type theory regarding the effect of preferences on occupational choice. One

would expect, therefore, that type and interests would be related.

The tenets of Holland's theory of vocational interests include the idea that both people and environments can be classified into six vocational types: Realistic, Investigative, Artistic, Social, Enterprising, and Conventional (RIASEC); these are often displayed on a hexagon. Figure 9.9 shows the hexagon and gives a brief definition of each vocational theme. The interests that are most closely related are next to one another on the hexagon, and those that are situated opposite each other are least related. Thus Realistic and Investigative are closely related, whereas Realistic and Social are not. Within the Holland codes, individuals are more likely a combination of vocational types rather than just one; for example, a counselor may have a Social-Enterprising-Artistic (SEA) code.

A number of instruments measure interests, and several have been correlated with the MBTI Step I assessment. Tables showing correlations of the MBTI preference scales with the scales of the *Strong Interest Inventory*® (Strong) assessment; the *Kuder Occupational Interest Survey*; the *Opinion, Attitude, and Interest Scales* (OAIS); and the *Vocational Preference Inventory* (VPI) are included in chapter 11 of the 1985 *MBTI® Manual* (Myers & McCaulley, 1985). A summary and discussion of these studies and additional research also appear in Hammer (1996a), Hammer and Kummerow (1996), and Hammer and Macdavid (1992a). The latter also summarizes a study correlating type with the *Career Occupational Preference System Interest Inventory* (COPS).

Table 9.9 shows differences between individuals with opposite MBTI preferences on the General Occupational Themes, Basic Interest Scales (BISs), and Personal Style Scales (PSSs) of the *Strong Interest Inventory* assessment for MBTI Form M, rescored using LCA, for a very large database ($N = 42,497$) compiled by the publisher. Mean differences on the Strong Occupational Scales are not included here due to space limitations. Following is a summary of the results in the table.

EXTRAVERSION—INTROVERSION

Extraverts tend to score higher on the Social and Enterprising Themes and on Basic Interest Scales (BISs) relating to interacting with others (e.g., Counseling & Helping, Teaching & Education) and attempting to influence or manage others (e.g., Marketing & Advertising). Extraverts also score higher on the Personal Style Scales (PSSs) in the direction of working with others, taking leadership positions, taking risks, and working on teams. Although not obvious from the table, the PSSs are bidirectional, which also means that Introverts score more toward the poles of these scales that indicate a preference for working alone, being less inclined to take leadership positions, less likely to take risks, and more

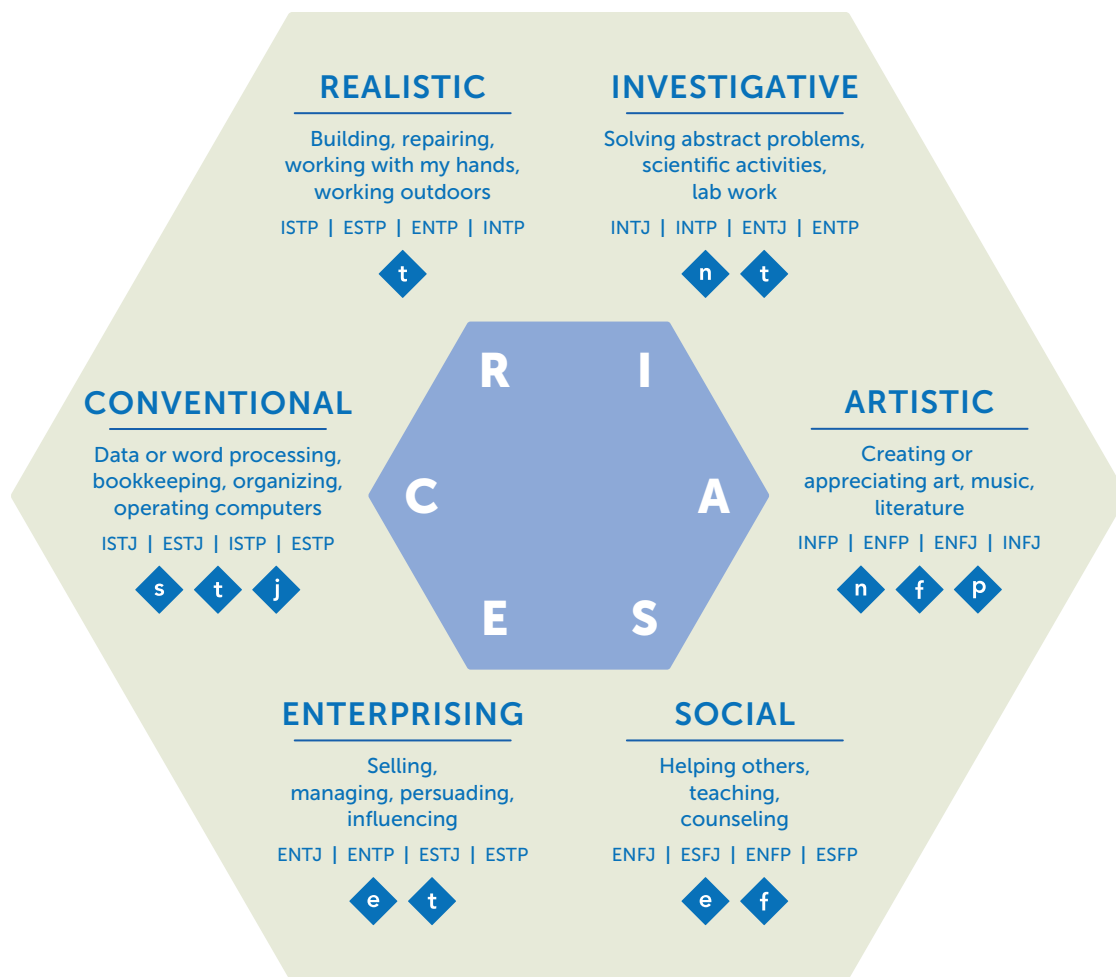


Figure 9.9 | Holland's vocational types associated with MBTI® preferences and whole types

Note: The associated preferences are those with $d \geq .20$; associated whole types are those with the four highest scores for that vocational type.

likely to want to work independently rather than on a team than do Extraverts.

SENSING–INTUITION

Intuitive types tend to score higher on the Investigative and Artistic Themes, while Sensing types score higher on the Conventional Theme. On the BISs, Intuitive types score higher on the artistic scales (e.g., Visual Arts & Design), while Sensing types show greater interest in activities requiring attention to detail (e.g., Office Management, Taxes & Accounting). On the PSSs, Intuitive types prefer more abstract learning environments and indicate an interest in leadership.

THINKING–FEELING

Thinking types tend to score higher on the Realistic, Investigative, Enterprising, and Conventional Themes, while Feeling types score higher on Artistic and Social. On the BISs, Thinking types show more interest in technical (e.g., Computer Hardware & Electronics) and research (e.g., Science) activities, while Feeling types

show more interest in helping others (e.g., Counseling & Helping). On the PSSs, Feeling types show interest in working with others, while Thinking types are more inclined toward leadership and risk taking.

JUDGING–PERCEIVING

Perceiving types tend to score higher on the Artistic Theme, while Judging types score higher on Conventional. On the BISs, Judging types score higher on activities related to managing others (e.g., Office Management). On the PSSs, the only significant difference is for Perceiving types, who are more inclined to take risks.

As can be seen from this table, there were fewer significant results for the J–P dichotomy. Analyzing the correlations between the Strong Occupational Scales and type preferences, Hammer (1996a) noted that the J–P scale also showed the fewest and smallest correlations with Strong Occupational Scales. Hammer and Macdaid (1992b) suggest that the J–P scale is the one least likely to reflect an attraction to a particular occupational field.

Table 9.9 | MBTI® opposite-preference differences on *Strong Interest Inventory®* scales

	Extraversion (n = 23,491)		Introversion (n = 19,006)			
Strong scale	M	SD	M	SD	Cohen's d	Summary
General Occupational Themes (GOTs)						
Realistic	51.25	9.67	51.45	9.71	0.02	
Investigative	50.65	9.88	51.93	9.83	0.13	
Artistic	54.04	9.06	52.59	9.14	−0.16	
Social	55.73	9.44	52.51	9.65	−0.34	E
Enterprising	55.23	9.44	49.05	9.48	−0.65	E
Conventional	50.56	10.17	51.33	10.22	0.08	
Basic Interest Scales (BISs)						
Mechanics & Construction	50.59	9.63	51.66	9.70	0.11	
Computer Hardware & Electronics	47.34	9.55	49.23	9.82	0.20	I
Military	50.20	10.01	49.98	9.82	−0.02	
Protective Services	51.28	9.77	50.56	9.69	−0.07	
Nature & Agriculture	52.22	9.69	52.31	9.90	0.01	
Athletics	54.34	9.62	51.22	9.82	−0.32	E
Science	50.62	9.78	51.98	9.78	0.14	
Research	51.43	10.20	51.95	10.14	0.05	
Medical Science	51.93	9.97	51.81	9.74	−0.01	
Mathematics	50.18	10.01	51.08	10.09	0.09	
Visual Arts & Design	52.87	9.05	52.61	9.10	−0.03	
Performing Arts	54.25	9.20	51.76	9.36	−0.27	E
Writing & Mass Communication	53.11	9.29	52.06	9.37	−0.11	
Culinary Arts	55.11	8.59	51.04	9.48	−0.45	E
Counseling & Helping	55.28	9.28	52.54	9.66	−0.29	E
Teaching & Education	55.17	9.73	52.38	9.81	−0.29	E
Human Resources & Training	55.03	8.81	50.50	9.49	−0.50	E
Social Sciences	54.27	9.27	52.67	9.30	−0.17	
Religion & Spirituality	52.62	10.60	51.39	10.50	−0.12	
Healthcare Services	52.88	9.59	52.13	9.40	−0.08	
Marketing & Advertising	54.55	9.18	49.26	9.44	−0.57	E
Sales	55.04	10.72	50.14	9.67	−0.48	E
Management	54.17	9.29	50.43	9.76	−0.39	E
Entrepreneurship	50.87	9.58	48.02	9.98	−0.29	E
Politics & Public Speaking	54.06	9.41	49.00	9.31	−0.54	E
Law	52.46	9.71	50.05	9.52	−0.25	E
Office Management	51.45	9.52	51.65	9.67	0.02	
Taxes & Accounting	50.18	10.22	51.05	10.35	0.08	
Programming & Information Systems	47.32	9.72	49.18	9.84	0.19	
Finance & Investing	52.06	10.15	50.30	10.19	−0.17	
Personal Style Scales (PSSs)						
Work Style	55.43	8.87	49.55	9.55	−0.64	E
Learning Environment	53.49	9.27	51.71	9.42	−0.19	
Leadership Style	55.71	8.06	48.60	8.89	−0.84	E
Risk Taking	52.71	9.66	50.32	9.64	−0.25	E
Team Orientation	55.75	8.71	50.33	9.74	−0.59	E

CONTINUED

Table 9.9 | MBTI® opposite-preference differences on *Strong Interest Inventory*® scales *CONTINUED*

	Sensing (n = 22,240)		Intuition (n = 20,257)			
Strong scale	M	SD	M	SD	Cohen's d	Summary
General Occupational Themes (GOTs)						
Realistic	51.41	10.00	51.25	9.33	−0.02	
Investigative	50.28	9.88	52.25	9.78	0.20	N
Artistic	49.82	8.68	57.30	7.92	0.90	N
Social	53.78	9.93	54.84	9.35	0.11	
Enterprising	52.13	10.03	52.84	9.83	0.07	
Conventional	52.95	10.26	48.66	9.64	−0.43	S
Basic Interest Scales (BISs)						
Mechanics & Construction	50.96	9.92	51.19	9.40	0.02	
Computer Hardware & Electronics	48.95	9.79	47.35	9.56	−0.17	
Military	51.16	10.05	48.94	9.65	−0.22	S
Protective Services	51.36	9.94	50.51	9.50	−0.09	
Nature & Agriculture	51.37	9.94	53.24	9.51	0.19	
Athletics	53.65	9.81	52.17	9.80	−0.15	
Science	50.31	9.84	52.23	9.67	0.20	N
Research	50.78	10.24	52.63	10.02	0.18	
Medical Science	51.78	9.98	51.97	9.75	0.02	
Mathematics	51.04	10.05	50.08	10.04	−0.09	
Visual Arts & Design	49.67	8.79	56.14	8.13	0.76	N
Performing Arts	50.23	9.26	56.33	8.37	0.69	N
Writing & Mass Communication	50.02	9.26	55.52	8.55	0.62	N
Culinary Arts	52.37	9.37	54.29	8.95	0.21	N
Counseling & Helping	52.99	9.71	55.22	9.23	0.24	N
Teaching & Education	53.10	10.08	54.83	9.54	0.18	
Human Resources & Training	52.90	9.56	53.11	9.21	0.02	
Social Sciences	51.63	9.43	55.67	8.72	0.44	N
Religion & Spirituality	51.43	10.52	52.77	10.59	0.13	
Healthcare Services	53.19	9.74	51.84	9.20	−0.14	
Marketing & Advertising	51.34	9.62	53.10	9.62	0.18	
Sales	53.56	10.60	52.06	10.44	−0.14	
Management	53.36	9.64	51.56	9.65	−0.19	
Entrepreneurship	48.55	10.00	50.73	9.58	0.22	N
Politics & Public Speaking	50.26	9.64	53.49	9.48	0.34	N
Law	51.20	9.81	51.58	9.57	0.04	
Office Management	53.63	9.63	49.25	9.00	−0.47	S
Taxes & Accounting	52.36	10.51	48.61	9.66	−0.37	S
Programming & Information Systems	48.25	9.94	48.04	9.68	−0.02	
Finance & Investing	52.10	10.11	50.37	10.23	−0.17	
Personal Style Scales (PSSs)						
Work Style	52.86	9.65	52.75	9.62	−0.01	
Learning Environment	49.28	8.84	56.45	8.47	0.83	N
Leadership Style	50.99	9.26	54.23	8.72	0.36	N
Risk Taking	51.10	9.84	52.23	9.56	0.12	
Team Orientation	53.05	9.62	53.63	9.50	0.06	

CONTINUED

Table 9.9 | MBTI® opposite-preference differences on *Strong Interest Inventory®* scales *CONTINUED*

	Thinking (n = 23,888)		Feeling (n = 18,609)			
Strong scale	M	SD	M	SD	Cohen's d	Summary
General Occupational Themes (GOTs)						
Realistic	53.37	9.54	48.72	9.24	−0.49	T
Investigative	53.03	9.61	48.91	9.74	−0.43	T
Artistic	52.38	9.00	54.68	9.12	0.25	F
Social	52.60	9.50	56.45	9.46	0.41	F
Enterprising	53.46	9.84	51.19	9.93	−0.23	T
Conventional	52.65	9.88	48.67	10.17	−0.40	T
Basic Interest Scales (BISs)						
Mechanics & Construction	53.36	9.56	48.13	9.02	−0.56	T
Computer Hardware & Electronics	50.27	9.61	45.51	9.18	−0.51	T
Military	52.13	10.07	47.50	9.09	−0.48	T
Protective Services	52.07	9.52	49.54	9.85	−0.26	T
Nature & Agriculture	52.64	9.65	51.77	9.93	−0.09	
Athletics	53.60	9.72	52.11	9.92	−0.15	
Science	52.95	9.60	49.02	9.61	−0.41	T
Research	54.21	9.59	48.38	9.97	−0.60	T
Medical Science	52.68	9.77	50.85	9.90	−0.19	
Mathematics	53.03	9.78	47.44	9.51	−0.58	T
Visual Arts & Design	52.19	8.97	53.48	9.16	0.14	
Performing Arts	51.85	9.23	54.79	9.26	0.32	F
Writing & Mass Communication	52.55	9.11	52.75	9.63	0.02	
Culinary Arts	52.61	9.38	54.15	8.94	0.17	
Counseling & Helping	52.07	9.32	56.60	9.23	0.49	F
Teaching & Education	52.71	9.61	55.49	9.96	0.29	F
Human Resources & Training	53.08	9.34	52.91	9.47	−0.02	
Social Sciences	53.75	9.20	53.31	9.46	−0.05	
Religion & Spirituality	50.70	10.29	53.82	10.67	0.30	F
Healthcare Services	51.97	9.16	53.29	9.90	0.14	
Marketing & Advertising	52.66	9.47	51.57	9.87	−0.11	
Sales	53.74	10.70	51.70	10.24	−0.19	
Management	54.01	9.46	50.57	9.62	−0.36	T
Entrepreneurship	50.91	9.53	47.90	10.02	−0.31	T
Politics & Public Speaking	53.51	9.49	49.60	9.52	−0.41	T
Law	53.15	9.37	49.12	9.65	−0.42	T
Office Management	51.48	9.29	51.61	9.96	0.01	
Taxes & Accounting	52.98	9.95	47.48	9.87	−0.56	T
Programming & Information Systems	49.94	9.57	45.85	9.66	−0.43	T
Finance & Investing	54.07	9.84	47.69	9.51	−0.66	T
Personal Style Scales (PSSs)						
Work Style	50.14	9.25	56.23	9.03	0.67	F
Learning Environment	53.40	9.00	51.79	9.76	−0.17	
Leadership Style	53.57	9.02	51.20	9.14	−0.26	T
Risk Taking	53.53	9.47	49.21	9.51	−0.46	T
Team Orientation	53.52	9.49	53.08	9.66	−0.05	

CONTINUED

Table 9.9 | MBTI® opposite-preference differences on Strong Interest Inventory® scales *CONTINUED*

	Judging (n = 23,180)		Perceiving (n = 19,317)			
Strong scale	M	SD	M	SD	Cohen's d	Summary
General Occupational Themes (GOTs)						
Realistic	50.69	9.89	52.11	9.38	0.15	P
Investigative	51.09	9.97	51.39	9.76	0.03	
Artistic	52.15	9.06	54.88	8.98	0.30	
Social	54.57	9.72	53.94	9.60	−0.07	
Enterprising	52.50	9.98	52.43	9.89	−0.01	
Conventional	52.29	10.31	49.24	9.81	−0.30	J
Basic Interest Scales (BISs)						
Mechanics & Construction	50.68	9.83	51.54	9.47	0.09	P P <

Note: MBTI and Strong commercial sample, N = 42,497. Summary column shows the preference for which $d \geq .20$ in either direction.

Table 9.10 | Rank order of MBTI® types by mean score on Strong General Occupational Themes

Realistic			Investigative			Artistic		
Type	M	SD	Type	M	SD	Type	M	SD
ISTP	55.12	9.08	INTJ	55.75	9.17	INFP	58.21	7.60
ESTP	54.64	9.51	INTP	55.57	9.24	ENFP	58.11	8.22
ENTP	54.30	9.00	ENTJ	54.21	9.32	ENFJ	57.83	7.77
INTP	53.87	8.99	ENTP	54.07	9.23	INFJ	57.57	7.94
ENTJ	53.10	9.14	ISTP	52.35	9.30	ENTP	57.14	7.73
ESTJ	52.69	10.02	ISTJ	52.12	9.68	ENTJ	56.60	7.80
ISTJ	52.55	9.92	ESTJ	51.44	9.76	INTP	56.41	7.80
INTJ	52.42	9.20	ESTP	51.29	9.55	INTJ	55.74	7.92
ISFP	50.16	9.42	INFP	51.25	9.44	ESFP	50.85	8.84
INFP	50.16	8.99	INFJ	50.51	9.73	ESFJ	50.78	8.71
ESFP	49.99	9.43	ENFP	49.20	9.61	ESTJ	50.20	8.64
ENFP	49.61	8.88	ENFJ	49.15	9.70	ISFP	50.08	8.59
ENFJ	47.76	9.00	ISFP	48.44	9.42	ISFJ	49.97	8.82
INFJ	47.55	9.29	ISFJ	48.11	9.87	ESTP	49.89	8.69
ISFJ	47.39	9.50	ESFP	47.67	9.68	ISTP	49.39	8.70
ESFJ	47.07	9.20	ESFJ	46.99	9.71	ISTJ	48.61	8.44

Social			Enterprising			Conventional		
Type	M	SD	Type	M	SD	Type	M	SD
ENFJ	58.70	8.98	ENTJ	57.47	8.89	ISTJ	54.66	10.05
ESFJ	58.02	9.29	ENTP	56.63	8.87	ESTJ	54.55	10.21
ENFP	57.35	9.12	ESTJ	56.36	9.23	ISTP	52.90	9.75
ESFP	56.20	9.70	ESTP	55.99	9.38	ESTP	52.63	9.45
INFJ	55.78	9.09	ENFJ	54.06	9.44	ENTJ	51.87	9.44
ISFJ	55.20	9.80	ENFP	54.00	9.56	ISFJ	51.85	10.42
ENTJ	54.80	8.92	ESFJ	53.19	9.69	INTJ	51.32	9.37
ESTJ	54.66	9.56	ESFP	53.17	9.47	ESFJ	51.06	10.54
INFP	54.46	9.22	INTJ	50.50	9.26	ISFP	50.73	10.34
ISFP	53.77	9.79	INTP	50.16	9.10	INTP	49.80	9.13
ENTP	53.68	8.86	ISTP	49.87	9.74	ESFP	49.80	9.99
ESTP	52.49	9.55	ISTJ	49.78	9.55	ENTP	49.70	9.11
INTJ	51.84	8.96	INFJ	47.80	9.29	INFJ	47.76	9.95
ISTJ	51.23	9.74	INFP	47.73	9.21	ENFJ	47.59	9.63
INTP	50.61	8.95	ISFP	47.64	9.73	INFP	46.62	9.68
ISTP	49.69	9.52	ISFJ	47.45	9.34	ENFP	46.35	9.42

Note: MBTI and Strong commercial sample, $N = 42,497$.

It more often can be used to suggest how an individual might structure his or her job in any field.

WHOLE TYPES AND INTERESTS

The relationship between the MBTI Step I assessment and interests reflected in the Strong General Occupational Themes is even clearer when whole types are examined. Table 9.10 shows the 16 MBTI types in rank order for each of the Themes. The results can be summarized as follows:

- The four highest-ranking types for the Realistic Theme, which taps an interest in working with things,
- working with one's hands, or working outdoors, are the TP types and the four ranking lowest are the FJ types.
- For the Investigative Theme, indicating an interest in solving abstract problems, the four highest-ranking types are the NT types and the four ranking lowest are their opposites, the SF types.
- The four types scoring highest on the Artistic Theme are the NF types and three of the four ranking lowest are their opposites, the ST types.
- The four highest scorers on the Social Theme are the EF types and the four lowest are the IT types.

- The four types that score highest on the Enterprising Theme are the ET types and the four who score lowest are the IF types.
- Finally, for the Conventional Theme, the four highest scores are for the ST types and the four lowest are for the NF types.

The results by MBTI preferences and whole types are summarized in figure 9.9. The whole-type results for the most part follow from those of the preferences. It is interesting that for each Theme there is a different set of processes, or process-orientation combination, that yielded the highest score. No combination was repeated. The research on the relationship between type and interests strongly supports the predictions of type theory that different type preferences manifest in different occupational interests.

Table 9.11 shows correlations of Form M continuous scores with the scales of two additional career-related inventories, the *Career Factors Inventory* (Chartrand, Robbins, & Morrill, 1997) and the *Skills Confidence Inventory* (Betz, Borgen, & Harmon, 2005). The latter is a measure of self-efficacy, or the confidence an individual has in his or her abilities to perform various activities; skills confidence was measured in each of the six Holland Themes. For five of the Themes, the findings for MBTI type and confidence parallel those for MBTI type and interests. The pattern for the *Skills Confidence Inventory* data suggests that what one has confidence in, one tends to like, and vice versa. The exception is that confidence in exercising Realistic skills is related to a preference for Perceiving and not to scores on the Thinking–Feeling scale. This is different from the results for interests, where Thinking types scored higher than Feeling types and there was no difference for Perceiving types (although $d = .15$).

On the *Career Factors Inventory*, Self-Knowledge is related to a preference for Judging. Career Choice Anxiety and Indecisiveness are related to Introversion, and Indecisiveness is also related to a preference for Feeling.

Sources of Occupational Data by Type

This section turns from a consideration of interests to actual occupational membership. The MBTI research literature has a number of sources of occupational data, including the following:

- *MBTI[®] Type Tables for Occupations* (Schaubhut & Thompson, 2012)
- *Manual: A Guide to the Development and Use of the Myers-Briggs Type Indicator[®]*, appendix D (Myers & McCaulley, 1985)
- *MBTI[®] Career Report User's Guide* (Hammer, 2005a)
- *Journal of Psychological Type* articles
- *Association for Psychological Type Conference Proceedings*

The most up-to-date lists of occupational data sorted by type are published in *MBTI[®] Type Tables for Occupations* (Schaubhut & Thompson, 2012), which at the time of this publication reported type data for 404 occupations based on data gathered between 2002 and 2011. More than 421,000 individuals who took either MBTI Form M or Form Q were included, and their type's frequency in their occupation was compared against their type's frequency in the Step I or Step II US national representative sample.

When reviewing this or any other occupational database, the reader is urged to consider these issues:

- How *representative* is the career sample? When was the sample compiled? While probably representative

Table 9.11 | **Correlations of MBTI[®] Form M continuous scores with the *Skills Confidence Inventory* and *Career Factors Inventory* scales**

	E-I	S-N	T-F	J-P	Summary			
<i>Skills Confidence Inventory scale</i>								
Realistic	-.06	.12	-.02	.27**	—	—	—	P
Investigative	.04	.13*	-.23**	.02	—	—	T	—
Artistic	-.02	.47**	.17*	.29**	—	N	—	P
Social	-.34**	.19**	.20**	.18**	E	—	F	—
Enterprising	-.37**	.11	-.20**	.13*	E	—	T	—
Conventional	-.01	-.20**	-.26**	-.11	—	S	T	—
<i>Career Factors Inventory scale</i>								
Career Information	-.07	-.09	.01	-.13	—	—	—	—
Self-Knowledge	.07	-.05	.09	-.20**	—	—	—	J
Career Choice Anxiety	.20**	-.11	.08	-.13	I	—	—	—
Indecisiveness	.22**	-.09	.24**	.03	I	—	F	—

Note: *Skills Confidence Inventory* assessment data from Iowa State University, $n = 245$; *Career Factors Inventory* assessment data from Middle Tennessee State University, $n = 197$.

* $p < .05$; ** $p < .01$.

of the occupation if the numbers are large and if occupational requirements have not changed substantially since the data were collected, some occupational samples may reflect a bias toward individuals who are interested in personality type, since they have all taken the MBTI assessment.

- What are the *educational levels* represented in the occupations? This is particularly important when deciding which samples are appropriate for use as a base population. For example, if your interest were in which types are attracted to engineering, an appropriate base population for comparison would be college students or those with a college degree, since that is an educational requirement for an engineer.
- How *satisfied* are people of those types with their careers? This is important because to the extent that people are satisfied in their job or career, the relationship with their type preferences will be more evident.

UNDERSTANDING THE SELF-SELECTION RATIO

The sources listed above contain type tables on hundreds of occupations. The method primarily used to analyze distributions in type tables such as these is the self-selection ratio (SSR; McCaulley, 1985). SSR type table analysis can be used to test how the frequency of a type in an occupation compares with the frequency of that type in a relevant base population. It therefore provides an index of the magnitude of over- or underrepresentation of a given preference, combinations of preferences, or whole types in a group or occupation.

The SSR is a ratio, the numerator of which is the percentage of that type in the occupation; the denominator is the percentage of that type in the base population.³ The value of the SSR will be exactly 1.00 when the type is found in an occupation in exactly the same proportion as it is found in the base population. If type had no relationship to occupational choice or selection, all SSRs for all occupations would be 1.00, except for chance deviations. The SSR will be greater than 1.00 when that type is overrepresented in the occupation and less than 1.00 when that type is underrepresented. Thus a ratio of 2.00 indicates that about twice as many individuals of that type are found in the occupation as would be expected given the frequency of that type in the base population. Likewise, a ratio of 0.50 indicates that half as many of that type are found in the occupation as would be expected by chance.

These ratios provide evidence for the validity of the MBTI Step I assessment as an expression of type theory and are also meaningful in practice when one is working with

individuals to determine whether a given occupation or field is relatively more or less attractive to people with their type preferences. Note that SSRs can be calculated on individual preferences; on any combination of preferences, including combinations indicating type dynamics; and on whole types. This can be followed by a chi-square test of the contingency table of interest to determine statistical significance. The importance of using SSRs to draw conclusions either about validity or about the occupational choice of an individual is illustrated in the following example.

The Form M US national representative sample ($N = 3,009$, a base population) contains more ESTJs (8.7%) than INFPs (4.4%). Yet if one were looking at an occupational sample in which ESTJs and INFPs have somewhat similar frequencies, such as librarians (Scherdin, 1994), and looked only at the frequencies of these types in that occupation, the effect of type would be masked. Among librarians, 7.3% prefer INFP and 6.1% prefer ESTJ. From these frequencies, it looks like INFPs and ESTJs are about equally drawn to library work. However, if their frequencies in the sample are compared to those in the base population, the INFPs have an SSR of 1.66 (7.3% in the occupation/4.4% in the base population) and the ESTJs have an SSR of .70 (6.1% in the occupation/8.7% in the base population). So, INFPs are about 1.7 times more likely, and ESTJs about one-third less likely, to choose librarian as an occupation than would be expected from knowing their frequency in the population. There appears to be something about library work that disproportionately attracts INFPs. Given the nature of library work, this result supports the prediction of type theory.

In addition to SSRs, other methods for analyzing type tables, including log-linear analysis, can also be used to draw inferences about construct validity. O'Shea and Mamchur (1989) demonstrate how log-linear models can be analyzed to test the predictions of type theory about occupational distributions. McCaulley, Macdavid, and Magidson (1997) applied log-linear analysis to type distributions and offer a graphical display of the frequency of the types in a sample compared to their frequency in a base population. One of the advantages of the log-linear method is that different models can be fit to the type distribution and then tested statistically. For example, a main-effects model would predict that the distributions of individuals in occupations are a function of the four independent preference scales, with no interaction among them, while a "saturated" model attributes the distributions to the 16 types, that is, to the interaction of the preferences. Another advantage is that when various models are being tested, the technique controls for the effects of the type

3. Despite this clear definition of how to calculate an SSR, some reviewers who calculated SSRs reversed the numerator and denominator and then erroneously concluded that the data did not support the type hypothesis.

Table 9.12 | **Percentage of overlap in the top 50 occupations for opposite MBTI® types**

Types	% occupation overlap
ENTJ and ISFP	2
ESTJ and INFP	2
ENFJ and ISTP	4
ESFJ and INTP	4
ENFP and ISTJ	4
ENTP and ISFJ	14
ESFP and INTJ	6
ESTP and INFJ	4
<i>Median</i>	5

Note: CPP commercial database, $N = 60,000$.

Source: Hammer & Macdaid (1992a), p. 37. Used with permission.

characteristics that are not being tested. Salter (2003b) used log-linear analysis to examine the type distribution of computer programmers.

Another technique that has been used to analyze type tables is predictive configural frequency analysis (PCFA), which was used by Salter (2006) to determine the types most over- and underrepresented among college faculty.

An overview of the effect of type on attraction to various occupations is found in the analysis by Hammer and Macdaid (1992a), who compiled lists of the 50 occupations with the highest SSRs for each of the 16 types. Table 9.12 shows the percentage of overlap of the occupations on these top 50 lists for each pair of opposite types. As the table shows, there is almost no overlap. Even for the pair of opposite types with the highest percentage of overlap, ENTPs and ISFJs, there are only 7 occupations (14%) common to both top 50 lists. One suspects that the opposite types who share an interest in the same occupation fill different niches or perform different functions within that occupation. The other pairs of opposite types have only 1 or 2 occupations in common. The results in this table provide strong evidence for the validity of MBTI results as applied to occupational choice.

Examples of Occupational Selection

The hundreds of type tables on occupations reveal the connection between individuals' MBTI type and their choice of career. A few examples are discussed below.

Type theory predicts that Feeling types will be drawn to the helping professions. Sixteen studies reported in the 1985 *MBTI® Manual* (Myers & McCaulley, 1985) looked at characteristics of providers of psychological services. A higher percentage of therapists/counselors who preferred Feeling was found in all but 5 of the 32 samples analyzed separately in the 16 studies. In the 27 studies in which a Feeling preference predominated, the

percentage of therapists/counselors preferring Feeling over Thinking ranged from 52 to 84.

Managers represent another class of workers that has been extensively studied. The overrepresentation of managers who prefer the combination of Thinking and Judging is well documented in a wide variety of studies and in data from a wide variety of cultures (Fitzgerald & Kirby, 1997; Fleenor, 1997; Reynierse, 1997). For example, in a sample of 26,477 managers compiled by Fleenor (1997), 57.8% were Thinking Judging types, which was almost 2.5 times the frequency of these types in the Form M US national representative sample ($N = 3,009$). Kirby suggested the following explanation:

The structure and values of organizations seem to have favored the logical and decisive behaviors most comfortable to those preferring Thinking and Judging, but that are less natural or comfortable to the three other comparable combinations of preferences—Feeling and Judging, Feeling and Perceiving, and Thinking and Perceiving. Because Thinking and Judging types are so prevalent in organizational leadership, it may be that Thinking and Judging behaviors have become the accepted definition of what it means to lead, and, therefore, people displaying these behaviors are seen as “leadership material.” Other styles of leading may then not be seen as “leadership” because they do not fit the standard definition. (Kirby, 1997, p. 18)

It is interesting to note that individuals with TJ preferences are overrepresented in samples of managers and leaders not only in corporate settings but also in contexts where the work would seem to appeal to people of other types, such as in clinical settings (Craig, Craig, & Sleight, 1988) and in education (Lueder, 1986a, 1986b).

Two studies by Salter not only demonstrate the relationship of type to occupational choice but also provide models for additional analytical approaches for studying the relationship. One study (Salter, 2006) examined the types of college professors. Given that professors work at many different institutions and specialize in many different disciplines, this was a challenging test for type theory. The sample was drawn from a CAPT database of 44,428 employed individuals who took MBTI Form G; 556 of the participants in this sample indicated that they were university teachers. Salter used predictive configural frequency analysis (PCFA) to analyze the data. A preliminary log-linear analysis showed that all four MBTI scales had some relationship to choice of this occupation. This preliminary analysis revealed main effects for E–I and S–N, as well as for an interaction of these two scales. Furthermore, although T–F and J–P main effects were not significant, their interaction was. The follow-up PCFA identified INTJs, INFJs, and INFPs as overrepresented among university faculty, with the findings about INTJs being the most pronounced. ISFJs were significantly

underrepresented. Salter reported that while some estimates of types in the population show that INTJs range from 2% to 7% of the population, nearly 20% of faculty in this study were INTJs. He concluded that homogeneity of this magnitude is impressive.

Salter (2003b) drew from the same base sample to examine whether the field of computer programming attracted certain types. Using log-linear analysis, he found that INTPs were significantly overrepresented among programmers and ESTJs were significantly underrepresented. As Salter pointed out, what is important here is that both the overrepresented and the underrepresented types are dominant Thinking types, but with different auxiliary processes, which also means that they use their dominant Thinking in different orientations. He concluded, “Apparently, the complexity of these distinctions [between using the dominant in the inner or outer world, and between Intuition and Sensing as auxiliary processes] may be a key to understanding how these particular individuals made their occupational choices” (p. 116).

In another study of programmers, Capretz (2003) sampled 100 software engineers and found that NTs were overrepresented in this sample, with a frequency of over 2.5 times what would be expected by chance. Thinking types and Thinking Judging types were over twice as likely to be found in the sample. The most overrepresented types were INTJs and INTPs, with SSRs of 3.4 and 2.5, respectively. These large attraction ratios are further evidence for the validity of the MBTI Step I assessment in career choice.

Briggs, Copeland, and Haynes (2007) looked at the personality preferences of accounting students over a five-year period. The sample consisted of third-year accounting undergraduates from 1999 to 2003. The authors found that the percentage of STJ personality preferences among accounting students was consistent with that among practicing accountants and did not change significantly over time. ISTJ was consistently the most common type, followed by ESTJ. The least common types were INFJ, ENFJ, and INFP. Students with Sensing and Judging preferences composed 50% of the overall sample and were never less than 47% in any one year. The authors pointed out that while the accounting profession and the skills required of a successful accountant have changed over time due to factors that include the impact of technology, the global economy, new sources of competition, and demands for accountability, their results suggest that personality preferences in the accounting profession have not changed significantly in more than 20 years. They concluded that the “accounting profession has not lost its thirst for STJs” (p. 534). They believe that a broader mix of personalities needs to be attracted to and retained in the accounting profession.

The Importance of the Mental Processes in Occupational Choice

The examples above, as well as the low degree of overlap in occupations chosen by opposite types illustrated in table 9.12, show how whole types are attracted to occupations that are congruent with the predictions of type theory. Because occupations seem to both require and reward specific ways of taking in information and making decisions based on that information, the process pairs—ST, SF, NF, and NT (i.e., the columns of the type table; see chapter 5 for a description of type tables and how they are constructed)—are deemed to have the most impact on job selection. The data support this hypothesis. Hammer and Macdaid (1992a) examined lists of occupational titles for each type and determined the amount of overlap on the occupational lists within types that have the same process pair (ST, SF, NF, or NT) but different orientation pairs (IJ, IP, EP, or EJ). When the process pair was the same but the orientation pair was different, they found a median overlap of 41%. When the process pairs were opposite (ST and NF; SF and NT) but the orientation pair was the same (both IJ, for example), the mean overlap was only 4%. These results are shown in table 9.13. Thus those sharing the same two processes have far more in common in terms of occupational selection than do those with opposite process pairs. For example, SFs are likely to be found in jobs similar to one another and likely to be in different jobs than NTs.

The same theoretical logic regarding the attraction of different types to different occupations can also be seen in the choice of academic major. For example, DiRienzo, Das, Synn, Kitts, and McGrath (2010) followed more than 6,000 students at a medium-size private university over a nine-year period and found that the type distributions in academic majors were similar to those in related occupations and were consistent with type theory. The most comprehensive source of information about type

Table 9.13 | Percentage of occupation overlap for MBTI® types with opposite process pairs/same orientation pairs versus types with same process pairs/opposite orientation pairs

Types			% occupation overlap
ISTJ	and	INFJ	4
ISTJ	and	ESTP	36
ESFJ	and	ENTJ	2
ESFJ	and	ISFP	32
ENFJ	and	ESTJ	4
ENFJ	and	INFP	44
INTJ	and	ISFJ	4
INTJ	and	ENTP	52

Source: Hammer & Macdaid (1992a), p. 37. Used with permission.

distributions in academic majors is *MBTI® Type Tables for College Majors* (Schaubhut & Thompson, 2011b).

Type Diversity Within an Occupation

While most of the research attention has been on the differential attraction of types to occupations, analyses have also examined type diversity within an occupation. Many “umbrella” occupations are composed of identifiable specialties or niches that would be predicted to differentially attract individuals with different preferences and of different types.

McCaulley (1977) analyzed data on the specialties of physicians from a longitudinal sample of 5,355 medical students who were followed over a decade after their admission to medical school. She found that types were likely to specialize in medical fields as predicted by type theory. For example, ISTJs were overrepresented in the specialty of pathology and underrepresented in psychiatry; INFPs were twice as likely to be found in psychiatry as would be predicted from their frequency as medical students; ESFJs were more likely to be found in pediatrics. McCaulley also showed that when physicians changed specialties, they more likely moved into specialties aligned with their personality preferences. This finding was interpreted as evidence that it is the personality that shapes the occupational path, not the occupational path that shapes the personality. For example, pediatricians did not become ESFJs after entering that specialty; rather, it was their ESFJ preferences that drew them to pediatrics. In other words, according to these data and to type theory, individuals in the same occupation do not develop a particular personality as a result of their work. The occupation does not cause the personality; rather, personality characteristics draw individuals toward certain occupations. Thus, individuals in the same occupation are likely to share some similar personality characteristics.

As further support of within-occupation type influence, Scherdin (1994) provided data on librarians and on the selection of various roles within the field by librarians with different type preferences. The two types most overrepresented in a sample of 1,600 librarians were ISTJ and INTJ. However, as predicted by type theory, librarians in children’s public services were more likely to prefer INFJ, INFP, and ENFJ. Those serving as administrators preferred EN. In another study, Garden (1997) examined the career paths of 341 software technical professionals in small high-tech companies. In examining four possible career paths (managerial, technical, challenging project, own company), she found statistically significant type differences. For example, a technical path was chosen most often by INTJs. So, even within an occupation, choice of specialty, choice of role, and selection of career path are also congruent with type.

There is compelling evidence from hundreds of occupations and hundreds of thousands of people demonstrating that people choose occupations or roles within occupations that are congruent with their type.

Attraction-Selection-Attrition

The results discussed above demonstrated the validity of Myers’ hypothesis that, when possible, people tend to choose their occupation in order to exercise their preferences. From the perspective of the occupation or the organization, these findings suggest that the result of these individual choices may be homogeneity within occupations and organizations, which was a concern of Briggs, Copeland, and Haynes (2007) for the accounting profession. If such homogeneity of occupations and organizations can be demonstrated, it offers additional evidence for the validity of type theory and its manifestation in the MBTI Step I assessment, but from the opposite direction.

Schneider and his colleagues proposed an explanation for occupational or organizational homogeneity called the attraction-selection-attrition (ASA) model. Schneider summarized the model in an article titled “The People Make the Place” (1987). The ASA model posits that differential selection processes should result in organizations that are relatively homogeneous with respect to member personalities, attitudes, and values (Schneider et al., 1998; Schneider, Goldstein, & Smith, 1995). The process through which similar people end up in similar organizations and career fields is as follows. First, individual differences in personality and interests lead to differential attraction to organizations in which people believe there are opportunities to exercise those personality preferences. Next, organizational decision makers select applicants who are similar to individuals already in the organization. Finally, individuals who are dissimilar to other organizational members but make it past the gatekeepers eventually leave, either voluntarily or involuntarily. So, the key factor influencing the relationship between the individual and the organization is the fit of the individual’s personality with that of the modal personality of the organization. Organizational homogeneity is increased by selecting people who fit and ridding the organization of those who do not. In other words, the theory posits that people do not randomly end up in particular occupations or organizations. They actively choose them and in turn are selected or rejected by the organizational gatekeepers, either immediately or eventually. The parallel with type theory is obvious.

In one study of the homogeneity hypothesis (Schneider et al., 1998), the sample consisted of 12,739 managers from 142 organizations. Analyses of both MBTI continuous scores and MBTI categories were conducted, but because the results were very similar, only the results of the continuous analyses were reported. Organizational

membership was the categorical independent variable, and the four continuous scores on the MBTI preference scales were dependent variables. The index used to examine the data was a multivariate analog of the univariate effect size. On the basis of a literature review, Schneider et al. adopted a median value of the effect size as the cutoff to determine whether homogeneity of personality is evident in organizations. A significant main effect was found for organizational membership on each of the four MBTI preference scales ($p < .01$). This finding suggests that organizations do, in fact, differ with respect to the personality characteristics of their members. Canonical correlations showed that the relationship between each of the four canonical variates (which were the best linear combination of the MBTI preference scales) and organizational membership ranged from .34 to .16. The average squared canonical correlation was .24, substantially exceeding the median value of .12 used as the criterion. The T–F preference scale dominated the first function and best distinguished between organizations, followed by, in order, the S–N, E–I, and J–P preference scales (note that the judging and perceiving processes had the highest correlations, similar to what was found with careers above). Based on this analysis, the authors concluded that organizations are relatively homogeneous with regard to the personality characteristics of their managers.

Bradley-Geist and Landis (2012) tested the ASA model with 1,013 managers from 17 occupations. Both organizational membership and occupational membership served as categorical independent variables in two analyses predicting the four MBTI preference scales, treated as continuous scores. Average deviation values were calculated for each of the MBTI preference scales by summing the deviation of each manager's personality score within one organization or occupation from the mean score for that organization or occupation, and then dividing by the total number of managers. Homogeneity criteria were determined from a literature review. For T–F, 100% of the T–F values met the criteria for homogeneity; 92.9% for E–I, 88.1% for J–P, and 61.9% for S–N. Organizations and occupations were generally homogeneous (83% of organizations and 89% of occupations, respectively). In addition, a significant relationship was evident between organizational membership and the MBTI variables beyond the effects of occupational membership. Similarly, when organizational membership was controlled for, a significant relationship was found between occupational membership and the MBTI variables. So, organizations can be distinguished from other organizations and occupations can be differentiated from other occupations in terms of member personality.

Wallick, Cambre, and McClugage (2000) demonstrated an element of the ASA model in admission to medical school. All 175 members of the first-year class at one

medical school completed the MBTI Step I assessment, as did 22 members of the school's admission committee. The majority of the students' MBTI preferences were for Extraversion, Sensing, Thinking, and Judging. The majority of the admission committee members' preferences were the same, except that there were equal numbers of Extraverts and Introverts. The authors concluded that similarity of the candidates' personality preferences to those of the interviewers may have influenced an interviewer's impression of an applicant and may have affected his or her "perceptions and decisions." This study provides indirect support for the attraction element and direct support for the selection component of the ASA model.

Quintero, Segal, King, and Black (2009) attempted to replicate and extend the Wallick study to determine whether the same process was occurring in the selection of orthopedic residents. The sample consisted of 135 fourth-year students attending US medical schools who interviewed for an orthopedic surgery training spot. The interviewers consisted of 30 department faculty members who interviewed these candidates. After each interview, interviewers ranked each interviewee. The authors then examined the degree of similarity in personality preference between an interviewer and an interviewee in relation to the rankings. Similarities in certain types of personalities between clinical faculty and medical students were associated with higher rankings by the faculty. The authors concluded that among clinician interviewers of medical students, "like may be selecting like."

Thomas, Benne, Marr, Thomas, and Hume (2000) studied engineering students and concluded that psychological type was related to both attraction into and attrition from the engineering program, as well as to certain aspects of performance. The predominant preferences in the class were Introversion, Thinking, and Judging, suggesting that engineering attracted individuals with these preferences; the sample was about equal on Sensing and Intuition. While there were no type differences for overall class grade, there were four significant outcomes for the grade on the final exam: Introverts had higher grades than did Extraverts; there were two two-way interactions (E–I x S–N and E–I x T–F) indicating that the ISs and the ITs performed better on the final exam; and there was also a significant four-way interaction showing that ISFPs were overrepresented among those who failed. The authors also analyzed attrition and found that of the 26 individuals who withdrew from the class, 25 had a preference for Extraversion or Feeling or for both Extraversion and Feeling.

Gailbreath, Wagner, Moffett, and Hein (1997) suggested that in addition to selection and attrition, promotion is also a factor influencing organizational homogeneity. If the ASA model is operating, and if an organization relies heavily on internal promotion, then high levels

of homogeneity among its leadership may occur. The authors tested this hypothesis in the US Army, which, like other organizations with strong cultures, has in place myriad mechanisms that manage attraction, selection, and attrition at every level of leader education and promotion. The primary study consisted of 1,755 lieutenant colonels and colonels selected for battalion and brigade command from 1987 to 1989. A follow-up sample consisted of 380 lieutenant colonels and colonels selected for battalion and brigade command in 1993. The base population used for comparison was 4,446 male college graduates.

Participants with STJ preferences were significantly overrepresented in the primary study sample (57% in the sample vs. 32% in the base population), and those with a Feeling preference were underrepresented (9% vs. 29% in the base). Furthermore, individuals who preferred Intuition were underrepresented, while those who preferred Introversion were overrepresented. The authors' results revealed a high degree of homogeneity in the sample: two types, ISTJ and ESTJ, composed more than 50% of the sample, whereas by chance any two types would make up 12.5% of a sample. Analysis of the follow-up sample yielded results that mirrored those from the larger sample. Although these findings supported their hypothesis of homogeneity in army leadership, the authors also noted an alternative explanation to ASA for these findings. Organizational members may have learned to suppress their true behavioral preferences or to behave in ways that the organization defines as successful. Once such adaptation has occurred, employees may begin to report organizationally preferred behaviors as their own, and the longer they are with an organization, the more apt they may be to do so. For this hypothesis to work, the reporting of preferred organizational behaviors would have had to carry over to participants' responses to the MBTI items as well. The reasoning behind this explanation is akin to the theory of work adjustment (Dawis & Lofquist, 1984; Lofquist & Dawis, 1969), which proposes that over time an individual's work personality becomes more congruent with the work environment. The relationship between type and work environments is discussed in the section "Organizations" later in this chapter.

Career Decision Making

The discussions above were focused on outcomes—the occupations or organizations chosen by individuals of different types. Another way to approach the relationship between type and occupation choice is to focus on the career decision-making process of the individuals. Career decision making offers additional fertile ground for an examination of the validity of type theory. The core of type theory is perception and judgment, and any effective career decision depends on good information

and rational judgment. Ideally, all important decisions should involve each mental process: the facts from Sensing, the possibilities from Intuition, the objective analysis of Thinking, and the values focus of Feeling. Also needed are each orientation: some Introverted reflective time and some Extraverted action, as well as time to stay open with Perceiving and time to come to a conclusion with Judging. Thus career decision making provides many opportunities to test type theory.

Hammer (1996a) reviewed six studies relating career decision and indecision to type variables. Type theory predicts that Judging types would move toward closure and Perceiving types would move toward gathering options. These studies, mostly of college students, provided evidence that Judging types demonstrate more career decisiveness than do Perceiving types. The data on the other preference pairs did not show consistent relationships to career decisiveness.

Vaccaro (1988) tracked a sample of 210 outplaced managers and noted that on average it took the Introverted types one month longer to find a new position than it took the Extraverted types, at every level of compensation. The hypothesized explanation for these results was that Extraverted types are likely to have more contacts or at least to make more contacts in the career search process and thus connect more quickly with a new employer. The Introverted types are likely to have or make fewer contacts and want to take in information and ponder it, and thus are predicted to move more slowly and carefully through the search process.

Krumboltz (1992) offered another perspective on career indecision, focusing on the social pressure accompanying the word *indecision*. He believed that career indecision is treated almost as a mental disorder by some professional counselors, yet learning how to make a career decision is not necessarily a legitimate activity in US society. He suggested an alternate view whereby "Open-mindedness can be viewed as a virtue. Keeping one's options open may be far preferable to stating goals which may or may not be possible of attainment and which could lead to frustration and unhappiness" (p. 244). In his *Career Beliefs Inventory* (CBI) Krumboltz (1991) identified 25 scales measuring career beliefs that may keep people from attaining their career goals. In a sample of adults seeking career counseling, there were no significant correlations between MBTI personality type and the beliefs blocking career goal achievement as measured on the CBI (Hammer, 1996a). Thus type appears to be independent of behaviors blocking career goal achievement, at least in this study with this sample.

Tinsley, Tinsley, and Rushing (2002) looked at type differences in response to highly structured career interventions in a sample of 50 women who had reentered the university but who were undecided about

their future career. Only the E–I and J–P preference pairs were investigated. Two career interventions were used: a card sort and a structured audiotape. Preferences for Introversion and Judging were positively associated with increased occupational knowledge, increased self-knowledge, and the perception that the intervention was helpful. Furthermore, a significant relationship was found between E–I and job search activity subsequent to the intervention: Introversion was associated with less job search activity following the intervention.

Järllström and Tsuzuki (2008) examined career search activities in relation to type. Individuals with preferences for NJ were significantly more likely to engage in career planning. Extraverts and individuals with preferences for ES were more likely to actively use career strategies. INFPs and individuals with preferences for NP or IP reported low career planning activity. Introverts and individuals with preferences for IP and IN reported low use of career strategies. All these findings can be explained by type theory.

ORGANIZATIONS

The core of type theory is how individuals take in information (perception) and how they use that information to make decisions (judgment). Since these processes underlie most work tasks, using variables from the work environment offers many possibilities to examine the validity of MBTI results. Type theory and the MBTI assessment provide a logical, coherent structure for understanding normal differences among individuals in a host of work-related activities. Furthermore, type theory outlines a model of lifelong individual development that can be useful when coaching leaders and managers in an organization.

Type theory predicts that when there is congruence between the individual's psychological type and the task demands of an occupation, more satisfaction is likely to result. Myers and McCaulley (1985) stated,

When there is a mismatch between type and occupation, the client usually reports feeling tired and inadequate. According to type theory, the mismatch causes fatigue because it is more tiring to use less-preferred processes. A mismatch also causes discouragement, because despite the greater expenditure of effort, the work product is less likely to show the quality of products that would be developed if the preferred process were utilized. Tasks that call on preferred and developed processes require less effort for better performance and give more satisfaction. (p. 78)

Organizational Leadership

Research findings shown in various tables in this chapter have provided evidence for the relationship between type preferences and leadership. Analyses of the CPI 260 scales and the MBTI preference scales (see tables 9.4 and 9.5) indicated significant effect sizes for the Leadership and Managerial Potential scales in the direction of E, N, and T, but not for the J–P scale. The four types scoring highest on the Leadership scale were ENTJ, ENTP, ESTJ, and ENFJ; those scoring lowest were the four Introverted Feeling types (INFP, ISFP, INFJ, and ISFJ). Once again, the difference between the findings for individual preferences and for the whole types points to the need to understand the effects of whole type as more than an additive combination of preferences.

This same pattern on the MBTI preference scales holds for the Leadership Style Personal Style Scale (PSS) of the *Strong Interest Inventory* assessment, as seen in table 9.9. This scale was designed to measure a preference for “meeting, directing, persuading, and leading” others versus leading by example and doing the job oneself rather than directing others. Individuals scoring toward the “directs others” pole enjoy “moving readily and gregariously into interpersonal settings and like to take initiative and take charge” (Harmon, Hansen, Borgen, & Hammer, 1994, p. 159). A study by Buboltz, Johnson, Nichols, Miller, and Thomas (2000) with undergraduates found significant correlations of Extraversion and Intuition with the “directs others” pole of the Strong Leadership Style PSS but not of Thinking; they confirmed the lack of relationship with J–P. Using a regression analysis, they found that all four preference scales accounted for 37% of the variance in the Strong Leadership Style PSS. In the regression analysis, the beta coefficient was significant for T–F in the direction of Thinking, along with the E–I and S–N coefficients. The types scoring toward the “likes to direct others” pole were the four Extraverted Intuitive types (ENFP, ENTP, ENFJ, ENTJ), and the types scoring toward the “lead by example” pole were the four Introverted Sensing types (ISTJ, ISFP, ISTP, ISFJ). Data provided in the article (Buboltz et al.) showed that the effect size for the difference between the highest- and lowest-scoring types was very large ($d = 2.11$).

The Expressed Control scale of the FIRO-B assessment also measures a need to take charge and establish direction for others. The results shown in table 9.7 indicate that there are significant effects for Extraversion and Thinking. The research discussed below using the *Benchmarks 360* rating scales and other such scales also provides evidence of the effect of MBTI preference and facet scales on a variety of behaviors relevant to leadership. Extraverts tend to rate themselves significantly higher on many of these scales, although these high evaluations of their leadership skills are not always shared by those they work with. Also, on these various

scales Thinking types tend to rate themselves higher on leadership in task-oriented behaviors, while Feeling types tend to rate themselves higher on people-oriented leadership behaviors. Research discussed in the section “Education” later in this chapter shows that student leaders tend to favor Extraversion and Judging.

Hautala (2006a) studied transformational leadership using a modified version of the *Leadership Practices Inventory* and the Finnish translation of an earlier version of the MBTI Step I assessment with 439 leaders and their subordinates. Leaders who prefer Extraversion rated themselves higher on visioning, challenging, enabling, and modeling than did Introverts. Leaders who prefer Intuition rated themselves higher on visioning, challenging, and modeling. Leaders with a Feeling preference had higher self-ratings on enabling. Leaders with a Perceiving preference rated themselves higher on challenging and rewarding.

On the overall transformation profile, leaders with E, N, and P preferences rated themselves higher than did their opposites. Leaders tended to believe that they were more transformational than did their subordinates. Subordinates agreed with the Perceiving leaders that they were more challenging and with Feeling leaders that they were more enabling. The differences between leaders and their subordinates were greatest for leaders who prefer Sensing versus leaders who prefer Intuition. Sensing leaders were rated by their subordinates as higher on visioning, modeling, and overall transformational abilities than Intuitive leaders, whereas the opposite was true of the self-ratings by the leaders. This is the opposite of what would be expected based on type theory. Hautala hypothesized that if the subordinates were primarily Sensing types, the imaginative and future-oriented Intuitive leaders may have been experienced as confusing.

Brown and Reilly (2009) conducted a similar study using a different measure of transformational leadership, the *Multifactor Leadership Questionnaire* (MLQ). The questionnaire was given to a sample of 2,411 managers, each with a minimum of three direct reports. Similar to what was reported in the Hautala (2006b) study, leaders who prefer Extraversion and Intuition rated themselves higher on overall transformational leadership, as well as on intellectual stimulation, inspirational motivation, and idealized influence. There were no differences on T–F or J–P. Unlike with the previous study, no significant correlations were found between subordinates’ ratings and any of the transformational leadership scales.

Hautala (2005) studied the effect of 167 subordinates’ MBTI preferences on their ratings of a small sample of managers (38 leaders) and found that the E and F subordinates rated their leaders as more transformational than did the I and T subordinates. Overall, the results did not confirm the hypothesis that subordinates whose

preferences are similar to their leaders’ preferences will rate them higher, although this was not tested directly.

Sundstrom and Busby (1997) focused on leaders’ peers, who rated the leaders of a large manufacturing firm ($N = 487$ top-tier leaders) using the *System for the Multiple Level Observation of Groups* (SYMLOG®) tool. The peers’ ratings of each of the eight Thinking types and a composite group of all Feeling types combined were compared. An overall MANOVA with the nine groups on the 26 SYMLOG scales showed a significant difference, indicating that their co-workers observed type differences in the leaders. This test was followed by univariate tests, which showed significant differences on 14 of the 26 scales. These in turn were followed by post hoc tests to determine which types differed. For space reasons, only a few of the findings are summarized here; interested readers should consult Sundstrom and Busby (1997) for extensive details. As would be expected, ESTJ and ISTJ leaders were rated significantly higher than were ENTP leaders on active reinforcement of authority, rules, and regulations. The composite group of Feeling types were rated higher on protecting less able members and providing help when needed; on having a good time, releasing tension, and relaxing control; on friendship and mutual pleasure; and on trust in the goodness of others. ISTJs were rated significantly higher on conservative, established, correct ways of doing things than were ENTPs. As can be seen from these examples, most of the ratings corresponded to predictions from type theory. The authors organized the findings by whole types and provided portraits of the leadership style of each type based on the ratings, which are worth reading.

A study of the potential for *ailing* at leadership or management was conducted over a six-year period by Gentry, Mondore, and Cox (2007) with managers ($N = 6,124$) from 1,889 different companies over 16 industries. Ratings on the five *Benchmarks* derail or problem scales (see below and appendix E) were used as the criterion, and an overall score was also calculated. (*Note: Derailers* are defined as dysfunctional behaviors that can stall or derail a career.) The ratings from their boss, peers, and direct reports were aggregated for each of the managers. No differences in overall derailment were found between managers who preferred Extraversion and those who preferred Introversion. Across the individual derailment scales, results were mixed for managers who preferred Extraversion and those who preferred Introversion. Managers who preferred Intuition, Thinking, and Perceiving were considered by the raters more likely to display derailment characteristics than were their opposites. In terms of whole types, ENTPs were considered the most likely to display derailment behaviors, and INTPs and ESTPs were among the top five types on many of the comparisons. ISFPs were the least likely to be rated as exhibiting derailment behaviors, along with ESFJs, ESFPs, and ISFJs.

Hautala (2006b) asked subordinates with different process pairs how they experienced development conversations with their ST leaders. The NT subordinates differed the most from the others in perceiving the discussions most positively. The NF subordinates, who differed from their leaders on both processes, experienced the discussions most negatively. There was little difference between the SF and ST subordinates' experiences.

A study of promotion in a large manufacturing organization (Pearman, 1993) found that Intuitive types were significantly more likely to be promoted to leadership positions (higher levels of responsibility) than were Sensing types. Rytting, Ware, and Prince (1994) compiled a sample of CEOs of successful moderate-size companies (between 250 and 1,000 employees). The type table frequencies provided in the article were compared to those for a US national representative sample (Hammer & Mitchell, 1996). Intuitive types were overrepresented among the CEOs, and the frequency of NT types was almost 2.5 times what would be expected by chance. Hammett (2007) provided type data on C-level executives ($N = 70$) attending programs at the Center for Creative Leadership. Comparing the type distribution to that of the Form M US national representative sample ($N = 3,009$) showed that NTs were more than 4.5 times more likely to be found in the C-level group than in the US national sample ($N = 3,036$). The frequency of two types, ENTJ and INTJ, was more than seven times the frequency in the US national representative sample. Hackston (2015), using a sample from seven countries, also found that Intuitive types were more likely to be found at more senior levels of organizations. Intuitive types represented 80% of the top executives versus 50% of employees.

In general, the leadership studies reviewed here suggest that individuals who prefer Extraversion and Intuition tend to score higher on scales measuring leadership. This is in accordance with type theory, in that many leadership roles require an outward focus and the ability to step back and see the big picture. Results for the T–F scale are mixed, as it depends on whether the focus of the leadership scale is on tasks or people. For additional compilations of research, as well as applications on the MBTI assessments and leadership, see Fitzgerald and Kirby (1997), Fitzgerald (1994), and Hammett (2007).

Behaviors, Competencies, and Performance

Participants in programs at the Center for Creative Leadership were administered MBTI Form Q ($N = 4,442$), and each participant also rated him- or herself on the CCL *Benchmarks* assessment, which measures 16 leadership behaviors and six derailers. (See appendix E for a definition of the *Benchmarks* scales used in this analysis.) Each participant was also rated by his or her boss, peers, and direct reports. Mean differences

between opposite preferences were calculated for each set of raters. While there were many statistically significant differences, only those for which Cohen's $d \geq .20$ are discussed below.

Extraverts rated themselves significantly higher than did Introverts on the *Benchmarks* scales Resourcefulness, Doing Whatever It Takes, Decisiveness, Leading Employees, Confronting Problem Employees, Change Management, Building and Mending Relationships, Balance Between Personal Life and Work, Putting Others at Ease, Differences Matter, and Career Management. Introverts rated themselves higher on Difficulty Building and Leading a Team. What is most interesting about these results is that they were *not* duplicated when ratings by others were examined. These managers' bosses and peers rated the Extraverts higher on only one leadership behavior, Putting People at Ease, and there were no differences between Extraverts and Introverts in ratings by their direct reports.

Participants with a preference for Intuition rated themselves higher on Doing Whatever It Takes and Being a Quick Study, behaviors that would seem to take advantage of their ability to develop alternative approaches and quickly intuit the relevance of new information. There were no differences at the specified criterion level in ratings by bosses, peers, or direct reports. However, bosses did rate Intuitive types somewhat higher on Differences Matter (a measure of recognition of the positive effects of diversity) and on one derailer scale, Failure to Meet Business Objectives (both $d = .11$). Both peers and direct reports agreed with the latter rating, and the effect size was the same.

Managers who preferred Thinking rated themselves higher on Decisiveness, while managers who preferred Feeling rated themselves higher on Compassion and Sensitivity, Balance Between Personal Life and Work, and Putting People at Ease. Bosses also rated Feeling managers higher on Compassion and Sensitivity and Putting People at Ease. Thinking types were rated higher by their boss on one of the derailer scales, Problems with Interpersonal Relationships. However, the majority of the differences on T–F were between self-ratings and ratings by peers and direct reports. Peers rated managers who preferred Feeling significantly higher than managers who preferred Thinking on Participative Management, Building and Mending Relationships, Compassion and Sensitivity, Balance Between Personal Life and Work, Self-Awareness, Putting People at Ease, Differences Matter, and Career Management, and rated them somewhat higher on Leading Employees ($d = .17$). Managers who preferred Thinking were rated higher by their peers on two derailer scales: Problems with Interpersonal Relationships and Difficulty Building and Leading a Team. Their direct reports rated the Feeling types higher on the same leadership skills as did their peers, except for Self-Awareness. Their direct reports rated Thinking managers

higher than Feeling managers on the same two derailers scales. These findings are in accordance with type theory and provide evidence for the observation of type differences by others.

No differences at the criterion level were found between Judging and Perceiving managers by any of the raters. However, Judging types rated themselves higher on Confronting Problem Subordinates ($d = .18$) and Perceiving types were rated higher by all raters on Failure to Meet Business Objectives, although the effects were very small, with Cohen's d values ranging from .09 by direct reports to .13 by self and boss.

Johnson and Thompson (2013, 2014) conducted two studies of the relationship between MBTI preferences and a competency model that identifies three overarching themes of personality in the workplace—intrapersonal, interpersonal, and task-oriented. These themes are measured via 25 separate competencies. In both studies, participants were asked to rate each item making up the competency scales as to whether it was “like me,” or typically described them, or was “not like me.” Note that a rating of “not like me” was not a statement about the participants’ performance; it was about the frequency or typicality of the particular behavior. In the first study (Johnson & Thompson, 2013), the sample consisted of 743 individuals who had completed the MBTI Step I assessment and had agreed to participate in research; the age range was 19 to 71 years, with an average age of 44. Among the participants, 38% were nonsupervisory employees and 30% were in management; 13 industry sectors were represented. Findings showed that on all competencies participants with a preference for Extraversion endorsed more items as “like me” than did those with a preference for Introversion. (Recall that in the ACL findings reported earlier, Extraverts also used more adjectives, and more positive adjectives, to describe themselves.) The results for item-level descriptors endorsed at significantly different rates by each of the 16 types are shown in table 9.14. These results offer further validity evidence, as the significant items for each type, both those that were seen as “like me” and those assessed as “not like me,” conform quite well to what would be predicted by type theory.

In the second study by Johnson and Thompson (2014), the same competencies measure was administered to a wide range of individuals ($N = 166$). In addition, 37 of these individuals were able to solicit 119 co-workers, who rated the participants on the same competencies. The number of ratings by others per participant ranged from 1 to 13, with an average of 4. The self-ratings were very similar to those reported above for the first study. In both studies, there was clear separation between preference pairs on the competencies as well as clear separation between item-level descriptors for the 16 types. Perhaps most important, the significant differences found in both studies aligned with definitions of the preferences

and type descriptions, as well as with Jungian theory. However, results from the observer ratings obtained in this study were very different from the self-ratings in both studies, showing an inverse pattern of ratings between self and others. For example, on a number of competencies (e.g., Planning and Organizing), individuals who preferred Sensing rated themselves significantly higher than did those with a preference for Intuition. But the observer ratings on many of these competencies were just the opposite, with the managers who preferred Intuition being rated higher. Recall that in the study by Hautala (2006a), inverse ratings between managers and others was found for transformational leadership. The inverse pattern of differences between self-ratings and ratings by others requires further research before any conclusions can be drawn. It is likely that the type preferences of the observers must be taken into account, or the interaction between the type of the manager and the type of the observers.

Schullery, Knudstrup, Schullery, and Pfaff (2009) collected self-ratings from 423 managers, as well as ratings by their bosses, peers, and direct reports, using the *Management Leadership Practices Inventory* (MLPI). Given that the authors examined 52 different combinations of MBTI preferences across four sets of raters on 20 MLPI scales, plus four additional factors, their many significant results cannot be reported here due to space limitations. A broad summary will have to suffice; interested readers are encouraged to consult the original article. Managers with SJ preferences were evaluated highly by their bosses, peers, and direct reports on a task factor, while ENTP and ENFP managers were rated highly by their peers on a people-skill factor. SP managers were rated lower by peers on the task factor. NJs tended to be rated low by their boss on the task factor. One finding that runs counter to those discussed above for the other management rating assessments was that no significant differences were found on the E–I scale on any of the 20 MLPI scales, although some significant differences were found when Extraversion was paired with other preferences. SJ managers were the only group for which all four rating groups agreed. As the authors stated, “The SJs alone seemed to be able to please all of the people all of the time, at least concerning task accomplishment” (p. 147).

Cheng, Wang, Horng, and Huang (2007) investigated the relationship between personality-task congruence and performance in a sample of R&D managers. Different R&D tasks were grouped into four categories of creativity. The sample consisted of 205 R&D project leaders. Fifty leaders with the highest I, N, F, or P preference scores were compared to 50 leaders whose E, S, T, or J preference scores were the highest. Outside judges were used to rate performance. Researchers who preferred Intuition outperformed those who preferred Sensing on proactive-creativity projects, which were self-initiated and required goal definition. Sensing researchers

Table 9.14 | Item-level descriptors of competencies for the 16 MBTI® types

	ISTJ (n = 100)	ISFJ (n = 39)	INFJ (n = 25)	INTJ (n = 43)
Like me	Organize and structure Organize materials Analyze data Follow guidelines	Avoid confrontation Maintain organized records Avoid conflict	Avoid confrontation	
Not like me	Build relationships Relate to people Personalize the message Express emotions	Think about the big picture Debate alternatives Integrate information Relate to the bigger picture	Accept criticism Make quick decisions Make bold decisions Confront others	Have a sense of humor Empathize with others Express emotions Consider others' welfare
	ISTP (n = 35)	ISFP (n = 17)	INFP (n = 43)	INTP (n = 39)
Like me		Avoid confrontation	Consider others' welfare Avoid confrontation Consider feelings of others Demonstrate concern	Experiment Disregard limitations
Not like me	Communicate thought processes Identify opportunities Generate novel ideas Set clear goals	Self-confident Debrief unsuccessful events	Set clear goals Set clear standards Take action Guide team goals	Follow guidelines Express emotions Maintain organized records Network
	ESTP (n = 30)	ESFP (n = 13)	ENFP (n = 89)	ENTP (n = 47)
Like me	Handle technical challenges	Polite Listen objectively Embrace ambiguity Relate to people	Network Consider feelings of others Relate to people Express emotions	Embrace change Generate excitement Improvise Make quick decisions
Not like me	Organize materials Organize and structure		Organize materials Organize and structure Focus on time Handle technical challenges	Create schedules Avoid conflict Avoid confrontation Dignified
	ESTJ (n = 88)	ESFJ (n = 41)	ENFJ (n = 40)	ENTJ (n = 54)
Like me	Confront others Confront issues early Create schedules Network	Consider feelings of others Create detailed records Organize teams Follow guidelines	Express emotions Build relationships Provide feedback Provide encouragement	Provide clear vision Communicate vision Generate novel ideas Delegate tasks
Not like me	Avoid confrontation Consider feelings of others Avoid conflict Listen to others		Analyze data Handle technical challenges Explain technical information	Avoid conflict Avoid confrontation

Note: Research sample, N = 743.

Source: Johnson & Thompson (2013). Used with permission.

outperformed Intuitive researchers on expected-creativity projects, which were tasks demanded by others, but for which there was no known solution to the problem. Judging types outperformed Perceiving types on responsive projects, which were tasks demanded by others for which a solution had been defined previously. The distribution of types in the sample precluded a test of the T–F scale. The authors concluded that there was indeed a relation between the individual-project fit and research performance. Researchers who prefer Intuition, who are constantly looking for meaning beyond the surface and have a preference for originality, appear to perform better on self-initiated research with novel problems. In contrast, researchers who prefer Sensing, who attend primarily to sense-based, factual experiences, perform better on the externally assigned projects with novel problems. Researchers who prefer Judging do better with assigned and closed or bounded projects. (Other studies of the effects of personality-task fit are found in the section “Health, Stress, and Coping” later in this chapter.) This important study confirms one of the predictions of type theory: that people will perform better at tasks that fit their preferences.

Research and development managers performed better on research projects that fit their preferences.

Job Satisfaction

When occupational tasks are compatible with the preferences or inclinations of the individual performing them, higher satisfaction is likely to result than when the fit is less congruent (Dawis, 1996). Hammer (1996a) summarized studies of job satisfaction in the *MBTI® Career Report Manual* (Hammer & Macdavid, 1992a) and included a synopsis of more than a dozen studies on the subject. Among the occupations studied for job satisfaction were bank officers/financial managers, computer professionals, dietitians, elementary and secondary school teachers, intensive care nurses, healthcare managers and executives, lawyers, managers, occupational therapists, parish pastors, pediatric nurse practitioners, pharmacists, marketing teachers, secretaries, teachers, and vocational education administrators. Hammer also pointed out some of the difficulties in conducting research on satisfaction, such as a restriction in range of data on job satisfaction (most people tend to say they are satisfied with their job). He summarized the literature as follows:

When satisfaction is measured globally, its relationship with psychological type is equivocal. However, among those studies that do show a relationship, a pattern seems to emerge. Overall, Introverts and Perceiving types seem less satisfied with their work than do Extraverts and Judging types, although the one study that examined men and women separately suggested that

overall results may be misleading if gender is not accounted for. When specific facets or aspects of job satisfaction are employed instead of global measures, the picture becomes clearer. For example, the T–F scale seems to be important in identifying satisfaction with co-workers. Type theory would predict that different types will have different criteria for satisfaction, and this seems to be at least partially supported by the research. . . . Studies of person-environment fit . . . suggest that those who are dissatisfied in an occupation tend to be those types who are opposite from the modal type in the occupation. A number of studies have also suggested that those types who are less frequent or underrepresented in an occupation tend to be less satisfied or have higher intention to leave the occupation than do those types who are more frequent or whose fit with the occupation is judged to be better. (pp. 40–41)

Subsequent studies support Hammer’s 1996 summary. Hopkins (1997) surveyed 133 members of an association of type practitioners (the sample was highly educated, high in job tenure, middle-aged, and Caucasian) and concluded that people who felt their job matched their personality were more satisfied with their job than were those who did not. SFs were the least satisfied group. In personal telephone calls with some participants, Hopkins (personal communication) also noted that, of the four process pair groups, SFs seemed the most grateful that someone was listening to them and their concerns. In his study, co-workers, the work itself, and supervision were far more important to job satisfaction than were pay and promotions.

Ratings of work satisfaction and dissatisfaction were gathered from participants in the Form M US national sample. Participants were asked to indicate their responses on a 4-point scale of “very satisfied,” “somewhat satisfied,” “somewhat dissatisfied,” or “very dissatisfied” on 3 questions related to their overall work (the work itself, the company where they worked, and future work opportunities) and on 14 questions related to specific aspects of their work. Income data were gathered as well. The vast majority of participants in this sample, over 89%, were at least somewhat satisfied with the kind of work they were doing. The participants expressing the most satisfaction with their work were the EJ group. Regarding the organization for which they worked, 77.2% were at least “somewhat satisfied,” with ESFJs and ENFJs the most satisfied. Regarding future work opportunities, 67.2% were at least “somewhat satisfied,” with ENTJs clearly the most satisfied. The participants with the highest income level were the ENTJs, while their opposites, the ISFPs, had the lowest average income level. The most dissatisfied participants appeared to be the IP group, especially the INPs, who were dissatisfied with their work, their company, and their future work opportunities. Satisfaction of whole types with specific aspects of their work are summarized in table 9.15. Clearly, satisfaction with various aspects of work differs across the 16 types.

Table 9.15 | Percentage of satisfaction/dissatisfaction with aspects of work by MBTI® type

Aspect of work	Average % of "very dissatisfied" responses	Type with highest % of "very dissatisfied"	Average % of "very satisfied" responses	Type with highest % of "very satisfied"
Opportunities for promotion	22.3	INTJ 32.0	19.6	ENTJ 35.1
Salary	15.0	INFJ 25.8	19.3	ESTJ 28.9
Amount of stress	13.3	INFP 18.9	19.8	ESTP 29.7
Job security	11.8	INTP 19.7	39.0	ENFJ 56.9
Opportunities for accomplishment	11.2	INFP 15.3	30.8	ENTJ 51.4
Opportunities for learning	10.3	ESTP 14.4	36.1	ENFJ 56.9
Opportunities to contribute to society	10.3	ESTP 16.7	36.3	ENTJ 54.1
Predictability/stability of job	10.1	INTP 17.4	32.3	ESFJ 54.1
Working conditions (hours, vacations, benefits, etc.)	9.5	INTP 15.5	38.7	INFJ 48.4
Opportunity to use talents or training	9.5	ENTP 15.5	38.3	ENTJ 59.5
Respect I get	8.9	INFP 13.5	36.1	ESFJ 48.6
Responsibility	4.4	INTP 8.6	41.2	ENFJ 52.9
Day-to-day tasks	3.8	ESFP 5.9	32.0	ENTJ 41.7
People I work with	3.4	INTP 10.0	45.9	ENFJ 60.0

Note: MBTI® Form M US national sample, $N = 3,036$.

Participants in two samples drawn from a database compiled by the publisher were asked questions about job satisfaction. Participants in one sample ($n = 5,549$) were asked how satisfied they were with their current job. Participants in the other sample ($n = 11,204$) were asked how satisfied they were with their general line of work. Table 9.16 shows the results for both questions for whole types, ranked from most to least satisfied. For the general-line-of-work question, the Extraverts were more satisfied than the Introverts, although within each of those preference groups there seems to be little evidence of a pattern for the other preference pairs. It appears that characteristics of whole types combine to determine overall satisfaction. Regarding satisfaction with their current job, the results were somewhat different. The dominant extraverted Intuitive types (ENTP, ENFP) were the most satisfied, and three of the IP types (INFP, ISTP, ISFP) were the least satisfied. Again, the rankings seem to reflect differences in whole types rather than preference scales.

Retention/Turnover

Job satisfaction is one of the drivers of turnover and its counterpart, retention. Turnover is very costly to organizations, and while there are descriptions of the expected relationships between type and turnover (Hammer, 2003), much more research is needed on this important topic. Turnover was an initial topic of research on type in organizations. Myers and McCaulley (1985) summarized some of these very early studies, which were conducted by Laney as far back as 1949. These studies showed that types that were less frequent in the work environment or who were working at tasks that did not seem a good fit for their type were more likely to leave.

Turnover may be voluntary or involuntary and can result from many factors, including unsuccessful job performance, changes in the market leading to downsizing, or job dissatisfaction. Vocational research amply supports (1) the prediction of turnover from job dissatisfaction and (2) the fact that people perform better in a job when they are satisfied (Dawis, 1996). The ASA model, discussed in the section "Occupational Selection and Careers" earlier in this chapter, can also encompass turnover, since it is a manifestation of the attrition component of the process.

In reviewing the literature on type and turnover, the focus has been on the relationship between job fit and job dissatisfaction. Hammer (1996a) reviewed research from the 1985 *MBTI® Manual* (Myers & McCaulley, 1985) and other studies and concluded,

Although few studies have been conducted on turnover, those that are available provide some support for the proposition that types working in environments or jobs that are not a good match for their preferences are more likely to leave or to say they are going to leave than are those whose type provides a better fit for either the tasks or the environment. Future studies examining the relationship of type and turnover should heed Garden's (1989) finding that organizational size may be a mediating variable. (p. 47)

Individuals composing a portion of the global sample ($n = 11,216$) were asked about their intent to leave their job. Judging types were slightly less likely to indicate that they might leave their job than were Perceiving types ($d = .09$). The results for whole types are shown in table 9.16. The overall difference between the 16 types was significant at $p < .001$. The types least likely to want

Table 9.16 | Satisfaction with line of work, job satisfaction, and turnover intent of MBTI® types ranked by mean score

Satisfaction with line of work (n = 11,204)			Job satisfaction (n = 5,549)			Turnover intent (n = 11,216)		
Type	M	SD	Type	M	SD	Type	M	SD
ESFJ	4.71	1.11	ENTP	4.51	1.22	ESFJ	2.45	1.43
ENTJ	4.70	1.14	ENFP	4.49	1.30	ESTJ	2.47	1.35
ESTJ	4.64	1.16	INTJ	4.47	1.20	INFJ	2.49	1.42
ENTP	4.64	1.19	ENTJ	4.47	1.21	ESFP	2.52	1.41
ESFP	4.62	1.12	ESTJ	4.47	1.23	ISFP	2.52	1.31
ENFJ	4.58	1.12	INFJ	4.45	1.26	ISTJ	2.52	1.39
ESTP	4.54	1.20	ESFJ	4.43	1.26	ENFP	2.54	1.40
ENFP	4.52	1.21	ISFJ	4.40	1.28	ISFJ	2.54	1.43
INTJ	4.52	1.20	INTP	4.38	1.12	ENFJ	2.54	1.40
ISFJ	4.47	1.20	ENFJ	4.38	1.22	ENTJ	2.54	1.21
ISTJ	4.46	1.16	ESFP	4.38	1.27	ISTP	2.59	1.33
INTP	4.45	1.14	ESTP	4.26	1.31	ENTP	2.60	1.38
ISTP	4.44	1.15	ISTJ	4.25	1.25	INTJ	2.68	1.39
INFJ	4.42	1.15	ISFP	4.23	1.30	ESTP	2.69	1.38
ISFP	4.34	1.22	ISTP	4.23	1.23	INTP	2.74	1.33
INFP	4.27	1.24	INFP	4.00	1.37	INFP	2.81	1.38

Note: Portions of the global sample who responded to research questions. For turnover intent, higher scores indicate higher intent to leave.

to leave were the ESFJs and the ESTJs. The types most likely to want to leave were the INTPs and the INFPs. However, the difference between the highest- and lowest-ranking types, the ESFJs and INFPs, was in the small range ($d = .26$).

Garden (1989) studied computer professionals in large and small companies. Those with I, N, and P preferences expected to leave their employer if it was a larger company; perhaps they sought more autonomy, which is likely to come in a smaller company. Those with S and J preferences said they were more likely to leave smaller companies, perhaps seeking the more structured regimen of a larger company.

Hanbury, Sapat, and Washington (2004) examined retention among city managers. Those with higher scores toward Introversion and Perceiving were significantly more likely to have longer tenure. This finding runs counter to other research on job satisfaction and to type theory. Previous research suggested that IP types tend to be less satisfied with many aspects of their job.

Work Environments

Research has documented the preference for different work environments by different types. A set of work environment characteristics was presented to participants in the Form M US national sample, who were asked to indicate whether they liked, were indifferent to, or disliked each characteristic. Table 9.17 shows, in

descending order, the five characteristics that were most liked by participants of each of the 16 types.

For almost all the Sensing participants, the exceptions being the ESTPs and ESFPs, “clear structure” was the most important work environment characteristic. For the ESTPs and ESFPs, “clear structure” was either second or third in importance. “Variety of tasks” was the most important work environment characteristic for all Intuitive types except the INFPs, who placed it second; having a variety of tasks may lead to the possibilities often sought by Intuitive types. “Loyalty and security” was among the top five for all types except the NTs, who may see security as something that can’t be logically guaranteed. They chose instead working with “people from different backgrounds,” a characteristic appearing in their top five but in no other types’ top five. Perhaps people from different backgrounds help bring the desired variety and challenge to the work of NTs.

A different set of work environment characteristics, the Ideal scales from the *Work Environment Scale*, were correlated with Form M continuous scores in a sample of 98 employees in a variety of occupations at a large midwestern university. The results are summarized in table 9.18. For space reasons, the table includes only those work environment characteristics that correlated $\geq .20$ with one of the MBTI preferences or for which post hoc analyses were significant for the process pairs or the orientation pairs. Work environments favoring positive interpersonal interaction were more important to Extraverted and Feeling types and less so to ST types.

Autonomy was more important to the IPs than to the EJs; studies on values discussed later in this chapter also show this pattern. Innovation was important to Intuitive types, especially the NFs. Pleasant physical surroundings were more important to EPs than to IJs. These studies of work environments demonstrate that people of different types tend to prefer environments that allow them to exercise their preferences.

The work setting preferences of physicians were reported by Quenk and Albert (1975). Among the findings, Extraverted physicians favored active involvement with the outside world, including with their community. Feeling types emphasized involvement with the people in their life and practice. Thinking types seemed to

concentrate more on the nonpeople aspects of their work, such as their continued advancement. Judging types desired to maintain the boundaries in their practice between their personal and professional lives.

A wealth of research evidence relevant to organizational behavior; leadership; job performance; job satisfaction, retention, and turnover; and preferred work environment characteristics overwhelmingly affirms the relationship of these variables to predicted type characteristics.

Table 9.17 | **Top five most important work environment characteristics for each MBTI® type**

ISTJ	ISFJ	INFJ	INTJ
Clear structure	Clear structure	Variety of tasks	Variety of tasks
Independence and achievement	Loyalty/security	Clear structure	Clear structure
Loyalty/security	Independence and achievement	Independence and achievement	Independence and achievement
Variety of tasks	Variety of tasks	Loyalty/security	Teamwork
Teamwork	Teamwork	Teamwork	People from different backgrounds
ISTP	ISFP	INFP	INTP
Clear structure	Clear structure	Independence and achievement	Variety of tasks
Variety of tasks	Loyalty/security	Variety of tasks	Independence and achievement
Independence and achievement	Independence and achievement	Clear structure	Clear structure
Loyalty/security	Variety of tasks	Teamwork	Teamwork
Teamwork	Teamwork	Loyalty/security	People from different backgrounds
ESTP	ESFP	ENFP	ENTP
Variety of tasks	Variety of tasks	Variety of tasks	Variety of tasks
Independence and achievement	Clear structure	Independence and achievement	Independence and achievement
Clear structure	Independence and achievement	Teamwork	Clear structure
Teamwork	Loyalty/security	Clear structure	Teamwork
Loyalty/security	Teamwork	Loyalty/security	People from different backgrounds
ESTJ	ESFJ	ENFJ	ENTJ
Clear structure	Clear structure	Variety of tasks	Variety of tasks
Teamwork	Loyalty/security	Independence and achievement	Independence and achievement
Variety of tasks	Variety of tasks	Teamwork	Clear structure
Independence and achievement	Independence and achievement	Clear structure	Teamwork
Loyalty/security	Teamwork	Loyalty/security	People from different backgrounds

Note: Form M US national sample, N = 3,036. Work environment characteristics in bold type were “liked” by more than 80% of individuals of that type.

Table 9.18 | **Correlations of MBTI® Form M continuous scores with Ideal scales from the Work Environment Scale**

Work Environment Scale Ideal scale	Preference	Process pair	Orientation pair
Co-worker cohesion (friendly, supportive co-workers)	E, F	ST < SF, NF, NT	
Supervisor support (supportive management)	E	ST < SF, NF, NT	
Autonomy (self-sufficient employees)			IP > EJ
Task orientation (emphasis on good planning, efficiency, getting job done)	E	ST < SF, NF, NT	
Clarity (employees know what to expect in daily tasks and company policies)		SF > ST, NT	
Managerial control (use of rules to keep employees under control)	S, J	ST > NF, NT SF > N	EJ > IP EJ > EP IJ > IP
Innovation (variety, change emphasized)	N	NF > ST, SF NF > ST	
Physical comfort (pleasant physical surroundings)			EP > IJ

Note: $N = 98$. Preferences and pairs represent results for which correlations with preferences are $\geq .20$. All results are statistically significant at the .05 level or better. Pair results are based on MANOVAs and post hoc analyses.

Source: Data provided by the Employee Career Enrichment Program, University of Minnesota.

Decision Making

With the exception of career decision making, which is discussed above, most research on decision making has been conducted with samples of managers because making decisions is one of the primary actions they are expected to perform and on which they are evaluated. Isabel Myers described using each of the four processes to improve decision making (see Myers with Myers, 1980/1995, especially pp. 194–196). Because type is dynamic, attempting to find relationships between only one MBTI preference pair and some aspect of decision making may lead to frustration. For example, though ENTJs and ENFJs differ on only one preference, ENTJs extravert their dominant Thinking process and will make decisions very differently from ENFJs, who extravert their dominant Feeling process. Walck (1996) discussed additional problems in research on type and decision making and also noted that situational factors play an important role in the way decisions are made. Despite these inherent difficulties, research on type and decision making has progressed. See Walck (1996) for a summary.

Hough and Ogilvie (2005) conducted a test of the effect of cognitive style on strategic decision outcomes. In reviewing the literature, they observed that most of the studies that use cognitive style or personality as antecedents of strategic decision performance used the MBTI Step I assessment as the measure of cognitive style, for four reasons. First, the MBTI assessment is widely used and understood within organizations. Second, the use of MBTI preferences, as opposed to

absolute measures of cognitive ability, is less threatening to managers. Third, the MBTI personality types are conceptually related to the information-gathering and information evaluation aspects of decision making. Finally, the authors believed that the research evidence supported the conceptual, construct, and predictive validity of the MBTI Step I assessment.

In their study, Hough and Ogilvie recruited 749 senior managers and executives from programs at the Center for Creative Leadership who had taken the MBTI assessment. The dependent measures were decision quality, decisiveness, and decision effectiveness. Criteria for decision quality were determined a priori by a panel of experts, and then each participant was graded against these external criteria. Decisiveness was measured by the percentage of problems presented to participants that were addressed by them. Effectiveness was measured by asking each manager to rank every other manager, and him- or herself, on the “ability to get things done.”

For decision quality, the authors found that NTs made the highest-quality decisions, NFs the next-highest, SFs the third-highest, and STs the lowest. For decisiveness, they found that NT managers exhibited higher levels of decisiveness than did SF managers. NT managers seem to use the speed of their intuition to rapidly incorporate logical conclusions into choices, thereby making more decisions in a given time frame than managers with other combinations of processes. Hough and Ogilvie hypothesized that SF managers made the fewest decisions in a given time frame because “taking

into account the subjective and emotional values of numerous others can be a more time-consuming process than using logical analysis and would therefore lead to fewer decisions in a given time frame” (p. 441). There were no differences between Extraverts and Introverts or between Perceiving and Judging types. The lack of results for E–I and J–P recalls the study reported above using the *Benchmarks* assessment. In that study, Extraverts rated themselves higher than did Introverts on Decisiveness, whereas bosses saw no difference between Extraverts and Introverts on this behavior. In that same study, there were no differences between Judging and Perceiving types on Decisiveness, confirming the result here. It may be that E–I and J–P influence the process by which decisions are made and not the quality of the decision.

For decision effectiveness, managers with a preference for Thinking and those who preferred Extraversion were seen by others as more effective than those who preferred Feeling or Introversion. The J–P preference pair was not statistically related to effectiveness. Decision makers who preferred Judging or Perceiving were equally decisive and equally effective. Hough and Ogilvie hypothesized that this result might be a function of the sample. Participants were mid- and upper-level managers, whose job it is to make decisions. These results might be different in the general population. Based on these results, Hough and Ogilvie concluded that cognitive style, as measured by the MBTI Step I assessment, influences not only actual decision outcomes but also how others perceive decision performance.

Pencheva, Vassileva, and Bozhinva (2006) correlated the MBTI preference scales with two scales developed by Franken to measure decision-making style: the Active-Decisive scale, which identifies an active style that quickly adapts to new information, and the Individual Responsibility scale, which taps a desire for personal freedom in decision making without external interference. A preference for Perceiving correlated significantly with the Active-Decisive style. There were also smaller but significant correlations of Active-Decisive with Extraversion, Intuition, and Feeling. The Individual Responsibility scale correlated significantly with Thinking.

A study by Haley and Stumpf (1989) focused on the heuristics and biases used by managers when making decisions. Managers participated in a large-scale simulation in which they had to make multiple decisions and recommendations for action. Judges with no knowledge of the participants’ type evaluated their decisions using a taxonomy of common biases. Eighty percent of SF participants manifested availability biases (focused heavily on value-laden or emotional information, even when confronted with more objective information); 57% of NFs exhibited vividness biases

(focused on the idiosyncratic, memorable information instead of quantitative information); STs succumbed to anchoring (based decisions on initial quantitative data and did not search for additional information or ignored information presented by others); NTs were affected by perseverance input biases (they stayed with their initial viewpoint). In terms of recommendations, STs made more functional fixedness proposals (operationally focused, low risk, a response to immediate problems), and NTs made more positivity recommendations (placed a strong emphasis on opportunities and minimized potential threats). It is important to note that the heuristic biases, which conform closely to type theory predictions, were observed by judges who did not know the type preferences of the managers they were rating.

Schmitt, Shupp, Swope, and Mayer (2008) conducted an ultimatum game experiment to test the effects of type on decision making. The decision-making paradigm involved one person who makes an offer and one who responds to the offer. This interchange is conducted in the presence of a third player, who receives five dollars when an offer is accepted. The sample consisted of 80 students at the US Naval Academy. Feeling types made higher offers than did Thinking types. The authors concluded that Feeling was measuring, to some degree, the altruistic or “other-regarding” preferences of participants, since an accepted offer would benefit not only the receiver of the offer but also the third player. The difference was highly statistically significant and meaningful in magnitude in this kind of game. None of the other differences between preferences or interactions among preferences were statistically significant in explaining the magnitude of offers. Also, Extraverts indicated a willingness to accept lower offers than did Introverts. Again, the difference was statistically significant and meaningful.

The MBTI Step I assessment deals with perception and judgment, and numerous studies of decision making reveal predictable, nuanced relationships between Thinking and Feeling in their introverted and extraverted forms and in combination with Sensing and Intuition.

Kench, Beekman, Wynn, and Niman (2010) examined the effects of type on economic decision making in a public goods experiment. Participants played a game in which they were asked to contribute some of their experimental earnings to a public account, which was then divided equally among the players. At the end of the experiment, all of their earnings, both from their individual performance and from their share of the public account, were paid to them in real money. Results of these kinds of experiments consistently show that participants don’t “free ride” but do contribute to the public good. In this experiment, participants were allowed to “punish” free riders by reducing their own and the others’ earnings. Extraverts earned more in the game than did Introverts,

but there were no differences in contributions to the public account or in punishment. Thinking types earned more than Feeling types and also assigned significantly fewer punishment points. Feeling types behaved more altruistically, punishing people who did not contribute to the common good by reducing their earnings, than did Thinking types. Such punishment behavior is considered altruistic in behavioral economics because it reduces the punisher's earnings, in this case the Feeling types' earnings, as well. There were no S–N or J–P differences.

Gallen (2006) compared type preferences to a typology of strategic decision making (see Miles & Snow, 1978), which consists of three generic strategies: defender, prospector, and analyzer. Prospectors have a very broad product-market domain and are “first to market.” Defenders offer a stable set of products or services and compete primarily on the basis of value and/or cost. Analyzers have characteristics of both the defender and the prospector and pursue a “second-in” strategy. The sample consisted of 70 members of the top management teams in 13 companies in the spa industry. Overall, participants who preferred Intuition favored the analyzer or prospector strategies, while those who preferred Sensing favored the defender or analyzer strategies. Fifty-seven percent of the ST managers preferred a defender strategy. Interestingly, not even one ST manager's view of a viable strategy represented a prospector strategy.

These studies on decision making demonstrate the validity of MBTI results in describing differences in decision-making styles and performance that are in accordance with type theory.

Change

A common issue in organizations is how people deal with change; consequently, this is also a popular topic in the type literature (see Barger & Kirby, 1995, 1997a, 1997b). Huszycz and his colleagues (Huszycz & Bush, 2015; Huszycz & Claus, 2000) used this literature to develop questions about how people typically deal with change. The sample consisted of 317 undergraduate and graduate students in business. The results for which differences between opposite preferences were significant at $p < .001$ are shown in table 9.19, with some additional items added to round out the picture. Note that the questions or statements were not combined into scales, but were analyzed as individual items and so likely have attenuated reliabilities, which would minimize differences. The items endorsed as typical by those at opposite poles of each preference pair conform quite well to the theoretical descriptions appearing in the literature of how types deal with change. It is interesting to note that the fewest differences to meet the significance criterion were between Sensing and Intuitive types (none reached this level for the Sensing types, and only three did so for the Intuitive types). There were quite a few differences at $p < .01$, however, on

which the Sensing types scored higher. Some of these descriptions seem to involve other preferences. For example, Sensing types said that a typical response to change was to “be organized and involved in planning, scheduling, and setting deadlines” and to “be very logical and organized, and respond to the details/facts in a responsible/dependable manner . . . providing some stability within the change.” The former behavior seems related to Judging (and indeed Judging types also said this was typical of them) and the latter to Thinking. So some of these S–N results may be a function of the type distribution in the sample.

Lawrence (2007) administered MBTI Form Q and the *Change Style Indicator* (CSI; Musselwhite & Ingram, 2003) to 45 healthcare executives in five hospitals. The CSI identifies three kinds of change agents: Conserver, Pragmatist, and Originator. Conservers prefer gradual and incremental change; Pragmatists focus on practical outcomes and can serve as mediators; Originators prefer quick and transformational change. Sensing and Judging were related to the Conserver style. Preferences for E, N, T, and P correlated significantly with the Originator style. There were no significant relationships of MBTI preferences with the Pragmatist style.

Teams

The prior *MBTI® Manual* (Myers et al., 1998; table 13.13, p. 351) summarized a number of studies in which the team rather than the individual was the unit of analysis. In addition, Hammer and Huszycz (1996) summarized the extensive research on type and teams and reviewed additional studies (Bradshaw, 1986; James, 1984; Reynolds, 1990) that used the *Team Communication Adjustment Index* (Yeakley, 1982, 1983) as a tool for determining the homogeneity/heterogeneity of teams based on the communication patterns of people with different dominant and auxiliary processes. In brief, these two reviews can be summarized as follows:

Teams with *similar* communication styles

- Perform their tasks more quickly
- Experience less conflict
- Like each other more
- Listen to one another more

Teams with *diverse* communication styles

- Are more effective
- Produce better outcomes
- May take more time

Hammer and Huszycz (1996) also reviewed the literature on the interaction between team leaders and members and found that

- Managers or leaders with preferences for Extraversion and Feeling were generally rated higher on teamwork variables.

- Managers or leaders with NTP preferences, especially INTPs, were found to value self-oriented individualism, a value that research on teams indicates interferes with teamwork.
- Leaders who differed from team members, especially on the Thinking–Feeling preference pair, were rated by team members as more effective than were leaders who shared the same preferences as the team.
- After training in participative management, all types were able to learn the needed skills, but Feeling types were more able to actually use the participative leadership style.

Schullery and Schullery (2006) examined the effect of heterogeneous and homogeneous groups on a number of variables. Their sample consisted of 394 students divided into 102 groups of 2 to 6 (average = 3.8) students. The groups were assigned projects designed to develop

Table 9.19 | Self-ratings of behaviors regarding dealing with change by MBTI® preference

e Extraversion Be energized by generating and developing ideas through discussion Provide spontaneous, verbal information and feedback Be action oriented—willing to wing it Focus on the external, on the people and things surrounding the change Keep up with or even set the pace of the change Try to provide creativity and be innovative	i Introversion Be the “behind-the-scenes” person during the change Like to reflect deeply on the change before beginning Be reserved initially and need time to be comfortable with change Try to provide creativity and be innovative Provide innovative/creative strategies that will help me and the other people Be a visionary on how the change strategy could work ($p < .01$)
s Sensing No items at $p < .001$ Be focused on short-term effects and the step-by-step process to be used to implement the change ($p < .01$) Be organized and involved in planning, scheduling, and setting deadlines ($p < .01$) Seek fairness in rules/procedures associated with the change . . . equality for all ($p < .01$) Be very logical and organized and respond to the details/facts in a responsible/dependable manner . . . providing some stability within the change ($p < .01$)	n Intuition Try to provide creativity and be innovative Provide innovative/creative strategies that will help me and the other people Be intrigued by possibilities and patterns Be a visionary on how the change strategy could work ($p < .01$)
t Thinking Provide carefully thought out or maybe even written information and feedback Be logical and reasoning and ask why Appear to be impersonal without meaning to by being blunt or seen to be blunt Be questioning, skeptical, and analytical—keeping things as objective as possible Try to figure out the cause-and-effect aspects of the change situation Be process oriented and figure out how this might work	f Feeling Be relationship focused, concerned about the needs of others and maybe even smooth over any conflicts involved in the change Seek or give approval and individual recognition, appreciation, support, and positive feedback ($p < .01$) Be trusting and maybe even overly accepting of the change ideas ($p < .01$) Seek to include and gain input from all the people involved . . . helping people feel included ($p < .01$) Communicate to others about the views/values associated with the change ($p < .01$)
j Judging Be organized and involved in planning, scheduling, and setting deadlines Be very logical and organized and respond to the details/facts in a responsible/dependable manner, providing some stability within the change	p Perceiving Be action oriented—willing to wing it Be relatively comfortable with the unknown/ambiguity Be spontaneous and see where the change takes us

Note: $N = 317$. All items significant at $p < .001$, except as noted.

Source: Adapted from Huszycz & Claus (2000). Used with permission.

critical thinking and collaborative work skills. In addition to the MBTI assessment, the authors also used a scale measuring argumentativeness. Eighteen outcome variables were tested, including these:

- Self-perceived improvements on a variety of group skills
- Students' overall satisfaction with their group experiences
- Students' grades on the group projects
- The instructor's perception of how well the group functioned

Homogeneity was indexed for each preference scale using the percentage of people with the majority preference in the group (so a score of 75 on the E–I scale would mean either that 75% of the group members preferred Extraversion or that 75% preferred Introversion) and then defined the overall “average MBTI homogeneity” as the average of these four preference-based homogeneity indexes. The results indicated that outcomes depended on the particular dependent measure being examined, the specific homogeneity measure used, and the students' gender. Students in heterogeneous groups reported greater improvements in speaking up or arguing a point, organizing and presenting thoughts (females), and overall average skills at the individual (females) and group levels. Students in overall homogeneous groups earned higher grades, and males in such groups reported less shyness. For the preference scales, the E–I scale and, for men, the T–F scale were associated with higher grades. A group with high heterogeneity on the T–F scale benefited shy women but was associated with lower grades for men. The only significant finding for overall group functioning was for the percentage of Feeling types in the group. Groups rich in Feeling types had fewer problems, managed to work out their problems among themselves, and acquiesced rather than complained to the teacher. The instructor's perception of how well the groups functioned was unrelated to group composition. Because the above is a brief summary of very nuanced results, the reader is referred to the article for more detail.

Choi, Deek, and Im (2008) examined the effect of dyads composed of different type pairings on programming productivity. In pair programming, two individuals sit next to each other and share one keyboard and one display. Participants were placed into three different kinds of groups based on their type. Diverse groups were composed of pairs of programmers who shared one mental process (S, N, T, or F). Alike groups were composed of pairs of programmers who shared both processes (ST, SF, NF, or NT). Opposite groups were pairs of programmers who were completely

opposite on both processes (e.g., an NF and an ST).⁴ The sample consisted of 128 students who had no prior programming experience. The dependent variable was code productivity. The diverse-processes group ranked highest on productivity (and also had the lowest standard deviation of the three groups), followed by the opposite-processes group, followed by the alike-processes group. The difference between the diverse group and the other two was statistically significant. There was no significant difference between the opposite and alike groups. The finding—that pairing people with no programming experience who have one process alike is likely to lead to higher productivity—has implications for software development.

Katira et al. (2004) also conducted a study on pair programming. Matched and unmatched pairs of programmers in two different classes were asked to evaluate their compatibility on a one-item, 3-point scale. A pair was considered matched if they had the same MBTI type. The unmatched pairs could be any other combination of types. The results were mixed—the unmatched pairs in one class reported being significantly more compatible, but those in the other class did not.

Amato and Amato (2005) examined the effect of type compatibility within teams on various outcome measures. The sample consisted of 64 students placed into 17 groups—8 groups from a class of seniors with extensive prior experience working on teams and 9 from a class of sophomores with little group work experience. Compatibility scores were median values of Yeakley's (1982) communication index (CI) for each pair of students on each team. The higher the median CI, the more compatible the team. Teams with scores higher than the median for all teams were classified as compatible communication groups, whereas teams with scores lower than the overall median were classified as complementary groups. Outcome variables were specialization of labor, collective action and team skills development (what did I learn regarding teams?), and problems with group work. For specialization of labor, the groups composed of experienced students in compatible groups viewed the team experience more favorably than did those in complementary groups. For the less experienced students, the reverse was true: those in the complementary groups had a better overall impression. There were no significant differences between compatible and complementary groups for either experienced or naive students on team skills development or collective action. For problems with group work outcomes, the experienced students in the compatible groups indicated fewer problems on 14 of the items in this domain (e.g., members being unprepared, missing meetings, coordinating work, poor quality

4. Readers who may review this article are cautioned that the authors mistakenly label pairs of types as having similar or different dominant and auxiliary functions (processes). These erroneous labels have been removed from this discussion; this error had no influence on the results, as the analyses did not address dominant functions.

work). There were only two significant findings for the less experienced students, with those in the compatible group indicating more problems with scheduling and members being late for team meetings. This study adds an interesting wrinkle and should be researched further, as most of the previous team research has been with either very experienced or inexperienced participants and so did not use experience level as an independent or grouping variable.

Gaines (1998) looked at the climate of entire schools as a function of the personality of the administrative team. Teacher perceptions of school climate, measured with the subscales of the *Teacher Climate Survey*, were the dependent variable. Teachers ($N = 1,592$) rated 30 administrative teams. Significant effects of team type were found on seven of the nine school climate factors (the data were not reported in the article).

The importance of type to the behavior, success, and satisfaction of team members has been widely studied from many perspectives and with a very wide variety of variables, affirming the influence of type in this very important area.

In a study of 43 senior managers from 31 countries attending a four-week training program, Glaman, Jones, and Rozelle (1996) asked participants to rate each other on social liking; co-worker preference (willingness to work with them in the future); and attributions of orientation toward people (consideration, friendliness, courtesy), work habits (organization, hard work, capability), self-confidence (confident, outspoken), and aggression (based on a single item). Personality type similarity was measured by counting the number (0–4) of MBTI preferences in common in each dyad. MBTI similarity correlated negatively with social liking and attributions of work habits and self-confidence, and positively with attribution of aggression. The sign of the coefficient indicates that diversity on the MBTI preferences was related to social liking and to positive ratings of work habits and self-confidence, but similarity of MBTI preferences was related to higher ratings on aggression. Given that one purpose of the training was to bring together people of different cultures, differences, including personality differences, were identified as positive and/or complementary. Perhaps the higher rating for aggression in more similar dyads was a result of competition among the managers of similar types. In regression equations, the MBTI similarity variable added significantly to predictions of social liking and co-worker preference over and above demographic similarity, and to predictions of social liking over and above similarity on personal values. Demographic and MBTI similarity were most important in predicting attributed work habits and similarity of status values, and MBTI similarity contributed to attributed aggression ratings.

Huszczo (2013) asked 270 business students, both graduate and undergraduate, to indicate how they typically behaved as a member of a team, using behaviors described in popular type publications. Table 9.20 shows those behaviors for which there were significant differences ($p < .001$, except as noted) between people

Table 9.20 | Summary of typical team member behaviors by MBTI® preference

Extraversion

Ensure support and belongingness
Ensure positive challenge
Scan environment for opportunities
Talk more
Influence teammates
Focus on relationships
Play cheerleader role
Celebrate, capitalize on diversity
Build relations with management
Challenge for excellence
Creatively communicate
Seek fun, harmony, innovation
Help teammates grow
Make things people oriented
Inspire people to care

Introversion

Play avoider role ($p < .01$)

Sensing

Use proven methods for productivity ($p < .01$)

Intuition

Play initiator role
Creatively communicate

Thinking

Break down the complex for practical results
Play decision pusher role ($p < .01$)
Challenge for excellence ($p < .01$)

Feeling

Encourager/supporter role
Play cheerleader role
Seek fun, harmony, innovation
Play harmonizer role
Make things people oriented

Judging

Set goals ($p < .05$)
Distribute work equitably ($p < .05$)
Establish work procedures ($p < .05$)

Perceiving

Inspire people to care ($p < .05$)
Emphasize competency and achievement ($p < .05$)

Note: $N = 270$. All items significant at $p < .001$, except as noted.
Source: Adapted from Huszczo (2013). Used with permission.

of opposite preferences. Extraverts clearly believed that there were many behaviors typical of their performance on teams. The only behavior that Introverts endorsed significantly more was playing the role of an avoider (presumably of conflict, but perhaps of working with the team itself). Given the findings on the *Benchmarks* assessment discussed earlier, one wonders whether outside observers would agree that these behaviors are differentially typical of Extraverts. The results for the S–N and T–F preference pairs and for the Judging preference are consistent with type theory.

Hackston (2015), in an across-country study of 1,029 people, found that differences between types measured by the MBTI assessment had a greater effect on dimensions of cultural orientation than did country of origin, country of residence, age, gender, or level of seniority. He suggested that “individual differences in personality are likely to be one of the most important factors when building cross-cultural teams. While there will be national differences in how personality is expressed (Kirby et al., 2007), those of a similar personality will have a great deal in common, wherever they live and wherever they were born” (p. 25).

The research on type and teams demonstrates an interest in the effects of type homogeneity/heterogeneity on perceived compatibility and on the outcomes of working as a team. However, there is little homogeneity among researchers on how to define type homogeneity (e.g., alike on all four preferences, alike on only the processes) or on what outcome variables are of interest (perceived compatibility, actual team behaviors). Researchers also differ in the choice of moderator variables (gender, experience in groups). Given the importance of teams in modern organizations and the promising results when using type as one variable, this is an area of research that deserves more systematic attention.

Emotional Intelligence

Emotional intelligence is believed to affect how people relate to others. This has become a popular area for training and research with individual managers and teams in organizations. Correlations of the MBTI preferences in the Form M US national sample with the *Emotional Quotient Inventory* (EQI™) showed that Extraversion was significantly related to the following EQI scales: Assertiveness, Emotionality, Empathy, Flexibility, Happiness, Interpersonal Relations, Optimism, Self-Actualization, Self-Regulation, Total Adaptability, Total Emotional Quotient Index, General Mood, and Total Stress Management. The S–N preference pair did not correlate significantly with any EQI scales. A preference for Thinking correlated significantly with Assertiveness, Independence, and Stress Management. Feeling was positively related to Empathy, Interpersonal, Problem Solving, Social Responsibility, and Total Interpersonal.

Judging was associated with Impulse Control, Problem Solving, Reality Testing, and Total Stress Management.

Leary, Reilly, and Brown (2009) also examined the relationship between the MBTI preferences and the EQI scales, although they used only the five major scales. Employing a categorical analysis, they found that Extraverts scored higher than Introverts on all the scales, as well as on the overall EQI: Intrapersonal, Interpersonal, Adaptability, Stress Management, and General Mood scales. Feeling types scored higher on the Interpersonal scale. The authors tested the same hypotheses using MBTI continuous scores and found basically the same results.

Sitarenios (2004) used the EQI with a corporate sample ($n = 188$). His results were similar to those reported above. Extraverts scored higher than Introverts on Emotional Self-Awareness, Assertiveness, Self-Actualization, Empathy, Interpersonal Relationships, Flexibility, Optimism, and Happiness. Intuitive types scored higher than Sensing types on Emotional Self-Awareness and Interpersonal, while Sensing types scored higher on Reality Testing and Impulse Control. Feeling types scored higher than Thinking types on Emotional Self-Awareness, Self-Actualization, Empathy, and Interpersonal, while Thinking types scored higher on Independence. Perceiving types scored higher than Judging types on Empathy and Flexibility.

A different measure of emotional intelligence, the *Emotional Judgment Inventory* (EJI), was used with a sample of 178 participants in a training program (Bedwell, 2015). Extraverts scored significantly higher than Introverts on Being Aware of Emotions, Identifying Own Emotions, Identifying Others' Emotions, Managing Others' Emotions, Using Emotions in Problem Solving, and Expressing Emotions Adaptively. There were no differences between Extraverts and Introverts on Managing Own Emotions or Impression Management (tendency to present themselves in a positive light). Intuitive types scored significantly higher than Sensing types on Being Aware of Emotions, Managing Others' Emotions, Using Emotions in Problem Solving, and Expressing Emotions Adaptively. Feeling types scored significantly higher than Thinking types on Being Aware of Emotions, Identifying Others' Emotions, Managing Others' Emotions, Using Emotions in Problem Solving, Expressing Emotions Adaptively, and Impression Management. The effect sizes for the T–F differences were larger than those for any of the other comparisons; the effect size, with Feeling types scoring higher, for Being Aware of Emotions was $d = 1.4$. There were no differences between Judging and Perceiving types on any of the EJI scales. Differences in dominant processes were also analyzed. Extraverted Feeling and Introverted Feeling types scored highest on Being Aware of Emotions, Managing Others' Emotions, and Impression Management. Dominant extraverted Intuitive types

scored highest on Expressing Emotions Adaptively. Dominant introverted Thinking types scored lowest on Being Aware of Emotions, Identifying Others' Emotions, Managing Others' Emotions, and Using Emotions in Problem Solving.

The results of these studies, using four different samples and two different measures of EQ, provide evidence for the validity of the four preference scales of the MBTI Step I assessment.

Transfer of Learning

Research by Belling (2009) addressed a neglected topic in type research in organizations. She pointed out that while type is used extensively in management development programs, little research has been done on how type affects the transfer of learning from the program to the work setting. Using a combination of surveys and in-depth interviews, Belling studied 234 managers from 15 organizations who were attending management development programs. Surveys were administered before, immediately after, and then three to six months after the training. Belling also used the *Salter Environmental Type Assessment* (SETA) to type the work environment. The predominant preferences of the managers in the sample were E, N, T, and J. The SETA results indicated that these managers perceived their environments as E, S, F, and P. Of all the individual characteristics measured in the surveys, type had the highest frequency of significant associations with learning outcomes, as well as with perceived barriers and facilitators of transfer. Belling found that certain types were more likely to transfer particular kinds of learning. Type was also significantly associated with 15 of the 16 barriers to transfer (e.g., Introverts identified lack of time to introduce new ideas as a barrier; Feeling types identified feeling pressure to give priority to the bottom line instead of trying new behaviors). Type was also significantly associated with 9 of the 17 characteristics of the environment that were perceived as facilitators of learning transfer (e.g., Intuitive types liked learning a common language; Thinking types liked learning to appreciate their own strengths and weaknesses). The majority of the significant associations with barriers to transfer of learning were by Feeling types, while the majority of the significant associations with facilitators of transfer were by Intuitive types and Thinking types. The author concluded that type affected transfer of learning in three ways:

- Type directly influenced learning outcomes, that is, what was learned in the program that could be transferred.
- Type affected perceptions about what helps or hinders transfer of learning.
- Type influenced what individuals consider both critical and ideal elements in how learning is applied and transferred.

EDUCATION

The application of type in educational settings has a long history and, as with the career domain, was one of Briggs and Myers' original areas of interest in using the MBTI instrument. They realized that type theory was useful in predicting how students learn, the environments students prefer, how teachers teach, the interaction between students and teachers, reading and writing styles, persistence and retention, and the kind of extracurricular activities students prefer. To the extent that research with the MBTI assessment supports these theoretical predictions, the validity of the assessment for use in educational settings is established.

Extensive work has been done in this area, which can only be briefly summarized here. This research has been conducted primarily with college students, with a few studies of high school students as well. For information on type and education research with younger students, see Murphy & Meisgeier (2008). Also, see the section "Outcome Studies" later in this chapter for additional important studies on the effect of type knowledge in education.

Characteristics of Learners

This section highlights some of the more consistent research findings on each MBTI preference pair relating to characteristics of learners. Comprehensive reviews of these studies were conducted by Lawrence (1984) and DiTiberio (1996). What follows is a description of the patterns that have emerged relating each MBTI preference pair to aspects of learning styles, cognitive styles, brain patterns, and information processing. This research is summarized in table 9.21.

EXTRAVERSION—INTROVERSION

The series of studies discussed earlier on brain electrical activity (Wilson & Languis, 1989, 1990) found for both adults and adolescents that Extraverts have lower internal arousal than do Introverts under all conditions established by the experiment. Extraverted students have been found to seek stimulation from the environment, preferring approaches to learning that are collaborative and that often depend on the external world for suggestions on how to proceed (Elliott & Sapp, 1988). A number of studies have also found Extraverts to lean toward learning styles that involve active experimentation or concrete experience and sometimes both (Gordon, Coscarelli, & Sears, 1986; Hinkle, 1986; Luh, 1991; Penn, 1992). In contrast, Introverted students prefer reflective observation (Hinkle, 1986) and lecture formats (Fourqurean, Meisgeier, & Swank, 1990). Extraversion correlated positively with the Environmentally Open scale and Introversion with the Environmentally Restricted scale of the *Exam Preparation Inventory* (Williams, Staley, & Rudyk, 2000).

Table 9.21 | Summary of research on characteristics of learners by MBTI® type

Extraverted types	Introverted types	Sensing types	Intuitive types
Concrete experiential learning style ^{1, 2, 3, 4}	Reflective observational learning style ^{2, 6}	Concrete experiential learning style ⁶	Abstract conceptual learning style ⁶
Active experimental learning style ^{2, 5, 6, 4}	Visual learners ⁷	Learn in several ways ⁷	Visual learners ⁷
Learn in several ways ⁷	Auditory learners ⁷	Abstract sequential learning style ¹⁵	Auditory learners ⁷
Collaborative learners ⁸	Abstract sequential learning style ¹⁵	Concrete sequential learning style ¹⁵	Concrete random learning style ¹⁵
Dependent learners ⁸	Participant learners ⁸	Collaborative learners ⁸	Participant learners ⁸
Like projects, simulations, and peer teaching ⁷	Like lecture formats ⁷	Dependent learners ⁸	High conceptual level ²⁴
External decision makers ¹	Adaptive in creativity ⁹	High in fact retention, methodical study, and serialist learning ¹⁸	Holistic learners ¹⁸
Innovative in creativity ⁹	Postconventional decision makers ^{16, 17}	Field dependent ^{19, 20}	Internal decision makers ¹
High in goal orientation ¹⁰	High internal arousal of brain electrical activity ^{13, 14}	Left hemisphere learners ^{18, 21, 22, 23}	Field independent ^{19, 20}
High academic self-esteem ¹¹		Adaptive in creativity ⁹	Thin boundaries ²⁵
Connected knowers ¹²			Right hemisphere learners ^{18, 21, 22, 23}
Low internal arousal of brain electrical activity ^{13, 14}			Innovative in creativity ⁹
			Postconventional decision makers ¹⁷
			High in reflective judgment ²⁶
			High in goal orientation ¹⁰
			High academic self-esteem ¹¹
			High in academic comfort ²⁷
			Like self-directed learning ⁵
Thinking types	Feeling types	Judging types	Perceiving types
Abstract conceptual learning style ⁶	Concrete experiential learning style ⁶	Abstract conceptual learning style ⁶	Concrete experiential learning style ⁶
Abstract sequential learning style ¹⁵	Abstract random learning style ¹⁵	Like structure and motivation in learning ⁷	Active experimental learning style ²
Participant learners ⁸	Dependent learners ⁸	Concrete sequential learning style ¹⁵	Like tactile and loud noise learning stimulus ⁷
High in fact retention, methodical study, and serialist learning ¹⁸	Holistic learners ¹⁸	Participant learners ⁸	Abstract random learning style ¹⁵
Systematic decision makers ¹	Field dependent ^{19, 20}	High in fact retention, methodical study, and serialist learning ¹⁸	Concrete random learning style ¹⁵
Field independent ^{19, 20}	Thin boundaries ²⁵	Like drill and teaching games ⁷	Collaborative learners ⁸
Left hemisphere learners ^{18, 28}	Right hemisphere learners ^{18, 22}	Like independent study ⁷	Dependent learners ⁸
Adaptive in creativity ⁹	Adaptive in creativity ⁹	Left hemisphere learners ^{18, 22, 23}	Holistic learners ¹⁸
Seek self-justice in moral orientation ²⁹	Seek care and self-care in moral orientation ²⁹	Adaptive in creativity ⁹	Thin boundaries ²⁵
High in goal orientation among adults ¹⁰	High in goal orientation among junior high students ¹⁰	High in goal orientation ¹⁰	Right hemisphere learners ^{18, 23}
	Connected knowers ¹²	High academic self-esteem ¹¹	Innovative in creativity ⁹
		High academic comfort ²⁷	Postconventional decision makers ¹⁷

1. Gordon, Coscarelli, & Sears (1986)

2. Hinkle (1986)

3. Luh (1991)

4. Penn (1992)

5. Johnson, Sample, & Jones (1988)

6. Myers & McCauley (1985)

7. Fourqurean, Meisgeier, & Swank (1990)

8. Elliott & Sapp (1988)

9. Jacobson (1993)

10. Atman (1993)

11. Schaefer (1994)

12. Carter (1990)

13. Wilson & Languis (1989)

14. Wilson & Languis (1990)

15. Drummond & Stoddard (1992)

16. Catoe (1992)

17. Faucett, Morgan, Poling, & Johnson (1995)

18. Beyler & Schmeck (1992)

19. Canning (1983)

20. Holsworth (1985)

21. Hockersmith (1986)

22. Holtzman (1989)

23. Taggart, Kroeck, & Escoffier (1991)

24. Brown & DeCoster (1991)

25. Ehrman (1993)

26. Bowen (1990)

27. Apostol & Trontvent (1989)

28. Shiflett (1989)

29. Liddell, Halpin, & Halpin (1992)

SENSING–INTUITION

The S–N preference pair has consistently been related to distinct differences in the ways students view and respond to education, in accordance with type theory. Sensing types have been found to like sequential learning (Drummond & Stoddard, 1992). Their approach to creativity tends to be adaptive, while Intuitive types are innovative, on the *Kirton Adaption-Innovation Inventory* (Jacobson, 1993). Sensing types approach learning through fact retention, methodical study, and serialist learning (Beyler & Schmeck, 1992). Sensing was the clearest preference among both students and teachers of vocational education in three studies (Barrett, 1989; Johnson, Zimmerman, & Brooker, 1994; Vollbrecht, 1991). In the 1985 *MBTI® Manual*, Myers and McCaulley reported data showing Sensing types as valuing concrete experience on Kolb's *Learning Style Inventory*, while Intuitive types valued abstract conceptualization. Several studies using the *Human Information Processing Survey* (Beyler & Schmeck, 1992; Hockersmith, 1986; Taggart, Kroeck, & Escoffier, 1991) found that Sensing students favored the brain's left hemisphere, in contrast to Intuitive students, who favored the right hemisphere. A review of the literature on brain functioning and type (Power & Lundsten, 1997) confirmed these patterns and called attention not only to hemispheric preference but also to cerebral (favoring Intuition) versus limbic (favoring Sensing) thinking processes as measured by the *Herrmann Brain Dominance Instrument®*. In another study, Sensing was correlated with the Factual Practical scale and Intuition with the Theoretical scale of the *Exam Preparation Inventory* (Williams et al., 2000).

Intuitive types are described as holistic (Beyler & Schmeck, 1992) and conceptual and abstract (Holsworth, 1985) learners. McCaulley and Natter (1974) discovered that a preference for independent study was found among Intuitive high school students, a pattern confirmed in a later study of self-directed learning (Johnson, Sample, & Jones, 1988). Intuitive types were also found to identify with the philosophy of alternative high schools (Steele, 1986). Academic self-esteem (Schaefer, 1994) and academic comfort (Apostal & Trontvent, 1989) have been found to be higher among Intuitive types. The research cited in table 9.21 suggests that, consistent with type theory, Sensing types move first from the particular to the general, while Intuitive types move from the general to the particular. Both approaches are useful, although this is not recognized in some classroom contexts.

In foreign language learning, a number of quantitative studies have suggested that Intuitive and Perceiving types are more interested and perform better than do their counterparts (Ehrman, 1995; Ehrman & Oxford, 1990, 1995; Gautsch, 1993; Oxford & Ehrman, 1988). Sensing types were rated among the weakest language learners (Ehrman, 1994b).

THINKING–FEELING

Research has demonstrated that Thinking types take a systematic approach to learning (Gordon et al., 1986). Beyler and Schmeck (1992) found that Thinking types prefer a factual orientation, methodical study, and serialist learning, while Feeling types prefer holistic learning. Thinking types in this study were also found to be oriented to the brain's left hemisphere and Feeling types to the right hemisphere. Shiflett (1989) produced similar findings for Thinking types in terms of brain patterns. Approaches to moral decision making have been found to link Thinking types with concerns for self-justice and Feeling types with care and self-care (Liddell, Halpin, & Halpin, 1992). Some of what passes for gender differences may be conceptualized as a type difference, as was suggested by Oxford, Nyikos, and Ehrman (1988) in their study of foreign language learners. In a study of separate versus connected knowers inspired by the writings of researchers in women's studies, Feeling was preferred (along with Extraversion) more often by connected knowers (Carter, 1990), who are receptive to new ideas and seek to understand from a perspective of empathy, in contrast to the analytical approach of the separate knower. On the *Exam Preparation Inventory* (Williams et al., 2000), Thinking and Intuition correlated with the Analytical scale.

JUDGING–PERCEIVING

Judging types have been found to prefer learning settings with clear structure, external motivation, drill, teaching games, and independent study (Fourqurean et al., 1990); sequencing (Drummond & Stoddard, 1992); and fact retention, methodical study, and serialist learning (Beyler & Schmeck, 1992). Two studies (Beyler & Schmeck, 1992; Taggart, Kroeck, & Escoffier, 1991) found Judging types to be oriented to left-hemisphere kinds of learning and Perceiving types to right-hemisphere learning. Academic self-esteem was higher for Judging types (Schaefer, 1994), as was academic comfort among female Judging types (Apostal & Trontvent, 1989). Perceiving types liked tactile learning and loud noise (Fourqurean et al., 1990). They were innovative in creativity, while Judging types were adaptive (Jacobson, 1993). The Open-Ended and Spontaneous scale of the *Exam Preparation Inventory* correlated with Perceiving (Williams et al., 2000).

Teachers and Teacher-Learner Interactions

Learning takes place in a context, and the context usually involves interaction with instructors. Most of the available studies on the MBTI Step I assessment and teachers deal with one preference pair at a time. The following type patterns of teachers are based on a review of research by Lawrence (1993).

Teachers who prefer Extraversion like to give students choices and a voice in decisions, are easily attuned to

the attention levels of their students, and have classrooms with movement and noise. Teachers who prefer Introversion have quieter and more orderly classrooms and structured learning activities and are more attuned to the ideas they are teaching than to students' attention levels.

Teachers who prefer Sensing tend to emphasize facts, practical information, and concrete skills; keep learning centralized; provide a narrow range of student choices; and start a sequence of questions with a request for facts, seeking a predictable response. Teachers who prefer Intuition instead emphasize concepts, relationships, and implications; provide a wide range of choices for students and encourage their voice in decisions; like to form small groups; expect independence and creative behavior; move freely around the room (especially in elementary or middle school settings); have movement and noise in the classroom; and start their questioning of students with an invitation for synthesis and evaluation.

Teachers who prefer Thinking seldom make comments about student performance, and when they do, it is from an objective standard. They attend to the class as a whole and invite students to attend to what they (the teachers) are doing or saying. Teachers who prefer Feeling regularly provide both praise and criticism in their words and body language. They like students to spend time on individual work, and they move from student to student to assist them, often attending to more than one student at a time.

Teachers who prefer Judging tend to set and adhere to schedules and have quiet and orderly classrooms. Teachers who prefer Perceiving, in contrast, encourage movement around the classroom, independent work, open-ended discussion, and socializing in groups, and allow students to have a voice in decisions.

Lawrence (1993) also reported that when students get off task, they do so differently depending on the type of their teacher. The result is increased activity and noise with teachers who prefer Extraversion, Intuition, and Perceiving. With teachers who prefer Introversion, Sensing, and Judging, the result is daydreaming, doodling, or withdrawal.

MATCHING TEACHERS WITH LEARNERS BY TYPE

Efforts to match teachers with learners by MBTI profiles have met with mixed results at best. A full review of the literature in this area can be found in DiTiberio (1996). This review includes 9 studies on matching and learner satisfaction, 6 of them with no statistically significant findings, and 18 other studies on matching and learner outcomes, 9 of them lacking statistical significance, plus another with results opposite the hypothesized direction.

It appears that students may need different kinds of instruction at different points in their development. Indeed, some studies found that opposite-type teachers facilitated significant learning for their students (Donovan, 1994). Others found that matching students with same- or similar-type instructors led to better outcomes (Walter, 1984) or greater student satisfaction (Cooper & Miller, 1991; Lamphere, 1985). One study (Boyd, 1995) found that pairing students with teachers of their same type, while favorable overall, was less favorable than when one MBTI preference was different. The implication is that students at times need the natural support obtained from a teacher of their same type and at other times may benefit from the challenge of a different type. A careful assessment of the student's needs at any point in time is thus called for, rather than a rote assignment by type. The very mixed results from MBTI research suggest that matching teacher type with learner type may be too simple of a solution.

Reading and Writing

In studies of reading, the quantitative data are consistent and clear. As with academic aptitude and achievement in general, verbal fluency tends to favor Intuitive types. Studies have investigated the relationship of type to language competence and reading comprehension (Hester, 1990), self-reported numbers of books read (Hammer, 1985; Hicks, 1984, 1985, 1989; Levine, 1988; Moore & Bayne, 1997), reading comprehension scores on the *McGraw-Hill Basic Skills System Reading Test* (Thomason, 1983) and the *Nelson-Denny Reading Test* (Manske, 1988), and proofreading ability (Gordy & Thorne, 1994). In almost every case, Intuitive types had the advantage; the exception was in Hammer's study, where NFs were followed by STs and then NTs in number of books read. These results pertaining to the advantage of Intuition in reading confirm type theory and findings from earlier studies reported by Myers and McCaulley (1985). Thinking types performed better in comprehension when reading expository passages and Feeling types when reading narratives (Singer, 1989).

Research on writing is also fertile ground for testing type theory. Some of the patterns of preferred writing processes found by quantitative research are as follows: As students, Extraverts prefer to choose their own topics and to read papers aloud for peer response, while Introverts prefer to have the teacher lead the class (Severino, 1989). Introverts tend to score higher than Extraverts both on composition grades and on the *Test of Standard Written English* (Schurr, Houlette, & Ellen, 1986). Extraverts were also found to send more and longer e-mail messages and replies in a study of computer use (Bail, 1993). Sensing types want the teacher to lead the class (Severino, 1989), while Intuitive types prefer to choose their own topics and to read papers aloud (Severino, 1989), use more words per sentence (Carrell &

Monroe, 1993), receive high grades on reflective papers and lower grades on reporting (Fisher, 1994), and tend to present an overview with little focus when asked for a summary (Held & Yokomoto, 1983).

Academic Persistence and Student Retention

Retention in both college and professional education is an ongoing concern of administrators. Type differences seem to relate to how students move successfully through college. For example, a study by Schurr, Ruble, Palomba, Pickerill, and Moore (1997) found that while preferences for Introversion, Intuition, and Judging predicted entering scores on the *Scholastic Aptitude Test* (SAT®) and high school rank, and a preference for Judging predicted college grades, a higher overall persistence-to-graduation rate was found for Sensing and Judging types. What was most notable from this study was the fact that persistence to graduation was often the result of an indirect influence of type on different aspects of the college experience. For example, Judging types were more academically and career oriented; Extraverts valued extracurricular life. The authors concluded that “high levels for some aspects of the total college experience can offset lower levels for other aspects in determining whether students will persist to graduation” (p. 40).

The focus of the academic and social context with which students interact seems to be a critical factor. Often the types of students who are in the minority or whose preferences diverge from the focus of the curriculum tend to drop out. This was found for Feeling, NF, and FP types at the US Naval Academy (Roush, 1989), NF types in engineering (McCaulley, 1990), and Thinking types in nursing (Kalsbeek, 2003). Conversely, other researchers have called attention to those types most likely to succeed or to persist to graduation, whose preferences tend to be congruent with the focus of the major field: Introversion, Thinking, and Judging in engineering (Rosati, 1997); Thinking and Judging in law (Gilchrist, 1991); Sensing, Feeling, and Judging in family medicine and Sensing in obstetrics (Friedman & Slatt, 1988); Judging in dentistry (Erskine, Westerman, & Grandy, 1986); and Sensing and Judging in nursing (Schurr, Henriksen, Alcorn, & Dillard, 1992).

A study on performance and attrition in engineering (Felder, Felder, & Dietz, 2002) looked at attrition rates in chemical engineering. Students were classified as stronger or weaker based on predicted grade point average (GPA) at admission. Among the weaker students, by the end of the fifth year after matriculation 62% of the Introverted Sensing types had left chemical engineering, compared to 25% of the Extraverted Intuitive types, 27% of the Introverted Intuitive types, and only 11% of the Extraverted Sensing types. Women in engineering have historically earned lower grades and exhibited higher attrition rates than their male colleagues. The results of this study suggest that gender differences may be

strongly mediated by MBTI personality type. Among the Thinking types, which are predominant in engineering, men only slightly outperformed women. However, among the Feeling types, the male students considerably outperformed the female students. The authors conclude that “women with a preference for Feeling on the MBTI may be particularly vulnerable in engineering school” (p. 8). Attrition rates for Judging and Perceiving types were similar, although the Judging types earned higher grades.

Foster and Pashneh-Tala (2002) studied attrition at the US Naval Academy for five entering classes (1995–1999). They found that use of the continuous scores of the MBTI preference scales improved the prediction of voluntary and overall attrition, but not attrition for academic or performance/conduct reasons. Introverts, Feeling types, and Perceiving types were more likely to leave. The MBTI scores were a better predictor of voluntary attrition than any of the other predictors in the model.

Kahn, Nauta, Gailbreath, Tipps, and Chartrand (2002) examined retention in a sample ($n = 677$) of entering college freshmen at a southeastern university. Overall, 77% of the freshmen persisted into the sophomore year. Having a preference for Extraversion (versus Introversion) resulted in more than a 100% increase in the odds of persisting, and having a preference for Sensing (versus Intuition) resulted in 90% greater odds of persisting into the sophomore year, controlling for ACT/SAT scores.

The general atmosphere of higher education also appears to be more conducive to success for some types than for others. Macdaid, Kainz, and McCaulley (1984) analyzed data on attrition in a 10-year follow-up at the University of Florida. ISTJs and ISFJs had the highest rate of graduation, while ESTPs, ENTPs, and ISTPs significantly more often had not graduated. FJs were likely to have graduated early and TPs late; NTs were most likely and ESs least likely to have attended graduate school. Schurr, Ruble, and Henriksen (1989) called attention to the relatively low SAT scores of SJs, who, once admitted, persisted beyond expectation toward graduation, in large measure due to their finding applied major fields that channeled their interests and energies. Some of these findings about the varied education experiences of the different types may be related to differences between the types of students and the predominant type in the faculty, which Salter (2006) identified as INTJ.

Achievement

Type theory would predict that students who prefer Intuition and Perceiving would score higher on aptitude measures than Sensing types and Judging types. Aptitude measures tend to be more abstract and thus favor Intuitive types. Perceiving types tend to stay open to a wide range of information, which is also believed to have a positive effect on aptitude scores. However, Judging

students are hypothesized to get higher grades because they finish their work and turn it in on time.

DiRienzo and her colleagues (2010) studied the relationship between MBTI preferences, choice of major, and performance across academic disciplines in a sample of 6,280 students from 40 majors (grouped into six broad areas), using nine years of student data. Students tended to choose majors that were congruent with their type preferences. In terms of performance, the authors found that IFJ types had higher average GPAs, with the exception of business students (where the SJ types had higher GPAs). Perceiving types generally had lower GPAs. However, types that were overrepresented in a major did not necessarily perform better than other types. Some did and some did not. For example, ESFJs were overrepresented in education majors and had significantly higher GPAs. But in business ESTJs and ESTPs were overrepresented, with the former having significantly higher and the latter significantly lower GPAs than other types in that major.

Nyland, Ybarra, Sammut, Rienecker, & Kameda (2000) conducted a study of the relationship between type preferences, anxiety sensitivity, and academic achievement in college students. Anxiety sensitivity indicates an elevated fear of anxiety-related sensations, arising from the belief that these sensations have harmful somatic, psychological, or social consequences. There was a significant effect for the S–N and J–P preference pairs on achievement, with Intuitive and Judging types having higher GPAs. More interesting was the significant interaction between anxiety sensitivity and the J–P preference pair on GPA. Judging types with high anxiety sensitivity obtained higher GPAs than did Judging types with low anxiety sensitivity or Perceiving types with high anxiety sensitivity. Conversely, Perceiving types with low anxiety sensitivity obtained higher GPAs than did Perceiving types with high anxiety sensitivity or Judging types with low anxiety sensitivity. Anxiety was debilitating to Perceiving types. On the other hand, for Judging types anxiety sensitivity appeared to be associated with a mild facilitation of achievement. These results suggest that students who prefer Intuition and Perceiving may suffer more from the debilitating effects of anxiety sensitivity and those who prefer Sensing and Judging may be facilitated by anxiety. Differences were also found within preferences. The GPA mean difference between the high-anxiety-sensitivity group and the low-anxiety-sensitivity group within Extraverts (but not Introverts) suggests that one reason Introverts perform better academically in other studies may be the debilitating effect of anxiety on Extraverts.

Extracurricular Activities

Type theory has also been used to examine nonacademic factors that may be relevant to success or persistence. For example, one group of students (EP and TP types especially) appears at risk for academic probation or other disciplinary action unless their need for spontaneous play can be constructively channeled. This section summarizes some of this research.

Even though Introverted and Judging types seem to persevere under the academic demands of college, among students in general who persisted to graduation, Extraverts were shown to have been helped by active involvement in campus life, while Introverts, particularly IP and ISTP types, said they were helped by not being so involved (Provost, 1985). Extraverts also tended to develop a sense of autonomy and purpose and mature interpersonal relationships (Anchors & Robinson, 1992). On the other hand, Extraverts were also found frequently among campus judicial offenders (Griffin & Salter, 1993), as well as among those with substance abuse records, as were Intuitive and Perceiving types (Anchors & Dana, 1989; Barrineau, 1997; Provost, 1991).

Type theory would hypothesize that Sensing and Intuitive types would vary greatly in their need for extracurricular activity, especially given the primarily Intuitive environment of academia. Sensing students in college were found frequently among male scholarship athletes, as were Thinking and Perceiving types (Chesborough, 1994). In another study, Sensing types were found to participate more often in athletics in general (Ruble, Mahan, & Schurr, 1987). And as entering freshmen, they foresaw marriage as part of the college experience (Provost & Anchors, 2003).

Thinking types, like Extraverts and Perceiving types, were found frequently among judicial offenders (Griffin & Salter, 1993). Also, similar to findings for Extraverted, Intuitive, and Perceiving types, Thinking types were found frequently among those with substance abuse records (Anchors & Dana, 1989). Judging types were found to have higher grades among students who persisted in college (Anchors, Robbins, & Gershman, 1989). In contrast, even as they entered college as freshmen, Perceiving types reported that they considered the possibility of dropping out (Provost & Anchors, 2003). Among those on academic probation, EP types were the most frequently found, and TP and INTP types were the most frequently found among those with drug violations (Provost, 1991).

Type has been shown to be related to characteristics of students and teachers, retention and dropout rates, the influence of nonacademic factors, and the effect of classroom environments.

Student Leadership and Residence Life

Student leaders, including resident assistants (RAs), tend to prefer Judging, usually in combination with Extraversion and often also with Thinking (Anchors & Hay, 1990; Brush, 1989; Hardy-Jones & Watson, 1990; Petty, 1985). ENFJs, ESTJs, and ENTJs predominate among RAs. These patterns suggest that leadership roles in college settings tend to attract Judging types, who like to take charge, and Extraverts, who bring noticeable energy to their activities.

A number of studies have been completed on the use of MBTI results to match roommates (Carey, Hamilton, & Shanklin, 1985, 1986; Jackson, 1985; Kalsbeek, Rodgers, Marshall, Denny, & Nicholls, 1982; Schroeder & Jackson, 2003). The results have been mixed, at times identifying apparent success and at other times lacking significant findings. One interesting pattern that calls attention to the complexity of matching was seen in Jackson's (1985) study, which reported that matches in which the dominant and auxiliary processes of the roommate pair were the same, while successful, were superseded in terms of higher student grades by complementary pairings (dominant the same, auxiliary different). These latter findings are similar to those reported earlier on teacher-student matching by type (Boyd, 1995), where matching students with teachers of the same type was less successful than when a student and teacher had different preferences on one MBTI preference pair.

Type and Diversity Differences in Education

In college and university settings, a preference for Sensing and Thinking appears to characterize black students in higher percentages relative to white students (Hill & Clark, 1993; Johnson, 1990). Samples of black females especially tend to contain higher percentages of Thinking types (Harrison, 1995; Hill & Clark, 1993). ISTJs were found to be overrepresented and ENFPs underrepresented among black students (Hill & Clark, 1993). International students coming to US colleges were found by Fu (1992) to report better social and emotional adjustment if they preferred Extraversion. These findings suggest either that the overall distribution of types varies across cultural groups or that in order to meet the requirements for entering college, students from minority groups who report a preference for the practical (Sensing) and the logical (Thinking) survive best.

In high schools, Ross (1994) found that Extraversion predicted academic achievement among black students, contrary to findings for white students. Melear and Richardson (1994) found higher percentages of preferences for Perceiving, EP, SP, and TP among black males in high school, while May (1992) found higher percentages of SP types among black males in high school but not among black females.

Classroom Environment

Lynch and Sellers (1996) asked students to indicate the characteristics of their preferred learning environments from descriptions written to reflect ES, IS, EN, and IN patterns. Both adult learners and traditional-age college students tended to prefer environments consistent with their type preferences. However, there was an overall general preference for ES environments, with IN environments least preferred, especially for adult learners.

Kent and Fisher (1997) examined relationships between MBTI continuous scores and the *Secondary Colleges Classroom Environment Inventory* (SCCEI). The SCCEI has the following subscales: personalization (opportunities for individual students to interact with teacher and concern for students' personal welfare), informality (class behavior guided by formal rules vs. informal), student cohesiveness (students know, help, and are friendly toward each other), task orientation (activities are clear and well organized) and individualization (students are allowed to make decisions and are treated differentially). The sample consisted of 108 teachers and 1,883 college students. Perceptions of the classroom environment by both students and teachers were correlated with the MBTI scores of the teachers. Extraversion of the teacher was positively correlated with student cohesion, Sensing with task orientation and personalization, Intuition with informality, Perceiving with informality and individualization, and Judging with task orientation. No significant relationships were found with the Thinking–Feeling preference of the teacher. With the exception of the lack of results for the T–F preference pair, these results conform to type theory.

Salter (2003a) employed his *Salter Environmental Type Assessment* (SETA; Salter, 2000), which uses Jungian theory to describe environments, to explore the differential impact of educational environments on female students. Participants ($n = 421$ students) were asked to pick either a previous college-level course in which they had felt a high degree of fit as a student or one in which they perceived their fit with the classroom to be poor. Only the scores from the SETA Thinking–Feeling scale were used in an asymmetrical log-linear analysis. There were main effects for classroom climate and, to a lesser extent, for gender. Good fit was associated most strongly with Feeling-oriented classroom environments and poor fit with Thinking-oriented environments. Women reported more good-fit experiences than did men. A student's personality type alone did not predict whether she or he would report good or bad fit. However, there was a significant interaction between student type and classroom climate. For Thinking students, male or female, the nature of the classroom climate seemed to have little relationship to perceived fit. However, for Feeling women, the effect was pronounced, with significant positive residuals associated

with good fit in Feeling classrooms and poor fit in Thinking classrooms. For Feeling men, on the other hand, only the cells associated with fit in Thinking classrooms exceeded the criterion, with poor and good fit being overrepresented and underrepresented, respectively. Unlike their female peers, Feeling men did not indicate much preference for Feeling classrooms.

HEALTH, STRESS, AND COPING

The MBTI assessment is used widely in various domains within healthcare. Studies covered here include the effect of type on variables such as the reporting of heart disease and hypertension; positive and negative health-related behaviors; stress, coping, and burnout; physical work stress; and mental health. The review of research on health, stress, and coping provided by Shelton (1996) supports the differential experiences and responses of the 16 types and groupings of types in these important areas.

A study of satisfaction with one's health using a single item from the *World Values Survey* with a subset of participants in the MBTI global sample ($n = 5,477$), who used a 4-point rating scale from "very dissatisfied" to "very satisfied," showed that Extraverts were significantly more satisfied with the state of their health than were Introverts ($p < .001$, $d = .22$). Thinking types also had higher satisfaction with health than did Feeling types ($p < .001$, $d = .14$). The results for whole types are shown in table 9.22. The four types with the highest ratings on satisfaction with their health were those with preferences for Extraversion and Thinking (ENTJ, ESTJ, ENTP, ESTP), while those types preferring Introversion and Feeling rated their satisfaction lowest (INFP, INFJ, ISFP, ISFJ).

The Form Q US national sample ($N = 1,380$) participants rated their level of health on a 5-point scale from very poor to very good. Those with preferences for Extraversion and Intuition rated their health as significantly better than did Introverts and Sensing types.

Table 9.22 | Rank order of satisfaction with health by MBTI® type

Type	M	SD	Type	M	SD
ENTJ	3.07	0.79	INTJ	2.86	0.80
ESTJ	3.04	0.70	ESFP	2.85	0.81
ENTP	3.01	0.81	ISTJ	2.83	0.76
ESTP	3.01	0.74	ISTP	2.83	0.82
ENFJ	3.00	0.71	INFP	2.81	0.87
ESFJ	2.91	0.74	INFJ	2.75	0.85
ENFP	2.90	0.79	ISFP	2.73	0.78
INTP	2.86	0.83	ISFJ	2.65	0.79

Note: Portion of the global sample who responded to the question, $n = 5,477$.

Myers believed that Introverts were generally less satisfied than Extraverts in any number of areas, because Introverts are more focused internally and tend to be more aware of internal difficulties than are Extraverts. Thus Introverts would be predicted to rate satisfaction with their health lower than would Extraverts. This is confirmed by the two studies above, but with the added result that Feeling and Sensing types also may rate their satisfaction lower. It may be that Sensing types are more aware of bodily sensations that they associate with possible health problems.

Heart Disease

The several previous studies that looked at type and heart disease, as summarized by Shelton (1996), found that across studies, Introverts were found more than 1.5 times as frequently in coronary heart disease (CHD) groups as would be expected relative to their proportion in a base sample. Sensing types were also found to be overrepresented in CHD studies. This is not to suggest that a preference for Introversion and/or Sensing is necessarily a risk factor for heart disease. These studies used subjects who had been diagnosed with CHD and/or were in treatment programs for their illness. So an alternative hypothesis for these results is that having heart disease resulted in patients focused more inwardly and on their sensory experience.

Participants in the Form M US national sample ($N = 3,036$) were asked whether they had ever had heart disease or hypertension or suffered chronic pain or insomnia. Sensing types were significantly more likely to report having had heart disease, at a rate almost twice that of Intuitive types. Feeling types and Perceiving types were more likely to say they had experienced insomnia and chronic pain. Table 9.23 shows the rank order of the 16 types by percentage of each type who claimed to have had heart disease or hypertension. In general, the rankings in this table are in accordance with data that reveal specific over- and underrepresentations of types and two-letter combinations of preferences that were found in earlier research. For example, ENFP, INTJ, and INFP, which were underrepresented in the CHD groups in earlier research, occupy the 13th, 14th, and 15th ranks in table 9.23. Dominant introverted Sensing types (ISTJ and ISFJ) appear two to three times more frequently than other types, and SJ types in particular are overrepresented.

Health Behaviors

Martin (2011) examined the relationship between MBTI results and selected health-promoting behaviors among a randomly selected sample of 406 residential college students. The health variables were measured using the *Health-Promoting Lifestyle Profile II* with six subscales: Health Responsibility (accountability

Table 9.23 | Rank order of MBTI® types that answered “yes” to ever having heart disease or hypertension by percentage

Type	%	Type	%
ISFP	27.4	ESTJ	19.4
ISTP	27.2	INTP	18.2
ISTJ	26.6	ENTJ	17.0
ISFJ	24.7	ESTP	15.5
ESFJ	21.8	ENFP	15.4
ESFP	21.0	INTJ	15.4
ENFJ	20.7	INFP	11.7
INFJ	20.4	ENTP	10.7

Note: Form M US national sample, N = 3,036.

for one’s own well-being), Interpersonal Relations (communicating to produce intimacy), Physical Activity (regular participation), Nutrition (healthy foods), Spiritual Growth (development of inner resources), and Stress Management (using resources to reduce tension). A total score on health-promoting behaviors was also calculated. Regression analysis used MBTI preferences, age, gender, ethnicity, academic classification, parental social position, and mean household income as independent variables. Extraversion helped predict health behaviors associated with interpersonal relations, and Extraversion and Judging predicted the use of spiritual growth behaviors as well as the total score for health-promoting behaviors. Judging predicted the use of stress management behaviors.

Overall, ENTJs were significantly more likely to engage in the positive health-promoting behaviors of nutrition, health responsibility, and physical activity, and they had a high total score for overall health-promoting behaviors. INTPs and INFPs were the least likely to engage in health-promoting behaviors; these two types predicted significantly less participation in interpersonal relations, spiritual growth, and total health-promoting behaviors. Overall, 8 of the 16 personality types were predictors of at least one health-promoting behavior. It is interesting to note that while certain Extraverted types, particularly ENTJs but also ESTJs, ENFJs, and ESFPs, were more likely to engage in health-promoting behaviors, some of these same types (as well as other Extraverted types) were also more engaged in risky health behaviors, such as increased alcohol consumption, binge drinking, heavy drinking, and rapid drinking (all four EP types plus ESTJs) and smoking (ENFPs). Potentially positive and negative effects appear to be associated with a preference for Extraversion in combination with the mental processes. The author concluded that for a number of these analyses, the individual preferences were not significant predictors of health behaviors, but whole types were. And which types were significant predictors differed depending on the dependent variable that was the

subject of analysis. See the “Coping” section later in this chapter for more on this study.

Type differences are also found in eating and exercise habits. College students’ dietary habits were studied by Horacek and Betts (1998), who found that Extraverts exhibited positive eating habits but also took in a significantly higher percentage of calories from alcohol than did Introverts, a finding that mirrors the Martin (2011) study. Students with a preference for Intuition also displayed more effective eating patterns, such as frequent breakfasts, decreased snacking, and selection of diets providing a lower percentage of calories from sweets, than did their Sensing peers. Overall, those preferring Sensing had less desirable dietary habits, possibly as a result of more frequent snacking and consumption of desserts. Students preferring Judging chose diets providing a lower percentage of calories from alcohol and consumed more frequent breakfasts compared to those preferring Perceiving. This is an area in which research has outpaced theory, since with the exception of alcohol consumption, type theory at this point has little to say about specific eating behaviors.

Buckworth, Granello, and Belmore (2002) studied exercise self-efficacy in students, although only the E–I and J–P scales were used. Self-efficacy is a predictor of action; that is, those who believe they can do something are more likely to take action. Exercise self-efficacy in particular is an individual’s judgment of his or her ability to organize and execute actions needed to engage in exercise. Extraverts scored significantly higher on exercise self-efficacy than did Introverts. No significant differences were found on this variable between Judging and Perceiving types. However, among students who did not exercise regularly, Judging types scored significantly lower than Perceiving types on exercise self-efficacy. Among those who did exercise regularly, Extraverts scored significantly higher on exercise self-efficacy. So, among those who already exercise, Introverts may be more susceptible to relapse; among those who don’t already exercise, Judging types may be less likely to start.

Participants in the Form M US national sample were asked about whether they engaged in drinking alcohol and smoking. Extraverts, Intuitive types, and those with a Thinking preference were more likely to report drinking alcohol more frequently. There were no differences on any of the MBTI preference scales for frequency of smoking. Note that these are self-reports of health behaviors. An interesting line of research would be to test how well self-reports of these behaviors conform to actual behavior.

Stress

Participants in the Form M US national sample were asked to rate their typical level of stress about work, finances, children, intimate relationships, school, health,

caring for aging parents, balancing home and work, and “other.” Sample participants overall rated all of the areas as moderately stressful, so the focus here is on the more extreme responses—those rating an area as extremely stressful and those rating the area quite low as a source of stress.

Individual preference effects were studied using log-linear comparisons of degree of reported stress for each stress category. The following significant differences were found:

- Introverts scored significantly higher than Extraverts on work stress.
- Sensing types scored significantly higher than Intuitive types on stress associated with health, care for aging parents, and balancing home and work.
- Feeling types scored significantly higher than Thinking types on the stress of finances, children, health, care for aging parents, and balancing home and work.
- Judging types reported higher stress around health issues and balancing home and work.
- Perceiving types reported higher stress associated with work.

Table 9.24 shows the top- and bottom-ranked types in each stress category. ESFP is one of the types reporting the highest stress in seven of the nine categories, while ENTP and ESTP are among the types reporting the lowest stress in six categories. Sensing Feeling types are particularly likely to identify more potentially stressful areas, and this was true regardless of their preference for Extraversion or Introversion. It is interesting that ESFP experiences high stress while the other dominant extraverted Sensing type, ESTP, experiences low stress. Both INTJ and INTP report high stress in some areas and low stress in others. The types that appear less stressed are ESTP, ENTP, INTJ, and ENTJ; the most clearly stressed are ESFP, ISFP, ISFJ, and INFJ. It is also of note that of the “idealistic” NF types, who might be expected to be especially vulnerable to stress, only INFJs

reported numerous stress areas. As can be seen, some areas appear to be particularly stressful for some types but provide minimal stress for others. Not all of these associations would be clearly predicted by type theory.

Although some types appear to be more stressed in general than other types, what different types find particularly stressful and nonstressful probably involves a complex interaction among personality, the nature of the stress, and unknown other factors. These possible type-by-stressor relationships seem to suggest that the stresses of everyday life have different effects on different types and that some types are more naturally reactive to (or willing to acknowledge) stress than others. This is an area in which further theory and research are needed.

The association of type preferences with different kinds of stressful stimuli was explored by Short and Grasha (1995) in a sample of managers. They looked at sources of stress, coping strategies, and personality moderator variables in relation to MBTI continuous scores. They found that sensation seeking, assertiveness, and “social hardiness” were significantly associated with Extraversion; social stress was associated with Introversion; Thinking was correlated with Type A behavior, assertiveness, and perceptions of problem-solving skills; Judging was related to effective time management and Perceiving to sensation seeking. Based on various analyses of their data, they concluded that the E–I preference scale was the best predictor of stress and coping and was also related to perceptions of stress.

Maybee and Carter (1999) used regression analysis to examine the relationship between MBTI preference scales and the indexes of stress measured by the *Personal Stressor Profile*. They found that higher scores toward Introversion were a significant predictor of anxious reactivity (additional anxiety as a reaction to initial anxiety), control (fear of loss of control) and self-perception (self-consciousness). Sensing and Feeling both predicted control and Feeling also predicted self-perception.

Table 9.24 | MBTI® types reporting highest and lowest levels of stress in nine categories

Category	Types reporting highest stress				Types reporting lowest stress			
Work	INFJ	INFP	INTJ	ISTJ	ENTJ	ENFJ	ESFJ	ISFP
Finances	ISFP	ESFP	ENFP	INTJ	ENTJ	ISTP	ESTP	ESTJ
Children	ISFP	ESFP	ISFJ	INTP	ENTP	INTJ	ESTP	ENFP
Intimate relationship	INFJ	ESFP	INTJ	INTP	ENTP	ENFJ	ESTP	ENTJ
School	INFJ	ESFP	INTP	ISFJ	INTJ	ENFP	ENTJ	ESTJ
Health	ISFJ	ESFJ	ISFP	ESFP	ENTP	INTJ	ENTJ	INFP
Caring for aging parents	ISFJ	ESFP	ISFP	ESFJ	INTJ	ESTP	ENTP	INTP
Balancing home and work	ISFJ	INFJ	ESFJ	ENFJ	ISTP	ENTP	INTJ	ESTP
Other	INFJ	INTP	ISFP	ESFP	ENFP	ENTP	ENFJ	ESTP

Note: Form M US national sample, N = 3,036. Types reporting highest stress and lowest stress in each row are rank ordered from left to right.

Work-Related Physical Stress

Myers and McCaulley (1985) stated that “when there is a mismatch between type and occupation, the client usually reports feeling tired and inadequate. According to type theory, the mismatch causes fatigue because it is more tiring to use less-preferred processes” (p. 78).

Allread and Marras (2006) provided a partial test of this hypothesis by examining whether personality affected the risk of developing musculoskeletal discomfort. Musculoskeletal disability is reported to be the most frequent chronic condition causing long-term disability. The sample consisted of 133 individuals recruited from the warehouse distribution centers of two companies, who averaged three years in their current job. MBTI Form M was used to measure type and the *Salter Environmental Type Assessment* to measure perceptions of the environment. Employees in both companies overwhelmingly viewed their environment as Extraverted and Sensing. Company A employees also typed their workplace as Thinking and Judging, while more Company B employees viewed their work environment as Feeling and Judging, but these percentages were not statistically significant. Introverts, who were considered mismatched with their (Extraverted) work environment, reported higher mean anxiety levels than did their Extraverted co-workers. Anxiety and job future ambiguity scores were also higher for Introverts than for Extraverts. And Introverts had significantly more neck and shoulder discomfort. Similarly, Intuitive types in Company B, who were mismatched with their Sensing environment, reported significantly higher mean levels of role conflict and role ambiguity than did their Sensing colleagues. A mismatch between one’s personality and the nature of the work environment was also related to increased strain responses. Thinking types in the predominantly Feeling environment in Company B reported more thigh/knee discomfort; Feeling types in this company had higher job satisfaction. In one company with a perceived Sensing environment, Intuitive types had lower job satisfaction and reported far greater elbow/forearm discomfort.

Chany, Parakkat, Yang, Burr, and Marras (2006) wondered whether the biomechanical response of an individual worker to physical stressors might be dependent on the worker’s perceptions about stress, which in turn may be a function of the match between the worker’s personality and the tasks he or she performs. Specifically, they tested the hypothesis that a personality-job mismatch increases the level of perceived stress, which in turn increases the trunk muscle activities and thus increases loading on the lumbar spine. Twenty-four workers whose job involved lifting repetitively for eight hours a day participated in the study. Only the S–N and J–P preference pairs were examined due to the distribution of types in the sample. Three outcome measures were used: compression, anterior-posterior (A/P) shear, and lateral shear (a measure of stress or distortion in a

muscle). The J–P preference pair significantly affected all three outcomes, while the S–N preference pair significantly influenced A/P shear. The main effect of the J–P preference pair on compression was significant. Perceiving types had 16% higher compressive loads and 12% higher lateral shears than did Judging types. For the S–N preference pair, Intuitive types exhibited 12% higher A/P shears than did Sensing types. The authors’ explanation for these findings was that the Intuitive and Perceiving types were mismatched to the repetitive task, thus causing more measurable physical stress. Chany et al. (2006) also reported many interactions between S–N and other variables. While these results should be interpreted cautiously given the small sample size and the examination of only the S–N and J–P preference pairs, this seems like a promising line of inquiry for future research using the MBTI assessment.

Burnout

Job burnout as a response to stress has been productively studied using the MBTI assessment, particularly in Garden’s research, as reviewed in Shelton (1996). Garden found that Thinking types react more positively, and Feeling types more negatively, to people as they experience burnout.

The *Maslach Burnout Inventory* (MBI; Maslach, Jackson, & Leiter, 1996) measures the effects of job burnout on individuals. It contains three subscales: Emotional Exhaustion (the depletion of emotional energy, as distinct from physical or mental fatigue), Depersonalization (a response to work that requires personal sensitivity to clients), and Personal Accomplishment (an emphasis on one’s effectiveness and success in bringing benefit to clients). The MBI and Form M of the MBTI assessment were administered to a sample of 82 employees at a large hospital. Introverts scored significantly higher on Emotional Exhaustion and Depersonalization. These results are in accordance with the theoretically predicted differential availability of an “extraverting energy” for Extraverts and Introverts. Personal Accomplishment was correlated with Extraversion. Analyses for the other three preference pairs did not yield significant results on any of the MBI scales. In terms of whole types, ISFPs and ISFJs scored highest on Emotional Exhaustion and Depersonalization. ESTPs and ENTPs were among the four lowest-scoring types on Emotional Exhaustion and among the highest-scoring types on Personal Accomplishment.

Coping

In addition to types responding differentially to stressors, research also supports the hypothesis that types cope with stress differently and with greater or lesser facility. Genco and Mitchell (2006) conducted a correlational study using continuous scores of the MBTI preference

scales and scores of the *Coping Inventory for Stressful Situations*, which measures task-, emotion-, and avoidance-oriented coping. For the sample overall, higher scores toward Judging were significantly related to task-oriented coping. For women, higher scores toward Thinking correlated with task-oriented coping.

Golden (2009) examined the coping behaviors of high school students during the college search process using Form M of the MBTI assessment and a coping scale adapted from the *Responses to Stress Questionnaire* (RSQ; Connor-Smith, Compas, Wadsworth, Thomsen, & Saltzman, 2000) that measured three kinds of coping. *Primary control coping* is defined as goal-directed efforts to address the sources of stress; the efforts include problem solving, controlling one's emotions, and expressing emotions. *Secondary control coping* is defined as attempts to assimilate to a stressful situation in order to mitigate any negative impact; it is achieved through positive thinking, cognitive restructuring, acceptance, and distraction. Finally, *disengagement coping* refers to intentional efforts to insert psychological (or even physical) distance between the individual and the stressor as a means of escapism; the efforts include denial, wishful thinking, and avoidance. Perceiving types were more likely to engage in secondary control coping in the college search context. Introverts scored higher than Extraverts on disengagement coping, indicating that they were much more likely to seek an escape from the search process. Extraverts scored higher on secondary control and Judging types scored higher on primary control. Hierarchical regression was used to account for the effect of other variables, such as gender, ethnicity, first-generation student, SAT scores, and high school GPA. Type variables were treated as dichotomous variables in the regression. The Judging–Perceiving preference pair significantly increased the amount of variance explained on primary control. The Extraversion–Introversion preference pair significantly added to the prediction of disengagement, as did the Thinking–Feeling preference pair (with the Thinking types being more disengaged).

Hammer (1991) used the *Coping Resources Inventory* (CRI) to study type differences in a sample that included MBA students, residence hall workers, stress management program participants, and pain patients ($N = 270$). The CRI measures cognitive, social, emotional, spiritual/philosophical, and physical coping resources; a total resources score is also available. Table 9.25 shows the ranking of these five resources for each of the 16 types. The types found to have the highest total resources were ENFPs and ESTJs, and the types with the lowest total resources were INTPs and ISFPs. Each type showed a unique order of typical use of the five coping resources. For example, social resources had the top rank and physical resources the bottom rank for ENFJs; emotional resources was ranked highest and spiritual/

philosophical lowest for ESFPs. These whole-type differences could not be explained by knowledge of the preferences alone.

Physical, psychological, and emotional health have been found to be influenced in predictable ways by type differences, providing important information for laypeople and professionals in identifying vulnerabilities of and effective strategies for dealing with individuals of different types.

The results of Martin's (2011) study of health behaviors discussed above replicate many of Hammer's findings with the CRI. In Martin's study, INTPs, INFPs, and ISTJs used fewer interpersonal relations behaviors to maintain health; Hammer found that these three types ranked 15th, 12th, and 11th in social resources. In Martin's study, ESTJs reported significantly more interpersonal relations behaviors; Hammer found that ESTJs ranked 2nd on social resources. For spiritual growth behaviors, Martin found that INFPs and INTPs scored lower, which is matched by Hammer's findings for these types on spiritual resources, on which they ranked 13th and 14th, respectively. One point of difference between the two studies was for ESTJs, who scored high on spiritual growth behaviors in the Martin study but were ranked 11th on spiritual resources in Hammer's study.

Another set of coping behaviors were examined in the Form M US national sample. Participants were presented with a list of 12 coping strategies that had been created by interviewing people of different types about how they coped. Extraverts were significantly more likely to confront the problem, talk with others, rely on their beliefs, and exercise. Introverts were more likely to avoid the problem, get angry but hide it, develop physical symptoms, and sleep. Sensing types were more likely to avoid, while Intuitive types were more likely to confront the problem, think about solutions, develop physical symptoms, and exercise. Thinking types were more likely to confront the problem, whereas Feeling types were more likely to talk with others, develop physical symptoms, and rely on their beliefs to guide them. Judging types were more likely to rely on their beliefs, while Perceiving types were more likely to get angry but hide it. This is an area in which theory needs to guide further research. Some patterns of interest are these: FJs tend to rely on their religious beliefs, while NTs do not; SFs try to avoid stressful situations, whereas three of the ET types (and ENFJs) confront the problem; three of the EN types (and INTJs) use exercise, whereas the types who do not are all Sensing types (including three SF types). The data were also analyzed by whole type; the results are shown in table 9.26 for the four types with the highest and lowest percentage of endorsement of each strategy. The findings by preference, preference combinations, and whole type are generally congruent

Table 9.25 | Rank order of coping resources by total number for each MBTI® type

ISTJ	12	ISFJ	7	INFJ	4	INTJ	11
Cognitive	10	Spiritual/philosophical	4	Spiritual/philosophical	1	Spiritual/philosophical	3
Spiritual/philosophical	10	Cognitive	5	Cognitive	4	Cognitive	7
Social	11	Social	5	Physical	5	Emotional	11
Physical	11	Physical	7	Social	8	Physical	13
Emotional	13	Emotional	10	Emotional	9	Social	14
ISTP	9	ISFP	15	INFP	13	INTP	16
Physical	2	Social	10	Physical	9	Physical	12
Emotional	5	Spiritual/philosophical	12	Social	12	Spiritual/philosophical	14
Spiritual/philosophical	6	Cognitive	14	Cognitive	13	Cognitive	15
Cognitive	12	Emotional	15	Spiritual/philosophical	13	Social	15
Social	16	Physical	16	Emotional	14	Emotional	16
ESTP	14	ESFP	10	ENFP	1	ENTP	8
Physical	8	Emotional	3	Social	1	Physical	6
Emotional	12	Social	6	Emotional	1	Emotional	6
Social	13	Cognitive	9	Cognitive	2	Social	9
Spiritual/philosophical	15	Physical	15	Physical	4	Spiritual/philosophical	9
Cognitive	16	Spiritual/philosophical	16	Spiritual/philosophical	7	Cognitive	11
ESTJ	2	ESFJ	5	ENFJ	3	ENTJ	6
Cognitive	1	Spiritual/philosophical	2	Social	3	Physical	1
Emotional	2	Social	4	Cognitive	3	Cognitive	6
Social	2	Emotional	7	Emotional	4	Social	7
Physical	3	Cognitive	8	Spiritual/philosophical	5	Spiritual/philosophical	8
Spiritual/philosophical	11	Physical	10	Physical	14	Emotional	8

Note: $N = 270$. Number after each type indicates the ranking of that type on total resources. Number after each resource indicates the ranking of each type on that resource compared to other types.

Source: Hammer (1996b), p. 206. Copyright 1996 by CPP, Inc. Used with permission.

with type theory and with previous research on type and coping.

Du Toit, Coetzee, and Visser (2005) studied the relationship of coherence to MBTI preferences. Coherence, measured with the *Orientation to Life Questionnaire* (Antonovsky, 1993), is a sense that one can cope effectively due to a belief that potentially stressful events are structured, predictable, and understandable. EJ types scored significantly higher than IP types, and SJ types scored significantly higher than SP types. The category of respondents that obtained the highest mean score on the sense of coherence measure among the four orientation types was the EJs. Correlational analysis showed that E–I and J–P continuous scores contributed significantly to the prediction of sense of coherence, which is not surprising given the definition of this attitude toward coping. Due to small samples for many of the types, the test of whole types yielded a statistically nonsignificant result ($p < .18$). Nevertheless, there were sizable mean differences on coherence between some of the types, such as between

ISTP and ESTJ ($t = 4.30, p < .001$), between ISTP and ISTJ ($t = 3.05, p < .01$), and between ESTJ and ESTP ($t = 2.43, p < .01$). It is important to note that these differences are between types that differ by only one or two preferences, suggesting that there is something about the whole type or the dynamic combination of preferences that is relevant to a sense of coherence.

Mental Health

This section summarizes some of the research on type and mental health presented in Myers et al. (1998) and briefly comments on studies included in *MBTI® Applications* (see Quenk & Quenk, 1996). Topics that have received research attention include the types of providers of psychological services and the kinds of issues presented by clients of different types.

Type can illuminate many of the personality and behavior differences among providers of psychological services, and it raises issues about the delivery of mental health care. All 16 types are represented by

Table 9.26 | Top- and bottom-ranking MBTI® types by percentage of endorsement of strategies for coping with stress

Coping strategy	Type with highest %		Type with lowest %		Coping strategy	Type with highest %		Type with lowest %	
Try to avoid stressful situations	ISFP	59.8	ENTP	26.0	Get upset or angry but don't show it	ISFP	36.1	ENTJ	16.7
	ESFP	54.5	ENTJ	33.3		INFP	34.9	ESTP	20.3
	ESFJ	53.3	ENFP	34.4		ISFJ	34.5	ENFJ	20.7
	ISFJ	53.3	ESTJ	35.2		INTP	34.3	ESTJ	21.1
Confront the problem	ENTP	79.8	ISFP	35.7	Get upset or angry and show it	ISTP	42.9	ENTJ	18.8
	ENTJ	79.2	ISFJ	41.0		ISFP	39.1	ESTJ	27.7
	ESTJ	73.8	INFP	41.8		ESFP	38.3	INTJ	27.7
	ENFJ	68.3	INFJ	44.9		ENFJ	36.6	INTP	29.3
Talk to someone close	ESFJ	74.1	ISTP	39.0	Develop physical symptoms	ENFP	21.1	ESTP	5.9
	ENFJ	72.0	INTP	45.5		INFJ	20.4	ENTP	7.7
	ESFP	71.1	ISTJ	45.6		ENFJ	19.5	ISTP	7.8
	INFP	63.7	ESTP	45.8		ISFP	16.9	ESTJ	8.6
Talk to professional	INFJ	24.5	ISTP	3.2	Sleep	ISFP	23.3	ESTJ	14.1
	INTJ	16.9	ENTP	6.7		INTJ	23.1	ESTP	15.3
	ENFP	10.9	ESTP	7.6		ESFP	20.6	ENTP	15.4
	INFP	10.3	ESTJ	7.8		ISFJ	20.2	INTP	17.2
Rely on religious beliefs	ESFJ	55.5	INTP	20.2	Exercise	ENFJ	40.2	ISTP	13.0
	INFJ	53.1	ENTP	20.2		ENTJ	35.4	ISFP	16.2
	ENFJ	50.0	INTJ	24.6		ENTP	34.6	ESFP	17.8
	ISFJ	49.4	ENTJ	25.0		INTJ	30.8	ISFJ	20.9
Try to think of options	ENTJ	87.5	ISFP	54.5	Watch television	ISFP	40.6	INTJ	15.4
	INTP	80.8	ISFJ	56.2		ESFP	33.2	ESTP	18.6
	ENTP	79.8	ESFP	62.5		ISTP	31.2	ENFP	20.6
	INTJ	76.9	INFJ	63.3		ISFJ	31.1	ENFJ	20.7

Note: Form M US national sample, $N = 3,036$. Respondents could mark as many strategies as applicable; p values for all 2×16 chi-squares significant at .000 to .027, except for "Sleep."

providers of psychological services, and all 16 types become clients, but the distributions of both differ from the general population. In theory, helping people through psychological methods requires more intuitive than sensing skills, since Intuition is concerned with perception of patterns, recognition of inferred meanings, and intangible relationships. In a study by Newman (1979), Intuitive types scored significantly higher than Sensing types on ability to identify implied meanings. A study by DiTiberio (1977) found that Intuitive types have higher sensitivity for covert feeling messages.

Any career that deals mainly with people tends to attract more Feeling types than Thinking types. As reported in the "Occupational Selection and Careers" section of this chapter, 16 studies in the 1985 *MBTI® Manual* (Myers & McCaulley, 1985) that looked at characteristics of providers of psychological services confirmed this expectation. Across all these studies, Intuition and Feeling tended to predominate. Occupations that require practitioners to deal with large numbers of

people tended to have more Extraverts, and there were relatively more Sensing types in the occupations in which counseling is accompanied by more paperwork.

Quenk and Quenk (1996) and Myers et al. (1998) reported on several studies that found significant differences in therapeutic orientation among counselors of different types. Practitioners who preferred Thinking tended to apply linear, logical, analytic processes (as in behavioral and cognitive approaches); Feeling types preferred humanistic, affective approaches that emphasize the therapeutic relationship. Judging counselors liked operating in systematic ways, while Perceiving practitioners preferred more flexibility; Intuitive practitioners liked to understand how facts and ideas are interconnected, while Sensing types used more prescribed methods. There appeared to be little if any effect attributable to Extraversion–Introversion differences.

Carskadon (1979a) asked college students what qualities they valued in a counselor. Thinking types rated

behavioral characteristics higher, and Feeling types rated humanistic characteristics higher. In a study of college students, Weir (1976) found that Feeling types who listened to an audiotape of a counselor demonstrating high unconditional positive regard were significantly more likely to prefer this approach than were other types. A study by Arain (1968) found that among high school students seeking counseling, Thinking types preferred cognitive characteristics in prospective counselors and Feeling types preferred affective characteristics. In another study, undergraduates who volunteered for a small-group “self-understanding” research project were 84% N (and 51% NF) types (McCary, 1970).

Otis and Louks (1997) explored the relationship between personality type and psychological dysfunction in a sample of male veterans who had requested psychotherapy. Due to the type distribution, only data for the 158 Introverts in the sample were analyzed. An ANOVA was performed on participants’ scores on factors derived from the *Minnesota Multiphasic Personality Inventory*® (MMPI®) assessment, using age as a covariate and the four MBTI preference pairs as independent variables. The results showed significant main effects due to T–F and J–P and no significant interaction terms. A salience analysis of the case record variables was also performed in which the mean, median, mode, and variance for each type were compared to those of a large number of random samples taken from the total pool of scores. Analyses revealed that INFPs were most likely to have anxiety disorder, INTJs were most likely to exhibit major depression, ISTPs were most likely to exhibit combat-related post-traumatic stress disorder (PTSD), ITPs were most likely to display antisocial and avoidant personality disorders, ISTJs were most likely to be designated obsessive-compulsive, and IFJs were most likely to be characterized as “dramatic.” Analysis of life history variables found that ITPs were most likely to have been charged with a crime unrelated to substance abuse and to have had disciplinary problems at the domiciliary. ISTPs were most likely to have never married, while INTPs and ISFPs were least likely to have never married. INTPs tended to have a greater number of marriages than any other type, while ISTJs had the fewest. ISJs tended to stay in the same job longer than the other types, while INFJs and INTPs had the shortest mean time in their jobs.

This study of veterans highlights the need for studies of whole type and type dynamics.

In addition to the insight they provide into mental health problems of veterans, the findings of this study are important because of the distinct difference they point out between whole types in terms of diagnoses and life history variables that were not just summative effects of the separate preferences. For example, INTPs and ISTPs were almost identical in their Axis II profiles and in their crime and domestic trouble histories, which might argue for preference effects. However, none of the other Introverted, Thinking, or Perceiving types appear to share these effects. The outcomes observed for these two types seem to derive from the interactive combination of all four preferences rather than their separate summative effects. For another example, consider that INTJs were significantly more likely than any other type to be diagnosed with major depression, while INFPs were most likely to receive the diagnosis of anxiety disorder. These differences are clearly specific to each of these types. None of the other types had significant profiles involving these diagnoses. These examples illustrate that the study yielded very different results depending on the kind of analysis used; the qualitative descriptions described differences between types that were not captured in analyses of their quantitative data. The interested reader is referred to the Otis and Louks (1997) paper for the rich details and insights that cannot adequately be conveyed in a brief summary of their work.

Kameda and Nyland (2003) studied anxiety sensitivity (an elevated fear of anxiety-related sensations) in relation to type. Sensing types revealed significantly higher sensitivity than did Intuitive types, and Sensing and Thinking types exhibited significantly higher mean anxiety sensitivity than did Intuitive and Feeling types. There were no significant direct effects for E–I or J–P differences in anxiety sensitivity. However, there was a significant interaction of E–I with age. Younger participants (18- to 30-year-olds) with a preference for Extraversion exhibited a slightly higher mean anxiety sensitivity than did those with a preference for Introversion, whereas in the middle-aged group (31- to 54-year-olds) Introverted participants exhibited a considerably higher mean anxiety sensitivity than did Extraverted participants. In other words, older participants exhibited the classical significant relationship between Introversion and anxiety, while younger participants did not.

In another study showing an interaction, this time with gender, Quenk (1966) found that anxious females were more likely to be Introverts and Intuitive types; anxious males were more likely to be Feeling types. Future studies of type and anxiety would do well to look for interactions with other variables, such as age and gender.

VALIDITY SHOWN IN OTHER STUDIES OF TYPE DIFFERENCES

Earlier in this chapter, MBTI type distributions on a variety of scales from other assessments were examined, as well as studies of MBTI types in the areas of occupations, organizations, education, and health. This section presents an examination of studies in more specific fields or contexts. There is no claim that this review is comprehensive. Thousands of studies describe how particular behaviors or specific variables are related to type preferences, and space precludes more than brief coverage here. Instead, the text below samples studies from many domains with a goal of showing how type theory can help explain a great variety of behaviors across many domains. Since the MBTI assessment is used to test these theoretical predictions, positive results also confirm the validity of the instrument as a measure of the theory. An ancillary goal of this section is to spur further research in these areas.

Self and Rater Descriptions Associated with MBTI® Preferences

The staff of the Institute of Personality Assessment and Research (IPAR) included behavioral ratings by the staff as part of the standard assessment protocol. Data reported by Gough (1981) and Helson (1975) show significant correlations between rating assessments and MBTI preferences. Ratings are based on the *Adjective Check List* (ACL; Gough & Heilbrun, 1983) and the *California Q-Sort* (Block, 1978). Table 9.27 shows the staff ratings reported by Gough (1981) for 244 men and 186 women assessed at IPAR. An important paper by Gough (1965) discusses how data such as these can help provide a conceptual analysis of scale meaning. The descriptions are consistent with type theory, although some of the male Thinking terms fit STs better than NTs. What is important to note here is that the ratings were conducted by trained observers who had no knowledge of the type of the participants.

The ACL was also used to acquire self-ratings from participants in a portion of the MBTI global sample. The endorsement rate for each of the 369 adjectives on an adapted version of the ACL was calculated for individuals with each of the eight preferences. However, on the ACL, certain adjectives tend to be endorsed by everyone, regardless of type. For example, the most frequently endorsed adjective for participants with seven of the eight preferences was *honest*; the exception was Intuitive types, for whom it was second. Most participants also described themselves as *friendly* and *kind*, which were among the 10 most frequently endorsed adjectives for all preferences; the exception was Thinking types for *kind*. Even among these highly endorsed adjectives, however, there were differences in rate of endorsement between people with opposite preferences. To take

honest again, this adjective was endorsed by 62% of Extraverts and 55% of Introverts, by 56% of Sensing types and 63% of Intuitive types, and by 55% of Thinking types and 62% of Feeling types. It was endorsed equally by Judging and Perceiving types (58%).

Table 9.28 shows the 15 adjectives with the greatest difference in frequency of endorsement between participants with opposite preferences. The numbers in parentheses show the difference in frequency for the adjectives with the highest and lowest endorsement rates. For example, 26.9% more Extraverts than Introverts endorsed *sociable*, while 7.5% more Introverts than Extraverts described themselves as *plain*. These lists provide evidence for validity of the preferences. The adjectives unique to each preference overlap considerably with those used by Myers and others to describe people with opposite preferences. A few adjectives, such as *cautious* and *serious*, were endorsed by individuals at one pole more than those at the opposite pole on three of the four preference scales (both of these adjectives were endorsed by Introverts, Sensing, and Judging types). Although not analyzed by preference combinations, these adjectives may reflect how ISJs describe themselves. Some of the adjectives endorsed have a negative connotation (e.g., *thoughtless* and *intolerant* endorsed by Sensing types), but these descriptors were chosen by the participants themselves. Some of the adjectives that contrast the Sensing and Intuitive types must be interpreted with caution since the percentage of difference in endorsement rate was very small for this preference pair. This demonstrates a recurring problem of finding adjectives that describe well-developed Sensing.

Pearman and Fleenor (1996) designed a study to test whether the 16 types differ significantly from each other on observer ratings and self-report. The sample consisted of 2,398 managers and leaders who attended programs at the Center for Creative Leadership between 1985 and 1994. Using a large database, 150 individuals of each type were randomly selected for analysis, although for two types, only 149 cases were available. Most of the sample consisted of white males between 40 and 55 years old. Their average tenure in a managerial or executive position was eight years. The two assessments used for testing type differences were the *Leadership Style Indicator* (LSI) and the CPI® 434 assessment. The CPI 434 assessment yields t-scores on standardized scales. The LSI contains 48 adjectives used by observers to rate others. The individuals being rated also rated themselves on the same 48 adjectives in the LSI.

Twenty of the CPI 434 scales were analyzed as dependent variables in ANOVAs, with MBTI type as the independent variable. An ANOVA was also performed using the LSI scales as dependent variables. The Duncan Multiple Range Test was used post hoc to determine the ordering of types on the dependent variables. The

Table 9.27 | **Correlations of rater descriptions based on ACL and California Q-Sort associated with MBTI® preferences**

Extraversion	Men		Women	
ACL	outgoing	-.30	outgoing	-.45
	sociable	-.31	sociable	-.44
	enthusiastic	-.22	enthusiastic	-.49
	optimistic	-.25	optimistic	-.25
	talkative	-.28	talkative	-.47
	good-natured	-.26	good-natured	-.36
Q-Sort	emphasizes being with others	-.30	emphasizes being with others	-.50
	is talkative	-.24	is talkative	-.43
	behaves in assertive fashion	-.24	behaves in assertive fashion	-.35
	has a rapid personal tempo	-.17	has a rapid personal tempo	-.41
	is facially and/or gesturally expressive	-.28	is facially and/or gesturally expressive	-.29
Introversion	Men		Women	
ACL	quiet	.24	quiet	.42
	reserved	.24	reserved	.40
	aloof	.19	aloof	.36
	retiring	.20	retiring	.35
	inhibited	.19	inhibited	.33
	shy	.17	shy	.29
Q-Sort	keeps people at distance	.28	keeps people at distance	.48
	tends to ruminate and have preoccupying thoughts	.27	tends to ruminate and have preoccupying thoughts	.30
	reluctant to commit self to definite course of action	.20	reluctant to commit self to definite course of action	.41
	tends toward overcontrol of needs	.28	tends toward overcontrol of needs	.48
	is introspective and concerned with self as an object	.31	is introspective and concerned with self as an object	.17

CONTINUED

ANOVAs by type were significant for all CPI 434 scales and for 24 of the LSI scales. The authors also reported a hybrid analysis that incorporated both qualitative and quantitative differences. They compared all 16 types with respect to the scales on which the types scored highest. The result was a profile of the most important scales for each type that revealed how each type was characterized by the CPI 434 assessment or the LSI. The profiles based on each assessment showed that the types all differed from one another. While this kind of analysis does not yield the precision of more sophisticated statistical techniques, its advantage is in avoiding a narrow focus on purely quantitative results that too often obscures important differences in the qualities that distinguish the 16 types.

The self-reported evaluations consistently varied by type. Furthermore, each type could be characterized by a combined CPI 434 and LSI profile that matched the predictions of type theory and Myers' type descriptions. For example, type theory predicts that ETs are likely to be more dominant in social behavior than any other type,

and ENTJs had the highest score on Dominance. The theory also predicts that ISFPs are likely to score lowest on Dominance, which they did. This study found that all 16 types had scores in the normal t-score range on the CPI 434 assessment. From this the authors conclude that the MBTI Step I assessment measures normal and stable personality characteristics for the 16 types. Furthermore, within these normal score ranges, all 16 types can be distinguished from one another in ways predicted by type theory.

Communication

Type theory predicts that people with different preferences should communicate in different ways. Some of the most creative and productive studies in the area of communication and type preferences have been conducted by Thorne and her colleagues. These studies are discussed at some length here because of the richness of their observations about how Extraverts and Introverts, as measured by the MBTI Step I assessment, communicate.

Table 9.27 | **Correlations of rater descriptions based on ACL and California Q-Sort associated with MBTI[®] preferences** *CONTINUED*

Sensing	Men		Women	
ACL	conventional	-.41	conventional	-.43
	conservative	-.38	conservative	-.35
	practical	-.37	simple	-.33
	interests narrow	-.29	interests narrow	-.31
	simple	-.28	contented	-.16
	natural	-.27	shallow	-.25
	contented	-.27	natural	-.23
	moderate	-.21	practical	-.20
	commonplace	-.20	moderate	-.18
Q-Sort	favors conservative values	-.41	favors conservative values	-.39
	judges self and others in conventional terms	-.39	judges self and others in conventional terms	-.38
	uncomfortable with uncertainty and complexities	-.31	uncomfortable with uncertainty and complexities	-.26
	has a clear-cut internally consistent personality	-.31	has a clear-cut internally consistent personality	-.22
	subjectively unaware of self-concern, feels satisfied with self	-.29	subjectively unaware of self-concern, feels satisfied with self	-.20
Intuition	Men		Women	
ACL	original	.22	original	.22
	artistic	.20	artistic	.20
	imaginative	.25	imaginative	.25
	ingenious	.32	ingenious	.22
	complicated	.31	complicated	.27
	unconventional	.26	unconventional	.20
	reflective	.24	reflective	.19
	curious	.22	curious	.27
	individualistic	.21	individualistic	.31
	interests wide	.20	interests wide	.21
Q-Sort	associates to ideas in unusual ways	.43	associates to ideas in unusual ways	.36
	rebellious and nonconforming	.36	high degree of intellectual capacity	.32
	values intellectual/cognitive matters	.25	unpredictable in behavior and attitudes	.32
	high degree of intellectual capacity	.24	rebellious and nonconforming	.31
	unpredictable in behavior and attitudes	.24	values intellectual/cognitive matters	.31

CONTINUED

Thorne (1987) asked undergraduate women to chat with a stranger of the same gender. When Extraverts were paired with other Extraverts they served as catalysts for mutual sociability. They explored a broader range of topics, looked for a variety of ways they were similar to one another, and discussed more pleasurable topics. Introverts paired with other Introverts tended not to look for common ground with one another beyond their similarity as students, more often hedged their answers to one another, and had problem-oriented discussions.

Thorne, Shapiro, Cardilla, Korobov, and Nelson (2009) examined how a stressful experience is collaboratively managed by discussing it in the context of a close

relationship. Participants were 50 pairs of same-sex friends, chosen based on their responses to 10 items from the E-I scale (Form M). The participants were placed into dyads composed of pairs of Extraverts, pairs of Introverts, and mixed dyads. The three kinds of dyads did not differ in friendship closeness or length of friendship. The stressful situation was created by asking participants to engage in conversation while being observed and recorded. A variety of behaviors were coded that related to how the participants communicated about being "onstage." Comments about the situation emerged consistently earlier for pairs of Extraverts than for pairs of Introverts or mixed dyads.

Table 9.27 | **Correlations of rater descriptions based on ACL and California Q-Sort associated with MBTI® preferences** *CONTINUED*

Thinking	Men		Women	
ACL	conventional	–.27	hard-hearted	–.32
	conservative	–.25	fault-finding	–.30
	moderate	–.23	logical	–.27
	interests narrow	–.21	ambitious	–.26
	steady	–.21	opinionated	–.26
			severe	–.26
			cold	–.25
			hostile	–.24
Q-Sort	prides self on being objective	–.32	prides self on being objective	–.32
	rational	–.36	critical, skeptical, not easily impressed	–.30
	judges self and others in conventional terms	–.29	basically distrustful of people, questions their motives	–.30
	favors conservative values	–.25		
Feeling	Men		Women	
ACL	despondent	.27	trusting	.31
	artistic	.26	affectionate	.30
	original	.26	pleasant	.30
			sympathetic	.26
Q-Sort			soft-hearted	.22
	enjoys aesthetic impressions	.27	has warmth, capacity for close relationships	.35
	tends to be rebellious	.20	tends to arouse acceptance in people	.30
	has insight into motive and character	.19	is personally charming	.24

CONTINUED

The first onstage comments emerged 30 words into the conversation for pairs of Extraverts, whereas latencies were over five times longer for mixed dyads and almost seven times longer for pairs of Introverts. However, because of considerable variability in the latencies, these differences were not statistically significant ($p = .20$).

Pairs of Extraverts produced a greater prevalence of diverting communication than did pairs of Introverts or mixed dyads. In mixed dyads, the most prevalent pattern was that the Extravert usually initiated and ended the onstage episode. Communication designed to soothe or comfort was more prevalent in mixed than other dyads, and joking was more prevalent in pairs of Introverts than in other dyads; however, these patterns were not statistically significant. When participants in Extravert dyads were interviewed later they stated that acknowledging a feeling of awkwardness was sufficient to move them into a regular conversation. Asked about joking communication as a means of dealing with the discomfort, half of the Extravert dyads attributed the joking to what the friends usually do with each other, while the other half attributed the joking to feeling that the situation was silly. On the other hand, Introverts

usually attributed their joking to their experience of feeling discomforted, awkward, or nervous.

Also of interest in the study was whether mixed dyads would evidence “contagion,” or the tendency to adopt the other’s behaviors in an effort to achieve common ground and thus provide comfort. However, the contagion theory was not supported. In mixed dyads, instead of Introverts behaving in a more extraverted fashion and vice versa, a division of labor emerged in which the Extravert tended to initiate and end the communication. The extraverted partner was assumed to be the expert in moving the conversation along, a role that the Extravert did not always relish but still performed, according to post-interaction interviews. Extraverted and introverted friends were seen to be sensitive to each other’s inclinations and developed routines that sustained, rather than reduced, their differences. The authors concluded that “given the brevity of such episodes [discussions about being onstage], the fact that significant differences were found for dyad personality is quite remarkable” (p. 641).

Another study, by Thorne, Korobov, and Morgan (2007), focused on E–I differences in storytelling among friends.

Table 9.27 | **Correlations of rater descriptions based on ACL and California Q-Sort associated with MBTI® preferences** *CONTINUED*

Judging	Men		Women	
ACL	conservative	–.39	conservative	–.39
	conventional	–.29	conventional	–.37
	industrious	–.26	industrious	–.33
	dependable	–.22	dependable	–.32
	organized	–.24	organized	–.24
Q-Sort	favors conservative values	–.34	favors conservative values	–.34
	is fastidious	–.28	is fastidious	–.37
	tends toward overcontrol of needs	–.23	tends toward overcontrol of needs	–.38
	judges self and others in conventional terms	–.32	judges self and others in conventional terms	–.25
	is moralistic	–.25	is moralistic	–.30
Perceiving	Men		Women	
ACL	disorderly	.28	disorderly	.25
	careless	.30	careless	.29
	adventurous	.26	adventurous	.26
	pleasure-seeking	.21	pleasure-seeking	.31
	lazy	.29	lazy	.22
	changeable	.23	changeable	.27
	restless	.23	restless	.23
Q-Sort	tends to be rebellious	.40	tends to be rebellious	.32
	enjoys sensuous experiences	.32	enjoys sensuous experiences	.32
	is unpredictable in behavior and attitudes	.22	is unpredictable in behavior and attitudes	.40
	is self-indulgent	.18	is self-indulgent	.44
	various needs tend toward relatively direct expression	.24	various needs tend toward relatively direct expression	.25

Note: Men, $n = 244$; correlations of .13 are significant at $p < .05$, .16 at $p < .01$, .21 at $p < .001$. Women, $n = 186$; correlations of .14 are significant at $p < .05$, .19 at $p < .01$, and .24 at $p < .001$. *Adjective Check List* and *California Q-Sort* ratings are based on observations by at least five IPAR staff members.

Source: Gough (1981). Used with permission.

The sample consisted of college students placed into 39 same-sex pairs of Extraverts or pairs of Introverts. Extravert dyads used significantly more words than did Introvert dyads, by an average difference of over 100 words. Extravert dyads told a significantly greater proportion of stories about events that had happened during the college years, and Introvert dyads told a significantly greater proportion of stories about precollege times. Extraverted friends more often told stories about romantic relationships, nearly three times more often than the Introverts, whereas the latter told significantly more stories about family and hometown. Introverted dyads specialized in embedding the story in an ongoing topic, in contrast to Extravert dyads, who more often just changed the topic, departing from the theme of the ongoing dialogue and casting their sudden stories as news or gossip. The Introverts advanced their stories alone, making little explicit effort to invite the listener to co-construct the plot. Extraverted storytelling,

in contrast, showed more listener engagement in constructing the plot; Extraverted friends essentially invited and asserted claims on each other's experience. The dyads did not differ in the number of stories produced, in how long it took for the first story to emerge in the conversations, or in whether stories were about "me" or "we" experiences.

Extraverts and Introverts exhibit very different communication patterns depending on who they are talking to.

Finally, another study of E–I dyads, by Nelson, Thorne, and Shapiro (2011), examined the tendency to participate and take pleasure in social interaction. The authors examined the characteristic roles that each friend played in the various domains of friendship: locale (where the friendship generally took place—away from home or close to home), energy (who energized the friendship

and who steadied the friendship), dialogue (who did most of the talking and who did most of the listening), intimacy (who usually disclosed personal information and who usually contained personal information), outsiders (attitudes toward peers or outsiders), and content (the primary content of the conversations). Participants were 66 pairs of same-sex friends consisting of pairs of Introverts, pairs of Extraverts, and mixed pairs. Friends in matched dyads reported playing similar roles in almost all domains, indicating that role complementarity was quite rare for matched dyads. The sole exception to this mutual role reinforcement in matched dyads occurred in the energy domain, where almost two-thirds of the

friends in Introvert pairs took on different roles. Friends in mixed dyads reported playing different roles in four of the six domains: locale, energy, outsiders, and dialogue. A strong exception to the general trend toward role complementarity in mixed dyads was found in the content domain, where 81% of friends in mixed dyads reported taking on the same role. There was a significant difference in mixed dyads in the domain of intimacy, which was mainly driven by Extraverts, who showed a definite preference for being expressive (80%) rather than contained (20%).⁵ whereas Introverts in mixed dyads were more evenly split.

Table 9.28 | ACL adjectives with the 15 largest percentage differences in endorsement rate between individuals with opposite MBTI® preferences

Extraversion (n = 654)		Introversion (n = 896)		Sensing (n = 1,049)		Intuition (n = 501)	
	%		%		%		%
sociable	26.9	shy	22.0	conservative	6.8	imaginative	30.2
outgoing	23.6	quiet	20.3	obedient	4.7	artistic	29.2
cheerful	21.4	reserved	19.7	dull	4.0	interests wide	22.4
talkative	20.6	silent	14.3	conventional	4.0	curious	21.6
active	19.4	withdrawn	12.0	average	3.5	adventurous	20.5
outspoken	19.2	timid	10.8	stern	3.2	intelligent	20.0
vibrant	17.8	cautious	10.5	cautious	2.7	individualistic	19.2
energetic	15.9	inhibited	10.1	prejudiced	2.2	humorous	18.4
animated	15.8	serious	8.7	annoyed	1.8	idealistic	18.3
self-confident	15.5	pessimistic	8.4	dependent	1.7	clever	18.1
confident	15.1	anxious	8.3	interests narrow	1.4	bright	17.6
charming	15.0	calm	8.3	intolerant	1.3	resourceful	17.4
enterprising	14.6	hesitant	8.2	pessimistic	1.3	inventive	17.4
spontaneous	14.6	conservative	8.1	serious	1.3	sympathetic	17.2
enthusiastic	14.5	plain	7.5	thoughtless	1.2	original	16.9
Thinking (n = 858)		Feeling (n = 692)		Judging (n = 777)		Perceiving (n = 773)	
	%		%		%		%
rational	9.9	sensitive	22.4	organized	14.2	spontaneous	16.7
ingenious	9.6	affectionate	21.1	cautious	10.5	impulsive	11.0
methodical	9.4	emotional	18.4	punctual	10.4	humorous	9.9
logical	9.0	soft-hearted	17.8	planful	9.3	adventurous	9.2
self-confident	7.8	sympathetic	16.4	meticulous	7.7	individualistic	8.8
sarcastic	7.2	warm	16.3	methodical	7.5	rebellious	8.7
formal	7.2	kind	16.1	obedient	7.4	original	8.3
sharp-witted	7.2	sentimental	14.9	thorough	6.8	imaginative	8.0
inventive	7.1	helpful	14.1	serious	6.7	natural	7.9
realistic	6.9	feminine	13.8	conservative	6.4	interests wide	7.5
stern	6.5	forgiving	13.3	stable	6.4	versatile	7.4
demanding	6.4	nice	13.0	responsible	6.3	informal	6.8
intelligent	6.3	gentle	12.8	careful	5.6	inventive	6.8
brainy	6.3	friendly	12.8	clear-thinking	5.1	outgoing	6.8
shrewd	6.3	talkative	12.6	precise	5.1	idealistic	6.8

Note: Portion of the global sample that completed the ACL, n = 1,550.

5. This distinction was based on their coding of the conversations; it does not refer explicitly to the Expressive–Contained facet of the Step II assessment.

A series of studies of the relationship of MBTI preferences and types to various aspects of communication was undertaken by Opt and Loffredo. In one study, Opt and Loffredo (2000) examined communication apprehension in four contexts: group, dyad, meeting, and public. People high in communication apprehension tend to avoid communication more often and engage in less verbal behavior than do those low in apprehension. Introverts and Sensing types scored significantly higher in overall communication apprehension across all four contexts than did Extraverts or Intuitive types. Feeling types scored significantly higher in communication apprehension in the public context than did Thinking types. The authors hypothesized that Feeling types might find the public situation more anxiety producing because of the lack of interpersonal involvement with the audience. There were no differences for Judging versus Perceiving. The authors pointed out that from a type perspective “communication apprehension should not be treated as a ‘problem’ to be overcome but viewed simply as a preference not to communicate” (p. 557).

These same researchers (Opt & Loffredo, 2006) also looked at communication from the point of view of receiver apprehension, which is anxiety about being a listener. Intuitive types scored significantly higher than did Sensing types on a measure of receiver apprehension. However, gender was a confounding variable. Intuitive women evidenced more receiver apprehension than did Sensing women. There were no S–N differences for men and no overall E–I, T–F, or J–P differences. The authors hypothesized that Intuitive types are more apprehensive about listening to information because it encroaches on their imagination, whereas Sensing types accept the information as reality. They wondered if Intuitive types would be more receptive to abstract information. It is telling from the perspective of type theory that in a study of listening, the only dimension that showed any significant differences was the perceiving process.

Loffredo and Opt (2006) also examined a different aspect of communication—argumentativeness. Participants who preferred Extraversion, Intuition, and Thinking scored significantly higher than their opposites on this measure. The T–F difference was the strongest predictor of level of argumentativeness. In terms of process pairs, NTs scored higher than other pairs. There was also a significant difference among whole types, with ENTJs scoring highest.

These researchers also studied type differences on communicator image (Opt & Loffredo, 2003), which is an overall self-evaluation of an individual’s perception of whether he or she is a good communicator. Individuals preferring Extraversion scored significantly higher on communicator image than those preferring Introversion. There were no significant differences between communicator image and the S–N, T–F, or J–P preference pairs.

Table 9.29 | **Communication wants by MBTI® preference**

Extraversion Want to focus on a wide range of topics	Introversion Want to focus on one problem
Sensing Want practical, realistic messages Want well-thought-out, detailed plans	Intuition Want imaginative, creative messages Want challenges and possibilities
Thinking No significant relationships	Feeling No significant relationships
Judging Want supervisor who “doesn’t surprise them” Want detailed action plans “with time tables” Want focus on “completing planned projects”	Perceiving Want supervisor who is “spontaneous” Want “spontaneous” and “flexible” plans Want projects to be flexible and open to modification

Note: Convenience sample of adults, $N = 108$.

Source: Hermon (1993), p. 92. Used with permission.

Finally, Loffredo, Opt, and Harrington (2008) looked at the relationship between MBTI preferences and communicator style using a discriminant function analysis. The strongest relationship was between E–I preferences and communication style variables, although S–N and T–F differences were also significant. There were no significant differences for J–P preferences. The best predictor of whether an individual was classified accurately as having a preference for Extraversion or Introversion were the communication style variables: dominant, open, contentious/argumentative, and communicator image. The variables most related to predicting T–F preferences were friendly, precise, dominant, open, contentious, and attentive. S–N preferences were related to the variables dramatic, impression leaving, animated/expressive, and communicator image.

In a study by Hermon (1993), significant differences were found in preferred communication style by type preference based on a sample of 108 adults (primarily construction workers and technicians). Table 9.29 summarizes Hermon’s statistically significant findings, which correspond quite well to predictions from type theory. Carskadon (1979b) reported on 40 psychology undergraduates asked to give a five-minute talk before six judges, with only five minutes’ notice. Extraverts stood closer to judges, had fewer seconds of silence, and after the talk remembered more names of the judges to whom they were introduced at the onset of the experiment.

Conflict

Another area in which type has been studied is conflict. Table 9.30 shows mean differences between individuals at opposite poles of the MBTI preference scales on the scales of the *Thomas-Kilmann Conflict Mode Instrument* (TKI® assessment). The TKI assessment measures preferences for five different styles, or modes, of handling conflict: competing, collaborating, compromising, avoiding, and accommodating (Thomas & Kilmann, 1974; see appendix F for a description of the TKI scales). The sample consisted of 52,207 individuals who had taken the MBTI assessment and had agreed to participate in additional research. Extraverts were more likely to use collaborating. Those with preferences for Introversion, Sensing, and Feeling were more likely to use avoiding than were those who preferred Extraversion, Intuition, and Thinking. Thinking types preferred competing. There were no differences at the criterion level ($d \geq .20$) on the J–P scale, and from the TKI perspective there were no significant differences for use of the compromising mode.

Table 9.31 shows, in rank order, the means and standard deviations of the raw scores of the 16 types for each of the TKI conflict modes in this same large sample ($N = 52,207$). The overall ANOVA was significant for all five of the conflict modes. The differences among the types were greatest for competing, accommodating, and avoiding and somewhat smaller for collaborating and compromising. The Extraverted Thinking types ranked highest on competing and the Introverted Feeling types ranked lowest. All the Thinking types scored higher than the Feeling types on this conflict mode. The E–I pattern was more mixed for competing. The results on accommodating show the four Introverted Feeling

types scoring highest and the four Extraverted Thinking types scoring lowest. The pattern for avoiding was similar, with the four Introverted Feeling types scoring highest and three of the four Extraverted Thinking types scoring lowest; the exception was that ENFJs scored lower than ESTPs. On collaborating and compromising, Extraverts scored higher than Introverts, although the ranking was not strictly in this order, as INTJs scored higher than ESFJs and ESFPs on collaborating and INFJs and ISTJs scored higher than ESTPs and ESFPs on compromising. Comparing tables 9.30 and 9.31 suggests that combinations of preferences or whole types are more predictive of preferred conflict mode than are the individual preferences.

Early research on personality type and conflict (Kilmann & Thomas, 1975; Mills, Robey, & Smith, 1985) found, in general, that Thinking types tended to be assertive and competitive in conflict situations, Feeling types favored cooperation and accommodation, and Introverts avoided conflict. Later research (Percival, Smitheram, & Kelly, 1992) found that combinations of preferences showed more significant patterns of relationships with the TKI conflict modes. For example, ITPs scored higher on compromising, EFJs higher on collaborating, and IFJs higher on accommodating.

In a study of the effect of competition or cooperation on outcomes of conflict or negotiations, Insko et al. (2001) used the S–N scale of the MBTI assessment and the Openness–Intellect scale of the Big Five Inventory as measures of abstraction. Abstraction was believed to influence trust by allowing for a long-term perspective that reduced competition. The authors found that individuals became more competitive as their level of abstraction, or their ability to see the big picture,

Table 9.30 | MBTI® opposite-preference differences in use of TKI® conflict modes

Conflict mode	Extraversion		Introversion		Cohen's <i>d</i>	Summary	Sensing		Intuition		Cohen's <i>d</i>	Summary
	<i>M</i>	<i>SD</i>	<i>M</i>	<i>SD</i>			<i>M</i>	<i>SD</i>	<i>M</i>	<i>SD</i>		
Competing	4.93	2.89	4.64	2.97	–0.10		4.79	2.96	4.84	2.87	0.02	
Collaborating	6.56	2.16	5.81	2.27	–0.34	E	6.11	2.20	6.47	2.27	0.16	
Compromising	7.48	2.17	7.20	2.17	–0.13		7.37	2.15	7.36	2.21	–0.01	
Avoiding	5.74	2.37	6.87	2.44	0.47	I	6.44	2.44	5.84	2.45	–0.25	S
Accommodating	5.29	2.27	5.47	2.39	0.08		5.28	2.34	5.50	2.29	0.09	

Conflict mode	Thinking		Feeling		Cohen's <i>d</i>	Summary	Judging		Perceiving		Cohen's <i>d</i>	Summary
	<i>M</i>	<i>SD</i>	<i>M</i>	<i>SD</i>			<i>M</i>	<i>SD</i>	<i>M</i>	<i>SD</i>		
Competing	5.34	2.87	3.47	2.63	–0.67	T	4.85	2.94	4.74	2.90	–0.04	
Collaborating	6.35	2.23	5.98	2.22	–0.17		6.26	2.23	6.22	2.25	–0.02	
Compromising	7.38	2.20	7.32	2.11	–0.03		7.45	2.17	7.23	2.17	–0.10	
Avoiding	6.04	2.45	6.66	2.44	0.25	F	6.21	2.47	6.22	2.46	0.00	
Accommodating	4.89	2.17	6.57	2.24	0.76	F	5.22	2.31	5.58	2.32	0.16	

Note: CPP commercial database, $N = 52,207$. Summary column shows the preference for which $d \geq .20$ in either direction.

Table 9.31 | Rank order of MBTI® types by mean score in use of TKI® conflict modes

Competing			Collaborating			Compromising			Avoiding			Accommodating		
Type	M	SD	Type	M	SD	Type	M	SD	Type	M	SD	Type	M	SD
ENTP	5.61	2.68	ENTJ	6.97	2.16	ENFJ	7.69	2.09	ISFJ	7.70	2.23	ISFP	6.95	2.24
ENTJ	5.60	2.68	ENTP	6.84	2.19	ESTJ	7.59	2.19	ISFP	7.65	2.25	ISFJ	6.83	2.22
ESTP	5.45	2.89	ESTJ	6.62	2.11	ENTJ	7.58	2.25	INFP	7.09	2.47	INFP	6.82	2.21
ESTJ	5.41	2.88	ENFJ	6.47	2.17	ESFJ	7.44	2.05	INFJ	7.03	2.38	INFJ	6.64	2.22
INTJ	5.35	2.81	ESTP	6.44	2.16	ENFP	7.41	2.13	ISTP	7.00	2.37	ENFP	6.48	2.27
INTP	5.31	2.88	ENFP	6.41	2.19	ENTP	7.38	2.24	ISTJ	6.80	2.43	ESFP	6.46	2.23
ISTP	5.09	2.86	INTJ	6.29	2.36	INFJ	7.37	2.14	ESFP	6.73	2.39	ESFJ	6.45	2.21
ISTJ	5.05	3.00	ESFJ	6.11	2.11	ISTJ	7.34	2.17	ESFJ	6.54	2.32	ENFJ	6.22	2.25
ENFJ	3.84	2.71	INTP	6.10	2.29	ESTP	7.33	2.14	INTP	6.41	2.44	ISTP	5.17	2.28
ENFP	3.78	2.72	ESFP	6.00	2.14	INTJ	7.21	2.28	INTJ	6.18	2.48	INTP	5.15	2.21
ESFP	3.64	2.68	ISTJ	5.86	2.22	ISFJ	7.19	2.04	ENFP	5.91	2.37	INTJ	4.97	2.20
ESFJ	3.45	2.58	INFP	5.67	2.36	ESFP	7.18	2.09	ESTP	5.86	2.30	ISTJ	4.95	2.30
INFP	3.38	2.63	ISTP	5.66	2.24	ISTP	7.08	2.16	ENFJ	5.78	2.35	ESTP	4.92	2.13
INFJ	3.35	2.59	INFJ	5.61	2.31	INFP	7.04	2.20	ESTJ	5.68	2.35	ENTP	4.92	2.03
ISFP	2.99	2.36	ISFP	5.44	2.13	INTP	7.03	2.20	ENTP	5.25	2.33	ENTJ	4.75	2.02
ISFJ	2.95	2.46	ISFJ	5.33	2.17	ISFP	6.97	2.03	ENTJ	5.10	2.26	ESTJ	4.69	2.12

Note: CPP commercial database, N = 52,207.

decreased. While there were many other significant results, the relevant finding here is that the effect on trust was greater for the S–N scale than for the Openness–Intellect scale from the Big Five Inventory. The authors hypothesized that the forced-choice items on the S–N scale were an advantage because they required more careful judgments than did the 7-point ratings used by the Openness–Intellect scale.

Positive Psychology

In the early part of the 21st century, a “new wave” of psychological theory and research appeared under the rubric of positive psychology. While this approach was welcomed as a reaction to the primarily medical and deficit models that had previously characterized psychological research and practice, scant attention was given to the predecessors of this approach, which included, among others, the theories of Gordon Allport, Abraham Maslow, Carl Rogers, Carl Jung, and, of course, Isabel Myers. Myers’ work on type, and particularly on type development (Myers et al., 2009), preceded this new wave by decades. Type, like positive psychology, adopts a focus on the positive, adaptive aspects of personality, which are seen as arising from inborn predispositions that are developed through interaction with the environment. The Jung–Myers theory and positive psychology also share a belief in the universality of personality descriptors, which the MBTI Global Step I and Step II assessments validate. An entire issue of the *Bulletin of Psychological Type* (volume 34, no. 3) was devoted to examining the correspondence between type theory and positive psychology. Articles by Holm, Geyer, Soper

and Allen, and Martin discussed how Myers’ approach is philosophically and conceptually congruent with much of positive psychology. In the same issue, Pearman and Huszczo briefly introduced some of the research relating the two approaches. Some of the variables of interest to researchers in positive psychology include character strengths, optimism, happiness, life satisfaction, well-being, purpose in life, and actualization. The relationship of type to these variables is discussed below.

Hammer (2005b) investigated the relationship between MBTI preferences and whole types and character strengths in a national random sample of 697 US adults who completed the MBTI Step I assessment and the *Adjective Check List* (ACL). Using definitions found in Peterson and Seligman (2004), he created scales for each of the six character strengths from the list of 300 ACL adjectives. The six character strengths are Courage (bravery, integrity, vitality), Humanity (love, kindness, social intelligence), Temperance (forgiveness, mercy, self-regulation), Transcendence (spirituality, appreciation of beauty), Justice (fairness, citizenship), and Wisdom (open-mindedness, perspective). Extraverts scored significantly higher than Introverts on five of the strengths, the exception being Temperance, on which the Introverts scored higher, although the difference did not reach significance. Effect sizes for the E–I differences ranged from .18 for Wisdom to .31 for Transcendence. Likewise, Intuitive types scored significantly higher than Sensing types on all the strengths except Temperance, on which the Sensing types scored higher, although again the difference did not reach significance. Effect sizes ranged from .08 for Justice to .34 for Wisdom. Feeling

types scored significantly higher on Humanity, Justice, Temperance, and Transcendence, while Thinking types scored higher on Courage and Wisdom, although the differences on the latter were not significant. Effect sizes ranged from .08 for Courage to .27 for Humanity. The only significant J–P difference was on Transcendence, with Perceiving types scoring significantly higher.

Whole-type differences were also examined. Six one-way ANOVAs were performed with the 16 types as the independent variables. Table 9.32 shows the ranking of the 16 types for each of the six character strengths, as well as the eta (an index of the magnitude of the overall differences) for each of the analyses. Four of these analyses were significant at $p < .001$; the exceptions were Justice ($p < .03$) and Temperance ($p < .04$). The differences among the 16 types on Transcendence, Wisdom, Humanity, and Courage resulted in large effect sizes (for eta, large effect sizes = .40). On Humanity, five of the Feeling types had the highest scores and five of the Thinking types the lowest. For Courage, six of the top eight types were Extraverted types, although INTJs scored second highest and INTPs eighth. For Justice, the top half included seven Extraverted types, but with INFJs third. For Temperance, the four top types were Judging types, although another Judging type, ESTJ, scored 14th. For Transcendence, the top three types were Extraverted Intuitive types and six of the top eight were Intuitive types. For Wisdom, the top half was dominated by Intuitive types, although ESTJs were eighth. While this table shows that the individual preferences do influence self-descriptions associated with the six character strengths, there are some whole types that defy the pattern.

Choong and Britton (2007) looked at the relationship between 24 character strengths measured by the *Values in Action Inventory of Strengths* (VIA-IS) in a sample of adults ($n = 98$). Sixteen of the character strengths were related to one or more of the MBTI preference pairs or combinations of preferences: Vitality was related to Extraversion, love of learning to Introversion; integrity and fairness were related to Sensing and creativity to Intuition; open-mindedness was related to Thinking and love to Feeling; persistence was related to Judging, while none of the character strengths were associated with Perceiving. Love was the only character strength that was related to more than one preference pair, namely, to Extraversion and to Feeling. There were four character strengths—curiosity, perspective, bravery, and hope—that only showed differences for combinations of preferences and not for individual preferences. The authors concluded that the combination of individual preferences produces a dynamically unique type profile that is more than just the sum of the parts.

Munro, Chilimani, and O'Neill (2012) also used the VIA-IS measure, in a South African sample of 69 university students being trained as peer educators, 82.6% of whom identified themselves as black Africans. Extraverts scored significantly higher than Introverts on the character strengths of curiosity and humor. Intuitive types scored significantly higher than Sensing types on appreciation of beauty and excellence. Feeling types scored significantly higher than Thinking types on the strength of capacity to love and be loved; Feeling types also scored higher on gratitude than did those with a Thinking preference.

Table 9.32 | Rank order of MBTI® types by mean score on the six ACL character strengths

Courage		Humanity		Temperance		Transcendence		Justice		Wisdom	
ENTP	19.00	ENFJ	15.78	INTJ	2.00	ENFP	3.46	ENFP	2.78	ENTP	8.57
INTJ	18.29	ENFP	15.12	ENFJ	1.89	ENTP	3.43	ENTP	2.71	ENFJ	7.61
ENFP	17.06	ESFJ	14.16	ESFJ	1.74	ENFJ	3.06	ENFJ	2.61	ENFP	7.48
ENFJ	17.06	INFJ	13.50	INFJ	1.56	ESFP	2.92	INFJ	2.56	INFJ	7.11
ESTJ	15.74	ESFP	13.46	ESFP	1.50	INTJ	2.86	ESFP	2.54	INTP	7.08
ESTP	15.17	ESTJ	13.40	ISFP	1.38	INFJ	2.83	ESFJ	2.49	INTJ	7.00
ENTJ	14.75	ENTJ	13.25	INFP	1.35	ENTJ	2.83	ENTJ	2.42	ENTJ	6.83
INTP	14.25	INTJ	13.14	ISFJ	1.32	ESTJ	2.69	ESTJ	2.33	ESTJ	6.02
INFJ	13.89	ISFJ	12.85	ISTJ	1.32	INTP	2.42	INFP	2.26	INFP	5.89
ESFP	13.83	INFP	12.07	ISTP	1.11	ESTP	2.38	ISFJ	2.26	ESFP	5.31
ESFJ	12.75	ISFP	12.04	ENTJ	0.92	ESFJ	2.20	INTP	2.04	ESTP	5.28
ISTJ	11.72	ENTP	10.57	ENFP	0.88	INFP	2.02	ISFP	2.04	ESFJ	4.78
INFP	10.83	ESTP	10.48	INTP	0.83	ISFP	1.73	ESTP	2.03	ISTJ	4.60
ISTP	10.69	INTP	9.38	ESTJ	0.48	ISFJ	1.54	ISTP	2.03	ISFJ	4.30
ISFJ	10.41	ISTJ	9.26	ESTP	0.41	ISTJ	1.11	ISTJ	1.86	ISTP	4.11
ISFP	9.25	ISTP	8.00	ENTP	0.29	ISTP	0.94	INTJ	1.71	ISFP	3.89
Eta		.38		.19		.41		.20		.40	

Note: Portion of Form M US national sample that completed the ACL, $n = 697$.

Table 9.33 | Rank order of MBTI® types by mean score on life satisfaction, happiness, and optimism

Life satisfaction (<i>n</i> = 15,879)			Happiness (<i>n</i> = 15,824)			Optimism (<i>n</i> = 697)		
Type	<i>M</i>	<i>SD</i>	Type	<i>M</i>	<i>SD</i>	Type	<i>M</i>	<i>SD</i>
ESTJ	7.33	1.84	ESTJ	3.11	0.59	ESFJ	2.75	0.49
ESFJ	7.04	2.01	ENTJ	3.11	0.56	ENFP	2.69	0.55
ENTJ	7.03	1.81	ESFJ	3.10	0.60	ENTJ	2.58	0.52
ENFJ	6.99	1.85	ESFP	3.08	0.61	ENTP	2.57	0.79
ENTP	6.93	1.87	ENFJ	3.07	0.63	ESTP	2.55	0.74
ESFP	6.91	2.03	ENTP	3.06	0.62	ENFJ	2.50	0.79
ESTP	6.79	2.00	ESTP	3.03	0.64	INTP	2.42	0.78
ENFP	6.79	1.97	ENFP	3.01	0.64	ESTJ	2.39	0.80
INTJ	6.62	2.04	ISTJ	2.89	0.64	ESFP	2.36	0.74
ISTJ	6.52	2.01	INTJ	2.89	0.64	INTJ	2.29	1.11
ISTP	6.40	2.06	ISTP	2.89	0.67	INFJ	2.28	0.58
ISFJ	6.38	2.09	ISFJ	2.88	0.67	INFP	2.27	0.69
INTP	6.37	2.01	ISFP	2.88	0.64	ISFJ	2.18	0.84
INFJ	6.34	2.11	INFJ	2.84	0.69	ISTP	2.17	0.88
ISFP	6.31	2.11	INTP	2.83	0.68	ISFP	2.10	0.88
INFP	6.16	2.08	INFP	2.80	0.67	ISTJ	2.02	0.90

Note: For "life satisfaction" portion of the global sample, *n* = 15,879. For "happiness" portion of the global sample, *n* = 15,824. For "optimism" portion of Form M national sample, *n* = 697. Differences across the 16 types are significant at *p* < .001 for all three variables.

Happiness is another variable studied in positive psychology. A subset of participants in the MBTI global sample (*n* = 15,824) were asked to indicate, using a 4-point scale from "not at all happy" to "very happy," their general level of happiness. These results are shown in table 9.33. Extraverts were happier than Introverts (*d* = .31). The eight Extraverted types ranked higher on happiness than the eight Introverted types. The types with the highest scores on happiness were ESTJ and ENTJ; the lowest were INTP and INFP.

Research demonstrates the relationship between type results and the topics studied by positive psychology, including character strengths, optimism, life satisfaction, and happiness.

Huszczo and Endres (2013) studied college students' (*n* = 208) reasons for engaging in 12 "happiness activities." There were significant type differences, all between Thinking and Feeling types, on 7 of the 12 activities. Feeling types, compared to Thinking types, reported significantly more use of 6 activities because they enjoyed the activity or found it interesting or challenging: expressing gratitude, cultivating optimism, practicing kindness, nurturing relationships, savoring joy, and practicing religion/spirituality. They also reported more use of 5 activities because it seemed natural to them: practicing kindness, nurturing relationships, learning to forgive, savoring joy, practicing religion/spirituality. Feeling types engaged more in 3 activities because they valued the behavior and would keep doing

it even if it were not necessarily enjoyable: nurturing relationships, learning to forgive, practicing religion/spirituality. They also engaged in 1 behavior because not to do so would result in their feeling guilty or anxious: learning to forgive. Thinking types were significantly more likely than Feeling types to commit to goals because they would feel guilty otherwise.

There was also an interaction between gender and personality. An ANOVA revealed a significant interaction for 5 activities, namely, how individuals express gratitude, practice acts of kindness, nurture social relationships, savor life's joys, and practice religion/spirituality. The significant differences were primarily between female Feeling types and male Thinking types, with the former rating "enjoyment" significantly higher as a reason for using 4 of the activities: expressing gratitude, practicing kindness, nurturing relationships, and savoring joy. Female Feeling types also rated engaging in a behavior because it felt natural significantly higher than did male Thinking types for using 3 behaviors: practicing kindness, nurturing relationships, and savoring joy. Male Feeling types rated wanting to avoid feelings of guilt significantly higher as a reason for savoring life's joys than did female Feeling types. Male Feeling types were more likely than male Thinking types to engage in 1 activity because it felt natural, was enjoyable, and was valued: practicing religion/spirituality.

In another analysis, a subset of participants in the MBTI global sample (*n* = 15,879) were asked to rate, using a 4-point scale from "completely dissatisfied" to

"completely satisfied," their overall satisfaction with their life. These results are shown in table 9.33. Extraverts were significantly more satisfied with their life than were Introverts ($d = .29$). The four types most satisfied were the four Extraverted Judging types (ESTJ, ESFJ, ENTJ, and ENFJ) and the four least satisfied were INTP, INFJ, ISFP, and INFP. Myers believed that Introverts and Perceiving types were generally less satisfied than Extraverts and Judging types in any number of areas because Perceiving types tend to perceive external difficulties more fully than do Judging types, which is only natural because they are in the habit of using their perceiving process in the outer world. They assume that there is more to be seen than they have seen so far, that some of those things may not be pleasant or favorable, and they make a point of looking for those things. The E–I preference makes a difference, too. Introverts, having more interest in the world inside their head, tend to be more aware of internal difficulties than are Extraverts. Thus the perceiving Introverts, the IP types, are usually the most conscious of both kinds of difficulties, external and internal. It is important to note that Myers believed this tendency on the part of the IP types could be genuinely useful if it did not discourage them from coping. On the other hand, the judging Extraverts, the EJ types, are likely to see relatively fewer difficulties of either kind and to be accordingly more satisfied.

Another variable receiving attention from positive psychology researchers is optimism. Myers believed that Sensing is often related to a pessimistic attitude and Intuition to an optimistic attitude. In a difficult situation, the tendency of Sensing is to assume that what is will not change. Clinicians (e.g., von Franz, 1971) have noted that dominant Sensing types often find that when intuitions do occur to them, they are negative and pessimistic. (In the dynamics of type theory, the use of Intuition by a dominant Sensing type would be less conscious and differentiated and therefore more likely to yield vague, ungrounded, or negative possibilities.) The more optimistic tendency of Intuitive types is to assume that the situation may be difficult now but some hitherto unseen possibility will turn up. Extraversion and Introversion can also be related to optimism. In theory, Extraverts are more at home in the world, and thus they may also report themselves as more competent to cope with its demands, and therefore more optimistic.

To test these predictions, participants in the Form M US national sample indicated, using a 5-point scale, how optimistic they were. Those with preferences for Extraversion and Intuition, and to a somewhat lesser extent Feeling types, were significantly more optimistic than their opposites. There were no differences on the J–P preference pair. Whole types were also examined. Table 9.33 shows the ranking of the 16 types on optimism. The difference between the 16 types was significant ($p < .001$, $\eta^2 = .31$). Clearly the Extraverted

types scored higher, paralleling the E–I preference finding. However, the top five types on this simple measure of optimism are as follows: an Extraverted Feeling type (ESFJ), the two Extraverted Intuitive types (ENFP, ENTP), an Extraverted Thinking type (ENTJ), and an Extraverted Sensing type (ESTP). This ranking suggests that type dynamics or whole types may be more important in understanding optimism than E, I, S, or N as separate preferences.

Quenk (1966) looked at type differences as part of a study of fantasy and personal outlook in relation to optimism, pessimism, realism, and anxiety. Responses of the 25 male and 32 female participants who recorded daydreams for 10 days showed consistent gender differences. Significant MBTI findings were that optimists were more likely to be Extraverts for both genders; female optimists were more likely to be Intuitive types, and male optimists were more likely to be Thinking types.

Vaughan and Knapp (1963) reported a study of dimensions of pessimism in which a number of assessments, including the MBTI assessment, were administered to 75 male Wesleyan University undergraduates. Pessimism was significantly associated with Introversion and Thinking, confirming the E–I difference found by Quenk but not the male T–F difference.

Harrington and Loffredo (2001b), using a sample of college students, examined differences by MBTI preference on participants' life satisfaction, self-consciousness, and social anxiety, as well as on scales measuring overall and specific domains of personal well-being. Extraverts reported significantly higher scores on overall personal well-being and life satisfaction than did Introverts. However, when differences due to self-consciousness were removed through a MANCOVA analysis, the E–I differences in personal well-being were no longer significant. Harrington and Loffredo hypothesized that because Extraverts are oriented toward others, generally relaxed and confident, and "endowed with an ease of communication and sociability," they are more likely to get their needs met and thus would report significantly higher life satisfaction than would Introverted types. Myers had a somewhat different explanation for the relationship between type and satisfaction, centering on the likelihood that internal and external difficulties would be perceived differently, which is discussed above. Intuitive types reported higher personal well-being scores than did Sensing types. In particular, Intuitive types reported significantly higher scores on autonomy, personal growth, and positive relations on the personal well-being scale than did Sensing types. There were no differences in overall life satisfaction between Intuitive and Sensing types, leading Harrington and Loffredo to suggest that S–N differences are less sensitive to global and more sensitive

to particular dimensions of life satisfaction. Judging types reported significantly higher personal well-being, but only on the purpose in life dimension. Extraverts reported significantly lower scores on social anxiety than Introverts. Sensing types reported significantly higher social anxiety than did Intuitive types. The authors speculate that the need of Sensing types for predictability and explicit signs of commitment in relationships leads them to experience more social anxiety than Intuitive types.

Doerries and Ridley (1998) believed that a preference for Judging or Perceiving would be unrelated to a measure of purpose in life. They used the *Purpose in Life Test*, which measures how strongly an individual had developed a sense of meaning and purpose in life, with 100 college students. The scores of Judging types and Perceiving types showed no difference. This finding is contrary to that of Harrington and Loffredo (2001b) using a different measure of purpose of life, although both used samples of college students.

Schadt (1997) used the MBTI assessment in an eight-week individual and group counseling project that involved 50 midlife women describing themselves as being in personal and career transition. A significant correlation was found between participants' clarity of preference for either E or I and psychosocial well-being and self-esteem.

Creativity

According to type theory, the creation of something completely new or the perception of a unique innovative possibility should be related to a preference for Intuition, which is the mode of perception that is oriented to possibilities and to seeing hitherto unknown patterns. Creativity therefore is expected to be associated primarily with a preference for Intuition and secondarily with a preference for the Perceiving orientation, which supplies curiosity and receptiveness as well as flexibility in adapting to change.

The earliest research with the MBTI assessment and creative people was conducted at the Institute of Personality Assessment and Research (IPAR) by Donald MacKinnon and his colleagues (Gough, 1976, 1981; Hall & MacKinnon, 1969; Helson, 1965, 1968, 1971, 1975; Helson & Crutchfield, 1970; MacKinnon, 1960, 1962a, 1962b, 1965, 1971). The IPAR samples of creative people were selected by peer nomination from professions judged to be creative; thus they represent highly creative people in creative fields. It is useful, therefore, to compare the type distributions of highly creative people with one another, with those of samples of others in creative fields, and with those of the general public.

Type distributions of samples of creative people confirm a preference for Intuition. The field in which they work makes little difference, although writers tend to prefer NF and mathematicians and scientists tend to prefer NT. In the creative samples, all but three of the highly creative men in the four IPAR samples preferred Intuition. Comparison of this distribution with the selected college base population yields a chi-square statistic of more than 50, an occurrence that could happen by chance less than one in a million times.

If Intuition is a key factor in creativity, as theory would predict, then groups rated at different levels of creativity should also differ in the proportion of Intuitive types in the group. In Hall and MacKinnon's (1969) data, architects were divided into three groups. Those in the first group were judged by other architects to be highly creative. Those in the second group were judged to be less so, while the third group was a cross-section sample of members of the American Institute of Architects, matched to the creative architects on age and geographic location. The proportion of Intuitive types decreased significantly across the first, second, and third groups. The contrast between the first and third groups yielded a chi-square of 17.0 ($p < .001$). Not only did the proportion of Intuitive types increase with level of creativity, but there was also an increase in the Intuition preference score, which measures clarity of the preference for Intuition (not the amount or degree of Intuition).

Myers hypothesized that creativity would also be associated with the open, curious receptivity of the Perceiving orientation and with the interest in possibilities of the Intuitive type. These predictions are mainly upheld. Indeed, in the samples IPAR grouped by levels of creativity, greater creativity was associated with a higher proportion of Perceiving types and Intuitive types.

Based on 20 years of creativity research at IPAR, Gough (1981) devised an experimental MBTI Creativity Index (CI) as follows:

$$\text{MBTI Creativity Index} = 3\text{SN} + \text{JP} - \text{EI} - .5\text{TF}$$

The size and the signs of the coefficients in the CI indicate that creativity was strongly related to a preference for Intuition, somewhat less to Perceiving and Extraversion and least to Thinking. The original CI was based on MBTI continuous scores from Form G. Based on data from the Global Step I form, a CI was created that employed both the IRT scoring used on Form M and the latent class scoring used with the global form and tested in two samples.⁶ The Form M CI was applied to a very large sample ($N = 136,220$) from 384 occupations from a commercial database. The CI derived from the global form was applied to a sample ($n = 5,267$) of people from

6. To compute the MBTI Creativity Index from the scores of the MBTI Global Step I assessment, use the logit scores for the four preference scales.

141 occupations who had completed MBTI Form Q. The criterion for both samples was ratings of occupational creativity based on three variables used in the US Department of Labor's O*NET database of careers: how much originality, fluency of ideas, and creative thinking were required for a particular occupation. For the occupations in the two samples, each of these three variables was rated on "importance" and "level" required for the job. Since the three variables correlated highly, they were combined into one unit-weighted composite. The Form M CI correlated $r = .49$ with the creativity criterion in the commercial sample; the Global Step I form scoring correlated $r = .48$ with the creativity criterion in the Form Q sample. Results were in line with those of Gough (1981) and indicate a strong relationship between scores on the MBTI Creativity Index and the creativity requirements of an occupation. Taken together, this evidence suggests that the MBTI assessment provides insight into how creativity is manifested in individuals of different types.

The MBTI assessment can be used to gain perspective on an individual's creativity.

Houtz et al. (2003) examined the relationship of creativity to MBTI Form G continuous scores in a sample composed of 62 education students. The assessment used to measure creativity was the *Kirton Adaption-Innovation Inventory* (KAI; Kirton, 1987), which yields a total score indicating a preference for an adaptor or innovator style, as well as scores on the subscales of Sufficiency of Originality (prefer a few very good ideas to generating many ideas), Efficiency (prefer small-scale, specific ideas to inclusive, global ideas), and Rule Conformity (prefer working within the rules to ignoring the rules). As predicted from type theory, and consistent with the results discussed in the creativity studies above, an Innovator style was associated with a preference for Extraversion, Intuition, and Perceiving. Higher scores on Sufficiency of Originality indicated a preference for generating many ideas and were associated with the same three MBTI preferences plus Feeling. Higher scores on Efficiency, indicating a global approach, were correlated with a preference for Perceiving. Rule Conformity was not significantly correlated with any of the MBTI preference scales.

Pencheva, Vassileva, and Bozhinva (2006) also found that the Creativity Index correlated significantly with the Sufficiency of Originality score and the Total score of the KAI. They also found that the Sufficiency of Originality score and the Total score of the KAI were positively correlated with Extraversion, Intuition, and Perceiving and that the Total score was also correlated with Feeling. Van Rooyen (1994), in a small sample ($N = 87$) of female managers in South Africa, found that the highest-scoring types on the KAI were the two dominant extraverted

Table 9.34 | **Correlations of MBTI® Global Step I™ continuous scores with ACL and CPI 260® creativity scales**

Creative scale	E-I	S-N	T-F	J-P
ACL Creative Personality ($n = 1,550$)	-.25	.38	-.07	.19
CPI Creative Temperament ($n = 1,728$)	-.34	.42	.00	.35

Note: Portion of global sample.

Intuitive types, ENFP and ENTP, followed by INFJ, a dominant introverted Intuitive type. Those types with the lowest scores (indicating an adaptor creativity style) were the dominant introverted Sensing types, ISFJ and ISTJ.

In another examination of creativity, the continuous logit scores for each MBTI preference scale were correlated with scores from two creativity scales available from the MBTI global sample: the Creative Personality scale of the *Adjective Check List*, and the Creative Temperament scale of the CPI 260 assessment. These correlations are shown in table 9.34. The coefficients in this table confirm the predictions embedded in the Creativity Index, with higher correlations toward Intuition, Extraversion, and Perceiving, although for these two creativity scales, there was no relationship to the T-F preference pair.

A study of divergent thinking (Furnham, Crump, Batey, & Chamorro-Premuzic, 2009) in a large sample ($N = 2,603$) of primarily middle managers found that Extraversion, Intuition, and Perceiving were related to creativity, with the largest regression weights for Intuition, consistent with Gough's index.

Fantasy and imagery, which are often associated with creativity, have also been studied in relation to type. Representative studies can be found in Edmunds (1982), Ireland and Kernan-Schloss (1983), O'Haire and Marcia (1980), and Palmiere (1972). These studies found that, as predicted, both fantasy and imagery were of more interest to Intuitive types. In addition, significant differences are found concerning the content and effect of images.

Orientation to Time

Type theory predicts that Sensing types are more oriented to the present, which is what can be perceived by the senses. Intuitive types are more attuned to the future, since Intuition is the process that sees possibilities beyond the immediate present. Thinking types are expected to be more concerned with the sequence of time from past through present to future, since Thinking is concerned with cause-effect sequences. Feeling types are expected to be more oriented toward the past, since values are transmitted as the heritage of the past.

For each type, the two processes may complement or conflict with each other. For example, SF is concerned with the present moment, which is colored by past values and experience. Thus one would expect SF types to be relatively more conservative. The NF group, however, is pulled toward the future by Intuition and toward the past by Feeling. These contrasting forces can be expected to differ in the four NF types as a result of the interplay of the dominant and auxiliary processes and depending on whether these are extraverted or introverted. Time perspective of the Jungian functions (processes) is discussed further in Mann, Siegler, and Osmond (1968).

There are fewer hypotheses about time and the orientations, though one would expect Extraverts to be more aware of time changes in the world around them and to see time in more discrete units. Introverts should be more oriented toward ideas that have more continuity and to be relatively less aware of changes in the world around them. Thus Introverts may be expected to be less accurate in evaluating the time of events. Myers hypothesized that Intuitive types are less accurate than Sensing types in projecting how long any given task will require. Seeing possibilities in a sudden flash, Intuitive types would underestimate the time it takes to work out necessary details and to bring the possibility to completion. Organized Judging types can be expected to take time more seriously than spontaneous Perceiving types and to be more effective in meeting deadlines.

Evans (1976) and Yang (1981) supported theoretical connections between Sensing and the present, Intuition and the future, Feeling and the past, and Thinking and no specific time orientation. Studies of type differences in ease of visualizing the future found longer extensions with Intuition and Thinking and shorter ones with Sensing and Feeling (Harrison, 1984; Nightingale, 1973; Seiden, 1970; Smith, 1976).

Seiden (1970) found that Extraverts related their identity to events or ceremonies that represented change. Extraverts were also more accurate in time perception (Veach & Touhey, 1971). Introverts, who are oriented to ideas that are more continuous than external events, see time as more continuous (Seiden, 1970) or oceanic (Knapp & Lapuc, 1965). Nightingale (1973) found that Introverts extended time farther back into the past; Extraverts extended time farther forward into the future.

Evidence for J–P differences in time management appeared in a study by Jaffe (1980), in which Sensing, Thinking, and Judging managers were more successful with management by objectives. In student populations, significant correlations are often seen between Judging and planning and control, which are essential elements of time management. For example, Williams, Verble, Price, and Layne (1995) examined MBTI Form G continuous scores with the *Time Management Questionnaire* in a sample of 204 students. The

correlation of time-management attitudes with the J–P scale was clear, with a preference for Judging correlating positively with both short- and long-range planning. Long-range planning was correlated with Extraversion and Sensing, as well as with Judging. In other studies, Judging students took fewer incompletes (Smith, Irey, & McCaulley, 1973) and Judging medical school applicants completed their applications earlier (Kainz & McCaulley, 1975). Gender differences also appear to be important in some samples (Marcus, 1976; Squyres, 1980). A qualitative study by Beckham (2012) provided rich descriptions of the profound differences in how Judging and Perceiving types view time and their relationship to it. This article offers many clues for researchers wishing to examine this area.

Harrison and Lawrence (1985) studied the time orientation of middle school students ($N = 302$), who were asked to write scenarios about their personal future, being free to decide how far ahead their description would extend. Based on previous theoretical work by Mann, Siegler, and Osmond (1968, 1971), the authors hypothesized that Intuitive types would be oriented toward the future; Thinking types would have a “linear” perspective incorporating the future, present, and past; Sensing types would be oriented toward the present; and Feeling types would focus more on the past. Using these hypotheses about time orientation, and considering the dominant and auxiliary processes for each type, the authors hypothesized a rank ordering of types based on how long into the future the scenarios of each type were predicted to extend.

Table 9.35 compares the observed with the predicted results based on type theory and trait theory. These findings are particularly impressive because the predicted rank ordering of types on the left side of the table was based on typological criteria, which predicts a different ordering than does the strictly linear or independent trait criteria used on the right side of the table. The differences between the two sets of predicted rankings illustrate that the typological predictions do not reflect a simple additive combination of the S–N and T–F preference pairs. The typological predictions Harrison and Lawrence made rank the types by their dominant process in combination with the time orientation associated with that process. This made maximum use of typological information and enabled the authors to make more precise predictions about rankings of the types.

From the trait perspective, only the additive effects of the preferences are important. The S–N preference pair is most salient for time orientation, with Intuitive types predicted to have a longer frame of reference than Sensing types. The T–F preference pair is second in importance, with Thinking types assumed to have a longer frame of reference than Feeling types. The summative predicted ordering on the right side of the

Table 9.35 | Rank order of MBTI® types based on prediction and observation of temporal extension

Typological predictions ($r_s = .98$) ¹				Trait/linear predictions ($r_s = .88$) ¹			
Type ²	Temporal orientation ³	Predicted rank	Observed rank	Type ⁴	Temporal orientation	Predicted rank ⁵	Observed rank
INTJ	<i>Future</i> and linear	1	1	INTJ	Future and linear	2.5	1
ENTP		2	2	ENTP		2.5	2
INFJ	<i>Future</i> and past	3	3	INTP	Future and linear	2.5	4.5
ENFP		4	4.5	ENTJ		2.5	6
INTP	Future and <i>linear</i>	5	4.5	INFJ	Future and past	6.5	3
ENTJ		6	6	ENFP		6.5	4.5
INFP	Future and <i>past</i>	7	7	INFP	Future and past	6.5	7
ENFJ		8	9	ENFJ		6.5	9
ISTP	Present and <i>linear</i>	9	8	ISTJ	Present and linear	10.5	12
ESTJ		10	10	ESTP		10.5	14
ISTJ	<i>Present</i> and linear	11	12	ISTP	Present and linear	10.5	8
ESTP		12	14	ESTJ		10.5	10
ISFJ	<i>Present</i> and past	11		ISFJ	Present and past	14.5	11
ESFP		14	13	ESFP		14.5	13
ISFP	Present and <i>past</i>	16		ISFP	Present and past	14.5	16
ESFJ		16	15	ESFJ		14.5	15

Note: $N = 302$. ¹Spearman's rho; ²dominant type function; ³dominant time reference italicized; ⁴most important preference for temporal orientation is in bold type; ⁵from the trait perspective, NT > NF > ST > SF is the only ranking that can be made; consequently, tied ranks were assigned to the four types within each of these groups.

Source: Harrison & Lawrence (1985), p. 11. Copyright 1985 by *Journal of Psychological Type*. Used with permission.

table, based on the trait perspective, is therefore NT > NF > ST > SF. From the trait perspective, all Intuitive types are predicted to rank first, followed by all Sensing types. Within the Intuitive and Sensing types, the linear ordering would then rank Thinking types above Feeling types. As illustrated, the linear model lumps the types together by process pairs because it considers only the S–N and T–F preference pairs in determining ranks.

The correlations of the trait predictions and the typological predictions with the observed rankings were .88 and .98, respectively, both of which are significant at $p < .001$. Nevertheless, a visual comparison of the actual and predicted rankings, as well as the difference in the sizes of these coefficients, makes it clear that the typological predictions are much closer to the observed rankings than are the linear predictions. This is because the linear model fails to account for characteristics that are important in understanding an individual's orientation to time, while these characteristics are accounted for by type theory. More such studies that contrast the two competing explanations—type versus trait—would be invaluable.

A comparison of linear trait predictions versus dynamic typological predictions of time orientation demonstrates the superiority of the type approach.

Hackston (2015) examined type differences on three dimensions related to time from the *Cultural Orientations Framework*, using a sample of 1,029 individuals from a wide range of countries. Extraverts were more likely to believe that they had an ability to concentrate simultaneously on multiple tasks or people (polychronic), whereas Introverts favored a monochronic approach, meaning that they preferred to focus on one activity or relationship at a time. Sensing types believed that they were more capable of dealing with scarce resources, more likely to carry out tasks in a monochronic manner, and more oriented toward the past and present. Intuitive types believed that they were more capable of dealing with plentiful resources, were more likely to take a polychronic approach, and were more oriented toward the future. Thinking types were more likely to treat time as a scarce resource, whereas Feeling types were more likely to see it as plentiful and to have a polychronic orientation. Judging types believed that they were more capable of dealing with scarce resources, were more likely to carry out tasks in a monochronic manner, and

were more oriented toward the past. Perceiving types felt that they were more capable of dealing with plentiful resources, behaved in a polychronic manner, and were more oriented toward the present.

Participants with different dominant and auxiliary processes also varied significantly in their perception of time. For example, participants with dominant extraverted Intuition were the most likely to take a polychronic approach. Dominant Sensing types were more oriented toward the past and present, while dominant introverted Intuitive types were most focused on the future. Across-country differences were also examined for these time variables, and while some significant differences were found, the author concluded that “type differences may be more important than culture in time management” (p. 30).

Technology and Social Media

Quite a few studies examining the relationship between type and technology have appeared since the publication of the 1998 *MBTI® Manual*. Some of these examined the performance of different types across different kinds of learning modalities (e.g., traditional classroom, online, computer-assisted instruction). In other studies, individuals with different preferences were exposed to user interfaces that were designed to have “personalities.” Other studies described the use of social media, different kinds of technology, and the internet by people with different MBTI preferences. This is another area in which research has leaped ahead of theory, although the results of many of the studies below are congruent with predictions that would be derived from the Jung-Myers theory. Other results raise more questions than they answer.

Lu and Hsiao (2010) examined the influence of E–I preferences on intention to pay for social networking sites. They sampled users of a large internet service provider in China; 37% of the respondents were students and 35% were office workers. The authors asked about different kinds of values received from the networking sites: social, monetary, and emotional. Perceived social value was a more important factor for Extraverts than for Introverts. Extraverts indicated a greater willingness to pay for the services if they believed that they would enhance their social self-concept. Introverts put more emphasis on the price or perceived value for money and also on emotional value. They were more cost-sensitive and cared more about the emotional pleasure that the services provided them. On sites where participants chose avatars for online interaction, Extraverts were more willing to pay for decorations used to adorn their avatars and for virtual rooms so as to encourage others to interact with them. Extraverts cared less about the price of the items and cared more about the interaction with others. In addition, Extraverts indicated an intention to use and pay extra for the photo albums service,

clearly indicating that they wanted to disclose their photos to others.

The publisher of the MBTI assessment conducted a survey (Weber, Schaubhut, & Thompson, n.d.; Schaubhut, Weber, & Thompson, 2011) on the use of social networking platforms by people who had taken the MBTI assessment ($N = 1,784$). Extraverts were significantly more likely to use Facebook, LinkedIn, Twitter, and blogs than were Introverts; there were no differences in the use of YouTube. Intuitive types were more likely to use all of the platforms compared to Sensing types. There was also a significant E–I x S–N interaction. Extraverted Intuitive participants used Facebook more than Extraverted Sensing types or any Introverted types. Feeling types were more likely to use Facebook and blogs, while Thinking types were more likely to use LinkedIn, which is business oriented. Thinking and Feeling types showed no significant differences in use of Twitter or YouTube. Judging and Perceiving types showed no significant differences either, although those who favored Perceiving were somewhat more likely to use Twitter ($p < .06$). Differences in the use of social media by whole types were also analyzed. There was a prevalence of Extraverted types in the highest four types for each platform, with the exception of YouTube, which is not an interactive platform. Another question asked of participants was whether they believed that use of social media was a waste of time. ISTJs were more likely to “strongly agree” with the statement.

Another survey on the use of and attitudes toward various kinds of technology was conducted with individuals ($N = 1,784$) who had taken the MBTI assessment and volunteered to participate in research (Hammer, 2014b). The average age of the participants was 24. Sensing types were more likely to use a GPS. Feeling types were more likely to use a camera and a music player. Introverts were twice as likely as Extraverts to use a game console. Perceiving types were more likely to want to use technology to keep connected with others; Introverts to listen to music or for visual entertainment, or to learn something new. Sensing types believed that use of technology helped them procrastinate or deal with boredom. While no one owned up to texting while driving, Introverts and Perceiving types were more likely to text during dinner with friends and Sensing types during class and while watching TV. Perceiving types were more likely to be distracted by incoming messages than were Judging types. SJs were the least distracted, believing that they could comfortably multitask.

Owen and Liles (1999) found that faculty who reported preferences for Extraversion and Intuition were more likely to use the internet for a variety of purposes. Intuition in particular was significantly related to use of the internet.

Lee and Nass (n.d.) conducted two experiments by asking undergraduate students to rate the social presence of computerized representations of voices. Social presence in this context means the sense that intelligent beings coexist and interact with you, even if they are machines. This variable was measured by asking participants to listen to different computerized voices describe a book. In one experiment, the voice designated as Extraverted had a higher rate of speech, was louder, and had higher frequency and pitch than did the voice labeled Introverted. In a similar experiment, the computerized voices differed on other features: the Extraverted voice spoke for a relatively longer period and used strong and friendly language expressed in the form of confident assertions. The Introverted voice spoke for a relatively shorter time and used “weaker” language expressed in the form of suggestions. In both experiments, respondents perceived the Extraverted voice as more extraverted than the Introverted voice. More important, participants had a stronger sense of social presence when they heard the computer voice that manifested a personality similar to their own.

Technology and social media have introduced a whole new area for study that requires attention by both theorists and researchers.

A study by Goby (2006) asked university students in Singapore to rate their preferences for online (e.g., Skype) or offline (e.g., face-to-face) means of interacting with others in four communication contexts: widening their social circle, looking for work, positive interactions with friends (e.g., sharing good news), and negative interactions with friends (e.g., sharing bad news, arguing, handling conflict). Across all four communication contexts, Extraverts, Feeling types, and Perceiving types were more likely to prefer offline modes of interaction, preferring to interact in face-to-face situations. Introverts, on the other hand, believed that online modes offered greater freedom of expression. The lack of a significant overall relationship between Sensing and Intuition and choice of interaction mode was due to differences across the contexts that cancelled each other out: Sensing types preferred looking for work online more than did Intuitive types. Intuitive types preferred online modes for positive interactions with friends, while Sensing types preferred to use online modes for negative interactions with friends.

Bail (1993) tracked the computer interaction of undergraduate students over three semesters. Extraverts sent more emails to peers than did Introverts. Extraverts and Intuitive types wrote more unrequired replies to class notes that their peers had posted online. Intuitive types posted longer class notes than did Sensing types.

Bowen, Ferguson, Lehmann, and Rohde (2003) examined the relationship of information technology students’

type preferences to the students’ performance in querying a database. Intuitive and Perceiving types made significantly fewer semantic errors in their queries than did those with opposite preferences. Da Cunha and Greathead (2007) assigned a code-review task to undergraduates in computer science. Intuitive types performed significantly better than Sensing types at the task. None of the other preference pairs showed a significant difference; however, there were differences between participants with different combinations of the four processes. Although the difference was not tested for significance due to low sample size, participants with NT preferences performed better than those with the other process pairs.

Ludford and Terveen (2003) examined how adult professionals performed two specific online tasks: handling email and using online reviews of media (music and movies). They found no differences across MBTI preferences in the handling of informational email, email of indeterminate status, and social invitation email. Task-related email management, however, was significantly related to participants’ J–P preference. When a task was completed, the majority of those who preferred Judging reported a preference for either deleting the email or moving it to a folder for reference. In contrast, most of those who preferred Perceiving kept emails related to completed tasks in their inbox for lengths of time varying from several days to more than a year. Perceiving types were apt to use their inbox as a storage place for completed task-related email, while Judging types used their inbox as a current task manager. No difference was found between those who favored Judging and those who favored Perceiving in regard to the number of folders they created, the amount of email in the folders, or the size of their inboxes. In regard to reading book reviews, Feeling types were more likely to read and use reviews by other customers, while Thinking types were more likely to use editorial reviews. Based on this finding, the authors speculate that Feeling types might have a general interest in collaboration interfaces and/or online communities as a resource for decision making. In regard to evaluating music, those with a Thinking preference were again more likely to read and use editorial reviews than were Feeling types. Feeling types in the study explained that they rarely considered either editorial or customer reviews when it came to choosing music. There were no differences across MBTI preferences for receiving movie recommendations.

Abrahamian, Weinberg, Grady, and Stanton (2004) designed two programs to teach a computer science concept. One program used an interface that was designed to appeal to Extraverted Intuitive (EN) types and the other to Introverted Intuitive (IN) types. EN and IN students were then assigned to the interface that matched their type. A control group used a randomly selected interface. Undergraduate computer science

students were given 30 minutes to learn the concept. The effect of matching personality-related interfaces on degree of learning was statistically significant. Students who preferred EN did better with an interface designed to appeal to EN preferences than did the control group, and the students who preferred IN did better with an interface designed to appeal to IN preferences than did the control group.

A number of studies have examined performance in and preferences for online versus traditional forms of learning. Elliott (2006) studied the effects of type differences on the achievement of adult learners in online versus traditional sections of an English composition course and a mathematical models course. Introverts were more likely to select the online sections of the courses, while Extraverts were more likely to select the traditional sections. There were no significant differences in achievement by personality type between online and traditional sections. However, a power analysis revealed that the power of many of the statistical tests used was low, which may have concealed some differences. Toole (2007) likewise found no difference in performance between Extraverts and Introverts within the same class or in online versus traditional business courses.

Another study (Bishop-Clark, Dietz-Uhler, & Fisher, 2006–2007) collected the reactions of a small sample ($N = 27$) of students with different preferences to a web-based distance learning introductory psychology course. While this was a qualitative study with no inferential statistics reported and a very small sample size, the differences in the means between preference pairs suggest that the variables used here should be tested with larger and more diverse samples. Extraverts found it more difficult to relate to other students in the class, due presumably to the online nature of the interactions. Introverts felt that working through course materials at their own pace was a high priority and reported that they enjoyed working on assignments and studying for the course more than did their Extravert counterparts. Sensing types were more likely to feel isolated from other students. Intuitive types reported that they enjoyed studying for the online course more than studying for face-to-face courses. Feeling types felt more isolated from the instructor and other students than did those with a Thinking preference, who enjoyed the course and said they would recommend it. Those with a Perceiving preference were more likely than Judging types to believe that students were given more opportunity to take responsibility for their own learning and to apply what they learned to real-world situations. Students with NF preferences were much more likely to feel isolated from other students than were ST students.

Preferences for online versus face-to-face instruction were also examined by Harrington and Loffredo (2010a) in a sample of psychology, nursing, and education students. A majority of Introverts preferred online

instruction and a minority preferred face-to-face instruction, whereas the opposite pattern occurred for Extraverts. The majority of Perceiving types preferred face-to-face instruction, while Judging types showed an opposite pattern; the effect size was small.

Harland (2000) was interested in whether type preferences were related to differences in participation in face-to-face versus asynchronous online discussions. The sample consisted of 18 adult students seeking an associate's degree who met online and face-to-face on a weekly basis. Even with this very small sample, a significant difference was found between Thinking and Feeling types: Thinking types preferred to answer difficult questions online versus face-to-face, while Feeling types preferred the opposite. Perceiving students perceived a greater equality of discussion (opportunity to participate equally) in the online communication than did the Judging types. In the online discussions, Extraverts and Perceiving types were more likely to initiate new topics than were Introverts and Judging types. In the face-to-face meetings, Extraverts initiated more peer-to-peer interaction and reported a stronger sense of community online than did Introverts.

Fowler (2009) found type differences in nursing students' preferences for various instructional methodologies: lecture, online, and simulation. Sensing types had a significantly higher liking than did Intuitive types for simulations as an instructional modality. These simulations reproduced actual events in a learning environment and allowed realistic and hands-on learning experience, so it was not surprising that these were preferred by students with a Sensing preference. Judging types exhibited a preference for both lecture and online learning more than did Perceiving types.

Risk Tolerance

The MBTI Step I assessment has been used to study tolerance for risk in people with different type preferences. Li and Liu (2008) examined individual differences in risk preference. The sample consisted of 200 first-year undergraduate students. As predicted by theory, students classified as risk averse were Sensing Judging types, whereas consistent risk seekers were Intuitive Perceiving types. While Judging types were risk averse compared to Perceiving types, among Judging types, those with clearer preferences for Judging were more risk seeking. This replicates the finding in the Filbeck, Hatfield, and Horvath (2005) study below.

Type differences can also be seen in the scores of individuals with opposite preferences on the Risk Taking Personal Style Scale of the *Strong Interest Inventory* assessment. As shown in table 9.9, participants in the MBTI global sample who preferred Extraversion, Thinking, and Perceiving scored higher on this scale than did their opposites. Buboltz et al. (2000) examined the relationship

between the Risk Taking scale of the Strong and MBTI preferences in a sample of undergraduates ($N = 426$) and found significant correlations with all four preference pairs in the direction of Extraversion, Intuition, Thinking, and Perceiving. The four preference scales accounted for 27% of the variance in the Risk Taking Personal Style Scale. The two types with the highest scores on the Risk Taking scale were the two dominant extraverted Intuitive types (ENTP, ENFP), and the three with the lowest scores had a dominant introverted perceiving process (ISTJ, INFJ, ISFJ), findings which are in accordance with type theory. The difference between the highest- and lowest-scoring types (ENTP and ISFJ) yielded a very large effect size ($d = 1.88$). Note that using whole types and considering type dynamics gives a more nuanced result than do the correlations. Although Intuition and Thinking correlated significantly with risk taking in this sample, the second-highest scoring type was a Feeling type. On a different dependent measure, the Risk scale of the *Values Scale* (see table 9.41), there were significant correlations with Intuition, Thinking, and Perceiving.

Filbeck, Hatfield, and Horvath (2005) examined risk tolerance for investing. Type theory might predict that the J–P, and perhaps the S–N, preference pair would be more likely to differentiate risk tolerance. The results indicated that the T–F preference pair had a significant effect on risk tolerance. Individuals with a clearer preference for Feeling had a lower tolerance for risk, while those with a clearer preference for Thinking had a higher tolerance. The J–P preference pair had a significant nonlinear relationship to risk tolerance, although in the opposite direction than was hypothesized. Individuals with the clearest preference for Judging exhibited an exponentially increasing tolerance for risk. Individuals with the clearest preference for Perceiving had virtually no tolerance. There was no direct relationship of the E–I or S–N preference pair to risk tolerance. In general, individuals who prefer Thinking and Judging have a higher tolerance for risk. The result of this study did not conform to theoretical expectations. Additional research in this area is needed to determine whether the results can be replicated or clarified.

Consumer Behavior

Consumer behavior offers a rich domain for type studies because the processes (Sensing, Intuition, Thinking, and Feeling) measured by the MBTI assessment refer to the way people take in information and arrive at decisions, which parallels consumer behavior theory in terms of searching for information and making decisions.

Research by McBride (1988) and Yorkston and LaBarbera (1997) showed that advertising copy is more effective when it provides information and uses words and images that are congruent with the type preferences of the audience. LaBarbera, Weingard, and Yorkston (1998) followed up and conducted an experimental study that

tested the appeal of advertisements to people with a preference for Sensing or Intuition and whether viewing the ads inspired an intent to purchase. The questions were whether visual images used in advertisements can be categorized according to personality type and, if so, whether such images make a difference in the effectiveness of advertising. Advertising professionals with knowledge of MBTI type chose 24 color images believed to be highly appealing to individuals with a preference for either Sensing or Intuition. In a pretest, 30 adults looked at each of the 24 images and were asked to choose as many words as they liked from a list of 20 words that best described the image. These words were based on descriptions used by experts to differentiate between Sensing and Intuitive types. The eight images—four Sensing images and four Intuition images—that showed the greatest difference were selected as the experimental stimuli. The ad copy for both images was exactly the same. One hundred participants viewed each of the eight ads and rated each on three scales: image appeal, overall evaluation of the ad, and purchase intention. Participants rated the two types of images significantly differently and, more important, favored the images that matched their MBTI preference. There were also significantly higher purchase intentions resulting from the advertising images that were congruent with their preference. The authors concluded with an interesting discussion of the implications of their findings by noting that 85% of advertising professionals are Intuitive types, compared to about 25% of consumers, suggesting that many ads may not be optimally tailored to their audience.

Leisure Activities

Research elsewhere in this chapter showing a strong relationship between MBTI type and career choice would suggest that a relationship between type and leisure choice would also be likely. McGuiggan (1999) was interested in whether demographic variables or personality variables were better predictors of leisure preferences. The MBTI Step I assessment and an 11-scale leisure questionnaire were administered to 908 adults. Regression analysis indicated that some attributes of leisure, such as planning, follow-through, variety, people, and the pace of leisure activities, were better explained by personality. Other attributes of leisure, such as household tasks, team sports, and traditional vs. modern activities, were better explained by demographics. Still other attributes of leisure, such as risk taking, were explained equally by personality and demographics. In all cases except the Planning scale, using a combination of personality and demographics led to a statistically significant improvement in explanatory power. For Planning, the J–P preference pair alone was the best predictor, accounting for 46.6% of the variability in planning leisure activities.

Participants in the Form M US national sample were asked about their leisure activities. Table 9.36 shows responses of the 16 types to these questions: (1) How do you spend your leisure time? (2) How many hours a day do you watch television? (3) How important are each of the following in your life (followed by a list of values)? The cells in table 9.36 show those activities and values for which the percentage of a given type endorsing the option was significantly overrepresented (+) or underrepresented (–) compared to the percentage of that type in the sample overall. Only activities in which the level of statistical significance was at least $p < .01$ are reported. Most notable are differences related to a preference for Sensing or Intuition, especially when one of these processes is dominant. The dominant introverted Sensing types, ISTJ and ISFJ, stand out by their underrepresentation in almost any kind of leisure pursuit or activity. Each indicated only one such activity as favored, but three activities for ISTJ and six for ISFJ were significantly underreported. The reverse finding was true for the dominant extraverted Intuitive types (ENTP, ENFP). Each endorsed six activities for either leisure or importance and were underrepresented on only one (for ENTP) or two (for ENFP) activities.

Other results showed that only NT types (INTJ, INTP, ENTP), who tend to be both conceptual and analytical in focus, listed taking classes and going to school as significantly favored activities; ISFPs did not. Two Intuitive types (ENFP, ENFJ) significantly listed education and learning as very important, while two Sensing types (ISFJ, ISFP) did not. Participants who endorsed writing as a leisure activity included overrepresentations of all four NF types plus ENTP and INTP, and underrepresentations of ISTJ, ISTP, ISFP, and ESFJ. Appreciating art was favorably listed for seven Intuitive types and no Sensing types; on the other hand, three Sensing types but no Intuitive types were significantly underrepresented on this item. Only Intuitive types favored reading, listening to music, or playing a musical instrument as leisure activities. Only Sensing types (ISTJ and ESTJ) were overrepresented in watching sporting events. Only SF types (ISFJ, ISFP, ESFP) reported watching three or more hours of television per day, while only Intuitive types (INTJ, ENTP, ENFP, ENFJ) were significantly underrepresented in this activity.

Thinking–Feeling differences in conjunction with orientations or other processes accounted for several interesting patterns. Only Thinking types with Extraversion (ESTJ, ENTP) significantly listed achievement and accomplishment as very important, and only Feeling types with Introversion (ISFJ, ISFP, INFP) did not. Playing video games drew INTP, ISTP, INTJ, and ENTP types; ISFJ types were underrepresented. Playing sports attracted ESTJ, ESTP, and ENTP types, with ISFJ again underrepresented. Working out/exercising was endorsed only by ENTJs and INTJs; it was shunned by ISFPs and

ESFPs. It appears that the interest in competition found in many video games and sports provides the challenge needed by Thinking types, and even the individual challenge of working out or exercising seems to appeal to them as well. For sports, the external activity required also appears to attract Extraverts. Though these results came from responses to single items on a questionnaire, the patterns offer broad confirmation for the predictions of type theory and the MBTI assessment.

Another point of interest seen in table 9.36 is that ESFJs engage in no leisure activities for which they are overrepresented, and six types (ISTJ, ISTP, ESTP, ISFJ, ESFP, and ENTJ) are overrepresented for only one activity each. This does not mean that these types do not engage in leisure activities. Rather, this finding could mean that either: (1) the percentage of these types who engage in any given leisure activity is about the same as the percentage who engage in that activity in the total sample, and thus these types are not overrepresented, or (2) the list of activities provided did not capture the leisure interests of some types or preferences.

Memory

Carlson (1980) tested the relationships between MBTI preferences and a number of variables associated with kinds of memory. Participants were asked to recall a significant personal experience. Judges familiar with type theory but blind to the type of the participants correctly identified the EF preferences of eight of nine EF participants and the IT preferences of five of six IT participants based on the participants' descriptions of their experiences. Introverts gave more intrapersonal and Extraverts more interpersonal memories of joy, excitement, and shame. Cognitive clarity and vividness of feeling were also rated blind by a new set of judges. Memories by the EF participants were rated significantly more vivid in expression of joy, excitement, or shame than were the memories of IT participants.

Carlson and Levy (1973) tested the relationship between the MBTI assessment and cognitive variables related to memory. They tested short-term memory using the Digit Span of the *Wechsler Adult Intelligence Scale*® and memory for faces with the Lightfoot Facial Expression Series. They found that ITs scored higher than EFs on digit span, and EFs scored higher than ITs on recognizing faces.

Religious Attitudes

Embree (2011) administered Form M of the MBTI assessment and the *Origins Attitude Survey* to 232 college students. Preferences for Extraversion and Feeling were significantly overrepresented among those endorsing creationism, with Introversion and Thinking underrepresented. Thinking types were significantly overrepresented among those endorsing evolution, and

Table 9.36 | Leisure time preferences for the 16 MBTI® types

ISTJ		ISFJ	
+ Watching sporting events	<i>I</i> = 1.21	+ Watching TV 3 hrs or more per day	<i>I</i> = 1.12
– Listening to music	<i>I</i> = 0.83*	– Education/learning very important	<i>I</i> = 0.84
– Writing	<i>I</i> = 0.52*	– Achievement/accomplishment very important	<i>I</i> = 0.80*
– Appreciating art	<i>I</i> = 0.49*	– Playing with computers or video games	<i>I</i> = 0.74*
		– Watching sport events	<i>I</i> = 0.69*
		– Appreciating art	<i>I</i> = 0.58*
		– Playing sports	<i>I</i> = 0.47*
ISTP		ISFP	
+ Playing with computers or video games	<i>I</i> = 1.39	+ Watching TV 3 hrs or more per day	<i>I</i> = 1.19*
– Listening to music	<i>I</i> = 0.78*	+ Watching TV for leisure	<i>I</i> = 1.12
– Writing	<i>I</i> = 0.39	– Reading	<i>I</i> = 0.80*
		– Achievement/accomplishment very important	<i>I</i> = 0.74*
		– Education/learning very important	<i>I</i> = 0.67*
		– Working out/exercising	<i>I</i> = 0.66*
		– Writing	<i>I</i> = 0.52
		– Appreciating art	<i>I</i> = 0.50*
		– Taking classes	<i>I</i> = 0.49
ESTP		ESFP	
+ Playing sports	<i>I</i> = 1.79*	+ Watching TV 3 hrs or more per day	<i>I</i> = 1.21*
		– Reading	<i>I</i> = 0.83*
		– Working out/exercising	<i>I</i> = 0.74
ESTJ		ESFJ	
+ Playing sports	<i>I</i> = 1.57*	– Writing	<i>I</i> = 0.62
+ Achievement/accomplishment very important	<i>I</i> = 1.35*		
+ Watching sporting events	<i>I</i> = 1.20		
– Listening to music	<i>I</i> = 0.85		

CONTINUED

Feeling types significantly underrepresented. There were also differences between dominant processes. Dominant Thinking types were significantly overrepresented among those endorsing evolution and significantly underrepresented among those endorsing creationism.

Ross and Francis (2010) examined religious orientation in a sample of Christians who attended church weekly. They identified categories of religious orientation: quest (readiness to face existential questions without reducing their complexity); intrinsic integration (the close relationship between religion and the rest of life, including public religion); and extrinsic compartmentalization (the separation of religion from the rest of life). Quest religious orientation was associated with preferences for Intuition and Introverted Intuition. Intrinsic religious orientation was associated with a preference for Feeling and with significant though less strong relationships with Sensing and Extraversion. Several items that compose the Intrinsic scale refer to

relationships and also address religion in a public context, which helps explain the association between intrinsic religiosity and Extraversion, despite the seemingly introverted label for the scale. There were no type differences for extrinsic compartmentalization. ESTJ was the only Thinking type ranked above the median on intrinsic integration. The ESTJ group is of interest also because ESTJs tended to score high on intrinsic integration (ranking 3rd out of 16 personality types) but low on quest religiosity (ranked 16th), which was in direct contrast to what is found for the two Thinking types who extravert their Intuition: ENTJs ranked 16th and INTJs 13th on intrinsic integration, and high on a quest orientation, on which they ranked 3rd and 1st, respectively.

Ross, Francis, and Craig (2005) were interested in the relationship between type and dogmatism as a religious attitude. They administered the Troidahl-Powell Dogmatism Scale along with MBTI Form G to 422

Table 9.36 | Leisure time preferences for the 16 MBTI[®] types *CONTINUED*

INFJ		INTJ	
+ Writing	<i>I</i> = 2.64*	+ Taking classes	<i>I</i> = 2.66*
+ Appreciating art	<i>I</i> = 1.87	+ Appreciating art	<i>I</i> = 1.78
– Watching sporting events	<i>I</i> = 0.45	+ Playing with computers or video games	<i>I</i> = 1.62
		+ Working out/exercising	<i>I</i> = 1.57
		– Watching TV 3 hrs or more per day	<i>I</i> = 0.58*
INFP		INTP	
+ Writing	<i>I</i> = 2.19*	+ Appreciating art	<i>I</i> = 2.16*
+ Appreciating art	<i>I</i> = 2.13*	+ Writing	<i>I</i> = 2.00*
+ Reading	<i>I</i> = 1.22*	+ Taking classes, going to school	<i>I</i> = 1.96
+ Listening to music	<i>I</i> = 1.20	+ Playing with computers or video games	<i>I</i> = 1.67*
– Achievement/accomplishment very important	<i>I</i> = 0.79		
ENFP		ENTP	
+ Writing	<i>I</i> = 2.03*	+ Taking classes	<i>I</i> = 1.87
+ Appreciating art	<i>I</i> = 1.97*	+ Writing	<i>I</i> = 1.74
+ Playing a musical instrument	<i>I</i> = 1.62 *	+ Appreciating art	<i>I</i> = 1.74
+ Listening to music	<i>I</i> = 1.22*	+ Playing sports	<i>I</i> = 1.56
+ Education/learning very important	<i>I</i> = 1.20	+ Playing with computers or video games	<i>I</i> = 1.48
+ Reading	<i>I</i> = 1.11	+ Achievement/accomplishment very important	<i>I</i> = 1.28
– Watching TV for leisure	<i>I</i> = 0.89	– Watching TV 3 hrs or more per day	<i>I</i> = 0.76
– Watching TV 3 hrs or more per day	<i>I</i> = 0.83		
ENFJ		ENTJ	
+ Appreciating art	<i>I</i> = 2.01*	+ Working out/exercising	<i>I</i> = 1.69
+ Writing	<i>I</i> = 1.89		
+ Education/learning very important	<i>I</i> = 1.54*		
+ Listening to music	<i>I</i> = 1.26		
+ Reading	<i>I</i> = 1.24		
– Watching TV 3 hrs or more per day	<i>I</i> = 0.66*		

Note: Portions of Form M US national sample; for "Watching TV 3 hrs or more per day," *n* = 2,976; for "Education/learning very important," *n* = 2,887; for "Achievement/accomplishment very important," *n* = 2,953; for all other items, *n* = 3,036.

I = index of attraction; + indicates significant overrepresentation of the type compared with those from the total sample that endorsed the item;

– indicates significant underrepresentation of the type compared with those from the total sample that endorsed the item.

**p* < .001; all other items *p* < .01.

female undergraduates. Higher dogmatism scores were associated with preferences for Sensing, Judging, and Extraversion. There were also process pair differences: STs scored significantly higher than NFs and NTs, as did SFs compared to NFs and NTs. NTs and NFs were not significantly different from each other, nor were SJs and SPs.

Relationships

A subset of participants in the MBTI global sample (*n* = 7,857) indicated their relationship status. More Introverts (38%) than Extraverts (28%) were single or never married. Sensing types were somewhat more likely to be married than were Intuitive types (55% to 46%, respectively). There was little difference between

Thinking and Feeling types. Judging types were more likely than Perceiving types to be married (56% to 48%, respectively) and less likely to be single or never married (30% to 38%, respectively). Results for whole types are shown in table 9.37. The four types most likely to be married were four Extraverted types: ESTJ, ENTJ, ESFJ, and ESFP. The four with the lowest frequency of being married were all Introverted types: ISTP, INFJ, INFP, and INTP. The two types with the highest marriage rate were dominant extraverted Thinking types, and two of the four with the lowest rate were dominant introverted Thinking types. The type with the third-highest marriage rate was an Extraverted Feeling type (ESFJ), and one of the lowest was an Introverted Feeling type (INFP). The ENTJs were about 1.75 times more likely to be married than were the INTPs. The INTPs were over 2.5 times more likely to be in

Table 9.37 | Rank order of marital status of each MBTI® type by percentage married

Type	% single/ never married	% married	% separated/ divorced/widowed
ESTJ	19.8	68.6	11.6
ENTJ	21.5	66.9	11.5
ESFJ	22.8	62.8	14.5
ESFP	25.8	58.7	15.5
ISTJ	33.3	54.0	12.7
ISFJ	31.8	51.9	16.4
ESTP	35.6	51.7	12.7
ISFP	33.0	51.2	15.8
ENFJ	34.1	50.0	15.9
ENFP	33.9	49.2	16.9
INTJ	40.5	45.5	14.0
ENTP	34.8	44.8	20.4
ISTP	42.6	44.7	12.7
INFJ	45.5	42.9	11.6
INFP	43.7	40.7	15.6
INTP	50.9	39.4	9.7
<i>Total in sample</i>	<i>33.9</i>	<i>52.0</i>	<i>14.1</i>

Note: Portion of the global sample, $n = 7,857$.

the single, never-married category than were the ESTJs. The types most likely to have been separated, divorced, or widowed were the ENTPs (20.4%) and the least likely were the INTPs (9.7%) because they tended not to be married to begin with. Comparing the frequency of the four preferences on married versus single, never-married, there were significant differences on E–I, S–N, and J–P such that those with preferences for Extraversion, Sensing, or Judging were more likely to be married. Given these results, we would expect that all the ESJ types would rank the highest on percentage married. Yet the top four included the ENTJs and ESFPs, and the next two most frequently married were the ISTJs and ISFJs. Clearly, the ranking of whole types does not conform to the predictions from the preferences alone.

Another study that demonstrated the importance of looking at whole types was by Marioles, Strickert, and Hammer (1996) and addressed two questions. The first was whether married and engaged couples are more likely to be of the same or similar type, or whether they are more likely to be of opposite or dissimilar types. The second question asked whether there was any relationship between type and satisfaction with the relationship. The sample consisted of 271 couples being seen in conjoint therapy and 155 couples who were not in therapy. Duration of marriage ranged from 0 to 47 years, with a mean of 11 years. Relationship satisfaction was measured using the items of the *Dyadic Adjustment Scale* (DAS) and by one additional research item.

There was little evidence that opposites attract or marry. There were differences, however, for satisfaction. An index of satisfaction was created that reflected discrepancies between partners' satisfaction ratings. A high index value meant there was a large percentage of a given type who were satisfied with their relationship while their partner was not. Interviews with couples suggested interpreting this ad hoc measure as an index of "obliviousness" because the partner who was very satisfied with the relationship seemed oblivious to his or her partner's lack of satisfaction. The obliviousness indexes for each type by gender are shown in table 9.38. Although not discussed here in the same way it is by the authors, this table provides an opportunity to observe whole type differences. In evaluating evidence for such differences, there are two competing explanations for the rankings shown in the table. If the obliviousness index for each type is produced by type characteristics that are greater than the sum of the separate preferences, then the indexes and the type rankings should be dissimilar across types having similar preferences. On the other hand, if the obliviousness index for each type is produced only by the summative effects of the separate preferences of the types, then the indexes and the type rankings should be strongly similar across types having similar preferences.

The simplest and clearest test of these two competing explanations is to consider the types that are ranked highest and lowest in obliviousness and compare

Table 9.38 | Rank order of MBTI® types by obliviousness to partner's satisfaction for men and women

Men ($n = 426$)			Women ($n = 426$)		
Type	Obliviousness index	Rank	Type	Obliviousness index	Rank
INTP	33	1	ENFJ	13	1
INFJ	31	2	ENFP	12	2
ISFP	22	3	INTJ	11	3.5
ESFP	21	4	ENTP	11	3.5
ENTJ	19	5	ISFJ	10	5
ISTJ	16	6	ESTJ	9	6
ISFJ	14	7	ESFJ	8	7
ESTJ	12	8	ISTJ	7	8
INFP	11	9.5	ESFP	5	9
ENTP	11	9.5	ISFP	4	10
ESFJ	10	11.5	INFJ	0	13.5
ENFJ	10	11.5	ISTP	0	13.5
ESTP	9	13	INFP	0	13.5
ENFP	7	14	INTP	0	13.5
INTJ	0	15.5	ESTP	0	13.5
ISTP	0	15.5	ENTJ	0	13.5

Note: $N = 852$ (426 couples).

Source: Marioles, Strickert, & Hammer (1996), p. 24. Copyright 1996 by *Journal of Psychological Type*. Used with permission.

Table 9.39 | Comparison of MBTI® types most similar to the type ranked highest in obliviousness to partner's satisfaction for men and women

Men (n = 426)			Women (n = 426)		
Type	Obliviousness index	Rank	Type	Obliviousness index	Rank
INTP	33.0	1.0	ENFJ	13.0	1.0
INTJ	0.0	15.5	ENFP	12.0	2.0
INFP	11.0	9.5	ENTJ	0.0	13.5
ISTP	0.0	15.5	ESFJ	8.0	7.0
ENTP	11.0	9.5	INFJ	0.0	13.5
<i>Mean for 4 similar types</i>			<i>Mean for 4 similar types</i>		
	5.5	12.5		5.0	9.0

Note: N = 852 (426 couples).

Source: Marioles, Strickert, & Hammer (1996), p. 24. Copyright 1996 by *Journal of Psychological Type*. Used with permission.

them to similar types that differ from them by only one preference. Table 9.39 shows such data for the types ranked highest in obliviousness among men and women. Here we see that INTPs were the most oblivious among men, with an index of 33 (rank of 1). The four types most similar to INTPs, differing by only one preference, are INTJ, INFP, ISTP, and ENTP. However, the mean index of these four types was only 5.5 and their mean rank was 12.5. Although it is normally meaningless to average ranks, in this case the mean rank is being used as an indicator of how close the four similar types are to the highest-ranked type. Among women, ENFJs were the most oblivious, with an index of 13 and a rank of 1. The four most similar types are ENFP, ENTJ, ESFJ, and INFJ. ENFPs were ranked second and had an index that nearly matches that of the ENFJs. But as the table shows, the remaining "similar" types are quite dissimilar in their indexes and their ranks. On average, the four most similar types have an index eight points lower than that of ENFJs.

Table 9.40 shows the same kind of comparisons for the types with the lowest obliviousness indexes. Here again, the differences between the lowest-ranked types and the types most similar to them are quite evident.

The differences between the highest- and lowest-ranked types for men and women provide additional evidence that the results are displaying typological rather than preference effects. First, there appear to be basic gender differences in obliviousness. The mean index for women (averaged across the 16 types, not across the total number of women) was 5.6, while for men the mean was 14.1. This is consistent with Gilligan's observation that women are more attentive to relationship issues than are men (Gilligan, 1982). Second, there are major differences between men and women as to the ranking of the types.

The two most oblivious types for men (INTP, INFJ) have indexes of 0 for women. The three highest-ranked types for women (ENFJ, ENFP, INTJ) are ranked in the bottom third of the list for men. So clearly there are gender differences in obliviousness at the level of whole type.

In summary, the patterns we see in these tables do not appear to be associated with the separate type preferences. With the one exception of ENFJ and ENFP women, the types having the highest and lowest ranks do not share their position with the four types that are most similar to them. Furthermore, gender seems to interact with whole type. We must conclude that the rankings appear to derive from the unique characteristics of whole types that are not explainable by the summative effects of the separate preferences. This finding provides strong support for the validity of whole types.

Values

The relationship between MBTI preferences and various measures of values has been examined in many studies. Table 9.41 shows correlations between MBTI Form M continuous scores and the scales of the *Values Scale* (Nevill & Super, 1986). Extraversion was correlated with valuing social interaction and social relations but also with authority. Introversion was associated with valuing economic security. Those who prefer Intuition valued the opportunity to utilize their abilities, lead an independent lifestyle, and engage in personal development. They also valued aesthetics, altruism, autonomy, creativity, risk, and variety. Thinking was associated with valuing ability utilization, authority, creativity, and risk. Feeling was related to valuing working conditions, economic security, and aesthetics. Judging was associated with valuing prestige and economic security, while Perceiving was associated with autonomy,

Table 9.40 | Comparison of MBTI® types most similar to the type ranked lowest in obliviousness to partner's satisfaction for men and women

Men (n = 426)			Women (n = 426)		
Type	Obliviousness index	Rank	Type	Obliviousness index	Rank
ISTP	0.0	15.5	ENTJ	0.0	13.5
ISTJ	16.0	6.0	ENTP	11.0	3.5
ISFP	22.0	3.0	ENFJ	13.0	1.0
INTP	33.0	1.0	ESTJ	9.0	6.0
ESTP	9.0	13.0	INTJ	11.0	3.5
<i>Mean for 4 similar types</i>			<i>Mean for 4 similar types</i>		
	20.0	5.7		11.0	3.5

Note: N = 852 (426 couples).

Source: Marioles, Strickert, & Hammer (1996), p. 24. Copyright 1996 by *Journal of Psychological Type*. Used with permission.

Table 9.41 | Correlations of MBTI® Form M continuous scores with the scales of the *Values Scale*

Values Scale scale	E–I	S–N	T–F	J–P	Summary			
Ability Utilization	–.08	.28**	–.26**	–.01	–	N	T	–
Achievement	–.09	.05	–.12	–.18	–	–	–	–
Advancement	.04	–.08	.03	–.07	–	–	–	–
Aesthetics	–.08	.26**	.20*	.00	–	N	F	–
Altruism	–.17	.33**	.08	.03	–	N	–	–
Authority	–.25*	.14	–.31**	.02	E	–	T	–
Autonomy	.05	.34**	–.13	.29**	–	N	–	P
Creativity	–.11	.62**	–.20*	.14	–	N	T	–
Economic Rewards	.18	–.09	.11	–.09	–	–	–	–
Lifestyle	.10	.24*	–.12	.24*	–	N	–	P
Personal Development	.03	.30**	–.05	.07	–	N	–	–
Physical Activity	–.08	–.03	.12	.11	–	–	–	–
Prestige	–.03	.02	–.16	–.25**	–	–	–	J
Risk	–.12	.30**	–.21*	.30**	–	N	T	P
Social Interaction	–.43**	.14	.00	–.05	E	–	–	–
Social Relations	–.25**	–.06	.11	–.17	E	–	–	–
Variety	–.11	.33**	.04	.25**	–	N	–	P
Working Conditions	.06	–.13	.19*	–.17	–	–	F	–
Cultural Identity	.12	–.03	.03	–.14	–	–	–	–
Physical Prowess	.03	.02	.01	.10	–	–	–	–
Economic Security	.23*	–.33**	.22*	–.33**	I	S	F	J

Note: $N = 110$. Summary column shows the preferences with correlations $\geq .20$ in either direction.

* $p < .05$; ** $p < .01$.

Source: Data provided by Fairview Health Services.

lifestyle, risk, and variety. These results generally accord with predictions from type theory.

Norris, Meisgeier, and Hermond (1998) examined correlations between the four MBTI preference scales and the *Allport Inventory of Values* in a sample of 117 college students. Extraversion was correlated with aesthetic values, Sensing with economic values, Thinking with theoretical values (empiricism, rationalism, and the discovery of truth), and Feeling with religious and social values. Combinations of preferences revealed that ST was correlated with political values, NT with theoretical values, and NF with religious values.

Mitchell (1981) studied work values in a sample of more than 800 employees in a banking organization. He identified eight organizational values. The types with these as high and low values are listed in table 9.42. By theory SFs value harmony, and for the SFs in this sample “happy family” was important. This same value did not appear important to their opposites, the NTs, who are more likely to value their personal autonomy. In general, when comparing opposite types, what is important to one type is not important to its opposite. The EP group does not seem to be particularly oriented toward climbing the corporate ladder; they are generally free-spirited people, open to the moment. Climbing a corporate ladder likely requires working within a system and planning one’s career moves in advance, a style that

likely does not appeal to EPs but does to ESTJs, who valued achievement within the system.

Participants in the Form M US national sample were asked to indicate how important 11 values were in their life. There was a very wide range of endorsements at the “very important” level across the 11 values in the total sample, ranging from 84.9% of the total sample claiming that home and family were very important to 7.6% who endorsed prestige and status at this level. For three of the values, the type with the highest percentage rating it “very important” did so at rates that were twice that of the average of the particular value: ENFPs for being creative, ENFJs for community service, and ESTJs for prestige and status. On the other hand, no INFJ rated prestige and status as very important. The ratings by INTJs for community service and relationships/friendships and ISFPs for being creative were around one-half of the average. These results are in line with what we would expect different types to see as important.

Routamaa and Heinäsuo (2006) compared students and adults on the scales of the *Swartz Values Survey* by whole type. There were significant differences between the students and adults on at least one of the values scales for 7 of the 16 types. The most differences were for ISTJs, with the adults scoring higher on Benevolence (enhancement of the welfare of people with whom one is in frequent personal contact), Universalism

Table 9.42 | Organizational values of the 16 MBTI® types

ISTJ		ISFJ		INFJ		INTJ	
High		Happy family				Financial analysis	
Low	Visible autonomy	Variety and challenge Visible autonomy				Business sociability	
ISTP		ISFP		INFP		INTP	
High		Happy family		Nurturing affiliation		Visible autonomy	
Low	Outgoing affiliation Financial analysis	Variety and challenge					
ESTP		ESFP		ENFP		ENTP	
High	Visible autonomy	Happy family				Variety and challenge Visible autonomy	
Low	Achievement within system Financial analysis	Achievement within system		Achievement within system		Happy family	
ESTJ		ESFJ		ENFJ		ENTJ	
High	Achievement within system	Happy family Outgoing affiliation Business sociability		Nurturing affiliation		Financial analysis	
Low						Nurturing affiliation	

Note: N = 805. Empty cells indicate that this type did not score among the highest or lowest types for any value.

Source: Mitchell (1981). Adapted with permission.

(understanding, appreciation, tolerance, and protection for the welfare of others and nature), Security, Conformity, and Tradition. The only difference for ENTPs was that adults had lower scores on Conformity than did students. Adult ENFPs were less interested in Achievement and Power and more interested in Tradition than were the students of that type. ISFJ and ESFJ adults scored higher on Tradition than did students. These results suggest that the values of each type may change and develop over time.

Culture

Morris and Cooley (2015) analyzed the MBTI global sample consisting of 16,773 individuals across 20 countries and 18 languages by comparing MBTI results to a variety of independent country-level measures. These analyses attempted to replicate the findings by Bartram (2013), who showed that country-level means and standard deviations (SD) using Big Five personality scales strongly predicted independent country-level measures of Hofstede's (2003, 2004) cultural dimensions and other indicators (e.g., cultural tightness, global competitiveness). The current study generally replicated Bartram's results for SDs. Because the standard deviations

of the continuous scores of the four MBTI preference scales were highly correlated, they were combined into a single index ($\alpha = .89$), which was then correlated with Hofstede's measures of cultural variables. The results showed that greater variance in MBTI scores was strongly associated with more positive results at the country-wide level on most variables. Specifically, the composite MBTI standard deviation index was significantly correlated with Hofstede's power distance ($r = -.78$), pragmatism ($-.70$), individualism (.76), and indulgence (.62), but not with masculinity/femininity ($-.10$) or uncertainty avoidance ($-.21$). Standard deviations on the MBTI assessment were also correlated with other variables: cultural tightness ($r = -.52$), a human development index (.74), overall life satisfaction (.76), economic freedom (.89), a global peace index ($-.50$; higher scores indicating less peace), globalization (.65), proportion atheist ($-.78$), and good governance (.74). The current findings suggest that MBTI scores can account for significant variance in a variety of country-level measures. Measures of variance on the MBTI scales, in particular, may be useful in explaining and understanding cultural differences.

Research suggests that MBTI scores can account for significant variance in a variety of country-level measures.

Hackston (2015) examined cultural and personality differences using a sample of 1,029 people from a wide range of countries. (Countries with the greatest frequency of respondents were the United States, India, the United Kingdom, Australia, South Africa, Germany, the Netherlands, and France.) The measure used was the *Cultural Orientations Framework*, which measures 17 dimensions of culture. The first analysis examined type differences on the 17 dimensions. All of the hypothesized relationships between cultural orientations and preference pairs were supported. For example,

As expected, Extraverts are more oriented towards *Sharing* than Introverts, preferring to build close relationships and sharing personal information and feelings with others. Extraverts also have a clearer preference for displaying emotions and warmth when communicating (*Affective*).

Intuitive types prefer to look towards the *Future*, and like to promote a far-reaching vision, more so than Sensing types. Intuitive types also rely on implicit communication (*High Context*) and appreciate the meaning of gestures, posture, voice and context, whereas Sensing types have a clearer preference for relying on explicit communication (*Low Context*) and favor clear and detailed instructions.

Thinking types are more *Direct* and in conflict they prefer to get their point across clearly even at the risk of offending or hurting someone, whereas Feeling types are more *Indirect* and would rather maintain a friendly relationship, at the risk of being misunderstood. When communicating, Feeling types prefer to display emotions and warmth (*Affective*), whereas Thinking types stress conciseness, precision and detachment (*Neutral*).

Judging types value *Stability* and prefer static and orderly environments, they are also *Monochronic* and prefer to concentrate on one activity and/or relationship at a time. Additionally, they are *Formal* in their communications and observe strict protocols and rituals. In contrast, Perceiving types value *Change* and prefer to work in dynamic and flexible environments, they are *Polychronic* and prefer to concentrate on multiple activities and/or relationships at a time. Additionally, they are *Informal* in their communications and favor familiarity and spontaneity. (p. 11)

Differences between participants with different dominant processes were also examined. All cultural orientations except for Direct vs. Indirect and Deductive vs. Inductive showed significant differences between dominant processes. For example, participants with dominant extraverted Thinking were more oriented toward Control (versus Harmony or Humility) and were more likely to see resources as scarce (rather than plentiful). Dominant extraverted Perceiving types were more likely to be polychronic (multitasking), while dominant introverted

Perceiving types were more monochronic (focus on single activity). Introverted Thinking and introverted Intuitive types were the most individualistic, whereas extraverted Sensing and extraverted Feeling types were more collectivist. Extraverted Feeling types were more likely to take an Affective approach.

The most frequent type in the sample was ENFP, so across-country differences were analyzed for this type in the largest samples. Hackston found many more similarities than differences in the cultural orientations of ENFPs across countries. Another across-country analysis compared ENFPs with their opposites, ISTJs. These two types were significantly different on all 17 of the cultural orientations. From these two analyses, Hackston concluded that ENFPs from different cultures may have more similarities than do ENFPs and ISTJs from the same culture.

In another test of the relative importance of personality versus culture, differences in cultural orientation were analyzed by country of residence, country of origin, job level, gender, age, and MBTI type. As reported above, for all orientations except Direct vs. Indirect and Deductive vs. Inductive, there were significant differences across type preferences. Furthermore, for 6 of the orientations, type preference was the only significant effect, and for the remaining 9, type preference was the most significant effect.

On 15 of 17 cultural orientations studied, there were significant differences across type preferences. Furthermore, for 6 of the orientations, type was the only significant effect, and for the remaining 9, it was the most significant effect.

Van Zyl and Taylor (2012) evaluated the MBTI Form M assessment in a South African context. The sample consisted of 10,705 South Africans (5,909 men and 4,651 women) who completed Form M between 2004 and 2010. The authors found excellent reliabilities, with alpha coefficients ranging between 0.88 and 0.92 for the total sample. They also used differential item functioning analysis to examine whether the performance of the items differed across groups. They concluded that the items did not appear to be biased across gender or ethnic groups and that the overall results demonstrated that the MBTI Form M assessment could be used reliably across a variety of ethnic and racial groups in South Africa and worked equally well for men and women.

Money and Finances

Schaubhut, Long, and Thompson (2011) examined financial attitudes and behaviors of individuals with opposite preferences as measured by the MBTI Form M assessment. A sample of 1,214 individuals (69% female) completed a financial survey with four scales: Financial

Worry, Financial Efficacy, Financial Tracking, and Financial Engagement. No significant differences were found between Extraversion and Introversion for any of the four financial scales. On the Financial Worry scale, participants with a preference for Sensing tended to worry more about finances. Significant differences between Thinking and Feeling were found on all four scales: participants with a preference for Feeling tended to worry more, while those with a preference for Thinking were more engaged, more likely to track their finances, and more likely to have higher financial self-efficacy. Significant differences between Judging and Perceiving were found on the Financial Tracking scale and the Financial Engagement scale: participants with a preference for Judging averaged higher scores on tracking and engagement. Effect sizes for these differences were in the small range. The authors discuss the implications of the results for financial advising.

VALIDITY OF WHOLE TYPES AND TYPE DYNAMICS

A number of the analyses and studies discussed earlier in this chapter provided validity evidence not just for the individual preferences but also for combinations of preferences, statistical interactions of preferences, interactions of preferences with other variables, and whole types. This section presents additional findings derived from studies or analyses specifically designed to examine the validity of whole types and/or combinations of preferences based on type dynamics theory.

The evidence for this category includes the following:

- Comparison of MBTI reported type with self-estimated type
- Descriptions of whole types by independent raters
- Interactions among preference scales
- Analyses of type dynamics using observer ratings, ANOVA, and Step II facet scales

Comparison of MBTI® Reported Type with Self-Estimated Type

One method for validating the MBTI Step I assessment is to compare MBTI results with self-assessment of type preferences. Two kinds of studies have been conducted in this area, differentiated on the basis of how the individual's self-assessment of type was determined. One category includes assessments that are based on agreement with written type descriptions by Myers and others. The other category includes self-estimates of type based on a verification process such as is described in chapter 6.

In the first category, the results provide evidence for the existence of the typology to the extent that the type

descriptions are based on the Jung-Myers theory of types. Comparisons with descriptions written by Myers and her colleagues are particularly important because the descriptions were purposely constructed to include statements based on type dynamics theory. Furthermore, these type descriptions were intentionally structured to parallel the dynamics of each type, as shown in chapter 4. When reviewing these results, it is important to keep in mind that the likelihood of an individual correctly picking his or her reported type by chance is 1 in 16, or 6.25%.

A sample of individuals ($N = 1,886$) who had completed the MBTI Step I assessment volunteered to take part in a study (Schaubhut, 2013a) in which they were asked to read type descriptions taken from the *Introduction to Type®* booklet (Myers with Kirby & Myers, 1998) and rate each one on a 5-point scale as to how well it described them, from "very little" to "very much." Participants read a type description that matched their reported type on all four preferences; descriptions that differed by one, two, and three preferences from their reported type; and a description that was the opposite of their reported type (i.e., differed on all four preferences). The data were analyzed separately for each of the 16 types. For all 16 types, the type description that matched the participants' reported type was rated higher on average than any other description, even the description that differed from their reported type by only one preference. For 13 of the 16 types, the average ratings increased incrementally from the description that was opposite the reported type to the description that matched the reported type. Sixteen ANOVAs were run using the number of matches as the independent variable and the ratings as the dependent variable. All the ANOVAs were significant. For 2 types, INTJ and INFJ, the effect sizes were small; for 9 types there was a medium effect size; and for 5 types (ISTJ, ESFP, ENFP, ENTP, and ESTJ) there were large effect sizes.

Anderson and Kulas (2013) also investigated the accuracy of type descriptions by creating 16 descriptions of hypothetical managers based on accounts of the 16 types of leaders available in the type literature. These descriptions included characteristics of a manager of a given type, his or her leadership style, and his or her potential contributions and pitfalls. Based on the descriptions, these hypothetical managers were rated on managerial skills using the *Benchmarks* assessment. *Benchmarks* ratings of over 4,000 real managers by bosses, peers, and subordinates were also obtained. The ratings of the hypothetical managers were then compared to the ratings of the real managers for each of the 16 types separately. The average correlation between the two sets of ratings was $r = .50$. The only outlier was INFJ, for which the two sets of ratings correlated at $r = .01$.

McPeck and Breiner (2013) administered MBTI Form M and the MMTIC assessment (Murphy & Meisgeier, 2008) to a sample of teenagers ($n = 123$). As part of the feedback session, the participants were given two type descriptions composed of 12 adjectives and short phrases drawn from the 16 type descriptions found in Lawrence (1998) and in Myers' *Introduction to Type*® booklet (1980). When the MBTI and MMTIC results disagreed, participants were given two type descriptions that corresponded to the results from the two assessments. When the results of the two assessments agreed (i.e., yielded the same four-letter type), participants received one description that matched their type and one that did not. The mismatched type description was randomly selected to be either a description of the opposite type (i.e., all four preferences were different from the reported type) or a description that differed from the reported type by only one preference. Participants were asked to rate the accuracy of each of the type descriptions they received. The mismatched descriptions were rated as much less descriptive than the matched ones, even when the mismatch involved only a single preference, although the difference was much greater when all four preferences differed from the reported type.

Carskadon (1982) and Carskadon and Cook (1982) conducted studies of 129 and 118 introductory psychology students, respectively, who had completed the MBTI assessment. In each study, participants were asked to rank five of Myers' type descriptions from the *Introduction to Type*® booklet: (1) the type reported for the participant by the MBTI assessment, (2) the type with the letter of the participant's least clear preference changed to the opposite preference, (3) the type with the participant's reported processes staying the same but E-I and J-P changed to their opposites (e.g., ESTJ changed to ISTP), (4) the type with processes the same but S-N and T-F changed to their opposites (e.g., ESTJ changed to ENFJ), and (5) the type with all four preferences opposite those in the reported type (e.g., ESTJ changed to INFP). The reported type was ranked first significantly more often than chance for both samples (35%, $p < .001$, and 50%, $p < .001$).

When choices for the descriptions of the reported type and the type with the least clear preference reversed were combined, the percentages of participants ranking one or the other of these descriptions first were 66% for the first study and 75% for the second study. The percentages of participants giving first rank to the opposite of the reported type were 4% and 13% in the two studies, respectively. Participants in both studies were significantly less likely ($p < .001$) to select as most accurate the description differing from the reported type on the processes (S-N and T-F) than they were to choose the type differing from the reported type on the orientations (E-I and J-P). The authors concluded that

(1) the idea that type descriptions other than one's reported type might be equally appealing had been refuted, and (2) their participants seemed to feel more confidence or certainty in their preferred processes (S-N and T-F) and/or more flexibility in their orientations (E-I and J-P). One critique of the MBTI assessment is based on the belief that people would claim that the content of any (untitled) horoscope accurately describes them and that the same would be true of any MBTI type description. The findings reported above strongly contradict this suggestion.

In a similar study, Ware and Yokomoto (1985) asked students who had completed the MBTI assessment to rate, on a scale of 0% to 100%, the accuracy of different type profiles developed by David Keirsey (Keirsey & Bates, 1978). Unbeknownst to the participants, one of these profiles matched their reported type as determined by the MBTI assessment, while the others varied in systematic ways. The type description based on participants' MBTI type was rated highest, at 62%. The description yielding the next-highest rating (47%) was the one that resulted from changing the preference associated with participants' lowest preference score to its opposite.

Another study that indirectly addresses the issue of self-perceived accuracy of type asked people to recall their whole type two weeks to three months after the MBTI Step I assessment had been administered to them (Thompson, Walling, et al., 2013). Sixty-five percent of the respondents accurately recalled their whole type. Within this group, the types that had the most accurate recall were types whose preferences included Intuition. The three types with the highest accuracy were INTJ, ENFP, and ENTJ. The three types with the lowest accuracy were ISFP, ESFP, and ISTP.

The second method for validating MBTI results via self-estimates focuses on studies that arrive at a self-estimate of type as the result of a verification process. This self-estimate, which has been referred to as the individual's "best-fit type," is then compared to the type reported by the MBTI assessment based on the individual's responses to the items. The results are reported as a percent agreement. Studies of Form G (Myers et al., 1998) showed that the agreement rate ranged from 53% to 85% with an average of 70%. The agreement rate in studies of Form M ranged from 58% to 78% with a mean of 67%. It is also interesting to note that in the few studies that reported agreement by clarity of preference, almost all of the disagreements occurred on scales for which the individual had a preference clarity index in the "slight" range (a preference score of 0–5). A number of those individuals may have reported in that range not because of a lack of measurement precision but because they were genuinely unclear about their preferences. If these individuals are excluded, the percent agreement is very

high indeed. Walck (1992) also found that over 80% of the disagreements between best-fit type and reported type occurred among individuals with “slight” preferences.

Table 9.43 shows agreement rates for a sample of 435 participants who were administered the global research version of the MBTI assessment in two countries, the United States and the United Kingdom, and then went through the online verification process embedded in MBTI[®]Complete to arrive at their best-fit type. Results are reported in terms of each possible level of agreement between reported and best-fit type, by country of residence, and for the sample combined. The combined agreement rate for participants matching on three and four preferences is also reported (78.6%). The combined agreement rate for whole type (agrees with all four preferences) was 63%, which is in line with the results from previous studies and is far above the chance value of 6.25%.

While these results help support the validity of the MBTI Step I assessment, they need to be interpreted with an understanding of their limitations, which have rarely if ever been acknowledged in discussions of best-fit type. An inherent limitation of such studies is the validity of the criterion. There was no standardized method in the verification process used to derive the best-fit type across the studies, despite the fact that the recommended process is described in manuals and taught in MBTI certification training. Differences in the verification process included, among other things, the time spent on verifying type; the materials used; whether the process occurred in a one-on-one session or in a group interpretation; and whether the respondent was given his or her reported type before, during, or after being asked to provide a self-estimate. Depending on the timing of the feedback, the reported type itself can influence the outcome in the direction of inflating the agreement rates (see chapter 7 for an in-depth discussion of this prompting effect). The materials used in the verification can also be questioned. The presentation of adjective checklists, phrases, or short descriptions to the respondent as aids to verification are essentially equivalent to the administration of “items” in

an assessment—items that have never been subjected to analyses of their reliability or validity. Estimates of best-fit type have also been shown to be subject to the influence of the type and gender of the professional doing the interpretation (Bathurst, 2000; Kummerow, 1988), characteristics of the respondent (McPeck & Breiner, 2014; Posey, Thorne, & Carskadon, 1999), and demand characteristics in the testing environment (Hammer & Treeger, 1984; Salter, Evans, & Forney, 1997). Bathurst (1999) also found that other factors affecting best-fit type estimates included the age of the respondent (older respondents had greater agreement rates), the setting of the MBTI administration (there was less agreement in corporate settings), and the type of the respondent (SJJs had higher agreement rates). So while encouraging, and useful as one piece of evidence in the nomological net, best-fit type estimates are certainly not the only or even the preferred criterion for determining the validity of MBTI Step I assessment results.

That being said, it is important to make a clear distinction between using a verification process with an individual to help him or her arrive at a best-fit type and using the agreement rate between best-fit and reported type in a sample as a criterion for validity. It is strongly recommended that a verification process be used as part of the interpretation of MBTI results to individual respondents. This has always been a unique feature of the MBTI assessment compared to other personality assessments. However, it is also incumbent on the professional to be aware of and address any factors that may influence the outcome. It is using sample agreement rates, and trying to compare them across studies, with all the innate limitations of the methodology, that requires caution.

Unique Descriptions of Whole Types

A study by Thorne and Gough (1991) was primarily focused on use of the MBTI Step I assessment in studying the interpersonal functioning of effective and creative people. The sample consisted of 614 individuals assessed at the Institute of Personality Assessment and Research (IPAR) between 1956 and 1984. The programs

Table 9.43 | MBTI[®] global sample letter agreement rates between reported and best-fit type

Letter agreement	UK sample		US sample		Combined	
	<i>n</i>	%	<i>n</i>	%	<i>N</i>	%
Agrees with 0 letters	2	1.0	1	0.4	3	0.7
Agrees with 1 letter	12	5.9	13	5.7	25	5.7
Agrees with 2 letters	36	17.6	29	12.6	65	14.9
Agrees with 3 letters	32	15.6	36	15.7	68	15.6
Agrees with 4 letters	123	60.0	151	65.7	274	63.0
Agrees with 3 or 4 letters	155	75.6	187	81.4	342	78.6

Note: Portion of the global sample having best-fit type estimates, *n* = 435.

involved architects, student writers, research scientists, mathematicians, engineering students, medical school students, law school students, college seniors and sophomores, community residents, and Irish business executives. Ratings of the participants in each program were conducted by 62 women and 127 men on the IPAR assessment staff using a variety of scales.

The MBTI Form F assessment was used to identify type. Although data were collected using a large battery of assessments, only data from the 300-item *Adjective Check List* (ACL) are presented here. Raters checked each adjective that applied to the individual they were assessing. Between 8 and 16 raters independently rated each participant. Ratings were standardized so that ratings for each individual on each ACL item could range from 0 to 10. Interrater reliability was .76. *None of the raters knew the types of the individuals they were assessing.* Thus their ratings were completely naive with respect to type.

The authors presented tables listing the 10 most and 10 least common adjectives associated with the MBTI types. Because the emphasis here is on evidence for whole-type differences, their data were reanalyzed. From the original tables listing the 10 most and least common adjectives associated with each type, the number of *unique* adjectives for each type was counted. These counts are an indicator of the degree to which each type was seen as different by independent observers who did not know the types of the individuals they were rating. Because of the type distribution in the sample, sufficient numbers were available to interpret the results for 10 of the 16 types.

The data showed that, as would be expected, types sharing several preferences tended to be described with the same adjectives. Types sharing one or no preferences tended to be described with very few of the same adjectives. Table 9.44 presents the number of unique adjectives used by raters to describe men and women of the 10 types for which sufficient data were available. When examining the last column in this table, it is important to remember that the maximum number of unique adjectives that can be associated with each type is 20 because Thorne and Gough reported only the 10 most frequently checked adjectives for their “most descriptive” and “least descriptive” categories.

These data reveal quite pronounced differences between the types. Even types sharing several preferences are seen as noticeably different by independent observers who do not know the types of the individuals they are describing. The number of unique descriptors for a single type ranged from a minimum of 12 for ESTJ women to a maximum of 20 for INFP women.

These results are especially significant because they illustrate what is possible when measurement is not constrained by using a narrow scale covering limited content. The ACL offers a great variety of qualitative descriptors that can be used to assess subtle differences in type behavior and is thus able to detect differences spanning an array of diverse qualities. This helps bring the differences between whole types into sharp relief. In fact, while the number of unique descriptors shown for each type in the table is impressive, the actual content of these lists of unique adjectives is even more so. For example, among men, ESTJs were described as practical, frivolous, opportunistic, boastful, tough,

Table 9.44 | Number of most descriptive and least descriptive adjectives uniquely ascribed to MBTI® types by IPAR raters

Type	Men (n = 374)			Women (n = 240)		
	Number of unique items		Total number of unique items	Number of unique items		Total number of unique items
	Most descriptive adjectives	Least descriptive adjectives		Most descriptive adjectives	Least descriptive adjectives	
INFP	8	5	13	10	10	20
INFJ	9	5	14	9	9	18
INTP	9	5	14	10	9	19
INTJ	9	4	13	10	8	18
ISTJ	8	6	14	4	9	13
ENFP	8	6	14	8	7	15
ENFJ	10	8	18	9	8	17
ENTP	10	6	16	8	6	14
ENTJ	7	8	15	8	6	14
ESTJ	8	8	16	5	7	12

Note: N = 614. The maximum number of unique adjective descriptors for each type is 20.
Source: Thorne & Gough (1991), pp. 83–100.

outgoing, smug, steady, self-confident, and unscrupulous (remember, these unique adjectives included those that were both most and least descriptive). ENTJ men were described as ambitious, planful, responsible, practical, thorough, cowardly, discreet, farsighted, industrious, and reliable. These are the two dominant extraverted Thinking types. In terms of single preferences, they differ only on S and N, which are their respective auxiliary processes. Yet among the 10 adjectives most frequently used to describe both types, raters saw them as having only practicality in common. For another example, INTP men (dominant introverted Thinking) were seen as original, imaginative, complicated, hasty, rebellious, high-strung, individualistic, restless, self-centered, and temperamental. INTJ men (auxiliary extraverted Thinking) were seen as formal, deliberate, logical, retiring, serious, aloof, methodical, painstaking, thorough, and reserved. Virtually all other pairs of types were described in equally diverse terms. When studies use assessments with a sufficiently broad content, even raters naive to type can clearly see and describe differences among individuals of different types. The reader is referred to Thorne and Gough (1991) for a listing of these adjectives for each type.

The IPAR study was significant because observers naive to the type of the participants detected meaningful differences between even individuals of similar types and those differences were in accordance with type dynamics theory.

Preference Scale Interactions

For the purposes of analysis, the four MBTI preference scales can be combined in many ways, but two broad categories can be defined. In the first category are preference combinations that might be called elementary interactions. These are interactions among preferences that can provide evidence for the typological nature of the theory because they represent the elements of whole types. Although these combinations of preferences are not derived specifically from the dynamic aspects of the theory, to the extent that significant interactions among preferences can be demonstrated, they are important as building blocks of the theory. Such combinations of preferences are also of interest to practitioners in applied situations. The elementary interactions used most frequently involve the process pairs, ST, SF, NF, and NT, although combinations of three preferences have also received attention. See chapter 5 and Hirsh, Hirsh, and Hirsh (2009) for labels assigned to various letter combinations, descriptions of the characteristics associated with each, and explanations of how they are used by practitioners.

The second category of preference combinations contains those explicitly defined by the theory. These

are the combinations of all four preferences that yield a complete type, as well as combinations that reflect the dynamic aspects of the types (e.g., dominant extraverted Thinking types).

In either case, type theory predicts that the interactions of preferences result in behaviors that cannot be predicted from each preference separately. Because each preference in a given type interacts with all the other preferences in that type, type theory posits that any combination of two or more preferences will yield behaviors that are more than the sum of the parts. Validity research at this level focuses on demonstrating that various combinations of two, three, and four preferences interact to produce behavior that cannot be accounted for by the same preferences considered separately.

Combinations of preferences derived from type dynamics are specific interactions between the E–I and J–P preference pairs that serve to identify the forms (dominant or auxiliary) and orientations (introverted or extraverted) of the perceiving and judging processes (S, N, T, F). The characteristics associated with the forms and orientations of the processes transcend the set of general qualities associated with each process.

Validity research on dynamics involves finding evidence that among and within types the dominant, auxiliary, tertiary, and inferior processes and their orientations can be distinguished from each other and that the characteristics associated with each are in accordance with those predicted by type theory.

Studies of Type Dynamics Theory

Validation of type dynamics goes a step beyond asking whether whole types can be distinguished from one another. In studies of type dynamics, the focus is on determining whether particular groups of types, defined by type theory, differ in observable ways. Evidence for the validity of the MBTI Step I assessment as a measure of type dynamics would include results indicating that the forms of the processes (e.g., dominant Feeling vs. auxiliary Feeling), the orientations of the processes (e.g., introverted Sensing vs. extraverted Sensing), or both (e.g., dominant introverted Intuition vs. dominant extraverted Intuition) are different from each other. For example, we might wish to see how extraverted Thinking when it is dominant, as in ESTJs and ENTJs, differs from extraverted Thinking when it serves as the auxiliary process, as in ISTJs and INTJs. We might also seek to find differences between the dominant introverted Intuition of INFJs and INTJs and the dominant extraverted Intuition of ENFPs and ENTPs. Other comparisons might involve only the dominant versus the auxiliary form of a process. This comparison would be used to determine whether the form of a process explains a difference in behaviors or characteristics associated with certain

types. For example, for Intuition the contrast would be between the dominant Intuitive types and those who have Intuition as their auxiliary process. At other times, the comparison may involve only the two different orientations (extraverted or introverted) of the processes; this would be a test of whether orientation alone can explain differences in behaviors or characteristics among certain types. With Intuition, for example, the contrast would be between the types with introverted Intuition (either dominant or auxiliary) and those with extraverted Intuition (either dominant or auxiliary). What comparisons are analyzed depends on what questions are being asked or what hypotheses about type dynamics are being tested.

The dominant form of a process is presumed to differ from its auxiliary form. Myers' method of identifying dominant versus auxiliary processes assumed an interaction between E–I and J–P preference pairs. The impact of this interaction shows up in its effect on the forms and orientations of the judging and perceiving processes. Interactive effects of the preferences produce something above and beyond the effects of separate preferences. The characteristics of dominant introverted Feeling, for example, are different and separate from those aspects of Feeling that are shared among all Feeling types. Few authors have spelled out such differences explicitly (an exception is Myers & Kirby, 1994). This is an area of type theory that is open for future development. Nevertheless, at present the theory presumes that such differences are real and can be observed.

Fortunately, there are several ways in which differences between dominant and auxiliary forms of the processes can be observed that remain true to the theory. For example, observer ratings of dominant and auxiliary Intuitive types could be examined. Certain characteristics

observed by the raters would be predicted to be more frequently associated with individuals whose Intuition is dominant compared to those for whom Intuition is auxiliary. The dominant and auxiliary uses of the processes would be expected to differ in some, but not necessarily all, ways. Another approach might be to use ANOVA to compare the mean scores of various groups who share the same process on measures of characteristics that are relevant to that process. Such measures might include vocational interests, values, and attitudes. A number of studies and analyses employing these approaches were reported in previous sections in this chapter; others are discussed below.

TESTING TYPE DYNAMICS WITH OBSERVER RATINGS

Although Thorne and Gough (1991) did not explicitly test predictions derived from type dynamics, their data can be analyzed to provide some answers to questions about this aspect of type theory. Recall that their studies involved having observers, who did not know the types of the individuals they were observing, rate participants on a variety of behavioral and psychological measures. The characteristics that the raters ascribed to individuals who use each process in its dominant and auxiliary forms can then be compared.

Information is not available on all 16 types because Thorne and Gough had sufficient numbers for comparing only 10 types; the tests of type dynamics with respect to Sensing are not available in this data set for some analyses. Table 9.45 compares the dominant and auxiliary forms of the Thinking, Feeling, and Intuition processes. The percentages in the table are the percentages of adjectives that were common to the dominant and auxiliary types being compared in that cell of the table. These percentages range from 0.0 to 13.8, suggesting that independent observers, who did not know the

Table 9.45 | **Percentage of ACL adjectives ascribed by IPAR raters common to MBTI® types with dominant or auxiliary Thinking, Feeling, and Intuition processes**

Processes		% of adjectives in common			
		Men (n = 374)		Women (n = 240)	
		Least descriptive	Most descriptive	Least descriptive	Most descriptive
Dominant Thinking	INTP, ESTJ, ENTJ	5.4	5.4	3.7	10.7
Auxiliary Thinking	ENTP, INTJ, ISTJ				
Dominant Feeling	INFP, ENFJ	5.3	5.4	0.0	2.6
Auxiliary Feeling	ENFP, INFJ				
Dominant Intuition	INTJ, INFJ, ENTP, ENFP	13.8	4.0	6.9	5.3
Auxiliary Intuition	ENTJ, ENFJ, INTP, INFP				

Note: N = 614.
Source: Thorne & Gough (1991), pp. 83–100.

types of the individuals they were describing, clearly did not see much similarity between types having the dominant versus the auxiliary forms of Thinking, Feeling, or Intuition.

McPeck and Martin (2012) examined the degree to which type experts and authors agreed about short descriptors (adjectives or short phrases) used to define each of the eight process-orientations of type theory (i.e., the four processes of Sensing, Intuition, Thinking, and Feeling, expressed in either an extraverted or introverted orientation). One hundred and fifty-two descriptors were drawn from books and monographs written about process-orientations and from descriptions used in past research. Nine experts rated each descriptor's applicability to each of the eight process-orientations. Results indicated that there were clearly descriptors that experts agree represent distinct process-orientations; there are 72 such words and phrases. Introverted expressions of processes generated fewer consensual descriptors than extraverted expressions. The fact that 72 descriptors emerged despite the small sample size was seen as indicative of a high degree of rater consensus for at least some descriptors of each process-orientation.

TESTING TYPE DYNAMICS USING ANOVA

Another method for testing hypotheses derived from type dynamics involves ANOVA. While log-linear models are equally appropriate, if not more so, for some data sets, ANOVA was used here because it is likely to be more familiar to a broader spectrum of readers. However, if anything, ANOVA is apt to be less sensitive in detecting differences. So the results reported below probably understate the differences among type dynamics groups compared with what might have been found with other, less familiar, analytic methods.

In these analyses, the scores of individuals grouped according to type dynamics for each process should differ on variables that are relevant to their judging (T–F) and perceiving (S–N) processes. Given the multifaceted nature of type constructs, however, we should not expect that any single variable will show differences among all type dynamics groupings, even for a single preference such as Thinking: the four Thinking groups, that is, dominant introverted Thinking, dominant extraverted Thinking, auxiliary introverted Thinking, and auxiliary extraverted Thinking, are both similar and different. The similarities will affect the scores just as much as the differences will. When a variable is mainly relevant to the orientation in which Thinking is used, then we do not necessarily expect that types with dominant introverted Thinking, for example, will differ from those with auxiliary introverted Thinking on that particular variable. However, across a diverse collection of variables relevant to the Thinking process, we do expect that all type dynamics groupings will differ in some respects on at least some of the variables.

In addition to selecting the proper variables, other challenges ensue because the requirements for validating the MBTI Step I assessment as a typology are more stringent than for trait-based personality assessments not based on a theory of interactions. Validating type dynamics using ANOVA with a given dependent variable requires the following two conditions: (1) the E–I x J–P interaction term is significant, and (2) the type dynamics groups for each process differ from one another. Both of these conditions are necessary; neither alone is sufficient.

Evidence for the first condition will be immediately available from the ANOVA results. Evidence for the second condition requires that we demonstrate that the types having different forms and orientations of the four processes can be distinguished from each other on the same variables for which the E–I x J–P interaction term was significant. For example, consider an analysis that compares individuals with dominant introverted Thinking (ISTP, INTP) with individuals whose extraverted Thinking is auxiliary (ISTJ, INTJ) on a dependent variable called “importance of friendships.” From a trait perspective, the only difference between these types is their preference for Judging or Perceiving, and therefore any differences observed between these groups should be only those attributable to the difference in the J–P preference. Thus in the ANOVA, only the J–P main effect would be expected to be significant. From a type perspective, however, we seek evidence that the differences between these two groups is greater than what can be explained by the J–P preference. If this is true, then in the ANOVA, the E–I x J–P interaction term should be significant among all Thinking types. When extended to all four processes and to all the type dynamics groups within each process, this analysis is a direct test of whether the MBTI Step I assessment accurately measures the form and orientations of the processes described by type dynamics.

To test these conditions, data from research items administered to the Form M US national sample were used. For each of the four MBTI processes, an ANOVA was conducted using the E–I and J–P preference pairs as the independent variables. For each process, the interaction between these preferences defines the four possible type dynamics groups (EJ, IJ, EP, and IP). The dependent variables were a series of items asking respondents about their satisfaction with various aspects of their work environment, the importance to them of various life values, and their preference for working in various organizational structures. Items for which the interaction term was significant were identified for further analysis. The second analysis involved computing Cohen's *d* for the type dynamics that were different. Recall that in all the ANOVAs a significant interaction means that the main effects for E–I and J–P have been accounted for. In other words, the E–I x J–P term indicates the amount of variance in the dependent

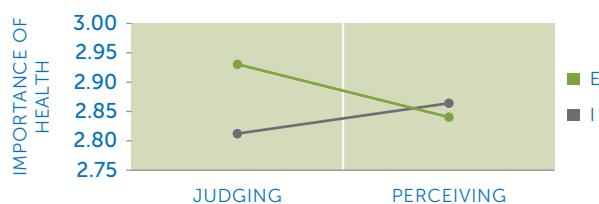


Figure 9.10 | Importance of health for Feeling types: E–I x J–P interaction

Note: Portion of Form M US national sample, $n = 428$.



Figure 9.11 | Importance of relationships/friendships for Feeling types: E–I x J–P interaction

Note: Portion of Form M US national sample, $n = 428$.

variable that is accounted for over and above any variance attributable to E–I or J–P independently.

Figures 9.10 and 9.11 show the results for the Feeling process on the variables “importance of health” and “importance of relationships/friendships” using a 4-point scale from “not important” to “very important.” Both analyses yielded significant results for the E–I x J–P interaction at $p < .05$. For the health variable, the ratings of importance were different for the extraverted and introverted Judging types. The EJs, the dominant extraverted Feeling types (ENFJ, ESFJ), differed from the IJs, those with auxiliary extraverted Feeling (INFJ, ISFJ). The effect on the importance of health of extraverted Feeling being dominant versus auxiliary was small to medium ($d = .33$). For the relationship/friendship variable, the difference was between the extraverted and introverted Perceiving types. The EPs with auxiliary introverted Feeling (ESFP, ENFP) said that friendships were more important than did the IPs with dominant introverted Feeling (INFP, ISFP). The effect of introverted Feeling being either dominant or auxiliary was in the medium range ($d = .57$). These two examples show that for both extraverted Feeling and introverted Feeling, Feeling being in the dominant or the auxiliary role makes a difference in the importance of health and friendships to the individual.

Figure 9.12 shows the results for individuals who prefer Thinking on the variable “satisfaction with salary” based on a 4-point scale from “very dissatisfied” to “very satisfied.” The E–I x J–P interaction term was significant at $p < .08$. The difference was between the EJs with dominant extraverted Thinking (ENTJ, ESTJ) and the IJs with auxiliary extraverted Thinking (ISTJ, INTJ). The effect of the different role for the Thinking process (either dominant or auxiliary) on satisfaction with salary was in the medium range ($d = .47$).

Figure 9.13 shows the results for Sensing on the variable “importance of religion/spirituality.” The E–I x J–P interaction term was significant at $p < .02$. The difference was between the EJs with auxiliary introverted Sensing (ESTJ, ESFJ) and the IJs with dominant introverted Sensing (ISTJ, ISFJ). The effect on the religion/spirituality variable of dominant versus auxiliary introverted Sensing was $d = .42$.

Figure 9.14 shows differences on both of the E–I and J–P combinations for Intuition. The variable was “satisfaction with opportunities for learning,” for which the E–I x J–P interaction was significant at $p < .04$. For the Intuitive Judging types, the difference was between those with dominant introverted Intuition (INTJ, INFJ) and those with auxiliary introverted Intuition (ENTJ, ENFJ), with the latter rating their satisfaction higher. The effect was in the medium range ($d = .48$). For the Perceiving types, the difference was between those with dominant extraverted Intuition (ENTP, ENFP) and those with auxiliary extraverted Intuition (INTP, INFP), with the latter group being more satisfied. The effect was in the small-to-medium range ($d = .33$). This is one of the rare cases where some Introverted types are more satisfied than the Extraverted types, and uncovering it required an analysis based on the dynamic theory.

Other results from this sample shed additional light on the ways in which type dynamics groups are both similar and different. In the first stage, there were generally more instances in which the main effects (due to the E–I and J–P preference pairs) were significant compared with the number of significant interactions (E–I x J–P). Of the five variables showing significant interaction effects, four also had significant main effects due to E–I, J–P, or both. This pattern of results tells us that (1) sometimes the differences among type dynamics groups are mainly due to E–I or J–P differences; (2) sometimes the differences among type dynamics groups are mainly due to interaction between E–I and J–P; and (3) sometimes the differences among type dynamics groups are due both to the effects of single preferences and to the interaction of those preferences. However, just comparing the number of significant main effects to the number of significant interaction terms can be misleading. As Cohen (1988) points out, tests of interactions have inherently lower power than do tests of main effects at the same level of



Figure 9.12 | Satisfaction with salary for Thinking types: E–I x J–P interaction

Note: Portion of Form M US national sample, $n = 181$.

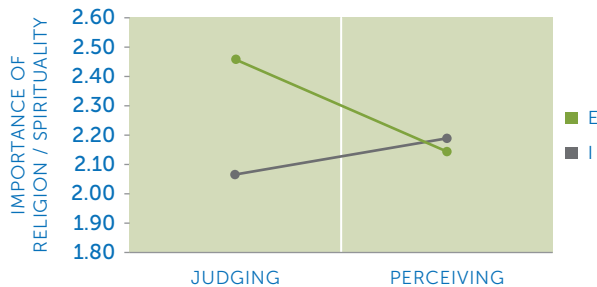


Figure 9.13 | Importance of religion/spirituality for Sensing types: E–I x J–P interaction

Note: Portion of Form M US national sample, $n = 502$.

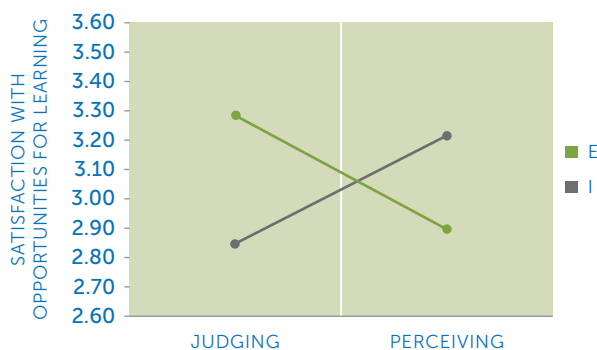


Figure 9.14 | Satisfaction with opportunities for learning for Intuitive types: E–I x J–P interaction

Note: Portion of Form M US national sample, $n = 143$.

significance. So it is possible that the above discussion underestimates the number of significant interactions.

Additional results of this study demonstrate both a statistically significant interaction term in the first-stage ANOVA and significant differences among the type dynamics groups in the multiple comparison tests. This result suggests that the effects of dissimilar aspects of type on the dependent variable are being observed within the groups that differ. The results for the variable “importance of health” demonstrate this. The E–I x J–P interaction term was significant for Thinking types on this variable. The mean for the auxiliary extraverted

Thinking types (ISTJ, INTJ) was 3.61, while the mean for the dominant extraverted Thinking types (ESTJ, ENTJ) was 3.83. The E–I main effect was significant, but because this effect is separate and independent from the significant interaction effect, it does not explain the difference in scores between these groups. From a strictly statistical viewpoint, it can be concluded that there are some characteristics of these types beyond the E or I preferences that explain how much importance one attributes to health in one’s life.

These findings make sense from the theory because Myers used the J–P preference pair to help identify or point to type dynamics groups. This is not to say that the E–I and J–P combinations bring about or cause type dynamics to appear. In this study, the results are not interpreted as the E–I x J–P interaction causing the importance one attaches to health. Rather, the interaction points to certain forms and orientations of Thinking (as defined by type dynamics) that influence how important health is in one’s life. In this example, the statistical interaction term corresponds to the characteristics of auxiliary extraverted Thinking and dominant extraverted Thinking.

The observed difference between dominant extraverted Thinking types and auxiliary extraverted Thinking types illustrates the working of a component that cannot be explained solely by differences in the preference pairs. This exemplifies a dynamic aspect of type.

In conclusion, Myers explicitly asserted that combinations of E–I and J–P preferences would reflect differences in personality that could not be explained from knowing the preferences separately. And this is precisely what the results in these examples demonstrate. These outcomes thus serve as partial evidence for the validity of dynamic and typological aspects measured by the MBTI Step I assessment. Finally, the results suggest that the method used to identify type dynamics (i.e., the E–I x J–P interaction) corresponds to aspects of type beyond the four simple preference pairs.

TESTING TYPE DYNAMICS USING STEP II™ FACET SCALES

Mitchell (1991) used facet scores from a prior form of the MBTI Step II assessment as dependent variables to test explicit hypotheses about type dynamics; thus this study is discussed here. (See chapter 10 for a discussion of the validity of the Step II facet scales.) Using type theory, Mitchell identified two groups of types defined by the dominant and auxiliary forms of the four processes, orientations in which the dominant and auxiliary forms of the processes are used, and orientations in which

processes are used among Extraverts and Introverts. He looked for mean differences on the facet scales in each group, using gender as another grouping variable and controlling for age.

In the first analysis, where the grouping was by dominant and auxiliary forms of the four processes, Mitchell found measurable differences on at least one Step II facet scale between types having dominant and auxiliary forms of all four processes: for example, men whose dominant process was Thinking (ISTP, INTP, ESTJ, ENTJ) scored more toward the Tough pole of Tough–Tender than did men for whom Thinking was auxiliary (ISTJ, INTJ, ESTP, ENTP). Women whose dominant process was Intuition (INFJ, INTJ, ENFP, ENTP) scored more toward the Abstract pole of Concrete–Abstract and more toward the Enthusiastic pole of Enthusiastic–Quiet than did women for whom Intuition was auxiliary (INFP, INTP, ENFJ, ENTJ).

For the second analysis, in which grouping was by orientations using dominant and auxiliary forms of the processes, Mitchell found significant differences for every process. For example, among men who are dominant Thinking types and women who are dominant Feeling types, the men who extravert their dominant Thinking (ESTJ, ENTJ) and the women who extravert their dominant Feeling (ESFJ, ENFJ) score more toward Questioning than do the men who introvert their dominant Thinking (ISTP, INTP) or the women who introvert their dominant Feeling (ISFP, INFP). In other words, extraverting a dominant judging process seems to be related to adopting a questioning stance when faced with differences of opinion.

Table 6.10 provides additional evidence of the dynamic differences. The table shows the Step II facet scale within each preference pair with the highest mean score for each type in the MBTI global sample. No two of the 16 types have the highest mean on the same four facet scales. But what about types that share a dominant process and an auxiliary process, but in a different orientation? Note that by definition, for each of the eight dominant processes, the two types that share that dominant process differ by only one preference, which is their auxiliary process. For four of the eight pairs of types that share the same dominant and auxiliary processes—dominant Sensing with auxiliary Feeling (ESFP, ISFJ), dominant Intuition with auxiliary Thinking (INTJ, ENTP), dominant Thinking with auxiliary Sensing (ISTP, ESTJ), and dominant Feeling with auxiliary Sensing (ISFP, ESFJ)—the Step II facet scales with the highest means differ for both of the preferences that they share (they also differ, of course, for the preferences that they do not share). So, for example, ESTJs (dominant extraverted Thinking with auxiliary Sensing) have their highest mean scores on the S–N facet scale Concrete–Abstract (toward the Sensing pole, Concrete) and the T–F facet scale Reasonable–Compassionate (toward the Thinking pole, Reasonable). But ISTPs (dominant introverted Thinking with auxiliary

Sensing) have their highest mean scores on the S–N facet scale Realistic–Imaginative (toward the Sensing pole, Realistic) and the T–F facet scale Logical–Empathetic (toward the Thinking pole, Logical). For the other four pairs—dominant Sensing with auxiliary Thinking (ISTJ, ESTP), dominant Intuition with auxiliary Feeling (INFJ, ENFP), dominant Thinking with auxiliary Intuition (ENTJ, INTP), and dominant Feeling with auxiliary Intuition (ENFJ, INFP)—the highest facet scale for the dominant process is the same but the highest facet scale for the auxiliary process differs, even though the auxiliary process is the same for each pair of types. For example, the two types with dominant Sensing and auxiliary Thinking (ISTJ and ESTP) both have Concrete–Abstract as their highest S–N facet scale, but ISTJs have Logical–Empathetic and ESTPs have Reasonable–Compassionate as their highest T–F facet scale.

A stronger test is to compare the two different types that share each of the eight dominant processes; these types differ only in their auxiliary process. For three of these eight dominant processes (introverted Sensing, extraverted Sensing, and extraverted Intuition), the two types within each dominant differ in terms of which facet scale has the highest mean score across the three preferences that these types have in common. For example, dominant introverted Sensing types, ISTJs and ISFJs, differ on which facet scale has the highest mean for the three preferences they have in common, I, S, and J. (They of course also differ on T–F, but that is because they have different preferences on this preference pair.) This is true even for the preference pair that defines their dominant process. For example, for these same two dominant introverted Sensing types, the highest mean facet scales in the S–N preference pair differ: Concrete–Abstract for the ISTJs and Traditional–Original for the ISFJs. For three more of the eight dominant processes (introverted Intuition, extraverted Feeling, and extraverted Thinking), the two types differ in which facet scale is highest for two of the three preferences they have in common. And for two of these three dominant processes, the preference with a different facet scale is the one that defines their dominant process. For example, the introverted Intuitive types differ on which facet scale is highest for their dominant Intuition: Concrete–Abstract for the INFJs and Experiential–Theoretical for the INTJs. For the remaining two dominant processes (introverted Thinking and introverted Feeling), the highest mean facet scale differs on one of the three common preferences. These results suggest that there is something unique about the types, even between those that share a dominant process and differ only on their auxiliary process. So, for example, dominant extraverted Intuition with Feeling types (ENFP) can be described by their highest facet scores as Enthusiastic, Abstract, Empathetic, and Casual; dominant extraverted Intuition with Thinking types (ENTP) as Initiating, Original, Reasonable, and Spontaneous.

Type dynamics is one of the most interesting, useful, and complex aspects of type theory. The research discussed above is suggestive and provides some support for the dynamic approach. For a much more in-depth treatment of this subject, which is beyond the scope of this manual, see Mitchell (2001, 2006). For a contrary viewpoint, see Reynierse (2009).

OUTCOME STUDIES

This section considers the effect on various outcome measures of using the MBTI Step I assessment as an intervention. Given the long history of applying the MBTI assessment in a variety of settings, it is surprising that there are so few published outcome studies. However, well-designed outcome studies are difficult to conduct in real-world settings. As with all outcome studies, certain factors have to be considered that may substantially influence interpretation of the results:

- How “strong” is the MBTI intervention? It is unlikely that significant measurable outcomes could be achieved on the basis of a brief and surface exposure to MBTI results.
- Was there a control group or a treatment group unrelated to the MBTI assessment used for comparison?
- Was there random assignment of participants to alternative groups that would serve to eliminate or mitigate other sources of systematic variance?
- Is it likely that the chosen dependent variable would be significantly affected by the MBTI information or materials?
- What is the power associated with the size of the sample and the criterion level used to determine statistical significance? In other words, what is the likelihood that the test will reveal predicted effects, if they exist?

In most settings outside the psychology laboratory, with employed adults instead of college students, it is difficult if not impossible to conduct a study with large numbers of individuals who have been randomly assigned to different groups and exposed to MBTI results in a substantial manner for a sufficient length of time, and then measure outcomes using dependent variables that are sensitive to the effect of the intervention as well as meaningful to the organization. Real-world settings provide natural limits on each of these factors.

Outcomes in Education

A series of studies by McPeck and his colleagues investigated the impact on a variety of educational variables of using the MBTI assessment in secondary schools. McPeck, Urquhart, Breiner, Holland, and Cavalleri (2011) examined the effect of type in the sixth

grade with 26 teachers and 253 students. Since random assignment was not possible, a pretest-posttest design was used in which the academic performance of two different student cohorts was measured before and after a group of teachers received type training. Students also received instruction in the basic concepts of type. The teachers took MBTI Form M and students completed the *Murphy-Meisgeier Type Indicator for Children* (MMTIC; Murphy & Meisgeier, 2008). The teachers received eight hours of training on their MBTI results and applications for the classroom. The type intervention resulted in significant grade improvements for sixth graders compared to prior-year students who took the same courses from the same teachers. These improvements were more apparent for on-level students; the advanced students evidenced a smaller increase, which may have been the result of a ceiling effect (i.e., the advanced students had little room to increase since they were already receiving top grades). Because of the small sample size, below-level students were excluded from the analysis. While the intervention did improve grades, there was no evidence of an impact on standardized test scores.

McPeck et al. (2013) tested the effect of teacher type training in a large-scale study in 10 schools in four states: 4 elementary, 4 middle, 1 high, and 1 K–8 school. Data were collected for 129 teachers and more than 2,600 students. In 4 schools, there were significant, albeit small, positive effects on student grades following the implementation of type training with teachers. Data from 6 other schools, which were suspect due to poor adherence to agreed-upon study protocols, showed neither grade improvements nor declines. There was no consistent post-training effect on standardized test scores. The study found some evidence for grade improvements in the wake of teacher training in type principles in schools where the training and study procedures were properly executed.

Reeder and McPeck (2011) randomly assigned 40 fourth-grade students to two reading instruction classes. Each class received 10 hours of instruction. One class was taught using a type-based curriculum first and the school’s standard instruction second. The order was reversed for the second class. Neither the teachers nor the students were administered the MBTI assessment (the students were below the age for which the assessment is recommended), as this study was designed as a test of the application of type theory to educational outcomes. The students’ mean reading comprehension scores were significantly higher after type-based instruction compared to the standard instruction, as was the change from before to after the instruction. In interviews after the study, teachers noted improvements in their students’ engagement, motivation, focus, retention, and confidence. They rated 35% of the students as

significantly affected and 63% of the type-related materials as very useful.

A school district in the Midwest created a freshman academy to house all ninth-grade students (Schaubhut, 2013b). All students ($N = 508$) were administered the MBTI assessment at the beginning of the year as well as a series of outcome measures, which were also administered at the end of the school year. Students received a 50-minute interpretation of their MBTI assessment results in groups. Over the course of the year students participated in two additional sessions, one focused on type and careers and one on type and learning styles, during which students were assisted in developing an action plan and a career pathway. Teachers and administrators also participated in separate in-service training on type concepts. Students with Extraversion and Thinking preferences reported being involved in more school activities than their counterparts. Significant differences between scores at the beginning of the year and those at the end were observed on measures of self-acceptance, social needs (indicating that more students' social needs were being met after one year), knowledge needs, and satisfaction with education. Preliminary achievement data, available after one semester, showed that the grade point average (GPA) for this cohort the year before (in eighth grade) was 2.85; the GPA after one semester in the academy was 3.09. Based on past experience at the school, the anticipated number of disciplinary actions for the year was estimated to be 500 for this class. After one semester, the number was 136, which, if the trend were to hold, would be a significant reduction. While these results are very promising, it is of course possible that other factors may have influenced the outcomes.

Fischetti and Mentore-Lee (2001) provided a brief report of the effect of using the MBTI assessment with tenth graders in a collaborative English and social studies class. What is worth noting is the extent and breadth of the use of the MBTI assessment and type information. Each teacher had previously taken the MBTI assessment and participated in a four-hour workshop on type, as well as a two-day summer workshop on how to teach their students about type and also how to use type in their day-to-day instruction. During the school year, meetings were held with the teachers to offer them support and further their development of teaching with type. Parent workshops were also offered. Teachers used their type knowledge and their knowledge of their students' types throughout the year. The authors claimed that student grades improved in English during the one-year period following use of the MBTI assessment, although they provided no data to support this claim.

Kise (2005) focused her intervention efforts on teachers rather than students. She coached them to recognize and meet the needs of their own type, as measured by the MBTI assessment, and to identify methods to reach

students who were of a different type. Language arts students were given two different assignments, one that used standard worksheets and one that used worksheets designed using type information. The percentage of students who received failing grades on the assignment was 26% for those who used the standard worksheets and 8% for those who used worksheets designed using type information. Likewise, the percentage of students in the two groups who received an A was 9% and 36%, respectively.

Kitchens and Marsh (2007) tested the effect of using the MBTI assessment in conjunction with college math classes. Students in two math classes who took a seminar focused on using MBTI results to identify learning styles were compared to students in six math classes who did not take the seminar. For example, in the MBTI seminar, the primarily SJ students were taught how to use Intuition to solve math problems. Also, Feeling types were encouraged to discuss their anxiety about math. All of the math classes were taught by the same teacher using the same materials. The classes linked with the MBTI seminar significantly outperformed the other classes. The percentage of As among those who took the seminar was 50%, compared to 34% among the other students. The percentage of F grades was 2% and 13%, respectively. Within the two math classes that were linked to the MBTI seminar were some students who did not take the seminar due to schedule conflicts or other reasons, so a within-class comparison was also available. Again, the differences were significant; the percentage of As among those who took the seminar was 58%, compared to 31% among the other students. The percentage of Ds was also less among the students who took the seminar (3% versus 25%), as was the percentage of Fs (0% versus 6%). The outcome was deemed so successful that the format was replicated with succeeding classes.

DeMoss, McCann, and Belcher (2000) reported on a freshman-year business course that used the MBTI assessment to examine learning styles, teamwork, communication, time management, choice of major, and career choice. Although they conducted no significance tests, they reported that the attrition rate for students in the class was 16% after one year compared to 23% for the university as a whole. After four years it was 29% for those who participated in the class versus 39% university-wide.

Burt (2005) used a one-group pretest-posttest design to investigate whether undergraduate students ($N = 16$) changed their problem-solving approach after a three-day workshop that included use of the MBTI assessment to help students identify their own problem-solving process—that is, the order in which they used the four mental processes and the amount of time they spent using each one. There was a significant relationship between students' MBTI type and the problem-solving approach they used. Interestingly, students had less

confidence in their problem-solving abilities after the intervention. The author attributes this to the fact that students were exposed for the first time to the complexity of problem solving as a result of the intervention. Another explanation for the results was that students may not have been able to sufficiently integrate the new problem-solving process with their existing process, which may have caused lower confidence. So while the MBTI results were related to the use of different problem-solving approaches, the approaches used by the students were fairly stable, possibly due to insufficient time for the intervention to become effective or lack of reinforced activities. It is also possible that this relationship may be difficult to manipulate. The students may have found that their preferred style of perception and judgment, and the problem-solving process that follows from these preferences, can be difficult to change when engaged in problem solving. A longer time frame or more intensive use of type-related problem-solving activities may be needed to affect the relationship between problem-solving style and outcomes. It is also quite likely that any significant effects that did exist were not found due to the very low power of significance tests applied to such a small sample.

The MBTI assessment has also been used in college as a tool to design residential environments (Schroeder & Jackson, 2003; Schroeder, Warner, & Malone, 1980). Students were grouped on dormitory floors by dominant process. This grouping provided the support that students needed to balance the challenge of the university. Outcome variables included retention at the university and amount of physical damage to the dormitory. The results were dramatic. In a three-year period, "retention from spring to fall terms increased by 30 percent and damages decreased by 360 percent" (p. 293). At another college, matching roommates by dominant process resulted in a first-year decrease in requests for roommate changes of over 65%, which likely resulted in considerable savings of both time and money for the residence hall staff.

One interesting pattern that calls attention to the complexity of roommate matching was seen in a study by Jackson (1985). He reported that identical matches (i.e., students had the same dominant and auxiliary processes), while successful, were superseded in terms of higher student grades by complementary pairings (dominant the same, auxiliary different). These findings are similar to those reported earlier on teacher-student matching by type (Boyd, 1995), whereby matching students with teachers of the same type was less successful than when one MBTI dichotomy was different.

Outcomes in Career Exploration

Katz, Joyner, and Seaman (1999) examined the effects of single or joint interpretations of the *Strong Interest Inventory* and MBTI Step I assessments on the career

choice variables of career goal, specificity, and certainty. The sample consisted of 427 college students who were administered career outcome measures both before and after interpretations of the MBTI and/or Strong assessment. Classes were randomly assigned to four treatment conditions: MBTI assessment and Strong joint, MBTI assessment only, Strong only, and a control group that received no interpretation. Although the assessments were interpreted (except in the control group), there was no further discussion of them or of careers during the courses, so this would be considered a very "weak" treatment. Those in the joint MBTI/Strong group were more likely to broaden their career goal. There were no differences among treatment groups in change of career goal or level of certainty. The joint group perceived the assessments as more helpful than did the two groups that received results from only one assessment. When asked which assessment they would have preferred if they could have taken only one, 53% of those in the joint group indicated the MBTI assessment and 48% the Strong assessment. Men in the joint group and in the group that took only the MBTI assessment changed their level of certainty significantly more than did men in the other two groups.

There were only minor differences in outcomes between the two one-assessment-only groups, but to summarize the results for the group that took only the MBTI assessment: 84% said using the MBTI assessment helped very much or helped some to increase their self-esteem relative to their career choice; 83% said it increased their understanding of themselves relative to their career choice; 62% said it helped them make a career choice; 73% believed that it encouraged them in following the career choice they had already made; 78% felt that it had increased their confidence that they could succeed in their given career choice; 55% believed that it helped them further clarify the type of job they might want in their chosen field; 59% felt that it encouraged exploration of career options through reading; 73% felt that it encouraged exploration through talking with others; 63% said it helped them in making a decision about their college major; 42% said it gave them confidence to consider a career that they hadn't seriously considered before.

Outcome Studies with Adults

Only a few outcome studies have been conducted with older students and adults. Bell (2005) asked executives who were receiving coaching and using the MBTI Step I assessment as part of the coaching process about their experiences. The sample consisted of 67 executives who represented 10 MBTI types. The length of each coaching meeting ranged from 1 hour to 3 hours, with an average of about 1 hour, 20 minutes. The number of coaching sessions per month ranged from 1 to 4.5, and the coaching engagements ranged from 5.5 months

to 18 months, with an average of 7.5 months. The executives rated various components of the coaching process on a 5-point scale ranging from “not helpful” (1) to “exceedingly helpful” (5). They were asked to rate their overall experience of coaching, as well as their reactions to the use of the MBTI assessment. The overall average rating for six questions asking about their experience with the MBTI assessment was 3.6. The ratings for these questions are shown in table 9.46. All of the MBTI ratings averaged at least moderately helpful, with four of them closer to the very helpful rating. The highest rating was for the item that tapped how helpful MBTI results were in changing their behavior; the lowest rating was for the effect MBTI results had on helping them balance career and family.

The table also shows the types who gave the highest and lowest ratings on each of the six questions. The 10 types represented in this study had varying reactions to how the MBTI assessment had helped them. Those who thought insights or information learned from the MBTI assessment were most helpful were ENFPs, whereas ENTPs gave the lowest ratings on four of the six questions. Coaches may have to change their approach to ENTP executives. The author also reported that robust changes were noticed very early in the executive coaching process, at somewhere between the first and second months. Changes resulting from MBTI feedback were noticed earlier than those from executive coaching that took place without MBTI feedback. The findings suggest that executive coaches who use MBTI results throughout the coaching process can expect their clients to experience early and significant gains.

Sedlock (2005) surveyed perceptions of the accuracy of various sections of the MBTI® Team Report and of whether using the report led to changed actions by the team. The sample consisted of 51 adults in a variety of organizations and industries. All had received a Team Report customized for their team that identified their individual MBTI type and the overall team type.

Participants viewed the statements in the Team Report describing team strengths and potential blind spots and the predicted problem-solving process as mostly accurate. Participants also said that there had been “some” specific changes in how their teams solved problems due to the information on the problem-solving process in the report. Some also said that they had taken steps based on the suggested action plan in the report. Whether the changes and action steps were successful was unknown. Nine of the participants were team leaders. There were differences between these team leaders and team members on five of the survey questions, with the former giving significantly higher ratings as to the usefulness of the Team Report. There were also significant differences on four of the questions between those who responded to the survey soon after receiving their Team Report results and those with a longer interval, with the former rating these four questions higher.

Brennan (2010) randomly assigned active-duty Coast Guard personnel to two different conditions. The treatment condition consisted of administering MBTI® Complete, the MBTI Career Report, the *Strong Interest Inventory* assessment, and the *Skills Confidence Inventory* and then providing the participants ($n = 77$) with their results as well as follow-up (phone) counseling. A second group was a control group ($n = 88$) that received no treatment. The *Career Decision Self-Efficacy* (CDSE) scale was used before and six weeks after treatment to measure outcomes. The CDSE measures respondents’ level of confidence in completing five domains of career-related tasks: (1) accurate self-appraisal, (2) occupational information gathering, (3) goal selection, (4) making plans for the future, and (5) problem solving. The treatment group exhibited a statistically significant increase in the overall CDSE postmeasure score over the premeasure score. The test of the control group resulted in no statistically significant difference between the means of their pre- and postmeasure scores. Multiple regression analysis indicated that

Table 9.46 | Ratings of MBTI® assessment effectiveness from executives receiving coaching, by type

Survey question	Overall average rating	Type with highest rating	Type with lowest rating
How helpful was insight gained or information learned from the Myers-Briggs Type Indicator® (MBTI®) assessment on			
changing your behavior?	3.93	ENFP	INFP
making overall changes?	3.82	ENFP	INFJ, INTJ, INFP, ENTP
attaining your goal or goals?	3.72	ENFP	ENTP
changing your level of satisfaction with your professional life?	3.63	ENTJ	ISTP
changing your level of satisfaction with your personal life?	3.33	INFP	ENTP
making changes to your sense of balance between career and family?	3.00	INTJ	ENTP

Note: $N = 67$. Rating scale from 1 = not helpful to 5 = exceedingly helpful.
Source: Adapted from Bell (2005).

administration of career assessments with follow-up counseling to the treatment group raised CDSE among active-duty Coast Guard personnel compared to the control group that received no treatment.

Ang (2002) designed a senior seminar for medical students that used the MBTI assessment as a framework for understanding communication styles. The MBTI assessment was used in two of the four seminar modules. The first module focused on insight into one's own communication style and styles the student might find problematic. The second module was devoted to ways different types might prefer to receive bad news. Although no data or statistics were reported, the author claimed that a pretest-posttest comparison showed that the course was effective in changing confidence and attitudes about communication.

Leong, Hardin, and Gaylor (2005) examined the effects of using the MBTI assessment along with other assessments in a career development workshop offered to three consecutive classes of 117 second-year students at a medical school. Each workshop consisted of two three-hour sessions. In addition to the MBTI Step I assessment, the *Gough Medical Specialty Preference Scale* (GMSPS) and the *Values Scale* were used. Assessment results were given to students in the second session as part of a discussion of the career choice process. Note that this can be considered a "weak" treatment in terms of the MBTI application. Despite this, students who did not change their specialty choice throughout the year found the workshop, the MBTI results, and the *Values Scale* all more helpful than did the students who changed their choice. It may be that for these students the assessment results and the discussion of careers helped to confirm their choice and lessen any concerns they may have had. Overall, the MBTI assessment was perceived as more useful than the other two assessments.

Outcome studies of the MBTI assessment, which are highly desirable, are limited due to difficulties in conducting research in natural settings.

Cianci (2007) investigated the effects of using the MBTI Step I assessment in substance abuse treatment planning. The sample consisted of 34 adults entering a treatment facility who were randomly assigned either to a treatment group that used the MBTI assessment in the treatment protocol or to a control group that used a standard protocol. The outcome measure was the Abstinence Self-Efficacy scale, which is designed to assess self-efficacy to abstain from substances and uses four subscales designed to tap how confident the individual is in being able to handle negative affect, take part in positive social engagements, deal with physical reactions, and cope with withdrawal urges. Twenty-six percent of the participants were ISJs. For the Negative Affect

subscale, the MBTI treatment group and the control group had similar pretest scores, and the MBTI treatment group had higher posttest scores than the control group. For the Social/Positive subscale, there was a significant interaction. The MBTI treatment group had lower pretest scores and significantly higher posttest scores than the control group. This scale measures an individual's confidence in his or her ability to remain abstinent in high-risk social situations. For the Physical and Other Concerns subscale, the MBTI treatment group's scores again increased more than those of the control group, but the interaction was nonsignificant at $p = .12$ (likely due to the very low power associated with an interaction in such a small sample). There were no significant differences for the Withdrawal and Urges subscale. So, despite the low power of these statistical tests, the effect was large enough to yield significant results on some of the subscales.

A sample of 1,516 individuals who had completed the MBTI Step I assessment (Thompson, Walling, et al., 2013) were asked within three months of taking the assessment about the impact of knowing their type. Respondents indicated that they had experienced the following benefits from learning about their MBTI type (numbers in parentheses indicate the percentage of respondents who rated the extent of the benefit *great* or *very great*):

- Achieved greater understanding of self (58.8%)
- Gained greater understanding of preferred work style (58.6%)
- Gained an understanding of how to communicate with others (56.4%)
- Identified personal strengths (55.6%)
- Identified personal weaknesses (54.1%)
- Identified ways to enhance personal strengths (51.2%)
- Obtained greater insight into how others perceive you (53.6%)

At least 40% of respondents also indicated that the MBTI assessment had helped them (to a great or very great extent) in the following ways: improved your relationship with others, gained greater understanding of how you overcome obstacles in your work life, identified ways to work on personal weaknesses, gained greater understanding of how you achieve success, improved understanding of where your personal interests lie, improved understanding of your career preferences, helped you to work effectively as part of a team, helped you to lead others effectively. Another group of respondents indicated that they were helped *some* or to a *moderate* extent to communicate with co-workers, make decisions at work, coach co-workers, innovate at work, work effectively with customers, resolve conflicts, manage stress, lead change efforts, appreciate their career choices, manage projects, and learn new information. Employing a widely used consumer

satisfaction question, 95.8% of the respondents indicated that they would recommend the MBTI assessment to a friend or colleague.

Five of the studies discussed above used random assignment to a control group or an alternative treatment group. Some of the studies employed what could be considered a “strong” intervention, in which there was either a long exposure or multiple exposures to MBTI assessment results and their application. However, as with much research conducted in real-life settings, there are many factors that were not or could not be controlled that represent potential threats to validity. For example, lack of a control group or the extremely low power in some of the studies mitigated interpretation of the results. Despite the challenges, these studies are encouraging and suggest that practitioners can anticipate valid results and positive outcomes from appropriate use of MBTI results. It is hoped that the studies will motivate more research in this important area.

CONCLUSION

This chapter examined the validity of the MBTI Step I assessment by providing evidence for the validity of the four preference scales, preference interactions, whole types, and type dynamics.

While the great majority of the studies reported above concern the validity of the MBTI Step I assessment as a measure of type theory, a collection of studies that address the impact of using the MBTI assessment as an intervention were also presented. These studies suggest that use of type knowledge may help increase students’ grades and other educational outcome and process variables. In college settings, there was some evidence that use of the MBTI instrument may help reduce attrition, damage to residence halls, and requests for roommate changes. In organizations, executives who received coaching with the MBTI assessment believed that doing so helped them make changes in their behavior. While these studies are a first step and the results must be interpreted with caution, given the possibility of alternative explanations, this is an area of research with the MBTI assessment that should receive increased attention. It should be noted that outcome studies of the MBTI assessment such as described here are unique among personality instruments. These studies demonstrate that the MBTI assessment can be used not only to test predictions derived from type theory but also to increase understanding and to change the attitudes and behaviors of those who receive an interpretation of the results.

More traditional methods of evaluating the validity of the MBTI Step I assessment included examining differences between opposite poles of the four preferences scales

on a wide variety of scales from other assessments. The MBTI Step I preference scales have been tested with over a hundred scales from over a dozen other assessments in a wide variety of samples. The results strongly support the predictions of type theory regarding the meaning of, and behaviors believed to be associated with, the four preference pairs.

Evidence for the dichotomous nature of the scales was seen in plots of preference scores against external variables. Analysis of these plots demonstrated that at least on some dependent variables, preferences are not continuous linear functions of MBTI continuous scores but rather exhibit a complex relationship with the external variable. An exciting line of research was presented that uses topographic mapping of brain activity patterns and provides strong evidence for the biological basis of the preference pairs.

Studies of occupational selection showed that MBTI results are related to career choice in predictable ways. Type tables of hundreds of occupations were referenced that show that individuals of different types as measured by the MBTI assessment are attracted to occupations and to different niches within occupations, supporting the claim that they do so as a means of finding opportunities to express their type preferences. A particular theory about the process by which types get sorted into occupations was presented and supported by studies across hundreds of occupations and organizations, some with very large samples, that provided evidence for the homogeneity of type within both. This evidence in support of the validity of the MBTI Step I assessment is abundant and compelling.

Research on the use of type in organizations showed that MBTI results are related to behaviors and competencies in ways predicted by the Jung-Myers theory. Evidence on the importance of type in understanding preferred work environments, job satisfaction, and retention was presented. The relationship of type to other organizational variables, such as teamwork, change, decision making, emotional intelligence, and transfer of learning, was also presented. These studies showed that the MBTI assessment was useful in predicting outcomes on many issues studied in organizations.

Extensive evidence on the validity of the MBTI Step I assessment with educational variables was discussed, including the types of learners and teachers, and their interaction. The effect of type on learning, reading, and writing was presented. Evidence also showed how type preferences can predict achievement and participation in extracurricular activities. The use of the MBTI assessment in healthcare—including in relation to health-related behaviors, sensitivity to stressors, and ways of coping—showed predictable type effects. Type preferences were also shown to predict physical stress when there was a

lack of fit between preferences and the task demands of the job. These studies support the validity of the MBTI Step I assessment in this domain.

A series of studies specifically designed to test the validity of the interactions among the preferences measured by the MBTI assessment, including whole types and type dynamics theory, were discussed. Self-descriptions as well as ratings by outside observers who were blind to the type of the participants provide evidence for the differences between the 16 types as measured by the MBTI assessment. While not conclusive, these studies provide some support for the claim that whole types are more than the sum of their preference parts.

Finally, a number of exploratory factor analyses of the MBTI scales have demonstrated very close correspondence with the hypothesized four-factor structure. More important, rigorous confirmatory factor analyses provide even stronger support for the four-factor model and its superiority over competing models.

The results of over 500 studies were discussed or cited in this chapter (and this represents only a small portion of the studies that are available in the literature). While some of these studies have raised additional questions for further research, the overall conclusion is that the validity of the MBTI Step I assessment as a measure of type theory is strongly supported.

CHAPTER 10

Validity of the MBTI® Step II™ Assessment

CHAPTER OVERVIEW

Correlations Among Step II™ Facet Scales and with Step I™ Preference Scales

Factor Analysis

Correlations with Other Instruments

Gender Differences

Correlations with Behavioral Ratings by Others

The Relationship of Step II™ Facet Scales to Step I™ Preference Scales

The Relationship of Step II™ Facets to Whole Types

Validity Shown in Studies of Step II™ Applications

Conclusion

This chapter presents validity evidence for the facet scales of the MBTI® Global Step II™ assessment, *validity* meaning that the scales are measuring what they are intended to measure. A number of approaches to validity are presented, including confirmatory factor analysis, correlation of Step II facet scales with the scales of other instruments, correlation with behavioral ratings by others, the contribution of the Step II facet scales in relation to the Step I™ preference scales and to whole types, and studies of the use of the Step II facets in various settings.

The chapter includes validity studies primarily employing Step II facet scales from the Global Step II assessment and from Form Q. Correlations between the facet scales of the Global Step II assessment and those of Form Q and the European Step II assessment are shown in table 10.1. The mean correlation between the Global Step II scales and the Form Q scales is .92, with the highest correlations (.99) occurring on the Expressive–Contained, Enthusiastic–Quiet, and Realistic–Imaginative facet scales. The lowest correlation (.55) occurs on Questioning–Accommodating. Excluding Questioning–Accommodating, the correlations between the scales of the two forms range from .80 to .99, with a mean of .93. Given these correlations and the fact that in most cases there is considerable item overlap between the two forms (see table 7.7), it is reasonable to assume that validity results will generalize across forms. A few studies reported in this chapter used a prior form of the Step II assessment, Form J or Form K, both of which contain the same 20 facet scales. For reasons of simplicity and continuity, the Form J and Form K scales are referred to here and elsewhere in this manual as earlier versions of the Step II facet scales.¹

CORRELATIONS AMONG STEP II™ FACET SCALES AND WITH STEP I™ PREFERENCE SCALES

Evidence for the validity of the Global Step II facet scales can be observed in these scales' correlations with the Global Step I preference scales (shown in table 10.2) and with each other (shown in table 10.3). Based on both theory and previous research, it was expected that each Global Step II facet scale would correlate highly with the Global Step I preference scale with which it is by definition associated and considerably lower with the other three preference scales. An examination of table 10.2 confirms that each of the facet scales within a particular preference scale correlates significantly higher with that preference scale than with any of the other

Table 10.1 | Correlations of MBTI® Global Step II™ facet scales with Form Q and European Step II™ facet scales

Global Step II™ facet scale	Form Q facet scale	European Step II™ facet scale
E–I facet scales		
Initiating–Receiving	.98	.96
Expressive–Contained	.99	.94
Gregarious–Intimate	.97	.99
Active–Reflective	.86	.88
Enthusiastic–Quiet	.99	.97
S–N facet scales		
Concrete–Abstract	.96	.95
Realistic–Imaginative	.99	.99
Practical–Conceptual	.84	.86
Experiential–Theoretical	.94	.98
Traditional–Original	.96	.95
T–F facet scales		
Logical–Empathetic	.94	.94
Reasonable–Compassionate	.93	.97
Questioning–Accommodating	.55	.69
Critical–Accepting	.80	.81
Tough–Tender	.97	.96
J–P facet scales		
Systematic–Casual	.94	.97
Planful–Open-Ended	.97	.98
Early Starting–Pressure-Prompted	.93	.93
Scheduled–Spontaneous	.94	.92
Methodical–Emergent	.96	.89
Mean	.92	.93

Note: Global sample, $N = 16,773$.

Table 10.2 | Correlations of MBTI® Global Step II™ facet scales with Global Step I™ preference scales

Global Step II™ facet scale	Global Step I™ preference scale			
	E–I	S–N	T–F	J–P
E–I facet scales				
Initiating–Receiving	.87	–.15	–.09	–.12
Expressive–Contained	.77	–.12	–.21	–.09
Gregarious–Intimate	.74	–.13	–.09	–.12
Active–Reflective	.83	–.14	–.11	–.12
Enthusiastic–Quiet	.80	–.28	–.17	–.20
S–N facet scales				
Concrete–Abstract	–.15	.87	.31	.42
Realistic–Imaginative	–.20	.85	.29	.38
Practical–Conceptual	–.15	.76	.11	.30
Experiential–Theoretical	–.04	.63	.14	.24
Traditional–Original	–.19	.78	.08	.47
T–F facet scales				
Logical–Empathetic	–.19	.28	.90	.25
Reasonable–Compassionate	–.12	.26	.89	.22
Questioning–Accommodating	.00	.01	.67	.06
Critical–Accepting	–.16	.20	.71	.14
Tough–Tender	–.08	.20	.81	.15
J–P facet scales				
Systematic–Casual	–.22	.55	.39	.82
Planful–Open-Ended	–.12	.35	.17	.86
Early Starting–Pressure-Prompted	–.08	.28	.06	.62
Scheduled–Spontaneous	–.14	.45	.19	.93
Methodical–Emergent	–.05	.19	.13	.63

Note: Global sample, $N = 16,773$. Coefficients in bold type indicate the correlation of a facet scale with its underlying preference scale.

three preference scales. There is no exception to this pattern, which is an impressive result. Correlations with the other three preference scales are generally very low or even negative, although the correlations of the S–N facet scales with the J–P preference scale, and of the J–P facet scales with the S–N preference scale, are small to moderate. These correlations are to be expected given the correlation between the S–N and J–P preference scales shown in chapter 9.

The intercorrelations among the 20 facet scales, shown in table 10.3, provide supporting validity evidence. Of the 190 intercorrelations, 185 demonstrate that a facet scale within a preference scale correlates higher with the other four facet scales within that same preference scale than it does with a facet scale within any of the other three preference scales. A facet scale within one preference

1. The names of the facet scales in the Global Step II assessment and the Form Q assessment are the same. Prior Step II forms may have used slightly different names for a few facet scales. In the interest of more easily interpreting validity evidence, the Global Step II facet scale names are used throughout the discussion in this chapter, regardless of the Step II form used in the original studies.

Table 10.3 | Intercorrelations of MBTI® Global Step II™ facet scales

Global Step II™ facet scale	1.	2.	3.	4.	5.	6.	7.	8.	9.	10.	11.	12.	13.	14.	15.	16.	17.	18.	19.	20.
E–I facet scales																				
1. Initiating–Receiving	–																			
2. Expressive–Contained	.61	–																		
3. Gregarious–Intimate	.60	.54	–																	
4. Active–Reflective	.72	.59	.57	–																
5. Enthusiastic–Quiet	.62	.56	.61	.63	–															
S–N facet scales																				
6. Concrete–Abstract	–.11	–.10	–.09	–.10	–.22	–														
7. Realistic–Imaginative	–.13	–.13	–.13	–.14	–.27	.68	–													
8. Practical–Conceptual	–.11	–.07	–.10	–.09	–.21	.58	.65	–												
9. Experiential–Theoretical	–.03	–.03	–.01	–.03	–.07	.55	.45	.38	–											
10. Traditional–Original	–.18	–.09	–.12	–.13	–.25	.60	.58	.60	.37	–										
T–F facet scales																				
11. Logical–Empathetic	–.11	–.22	–.11	–.14	–.21	.31	.31	.15	.12	.11	–									
12. Reasonable–Compassionate	–.07	–.18	–.06	–.08	–.14	.31	.26	.11	.17	.08	.76	–								
13. Questioning–Accommodating	.06	–.09	–.02	.02	.00	.08	.08	–.09	.01	–.20	.50	.56	–							
14. Critical–Accepting	–.12	–.20	–.10	–.11	–.16	.24	.20	.09	.08	.06	.55	.63	.70	–						
15. Tough–Tender	–.03	–.15	–.04	–.04	–.09	.25	.22	.08	.10	.04	.62	.66	.67	.67	–					
J–P facet scales																				
16. Systematic–Casual	–.18	–.16	–.17	–.17	–.28	.50	.46	.34	.26	.51	.39	.36	.15	.28	.27	–				
17. Planful–Open-Ended	–.11	–.07	–.10	–.09	–.15	.31	.26	.22	.17	.37	.17	.16	.03	.12	.11	.64	–			
18. Early Starting–Pressure-Prompted	–.04	–.03	–.09	–.08	–.13	.25	.23	.18	.20	.26	.07	.07	–.01	–.01	.02	.44	.49	–		
19. Scheduled–Spontaneous	–.11	–.08	–.12	–.12	–.18	.40	.37	.29	.25	.46	.21	.17	.02	.08	.11	.73	.73	.54	–	
20. Methodical–Emergent	–.02	–.04	–.07	–.04	–.10	.17	.16	.10	.10	.17	.14	.12	.06	.04	.08	.48	.50	.52	.57	–

Note: Global sample, N = 16,773.

scale correlates with a facet scale within another preference scale at a level higher than it does with the other facet scales within its own preference scale only five times, which is about half the number that would be expected by chance. These across-preference scale correlations are as follows:

- Realistic–Imaginative (S–N) correlates .46 with Systematic–Casual (J–P) but .45 with Experiential–Theoretical (S–N).
- Traditional–Original (S–N) correlates .51 with Systematic–Casual (J–P) and .46 with Scheduled–Spontaneous (J–P) but .37 with Experiential–Theoretical (S–N).
- Systematic–Casual (J–P) correlates .50 with Concrete–Abstract (S–N) but .44 with Early Starting–Pressure–Prompted (J–P) and .48 with Methodical–Emergent (J–P).

These across-preference scale correlations between an S–N facet scale and a J–P facet scale mirror the results above from the correlations of the facet scales with the preference scales.

FACTOR ANALYSIS

To test whether the five facet scales associated with each of the four underlying dichotomous factors, E–I, S–N, T–F, and J–P, load on the associated factor and not on the other factors, a confirmatory factor analysis was conducted. As discussed in chapter 9, MBTI theory allows for respondents with the same preference to differ in the degree of clarity with which they chose that preference. Different levels of clarity were taken into account by examining the loadings of the 20 facet scales on the four preference scales for each of three clarity classes: low, moderate, and high. The null hypothesis was formalized using the traditional exploratory factor model wherein each facet scale loads on each of the four factors. The results support the confirmatory factor model over the exploratory model when different subsets of respondents are allowed to have different levels of clarity. The final model consists of three clarity classes with the estimated loadings shown in table 10.4. For technical details, see Magidson (2015).

Table 10.4 | Factor loadings for MBTI® Global Step II™ facet scales for each clarity class

Global Step II™ facet scale	Low clarity				Medium clarity				High clarity			
	dfac1	dfac2	dfac3	dfac4	dfac1	dfac2	dfac3	dfac4	dfac1	dfac2	dfac3	dfac4
E–I facet scales												
Initiating–Receiving	0.51	0.00	0.00	0.01	0.77	–0.01	0.00	0.01	0.88	–0.01	0.00	0.01
Expressive–Contained	0.41	0.00	0.00	0.00	0.67	0.00	0.00	0.01	0.82	–0.01	0.00	0.01
Gregarious–Intimate	0.39	0.00	0.00	0.00	0.65	0.00	0.00	0.01	0.80	–0.01	0.00	0.01
Active–Reflective	0.46	0.00	0.00	0.00	0.73	–0.01	0.00	0.01	0.86	–0.01	0.00	0.01
Enthusiastic–Quiet	0.44	0.00	0.00	0.00	0.71	–0.01	0.00	0.01	0.85	–0.01	0.00	0.01
S–N facet scales												
Concrete–Abstract	0.00	0.49	0.00	0.00	0.00	0.75	0.00	0.01	0.00	0.87	0.00	0.01
Realistic–Imaginative	0.00	0.47	0.00	0.00	0.00	0.74	0.00	0.01	0.00	0.86	0.00	0.01
Practical–Conceptual	0.00	0.38	0.00	0.00	0.00	0.65	0.00	0.01	0.00	0.80	0.00	0.01
Experiential–Theoretical	0.00	0.30	0.00	0.00	0.00	0.54	0.00	0.00	0.00	0.71	0.00	0.01
Traditional–Original	0.00	0.39	0.00	0.00	0.00	0.65	0.00	0.01	0.00	0.80	0.00	0.01
T–F facet scales												
Logical–Empathetic	–0.01	–0.01	0.53	0.02	–0.01	–0.02	0.79	0.03	–0.01	–0.02	0.90	0.03
Reasonable–Compassionate	–0.01	–0.01	0.54	0.02	–0.01	–0.02	0.79	0.03	–0.01	–0.02	0.90	0.03
Questioning–Accommodating	0.00	–0.01	0.33	0.01	–0.01	–0.02	0.59	0.02	–0.01	–0.02	0.75	0.03
Critical–Accepting	0.00	–0.01	0.37	0.01	–0.01	–0.02	0.64	0.02	–0.01	–0.02	0.79	0.03
Tough–Tender	0.00	–0.01	0.45	0.02	–0.01	–0.02	0.72	0.03	–0.01	–0.02	0.85	0.03
J–P facet scales												
Systematic–Casual	–0.01	–0.04	0.00	0.53	–0.02	–0.06	0.00	0.83	–0.03	–0.07	0.00	0.97
Planful–Open–Ended	–0.02	–0.04	0.00	0.56	–0.02	–0.06	0.00	0.86	–0.03	–0.07	0.00	0.99
Early Starting–Pressure–Prompted	–0.01	–0.03	0.00	0.34	–0.02	–0.05	0.00	0.62	–0.02	–0.06	0.00	0.81
Scheduled–Spontaneous	–0.02	–0.05	0.00	0.65	–0.03	–0.07	0.00	0.93	–0.03	–0.08	0.00	1.00
Methodical–Emergent	–0.01	–0.03	0.00	0.35	–0.02	–0.05	0.00	0.63	–0.02	–0.06	0.00	0.82

Note: Global sample, $N = 16,773$. The formal statistical analysis conducted used multivariate latent profile models rather than traditional factor analysis to account for the dichotomous nature of the underlying theoretical factors E–I, S–N, T–F, and J–P. These dichotomous factors are labeled dfac1, dfac2, dfac3, and dfac4.

Ball and Geyer (2006) conducted a principal components analysis of the facet scores from a prior Step II form using a sample of 637 subjects from Australia. As predicted, the factor loadings clearly match the hypothesized structure for the set of five facet scales corresponding to each of the four preference scales. There are four relatively minor cross loadings: one J–P facet scale (Scheduled–Spontaneous) loads on the S–N preference scale, and another J–P facet scale (Systematic–Casual) loads on the T–F preference scale. One T–F facet scale (Questioning–Accommodating) loads on the S–N preference scale, and one S–N facet scale (Traditional–Original) loads on the J–P preference scale. The factor loadings for these four facet scales are considerably lower than are those for the other facet scales on their respective preference scales.

CORRELATIONS WITH OTHER INSTRUMENTS

A common method for examining the validity of an instrument is to correlate its scales with those of other instruments. In addition to responding to the MBTI Global Step II items, a subset of participants in the global sample also completed the CPI 260® assessment (Gough & Bradley, 2005) and another subset completed the *Adjective Check List* (ACL; Gough & Heilbrun, 1983). Correlations of the Step II scales with the scales of the *Fundamental Interpersonal Relations Orientation–Behavior*® (FIRO–B®) assessment (Schutz, 1978), the *Strong Interest Inventory*® assessment (Donnay, Morris, Schaubhut, & Thompson, 2005), the *Benchmarks*® assessment (Lombardo & McCaulley, 1994), the *Birkman Method*® assessment (Birkman, Elizondo, Lee, Wadlington, & Zamzow, 2008), the *Thomas–Kilmann Conflict Mode Instrument* (TKI®) assessment (Thomas & Kilmann, 1974), and a measure of the Big Five were also available from various sources. These analyses were conducted using data from a variety of samples, including the global sample, a US national representative sample, and a large sample of managers. Three validity questions apply to correlations of the Step II facet scales with other instruments' scales.

- Do the Step II facet scales within a preference scale generally correlate (or not) with another assessment's scales in a manner predicted by theory?
- Do certain facet scales within a preference scale correlate higher (or lower, depending on the scale) with the other assessment's scales than do other facet scales within that preference scale? This question deals more narrowly with the individual facet scales.

- Do certain facet scales within a preference scale correlate with the other assessment's scales in the opposite direction from the other facet scales within that preference scale? What do these crossover patterns suggest about the validity of the facet scales?

CPI 260® Assessment

Correlations of the Global Step II facet scales with the scales of the CPI 260 assessment are shown in table 10.5 for a subset of participants in the global sample who completed the CPI 260 assessment ($n = 1,728$). Since there are 640 correlations in this table, coefficients with $r \geq .20$ in either direction, indicating the more important relationships, are in bold type.² To better understand the relationships described below between Step II facet scales and CPI 260 scales, the reader can refer to the brief definitions of the CPI 260 scales presented in appendix C.

OVERALL PATTERN OF STEP II® FACET SCALE AND CPI 260® SCALE CORRELATIONS

To answer the first question (that is, to examine the correlations between the two instruments at the level of overall patterns), the CPI 260 scales are discussed by sector as defined by their location on the CPI 260 profile, titled the *CPI 260® Client Feedback Report*.

Dealing with others. This first sector is made up of seven CPI 260 scales described as measures of poise, self-assurance, and interpersonal proclivities. These scales are Dominance (Do), Capacity for Status (Cs), Sociability (Sy), Social Presence (Sp), Self-acceptance (Sa), Independence (In), and Empathy (Em).

All of the Step II E–I facet scales correlate with these CPI 260 scales in the direction of Extraversion. They have their highest correlation with Sociability, and then with Dominance.

Among the S–N facet scales, the Experiential–Theoretical scale does not correlate with any of the CPI 260 scales in this sector. All correlations of Practical–Conceptual and Traditional–Original with the CPI 260 scales are toward the Intuition poles of these facets, although the correlation of Practical–Conceptual with Independence does not reach the cutoff of .20. The correlations of Concrete–Abstract and Realistic–Imaginative with Capacity for Status, Social Presence, and Empathy are also all toward the Intuition poles.

Among the T–F facet scales, almost all correlations with the CPI 260 scales in this sector are in the direction of Thinking, with the exception of those with Empathy, although many do not reach the cutoff.

2. As with the preferences, negative correlations point toward the E, S, T, and J poles of the facet scales; positive correlations point toward the I, N, F, and P poles.

The Logical–Empathetic, Critical–Accepting, and Tough–Tender facet scales have no correlations that meet the cutoff with the CPI 260 scales in this sector. A score toward the Questioning facet pole is related to higher scores on Dominance, Self-acceptance, and Independence. The lack of correlation ($r = .02$) between the Step II facet scale Logical–Empathetic and the CPI 260 Empathy scale might at first glance call into question the validity of the Logical–Empathetic facet scale. However, despite the similar name, the meaning of Empathetic as a facet pole of the MBTI judging process differs considerably from the intended purpose of the CPI 260 Empathy scale. People who score toward Empathetic in their Step II results tend to “assume that honoring everyone’s values will lead to desired harmony, use empathy to understand others’ viewpoints and needs, have difficulty separating the argument from the arguer, see feelings, values, and outcomes for individuals as primary decision-making criteria” (Quenk & Kummerow, 2011, p. 17). In contrast, the CPI 260

Empathy scale was designed to “identify people with a talent for understanding how others feel and think, and who display warmth and tactfulness in their dealings with others” (Gough & Bradley, 2005, p. 6). It can be readily observed from these contrasting definitions why the correlation between the two scales would be low.

Among the J–P facet scales, only two correlations with CPI 260 scales reach the cutoff, indicating that a score toward the Casual facet pole is related to higher scores on Social Presence and Empathy.

Self-management. This sector of CPI 260 scales, described as measures of normative orientation and values, is made up of Responsibility (Re), Social Conformity (So), Self-control (Sc), Good Impression (Gi), Communitary (Cm), Well-being (Wb), and Tolerance (To). There are fewer significant correlations between Global Step II facet scales and the CPI 260 scales in this sector compared to the previous sector. No correlations of E–I facet scales with the CPI 260 Responsibility,

Table 10.5 | Correlations of MBTI® Global Step II™ facet scales with CPI 260® scales

Global Step II™ facet scale	CPI 260® scale													
	Dealing with others							Self-management						
	Do	Cs	Sy	Sp	Sa	In	Em	Re	So	Sc	Gi	Cm	Wb	To
E–I facet scales														
Initiating–Receiving	–.61	–.52	–.67	–.50	–.58	–.43	–.49	–.10	–.11	.13	–.09	–.04	–.29	–.15
Expressive–Contained	–.44	–.41	–.53	–.42	–.43	–.31	–.39	–.10	–.15	.12	–.10	–.03	–.25	–.16
Gregarious–Intimate	–.40	–.39	–.50	–.38	–.37	–.24	–.38	–.05	–.13	.14	–.08	.05	–.20	–.14
Active–Reflective	–.53	–.46	–.59	–.47	–.49	–.32	–.42	–.04	–.09	.24	–.01	.01	–.20	–.10
Enthusiastic–Quiet	–.46	–.44	–.53	–.46	–.44	–.31	–.41	–.04	–.07	.23	.01	.00	–.18	–.13
S–N facet scales														
Concrete–Abstract	.08	.25	.14	.20	.16	.08	.28	.04	–.11	–.17	–.17	–.06	–.04	.10
Realistic–Imaginative	.09	.27	.16	.23	.15	.07	.27	.02	–.12	–.21	–.16	–.09	–.07	.07
Practical–Conceptual	.20	.32	.21	.24	.25	.17	.30	.08	–.03	–.20	–.12	–.06	.00	.09
Experiential–Theoretical	.04	.16	.05	.11	.09	.05	.18	–.01	–.11	–.12	–.14	–.13	–.06	.04
Traditional–Original	.24	.33	.25	.31	.27	.26	.32	.05	–.10	–.23	–.17	–.05	.04	.08
T–F facet scales														
Logical–Empathetic	–.10	–.05	.02	.00	–.07	–.18	.02	–.06	–.08	–.04	–.07	–.01	–.12	–.05
Reasonable–Compassionate	–.16	–.08	–.06	–.07	–.11	–.21	.00	–.01	–.05	.06	–.01	–.01	–.09	.01
Questioning–Accommodating	–.28	–.19	–.15	–.14	–.26	–.27	–.10	–.03	.04	.13	.09	.01	–.06	.00
Critical–Accepting	–.05	.02	.06	.03	–.04	–.05	.13	.11	.10	.14	.13	.16	.10	.16
Tough–Tender	–.15	–.06	–.03	–.05	–.15	–.17	.04	.10	.04	.13	.09	.10	–.03	.08
J–P facet scales														
Systematic–Casual	.05	.13	.12	.22	.14	.08	.20	–.09	–.18	–.20	–.22	–.06	–.04	.02
Planful–Open-Ended	.04	.09	.06	.17	.12	.11	.15	–.11	–.16	–.16	–.18	–.08	–.01	–.01
Early Starting–Pressure-Prompted	.06	.15	.06	.17	.14	.07	.16	–.09	–.14	–.25	–.23	–.18	–.06	.01
Scheduled–Spontaneous	.02	.08	.04	.15	.11	.04	.14	–.16	–.24	–.24	–.26	–.19	–.11	–.08
Methodical–Emergent	–.06	–.01	–.05	.06	.01	–.03	.04	–.14	–.18	–.18	–.20	–.17	–.09	–.08

CONTINUED

Social Conformity, Good Impression, Communality, or Tolerance scale reach the cutoff. Individuals who scored toward the Reflective and Quiet facet poles scored high on CPI 260 Self-control. All but one of the E–I facet scales correlates above the cutoff with Well-being in the direction of Extraversion; the exception is Enthusiastic–Quiet. Among the S–N facet scales, the Realistic, Practical, and Traditional facet poles correlate with Self-control. No correlations between T–F facet scales and CPI 260 scales in this sector meet the cutoff. Among the J–P facet scales, scores toward the Systematic, Early Starting, and Scheduled facet poles are related to higher scores on CPI 260 Self-control and Good Impression. A score toward the Scheduled facet pole is also related to a higher score on Social Conformity, which describes people who are conscientious and easily accept and conform to normative rules; and a score toward Methodical is related to a higher score on Good Impression.

Motivations and thinking style. The next sector of CPI 260 scales is described as measures of cognitive and intellectual functioning. These scales include Achievement via Conformance (Ac), Achievement via Independence (Ai), and Conceptual Fluency (Cf). All five E–I facet scales correlate with Conceptual Fluency in the direction of Extraversion. Among the S–N facet scales, scores toward the Abstract, Conceptual, and Original facet poles are related to higher scores on Achievement via Independence, and scores toward the Conceptual and Original facet poles correlate with Conceptual Fluency. No correlations of the T–F facet scales with the CPI 260 scales in this sector reach the cutoff. All of the J–P facet scales correlate with Achievement via Conformance in the direction of Judging.

Personal characteristics. This sector contains three CPI 260 scales described as measures of personal style. These scales are Insightfulness (Is), Flexibility (Fx), and Sensitivity (Sn). All five E–I facet scales correlate with

Table 10.5 | **Correlations of MBTI® Global Step II™ facet scales with CPI 260® scales** *CONTINUED*

Global Step II® facet scale	CPI 260® scale														
	Motivations and thinking style			Personal characteristics			Work-related measures						Vector		
	Ac	Ai	Cf	Is	Fx	Sn	Mp	Wo	Ct	Lp	Ami	Leo	v.1	v.2	v.3
E–I facet scales															
Initiating–Receiving	–.19	–.17	–.33	–.19	–.02	.36	–.37	–.18	–.32	–.54	–.12	–.22	.44	–.11	–.14
Expressive–Contained	–.19	–.12	–.24	–.13	.00	.24	–.29	–.20	–.24	–.41	–.15	–.17	.35	–.15	–.15
Gregarious–Intimate	–.11	–.13	–.20	–.14	–.03	.24	–.27	–.12	–.24	–.36	–.14	–.15	.36	–.11	–.11
Active–Reflective	–.11	–.10	–.23	–.10	.00	.31	–.31	–.11	–.27	–.44	–.05	–.18	.49	–.13	–.09
Enthusiastic–Quiet	–.11	–.15	–.24	–.15	–.09	.23	–.27	–.11	–.32	–.39	–.07	–.12	.42	–.03	–.13
S–N facet scales															
Concrete–Abstract	–.06	.21	.13	.10	.31	.11	.04	–.03	.34	.07	–.07	–.18	–.16	–.17	.10
Realistic–Imaginative	–.07	.17	.12	.08	.27	.10	.02	–.05	.33	.07	–.08	–.18	–.20	–.15	.06
Practical–Conceptual	.03	.20	.20	.14	.17	–.02	.10	–.01	.32	.18	–.05	–.07	–.28	–.06	.05
Experiential–Theoretical	–.07	.16	.07	.08	.23	.06	.02	–.05	.24	.03	–.07	–.16	–.12	–.11	.04
Traditional–Original	–.06	.22	.21	.18	.25	–.09	.11	–.01	.42	.21	–.09	–.09	–.28	–.18	.08
T–F facet scales															
Logical–Empathetic	–.13	–.11	–.11	–.16	.12	.27	–.16	–.11	.01	–.10	–.06	–.14	.06	–.14	–.04
Reasonable–Compassionate	–.10	–.02	–.10	–.12	.19	.31	–.13	–.05	.02	–.13	.00	–.15	.15	–.14	.04
Questioning–Accommodating	–.05	–.09	–.16	–.17	.05	.30	–.16	.00	–.14	–.22	.11	–.14	.24	–.02	.02
Critical–Accepting	.08	.11	.05	.01	.14	.21	.04	.14	.06	.03	.18	–.02	.10	–.02	.16
Tough–Tender	.01	.04	–.04	–.08	.15	.29	–.06	.04	.02	–.10	.11	–.11	.17	–.06	.10
J–P facet scales															
Systematic–Casual	–.25	.06	.01	–.01	.37	.03	–.05	–.07	.36	.00	–.09	–.18	–.08	–.39	.07
Planful–Open-Ended	–.26	.04	–.01	–.01	.29	–.06	–.06	–.05	.30	.00	–.06	–.16	–.06	–.33	.05
Early Starting–Pressure-Prompted	–.22	.09	.00	.02	.33	–.08	.01	–.09	.31	–.01	–.09	–.16	–.17	–.27	.06
Scheduled–Spontaneous	–.33	–.01	–.06	–.06	.31	–.03	–.11	–.15	.29	–.06	–.16	–.23	–.11	–.36	–.01
Methodical–Emergent	–.27	–.06	–.11	–.09	.26	–.01	–.10	–.11	.16	–.12	–.10	–.16	–.01	–.31	.00

Note: Portion of global sample that completed the CPI 260 assessment, $n = 1,728$. Coefficients in bold type indicate correlations $\geq .20$ for I, N, F, and P and $\leq -.20$ for E, S, T, and J.

Sensitivity in the direction of Introversion. Four of the S–N facet scales correlate with Flexibility in the direction of Intuition, and the fifth, Practical–Conceptual, comes close ($r = .17$). All of the T–F facet scales correlate with Sensitivity in the direction of Feeling, and all of the J–P facet scales correlate with Flexibility in the direction of Perceiving.

Work-related measures. This sector contains a number of CPI 260 scales developed to describe behavior in work settings. These include Managerial Potential (Mp), Work Orientation (Wo), Creative Temperament (Ct), Leadership (Lp), Amicability (Ami), and Law Enforcement Orientation (Leo). Among the E–I facet scales, scores toward the Initiating, Expressive, Gregarious, Active, and Enthusiastic facet poles correlates with Managerial Potential, Creative Temperament, and Leadership. The Expressive facet pole also correlates with Work Orientation, and Initiating comes close ($r = -.18$). A score toward Initiating is also related to a higher score on Law Enforcement Orientation, indicating a favorable view of social rules. Among the S–N facet scales, all five correlate with Creative Temperament in the direction of Intuition. A score toward the Original facet pole is also related to a higher score on Leadership. For the T–F facet scales, the only correlation that reaches the cutoff indicates that a higher score toward the Questioning facet pole is related to a higher score on the CPI 260 Leadership scale. Given that Thinking types are overrepresented in samples of managers, it is interesting that more of the T–F facet scales do not correlate with the CPI 260 Leadership scale. Among the J–P facet scales, all but one correlate with the Creative Temperament scale in the direction of Perceiving. The exception is Methodical–Emergent, for which the correlation is .16. Note that the relationships between the Step II facet scales and the CPI 260 Creative Temperament scale mirror those between the facet scales and the four overall preference scales and are consistent with the MBTI Creativity Index discussed in chapter 9.

Higher-order measures. This sector of CPI 260 scales contains the structural, or vector, scales:

- **Vector 1, Orientation Toward Others**, is basically an Extraversion–Introversion scale, although it is not necessarily derived from Jung’s definition. All five E–I facet scales correlate with the vector 1 scale (indicating an internal orientation) in the direction of Introversion, as expected. Among the S–N facet scales, scores toward the Realistic, Practical, and Traditional poles correlate with the vector 1 scale. Among the T–F facet scales, a score toward the Accommodating pole correlates with vector 1. None of the J–P facet scales correlates significantly with the vector 1 scale.
- **Vector 2, Orientation Toward Societal Values**, is a scale designed to assess a continuum wherein a low score represents a rule-questioning perspective and

a high score represents a rule-favoring perspective. None of the E–I or T–F facet scales correlate significantly with the vector 2 scale. Among the S–N facet scales, the Traditional pole comes close to correlating with the vector 2 scale ($r = -.18$). Among the J–P facet scales, scores toward the Systematic, Planful, Early Starting, Scheduled, and Methodical poles correlate with the vector 2 scale.

- **Vector 3, Orientation Toward Self**, is a measure of self-realization. None of the Global Step II facet scales correlates with the vector 3 scale.

This general pattern of correlations of Global Step II facet scales within preference scales with CPI scales in the different sectors clearly demonstrates that the Global Step II facet scales are measuring their intended constructs.

DIFFERENTIAL CORRELATIONS OF STEP II™ FACET SCALES WITHIN PREFERENCE SCALES WITH CPI 260® SCALES

The second question is whether certain facet scales within a preference scale correlate higher (or lower, depending on the scale) with a CPI 260 scale than do other facet scales within that preference scale. Scales are not mentioned here if they are part of a general pattern that was already discussed above. Among the E–I facet scales, the Reflective and Quiet facet poles correlate higher with the CPI 260 Self-control scale (.24 and .23, respectively) than do the other facet poles. Among the S–N facet scales, the Original pole correlates higher with Leadership (.21) than do the other poles, although Conceptual comes close (.18). Among the T–F facet scales, the Questioning pole correlates higher with Dominance (–.28), Self-acceptance (–.26), and Leadership (–.22) than do the other facet poles. Among the J–P facet scales, the Casual pole correlates higher with Social Presence (.22) and Empathy (.20) than do the other facet poles, and the Scheduled facet pole correlates higher with Social Conformity (–.24) and Law Enforcement Orientation (–.23) than do the other facet poles.

Correlations of facet scales within the same preference scale with scales of the CPI, ACL, and other instruments differ from one another, revealing the nuances of meaning within the facet scales.

OPPOSITE CORRELATIONS OF STEP II™ FACET SCALES WITHIN PREFERENCE SCALES WITH CPI 260® SCALES

The third question involves examining a number of crossover correlations within a preference scale—that is, is the sign (negative or positive) of the coefficient for one or more facet scales within a preference scale opposite that of the other facet scales within that preference

scale? This analysis considers some correlations of less than .20 because the sign of the coefficient is of primary interest; however, those close to 0 are not analyzed, regardless of sign.

Among the T–F facet scales, correlations with the CPI 260 Empathy scale are toward the Questioning pole (–.10; Thinking direction) and the Accepting pole (.13; Feeling direction). It may be that questioning is employed as a way of understanding the perspective of others, which may lead to empathy. There are also opposite correlations among the T–F facet scales with Well-being, Achievement via Independence, and Work Orientation, with correlations indicating scores toward the Logical facet pole (Thinking direction) and the Accepting facet pole (Feeling direction) being associated with higher scores on these CPI 260 scales.

CORRELATIONS OF STEP II® FACET SCALES WITH CPI® 434 SCALES IN A SAMPLE OF MANAGERS

In another study (Hammer, 2014a), the Step II Form Q facet scales were correlated with the scales of the CPI® 434 assessment using data from a large sample ($N = 5,122$) of participants attending leadership programs at the Center for Creative Leadership® (CCL®).

All five E–I facet scales correlate with the CPI 434 scales Dominance, Capacity for Status, Sociability, Social Presence, Self-acceptance, Independence, Empathy, Creative Temperament, Managerial Potential, and Leadership in the direction of Extraversion. All five E–I facet scales correlate with Self-control and Sensitivity (using the CPI 260 name for the latter scale) in the direction of Introversion.

All five S–N facet scales correlate with Capacity for Status, Social Presence, Independence, Empathy, Achievement via Independence, Intellectual Efficiency, Flexibility, and Creative Temperament in the direction of Intuition. All five S–N facet scales correlate with Law Enforcement Orientation in the direction of Sensing.

Four of the T–F facet scales correlate with Sensitivity in the direction of Feeling; the exception is the Questioning–Accommodating facet scale.

All five J–P facet scales correlate with Social Conformity, Self-control, Good Impression, Communality, and Achievement via Conformance in the direction of Judging. All five J–P facet scales correlate with Flexibility and Creative Temperament in the direction of Perceiving.

OVERALL STEP II® FACET SCALE AND CPI® SCALE CORRELATION RESULTS

Comparing the two sets of correlations of the facet scales with CPI scales described above, one using the global sample (shown in table 10.5) and one using a sample of managers (described in the preceding section), requires some caution because two different Step II

forms of the MBTI assessment were used, as were two different forms of the CPI assessment. Despite these variations, which might be expected to lead to some differences, the two sets of correlations are almost identical. The average difference between the two sets of correlations ranges from .02 for Critical–Accepting to .08 for Questioning–Accommodating. The sample of managers has slightly lower correlations across the Step II facet scales than does the global sample, likely due to a restriction in ranges for the former. The almost identical correlations of the Step II facet scales with the CPI scales in the two samples verify that the scales are indeed valid measures of the facets.

Adjective Check List Assessment

The *Adjective Check List* (ACL) assessment (Gough & Heilbrun, 1983) was administered to a subset of participants in the global sample ($n = 1,550$). The same three questions that applied to the interpretation of CPI 260 scales are also relevant to the interpretation of ACL scales. These questions relate to (1) general patterns, (2) a facet scale that is more highly correlated with an ACL scale than are the other facet scales within its preference scale, and (3) a facet scale that correlates in the direction opposite that of other facet scales within its preference scale. The correlations of the Step II facet scales with the ACL scales are shown in table 10.6. Because there are 720 coefficients presented in this table, the discussion in the following section focuses on those for which $r \geq .20$ in either direction, which is midrange between a small and medium effect size for correlation coefficients (Cohen, 1988). To better understand the relationships described below, the reader can refer to the brief definitions of the ACL scales presented in appendix B.

OVERALL PATTERN OF STEP II® FACET SCALE AND ACL SCALE CORRELATIONS

All of the E–I facet scales correlate with the ACL scales Dominance, Exhibition, Aggression, Change, Self-Confidence, and Ideal Self in the direction of Extraversion. They also correlate close to or above the .20 cutoff with Achievement, Nurturance, Affiliation, and Personal Adjustment in the direction of Extraversion. They correlate with Self-Control in the direction of Introversion, with correlations ranging from .34 for Active–Reflective to .18 for Gregarious–Intimate. Four of the S–N facet scales correlate with the ACL Change and Creative Personality scales in the direction of Intuition; the exception is Experiential–Theoretical. All of the T–F facet scales correlate with Nurturance in the direction of Feeling. In addition, four of the T–F facet scales correlate with Deference in the direction of Feeling; the exception is Logical–Empathetic. Four J–P facet scales correlate with the ACL Order scale in the direction of Judging, and the fifth, Planful–Open-Ended, comes close (–.19). Three

Table 10.6 | Correlations of MBTI® Global Step II™ facet scales with *Adjective Check List* scales

Global Step II™ facet scale	ACL scale											
	Number of adjectives checked	Number of favorable adjectives checked	Number of unfavorable adjectives checked	Communality	Achievement	Dominance	Endurance	Order	Intraception	Nurturance	Affiliation	Exhibition
E–I facet scales												
Initiating–Receiving	–.08	–.19	.12	–.04	–.27	–.45	–.12	–.02	–.09	–.18	–.25	–.49
Expressive–Contained	–.08	–.18	.10	–.02	–.18	–.32	–.06	.01	–.09	–.24	–.26	–.40
Gregarious–Intimate	–.03	–.13	.12	–.01	–.16	–.27	–.07	.00	–.03	–.16	–.19	–.32
Active–Reflective	–.08	–.17	.08	–.02	–.24	–.40	–.08	.01	–.05	–.16	–.23	–.48
Enthusiastic–Quiet	–.10	–.18	.05	–.02	–.19	–.33	–.05	.03	–.07	–.18	–.24	–.46
S–N facet scales												
Concrete–Abstract	.12	.14	.05	.06	.02	.02	–.09	–.13	.13	.12	.13	.17
Realistic–Imaginative	.16	.16	.07	.06	.06	.03	–.07	–.10	.14	.11	.15	.18
Practical–Conceptual	.19	.19	.10	.07	.14	.08	.00	–.01	.16	.06	.15	.17
Experiential–Theoretical	.04	.03	.06	–.02	–.04	–.03	–.11	–.10	.05	.00	.02	.09
Traditional–Original	.21	.24	.06	.12	.21	.19	.05	.02	.19	.10	.21	.27
T–F facet scales												
Logical–Empathetic	.03	.05	–.03	.00	–.09	–.08	–.11	–.20	.02	.27	.12	.03
Reasonable–Compassionate	.00	.04	–.06	.03	–.14	–.12	–.14	–.21	.06	.29	.10	.00
Questioning–Accommodating	–.08	–.03	–.13	.02	–.21	–.25	–.14	–.17	.00	.24	.04	–.19
Critical–Accepting	.02	.15	–.19	.18	–.03	–.03	–.02	–.11	.16	.40	.21	.00
Tough–Tender	.03	.09	–.11	.08	–.12	–.18	–.07	–.12	.12	.36	.14	–.08
J–P facet scales												
Systematic–Casual	.10	.09	.07	.05	–.04	.02	–.19	–.26	.03	.12	.13	.21
Planful–Open-Ended	.07	.06	.06	.02	–.02	.02	–.16	–.19	.03	.04	.08	.14
Early Starting–Pressure-Prompted	.02	–.04	.14	–.07	–.10	–.04	–.22	–.21	–.07	–.11	–.03	.15
Scheduled–Spontaneous	.04	–.01	.11	–.06	–.08	–.03	–.23	–.26	–.05	–.03	.01	.15
Methodical–Emergent	–.03	–.09	.07	–.10	–.14	–.08	–.23	–.27	–.12	–.10	–.05	.06

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J–P facet scales also correlate with the ACL Endurance and Self-Control scales in the direction of Judging. Three J–P facet scales correlate with the ACL Change scale in the direction of Perceiving.

Thus, there is evidence that the Step II facet scales within a preference scale do indeed generally correlate with ACL scales, as would be predicted by theory.

DIFFERENTIAL CORRELATIONS OF STEP II™ FACET SCALES WITHIN PREFERENCE SCALES WITH ACL SCALES

Among the E–I facet scales, the Receiving and Reflective facet poles correlate higher with the ACL Deference scale than do the other three facet poles. Their opposites, Initiating and Active, correlate higher with the ACL

Achievement scale than do the other three E–I facet poles. Among the S–N facet scales, the Original facet pole correlates higher with the number of adjectives checked, the number of favorable adjectives checked, and the Achievement, Affiliation, Exhibition, and Self-Confidence scales than do the other facet poles. Its opposite, the Traditional facet pole, correlates higher with Deference and Self-Control than do the other S–N facet poles. Among the T–F facet scales, the Questioning facet pole correlates higher with Achievement and Dominance (and somewhat higher with Exhibition) than do the other facet poles, while its opposite, Accommodating, correlates higher with Self-Control than do the other T–F facet poles. Questioning and Tough correlate higher with the ACL Autonomy scale than do the other T–F facet poles. Among the J–P facet scales, the Casual facet pole

Table 10.6 | Correlations of MBTI® Global Step II™ facet scales with *Adjective Check List* scales *CONTINUED*

Global Step II™ facet scale	ACL scale											
	Autonomy	Aggression	Change	Succorance	Deference	Self-Control	Self-Confidence	Personal Adjustment	Ideal Self	Creative Personality	Military Leadership	Adult
E–I facet scales												
Initiating–Receiving	–.19	–.34	–.28	.21	.23	.31	–.44	–.24	–.30	–.25	–.15	–.12
Expressive–Contained	–.08	–.26	–.22	.08	.08	.23	–.34	–.23	–.22	–.16	–.13	–.05
Gregarious–Intimate	–.04	–.20	–.21	.10	.08	.18	–.30	–.18	–.22	–.13	–.09	–.04
Active–Reflective	–.21	–.36	–.30	.16	.23	.34	–.40	–.20	–.24	–.24	–.10	–.05
Enthusiastic–Quiet	–.16	–.33	–.29	.09	.17	.29	–.36	–.18	–.21	–.21	–.09	–.02
S–N facet scales												
Concrete–Abstract	.15	.11	.32	.00	–.12	–.19	.11	.04	.12	.30	.02	–.05
Realistic–Imaginative	.19	.10	.33	.01	–.12	–.19	.14	.05	.17	.32	.02	–.06
Practical–Conceptual	.21	.08	.28	–.06	–.17	–.18	.17	.05	.25	.34	.07	.01
Experiential–Theoretical	.12	.06	.15	.00	–.08	–.11	.02	–.05	.05	.19	–.05	–.06
Traditional–Original	.31	.18	.39	–.14	–.26	–.27	.26	.13	.24	.41	.13	.07
T–F facet scales												
Logical–Empathetic	–.09	.01	.11	.20	.16	–.03	–.02	.05	–.09	–.06	–.05	–.17
Reasonable–Compassionate	–.15	–.04	.09	.17	.22	.01	–.06	.04	–.09	–.05	–.07	–.13
Questioning–Accommodating	–.31	–.24	–.07	.17	.35	.20	–.15	.04	–.11	–.18	–.07	–.10
Critical–Accepting	–.17	–.08	.11	.07	.25	.04	.07	.19	.03	.01	.10	.01
Tough–Tender	–.23	–.16	.02	.23	.32	.10	–.07	.11	–.06	–.06	.01	–.08
J–P facet scales												
Systematic–Casual	.19	.15	.35	.02	–.14	–.27	.09	.01	.01	.21	–.05	–.15
Planful–Open-Ended	.18	.10	.27	–.03	–.15	–.22	.07	.02	.01	.19	–.03	–.08
Early Starting–Pressure-Prompted	.17	.13	.18	.01	–.16	–.19	–.01	–.10	–.03	.16	–.13	–.15
Scheduled–Spontaneous	.19	.12	.27	.01	–.18	–.23	.02	–.07	–.05	.15	–.12	–.17
Methodical–Emergent	.10	.04	.14	.01	–.10	–.13	–.05	–.09	–.09	.04	–.14	–.18

Note: Portion of global sample that completed the ACL, $n = 1,550$. Coefficients in bold type indicate correlations $\geq .20$ for I, N, F, and P and $\leq -.20$ for E, S, T, and J.

correlates higher with Exhibition and Creative Personality than do the other facet poles.

OPPOSITE CORRELATIONS OF STEP II™ FACET SCALES WITHIN PREFERENCE SCALES WITH ACL SCALES

There are no instances of crossover correlations, which provides evidence regarding the question of opposite correlations of Step II scales within preference scales (excluding scales with opposite signs that have coefficients very close to 0). The ACL scale that comes closest to having a crossover correlation is the number of favorable adjectives checked, with the Casual facet pole (Perceiving direction) correlating .09 and the Methodical facet pole (Judging direction) correlating $-.09$. Another is the ACL Self-Confidence scale, which is associated with higher scores toward the Questioning

facet pole ($-.15$; Thinking direction) and the Accepting facet pole (.07; Feeling direction).

ACL ADJECTIVES

Using the same subset of participants in the global sample ($n = 1,550$), endorsement rates were calculated for each of the 300 adjectives on the ACL plus an additional 69 adjectives that were included to round out the pool of descriptive terms. Endorsement rates were calculated separately for individuals who scored toward the poles of each facet scale and for those who scored in the midzone. However, as discussed in chapter 9 regarding the Step I preference scales, certain adjectives are endorsed by most people; that is, they do not show type differences, or in this case differences between the facet scales. For example, *honest* was endorsed by 58%

of the sample and so provides little information about the differences among the facet scales. Consequently, the focus of the analysis was on *differences* in endorsement rates for each adjective between respondents who scored at the poles of each facet scale (due to space limitations, differences in endorsement rates between respondents who scored in the midzone and those who scored at each pole are not described here). The 10 adjectives with the highest differences in endorsement rate between respondents at each pole are shown in table 10.7, along with the difference in the endorsement rate. For example, the adjective *sociable* was endorsed

by 34.9% more Initiating respondents than Receiving respondents. And *shy* was endorsed by 32.6% more Receiving respondents than Initiating respondents. When reading table 10.7, it is important to keep in mind that the percentages shown are *differences* in endorsement rates, not the endorsement rates themselves.

The table leads to a number of interesting observations. First, the adjectives that show the greatest differences in endorsement rates between respondents who scored toward opposite poles on the facet scales conform closely to the descriptions of the facets (Quenk &

Table 10.7 | **Top 10 ACL adjectives with highest differences in endorsement rates between respondents who scored at opposite poles on each facet**

E–I facet scales							
Extraversion	% difference	Introversion	% difference	Extraversion	% difference	Introversion	% difference
Initiating		Receiving		Active		Reflective	
sociable	34.9	shy	32.6	sociable	30.4	quiet	26.6
outgoing	30.4	quiet	25.9	outgoing	29.1	shy	25.8
talkative	27.6	reserved	24.3	talkative	26.9	reserved	20.2
active	27.0	withdrawn	17.9	outspoken	24.8	silent	17.6
outspoken	26.4	timid	16.6	active	20.5	withdrawn	15.2
cheerful	25.2	silent	16.4	cheerful	20.2	cautious	15.0
self-confident	23.1	inhibited	13.5	vibrant	19.5	timid	12.5
confident	21.6	cautious	13.2	confident	19.2	inhibited	12.0
energetic	20.9	anxious	12.5	energetic	18.5	plain	11.7
vibrant	20.2	nervous	12.3	animated	18.1	retiring	10.1
Expressive		Contained		Enthusiastic		Quiet	
sociable	30.7	shy	25.7	outgoing	31.7	quiet	22.7
outgoing	27.2	reserved	24.0	sociable	29.4	shy	22.5
cheerful	27.2	quiet	22.6	talkative	28.0	reserved	20.8
talkative	23.9	silent	18.4	vibrant	23.3	silent	17.0
warm	21.5	withdrawn	15.6	active	20.2	conservative	13.1
outspoken	20.1	cautious	13.2	energetic	19.9	withdrawn	11.9
vibrant	20.1	timid	12.8	cheerful	19.8	serious	11.6
active	19.9	serious	11.6	spontaneous	18.9	cautious	11.0
animated	19.5	pessimistic	11.4	outspoken	17.9	calm	9.6
affectionate	18.8	complicated	11.1	animated	17.6	inhibited	8.9
Gregarious		Intimate					
sociable	23.4	shy	21.2				
outgoing	21.6	quiet	21.0				
active	19.6	reserved	19.3				
talkative	17.6	silent	15.5				
animated	15.9	serious	13.9				
vibrant	15.8	withdrawn	13.1				
cheerful	15.6	anxious	11.2				
adventurous	14.2	defensive	11.1				
jolly	13.5	individualistic	10.8				
energetic	13.4	inhibited	10.8				

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Kummerow, 2011) and thus support the validity of the individual facet scales. For example, respondents who scored toward the Logical pole endorsed *rational* and *logical* (at a higher rate than did those who scored toward the Empathetic pole); respondents who scored toward the Quiet pole endorsed *quiet*; and respondents who scored toward the Tough pole endorsed *tough*. Second, many adjectives that differentiate individuals who scored at opposite poles on a given facet scale are similar within a preference scale. This is to be expected since the facet scales within a preference scale are aspects of that overall construct and are correlated with

one another and with the overall preference scale. For example, for all five of the E–I facet scales, *sociable* and *outgoing* are the two adjectives that show the greatest differences in endorsement rate between respondents who scored at the Extraversion pole and those who scored at the Introversion pole. Third, while there are many such similarities, there are also important differences. For example, while Expressive respondents endorsed *warm* and Contained respondents endorsed *complicated*, neither of these adjectives appears in the top 10 for any of the other E–I facet scales. Differences in the adjectives associated with the facet scales within

Table 10.7 | **Top 10 ACL adjectives with highest differences in endorsement rates between respondents who scored at opposite poles on each facet** *CONTINUED*

S–N facet scales							
Sensing	% difference	Intuition	% difference	Sensing	% difference	Intuition	% difference
Concrete		Abstract		Experiential		Theoretical	
conservative	9.4	artistic	35.1	realistic	11.9	artistic	16.9
conventional	7.0	imaginative	30.1	practical	10.3	imaginative	14.0
average	6.7	individualistic	23.5	outspoken	9.5	individualistic	13.9
cautious	6.6	adventurous	21.8	painstaking	8.7	deep	12.9
planful	6.3	interests wide	20.3	demanding	7.7	clever	12.5
serious	5.9	curious	19.8	punctual	7.6	idealistic	12.1
formal	5.5	bright	19.3	simple	7.4	adventurous	10.9
industrious	5.4	spontaneous	18.8	planful	7.2	progressive	10.3
stern	5.3	original	18.6	cautious	6.9	intelligent	10.1
punctual	5.1	clever	18.4	honest	6.8	inquisitive	9.9
Realistic		Imaginative		Traditional		Original	
conservative	6.7	artistic	34.1	conservative	7.7	imaginative	33.8
cautious	6.6	imaginative	33.7	cautious	6.0	adventurous	29.3
stable	5.7	curious	22.3	obedient	5.2	artistic	29.3
conventional	5.6	interests wide	21.7	conventional	4.8	interests wide	28.4
practical	5.2	idealistic	20.8	average	4.3	curious	27.4
stern	5.1	inventive	20.6	interests narrow	3.9	inventive	27.2
average	4.8	individualistic	20.5	pessimistic	3.3	original	25.9
serious	4.5	spontaneous	20.3	submissive	2.7	individualistic	25.6
dull	4.3	intelligent	19.6	prejudiced	2.5	humorous	25.1
proper	3.9	original	19.5	meek	2.3	independent	24.9
Practical		Conceptual					
average	6.2	imaginative	34.7				
cautious	3.4	artistic	27.5				
conservative	3.2	interests wide	25.0				
careful	2.4	inventive	24.4				
retiring	2.2	clever	23.3				
dull	1.9	original	22.4				
irritable	1.8	intelligent	22.2				
obedient	1.4	insightful	21.3				
interests narrow	1.1	brainy	21.0				
proper	1.0	curious	20.7				

CONTINUED

a preference pair demonstrate that the facets are unique and therefore fulfill their purpose of describing individual differences. Fourth, in some cases the same adjectives that differentiate individuals who scored at the facet poles within one preference scale play the same role for facet scales within a different preference scale. For example, *warm* appears in the list of top 10 adjectives for respondents who scored at four of the Feeling facet poles and also for those who scored toward Expressive, an Extraversion facet pole. And *punctual* was endorsed by respondents who scored toward all five of the Judging facet poles and also by those who scored toward

Concrete, a Sensing facet pole. This would also be predicted by the intercorrelations among the facet scales.

An analysis of differences in endorsement rates of *Adjective Check List* adjectives by respondents who scored at the poles of each facet scale supported the validity of individual facets as independently described in the *MBTI® Step II® User's Guide*.

A final observation is that the difference in endorsement rates for the top 10 adjectives is much lower for respondents who scored at the Sensing poles of the

Table 10.7 | **Top 10 ACL adjectives with highest differences in endorsement rates between respondents who scored at opposite poles on each facet** *CONTINUED*

T–F facet scales							
Thinking	% difference	Feeling	% difference	Thinking	% difference	Feeling	% difference
Logical		Empathetic		Critical		Accepting	
rational	17.5	sensitive	26.5	forceful	9.4	affectionate	32.7
logical	16.8	affectionate	25.2	aggressive	8.7	warm	29.0
methodical	14.2	emotional	22.1	cynical	8.7	kind	26.2
clear thinking	13.6	softhearted	21.7	sarcastic	8.7	sensitive	23.9
intelligent	12.2	sentimental	19.5	shrewd	8.6	emotional	23.7
sharp-witted	11.9	feminine	18.8	demanding	8.5	friendly	23.7
brainy	11.5	warm	18.8	stern	8.4	softhearted	22.4
clever	10.9	kind	18.6	meticulous	8.2	pleasant	22.2
realistic	10.5	nice	17.6	suspicious	8.2	sympathetic	21.9
self-disciplined	9.7	sympathetic	16.9	unemotional	8.1	cheerful	20.9
Reasonable		Compassionate		Tough		Tender	
methodical	13.2	sensitive	25.2	sarcastic	11.5	affectionate	29.3
outspoken	12.3	affectionate	22.2	stern	10.8	sensitive	28.3
rational	12.2	kind	20.5	self-confident	10.6	softhearted	26.3
self-disciplined	12.1	sympathetic	19.7	dominant	10.0	sympathetic	24.7
ingenious	11.6	emotional	19.3	forceful	9.5	kind	23.1
logical	11.4	forgiving	17.8	bossy	8.9	emotional	21.6
realistic	11.1	warm	17.8	cynical	8.4	friendly	21.3
demanding	10.5	gentle	17.3	aggressive	8.1	warm	21.1
formal	10.4	trusting	16.8	tough	7.6	gentle	21.1
shrewd	9.8	softhearted	15.5	self-disciplined	7.3	sentimental	19.0
Questioning		Accommodating					
rational	19.1	affectionate	15.8				
self-confident	19.0	sensitive	14.7				
sarcastic	17.1	kind	14.2				
logical	17.1	shy	13.7				
realistic	16.8	nice	12.2				
intelligent	16.3	softhearted	12.1				
clever	16.3	modest	11.9				
sharp-witted	16.2	sympathetic	11.0				
ingenious	16.1	easygoing	10.7				
thorough	15.9	average	9.7				

CONTINUED

S–N facet scales than for those who scored at any of the other facet poles. This is particularly true of individuals who scored at the Practical pole. *Average*, the adjective with the greatest difference in endorsement rate for people who scored toward Practical, was endorsed by only 6.2% more Practical respondents than Conceptual respondents; and 5 of the top 10 adjectives show a difference in endorsement rates of less than 2%. The same pattern does not hold for individuals who scored at the Intuition poles of the S–N facet scales. What this means is that while respondents who scored at the Practical pole endorsed some adjectives at a high rate,

those who scored at the Conceptual pole endorsed the same adjectives at a similar rate. For example, *practical* is one of the adjectives most highly endorsed by Practical respondents at 45.2%, but 44.6% of Conceptual respondents also endorsed this adjective, and so the 0.6% difference does not place it in the top 10. However, the reverse is not the case. People who scored at the Practical pole did not endorse adjectives that were also highly endorsed by those who scored at the Conceptual pole at anywhere near the same rate. For example, 54.0% of respondents who scored toward Conceptual endorsed *imaginative*, while only 19.3% of those who

Table 10.7 | **Top 10 ACL adjectives with highest differences in endorsement rates between respondents who scored at opposite poles on each facet** *CONTINUED*

J–P facet scales							
Judging	% difference	Perceiving	% difference	Judging	% difference	Perceiving	% difference
Systematic		Casual		Scheduled		Spontaneous	
organized	17.8	spontaneous	28.9	organized	21.4	spontaneous	21.5
punctual	13.6	humorous	20.6	cautious	17.3	impulsive	14.0
planful	12.6	adventurous	20.5	punctual	13.7	adventurous	11.7
methodical	10.6	imaginative	19.3	careful	12.2	sharp	11.2
serious	9.6	artistic	19.1	methodical	12.1	individualistic	10.9
self-disciplined	9.4	impulsive	18.4	planful	11.4	rebellious	10.9
precise	9.1	curious	16.9	thorough	11.1	natural	10.3
meticulous	8.9	natural	15.6	dependable	9.9	humorous	10.2
conservative	8.3	relaxed	14.5	obedient	9.6	interests wide	9.9
cautious	8.0	stubborn	14.4	meticulous	9.3	disorderly	9.6
Planful		Open-Ended		Methodical		Emergent	
organized	13.9	spontaneous	22.2	organized	20.6	impulsive	9.5
planful	11.7	natural	14.5	punctual	14.8	spontaneous	9.0
cautious	10.6	humorous	13.3	thorough	13.8	lazy	9.0
punctual	9.4	impulsive	13.2	cautious	13.8	infantile	7.6
obedient	7.6	adventurous	12.5	planful	13.3	absent-minded	7.0
stable	7.1	individualistic	12.4	methodical	12.6	distractible	6.1
meticulous	6.6	interests wide	12.2	dependable	12.1	forgetful	6.0
conservative	5.6	sharp	11.8	reliable	12.0	peculiar	6.0
serious	5.5	original	11.4	tactful	11.8	disorderly	5.9
dependable	5.3	absent-minded	11.1	clear thinking	11.6	careless	5.8
Early Starting		Pressure-Prompted					
punctual	17.7	lazy	16.1				
cautious	14.6	brainy	11.0				
organized	12.3	awkward	8.8				
careful	12.2	sarcastic	8.8				
responsible	10.2	clever	8.7				
dependable	9.1	distractible	8.6				
trustworthy	8.9	faultfinding	8.4				
thorough	8.9	exuberant	7.8				
stable	8.4	informal	7.7				
planful	8.3	sharp-witted	7.7				

Note: Portion of global sample that completed the ACL, $n = 1,550$. Adjectives listed in each cell are those that differentiate the most between individuals at opposite poles of the facet scale. The percentages represent the difference between the percentage endorsement of the adjective by individuals at the indicated pole compared to those at the opposite pole.

scored toward Practical did so. This same pattern holds for the other facet scales in the S–N preference pair. As shown in table 6.9, individuals who prefer Intuition appear to have endorsed a fair number of adjectives that would in theory be associated with Sensing. This finding offers an opportunity for further research. Does the endorsement of these “Sensing” adjectives by respondents who scored at the Intuition poles of the S–N facet scales indicate a lack of self-knowledge, wishful thinking on the part of individuals who favor Intuition, or evidence of adaptation or development by Intuitive types who must fit into a predominantly Sensing population? Or does it reflect a paucity of adjectives available to describe well-developed Sensing?

FIRO-B® Assessment

The Form Q facet scales were correlated with the scales of the *Fundamental Interpersonal Relations Orientation–Behavior* (FIRO-B) assessment using data from a sample of CCL program participants ($N = 4,977$). (See appendix D

for a description of the FIRO-B scales.) Given that the FIRO-B assessment is a measure of interpersonal needs, correlations are to be expected between FIRO-B scales, particularly those measuring Inclusion and Affection, and Step II facet scales within the E–I and T–F preference scales. There is no reason to believe that scales measuring interpersonal needs would correlate highly with personality scales designed to measure modes of perception (S–N) or styles of living in the outside world (J–P). The correlations are shown in table 10.8.

All five facet scales within the E–I preference scale correlate with the FIRO-B scales Expressed Inclusion and Wanted Inclusion in the direction of Extraversion. The coefficients range from $-.23$ for Initiating–Receiving with Wanted Inclusion to $-.52$ for Gregarious–Intimate with Expressed Inclusion. Similarly, the E–I facet scales correlate with Expressed and Wanted Affection in the direction of Extraversion. The range is from $-.20$ for Initiating–Receiving with Wanted Affection to $-.47$ for Expressive–Contained with Expressed Affection. Also,

Table 10.8 | Correlations of MBTI® Step II™ Form Q facet scales with FIRO-B® scales

Step II™ Form Q facet scale	FIRO-B® scales						
	Expressed Inclusion	Wanted Inclusion	Expressed Control	Wanted Control	Expressed Affection	Wanted Affection	Total Needs
E–I facet scales							
Initiating–Receiving	–.45	–.23	–.15	.04	–.39	–.20	–.37
Expressive–Contained	–.42	–.26	–.14	–.07	–.47	–.31	–.44
Gregarious–Intimate	–.52	–.36	–.17	–.03	–.38	–.30	–.47
Active–Reflective	–.45	–.27	–.15	.03	–.34	–.23	–.38
Enthusiastic–Quiet	–.47	–.30	–.21	.01	–.36	–.27	–.43
S–N facet scales							
Concrete–Abstract	.04	.04	.00	–.02	.06	.03	.04
Realistic–Imaginative	.08	.11	.04	–.01	.10	.11	.12
Practical–Conceptual	.01	.03	–.02	–.02	.05	.03	.02
Experiential–Theoretical	.02	.04	.03	–.02	.04	.03	.04
Traditional–Original	.06	.04	.08	–.06	.07	.01	.06
T–F facet scales							
Logical–Empathetic	.10	.09	–.11	.10	.24	.20	.15
Reasonable–Compassionate	.09	.07	–.15	.12	.21	.18	.13
Questioning–Accommodating	.04	.04	–.13	.14	.05	.11	.05
Critical–Accepting	.09	.08	–.22	.12	.14	.18	.09
Tough–Tender	.06	.09	–.21	.16	.18	.20	.12
J–P facet scales							
Systematic–Casual	.08	.07	–.09	.04	.10	.10	.08
Planful–Open-Ended	–.04	–.03	–.06	.00	–.01	–.02	–.05
Early Starting–Pressure-Prompted	.06	.07	.08	.01	.04	.02	.08
Scheduled–Spontaneous	.00	.01	–.05	–.02	.02	.01	–.01
Methodical–Emergent	–.02	.02	–.02	.01	.00	.00	.00

Note: Program participants at the Center for Creative Leadership who completed the MBTI Form Q and FIRO-B assessments, $N = 4,977$. Coefficients in bold type indicate correlations $\geq .20$ for I, N, F, and P and $\leq -.20$ for E, S, T, and J.

Source: Data provided by the Center for Creative Leadership®. Used with permission.

the E–I facet scales correlate on average $r = -.46$ with Total Expressed and $r = -.28$ with Total Wanted (not shown in the table).

Within the T–F preference scale, the Empathetic and Compassionate facet poles correlate with Expressed Affection and Wanted Affection (although Compassionate with Wanted Affection is just below the cutoff, $r = .18$). In addition, the Accepting facet pole correlates with Wanted Affection (also just below the cutoff, $r = .18$), and Critical and Tough correlate with Expressed Control. Another interesting result is that all of the correlations of the T–F facet scales with Expressed Control are in the direction of Thinking, while all of the correlations with Wanted Control are in the direction of Feeling.

Within the S–N and J–P preference scales, correlations of facet scales with FIRO-B scales are much lower, and none reach the cutoff. Within the S–N preference scale, the highest correlations are for Realistic–Imaginative with Wanted Inclusion (.11), Expressed Affection (.10), and Wanted Affection (.11), all in the direction of the Imaginative facet pole (Intuition preference). Individuals who score toward Imaginative may want to be included in the activities of others and also tend to be more open with others and want others to reciprocate, all of which may give them opportunities to express and expand upon the possibilities that occupy their attention. Within the J–P preference scale, the highest correlations indicate that individuals who scored toward the Casual facet pole tended to score higher on Expressed and Wanted Affection ($r = .10$). This is consistent with a number of findings reported later showing that individuals who score toward Casual tend to rate themselves higher on making other people feel at ease.

Correlations of the Form Q facet scales with the scales of the FIRO-B assessment were also analyzed using data from a sample of 492 participants (Schaubhut & Thompson, 2011a). The correlations are almost identical to those reported above. The largest difference is the correlation of Realistic–Imaginative with Expressed Inclusion, which is .08 in the CCL sample and .20 in this much smaller sample.

The pattern of correlations reported here is congruent with the intended meanings of the scales of the two assessments and with the theory that underlies each assessment. Some people might question why the Judging poles of the J–P facet scales are not more highly correlated with the Expressed Control scale of the FIRO-B assessment. However, the constructs measured by the MBTI and FIRO-B assessments are only superficially related. While individuals who score toward the Judging poles of the facet scales and those who score high on Expressed Control may both like structure and consistency, there are more differences between them than similarities. The Judging poles of the facet scales are concerned with an individual's preferences

for dealing with the outside world as manifested in a desire for a schedule, liking long-range plans, and taking a methodical approach to a project, while the FIRO-B scale, as a measure of interpersonal need, focuses on the need to control others.

Strong Interest Inventory® Assessment

A sample of 4,470 participants was administered both Form Q of the MBTI assessment and the *Strong Interest Inventory* assessment (Schaubhut & Thompson, 2011a). The correlations of the Step II facet scales with the General Occupational Themes (GOTs) and Personal Style Scales of the Strong assessment are shown in table 10.9.

Among the E–I facet scales, Initiating–Receiving correlates with the Social GOT in the direction of the Initiating pole. All five E–I facet scales correlate with higher scores on the Enterprising GOT in the direction of Extraversion (the Initiating, Expressive, Gregarious, Active, and Enthusiastic facet poles). Individuals in occupations in the Enterprising interest area (e.g., salesperson, lawyer, politician) like to actively engage others.

As predicted by type theory, all five S–N facet scales are related to the Artistic GOT in the direction of Intuition (the Abstract, Imaginative, Conceptual, Theoretical, and Original facet poles). In addition, four of the S–N facet scales are related to the Conventional GOT in the direction of Sensing (the Concrete, Realistic, Experiential, and Traditional facet poles), and Practical–Conceptual just misses the cutoff ($-.19$). Individuals who score high on the Conventional GOT tend to enjoy consistent, fact-based procedures and logically constructed environments.

Among the T–F facet scales, all but one (the exception being Questioning–Accommodating) correlate with the Realistic GOT in the direction of Thinking (the Logical, Reasonable, Critical, and Tough poles). Given that individuals who score high on the Realistic GOT tend to prefer occupations that emphasize logical decision making (e.g., engineer, electrician, auto mechanic), these correlations are meaningful. Scores toward the Feeling pole on these same four T–F facet scales (Empathetic, Compassionate, Accepting, Tender) are related to higher scores on the Social GOT, as would be predicted given that people in Social occupations (e.g., counseling, education, training) like to work with people. Two of the T–F facet scales, Logical–Empathetic and Reasonable–Compassionate, are related to the Investigative GOT in the direction of Thinking (the Logical and Reasonable facet poles). Most individuals who score high on the Investigative GOT like to work in occupations that require an objective, logical, analytical approach to problem solving (e.g., research and development and scientific occupations). Scores toward the Logical facet pole are also related to higher scores on the Conventional GOT.

Table 10.9 | Correlations of MBTI® Step II® Form Q facet scales with Strong Interest Inventory® scales

Step II® Form Q facet scale	Strong Interest Inventory® scales										
	General Occupational Themes (GOTs)					Personal Style Scales (PSSs)					
	Realistic	Investigative	Artistic	Social	Enterprising	Conventional	Work Style	Learning Environment	Leadership Style	Risk Taking	Team Orientation
E–I facet scales											
Initiating–Receiving	.00	.08	–.10	–.21	–.35	.05	–.34	–.13	–.45	–.13	–.29
Expressive–Contained	.07	.10	–.07	–.18	–.24	.07	–.30	–.07	–.31	–.02	–.26
Gregarious–Intimate	–.04	.07	–.03	–.14	–.33	.00	–.28	–.04	–.34	–.17	–.27
Active–Reflective	–.04	.09	.01	–.12	–.34	.03	–.28	–.01	–.37	–.18	–.25
Enthusiastic–Quiet	.03	.09	–.10	–.17	–.34	.10	–.32	–.12	–.41	–.15	–.26
S–N facet scales											
Concrete–Abstract	–.07	.07	.46	.10	.03	–.27	.06	.41	.19	.01	.04
Realistic–Imaginative	–.02	.11	.47	.09	.05	–.23	.02	.41	.19	.06	.03
Practical–Conceptual	–.08	.18	.47	.13	–.03	–.19	.00	.51	.18	–.04	.02
Experiential–Theoretical	–.04	.12	.38	.11	.00	–.25	.01	.45	.21	.03	.03
Traditional–Original	.03	.18	.41	.05	.07	–.22	–.05	.46	.28	.12	.04
T–F facet scales											
Logical–Empathetic	–.29	–.28	.15	.21	–.07	–.23	.37	–.09	–.10	–.24	–.01
Reasonable–Compassionate	–.26	–.23	.14	.26	–.12	–.17	.35	–.06	–.11	–.27	.01
Questioning–Accommodating	–.09	–.14	–.10	.08	–.05	.07	.15	–.24	–.22	–.17	.03
Critical–Accepting	–.20	–.17	.15	.26	–.03	–.09	.31	.01	–.05	–.24	.13
Tough–Tender	–.25	–.14	.18	.22	–.15	–.14	.27	.00	–.16	–.31	.02
J–P facet scales											
Systematic–Casual	–.02	–.07	.19	.04	–.02	–.23	.09	.03	–.01	.06	–.05
Planful–Open-Ended	.12	.04	.12	–.05	–.04	–.12	–.10	.03	–.03	.16	–.09
Early Starting–Pressure-Prompted	.06	.05	.11	.01	.06	–.11	.00	.11	.12	.16	–.01
Scheduled–Spontaneous	.06	.02	.15	–.05	–.03	–.18	–.07	.06	–.01	.12	–.12
Methodical–Emergent	.01	–.03	.06	–.05	–.04	–.13	–.04	.00	–.03	.07	–.11

Note: CPP commercial database, $N = 4,470$. Coefficients in bold type indicate correlations $\geq .20$ for I, N, F, and P and $\leq -.20$ for E, S, T, and J.

Many occupations in the Conventional interest area (e.g., accounting, banking) also require a logical decision-making style.

Among the J–P facet scales, Systematic–Casual correlates with the Conventional GOT in the direction of the Casual pole.

Correlations of the Form Q facet scales with the Strong Personal Style Scales are also shown in table 10.9. All of the E–I facet scales correlate in the direction of Extraversion with the “Works with people” pole of the Work Style scale, the “Directs others” pole of the Leadership Style scale, and the “Accomplishes tasks as a team” pole of the Team Orientation scale. All of the S–N facet scales correlate in the direction of Intuition with the “Academic” pole of the Learning Environment scale. Scores toward Theoretical and Original correlate with the “Directs others” pole of the Leadership Style scale, and

the other three S–N facet scales just miss the cutoff (all also in the direction of Intuition). Questioning is related to the “Academic” pole of the Learning Environment scale and the “Directs others” pole of the Leadership Style scale. Four of the five T–F facet scales correlate in the direction of Feeling with the “Works with people” pole of the Work Style scale and in the direction of Thinking with the “Takes chances” pole of the Risk Taking scale; the exception in both cases is Questioning–Accommodating. None of the J–P facet scales correlates significantly with the Strong Personal Style Scales.

Benchmarks® Assessment

A sample of managers ($N = 5,087$) who attended CCL programs and were administered a prior Step II form of the MBTI assessment also rated themselves on the Benchmarks® scales (Lombardo & McCauley, 1994; see

appendix E for brief definitions of the *Benchmarks* scales). The *Benchmarks* assessment solicits ratings (both self-ratings and ratings by others) on 21 managerial scales, 16 that tap managerial skills and 5 that are designed to measure potential flaws. The latter are sometimes referred to as “derailer” scales, as the flaws being tapped may derail a career if not addressed. Correlations of the Step II facet scales with self-ratings on the *Benchmarks* scales are shown in table 10.10. For the purpose of this discussion, $r = .10$ and above is used. This cutoff is used because the restriction in range in the sample (mostly middle-aged managers) and the narrowness of the rating scale (1–5) would tend to reduce the correlation coefficient.

Among the E–I facet scales, scores toward the Initiating and Expressive facet poles correlate with higher

self-ratings on 13 of the 16 skill scales, scores toward Gregarious correlate with higher self-ratings on 11 skill scales, scores toward Active correlate with higher self-ratings on 10 skill scales, and scores toward Enthusiastic correlate with higher self-ratings on 7 skill scales. For each of these facet poles, the highest correlations are with the skill of Putting People at Ease; the coefficients range from $-.20$ for Enthusiastic to $-.30$ for Expressive. For each of the E–I facet scales, respondents who scored toward the Introversion pole of the facet had higher self-ratings on Difficulty Building and Leading a Team, with coefficients that ranged from $.16$ for the Receiving facet pole to $.11$ for the Quiet facet pole.

All five S–N facet scales correlate with higher self-ratings on the *Benchmarks* scales Doing Whatever It Takes and Being a Quick Study in the direction of Intuition.

Table 10.10 | **Correlations of MBTI® Step II™ facet scales with *Benchmarks*® scales self-ratings**

Step II™ E–I facet scales	<i>Benchmarks</i> ® scales self-ratings			
Initiating–Receiving	Resourcefulness	–.16	Compassion and Sensitivity	–.12
	Doing Whatever It Takes	–.20	Balance Between Personal Life and Work	–.11
	Decisiveness	–.17	Putting People at Ease	–.27
	Leading Employees	–.13	Differences Matter	–.14
	Confronting Problem Employees	–.18	Career Management	–.27
	Participative Management	–.10	Difficulty Building and Leading a Team	.16
	Change Management	–.14	Difficulty Changing or Adapting	.11
	Building and Mending Relationships	–.16	Too Narrow Functional Orientation	.11
Expressive–Contained	Resourcefulness	–.11	Compassion and Sensitivity	–.15
	Doing Whatever It Takes	–.13	Balance Between Personal Life and Work	–.13
	Decisiveness	–.11	Self-Awareness	–.12
	Leading Employees	–.13	Putting People at Ease	–.30
	Confronting Problem Employees	–.11	Career Management	–.23
	Participative Management	–.10	Difficulty Building and Leading a Team	.13
	Change Management	–.10	Difficulty Changing or Adapting	.10
	Building and Mending Relationships	–.12		
Gregarious–Intimate	Resourcefulness	–.12	Change Management	–.12
	Doing Whatever It Takes	–.14	Building and Mending Relationships	–.15
	Decisiveness	–.12	Balance Between Personal Life and Work	–.10
	Leading Employees	–.13	Putting People at Ease	–.24
	Confronting Problem Employees	–.10	Career Management	–.21
	Participative Management	–.11	Difficulty Building and Leading a Team	.13
Active–Reflective	Resourcefulness	–.12	Building and Mending Relationships	–.12
	Doing Whatever It Takes	–.15	Balance Between Personal Life and Work	–.10
	Decisiveness	–.15	Putting People at Ease	–.22
	Leading Employees	–.10	Career Management	–.19
	Confronting Problem Employees	–.14	Difficulty Building and Leading a Team	.13
	Change Management	–.12	Too Narrow Functional Orientation	.10
Enthusiastic–Quiet	Resourcefulness	–.11	Change Management	–.10
	Doing Whatever It Takes	–.17	Putting People at Ease	–.20
	Decisiveness	–.14	Career Management	–.18
	Confronting Problem Employees	–.10	Difficulty Building and Leading a Team	.11

CONTINUED

Among the T–F facet scales, scores toward the Feeling pole on all five are related to higher self-ratings on the *Benchmarks* scales Compassion and Sensitivity and Putting People at Ease. Four of the T–F facet scales correlate in the direction of Thinking with higher self-ratings on Problems with Interpersonal Relationships; the exception is Logical–Empathetic.

Among the J–P facet scales, there are no correlations at or above .10 for Planful–Open-Ended, Early Starting–Pressure–Prompted, or Scheduled–Spontaneous, which is unexpected. Respondents who scored toward Systematic and Methodical had higher self-ratings on Confronting Problem Employees. Also, respondents who scored toward Methodical had higher self-ratings on Participative Management, and those who scored toward

Emergent had higher self-ratings on Failure to Meet Business Objectives.

Most of these correlations support the validity of the Step II facet scales, although there are also a few surprises. One interesting and surprising result is that scores toward Methodical are related to higher self-ratings on Participative Management ($r = .10$). Perhaps people who score toward Methodical involve others at each step as they methodically work their way through a task. Also, one might question why there are not more results showing a relationship between the J–P facet scales and the *Benchmarks* Decisiveness scale. The clue lies in the definition of the Decisiveness scale, which describes this behavior: “prefers quick and approximate actions to slow and precise ones in many management

Table 10.10 | Correlations of MBTI® Step II™ facet scales with *Benchmarks*® scales self-ratings *CONTINUED*

Step II™ S–N facet scales	<i>Benchmarks</i> ® scales self-ratings			
Concrete–Abstract	Doing Whatever It Takes	.12	Being a Quick Study	.11
Realistic–Imaginative	Doing Whatever It Takes	.16	Being a Quick Study	.11
Practical–Conceptual	Doing Whatever It Takes	.12	Being a Quick Study	.13
Experiential–Theoretical	Doing Whatever It Takes	.12	Being a Quick Study	.10
Traditional–Original	Doing Whatever It Takes	.18	Being a Quick Study	.15
Step II™ T–F facet scales	<i>Benchmarks</i> ® scales self-ratings			
Logical–Empathetic	Being a Quick Study	–.13	Compassion and Sensitivity	.20
	Decisiveness	–.12	Putting People at Ease	.22
Reasonable–Compassionate	Compassion and Sensitivity	.24	Decisiveness	–.12
	Putting People at Ease	.21	Balance Between Personal Life and Work	.12
	Being a Quick Study	–.11	Problems with Interpersonal Relationships	–.13
Questioning–Accommodating	Compassion and Sensitivity	.12	Problems with Interpersonal Relationships	–.11
	Putting People at Ease	.12		
Critical–Accepting	Decisiveness	–.10	Balance Between Personal Life and Work	.12
	Participative Management	.10	Putting People at Ease	.18
	Compassion and Sensitivity	.19	Problems with Interpersonal Relationships	–.16
Tough–Tender	Doing Whatever It Takes	–.10	Balance Between Personal Life and Work	.12
	Decisiveness	–.16	Putting People at Ease	.20
	Confronting Problem Employees	–.11	Problems with Interpersonal Relationships	–.11
	Compassion and Sensitivity	.20		
Step II™ J–P facet scales	<i>Benchmarks</i> ® scales self-ratings			
Systematic–Casual	Confronting Problem Employees	–.11		
Methodical–Emergent	Confronting Problem Employees	–.10	Failure to Meet Business Objectives	.10
	Participative Management	–.10		

Note: Program participants at the Center for Creative Leadership who completed a Step II form of the MBTI assessment and the *Benchmarks* assessment ($N = 5,087$).

Source: Data provided by the Center for Creative Leadership. Used with permission.

situations.” Clearly, there are many managers who score toward the Judging poles of the J–P facet scales who would not be comfortable saying that they prefer “quick and approximate” actions, maybe preferring quick and precise actions instead. In contrast, the Confronting Problem Employees scale is defined as “acts decisively and with fairness,” and it is this scale that is related to the two J–P facet poles Systematic and Methodical.

Birkman Method® Assessment

The *Birkman Method*® personality assessment (Birkman et al., 2008) measures occupational preferences (Interests), effective behaviors (Usual behaviors), interpersonal and environmental preferences (Needs), and ineffective behaviors (Stress behaviors). The Form Q facet scales were correlated with the Birkman scales (Schaubhut & Thompson, 2011a) using data from a sample of 216 adults who had completed both assessments. Correlations between the two sets of scales produced a large number of coefficients; only some of the notable correlations for each set of facet scales are discussed here.

The E–I facet scales correlate in the direction of Extraversion with Birkman’s Social Orientation (Usual) scale, which measures the degree of social interaction sought by an individual; Change Orientation (Usual), which measures an individual’s degree of restlessness; Activity Preference (Usual), which measures the degree to which action, quick thinking, and physical energy are preferred; and Interaction Preference (Usual), which measures the extent to which an individual desires to be talkative, enjoys group parties, and is approachable. The E–I facet poles correlate in the direction of Introversion with Communication Preference (Usual), which measures sensitivity, including shyness, ability to say no, getting feelings hurt, and embarrassment; and Personal Autonomy (Usual), which measures the degree to which an individual’s pattern of responding to items is conventional or unconventional.

The S–N facet scales correlate in the direction of Intuition with Birkman’s Emotive Orientation (Usual and Needs), which measures an individual’s favored rate of action, as well as the extent to which an individual expresses emotions and makes decisions; Empathy Preference (Usual and Needs), which measures an individual’s comfort with expressing emotions and feelings; Personal Autonomy (Needs); and Thought Preference (Needs), which measures the extent to which conclusions and decisions are approached, the concern for making correct decisions the first time, and consequences of decisions.

The T–F facet scales correlate in the direction of Thinking with Birkman’s Activity Preference (Needs); Control Orientation (Usual), which measures the extent to which an individual approaches others; and Authority Preference (Usual), which measures how much an

individual desires to persuade others, speak out, and openly express opinions. The T–F facet scales correlate in the direction of Feeling with Communication Preference (Usual); Emotive Orientation (Needs); and Change Orientation (Needs).

The J–P facet scales correlate in the direction of Judging with Birkman’s Process Orientation (Usual), which measures the degree to which an individual wants accuracy and wishes to give or receive clear directions, use systems, finish tasks, and work with details; and Activity Preference (Usual). The J–P facet scales correlate in the direction of Perceiving with Emotive Orientation (Usual); Empathy Preference (Usual); Thought Preference (Usual); Personal Autonomy (Usual); and Control Orientation (Needs). The correlations of the Form Q facet scales with the scales of the *Birkman Method* assessment generally support the validity of the Step II facet scales, despite the fact that some of the Birkman scales seem to encompass multiple Step II facet scales.

Thomas-Kilmann Conflict Mode Instrument

The *Thomas-Kilmann Conflict Mode Instrument* (TKI® assessment) measures preferences for five different styles, or modes, of handling conflict that are combinations of different degrees of assertiveness and cooperation. The five modes are competing, collaborating, compromising, avoiding, and accommodating (Thomas & Kilmann, 1974). Relationships between the scales of the TKI and MBTI assessments have been examined by several researchers (Johnson, 1997; Kilmann & Thomas, 1975; Mathew & Bhatewara, 2006; Mills, Robey, & Smith, 1985; Percival, Smitheram, & Kelly, 1992; Schaubhut, Herk, & Thompson, 2009). More recently, the relationships were examined using Form Q (Schaubhut & Thompson, 2011a) with a sample of 4,344 adults. Correlations between Step II facet scales and TKI scales are presented in table 10.11. All of the E–I facet scales correlate in the direction of Extraversion with the TKI Collaborating scale and in the direction of Introversion with the TKI Avoiding scale. The S–N facet scales correlate in the direction of Sensing somewhat less, yet consistently, with the TKI Avoiding scale and in the direction of Intuition with the TKI Collaborating scale. All of the T–F facet scales correlate in the direction of Thinking with the TKI Competing scale and in the direction of Feeling with the Accommodating scale. None of the J–P facet scales highly correlates with any of the TKI scales, and the Compromising scale is not strongly related to any of the Step II facet scales. These results are congruent with the meaning of the Step II facet scales. (The same pattern of relationships is found in a comparison of MBTI Form M preference scales and TKI scales; Schaubhut et al., 2009.)

Table 10.11 | Correlations of MBTI® Step II™ Form Q facet scales with TKI® conflict modes

Step II™ Form Q facet scale	TKI® conflict modes				
	Competing	Collaborating	Compromising	Avoiding	Accommodating
E–I facet scales					
Initiating–Receiving	–.01	–.15	–.08	.21	.01
Expressive–Contained	.02	–.16	–.04	.19	–.03
Gregarious–Intimate	–.03	–.14	–.06	.19	.03
Active–Reflective	–.04	–.14	–.06	.20	.03
Enthusiastic–Quiet	–.08	–.16	–.04	.23	.04
S–N facet scales					
Concrete–Abstract	–.08	.10	.02	–.11	.08
Realistic–Imaginative	–.03	.11	.00	–.13	.08
Practical–Conceptual	–.02	.11	.01	–.11	.02
Experiential–Theoretical	–.02	.10	.00	–.11	.04
Traditional–Original	.08	.19	–.01	–.22	–.05
T–F facet scales					
Logical–Empathetic	–.30	–.07	–.04	.12	.35
Reasonable–Compassionate	–.37	–.08	–.02	.16	.39
Questioning–Accommodating	–.31	–.11	.03	.20	.25
Critical–Accepting	–.40	–.01	.09	.12	.31
Tough–Tender	–.41	–.08	.00	.18	.39
J–P facet scales					
Systematic–Casual	–.13	.00	–.03	.02	.17
Planful–Open-Ended	–.03	.02	–.05	.00	.07
Early Starting–Pressure-Prompted	.07	–.02	–.03	–.07	.02
Scheduled–Spontaneous	–.01	.01	–.07	.00	.08
Methodical–Emergent	.00	–.04	–.06	.02	.08

Note: CPP commercial sample, $N = 4,344$. Coefficients in bold type indicate correlations $\geq .20$ for I, N, F, and P and $\leq -.20$ for E, S, T, and J.

A preponderance of evidence indicates that the Step II facet scales correlate in predicted ways with the relevant scales of other instruments.

The Big Five Factors

In order to integrate different interpretations of the Big Five factors, John (1989, 1990) mapped them into a common language using the *Adjective Check List*. Using the findings from this research, the adjectives from the ACL were scored to represent the Big Five factors. The Step II facet scales were then correlated with these measures using data from a subset of participants in the global sample who completed the ACL ($n = 1,550$). The results are presented in table 10.12. All five of the E–I facet scales correlate in the direction of Extraversion with the Big Five factor Extraversion. Expressive–Contained also correlates in the direction of Extraversion with Big Five Agreeableness. Four of the S–N facet scales (the exception being Experiential–Theoretical) correlate in the direction of Intuition with Big Five Openness. All of the T–F facet scales correlate in the direction of Feeling

with Big Five Agreeableness. Three of the J–P facet scales—Early Starting–Pressure–Prompted, Scheduled–Spontaneous, and Methodical–Emergent—correlate in the direction of Judging with Big Five Conscientiousness. The Big Five ACL scale Neuroticism is unrelated to any of the Step II facet scales. These results offer strong evidence for the validity of the Step II facet scales. It is important to note that while the correlations between the two sets of scales are in the expected directions, the magnitude of the coefficients indicates that the parallel scales are not measuring exactly the same constructs.

GENDER DIFFERENCES

The Global Step II facet scales were examined for gender differences. The results are shown in table 10.13. Women in the global sample scored significantly more in the direction of Feeling on all five of the T–F facet scales, with effect sizes (Cohen’s d) that range from 0.42 to 0.59. Effect sizes for the other facet scales were trivial to small, the largest being that men scored more toward the Concrete facet pole ($d = -0.17$), the Systematic pole

Table 10.12 | Correlations of MBTI® Global Step II™ facet scales with ACL Big Five Factors

Global Step II™ facet scale	ACL Big Five Factors				
	Extraversion	Agreeableness	Conscientiousness	Openness	Neuroticism
E–I facet scales					
Initiating–Receiving	–.54	–.13	–.03	–.18	.13
Expressive–Contained	–.45	–.20	–.02	–.12	.08
Gregarious–Intimate	–.40	–.10	–.01	–.11	.10
Active–Reflective	–.52	–.12	–.01	–.17	.07
Enthusiastic–Quiet	–.48	–.16	.00	–.18	.05
S–N facet scales					
Concrete–Abstract	.11	.14	–.09	.24	.06
Realistic–Imaginative	.14	.15	–.09	.29	.08
Practical–Conceptual	.15	.09	–.04	.35	.05
Experiential–Theoretical	.00	.03	–.11	.13	.03
Traditional–Original	.24	.12	.01	.36	.02
T–F facet scales					
Logical–Empathetic	.04	.31	–.08	–.07	.16
Reasonable–Compassionate	–.03	.31	–.10	–.07	.11
Questioning–Accommodating	–.16	.23	–.09	–.17	.02
Critical–Accepting	.02	.39	–.01	–.01	.03
Tough–Tender	–.08	.37	–.03	–.03	.10
J–P facet scales					
Systematic–Casual	.13	.15	–.19	.13	.08
Planful–Open-Ended	.08	.08	–.18	.11	.01
Early Starting–Pressure-Prompted	.04	–.08	–.21	.08	.04
Scheduled–Spontaneous	.07	.00	–.24	.08	.06
Methodical–Emergent	.00	–.09	–.24	–.02	.01

Note: Portion of global sample that completed the ACL, $n = 1,550$. Coefficients in bold type indicate correlations $\geq .20$ for I, N, F, and P and $\leq -.20$ for E, S, T, and J.

(–0.15), the Realistic pole (–0.13), and the Quiet pole (0.12), and women scored more toward the Gregarious pole (–0.10). These findings are consistent with the distribution of the Step I preferences in the global sample (as shown in tables 7.10 and 7.12) and thus confirm the validity of the Step II facet scales.

CORRELATIONS WITH BEHAVIORAL RATINGS BY OTHERS

The correlations presented earlier in this chapter are all based on self-reports on the various criterion measures. Examined here is the relationship of the Step II facet scales to ratings by others—superiors, peers, direct reports, and spouses, among others—whose results come from a variety of instruments. Overall, these studies of ratings by others of behaviors associated with the facet scales offer strong support for the meaning of the scales.

Benchmarks Ratings by Others

The same sample of managers attending CCL programs ($N = 5,087$) whose self-ratings on the *Benchmarks* scales were discussed above (see table 10.10) were also rated on the same *Benchmarks* scales by their superiors, peers, and direct reports. While all five E–I facet scales correlated in the direction of Extraversion with higher self-ratings on a range of *Benchmarks* scales and in the direction of Introversion with higher self-ratings on some of the derailer scales, almost none of these relationships were observed in the ratings by others. For the ratings by others the *Benchmarks* scale Putting People at Ease was related to scores toward the Expressive and Gregarious facet poles for all ratings by others, and toward those two facet poles plus Initiating and Active for ratings by superiors and peers but not for ratings by direct reports. This *Benchmarks* scale was not related to Enthusiastic–Quiet for superiors, peers, or direct reports. These results mirror those reported in chapter 9 for the E–I preference scale, which use mean differences rather than correlations. Those results show that Extraverts tend to rate themselves higher on skills, and Introverts

Table 10.13 | Means, standard deviations, and gender effect sizes (Cohen's *d*) of MBTI® Global Step II™ continuous scores

Global Step II [®] facet scale	Men		Women		Cohen's <i>d</i>
	<i>M</i>	<i>SD</i>	<i>M</i>	<i>SD</i>	
<i>E–I facet scales</i>					
Initiating–Receiving	0.10	0.85	0.08	0.92	0.03
Expressive–Contained	0.09	0.81	0.07	0.88	0.02
Gregarious–Intimate	0.02	0.79	0.10	0.78	–0.10
Active–Reflective	0.07	0.81	0.07	0.83	0.00
Enthusiastic–Quiet	0.17	0.82	0.06	0.86	0.12
<i>S–N facet scales</i>					
Concrete–Abstract	–0.31	0.85	–0.16	0.89	–0.17
Realistic–Imaginative	–0.30	0.84	–0.19	0.87	–0.13
Practical–Conceptual	–0.21	0.80	–0.19	0.82	–0.03
Experiential–Theoretical	–0.13	0.82	–0.19	0.80	0.08
Traditional–Original	–0.20	0.85	–0.17	0.87	–0.03
<i>T–F facet scales</i>					
Logical–Empathetic	–0.35	0.83	0.15	0.87	–0.59
Reasonable–Compassionate	–0.30	0.83	0.12	0.87	–0.49
Questioning–Accommodating	–0.19	0.77	0.13	0.78	–0.42
Critical–Accepting	–0.28	0.77	0.12	0.75	–0.53
Tough–Tender	–0.29	0.85	0.10	0.82	–0.47
<i>J–P facet scales</i>					
Systematic–Casual	–0.14	0.87	–0.01	0.90	–0.15
Planful–Open-Ended	0.07	0.84	–0.01	0.89	0.09
Early Starting–Pressure-Prompted	0.05	0.79	–0.02	0.80	0.08
Scheduled–Spontaneous	0.03	0.88	–0.03	0.90	0.06
Methodical–Emergent	0.08	0.79	0.01	0.81	0.09

Note: Global sample, *N* = 16,773 (men, *n* = 7,771; women, *n* = 9,002).

higher on problems, to a greater extent than do outside observers, and these same tendencies are reflected in the facet scales as well. The difference, however, is that for the E–I preference scale there are no significant differences for ratings by direct reports, whereas for the facet scales, the correlations show that two of the E–I facet scales (Expressive–Contained and Gregarious–Intimate) do exhibit a relationship (with higher ratings toward Expressive and Gregarious). Also, the fact that the score on Enthusiastic–Quiet is not related to any ratings by others and correlates with higher self-ratings on fewer *Benchmarks* scales than do other E–I facet scales points to the distinct meanings of the facet scales, even those within the same preference scale.

Among the S–N facet scales, no correlations with *Benchmarks* scales meet the criteria ($r \geq .10$) for any of the ratings by others.

The T–F facet scales show the greatest variation across scales both within and across raters. Two *Benchmarks* scales, Compassion and Sensitivity and Putting People at Ease, correlate with the Feeling pole on four of the T–F

facet scales for all raters; the exception is Questioning–Accommodating. One of the scales tapping potential flaws, Problems with Interpersonal Relationships, correlates with the Thinking pole on four of the T–F facet scales for all raters; again, the exception is Questioning–Accommodating. The study also shows that superiors tended to see fewer differences on the T–F facet scales than did any of the other raters. Another interesting observation is that the self-ratings on the *Benchmarks* “cognitive” scales (Being a Quick Study, Decisiveness) correlate with Logical–Empathetic and Reasonable–Compassionate in the direction of Thinking, whereas these scales do not correlate with ratings by others, who tended to focus on observable interpersonal behaviors. The Compassionate and Accepting facet poles are related to higher ratings by peers and direct reports on Participative Leadership, and these two facet poles, plus Tender, also correlate with higher ratings by peers and direct reports on Balance Between Personal Life and Work.

Among the J–P facet scales, there are no significant relationships for any raters for three facet scales: Planful–Open–Ended, Early Starting–Pressure–Prompted, and Scheduled–Spontaneous. The Casual facet pole is related to higher ratings by peers on Putting People at Ease.

Studies of management behaviors show that relationships between individual facet scales within each preference scale and also with facet scales within other preference scales may vary depending on who does the rating.

Van Velsor and Fleenor (1997) conducted a similar study correlating *Benchmarks* scales with the facet scales of an earlier Step II form of the MBTI assessment and concluded that the results “do show evidence of the added value of the . . . subscales [facet scales] in the study of leadership skills” (p. 157). The results of the study reported above confirm that conclusion and also provide evidence for the validity of the individual facet scales.

Survey of Management Practices Ratings of Managers by Others

Johnson and Golden (1994) studied the relationship between the facet scales of a prior Step II form and the scales of the *Survey of Management Practices* (SMP). The SMP is a 360-degree feedback instrument with scales measuring organizational skills and workplace climate variables. A sample of 208 managers from companies in the healthcare, grocery distribution, construction engineering, and pharmaceutical industries completed a prior Step II form and also were rated on the SMP by 1,766 managers, peers, and direct reports, whose ratings were combined. Ten Step II facet scales showed correlations with SMP scales, ranging from .05 to .22: Initiating–Receiving, Expressive–Contained, Enthusiastic–Quiet, Concrete–Abstract, Realistic–Imaginative, Logical–Empathetic, Questioning–Accommodating, Critical–Accepting, Tough–Tender, and Methodical–Emergent. Those correlations of $r \geq .10$ are shown in table 10.14. This threshold was chosen because the SMP ratings are composite ratings by managers, peers, and direct reports, which tend to lower correlations.

The significant correlations, although they appear low, are in the range typically obtained from composite ratings by others. Many of the management practices describing people-oriented behaviors (e.g., *communication, feedback, team building, approachability*) are related to facet scales in the direction of Extraversion (the Initiating, Expressive, and Enthusiastic poles) and Feeling (the Empathetic, Accommodating, Accepting, and Tender poles) and, to a lesser extent, Intuition (Abstract, Imaginative). In contrast, management practices describing

achievement-oriented behaviors, such as *goal pressure* and *control of details*, are related to the Critical and Methodical facet poles, respectively. A somewhat unexpected finding is that scores toward the Methodical pole are also related to higher ratings on three of the people-oriented behaviors: *upward communication, recognizing good performance, and interest in growth of subordinates*. This is congruent with the results above for the *Benchmarks* ratings, which show that scores toward Methodical are related to higher self-ratings on Participative Management. Furthermore, it can be seen in the table that Step II facet scales from all four MBTI preference pairs are represented, but in no instance do all five facet scales within a preference scale correlate significantly with any SMP scale. This suggests that the characteristics represented by individual Step II facet scales make an important contribution to understanding management behavior. These results provide strong support for the validity of the facet scales.

PROFILOR® Ratings of Managers by Direct Reports

Quast and Hansen (1994) examined the relationship of the *PROFILOR*, an instrument designed to measure managerial skills shown to be critical to managerial effectiveness, to a prior Step II form. A sample of 79 managers was rated on 25 dimensions by their direct reports, and those ratings were correlated with the managers’ scores on the Step II facet scales. The four facet scales that yield the most significant ($p < .05$) correlations with the skill scales are Questioning–Accommodating (all 25 skill scales, all in the direction of Accommodating), Active–Reflective (20 skill scales, all in the direction of Active), Experiential–Theoretical (14 skill scales, all in the direction of Experiential), and Practical–Conceptual (9 skill scales, all in the direction of Practical). For example, the Questioning–Accommodating facet scale correlates, in the direction of the Accommodating pole, with *PROFILOR* scales such as Champion Change and Influence Others; the Active facet pole is related to *PROFILOR* scales such as Manage Disagreements and Build Relationships. There are no significant correlations for nine of the Step II facet scales (Initiating–Receiving, Expressive–Contained, Reasonable–Compassionate, Critical–Accepting, Tough–Tender, Systematic–Casual, Scheduled–Spontaneous, Planful–Open–Ended, and Methodical–Emergent). The remaining seven facet scales correlate significantly with from one to four *PROFILOR* skill scales. The specific skill scales and the direction of the correlations support the validity of the Step II facet scales.

Adjective Ratings from Family and Friends

Harker and Reynierse (1999) examined the relationship between facet scale scores and independent observer ratings using a 276-item list of adjectives and descriptive

Table 10.14 | Correlations of MBTI® Step II™ facet scales with composite ratings of managers by others on the Survey of Management Practices

Step II™ facet scale	Survey of Management Practices scale			
Initiating–Receiving	Clarification of goals	–.12	Approachability	–.13
	Upward communication	–.18	Team building	–.17
	Work facilitation	–.13	Building trust	–.10
	Feedback	–.16	Interest in growth of subordinates	–.20
	Recognizing good performance	–.21		
Expressive–Contained	Clarification of goals	–.10	Approachability	–.12
	Upward communication	–.12	Team building	–.13
	Recognizing good performance	–.14	Opportunity for growth	–.13
Enthusiastic–Quiet	Clarification of goals	–.12	Team building	–.10
	Upward communication	–.10	Interest in growth of subordinates	–.10
	Recognizing good performance	–.11	Opportunity for growth	–.12
Concrete–Abstract	Upward communication	.10	Interest in growth of subordinates	.12
	Work facilitation	.12	Building trust	.11
Realistic–Imaginative	Upward communication	.11	Recognizing good performance	.10
	Feedback	.11		
Logical–Empathetic	Upward communication	.16	Team building	.16
	Work facilitation	.10	Interest in growth of subordinates	.12
	Goal pressure	–.21	Building trust	.14
	Delegation (permissiveness)	.16	Team atmosphere	.13
	Recognizing good performance	.10	Tension level	–.11
	Approachability	.19		
Questioning–Accommodating	Goal pressure	–.19	Approachability	.10
	Permissiveness	.13		
Critical–Accepting	Upward communication	.16	Team building	.15
	Work facilitation	.11	Interest in growth of subordinates	.17
	Goal pressure	–.17	Building trust	.12
	Recognizing good performance	.15	Opportunity for growth	.11
	Approachability	.15		
Tough–Tender	Upward communication	.12	Interest in growth of subordinates	.10
	Goal pressure	–.22	Building trust	.12
	Permissiveness	.14	Team atmosphere	.11
	Approachability	.16	Organizational climate	.10
	Team building	.10		
Methodical–Emergent	Upward communication	–.11	Recognizing good performance	–.11
	Orderly work planning	–.10	Interest in growth of subordinates	–.10
	Control of details	–.11		

Note: Managers who completed an MBTI Step II form and were rated by 1,766 supervisors, subordinates, and peers, $N = 208$. Positive correlations are in the direction of the I, N, F and P poles of the facet scales; negative correlations are in the direction of the E, S, T and J poles.

Source: Adapted from Johnson and Golden (1994).

phrases that they constructed. The subjects were 641 individuals who completed a prior Step II form. Each subject selected one person (a family member, friend, or co-worker) who knew him or her well and who was asked to complete the rating questionnaire with the subject in mind. The raters did not know the Step II results of their subject. Table 10.15 shows the top descriptors chosen by the raters for subjects who scored at each pole of the Step II facet scales. The results show an impressive match between observer ratings and facet pole preferences for the majority of the Step II facet scales. In many instances, the adjective used to describe the subject is the same as the facet pole name (e.g., *concrete* was the descriptor chosen most consistently for individuals who scored toward the Concrete pole and *abstract* for individuals who scored toward the Abstract pole). In other cases, the adjectives used by the independent raters are descriptors of the facet poles (e.g., *good mixer* and *social facilitator* for the Initiating pole). In rare cases, the adjectives seem not to align with the meaning of the pole (e.g., *enjoys pressure* for Reasonable), and some adjectives are surprisingly absent (e.g., Gregarious individuals were not described as *gregarious*). In some of these instances, perhaps few descriptors were available to the raters that captured the intent of the poles. Or the preferences manifested by these poles may not be apparent to outside observers, as are some of the behaviors associated with other poles. Yet overall the results demonstrate that behaviors of individuals who score at the poles of the facet scales are observable by others in predictable ways.

THE RELATIONSHIP OF STEP II™ FACET SCALES TO STEP I™ PREFERENCE SCALES

The discussion of the validity of the Step II facet scales to this point has focused on the relationships between the facet scales and the scales of other instruments and with ratings by others. These correlational analyses dealt only with the meaning of the facet scales themselves. In practice, however, the Step II facet scales are not interpreted in isolation. As discussed in chapter 6, the meaning of the facets is embedded, by definition and practice, in the meaning of the MBTI Step I preferences. Indeed, correlations between facet scales and their associated preference scales range from .62 to .93 (see table 10.2), and so it may be asked what additional information about an individual is gained from the Step II facet scales. The next stage in establishing the validity of the Global Step II assessment requires an approach that examines Step II results in relation to Step I results.

Contribution of the Step II™ Facet Scales

A primary question related to the validity of the Step II facet scales is whether they add to our understanding of dependent variables beyond what can be learned

about those variables from the four preference scales alone. To answer this question, a series of hierarchical regressions was performed using two sets of survey questions, one administered to the Form Q US national sample and one to the global sample, as the dependent variables. The surveys contained questions designed to elicit self-descriptions of attitudes and behaviors in the domains of work, health, stress and coping, relationships, satisfaction, happiness, and values. Before proceeding to the results, it is important to point out that this kind of analysis is a blunt instrument. The only purpose was to determine whether the addition of the 20 facet scales to a regression equation increased the amount of variance explained by the dependent variables over and above the variance explained by the four preference scales. No attempt was made to determine whether specific facet scales serve as significant predictors of particular dependent variables. (More targeted analyses of the effect of specific facet scales on specific dependent variables are described below.) The hierarchical regression analysis can help answer the question of how much additional explanatory power resides in the Step II facet scales.

This analysis was conducted on dependent variables available from two samples: the global sample and the Form Q US national sample. For the variables from the global sample, the logits derived from latent class analysis (LCA) scoring for the Step I preference scales were entered first. In the second step, the 20 Step II facet scores (thetas) were entered. For the variables from the Form Q US national sample, the four Step I preference clarity index (pci) scores were entered first, as the independent variables, followed by the scores for the 20 Step II facet scales. For each of the variable sets in the two samples, the results of the two models were then compared. Since the second regression equation contains both Step I and Step II scores, this analysis allows an examination of how much additional variance is explained by the facet scales, holding constant the variance explained by the Step I scales.

Table 10.16 shows the results for the 6 variables from the global sample and the 25 variables from the Form Q US national sample that showed significant improvements in R-squared change, which is the statistic used to quantify the amount of additional variance explained. The adjusted R-squared is reported in the table because the adjusted value takes into account the number of predictor variables in the regression equation. It increases only if the new variables, in this case the scores from the 20 facet scales, improve the model more than would be expected by chance. The adjusted value will always be lower than the original R-squared. The significance of R-squared change is tested with an F-test. The last column in the table shows the significance of the change in the F statistic, labeled *F-change*. A significant F-change

Table 10.15 | Top descriptors chosen by independent observers for people who scored at each of the MBTI® Step II™ facet poles

E-I facet scales			S-N facet scales			T-F facet scales			J-P facet scales		
Initiating	Receiving	assertively outgoing good mixer gregarious social facilitator talkative	Concrete	Abstract	Logical	Empathetic	Systematic	Casual			
Expressive	Contained	expresses feelings easily easy to know engaging	Realistic	Imaginative	Reasonable	Compassionate	Playful	Open-Ended			
Gregarious	Intimate	— prefers one-on-one	Practical	Conceptual	Questioning	Accommodating	Early Starting	Pressure-Prompted			
Active	Reflective	enjoys being with others fun loving easy to know	Experiential	Theoretical	Critical	Accepting	Scheduled	Spontaneous			
Enthusiastic	Quiet	flirtatious amusing gregarious	Traditional	Original	Tough	Tender	Methodical	Emergent			

Note: Individuals who completed either Form M or Form K and selected one rater to complete a 276-item rating questionnaire, N = 641.
Source: Descriptions by others adapted from Harker and Reynierse, 1999. Used with permission.

Table 10.16 | Predictive power of MBTI® Step I™ preference scales and Step II™ facet scales

Global sample	Dependent variable	Independent variables	R	Adjusted R ²	Sig. of F-change
	Life satisfaction	4 preference scales	.200	.040	
		4 preference scales + 20 facet scales	.223	.048	.000
	Health satisfaction	4 preference scales	.174	.029	
		4 preference scales + 20 facet scales	.208	.039	.000
	Happiness	4 preference scales	.207	.042	
		4 preference scales + 20 facet scales	.227	.050	.000
	Work satisfaction	4 preference scales	.112	.012	
		4 preference scales + 20 facet scales	.155	.022	.000
	Job satisfaction	4 preference scales	.093	.008	
		4 preference scales + 20 facet scales	.156	.020	.000
	Intent to leave job	4 preference scales	.066	.004	
		4 preference scales + 20 facet scales	.105	.009	.000
Form Q US national sample	Dependent variable	Independent variables	R	Adjusted R ²	Sig. of F-change
Preferred organizational environment	Clear structures	4 preference scales	.235	.046	
		4 preference scales + 20 facet scales	.397	.110	.000
	Everyone must toe the line	4 preference scales	.164	.018	
		4 preference scales + 20 facet scales	.366	.084	.000
	Variety of tasks	4 preference scales	.119	.005	
		4 preference scales + 20 facet scales	.318	.049	.006
Organizational behaviors	Part of a team	4 preference scales	.225	.042	
		4 preference scales + 20 facet scales	.354	.075	.022
	Punctuality	4 preference scales	.165	.018	
		4 preference scales + 20 facet scales	.399	.107	.000
	Finish the job	4 preference scales	.187	.025	
		4 preference scales + 20 facet scales	.344	.064	.014
	Self-control at work	4 preference scales	.157	.015	
		4 preference scales + 20 facet scales	.315	.044	.044

CONTINUED

means that the variables added in that step significantly improved the prediction.

For example, in predicting whether individuals would like to work in an organization that has clear structures for assigning responsibility, the Adjusted R² column shows that the four preference scales account for about 4.6% of the variance in this variable. When scores from the 20 facet scales are added to the regression equation, the amount of variance accounted for (adjusted for the number of variables) increases to 11%, an increase that is significant at $p < .001$.

These analyses show that the Step II facet scales significantly increase the amount of variance explained for a wide range of dependent variables in two different samples, over and above that explained by the four Step I preference scales. To people unfamiliar with interpreting R-squared, the magnitudes reported in the table may appear small. However, as Cohen (1988) points out, in the behavioral sciences R-squared is rarely larger than

.25, which is defined as a large effect size. A variable with R² = .09, which accounts for 9% of the variance in a dependent variable, is said to have a medium effect size, and a small effect size is defined as R² = .01. Cohen writes that such values “may represent stronger degrees of association than they seem” (p. 532).

While these analyses show the added effect of all 20 facet scales as a set, the next question is, what is the role of individual facet scales in understanding particular dependent variables?

DIFFERENCES BETWEEN FACET POLES AND THE OVERALL PREFERENCE

The analyses reported above show that for a large number of variables, the Step II facet scales as a group provided information over and above that provided by the Step I preference scales as a group. The individual facet scales within each preference scale are discussed next, moving from the use of correlations of continuous

Table 10.16 | Predictive power of MBTI® Step I™ preference scales and Step II™ facet scales *CONTINUED*

Form Q US national sample	Dependent variable	Independent variables	R	Adjusted R ²	Sig. of F-change
Coping with stress	Be upset, angry, and show it	4 preference scales	.125	.007	
		4 preference scales + 20 facet scales	.355	.077	.000
	Be optimistic	4 preference scales	.252	.055	
		4 preference scales + 20 facet scales	.396	.108	.001
	Talk to someone close	4 preference scales	.223	.041	
		4 preference scales + 20 facet scales	.349	.072	.024
	Avoid stressful situations	4 preference scales	.185	.026	
		4 preference scales + 20 facet scales	.321	.052	.043
Values	Religion/spirituality	4 preference scales	.281	.070	
		4 preference scales + 20 facet scales	.467	.173	.000
	Intellectual stimulation	4 preference scales	.413	.163	
		4 preference scales + 20 facet scales	.511	.218	.000
	Power	4 preference scales	.251	.054	
		4 preference scales + 20 facet scales	.444	.150	.000
	Prestige/status	4 preference scales	.247	.053	
		4 preference scales + 20 facet scales	.422	.131	.000
	Excitement	4 preference scales	.221	.040	
		4 preference scales + 20 facet scales	.439	.146	.000
	Education/learning	4 preference scales	.257	.058	
		4 preference scales + 20 facet scales	.382	.097	.009
	Health	4 preference scales	.221	.040	
		4 preference scales + 20 facet scales	.351	.073	.022
	Recreation	4 preference scales	.162	.017	
		4 preference scales + 20 facet scales	.320	.051	.023
Health	Achievement/accomplishment	4 preference scales	.247	.052	
		4 preference scales + 20 facet scales	.361	.081	.035
	Relationships/friendships	4 preference scales	.402	.154	
		4 preference scales + 20 facet scales	.471	.177	.043
	Smoking	4 preference scales	.083	.001	
		4 preference scales + 20 facet scales	.321	.051	.002
	Hypertension	4 preference scales	.089	.001	
		4 preference scales + 20 facet scales	.302	.038	.012
	Heart problems	4 preference scales	.123	.006	
		4 preference scales + 20 facet scales	.310	.043	.016

Note: Global sample, $N = 16,773$. Form Q US national sample, $N = 1,378$.

scores to the use of mean differences on categorical variables for both the facet scales and the overall preference scales. For each of the four preference scales, the difference in the means of respondents who scored at each preference pole were compared. The same was done for the facet scales (i.e., examining the differences in the means of respondents at each pole of the facet scale, those who scored 1 through 5 toward

that pole, eliminating only those respondents whose facet score was 0).³ In this analysis, the effect on a dependent variable of being at the pole of a facet scale was compared to the effect of being at the pole of the overall preference scale. The dependent variables were 26 of the scales of the CPI 434 assessment (Gough & Bradley, 1996). The sample consisted of 4,924 managers attending leadership programs at the Center for Creative

3. A facet score of 0 captures a very small range of continuous scores around the point where theta $[\Theta] = 0$. Therefore, the analyses here may very slightly overestimate the effect of the facet scales compared to the effect of the preference scales.

Leadership. The results are shown in table 10.17. The numbers in the cells of this table are effect sizes indexed by Cohen's *d*. For example, the value in the first row of the T–F section of the table, $d = -0.05$, shows that there is very little overall effect of Thinking versus Feeling on the Dominance scale. The negative sign indicates that the mean of Thinking types is higher than the mean of Feeling types. (As always, negative signs indicate higher means for the E, S, T, and J poles of the preference scales or the facet scales, and positive signs point to higher means for the I, N, F, and P poles of the preference scales or the facet scales.) Moving down to the Tough–Tender row in the first column, the effect on Dominance of being at the Tough versus the Tender pole is greater ($d = -0.24$) than is the effect of Thinking versus Feeling. The negative sign points to the fact that the mean of respondents who scored toward the Tough pole is higher than the mean of those who scored toward the Tender pole. In this example, the facet scale has more effect on Dominance than does the overall preference scale.

The effect sizes for the facet scales will now be compared to the effect sizes for the overall preference scales. The discussion below calls out those results in which the difference in the two effect sizes as measured by Cohen's *d* is greater than or equal to 0.05. This is a somewhat arbitrary value, but it allows the results in the table to be summarized.

For every CPI 434 scale, there is at least one Step II facet scale for which the opposite poles of the facet scale have a greater effect than do the opposite poles of the overall preference scale. In other words, the difference (in units of standard deviation, which is what Cohen's *d* reflects) between respondents who scored toward the opposite poles of the facet scale is greater than the difference between respondents who scored toward the opposite poles of the preference scale for that dependent variable. Likewise, from the perspective of the facet scales, in almost every case there is at least one CPI 434 scale for which there is a facet scale effect that is greater than the preference scale effect; the one exception is Experiential–Theoretical.

There are too many results to review facet by facet, but a few patterns stand out. One is that there are some facet scales with many effect sizes larger than that for the overall preference scale. Within E–I, the facet scales with the most effect sizes larger than the effect sizes for the E–I preference scale are Initiating–Receiving and Enthusiastic–Quiet. Within S–N, the Traditional–Original facet scale has larger effects than does the S–N preference scale on 12 of the 26 CPI 434 scales. Within T–F, Questioning–Accommodating has larger effects than does the T–F scale on 16 of the CPI 434 scales;

Logical–Empathetic has larger effects than does T–F on 13 CPI 434 scales; and Critical–Accepting has larger effects than does T–F on 11 CPI 434 scales. Within J–P, Systematic–Casual has larger effects than does J–P on 8 of the CPI 434 scales, and Methodical–Emergent has larger effects than does J–P on 7 of the CPI 434 scales.

The largest difference in effect size between a facet scale and the overall preference scale within E–I is that for the effect of Enthusiastic–Quiet on the Self-control scale (respondents who scored toward Quiet had higher scores on Self-control). Within S–N, the largest effect difference is that for Traditional–Original on the Independence scale (in the direction of Original). Within T–F, the largest effect difference is that for Questioning–Accommodating on the Good Impression scale (in the direction of Accommodating). Within J–P, it is that for Systematic–Casual on the Sensitivity scale (in the direction of Casual).⁴

For every CPI 434 scale, there is at least one Step II facet scale for which the opposite poles of that facet scale have a greater effect than do the opposite poles of the overall preference scale, i.e., the effect difference between the facet poles is greater than the effect difference between the preference poles for that particular dependent variable.

Also of interest are crossover effects, which are facet pole effects that are in the direction opposite that of the preference effects. For example, the effect of Questioning–Accommodating on Self-control has a sign indicating that respondents who scored at the Feeling pole, Accommodating ($d = 0.29$), have more self-control and the effect is much larger than was the T–F effect (-0.08), the sign of which indicates that Thinking types have slightly more self-control. In contrast, the effect for another T–F facet scale, Logical–Empathetic, is toward the Thinking pole Logical (-0.27), which is in the same direction as the overall preference. The fact that the effects of these two facet scales point to opposite poles of the underlying preference scale explains why the overall effect of the T–F preference scale is so low.

One of the largest crossover effects is that of Questioning–Accommodating on Creative Temperament. The difference between respondents who scored at the poles of this facet scale for the CPI 434 Creative Temperament scale indicates a medium effect size ($d = -0.42$) and points toward Questioning as the highest mean. The effect size for the T–F preference scale is 0.23, which is about half that size and points to the Feeling pole of the overall preference scale. The facet scales with the largest number of crossover effects are Questioning–Accommodating, for which this result

4. *Sensitivity* is the name used on the CPI 260 assessment for what is the Masculinity/Femininity scale on the CPI 434 assessment.

Table 10.17 | Differences in effect sizes (Cohen's *d*) between poles of the MBTI® preference scales and poles of the facet scales on CPI™ 434 scales

	Cohen's <i>d</i>												
	Do	Cs	Sy	Sp	Sa	In	Em	Re	So	Sc	Gi	Cm	Wb
E–I scale	–1.33	–1.10	–1.54	–0.90	–1.14	–0.67	–0.88	–0.10	–0.11	0.40	–0.06	0.17	–0.35
Initiating–Receiving	–1.35	–1.09	–1.54	–0.80	–1.18	–0.69	–0.89	–0.18	–0.10	0.27	–0.13	0.14	–0.37
Expressive–Contained	–0.83	–0.71	–0.97	–0.63	–0.74	–0.41	–0.61	–0.07	–0.14	0.29	–0.08	0.12	–0.26
Gregarious–Intimate	–0.91	–0.75	–1.10	–0.68	–0.81	–0.46	–0.63	–0.05	–0.19	0.33	–0.07	0.10	–0.34
Active–Reflective	–1.08	–0.81	–1.28	–0.76	–0.98	–0.56	–0.67	0.01	–0.13	0.37	–0.07	0.17	–0.34
Enthusiastic–Quiet	–1.01	–0.98	–1.20	–0.86	–0.92	–0.62	–0.77	–0.02	–0.05	0.55	0.08	0.22	–0.25
S–N scale	0.26	0.64	0.32	0.55	0.31	0.49	0.65	0.23	–0.27	–0.31	–0.26	–0.27	0.06
Concrete–Abstract	0.19	0.62	0.28	0.51	0.27	0.42	0.63	0.24	–0.28	–0.28	–0.25	–0.28	0.05
Realistic–Imaginative	0.23	0.64	0.33	0.54	0.33	0.43	0.61	0.19	–0.24	–0.37	–0.29	–0.26	0.03
Practical–Conceptual	0.19	0.60	0.28	0.40	0.24	0.33	0.58	0.27	–0.21	–0.19	–0.17	–0.16	0.02
Experiential–Theoretical	0.21	0.50	0.25	0.41	0.26	0.40	0.53	0.23	–0.25	–0.23	–0.21	–0.20	0.03
Traditional–Original	0.34	0.66	0.36	0.60	0.41	0.61	0.65	0.18	–0.34	–0.37	–0.32	–0.35	0.07
T–F scale	–0.05	0.16	0.17	0.12	0.02	–0.19	0.30	0.12	–0.05	–0.08	–0.04	–0.09	–0.15
Logical–Empathetic	–0.09	0.11	0.17	0.16	0.07	–0.25	0.25	–0.09	–0.17	–0.27	–0.21	–0.21	–0.24
Reasonable–Compassionate	–0.16	0.06	0.09	–0.01	–0.08	–0.28	0.24	0.16	–0.04	0.05	0.04	–0.08	–0.15
Questioning–Accommodating	–0.21	–0.28	–0.11	–0.30	–0.26	–0.38	–0.20	0.04	0.22	0.29	0.37	0.12	0.01
Critical–Accepting	–0.04	0.07	0.15	–0.01	–0.04	–0.18	0.25	0.23	0.13	0.14	0.19	0.16	0.11
Tough–Tender	–0.24	0.04	0.00	–0.01	–0.17	–0.36	0.20	0.15	–0.01	0.10	0.07	0.05	–0.15
J–P scale	0.01	0.23	0.09	0.37	0.14	0.21	0.37	–0.18	–0.34	–0.39	–0.40	–0.36	–0.06
Systematic–Casual	0.04	0.36	0.22	0.50	0.23	0.21	0.52	–0.18	–0.30	–0.42	–0.39	–0.36	–0.07
Planful–Open-Ended	–0.06	0.13	0.01	0.26	0.07	0.15	0.29	–0.15	–0.28	–0.26	–0.29	–0.27	–0.02
Early Starting–Pressure-Prompted	0.08	0.26	0.12	0.32	0.21	0.24	0.33	–0.16	–0.27	–0.40	–0.34	–0.36	–0.07
Scheduled–Spontaneous	–0.05	0.19	0.03	0.33	0.12	0.20	0.31	–0.17	–0.36	–0.40	–0.47	–0.38	–0.11
Methodical–Emergent	–0.09	0.02	–0.07	0.16	0.00	0.06	0.13	–0.28	–0.36	–0.37	–0.36	–0.38	–0.18

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occurs on 17 CPI 434 scales, and Critical–Accepting (11 scales). This partially reflects the relatively low correlations of these facet scales with the T–F preference scale (.29 and .55, respectively, for Form Q facet scales with Form M preference scales) and the fact that the overlap of items between these facet scales and the T–F preference scale is very low (0 and 1, respectively). So, it is not surprising that these two facet scales have the most crossover effects. (This pattern may change for the Global Step II form, as the correlations of these two facet scales with the T–F preference scale are larger and the item overlap is somewhat greater.) There are no crossover effects within the E–I and S–N facet scales. Among the J–P facet scales, Methodical–Emergent has seven crossover effects and Systematic–Casual has one.

These results can also be examined from the perspective of individuals who score high on a given CPI 434 scale. For example, the CPI 434 Dominance scale has a very large effect for E–I ($d = -1.33$) and a small effect for S–N ($d = 0.26$). Given the signs of these d values, individuals who score high on Dominance would be described as

extraverted and intuitive. But when the facet scales that have greater effects than the preference scales are taken into account, individuals who score high on Dominance can be more precisely described as extraverted, but also original, reasonable, questioning, tough, planful, pressure-prompted, and methodical. This greatly adds to the understanding of those who score high on Dominance.

Another analysis used to examine differences between outcomes for the preference scales and the facet scales, based on data from the Form Q US national sample, was performed for a set of variables measuring job satisfaction. Respondents were asked how satisfied they were with various aspects of their job. Respondents were offered a four-point response scale from “very satisfied” to “very dissatisfied.” Because the distribution of responses was so skewed toward being satisfied with one’s job, the two satisfaction categories were collapsed, as were the two dissatisfaction categories. The two resultant categories were then used as independent variables in an analysis of variance. For some of the job

Table 10.17 | Differences in effect sizes (Cohen's *d*) between poles of the MBTI® preference scales and poles of the facet scales on CPI™ 434 scales *CONTINUED*

	Cohen's <i>d</i>												
	To	Ac	Ai	Ie	Pm	Fx	Sn*	Mp	Wo	Ct	Lp	Ami	Leo
E–I scale	–0.20	–0.23	–0.22	–0.41	–0.12	–0.21	0.38	–0.54	–0.11	–0.59	–0.97	–0.03	–0.11
Initiating–Receiving	–0.24	–0.28	–0.23	–0.49	–0.15	–0.18	0.32	–0.56	–0.14	–0.58	–1.00	–0.07	–0.13
Expressive–Contained	–0.17	–0.19	–0.13	–0.24	–0.07	–0.17	0.23	–0.40	–0.11	–0.39	–0.65	–0.11	–0.08
Gregarious–Intimate	–0.16	–0.20	–0.18	–0.31	–0.07	–0.09	0.36	–0.42	–0.10	–0.37	–0.73	–0.10	–0.18
Active–Reflective	–0.13	–0.10	–0.10	–0.28	–0.05	–0.12	0.45	–0.44	–0.07	–0.46	–0.80	–0.04	–0.22
Enthusiastic–Quiet	–0.16	–0.11	–0.24	–0.34	–0.11	–0.31	0.32	–0.40	0.01	–0.64	–0.74	0.04	0.00
S–N scale	0.37	–0.09	0.69	0.54	0.39	0.79	0.11	0.20	0.05	0.92	0.22	–0.06	–0.51
Concrete–Abstract	0.38	–0.14	0.70	0.54	0.38	0.83	0.21	0.19	0.06	0.90	0.18	–0.03	–0.57
Realistic–Imaginative	0.36	–0.10	0.63	0.47	0.32	0.75	0.11	0.19	0.04	0.87	0.19	–0.08	–0.48
Practical–Conceptual	0.36	0.01	0.66	0.51	0.42	0.61	0.19	0.18	0.07	0.72	0.20	–0.08	–0.47
Experiential–Theoretical	0.28	–0.08	0.61	0.46	0.33	0.71	0.08	0.15	0.02	0.75	0.19	–0.09	–0.42
Traditional–Original	0.30	–0.19	0.62	0.55	0.42	0.74	–0.03	0.19	–0.02	1.00	0.30	–0.13	–0.55
T–F scale	0.19	–0.07	0.11	–0.09	–0.16	0.35	0.54	–0.01	–0.03	0.23	–0.08	0.09	–0.24
Logical–Empathetic	0.08	–0.25	–0.05	–0.25	–0.30	0.37	0.58	–0.15	–0.19	0.23	–0.18	0.01	–0.32
Reasonable–Compassionate	0.22	–0.05	0.19	–0.10	–0.14	0.39	0.63	–0.05	0.01	0.19	–0.15	0.17	–0.25
Questioning–Accommodating	–0.02	0.18	–0.23	–0.26	–0.22	–0.29	0.16	–0.01	0.14	–0.42	–0.14	0.27	0.21
Critical–Accepting	0.30	0.13	0.21	–0.01	–0.03	0.22	0.30	0.15	0.20	0.07	0.02	0.31	–0.03
Tough–Tender	0.20	–0.02	0.17	–0.08	–0.10	0.35	0.57	–0.04	0.08	0.10	–0.22	0.20	–0.25
J–P scale	0.12	–0.70	0.21	0.08	0.05	0.88	–0.01	–0.17	–0.16	0.81	–0.15	–0.17	–0.46
Systematic–Casual	0.15	–0.52	0.27	0.08	0.04	0.89	0.19	–0.12	–0.15	0.82	–0.09	–0.13	–0.51
Planful–Open-Ended	0.11	–0.65	0.20	0.08	0.05	0.70	–0.06	–0.16	–0.10	0.66	–0.17	–0.10	–0.33
Early Starting–Pressure-Prompted	0.02	–0.46	0.20	0.09	0.02	0.69	–0.06	–0.10	–0.19	0.61	–0.06	–0.20	–0.34
Scheduled–Spontaneous	0.10	–0.76	0.21	0.05	0.01	0.87	0.04	–0.21	–0.22	0.79	–0.19	–0.19	–0.51
Methodical–Emergent	–0.04	–0.64	–0.01	–0.12	–0.15	0.64	0.05	–0.24	–0.31	0.44	–0.28	–0.20	–0.37

* Sensitivity (Sn) is the name for the original CPI scale formerly called Masculinity/Femininity.

Note: *N* = 4,924. Negative signs indicate higher means for the E, S, T and J poles of the preference scales or the facet scales; positive signs indicate higher means for the I, N, F and P poles of the preference scales or the facet scales.

Source: Data provided by the Center for Creative Leadership, re-analyzed for this manual.

satisfaction questions, the results are significant for both a Step I preference scale and all five facet scales within it. For example, for satisfaction with job security, the overall E–I preference is significant ($p < .001$), with Extraverts being more satisfied, and all five of the E–I facet scales are also significant (all $p < .01$), with respondents who scored toward the Extraversion poles of the facet scales being more satisfied. Likewise, for the question on satisfaction with co-workers, there are significant differences for the overall E–I preference scale ($p < .001$) and the T–F preference scale ($p < .001$), with the satisfied group having mean scores toward Extraversion and Feeling. Furthermore, four of the five E–I facet scales and all five of the T–F facet scales show the same pattern as the overall preference scales. Only Enthusiastic–Quiet does not fit the pattern, reaching significance at $p < .02$.

For other aspects of work satisfaction, however, the Step I and Step II results differ. With some variables, the Step I preference scale is significant but only some of

the Step II facet scales associated with it show significant differences between respondents who were satisfied and those who were not. For example, when asked about satisfaction with day-to-day tasks, the differences between the satisfied and dissatisfied groups are significant for three of the Step I preference scales. For the E–I preference scale, the mean of the group that was satisfied with day-to-day tasks is toward Extraversion ($p < .04$). Of the E–I facet scales, however, only the Expressive–Contained scale shows significant differences ($p < .005$). For the S–N preference scale, both groups have a mean toward Sensing, reflecting the distribution of types in the US national sample, but the group that was satisfied with day-to-day tasks has a mean score significantly more toward Sensing ($p < .04$) than does the group that was not satisfied. Of the S–N facet scales, only Traditional–Original shows a significant difference ($p < .006$) between the satisfied and dissatisfied groups, toward the Traditional pole. The difference for the J–P preference scale is also significant ($p < .03$), with the

dissatisfied group having a mean score toward Perceiving and the satisfied group a mean score toward Judging. Of the J–P facet scales, only Scheduled–Spontaneous shows a significant difference ($p < .005$); both the satisfied and dissatisfied groups have a mean score toward the Judging pole of the J–P preference scale, but those who were satisfied with day-to-day tasks scored significantly closer to Scheduled, the Judging pole of the facet scale.

Another kind of difference between the Step I and Step II scales was also observed. In some instances, the Step I preference scale is not significant but one or more of the Step II facet scales within it are. For example, respondents were asked to gauge their satisfaction with the amount of responsibility they had at work. No significant differences exist between the satisfied and dissatisfied groups for any of the four Step I preference scales. However, there are significant differences for the Traditional–Original and Questioning–Accommodating facet scales in the direction of the Traditional and Accommodating poles.

The results discussed above demonstrate that the facet scales provide more information about, or have a greater effect on, selected variables than do the overall preference scales. Thus both the intention and the validity of the Step II facet scales are supported.

Midzone Scores

Midzone scores (–1, 0, and 1) are routinely discussed with respondents during a Step II interpretation, and thus respondents who score in the midzone deserve attention. In this section validity evidence for the midzone scores will first be examined and then related to the overall preferences. Specifically, two questions can be asked about midzone scores: (1) Do individuals who score in the midzone on a facet scale differ from those who score at either pole of that scale? And (2) Do individuals with opposite preferences who score in the midzone on the same facet scale differ from one another depending on their overall preference (e.g., does an Introvert with a midzone score on, say, Enthusiastic–Quiet, differ from an Extravert with a midzone score on the same facet scale)?

To answer the first question, five possible patterns can be visually summarized by the shape that results when a facet scale is graphed against a dependent variable:

- No differences (a flat line)
- A monotonically increasing or decreasing line as one moves from the left pole of the facet scale, through the midzone, to the right pole
- A “hockey stick” (facing either left or right, up or down) pattern, wherein individuals in the midzone have scores similar to those of individuals at one pole, and the scores of these two groups differ from those of the group at the other pole

- A V shape, indicating that individuals who score in the midzone have lower scores than do those who score at either pole
- An inverted V, indicating that individuals who score in the midzone have higher scores than do those who score at either pole

Note that for the V or inverted V pattern, the individuals at the poles may or may not have equal scores. The results for both *Benchmarks* ratings and CPI 434 scores show all five of these patterns. Examples follow.

Analyses revealed that for some managerial behaviors, individuals who score in the midzone on a facet scale differ from those who score at either pole and from each other, depending on their overall preference.

For example, on the *Benchmarks* scale Being a Quick Study, very little difference is found between the self-ratings of respondents who scored in the midzone on the Expressive–Contained facet scale and those who scored toward Expressive ($d = -0.02$) or Contained ($d = 0.02$). If average self-ratings in these three groups (Expressive, midzone, and Contained) were graphed, they would show a flat line. In contrast, an example of a decreasing difference can be seen in figure 10.1, which shows the scores on CPI 434 Dominance for the Gregarious–Intimate facet scale. The difference between respondents who scored in the midzone and those who scored at either pole was the same and represented a medium effect ($p < .001$, $d = 0.50$). An example of the third pattern, the “hockey stick” pattern, can be seen in figure 10.2, which shows scores on CPI 434 Leadership for the Practical–Conceptual facet scale. There was no difference between respondents who scored in the midzone and those who scored toward the Conceptual pole on this facet scale ($d = 0.0$), while respondents who scored toward the Practical pole differed from the midzone and Conceptual groups ($d = 0.21$). There are many *Benchmarks* and CPI 424 scales that fall somewhere between the two patterns illustrated in figures 10.1 and 10.2, in that there is a slight difference between respondents who scored in the midzone and

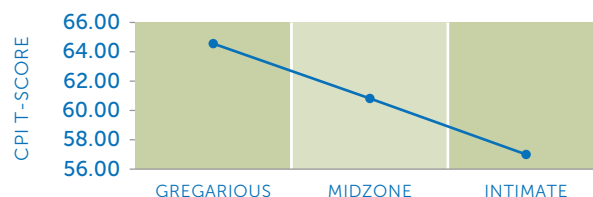


Figure 10.1 | CPI™ Dominance scale by MBTI® Step II™ Gregarious–Intimate facet scale

Note: $N = 5,122$. Gregarious–Intimate, $p < .001$.

Source: Data provided by the Center for Creative Leadership. Used with permission.

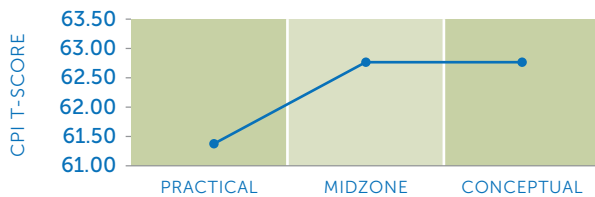


Figure 10.2 | CPI® Leadership scale by MBTI® Practical–Conceptual facet scale

Note: $N = 5,122$. Practical–Conceptual, $p < .001$.

Source: Data provided by the Center for Creative Leadership. Used with permission.

those who scored at one of the poles, which is reflected in an inflection in the line, with these two groups differing significantly from those at the other pole.

The fourth kind of result, the V pattern, can be seen in figures 10.3 and 10.4. Figure 10.3 shows the CPI Amicability scale graphed against the three scoring categories of the Logical–Empathetic facet scale. Respondents in the midzone scored lower on Amicability than did those at either facet pole. This pattern may represent some degree of unpredictability in individuals who score in the midzone. In work settings, these individuals are often described as challenging because their colleagues don't know how they will behave from one day to the next.

A similar pattern can be seen in figure 10.4 for one of the *Benchmarks* derail scales. This figure shows differences on the Questioning–Accommodating facet scale for the *Benchmarks* scale Difficulty Building and Leading a Team. On this facet scale, respondents at the poles tended to rate themselves higher, that is, as having more difficulty. Those in the midzone tended to rate themselves lower than did those at the Questioning pole ($d = 0.15$) and at the Accommodating pole ($d = -0.13$), which indicates less difficulty (overall $p < .001$). It may be that individuals who score in the midzone on this facet are better able to balance or alternate between questioning and accommodating behaviors, and that such balance is an advantage in successfully building and leading teams.

Finally, the opposite pattern, the inverted V, can be seen in figure 10.5, which illustrates a case in which respondents who scored in the midzone on the Systematic–Casual facet scale rated themselves higher on the *Benchmarks* scale Balance Between Personal Life and Work than did those who scored at the poles of the facet scale (overall $p < .004$). One might hypothesize that individuals at the Systematic pole tend to emphasize work and those at the Casual pole tend to emphasize personal life, but that remains to be tested. Regardless, individuals in the midzone appear to have achieved a balance between the two contexts.

The second question that can be asked about the midzone is whether significant differences exist between individuals who score in the midzone of a particular facet scale but at opposite poles of the overall preference scale. For example, are there differences on some dependent variables between individuals who are “midzone Extraverts” and those who are “midzone Introverts” on the same facet? Thirteen facet scales show such a difference on at least one *Benchmarks* scale. There is one such difference for Concrete–Abstract, Realistic–Imaginative, Practical–Conceptual, Tough–Tender, Systematic–Casual, and Planful–Open-Ended. With the exception of Concrete–Abstract and Practical–Conceptual, all of these midzone differences are on different *Benchmarks* scales. There

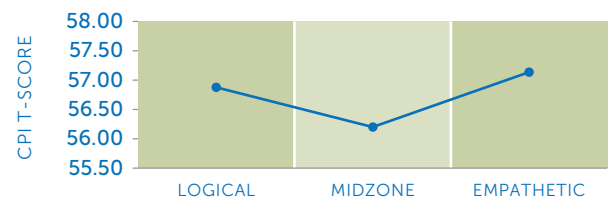


Figure 10.3 | CPI® Amicability scale by MBTI® Logical–Empathetic facet scale

Note: $N = 5,122$. Logical–Empathetic, $p < .06$.

Source: Data provided by the Center for Creative Leadership. Used with permission.

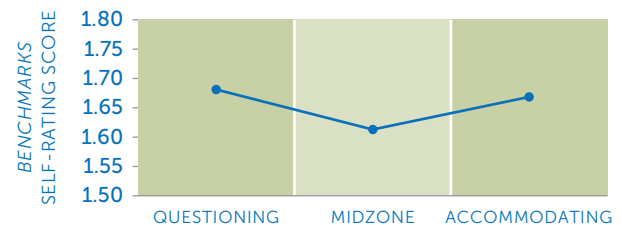


Figure 10.4 | *Benchmarks*® Difficulty Building and Leading a Team scale by MBTI® Questioning–Accommodating facet scale

Note: $N = 4,429$. Questioning–Accommodating, $p < .001$.

Source: Data provided by the Center for Creative Leadership. Used with permission.

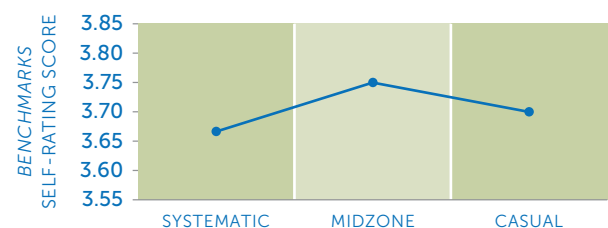


Figure 10.5 | *Benchmarks*® Balance Between Personal Life and Work scale by MBTI® Systematic–Casual facet scale

Note: $N = 4,438$. Systematic–Casual, $p < .004$.

Source: Data provided by the Center for Creative Leadership. Used with permission.

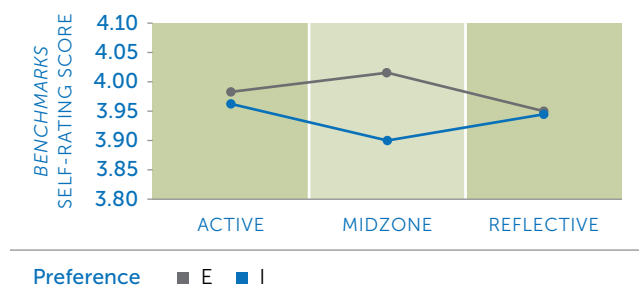


Figure 10.6 | *Benchmarks® Being a Quick Study scale by MBTI® Extraversion–Introversion preference scale and Active–Reflective facet scale*

Note: $N = 4,440$. $E-I$, $p < .001$; Active–Reflective, $p < .10$.

Source: Data provided by the Center for Creative Leadership. Used with permission.

are differences between midzone respondents with opposite preferences on three *Benchmarks* scales each for Expressive–Contained, Logical–Empathetic, and Questioning–Accommodating; on four *Benchmarks* scales for Critical–Accepting; and on five *Benchmarks* scales for Gregarious–Intimate. The differences for the T–F facet scales are primarily on the *Benchmarks* scales Compassion and Sensitivity and Putting People at Ease. For these scales, higher self-ratings are obtained by midzone Feeling types than by midzone Thinking types. The two facet scales with by far the greatest number of midzone differences are Enthusiastic–Quiet (11 *Benchmarks* scales) and Active–Reflective (16 *Benchmarks* scales). For Active–Reflective, three *Benchmarks* scales with midzone differences exhibit a larger effect than does any other comparison (i.e., any comparison of preferences, facet poles, or any combination thereof): Being a Quick Study ($d = -0.21$), Compassion and Sensitivity (-0.24), and Self-Awareness (-0.28). This can be seen in figure 10.6 for Being a Quick Study. Extraverts in the midzone on Active–Reflective rated themselves higher on these three *Benchmarks* scales than did Introverts in the midzone or respondents at the poles. The main effect for E–I is significant ($p < .001$), the difference between the facet categories approached significance ($p < .10$), and the interaction effect is significant at $p < .08$, using, per Cohen’s advice (1988), a less stringent criterion for interaction effects.

From the perspective of the *Benchmarks* scales, those with the greatest number of significant effects for respondents in the midzone are Doing Whatever It Takes (four E–I facet scales and two S–N facet scales), Compassion and Sensitivity (four T–F facet scales and two E–I facet scales), Putting People at Ease (three T–F facet scales and three E–I facet scales), and Career Management (four E–I facet scales and one T–F facet scale), which is about four times the number that would be expected by chance.

In-Preference or Out-of-Preference Scores

Another way to look at facet scales in relation to the overall preferences is to examine whether it makes a difference that a score is in-preference or out-of-preference. (See chapters 2 and 6 for a definition and explanation of out-of-preference scores.) While such a designation clearly has interpretive utility, it presents some problems for analysis because both scoring groups (in-preference and out-of-preference) contain a combination of individuals from opposite poles of the facet scale. For example, looking at individuals who are in-preference and out-of-preference on Tough–Tender, the in-preference group lumps together Tough Thinking types and Tender Feeling types, while the out-of-preference group consists of Tough Feeling types plus Tender Thinking types. To the extent that there are significant differences between individuals at the poles, as there is on, say, the *Benchmarks* Decisiveness scale ($d = -0.34$), and to the extent that there are more individuals in the in-preference group than in the out-of-preference group, as is always the case, then it would be expected that mean self-ratings for the in-preference group would be higher than those for the out-of-preference group, and this is indeed the outcome ($d = -0.25$).

In most cases, however, the in-preference vs. out-of-preference comparison will tend to cancel out differences, and this is what was observed. Among the 440 possible comparisons on the *Benchmarks* scales, only two in-preference vs. out-of-preference comparisons within the E–I facet scales reached the $d \geq 0.20$ cutoff: Enthusiastic–Quiet on Putting People at Ease and on Career Management. The in-preference respondents rated themselves higher on both of these *Benchmarks* scales than did the out-of-preference respondents. No comparisons within the S–N or J–P facet scales reached the cutoff. For the T–F facet scales, five such effects were found: Reasonable–Compassionate on Decisiveness and Tough–Tender on Confronting Problem Employees, for which the in-preference respondents rated themselves higher than did the out-of-preference respondents, and three instances in which the out-of-preference respondents rated themselves higher than did the in-preference respondents (Logical–Empathetic on Compassion and Sensitivity, and Reasonable–Compassionate on two scales, Putting People at Ease and Career Management). Taking one result as an illustration, the fact that the out-of-preference respondents rated themselves higher on Compassion and Sensitivity than did the in-preference respondents indicates that the combination of Empathetic Thinking types and Logical Feeling types had higher self-ratings than did the combination of Logical Thinking types and Empathetic Feeling types. This difference was driven primarily by the very low self-ratings by the in-preference Logical Thinking types on

the Compassion and Sensitivity scale, which is consistent with the theory.

Bringing the midzone into this kind of analysis, individuals who score in the midzone on a facet scale are compared to those who are in-preference or out-of-preference on that facet scale. There were no E–I or S–N facet scales on which respondents in the midzone rated themselves significantly differently than did those who were in-preference or out-of-preference. Two significant differences were found among the T–F facet scales: On both Logical–Empathetic and Reasonable–Compassionate, respondents who scored in the midzone had higher self-ratings on Compassion and Sensitivity and Putting People at Ease than did those who were in-preference. Again, this difference was driven by the very low self-ratings by the Logical Thinking types and the Reasonable Thinking types on these two people-oriented *Benchmarks* scales. Among the J–P facet scales, respondents in the midzone on Planful–Open-Ended rated themselves higher on Resourcefulness than did those who were out-of-preference (the Planful Perceiving types combined with the Open-Ended Judging types).

All Combinations of Facet Scale Groups with Preference Scales

All of the analyses discussed above grouped together certain individuals who might be expected to differ on some dependent variables. To examine further the contribution of the facet scales in relation to the overall preferences, it is helpful to disaggregate these groupings and compare all possible combinations of preference, facet pole, in- and out-of-preference scores, and scores in the midzone.

For any given dependent variable there are 17 comparisons that can be analyzed:

- The opposites of the preference pair, for example, Sensing vs. Intuition.
- The opposite poles of a facet scale within that preference pair, for example, Concrete vs. Abstract.
- The opposites of the preference pair combined with the opposite poles of the facet scale. There are six such combinations. To use two examples, individuals who prefer Intuition and score toward Abstract (in-preference) can be compared with individuals who prefer Sensing and score toward Abstract (out-of-preference). Or, individuals who prefer Intuition and score toward Abstract (in-preference) can be compared with individuals who prefer Sensing and score toward Concrete (in-preference). Note that the latter comparison is different from the overall comparison of Concrete vs. Abstract, which contains some individuals who have out-of-preference scores.

- Midzone results. These add an additional nine comparisons. For example, individuals who prefer Intuition and score in the midzone on a given facet scale can be compared to individuals who prefer Sensing and score in the midzone on that facet scale. Or, Intuitive types who score in the midzone can be compared to Intuitive types who score toward Abstract (in-preference).

To present all 17 comparisons for all 20 facet scales for all 21 *Benchmarks* scales is beyond the scope of this manual, so a few examples will be used as illustrations. The illustrations provide further evidence for the validity of the Step II facet scales.

Table 10.18 shows, for the Logical–Empathetic facet scale, the groups with the highest and lowest self-ratings on each of the *Benchmarks* scales and Cohen’s *d* for (1) the difference between these highest and lowest groups, (2) the overall T–F difference, and (3) the overall difference between respondents at the Logical and Empathetic poles. There are four *Benchmarks* scales with significant T–F effects and six with significant effects for the Logical vs. Empathetic difference. But looking at the fourth column in the table, which shows Cohen’s *d* for the difference between the groups with the highest and lowest self-ratings, 16 *Benchmarks* scales have a more specific combination of preference, facet pole, and/or midzone score that has a significant effect. In other words, for 16 *Benchmarks* scales the effects of various combinations of preference and facet category are greater than or equal to the effect of either the difference between the preferences in a pair or the difference between facet poles.

For example, there are two *Benchmarks* scales that reached the cutoff, Resourcefulness and Straightforwardness and Composure, on which Logical Thinking types rated themselves highest and Empathetic Thinking types rated themselves lowest. The results for Resourcefulness are shown in figure 10.7. There were also two *Benchmarks* scales, Doing Whatever It Takes and Decisiveness, on which Empathetic Thinking types rated themselves highest and Empathetic Feeling types rated themselves lowest. Decisiveness is shown in figure 10.8. An ANOVA of this breakdown shows a significant main effect for T–F ($p < .005$) and for the interaction of Thinking–Feeling with Logical–Empathetic ($p < .04$). On four *Benchmarks* scales (Balance Between Personal Life and Work, Difficulty Building and Leading a Team, Difficulty Changing or Adapting, and Failure to Meet Business Objectives—i.e., problems with follow-through), respondents in one of the midzone groups on the Logical–Empathetic facet scale are among those who rated themselves highest. And on one *Benchmarks* scale, Participative Management, midzone scorers rated themselves lowest. So being in the midzone apparently may help one achieve some balance between personal

Table 10.18 | Differences between highest and lowest self-ratings on *Benchmarks* scales for combinations of the MBTI® Thinking–Feeling preference scale and the Logical–Empathetic facet scale

<i>Benchmarks</i> ® scale	Self-rating		Cohen's <i>d</i>		
	Highest	Lowest	Highest vs lowest	Thinking vs Feeling	Logical vs Empathetic
Resourcefulness	Logical T	Empathetic T	−0.30	−0.09	−0.12
Doing Whatever It Takes	Empathetic T	Empathetic F	−0.34	−0.13	−0.16
Being a Quick Study	Logical T	Empathetic F	−0.32	−0.19	−0.32
Decisiveness	Empathetic T	Empathetic F	−0.40	−0.22	−0.25
Leading Employees	Empathetic T	Midzone T	0.18	0.04	0.03
Confronting Problem Employees	Logical T	Midzone F, Logical F	−0.11	−0.09	−0.06
Participative Management	Empathetic F	Midzone T	−0.23	0.14	0.18
Change Management	Logical T	Empathetic T	−0.16	−0.03	−0.04
Building and Mending Relationships	Empathetic F	Empathetic T	0.30	0.16	0.17
Compassion and Sensitivity	Empathetic F	Logical T	0.55	0.44	0.52
Straightforwardness and Composure	Logical T	Empathetic T	−0.30	−0.07	−0.15
Balance Between Personal Life and Work	Midzone F	Logical T, Midzone T, Empathetic T	0.29	0.23	0.19
Self-Awareness	Empathetic T	Logical F	0.25	0.05	0.13
Putting People at Ease	Empathetic F	Logical T	0.62	0.43	0.58
Differences Matter	Logical F	Empathetic T	0.34	0.13	0.10
Career Management	Empathetic F	Logical F	0.27	0.16	0.23
Problems With Interpersonal Relationships	Logical T	Empathetic F	−0.24	−0.18	−0.23
Difficulty Building and Leading a Team	Logical T, Midzone T	Empathetic T	−0.31	−0.12	−0.13
Difficulty Changing or Adapting	Logical F, Midzone T	Empathetic T, Empathetic F, Midzone F	−0.11	−0.01	−0.04
Failure to Meet Business Objectives	Midzone T, Empathetic F	Empathetic T	−0.29	0.08	0.13
Too Narrow Functional Orientation	Empathetic F	Logical T	0.13	0.08	0.12

Note: *N* = 4,924. Numbers in bold type indicate Cohen's *d* ≥ .20 for I, N, F, and P and ≤ −.20 for E, S, T, and J.

Source: Data provided by the Center for Creative Leadership. Used with permission.

life and work and may lead to fewer difficulties with teamwork, dealing with change, or following through.

Continuing the examination of table 10.18, we see five *Benchmarks* scales for which respondents who were in-preference at one pole of a facet scale rated themselves higher than did those who were in-preference at the opposite pole: two (Being a Quick Study, Problems with Interpersonal Relationships) on which Logical Thinking types rated themselves higher than did Empathetic Feeling types and three (Compassion and Sensitivity, Putting People at Ease, Too Narrow Functional Orientation) on which Empathetic Feeling types rated themselves higher than did Logical Thinking types. On two *Benchmarks* scales respondents in the two out-of-preference facet groups rated themselves highest or lowest: Empathetic Thinking types rated themselves higher than did Logical Feeling types on Self-Awareness, while the opposite was true for Differences Matter, where Logical Feeling types rated themselves higher than did Empathetic Thinking types. Finally, there is one scale (Building and Mending Relationships) on which

Empathetic Feeling types rated themselves highest and Empathetic Thinking types rated themselves lowest, and one (Career Management) on which Empathetic Feeling types had the highest self-ratings and Logical Feeling types had the lowest.

This very diverse collection of results for Thinking–Feeling by Logical–midzone–Empathetic respondents was not replicated when ratings by direct reports were examined for the same facet scale. In this case, on seven *Benchmarks* scales significant effects between the ratings of Thinking and Feeling types are apparent, and on an additional four *Benchmarks* scales significant effects between the ratings of respondents at the Logical and Empathetic facet poles are apparent. But the most prevalent pattern of high and low ratings by direct reports is that for Empathetic Thinking types, who were most often rated lowest, and Empathetic Feeling types, who are most often rated highest. This pattern is evident for 20 of the *Benchmarks* scales. The one exception is Being a Quick Study, on which Logical Thinking types were most often rated highest and Empathetic Thinking types were most often rated lowest. An interesting idea for future research would be to examine in more depth how Empathetic Thinking types are perceived by direct reports.

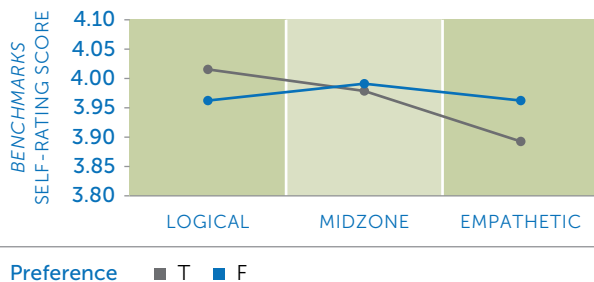


Figure 10.7 | *Benchmarks*® Resourcefulness scale by MBTI® Thinking–Feeling preference scale and Logical–Empathetic facet scale

Note: $N = 4,442$. T–F, $p < .015$; Logical–Empathetic, $p < .016$.
Source: Data provided by the Center for Creative Leadership. Used with permission.

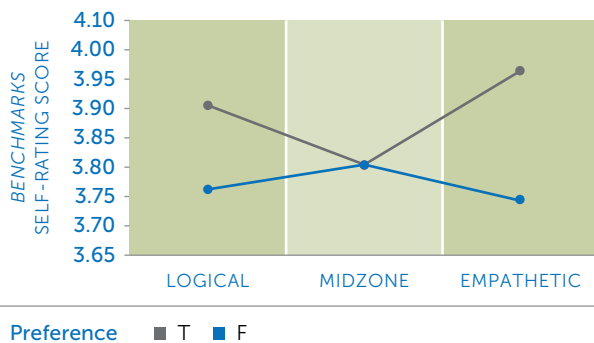


Figure 10.8 | *Benchmarks*® Decisiveness scale by MBTI® Thinking–Feeling preference scale and Logical–Empathetic facet scale

Note: $N = 4,441$. T–F, $p < .001$; Logical–Empathetic, $p < .001$.
Source: Data provided by the Center for Creative Leadership. Used with permission.

The most prevalent pattern of high and low ratings by direct reports is that for Empathetic Thinking types, who were most often rated lowest, and Empathetic Feeling types, who were most often rated highest. This pattern is evident for 20 of the *Benchmarks* scales.

Other facet scales show very different patterns. For example, for Initiating–Receiving, on 10 *Benchmarks* scales a specific combination of preference and facet pole has more of an effect than does either the overall E–I difference or the difference in ratings between respondents at the poles of the facet. For 6 of these 10 scales, the comparison between the ratings of the two in-preference groups has the maximum separation—the Initiating Extraverts rated themselves highest and the Receiving Introverts rated themselves lowest. For two scales, it is the Receiving Extraverts who rated themselves highest and the Receiving Introverts who rated themselves lowest. For one scale (Problems with Interpersonal Relationships), it is the Receiving Extraverts who had the highest self-ratings and the Initiating Extraverts who had the lowest (i.e., the out-of-preference respondents rated themselves higher than did those who were in-preference; perhaps this out-of-preference behavior confuses people or otherwise creates problems with their relationships). This result is shown in figure 10.9. The effect for E–I is not significant; the effect for the facet scale is significant at $p < .02$, but, more important, the interaction between Extraversion–Introversion and Initiating–Receiving is significant at

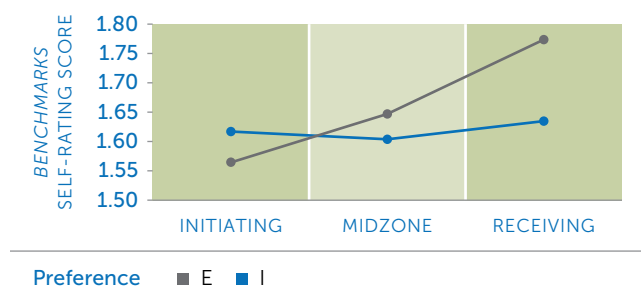


Figure 10.9 | **Benchmarks® Problems with Interpersonal Relationships scale by MBTI® Extraversion–Introversion preference scale and Initiating–Receiving facet scale**

Note: $N = 4,434$. E–I, ns; Initiating–Receiving, $p < .02$.

Source: Data provided by the Center for Creative Leadership. Used with permission.

$p < .03$. Finally, for one scale it is the out-of-preference groups that are at the extremes—the Initiating Introverts had the highest self-ratings and the Receiving Extraverts had the lowest. In regard to bosses' ratings, these same two groups are rated highest and lowest on five of the nine scales on which some combination of preference by facet has greater effects than do the overall E–I preference or the Initiating–Receiving facet pole preference alone.

THE RELATIONSHIP OF STEP II™ FACETS TO WHOLE TYPES

In addition to considering the relationship of the facets to the preferences, researchers have examined the relationship of the facets to whole types. Mitchell (1993), using a prior Step II form, tested for the existence of regression discontinuities in the facet scales by comparing opposite types. He gave two examples to illustrate his findings. For ISTJs and ENFPs, he found regression discontinuities on Concrete–Abstract, Logical–Empathetic, and Early Starting–Pressure–Prompted. For ISTPs and ENTJs, the discontinuities were on Enthusiastic–Quiet, Concrete–Abstract, Critical–Accepting, and Early Starting–Pressure–Prompted. He wrote that similar results were obtained for other pairs of opposites, but he did not list them.

Mitchell (1993) also examined the relationship of facets to whole types using 32 pairs of types that differed by only one preference—for example, ENFJs were compared to ENTJs. Using a balanced sample of 110 participants of each MBTI type, he compared upper and lower boundaries on the facet scales for each of the 32 pairs. Boundaries were defined as above (or below) the scores of 95% of the participants of that type. For example, if 95% of the ISTJs in the sample had a score of 3 or less on the Abstract side of the Concrete–Abstract facet scale, then 3 was the upper boundary for that type on that facet scale. Upper and lower boundaries were then

also determined for ESTJs, and these two boundaries were compared. Boundaries for the two types were not considered different unless they varied by 2 or more facet scale points. With the exception of the INTJ–ENTJ pair, every pair of similar types exhibited differences in their boundaries on one or more facet scales belonging to their shared preferences. No comparisons were made on the facet scales of opposite preferences, since the types would obviously be different on those facet scales. Additional data for this approach are available for the global sample and can be seen in table 6.10. This table shows, for each of the 16 types, the facet scales with the highest means within each of the four preference scales. For example, for INTPs the highest mean on the E–I facet scales was for Contained. Every one of the 16 types has a different pattern of highest facet scores.

These three studies add to the validity evidence that the facet scales demonstrate individual differences within whole types. And, in fact, all of the results in this section indicate that the facet scales are performing their function and contributing to the understanding of individual differences within type. The significant interaction effects shown also demonstrate the complex relationship between the overall preference scales and the individual facet scales and provide evidence for the need to interpret them in relation to one another.

It is clear from the many studies of the facet scales that the facet scales achieve their intended goal of demonstrating individual differences within whole types.

VALIDITY SHOWN IN STUDIES OF STEP II™ APPLICATIONS

Additional evidence for the validity of the Step II facet scales comes from research on applications of the Step II assessment in various settings. The following is a brief review of these studies in five different areas: organizations; health, stress, and coping; relationships; life satisfaction; and foreign language learning.

Organizations

Research on the contribution of the Step II assessment to understanding behavior in organizations sheds light on specific facet scales. The research also details instances in which whole type and type dynamics provide additional insight into organizational behavior.

PREFERRED FEATURES OF ORGANIZATIONS

Participants in the Form Q US national sample were presented with 11 features of organizations and asked to indicate their degree of liking for each feature on a 3-point scale: "dislike," "indifferent," or "like." The

correlations of Step II facet scales with the degree of liking for each feature are presented in table 10.19. Because the response scale consists of only three options, which attenuates the correlations, coefficients that are greater than or equal to .15 are considered meaningful.

As shown in the table, within the E–I facet scales, scores toward the Gregarious and Active facet poles correlate with enjoyment of teamwork. Respondents who scored toward the Initiating, Expressive, and Enthusiastic facet poles reported liking to work in organizations that employ people from different backgrounds. Those who scored toward the Enthusiastic facet pole also reported liking to perform a variety of tasks.

The facet scales within the S–N preference scale are related to many of the organizational features shown in the table. Respondents who scored toward the Sensing pole of all five S–N facet scales reported liking an organization that rewards employee loyalty, provides lifetime job security, does everything “by the book,” and simplifies jobs. Those who scored toward the Intuition pole of all five S–N facet scales reported liking opportunities to work in other countries. Respondents who scored toward Traditional reported liking clear structures and responsibilities and liking to work in an environment in which everyone must toe the line (i.e., follow the rules). In contrast, those who scored toward Original reported liking opportunities for advancement and high pay, even when job security is

Table 10.19 | **Correlations of MBTI® Step II™ Form Q facet scales with degree of liking for various features of organizations**

Step II™ Form Q facet scale	Feature of organizations										
	Clear structures and responsibilities	Independence and individual achievement	Employee loyalty and lifetime job security	Everything done by the book	Teamwork	Opportunities to work in other countries	Simplify jobs	Employees from different backgrounds	Variety of tasks	Everyone must toe the line	Advancement, high pay, but less security
E–I facet scales											
Initiating–Receiving	.08	.02	.01	.02	–.10	–.09	.06	–.15	–.06	.04	–.03
Expressive–Contained	.06	–.02	.03	.05	–.13	–.02	.09	–.16	–.03	.03	.07
Gregarious–Intimate	.00	.02	–.05	–.02	–.17	.05	–.06	–.02	–.01	–.05	–.02
Active–Reflective	–.02	.06	–.06	–.05	–.19	–.02	–.01	–.06	–.04	–.03	–.02
Enthusiastic–Quiet	–.01	–.02	.00	.00	–.10	–.08	.05	–.15	–.16	–.07	.00
S–N facet scales											
Concrete–Abstract	–.14	.09	–.19	–.24	–.05	.21	–.25	.14	.13	–.14	.08
Realistic–Imaginative	–.09	.08	–.16	–.25	–.13	.22	–.24	.12	.12	–.12	.02
Practical–Conceptual	–.07	.06	–.16	–.15	–.09	.24	–.20	.09	.04	–.10	.07
Experiential–Theoretical	–.12	.09	–.20	–.24	–.05	.21	–.22	.11	.07	–.17	.08
Traditional–Original	–.19	.12	–.20	–.24	–.14	.21	–.24	.09	.09	–.15	.16
T–F facet scales											
Logical–Empathetic	.10	–.08	.09	–.02	.08	–.06	.09	.13	.03	.06	–.23
Reasonable–Compassionate	.12	–.10	.11	.02	.13	–.11	.14	.13	–.04	.01	–.29
Questioning–Accommodating	.06	–.07	.09	.14	.15	–.16	.16	.04	–.12	.04	–.16
Critical–Accepting	.03	–.04	.00	.00	.09	–.04	.08	.10	–.03	–.02	–.11
Tough–Tender	.12	–.08	.06	–.05	.10	–.12	.10	.09	–.07	–.08	–.20
J–P facet scales											
Systematic–Casual	–.17	–.02	–.01	–.12	–.06	–.04	–.03	.02	.03	–.09	–.07
Planful–Open-Ended	–.08	–.06	.01	–.05	–.09	.03	.01	–.06	.02	–.08	.06
Early Starting–Pressure-Prompted	–.17	.02	–.10	–.11	–.13	.08	–.13	–.07	.02	–.03	.13
Scheduled–Spontaneous	–.18	.00	–.02	–.12	–.15	.00	–.05	–.10	.02	–.14	.05
Methodical–Emergent	–.07	–.06	–.01	–.11	–.07	–.03	.00	–.08	.04	–.12	.06

Note: Form Q US national sample, N = 1,378. Coefficients in bold type indicate correlations $\geq .15$ for I, N, F, and P and $\leq -.15$ for E, S, T, and J.

limited. Those who scored toward Experiential also liked organizations where everyone must toe the line.

Within the T–F preference scale, scores toward all but one Thinking facet pole (the exception being Critical, which does not reach the cutoff) correlate with liking advancement and high pay, even at the expense of job security. In addition, respondents who scored toward Questioning reported liking opportunities to work in other countries, while their opposites at the Accommodating pole reported liking teamwork and simplification of jobs.

Within the J–P preference scale, respondents who scored toward the Systematic, Early Starting, and Scheduled facet poles reported liking clear structures and responsibilities. Those who scored toward Scheduled reported liking to work as part of a team. Overall, the direction of the correlations reported above support the validity of the facet scales.

LEADERSHIP AND MANAGEMENT

Selected correlations presented in various tables in this chapter demonstrate a relationship between the Step II facet scales and leadership and management behaviors. Individuals who scored high on the CPI 260 Leadership scale (good leadership skills, aspire to positions of leadership, are accepted as leaders by others) tended to score toward the Extraversion pole on all of the E–I facet scales and also toward the Original and Questioning facet poles. On the CPI 260 Managerial Potential scale (effective interpersonal skills, good judgment, interest in managing others), high scores were related to scores toward Extraversion on all of the E–I facet scales and to some extent toward Questioning and Logical ($r = .16$ for both). Individuals who scored high on the FIRO-B Expressed Control scale (a need to take control, set direction, manage others) tended to score toward the Enthusiastic, Critical, and Tough facet poles. Individuals who scored toward the “Directs others” pole of the *Strong Interest Inventory* Leadership Style scale tended to score toward the Extraversion pole on all of the E–I facet scales and also toward the Theoretical, Original, and Questioning facet poles. Higher self-ratings on the *Benchmarks* Leading Employees scale correlated with all but one of the E–I facet scales in the direction of Extraversion; the exception was Enthusiastic–Quiet. There are some similarities across these results. Leaders and managers tended to favor the Extraversion pole of the E–I facet scales and also the Original and Questioning facet poles. The surprising discrepancy is that in the global sample, top executives compared to other executives were more likely to score toward the Reflective and Quiet facet poles. This discrepancy may be simply a function of the different samples, as the global sample is heterogeneous and characteristics of leadership may differ across cultures.

Using a prior Step II form of the assessment, Johnson (1992b) compared the results of 22 managers identified as top achievers who had been promoted to division-level managers in the wholesale grocery industry to the results of fellow employees ($N = 338$) who were attending a career development program. The findings were cross-validated by comparing the same sample of top achievers ($N = 22$) to a group of employees ($N = 315$) in a competing firm. The top achievers had higher mean scores on all five S–N facet scales, in the direction of the Intuition poles, than did others in their company and in the competing company. In addition, the top achievers scored significantly higher on Questioning and Spontaneous and also had significantly higher polarity indexes, indicating greater clarity of preferences across the Step II facet scales. The cross-validation study replicated the above results and also revealed higher scores on Reasonable for the top achievers.

SUPPORT FOR DECISIONS OF MANAGER

A question on the Form Q US national survey asked how often the respondent supported the decisions of his or her supervisor or manager. The response format used a four-point scale: “almost always,” “often,” “sometimes,” and “almost never.” Within the E–I preference scale, respondents who scored toward Receiving, Contained, and Intimate were less likely to support their manager’s decisions. Within the T–F preference scale, those who scored toward Accommodating and Accepting were significantly more supportive. Within the J–P preference scale, the significant differences were on the Early Starting–Pressure–Prompted scale, with those who were supportive scoring more toward Early Starting.

CHANGE

Lawrence (2007) examined Form Q facet scales in relation to the scales of the *Change Style Indicator*: Conserver, Pragmatist, and Originator. Conservers prefer gradual and incremental change; Pragmatists focus on practical outcomes and can serve as mediators; Originators prefer quick and transformational change. The sample consisted of 45 healthcare executives in five hospitals; 84% were female. The results were as follows: The Step II poles Expressive, Gregarious, Abstract, Imaginative, Theoretical, Open-Ended, and Emergent correlated with the Originator change style. Concrete, Realistic, Experiential, Planful, and Methodical correlated with the Conserver style. There were no correlations of Step II facet scales with the Pragmatist style.

Health, Stress, and Coping

Participants in the Form Q US national sample were asked to indicate their “typical level of stress” on a five-point scale: “very low,” “low,” “moderate,” “high,” or “very high.” The two high categories were combined, as were the two low categories; these two combined categories,

along with “moderate,” were then subjected to an analysis of variance (ANOVA) on the 20 facet scales and a separate ANOVA on the four overall preference scales.

Analysis of the preferences showed that Introverts reported significantly higher levels of stress ($p < .001$) than did Extraverts. However, analyzing the facet scales indicated that only Expressive–Contained and Gregarious–Intimate showed significant differences across the three stress categories, with respondents who scored toward Contained and Intimate reporting higher levels of stress than did those who scored toward Expressive and Gregarious. Those who scored toward Critical also revealed significantly higher levels of stress than did those who scored toward the opposite facet pole, Accepting.

Out-of-preference facet scale scores were also examined to determine whether they might be related to stress. Given the findings above, the following question was asked: Were Introverts whose scores were out-of-preference on any of the E–I facet scales more likely to have higher levels of stress than were Introverts whose scores were in-preference? The answer was no. Regardless of whether the scores were in-preference or out-of-preference, a preference for Introversion was the determining factor in regard to respondents’ level of stress. In performing this analysis, however, it appeared that having an in-preference or out-of-preference score on an E–I facet scale did make a difference for Extraverts. On the Enthusiastic–Quiet scale, participants whose scores were out-of-preference (Quiet Extraverts) had significantly lower levels of stress than did those whose scores were in-preference (Enthusiastic Extraverts) or in the midzone.

The Form Q US national survey asked whether participants had ever experienced hypertension, heart problems, insomnia, or chronic pain. Response options for the first two conditions were “yes” or “no.” Responses for the latter two conditions used a four-point scale of “never,” “rarely,” “occasionally,” and “frequently.” For the purposes of analysis, the first two and last two response categories were combined. ANOVAs were used to analyze the 20 facet scales, and $p < .01$ was used as the cutoff. Of the sample, 28% answered “yes” to experiencing hypertension (the term *hypertension* was not defined on the survey). Compared to those who answered “no,” these respondents had means more toward Reflective ($p < .001$), Quiet ($p < .01$), and Realistic ($p < .01$). Mitchell (1993) compared a sample of 110 participants with hypertension to a convenience control sample and found that when the two groups were compared on the basis of deviations from the norm for their MBTI type, the hypertension patients’ scores differed from those of the control group on 10 facet scales (the scales were not specified).

Regarding having heart problems, 15% of the Form Q US national sample responded in the affirmative. All of the S–N facet scales showed significant differences ($p < .01$), with those who reported experiencing such problems scoring more toward the Sensing pole of these scales. In addition, those reporting heart problems scored more toward Systematic and Early Starting (both $p < .01$). A pattern on the Gregarious–Intimate scale, although it did not satisfy the significance criteria, was nonetheless interesting. Those reporting heart problems had a mean toward Gregarious ($p < .03$), while the trend for the other E–I facet scales was toward the Introversion poles. This suggested the possibility that the finding toward Gregarious was due to Introverts whose scores were out-of-preference on this facet scale. However, further analysis revealed that the individuals who experienced heart problems were not Gregarious Introverts, but rather Gregarious Extraverts, those whose scores were in-preference.

Of the Form Q US national sample, 38% reported experiencing frequent or occasional insomnia. There were no significant differences on any of the S–N or J–P facet scales. Respondents who reported insomnia scored significantly closer to these E–I and T–F facet poles: Contained, Intimate, and Reflective (Introversion facet poles) and Empathetic, Accommodating, and Tender (Feeling facet poles). When asked about chronic pain, 33% of the sample reported frequent or occasional occurrences. On the T–F facet scales, these individuals scored significantly closer to four of the five Feeling poles: Empathetic, Compassionate, Accepting, and Tender.

Participants were also asked how frequently they drank alcohol or smoked. Those who reported drinking frequently or occasionally had means significantly closer to the Extraversion poles on four of the E–I facet scales (the exception was Gregarious, at $p < .03$) and toward the Intuition poles on all five S–N facet scales. In addition, on the T–F facet scales, those who reported drinking frequently or occasionally scored toward Reasonable and Questioning; on the J–P facet scales they scored toward Casual, Pressure–Prompted, and Spontaneous. As was discussed in chapter 9, Martin (2011) found that all four EP types plus ESTJs engaged in frequent alcohol consumption. In the Form M US national sample it was found that the EN types were more likely to drink alcohol more frequently. The Step II facet scales add another perspective to that provided by the Step I scales.

Participants who said they smoked more frequently scored toward Original, Questioning, Open–Ended, and Spontaneous. This would not have been predicted by the preferences alone, as only the J–P preference scale showed significance ($p < .02$). In Martin’s study (2011), it was the ENFPs who were the most likely to smoke. The results suggest that Step II facet scale scores can be very useful in forming hypotheses about behaviors

that may play a part in physical symptoms and health-related behaviors.

Hammer (1993) investigated the contribution of the Step II facet scales from a prior Step II form to prediction of coping resources in a sample of 225 chronic pain patients. The measure used was the *Coping Resources Inventory* (CRI; Hammer & Marting, 1987, 1988), a 60-item instrument composed of five scales (cognitive, emotional, social, spiritual/philosophical, and physical) designed to measure a person's resources for handling stress. The CRI scales have proved to be significant predictors of physical and psychological symptoms in adolescents (Zeidner & Hammer, 1990), of reactions to the highly stressful situations of war (Zeidner & Hammer, 1992), and of responses to unemployment (Wanberg, 1992). For this study, however, only the CRI total resource score was used as the dependent variable in a regression analysis. In the sample of chronic pain patients, six of the Step II facet scales were significant in predicting total resources, as evidenced by beta weights with significance levels at $p < .01$. The signs of the coefficients indicated that the following poles were associated with having greater coping resources: Initiating, Expressive, Active, Realistic, Accepting, and Systematic. All four preference pairs are represented by these six Step II facet scales.

The results for the three E–I facet scales mentioned above (Initiating–Receiving, Expressive–Contained, and Active–Reflective) suggest that individuals who initiate social contact, express their thoughts and feelings, and participate in group activities tend to have greater coping resources than do those at the opposite poles of these scales. Interestingly, previous research has demonstrated that Extraverts are likely to have greater coping resources than are Introverts (Hammer, 1991), but the results for the Step II facet scales reveal that not all of the Extraversion facet poles are equally useful in identifying respondents with greater resources. Scores toward Gregarious and Enthusiastic did not seem to contribute as much.

Among the S–N facet scales, scores toward Realistic suggest that individuals who pride themselves on their common sense (Kummerow & Quenk, 1992) tend to have greater coping resources. This, along with their efficiency and practicality, may help them cope more successfully than individuals who do not have these characteristics.

Among the T–F facet scales, the tolerance of individuals who score toward Accepting, and their tendency to look for win-win solutions to problems, may be why they tend to have greater coping resources. And among the J–P facet scales, those who score toward Systematic tend to have greater resources, perhaps because they use structure and planning to avoid stress in the first place.

Using the same data, a hierarchical regression was conducted to determine the unique contribution of the Step II facet scales. First, the four Step I preference

scales were entered as independent variables, and then the 20 Step II facet scales were added and subsequent changes in R-squared were noted. The Step I scales accounted for 34% of the variance in coping resources, while the Step I scales and Step II scales together accounted for 41% of the variance. Thus, the Step II facet scales represent a significant increase in explained variance—in other words, they provide more information for the purpose of predicting coping resources than do the four Step I scales alone.

Holland and VanBenshoten (1991) also investigated the relationship of an earlier version of the Step II facet scales to coping, using the *COPE Inventory* with a sample of 202 participants from a wide variety of settings. The COPE instrument was used as a measure of dispositional coping and was designed to tap the extent to which individuals engage in various coping behaviors when they are stressed. Four Step II facet scales, one from each of the four preference pairs, proved to be significant predictors: Expressive–Contained, Traditional–Original, Logical–Empathetic, and Systematic–Casual. The Expressive pole was related to Emotional and Instrumental Social Support, Venting, and Interpretation, while the Contained pole was associated with higher scores on Mental Disengagement. The Traditional pole correlated with Denial, Mental Disengagement, Behavioral Disengagement, and Use of Alcohol, while the Original pole was related to Active Coping, Planning, and Interpretation. The Logical pole was associated with Planning, while the Empathetic pole correlated with Emotional and Instrumental Social Support, Venting, and Interpretation. The Systematic pole was associated with Active Coping, Planning, Suppression, and Mental Disengagement, while the Casual pole was related to Use of Alcohol.

A subset of participants in the global sample ($n = 5,447$) were asked to rate their satisfaction with their health. Correlations with their facet scale scores showed that all of the E–I facet scales correlated with health satisfaction in the direction of Extraversion. The coefficients were small, ranging from $-.10$ for Enthusiastic–Quiet to $-.15$ for Gregarious–Intimate. No other correlations of facet scale scores with health satisfaction exceeded $.06$. The E–I facet scale results are partially consistent with results from the overall preferences in that Extraverted types, and to some extent Judging types, tend to say that they are more satisfied with most aspects of their lives. But there would be no other reason to expect that facet scales should be related to a global measure of satisfaction with one's health.

These studies indicate that the Step II facet scales provide a rich source of information about a wide variety of health and coping behaviors and that they contribute additional information over and above that provided by the Step I preference scales. However, in the health domain the research has tended to outpace the theory.

Theoretical explanations of the findings reported here are needed.

Relationships

Marital satisfaction was examined by Marioles, Strickert, and Hammer (1996), who found that couples in which both partners scored toward Enthusiastic and Expressive were more likely to have greater combined satisfaction as measured by the Dyadic Adjustment Scale. When marital and premarital samples were combined, the Step II facet poles that were associated with combined satisfaction were Expressive, Accommodating, and Planful.

Marital satisfaction was also examined in the Form Q US national sample, although by surveying responses from individuals rather than those from couples. Participants used a four-point response scale to indicate how satisfied they were with their “marriage/intimate relationship.” For purposes of analysis, the response categories “very satisfied” and “somewhat satisfied” were combined, as were “very dissatisfied” and “somewhat dissatisfied.” ANOVAs were run on all 20 facet scales, and $p < .01$ was used as the criterion for significance.

Among the E–I facet scales, all but Enthusiastic–Quiet showed significant differences between the facet poles, with the Extraversion facet poles relating to a higher marital satisfaction rating. There were no significant differences between the poles of the S–N facet scales. The overall T–F preference scale was not significant. Among the T–F facet scales, only Questioning–Accommodating showed a mean difference, with those who scored toward Accommodating reporting more satisfaction. Three J–P facet scales showed significant differences, with those who scored toward Planful, Early Starting, and Scheduled reporting being significantly more satisfied in their marriage than those who scored toward the opposite poles (Open-Ended, Pressure-Prompted, and Spontaneous).

Participants were also asked how many times they had been married and were offered choices of 0, 1, 2, and 3 or more. All of the S–N facet scales showed significant differences between the poles of the facet scales, with those who had been married more often having mean scores toward the Sensing poles of these scales. Stated conversely, those who had never married scored more toward the Intuition poles of these scales. Although none of the T–F scales met the significance criterion, an interesting pattern was observed on the Critical–Accepting scale ($p < .02$). Participants who had never married or had been married once or twice scored more toward Accepting, while those who had been married three times or more scored more toward Critical. Four of the five J–P facet scales showed significant differences. Participants who had never married scored toward the Casual, Pressure-Prompted, and Spontaneous poles. The Planful–Open-Ended scale showed a different pattern,

with those who had never married and those who had been married three times or more scoring more toward the Open-Ended pole, and those who had been married once or twice scoring toward Planful.

The many applications areas studied show that the 20 facet scales provide valuable differential information over and above that available from the preferences alone or in combination.

Satisfaction with Life and Level of Happiness

A subset of participants in the global sample ($n = 15,824$) were asked to indicate (1) how satisfied they were with their life and (2) their level of happiness. Correlations with their facet scale scores showed that all of the E–I facet scales correlated with both variables in the direction of Extraversion. The coefficients for both variables were small, ranging from $-.11$ for Enthusiastic–Quiet to $-.16$ for Expressive–Contained for the life satisfaction question. For happiness, the range was from $-.14$ for Enthusiastic–Quiet to $-.17$ for Initiating–Receiving and Expressive–Contained. No correlations with any of the other facet scales exceeded $.04$. As mentioned in the “Health, Stress, and Coping” section of this chapter, the E–I facet results are partially consistent with results from the overall preferences in that Extraverted types, and to some extent Judging types, tend to say that they are more satisfied with most aspects of their life. But there would be no other reason to expect that facet scales would be related to global measures of satisfaction with one’s life or degree of happiness.

Adult Success with Learning a Foreign Language

Ehrman (1994a) used a prior Step II form to study nearly 800 adults employed by the US Foreign Service, US Information Agency, US Department of Defense, and other US federal agencies who were undergoing intensive foreign language training. For the purposes of post-training analysis, students were grouped into three proficiency levels (high, medium, and low) for each of three outcome variables (speaking, reading, and faculty rankings). Those with high proficiency in speaking scored significantly higher on the Questioning facet pole; students with a medium level of proficiency in speaking had higher scores on the Accepting facet pole; and those with low proficiency in speaking had higher scores on the Concrete, Realistic, and Practical facet poles. Students graded as having high proficiency in reading scored significantly higher on Critical, Tough, Questioning, and Logical. Those ranked lowest by the faculty had higher scores on Realistic and Practical. Ehrman also reported more than 100 significant correlations of Step II facet scales with measures such as language aptitude, proficiency, anxiety, learning strategies, verbal risk, and

a host of other variables. Although the correlations are too numerous to report here, the overall pattern provides compelling evidence for the differential relationship of Step II facet scales to other measures and for relationships that are distinct from those found for the four Step I preference scales. Ehrman also found that a higher polarity index for the 20 Step II facet scales (i.e., scores more consistently toward the poles of the scales) was related to lower anxiety about training outcomes.

CONCLUSION

The data presented in this chapter provide evidence in support of the validity of the 20 Global Step II facet scales. Confirmatory factor analysis showed that the relationship of the Step II facet scales to the Step I preference scales is as predicted. Correlations of the facet scales with many scales from other assessments are consistent with the intended meaning of the facet scales. These analyses were conducted using data from a variety of samples, including the global sample, a Form Q US national sample, and a large sample of managers. These correlations reveal two patterns consistent with the proposed uses of the facet scales. First, the overall pattern of correlations of Step II scales within a preference scale with external variables is as expected. Second, there are differential results for facet scales within the same preference scale (i.e., correlations that are higher than, lower than, or opposite in sign from those of the other facet scales within that preference scale) that support the uniqueness of the facet scales.

The results provide clear evidence of the intended meaning and uniqueness of the facet scales.

Data were also presented from studies that showed meaningful correlations between the facet scales and behavioral descriptors. These included ratings by superiors, peers, direct reports, and others using a variety of rating scales and lists of adjectives.

The facet scales were also shown to explain additional variance—beyond that explained by the four Step I preference scales—in variables that cover several domains of organizational behaviors. The additional variance was observable for the entire set of 20 facet scales over the set of four preference scales as well as for individual facet scales compared to the preference scale with which each is associated. Analyses conducted using two large and very different samples showed that many facet scales demonstrated greater effects on the dependent variables than did the overall preference scale and also revealed the existence of a number of crossover results wherein the direction of the facet effect was opposite that of the overall preference. Analyses of all combinations of individual facet scale categories in relation to specific dependent variables confirmed both the unique meaning of the Step II facet scales and their contribution to understanding other variables when used in conjunction with the Step I results.

Finally, data were presented on the use of the Step II facet scales in applications using a wide variety of samples and dependent variables. These studies support the validity of the Global Step II facet scales.

APPENDIX A

MEANS, STANDARD DEVIATIONS, AND AVERAGE RANGE OF FACET SCORES FOR THE 16 MBTI® TYPES

The information in this appendix is arrayed for ease of use by practitioners. For each of the 16 types, the following data are provided:

- The number of individuals of the MBTI type in the global sample
- The average (mean) score for each facet scale
- The standard deviation for each facet scale
- The range of scores that vary from -1 to $+1$ standard deviation from the mean score

The types are presented in the order that approximates the familiar type table, with IJ types (the top row of the type table) listed first, followed by the IP types (the second row of the type table), then the EP types (third row) and the EJ types (fourth row).

Mean scores that are to the left of the center point are designated with a minus (–) sign (i.e., scores on the E, S, T, J side) in order to distinguish them from scores to the right of the center point (i.e., scores on the I, N, F, P side). When subtracting or adding the standard deviation to the mean results in a score greater than 5 or less than -5 , the result is recorded as a 5 or -5 , respectively, since these scores designate the endpoints of each scale. It is possible for such hypothetical scores to fall beyond the endpoints of the scales because in most cases the facet scores do not assume a normal distribution. Some types were more likely to score toward the endpoints of the scales than were other types.

Table A.1 | Means, standard deviations, and average ranges of scores for IJ types

Global Step II™ facet	ISTJ (n = 2,671)			ISFJ (n = 1,404)		
	M	SD	Range	M	SD	Range
E–I facets						
Initiating–Receiving	2.28	2.28	0.00 to 4.56	2.46	2.13	0.33 to 4.59
Expressive–Contained	2.22	2.13	0.09 to 4.35	1.60	2.21	–0.61 to 3.81
Gregarious–Intimate	2.47	2.24	0.23 to 4.71	2.33	2.30	0.03 to 4.63
Active–Reflective	2.04	2.08	–0.04 to 4.12	1.98	1.86	0.12 to 3.84
Enthusiastic–Quiet	2.71	2.21	0.50 to 4.92	2.36	2.32	0.04 to 4.68
S–N facets						
Concrete–Abstract	–2.63	1.87	–4.50 to –0.76	–1.90	1.97	–3.87 to 0.07
Realistic–Imaginative	–2.86	1.94	–4.80 to –0.92	–2.21	2.14	–4.35 to –0.07
Practical–Conceptual	–2.12	2.62	–4.74 to 0.50	–2.38	2.43	–4.81 to 0.05
Experiential–Theoretical	–2.24	2.51	–4.75 to 0.27	–2.06	2.45	–4.51 to 0.39
Traditional–Original	–2.05	2.06	–4.11 to 0.01	–2.28	1.79	–4.07 to –0.49
T–F facets						
Logical–Empathetic	–2.94	1.91	–4.85 to –1.03	1.79	1.93	–0.14 to 3.72
Reasonable–Compassionate	–2.85	1.95	–4.80 to –0.90	1.86	1.82	0.04 to 3.68
Questioning–Accommodating	–1.09	2.31	–3.40 to 1.22	2.04	1.88	0.16 to 3.92
Critical–Accepting	–1.49	2.47	–3.96 to 0.98	2.24	2.28	–0.04 to 4.52
Tough–Tender	–1.90	2.43	–4.33 to 0.53	2.75	2.27	0.48 to 5.00
J–P facets						
Systematic–Casual	–3.31	1.80	–5.00 to –1.51	–2.25	1.92	–4.17 to –0.33
Planful–Open-Ended	–2.60	2.25	–4.85 to –0.35	–2.49	2.21	–4.70 to –0.28
Early Starting–Pressure-Prompted	–2.69	2.76	–5.00 to 0.07	–2.92	2.53	–5.00 to –0.39
Scheduled–Spontaneous	–2.95	1.87	–4.82 to –1.08	–2.76	1.81	–4.57 to –0.95
Methodical–Emergent	–2.30	2.97	–5.00 to 0.67	–1.78	2.98	–4.76 to 1.20

CONTINUED

Table A.1 | Means, standard deviations, and average ranges of scores for IJ types *CONTINUED*

Global Step II* facet	INFJ (<i>n</i> = 387)			INTJ (<i>n</i> = 429)		
	<i>M</i>	<i>SD</i>	Range	<i>M</i>	<i>SD</i>	Range
<i>E–I facets</i>						
Initiating–Receiving	2.36	2.16	0.20 to 4.52	2.13	2.29	–0.16 to 4.42
Expressive–Contained	1.67	2.32	–0.65 to 3.99	2.26	2.12	0.14 to 4.38
Gregarious–Intimate	2.29	2.33	–0.04 to 4.62	2.27	2.32	–0.05 to 4.59
Active–Reflective	1.85	2.08	–0.23 to 3.93	2.01	2.00	0.01 to 4.01
Enthusiastic–Quiet	1.61	2.46	–0.85 to 4.07	2.08	2.31	–0.23 to 4.39
<i>S–N facets</i>						
Concrete–Abstract	2.26	1.88	0.38 to 4.14	1.49	2.02	–0.53 to 3.51
Realistic–Imaginative	2.20	2.05	0.15 to 4.25	1.95	2.19	–0.24 to 4.14
Practical–Conceptual	2.11	2.42	–0.31 to 4.53	2.36	2.45	–0.09 to 4.81
Experiential–Theoretical	1.26	2.70	–1.44 to 3.96	1.58	2.72	–1.14 to 4.30
Traditional–Original	0.78	1.86	–1.08 to 2.64	1.43	1.76	–0.33 to 3.19
<i>T–F facets</i>						
Logical–Empathetic	2.44	2.00	0.44 to 4.44	–2.60	1.90	–4.50 to –0.70
Reasonable–Compassionate	2.30	1.82	0.48 to 4.12	–2.44	2.02	–4.46 to –0.42
Questioning–Accommodating	1.42	1.88	–0.46 to 3.30	–1.32	2.15	–3.47 to 0.83
Critical–Accepting	2.25	2.28	–0.03 to 4.53	–0.91	2.58	–3.49 to 1.67
Tough–Tender	2.95	0.54	0.69 to 5.00	–1.46	2.29	–3.75 to 0.83
<i>J–P facets</i>						
Systematic–Casual	–0.80	0.59	–2.87 to 1.27	–2.13	1.93	–4.06 to –0.20
Planful–Open-Ended	–2.61	0.55	–4.73 to –0.49	–2.45	2.24	–4.69 to –0.21
Early Starting–Pressure-Prompted	–2.77	0.64	–5.00 to –0.04	–2.45	2.94	–5.00 to 0.49
Scheduled–Spontaneous	–2.50	0.50	–4.27 to –0.73	–2.28	1.93	–4.21 to –0.35
Methodical–Emergent	–2.69	0.61	–5.00 to 0.10	–2.79	2.78	–5.00 to –0.01

Note: Global sample, *N* = 16,773.

Table A.2 | Means, standard deviations, and average ranges of scores for IP types

Global Step II™ facet	ISTP (n = 1,651)			ISFP (n = 1,106)		
	M	SD	Range	M	SD	Range
E–I facets						
Initiating–Receiving	1.72	2.11	–0.39 to 3.83	1.86	2.06	–0.20 to 3.92
Expressive–Contained	1.68	2.20	–0.52 to 3.88	1.28	2.27	–0.99 to 3.55
Gregarious–Intimate	1.55	2.45	–0.90 to 4.00	1.52	2.36	–0.84 to 3.88
Active–Reflective	1.36	2.18	–0.82 to 3.54	1.50	1.96	–0.46 to 3.46
Enthusiastic–Quiet	1.90	2.33	–0.43 to 4.23	1.77	2.36	–0.59 to 4.13
S–N facets						
Concrete–Abstract	–1.66	1.84	–3.50 to 0.18	–1.11	1.89	–3.00 to 0.78
Realistic–Imaginative	–1.99	2.05	–4.04 to 0.06	–1.50	2.21	–3.71 to 0.71
Practical–Conceptual	–1.56	2.59	–4.15 to 1.03	–1.82	2.49	–4.31 to 0.67
Experiential–Theoretical	–1.48	2.61	–4.09 to 1.13	–1.68	2.53	–4.21 to 0.85
Traditional–Original	–0.87	2.03	–2.90 to 1.16	–1.26	1.86	–3.12 to 0.60
T–F facets						
Logical–Empathetic	–1.97	2.02	–3.99 to 0.05	1.95	1.91	0.04 to 3.86
Reasonable–Compassionate	–1.98	2.11	–4.09 to 0.13	1.66	1.81	–0.15 to 3.47
Questioning–Accommodating	–0.83	2.23	–3.06 to 1.40	1.70	1.95	–0.25 to 3.65
Critical–Accepting	–1.29	2.38	–3.67 to 1.09	1.75	2.47	–0.72 to 4.22
Tough–Tender	–1.65	2.26	–3.91 to 0.61	2.31	2.39	–0.08 to 4.70
J–P facets						
Systematic–Casual	0.23	2.07	–1.84 to 2.30	1.06	1.97	–0.91 to 3.03
Planful–Open-Ended	1.78	2.17	–0.39 to 3.95	1.89	2.19	–0.30 to 4.08
Early Starting–Pressure-Prompted	0.60	2.86	–2.26 to 3.46	0.03	2.91	–2.88 to 2.94
Scheduled–Spontaneous	1.37	1.53	–0.16 to 2.90	1.24	1.54	–0.30 to 2.78
Methodical–Emergent	1.35	2.73	–1.38 to 4.08	1.31	2.81	–1.50 to 4.12

CONTINUED

Table A.2 | Means, standard deviations, and average ranges of scores for IP types *CONTINUED*

Global Step II® facet	INFP (<i>n</i> = 1,065)			INTP (<i>n</i> = 809)		
	<i>M</i>	<i>SD</i>	Range	<i>M</i>	<i>SD</i>	Range
<i>E–I facets</i>						
Initiating–Receiving	2.14	2.21	–0.07 to 4.35	1.81	2.29	–0.48 to 4.10
Expressive–Contained	1.65	2.36	–0.71 to 4.01	2.06	2.13	–0.07 to 4.19
Gregarious–Intimate	2.03	2.36	–0.33 to 4.39	1.86	2.40	–0.54 to 4.26
Active–Reflective	1.73	1.87	–0.14 to 3.60	1.65	2.15	–0.50 to 3.80
Enthusiastic–Quiet	1.24	2.50	–1.26 to 3.74	1.40	2.49	–1.09 to 3.89
<i>S–N facets</i>						
Concrete–Abstract	2.64	1.97	0.67 to 4.61	2.20	2.08	0.12 to 4.28
Realistic–Imaginative	2.66	2.06	0.60 to 4.72	2.04	2.31	–0.27 to 4.35
Practical–Conceptual	2.47	2.50	–0.03 to 4.97	2.52	2.48	0.04 to 5.00
Experiential–Theoretical	1.40	2.75	–1.35 to 4.15	1.65	2.70	–1.05 to 4.35
Traditional–Original	1.84	1.99	–0.15 to 3.83	2.23	1.94	0.29 to 4.17
<i>T–F facets</i>						
Logical–Empathetic	2.67	1.98	0.69 to 4.65	–2.07	1.98	–4.05 to –0.09
Reasonable–Compassionate	2.39	1.83	0.56 to 4.22	–1.87	2.02	–3.89 to 0.15
Questioning–Accommodating	1.25	1.96	–0.71 to 3.21	–1.27	2.17	–3.44 to 0.90
Critical–Accepting	2.29	2.31	–0.02 to 4.60	–0.72	2.43	–3.15 to 1.71
Tough–Tender	2.71	2.38	0.33 to 5.00	–1.32	2.28	–3.60 to 0.96
<i>J–P facets</i>						
Systematic–Casual	2.73	1.85	0.88 to 4.58	1.46	2.23	–0.77 to 3.69
Planful–Open-Ended	2.59	2.40	0.19 to 4.99	2.16	2.37	–0.21 to 4.53
Early Starting–Pressure-Prompted	0.65	3.11	–2.46 to 3.76	1.01	3.13	–2.12 to 4.14
Scheduled–Spontaneous	2.16	1.78	0.38 to 3.94	2.04	1.66	0.38 to 3.70
Methodical–Emergent	1.36	3.09	–1.73 to 4.45	1.15	3.04	–1.89 to 4.19

Note: Global sample, *N* = 16,773.

Table A.3 | Means, standard deviations, and average ranges of scores for EP types

Global Step II™ facet	ESTP (n = 1,028)			ESFP (n = 1,001)		
	M	SD	Range	M	SD	Range
E–I facets						
Initiating–Receiving	–1.93	2.00	–3.93 to 0.07	–2.22	1.95	–4.17 to –0.27
Expressive–Contained	–1.19	2.19	–3.38 to 1.00	–1.97	2.16	–4.13 to 0.19
Gregarious–Intimate	–1.33	2.23	–3.56 to 0.90	–1.46	2.11	–3.57 to 0.65
Active–Reflective	–2.11	2.13	–4.24 to 0.02	–2.07	2.20	–4.27 to 0.13
Enthusiastic–Quiet	–1.60	2.32	–3.92 to 0.72	–2.00	2.24	–4.24 to 0.24
S–N facets						
Concrete–Abstract	–1.49	1.76	–3.25 to 0.27	–1.10	1.76	–2.86 to 0.66
Realistic–Imaginative	–1.43	2.09	–3.52 to 0.66	–1.29	2.14	–3.43 to 0.85
Practical–Conceptual	–1.30	2.68	–3.98 to 1.38	–1.84	2.41	–4.25 to 0.57
Experiential–Theoretical	–1.50	2.56	–4.06 to 1.06	–1.89	2.37	–4.26 to 0.48
Traditional–Original	–0.43	1.99	–2.42 to 1.56	–0.95	1.84	–2.79 to 0.89
T–F facets						
Logical–Empathetic	–1.62	1.92	–3.54 to 0.30	2.05	1.82	0.23 to 3.87
Reasonable–Compassionate	–1.96	1.89	–3.85 to –0.07	1.63	1.79	–0.16 to 3.42
Questioning–Accommodating	–0.86	2.12	–2.98 to 1.26	1.44	1.79	–0.35 to 3.23
Critical–Accepting	–0.88	2.36	–3.24 to 1.48	2.09	2.30	–0.21 to 4.39
Tough–Tender	–1.53	2.14	–3.67 to 0.61	2.04	2.40	–0.36 to 4.44
J–P facets						
Systematic–Casual	0.66	2.04	–1.38 to 2.70	1.45	1.89	–0.44 to 3.34
Planful–Open-Ended	1.79	2.19	–0.40 to 3.98	1.91	2.20	–0.29 to 4.11
Early Starting–Pressure-Prompted	0.64	2.80	–2.16 to 3.44	0.10	2.99	–2.89 to 3.09
Scheduled–Spontaneous	1.46	1.48	–0.02 to 2.94	1.32	1.45	–0.13 to 2.77
Methodical–Emergent	1.27	2.71	–1.44 to 3.98	1.26	2.80	–1.54 to 4.06

CONTINUED

Table A.3 | Means, standard deviations, and average ranges of scores for EP types *CONTINUED*

Global Step II® facet	ENFP (n = 1,370)			ENTP (n = 722)		
	M	SD	Range	M	SD	Range
E–I facets						
Initiating–Receiving	–2.86	1.95	–4.81 to –0.91	–2.57	1.94	–4.51 to –0.63
Expressive–Contained	–2.25	2.36	–4.61 to 0.11	–1.55	2.37	–3.92 to 0.82
Gregarious–Intimate	–1.62	2.28	–3.90 to 0.66	–1.52	2.29	–3.81 to 0.77
Active–Reflective	–2.42	2.15	–4.57 to –0.27	–2.03	2.18	–4.21 to 0.15
Enthusiastic–Quiet	–2.98	1.99	–4.97 to –0.99	–2.50	2.18	–4.68 to –0.32
S–N facets						
Concrete–Abstract	2.57	2.00	0.57 to 4.57	1.89	2.02	–0.13 to 3.91
Realistic–Imaginative	2.80	2.02	0.78 to 4.82	2.17	2.23	–0.06 to 4.40
Practical–Conceptual	2.56	2.50	0.06 to 5.00	2.64	2.44	0.20 to 5.00
Experiential–Theoretical	1.10	2.80	–1.70 to 3.90	0.88	2.83	–1.95 to 3.71
Traditional–Original	2.16	1.85	0.31 to 4.01	2.36	1.90	0.46 to 4.26
T–F facets						
Logical–Empathetic	2.97	1.82	1.15 to 4.79	–1.68	2.01	–3.69 to 0.33
Reasonable–Compassionate	2.29	1.83	0.46 to 4.12	–1.87	1.93	–3.80 to 0.06
Questioning–Accommodating	0.91	1.74	–0.83 to 2.65	–1.36	2.04	–3.40 to 0.68
Critical–Accepting	2.56	2.16	0.40 to 4.72	–0.39	2.45	–2.84 to 2.06
Tough–Tender	2.65	2.40	0.25 to 5.00	–1.43	2.16	–3.59 to 0.73
J–P facets						
Systematic–Casual	3.09	1.74	1.35 to 4.83	1.94	2.05	–0.11 to 3.99
Planful–Open-Ended	2.74	2.29	0.45 to 5.00	2.45	2.33	0.12 to 4.78
Early Starting–Pressure-Prompted	0.83	3.14	–2.31 to 3.97	1.11	3.00	–1.89 to 4.11
Scheduled–Spontaneous	2.27	1.69	0.58 to 3.96	2.11	1.70	0.41 to 3.81
Methodical–Emergent	1.60	3.00	–1.40 to 4.60	1.25	2.90	–1.65 to 4.15

Note: Global sample, N = 16,773.

Table A.4 | Means, standard deviations, and average ranges of scores for EJ types

Global Step II™ facet	ESTJ (n = 1,509)			ESFJ (n = 951)		
	M	SD	Range	M	SD	Range
E–I facets						
Initiating–Receiving	–2.49	1.93	–4.42 to –0.56	–2.49	1.90	–4.39 to –0.59
Expressive–Contained	–1.71	2.20	–3.91 to 0.49	–2.32	2.06	–4.38 to –0.26
Gregarious–Intimate	–1.59	2.20	–3.79 to 0.61	–1.25	2.37	–3.62 to 1.12
Active–Reflective	–2.16	2.20	–4.36 to 0.04	–2.26	2.22	–4.48 to –0.04
Enthusiastic–Quiet	–1.58	2.39	–3.97 to 0.81	–1.87	2.30	–4.17 to 0.43
S–N facets						
Concrete–Abstract	–2.30	1.80	–4.10 to –0.50	–1.73	1.81	–3.54 to 0.08
Realistic–Imaginative	–2.26	2.03	–4.29 to –0.23	–1.72	2.28	–4.00 to 0.56
Practical–Conceptual	–1.63	2.80	–4.43 to 1.17	–1.91	2.54	–4.45 to 0.63
Experiential–Theoretical	–2.23	2.47	–4.70 to 0.24	–2.02	2.40	–4.42 to 0.38
Traditional–Original	–1.41	2.06	–3.47 to 0.65	–1.90	1.86	–3.76 to –0.04
T–F facets						
Logical–Empathetic	–2.38	1.96	–4.34 to –0.42	2.05	1.87	0.18 to 3.92
Reasonable–Compassionate	–2.58	1.93	–4.51 to –0.65	1.78	1.82	–0.04 to 3.60
Questioning–Accommodating	–1.20	2.18	–3.38 to 0.98	1.50	1.87	–0.37 to 3.37
Critical–Accepting	–0.86	2.46	–3.32 to 1.60	2.41	2.22	0.19 to 4.63
Tough–Tender	–1.89	2.16	–4.05 to 0.27	2.37	2.39	–0.02 to 4.76
J–P facets						
Systematic–Casual	–2.81	1.88	–4.69 to –0.93	–1.71	2.03	–3.74 to 0.32
Planful–Open-Ended	–2.57	2.17	–4.74 to –0.40	–2.58	2.22	–4.80 to –0.36
Early Starting–Pressure-Prompted	–2.62	2.76	–5.00 to 0.14	–3.03	2.49	–5.00 to –0.54
Scheduled–Spontaneous	–2.68	1.88	–4.56 to –0.80	–2.55	1.84	–4.39 to –0.71
Methodical–Emergent	–2.19	2.87	–5.00 to 0.68	–2.12	2.83	–4.95 to 0.71

CONTINUED

Table A.4 | Means, standard deviations, and average ranges of scores for EJ types *CONTINUED*

Global Step II® facet	ENFJ (n = 361)			ENTJ (n = 309)		
	M	SD	Range	M	SD	Range
E–I facets						
Initiating–Receiving	–2.86	1.92	–4.78 to –0.94	–2.82	1.93	–4.75 to –0.89
Expressive–Contained	–2.37	2.24	–4.61 to –0.13	–1.48	2.53	–4.01 to 1.05
Gregarious–Intimate	–1.46	2.47	–3.93 to 1.01	–1.61	2.45	–4.06 to 0.84
Active–Reflective	–2.17	2.20	–4.37 to 0.03	–2.06	2.32	–4.38 to 0.26
Enthusiastic–Quiet	–2.40	2.32	–4.72 to –0.08	–2.54	2.19	–4.73 to –0.35
S–N facets						
Concrete–Abstract	2.02	1.96	0.06 to 3.98	1.57	1.89	–0.32 to 3.46
Realistic–Imaginative	2.34	2.00	0.34 to 4.34	2.06	2.24	–0.18 to 4.30
Practical–Conceptual	2.24	2.52	–0.28 to 4.76	2.63	2.34	0.29 to 4.97
Experiential–Theoretical	0.98	2.82	–1.84 to 3.80	1.04	2.95	–1.91 to 3.99
Traditional–Original	0.98	1.92	–0.94 to 2.90	1.66	1.84	–0.18 to 3.50
T–F facets						
Logical–Empathetic	2.58	1.97	0.61 to 4.55	–2.24	1.99	–4.23 to –0.25
Reasonable–Compassionate	2.07	1.89	0.18 to 3.96	–2.28	2.01	–4.29 to –0.27
Questioning–Accommodating	1.09	1.86	–0.77 to 2.95	–1.49	2.03	–3.52 to 0.54
Critical–Accepting	2.53	2.14	0.39 to 4.67	–0.68	2.31	–2.99 to 1.63
Tough–Tender	2.59	2.31	0.28 to 4.90	–1.44	2.35	–3.79 to 0.91
J–P facets						
Systematic–Casual	–0.60	2.16	–2.76 to 1.56	–1.76	2.02	–3.78 to 0.26
Planful–Open-Ended	–2.53	2.15	–4.68 to –0.38	–2.30	2.17	–4.47 to –0.13
Early Starting–Pressure-Prompted	–2.67	2.79	–5.00 to 0.12	–2.03	3.12	–5.00 to 1.09
Scheduled–Spontaneous	–2.22	1.78	–4.00 to –0.44	–2.25	1.84	–4.09 to –0.41
Methodical–Emergent	–2.62	2.65	–5.00 to 0.03	–2.67	2.80	–5.00 to 0.13

Note: Global sample, N = 16,773.

APPENDIX B

ADJECTIVE CHECK LIST SCALES

The *Adjective Check List* (ACL) contains 300 adjectives that can be used to describe personality traits. Individuals mark the adjectives that describe themselves (when the assessment is used for self-evaluation) or mark the adjectives that they think describe another person (when the ACL is used to help another person gain insight). Several scales are described briefly below. The sequence of the scales follows the order of the scales in the ACL.

Sum of number checked: Total number of adjectives checked

Sum of favorable checked: Total number of favorable adjectives checked

Sum of unfavorable checked: Total number of unfavorable adjectives checked

Communality: An indicator of providing common or similar responses compared to the responses of people in general

Achievement: To strive to be outstanding in pursuits of socially recognized significance

Dominance: To seek and maintain a role as leader in groups, or to be influential and controlling in individual relationships

Endurance: To persist in any task undertaken

Order: To place special emphasis on neatness, organization, and planning in one's activities

Intracception: To engage in attempts to understand one's behavior or the behavior of others

Nurturance: To engage in behaviors that provide material or emotional benefits to others

Affiliation: To seek and maintain numerous personal friendships

Exhibition: To behave in such a way as to elicit the immediate attention of others

Autonomy: To act independently of others or of social values and expectations

Aggression: To engage in behaviors that attack or hurt others

Change: To seek novelty of experience and to avoid routine

Succorance: To solicit sympathy, affection, or emotional support from others

Deference: To seek and maintain subordinate roles in relationships with others

Self-Control: To control one's behaviors and emotions

Self-Confidence: Poise, self-assurance, and belief in one's ability to achieve one's goals

Personal Adjustment: The ability to cope with situational and interpersonal demands and a feeling of efficacy

Ideal Self: Strong sense of personal worth; or harmony between what one is and what one wants to be

Creative Personality: The desire to do and think differently from the norm, and a talent for originality

Military Leadership: Steadiness, self-discipline, and good judgment of the kind required in positions of military (or related) leadership

Adult: Attitudes of independence, objectivity, and industriousness associated with the concept of "mature adult"

Source: Adapted from Gough & Heilbrun (1983).

APPENDIX C

CALIFORNIA PSYCHOLOGICAL INVENTORY™ (CPI™) SCALES

The *California Psychological Inventory*™ (CPI™) assessment is often used in leadership development and selection. Its goal is “to assess individuals by means of variables and concepts that ordinary people use in their daily lives to understand, classify, and predict their own behavior and that of others” (Gough & Bradley, 1996). The scale groupings of the CPI form composed of 260 items (and known as the CPI 260® assessment) are listed below, along with the intention of each scale.

Dealing with Others

Dominance (Do): Assess prosocial interpersonal dominance, strength of will, and perseverance in pursuit of goals

Capacity for Status (Cs): Measure personal qualities that are associated with and that lead to high social status, including ambition and self-confidence

Sociability (Sy): Identify people who are outgoing and socially affiliative, and who enjoy social participation

Social Presence (Sp): Identify people who are self-assured, comfortable being the center of attention, and socially adroit

Self-acceptance (Sa): Identify people with high self-esteem, a strong sense of personal worth, and optimism

Independence (In): Assess the twin elements of psychological strength and interpersonal detachment, including self-sufficiency and self-direction

Empathy (Em): Identify people with a talent for understanding how others feel and think, and who display warmth and tactfulness in their dealings with others

Self-Management

Responsibility (Re): Identify people who are aware of societal rules, and who can and do comply with them when this is appropriate

Social Conformity (So): Assess the degree to which societal norms have been internalized and become autonomously operational within the individual

Self-control (Sc): Assess a continuum going from undercontrol and expressiveness at one pole to overcontrol and suppression of affect at the other

Good Impression (Gi): First, for very high scores, identify overly strong attempts to create a favorable impression; and second, identify people whose style of self-presentation emphasizes ingratiation and compliance

Communality (Cm): Assess a continuum going from erratic or random answering at one pole to close agreement with ordinary beliefs and conventions at the other

Well-being (Wb): Assess feelings of physical and psychological well-being

Tolerance (To): Assess attitudes of tolerance, forbearance, and respect for others, stemming from ethical convictions about the worth of all people

Motivations and Thinking Style

Achievement via Conformance (Ac): Assess achievement potential in well-defined and structured situations, joined to a general desire to do well

Achievement via Independence (Ai): Assess achievement potential in open, minimally defined situations, in which ingenuity and initiative are required for successful performance

Conceptual Fluency (Cf): Identify people who deal easily with abstract and complex concepts, and who believe in their own talent

Personal Characteristics

Insightfulness (Is): Identify people who can think analytically about themselves and others, who can see beyond surface cues, and who are aware of subtle meanings

Flexibility (Fx): Assess a continuum going from resistance to change and dislike of uncertainty at one pole to a liking for change and innovation at the other

Sensitivity (Sn): Assess a continuum going from tough-minded practicality and relative disinterest in personal feelings at one pole to sensitivity, solicitude for others, and a sense of vulnerability at the other

Work-Related Measures

Managerial Potential (Mp): Identify people with an interest in management and who have effective interpersonal skills and good judgment

Work Orientation (Wo): Identify people with a dutiful work ethic, a strong sense of commitment to their job, and little need for overt recognition

Creative Temperament (Ct): Identify people of an imaginative, creative temperament, with both the need and potential for visualizing new and different ways of doing things

Leadership (Lp): Identify people who have good leadership skills, who aspire to positions of leadership, and who will be accepted as leaders by others

Amicability (Ami): Identify people who are amicable, friendly, and considerate of others, who try to avoid conflicts, and who seldom become angry or irritated

Law Enforcement Orientation (Leo): Identify people who view law enforcement and societal rules favorably, who believe punishment for violation of such rules is deserved, and who are well-suited for work in the law enforcement field

Higher-Order Measures

Vector 1 (v.1), Orientation Toward Others: Define a basic dimension of personality going from involvement, participative inclinations, and a readiness to act at one pole to a need for privacy, reluctance to commit self to any irreversible course of action, and a desire to shelter feelings at the other

Vector 2 (v.2), Orientation Toward Societal Values: Define a basic personality dimension going from a rule-questioning, norm-doubting perspective at one pole to a rule-accepting, norm-favoring perspective at the other

Vector 3 (v.3), Orientation Toward Self: Define a basic personality dimension going from general dissatisfaction, feelings of psychological inadequacy, and poor ego integration at one pole to self-realization, feelings of psychological competence, and ego resilience at the other

Source: Gough & Bradley (2005), pp. 6–7.

APPENDIX D

FIRO-B® SCALES

The *Fundamental Interpersonal Relations Orientation–Behavior*™ (FIRO-B®) assessment is based on three interpersonal needs: *inclusion*, *control*, and *affection*.

Inclusion: The need to establish and maintain satisfactory interactions and associations with other people; describes the extent of contact one desires with others and the degree to which one wants to join others in shared activity.

Control: Describes an individual's behavior with respect to responsibility, power, influence, and decision making; reflects the degree to which one desires to influence or direct the behavior of others or the extent to which a person wants to be perceived as competent, decisive, and in charge.

Affection: Describes an individual's behavior in forming close, personal relationships and determines the degree of warmth and personal connections one seeks with others; applies primarily to one-on-one interaction rather than to group behavior.

The theory refers to the extent to which each of the above interpersonal needs is *expressed* and *wanted*. Expressed needs are behaviors demonstrated toward others, while wanted needs are those behaviors that individuals want exhibited toward them (Schutz, 1958).

The FIRO-B assessment combines these two components with the three interpersonal needs:

- Expressed Inclusion
- Wanted Inclusion
- Expressed Control
- Wanted Control
- Expressed Affection
- Wanted Affection

Source: Adapted from Hammer & Schnell (2000), pp. 3–4. Used with permission of the publisher

APPENDIX E

BENCHMARKS® COMPETENCY SCALES

Benchmarks® is a 360-degree tool created by the Center for Creative Leadership® (CCL®) for assessing leadership competencies. With *Benchmarks*, managers rate themselves and are also rated by supervisors, peers, and employees who report to them. The assessment measures 16 competencies needed for success by a manager or leader. Also defined are 5 behaviors that can stall or derail career advancement.

Section 1: Leadership Skills and Perspectives

Resourcefulness: Can both think strategically and make good decisions under pressure; can set up complex work systems and engage in flexible problem-solving behavior; can work effectively with higher management in dealing with the complexities of the management job.

Doing Whatever It Takes: Has perseverance and focus in face of obstacles; takes charge; is capable of standing alone yet is open to learning from others when necessary.

Being a Quick Study: Quickly masters new technical and business knowledge.

Decisiveness: Prefers quick and approximate actions to slow and precise ones in many management situations.

Leading Employees: Delegates to employees effectively, broadens employee opportunities, acts with fairness toward direct reports, and hires talented people for his/her team.

Confronting Problem Employees: Acts decisively and with fairness when dealing with problem employees.

Participative Management: Uses effective listening skills and communication to involve others, build consensus, and influence others in decision making.

Change Management: Uses effective strategies to facilitate organizational change initiatives and overcome resistance to change.

Building and Mending Relationships: Knows how to build and maintain working relationships with co-workers and external parties; can negotiate and handle work

problems without alienating people; understands others and is able to get their cooperation in non-authority relationships.

Compassion and Sensitivity: Shows genuine interest in others and sensitivity to employees' needs.

Straightforwardness and Composure: Is steadfast, relies on fact-based positions, doesn't blame others for mistakes, and is able to recover from troubled situations.

Balance Between Personal Life and Work: Balances work priorities with personal life so that neither is neglected.

Self-Awareness: Has an accurate picture of strengths and weaknesses and is willing to improve.

Putting People at Ease: Displays warmth and a good sense of humor.

Differences Matter: Demonstrates a respect for varying backgrounds and perspectives. Values cultural differences.

Career Management: Develops, maintains, and uses professional relationships, including mentoring, coaching, and feedback to manage own career.

Section 2: Problems That Can Stall a Career

Problems with Interpersonal Relationships: Difficulties in developing good working relationships with others

Difficulty Building and Leading a Team: Difficulties in selecting members for and building a team

Difficulty Changing or Adapting: Resistant to change, learning from mistakes, and developing

Failure to Meet Business Objectives: Difficulties in following up on promises and completing a job

Too Narrow Functional Orientation: Lacks depth to manage outside of one's current function

Source: Center for Creative Leadership (1987). Used with permission.

APPENDIX F

THOMAS-KILMANN CONFLICT-HANDLING MODES

The *Thomas-Kilmann Conflict Mode Instrument* (TKI® assessment) is used to assess how a person is likely to behave in a conflict situation. As described in *Introduction to Conflict Management* (Thomas, 2002), the conflict-handling modes of the TKI® assessment represent the following combinations of *assertiveness* and *cooperativeness*.

Competing: assertive and uncooperative. Individuals who use this mode try to satisfy their own concerns at the other person's expense.

Collaborating: assertive and cooperative. Individuals who use this mode try to find a win-win solution that completely satisfies both persons' concerns.

Compromising: intermediate in assertiveness and cooperativeness. Individuals who use this mode try to find an acceptable settlement that only partially satisfies both persons' concerns.

Avoiding: unassertive and uncooperative. Individuals who use this mode sidestep the conflict without trying to satisfy either person's concerns.

Accommodating: unassertive and cooperative. Individuals who use this mode attempt to satisfy the other person's concerns at the expense of their own.

Source: Adapted from Thomas (2002), pp. 5–6.

GLOSSARY

achievement measure. An assessment of competence in an area, such as the grade obtained after a course of study; a summary of the grades received in obtaining an educational degree, such as the grade point average; scores on tests that assess the degree to which a respondent has learned a particular skill or body of information.

adaptation. Regarding assessment items, an effective conversion from one language to another involving not a word-for-word translation but rather a modification intended to maintain the general sense or purpose of those items in a particular language.

analysis of variance (ANOVA). A statistical analysis used to determine the effects of one or more categorical independent variables on the variance of a continuous dependent variable. The analysis indicates which of the independent variables have a significant effect on the variance of the dependent variable and also identifies any statistical interactions among the independent variables that may affect the dependent variable.

ANOVA. See **analysis of variance**.

a posteriori. A probability that has been revised based on prior information; a probability that is conditional on a prior event or on prior knowledge.

aptitude measure. A test or other device designed to assess a person's potential, either for learning in general (e.g., an intelligence test) or for learning in a specific area (e.g., a test of mechanical aptitude). Aptitude measures are in contrast to achievement measures, which assess acquired knowledge.

attitude. Refers to the MBTI preference pairs Extraversion–Introversion and Judging–Perceiving. The E–I referent is to orientation of energy; the J–P referent is to orientation to the outside world. Equivalent to the term **orientation**. This manual and other more recent publications tend to use *orientation* rather than *attitude*.

auxiliary function (process). The function (process) that is second in importance to the dominant function (process) and that provides balance (1) between perception and judgment and (2) between Extraversion and Introversion. Also known as the **second process**.

best-fit type. The type that a respondent has verified as providing the most accurate description of him- or herself based on self-knowledge. *Contrast reported type and true type.*

best-fit type study. A study that compares reported type results obtained from scoring an administration of an MBTI assessment to a group of respondents' self-estimates of best-fit type.

bivariate residual. The residual correlation remaining between items once their associations with the latent construct are controlled for. Low bivariate residuals are evidence that all the items in a scale are locally independent and that each contributes unique variance to the latent construct.

CAPT. See **Center for Applications of Psychological Type, Inc.**

Center for Applications of Psychological Type, Inc. (CAPT).

A nonprofit public organization founded by Isabel Myers and Mary McCaulley for education, research, and services relating to the MBTI assessment, including training workshops and publications.

chi-square test. A statistical test used to determine the degree of independence or relationship between nominal and categorical variables; the test involves a comparison of the observed frequencies in each category with the expected frequencies and determines the probability that the difference between these frequencies occurred by chance.

classical test theory (CTT). A model used to construct, score, and evaluate instruments designed to measure individual differences. Classical test theory assumes that each individual has some hypothetical but unobservable true score on a given variable, estimated through an observed score on the measuring instrument, along with methods for determining its reliability. CTT was used on MBTI assessments that preceded Form M.

coefficient alpha. A measure of internal consistency reliability based on the average inter-item correlation and the number of items on the scale. Also known as **Cronbach's alpha**.

continuous score. On Form M, Form Q, and the MBTI Global Step II assessment, the continuous scores are the **theta** scores from the IRT scoring program. On the MBTI Global Step I assessment, continuous scores are the **logits** from **latent class analysis**. Such scores are used for research purposes; it is not recommended that they be interpreted to respondents.

continuum. A characteristic or property that varies in size or amount, cannot be subdivided except by arbitrary division, and is measured by a series of numbers that are ordered from small to large, or vice versa. The increments between numbers are the same size all along the continuum and represent equal amounts of the characteristics being measured. In personality measures, the categories of Likert scales are assumed to be a continuum, for example, a seven-point rating scale ranging from very dissatisfied to very satisfied.

CPP, Inc. Former name of The Myers-Briggs Company. Publisher of the MBTI assessment and related MBTI materials, as well as a number of other psychological instruments.

CTT. See **classical test theory**.

Cronbach's alpha. See **coefficient alpha**.

dependent variable. In a research study or statistical analysis, the variable that is acted upon (influenced by) other variables. Whether a given variable is considered to be independent or dependent may vary depending on the study design or the analysis being conducted.

dichotomy. Literally, a division of a concept, construct, or variable into two distinct parts. In type theory, the two parts are assumed to identify opposite mental processes or orientations. Dichotomous constructs differ qualitatively as well as quantitatively from continuous variables. The four dichotomies (also called **preference pairs**) of the MBTI assessment are Extraversion–Introversion, Sensing–Intuition, Thinking–Feeling, and Judging–Perceiving. Also known as *pair of opposites*.

DIF. See **differential item functioning**.

differential item functioning (DIF). An analysis based on **item response theory (IRT)** that is designed to determine whether a test item performs similarly or differently across subgroups (e.g., males versus females, or younger versus older subjects).

differentiation. The process of separation whereby distinct parts emerge from an intermingled single entity. A differentiated type function is one that is separate and exists by itself, so that it can operate on its own without being mixed up with or contaminated by any other function.

dominant function (process). The function that is assumed to be the first developed, most conscious, and most differentiated, and which becomes the governing force dominating and unifying an individual's life. Also known as the **favorite process** or **first process**.

dropout ratio (DOR). A selection ratio that compares the proportion of a type that drops out of an occupation, program, or other group being studied to the proportion in the original population.

E. See **Extraversion**.

EAR. See **Expanded Analysis Report**.

EIR. See **MBTI® Step II™ Interpretive Report**.

elementary interactions. Interactions between two or more of an individual's four type preferences, excluding the type dynamics interactions.

elementary school. In the United States, typically grades 1 through 4; may also include kindergarten and grades 5 and 6. The age range of elementary school children is approximately 6 to 10 years.

Expanded Analysis Report (EAR). An earlier version of the **MBTI® Step II™ Interpretive Report**.

Expanded Interpretive Report. See **MBTI® Step II™ Interpretive Report**.

Extraversion (E). The orientation (attitude) that identifies the direction and flow of energy to an individual's outer world.

Extravert, Extraverted type. An individual who has a preference for the Extraversion orientation over the Introversion orientation.

F. See **Feeling**.

facet. In the MBTI Step II assessment, one of the 20 aspects of the four basic preference pairs (e.g., the Concrete–Abstract facet or the Systematic–Casual facet).

facet scale. One of the 20 pairs of opposites measured by the MBTI Step II assessment.

favorite process. See **dominant function (process)**.

factor analysis. A method of creating groupings of items or factors that are highly related to each other and distinct from other groups of items (factors); an underlying theme or construct that unifies the items that make up a factor. There are two broad approaches to factor analysis: *exploratory* factor analysis makes no assumptions about the underlying structure among the variables; *confirmatory* factor analysis tests for a specific hypothesized structure in the item set.

Feeling (F). Of the two opposite judging functions (processes), the one by which decisions are made through ordering choices in terms of personal values.

Feeling type. A person who prefers Feeling over Thinking as a way of making decisions or coming to conclusions.

first process. See **dominant function (process)**.

forced-choice format. A style of asking questions that requires the respondent to choose one of the two options presented. It is the method used for all MBTI items.

Form G. The form that preceded Form M as the standard MBTI assessment.

Form J. A 290-item research form of the MBTI assessment that replaced Form F. It was used to develop the **Type Differentiation Indicator (TDI)** as well as the **Expanded Analysis Report (EAR)** and the **Expanded Interpretive Report (EIR)**, which were earlier versions of the MBTI Step II assessment.

Form K. The MBTI Step II form that preceded Form Q. It contained the items needed to score four-letter type based on Form G items as well as additional items needed to yield the MBTI® Step II™ Expanded Interpretive Report and Profile.

Form M. The MBTI Step I form that preceded the MBTI Global Step I form.

Form M US national sample (NRS). The sample drawn in the United States ($N = 3,036$) upon which MBTI Form M was based.

Form Q. The MBTI Step II form that preceded the Global Step II form. It contained the items needed to score for four-letter type based on Form M items as well as additional items needed to yield the **MBTI® Step II™ Interpretive Report** and MBTI® Step II™ Profile.

fourth process. See **inferior function (process)**.

function (process). One of the four basic mental functions (processes) of **Sensing, Intuition, Thinking, and Feeling.**

function (process) pair. One of the four pairs created by combining one of the two perceiving functions (processes) with one of the two judging functions (processes): ST, SF, NF, NT.

grade point average. A measure of scholastic achievement in the US that averages a student's grades over a designated time period, such as an academic semester or total years in a degree program.

high school. In the US, typically grades 9 through 12, for students approximately ages 14 to 18 years. Graduation from high school is required for admission to a college or university.

I. See **Introversion.**

ICC. See **item characteristics curve.**

idiographic approach. A way of explaining and predicting human behavior that focuses on discovering and measuring individual differences. *Contrast nomothetic approach.*

independent variable. In a research study or statistical analysis, the variable whose influence on another variable is being studied. Whether a given variable is considered to be independent or dependent may vary depending on the study design or the analysis being conducted.

inferior function (process). The function opposite the dominant function. The inferior function is assumed to be the most unconscious, least experienced, and least differentiated of the functions, serving as a potential source of difficulties as well as growth. Also known as the **fourth process.**

internal consistency reliability. A measure of the consistency with which a group of individuals respond to the items that make up a scale. See **coefficient alpha.**

Introversion (I). The orientation (attitude) that identifies the direction and flow of attention and energy to an individual's inner world.

Introvert, Introverted type. An individual who has a preference for the Introversion orientation over the Extraversion orientation.

Intuition (N). Of the two opposite perceiving functions (processes), the one that is concerned with meanings, relationships, patterns, and possibilities.

Intuitive type. A person who prefers Intuition over Sensing as a way of perceiving.

irrational functions (processes). Sensation (Sensing) and Intuition. Identified as irrational by Jung because they are attuned to the flow of events without being constrained by rational direction.

IRT. See **item response theory.**

IRT scoring. The method of scoring used with both Form M and Form Q, as well as with the MBTI Global Step II assessment. This method considers a person's response to each item on a scale, as well as two item parameters (difficulty and discrimination), and calculates

a continuous **theta**, or underlying "true," score. Theta for the Global Step II assessment is then converted to an 11-point scale from -5 to +5.

Isabel Briggs Myers Memorial Library. A collection of more than 10,500 MBTI materials, including books, scholarly articles, theses and dissertations, and more. The library is housed at the Center for Applications of Psychological Type (CAPT) in Gainesville, Florida. Library materials are available to interested persons. View Mary and Isabel's Library Online (MILO®) at www.capt.org/MILO.

item characteristic curve (ICC). A curve/function demonstrating the rate at which an item would hypothetically be endorsed by people who have varying levels of the characteristic in question. The rate of item endorsement varies from 0% to 100% and is plotted on the vertical axis of the curve. The horizontal axis shows levels of the psychological characteristic, for example, scores on a personality assessment.

item response theory (IRT). An approach to measurement that is concerned with modeling the relationship between item responses and the underlying characteristic that is assessed by the scale or test (assessment). It can be used to select items for a test (assessment) and/or to score the items. IRT is a tool of modern test theory and is often compared to and contrasted with **classical test theory (CTT).**

item weights. Weights applied to item responses to compute the scores. In the case of the MBTI assessment, item weights are applied to the items associated with each of the four preference pairs to compute a score, or **logit**, for each preference.

J. See **Judging.**

Judging (J). The orientation (attitude) that indicates whether either Thinking or Feeling is an individual's preferred way of dealing with the outside world and is likely to appear in observable behavior.

judging functions (processes). Thinking and Feeling.

Judging type. A person who has a judging orientation (J), that is, prefers to use a judging process (Thinking or Feeling) rather than a perceiving process (Sensing or Intuition) when dealing with the outside world.

judgment. The process of making a decision or coming to a conclusion.

Jungian types. The eight dominant types identified by Jung, each made up of one of the four functions (processes) (S, N, T, F) in its extraverted or introverted form (e.g., extraverted Thinking; introverted Intuition).

Jung-Myers theory, Jung-Myers approach. A theoretical and explanatory system based on the type theory of Carl G. Jung as interpreted and expanded by Isabel Briggs Myers and Katherine C. Briggs. The most significant enhancement by Myers and Briggs was the identification of the role of the auxiliary function (process) in each type.

latent class analysis (LCA). A statistical approach that assumes that people belong to one of some number of groups or classes, a view that is consistent with type theory. LCA was used to select and score items for the MBTI Global Step I assessment.

LCA. See **latent class analysis**.

Likert scale. An item or question format in which a respondent is presented with a statement and is then asked to select from a number of choices that indicate his or her degree of agreement with the statement (e.g., from strongly disagree to strongly agree). The number of choices can vary, and the inclusion of a neutral category is optional.

linear regression. In statistics, an approach for modeling the relationship between a scalar dependent variable *y* and one or more explanatory variables (or independent variables) denoted by *x*. The case of one explanatory variable is called simple linear regression.

local independence. An assumption underlying scale construction in latent class analysis and other statistical approaches, including **item response theory (IRT)**. Items are locally independent if there is no residual correlation between two items once their correlations with the underlying latent construct are controlled for. If items are locally independent, then each item is contributing unique variance to the latent construct. Violations of local independence can result in inflated internal consistency estimates, for instance. See also **bivariate residual**.

location parameter. See **three-parameter model**.

logit. A continuous score computed via latent class analysis for each preference scale on the Global Step I assessment for each respondent and used for analyses of continuous preference scores. Logits can be transformed into the more easily interpretable **posterior membership probabilities**. A logit is similar to the theta score used in IRT.

MBTI® Complete. Online administration option for obtaining Step I or Step II results. Includes an interactive learning session that provides an introduction to the MBTI preferences and verification of results.

MBTI® European Step I™ assessment. The form of the Step I assessment that replaced Form G in Europe.

MBTI® European Step II™ assessment. The form of the Step II assessment that replaced Form K in Europe.

MBTI® Global Step I™ assessment. The form of the Step I assessment that replaced Form M and the European Step I assessment.

MBTI® Global Step II™ assessment. The form of the Step II assessment that replaced Form Q and the European Step II assessment.

MBTI® Manual Global Supplement Series. Supplements to this MBTI manual that provide detailed information about each of the samples included in the global research on which this manual is based, including psychometric data, sample descriptions, and so on, that supplements the information included in the manual.

MBTI Step I assessment. Contains all the items needed to obtain a respondent's four-letter type (e.g., ISTP, ENFJ).

MBTI Step I type. The four-letter type resulting from an administration of the MBTI Step I assessment.

MBTI Step II assessment. Contains all the items needed to obtain a respondent's results on the 20 Step II facet scales, as well as four-letter type (e.g., ISTP, ENFJ).

MBTI® Step II™ Interpretive Report. A report using the Step II scoring system to provide a detailed interpretation of a respondent's results on the 20 facets of the four Step I preference pairs. The MBTI Global Step II assessment produces these facet results, as did the prior form, the MBTI Form Q assessment.

mental function. See **function**.

mental process. See **process**.

middle school. In the US, typically grades 5 or 6 through 8. The ages of children in middle school range from approximately 10 through 14.

midpoint. The division point on the scale for one of the four preference pairs that separates respondents into one of the two opposite categories; for example, the point that distinguishes Introverts from Extraverts or Sensing types from Intuitive types. Accurate identification of the midpoint is critical for the MBTI Step I assessment because the opposite categories are hypothesized to be qualitatively and not merely quantitatively different.

multifaceted construct. A construct that contains several component parts that can be measured separately, in contrast to a **unidimensional construct**. The MBTI Step I constructs are multifaceted, permitting the identification of the Step II facets.

N. See **Intuition**.

nomothetic approach. A way of explaining and predicting human behavior that focuses on discovering general laws or principles based on knowledge of specific variables. Contrast **idiographic approach**.

omissions. On the MBTI assessment, items for which a respondent chose neither of the two available responses. For both the MBTI Step I and Step II assessments, if the number of omissions exceeds four, the record will not be scored and no results will be generated.

orientation. The preferred term, equivalent to the term **attitude**, for referring to the MBTI preference pairs Extraversion–Introversion and Judging–Perceiving. The E–I referent is to orientation of energy; the J–P referent is to orientation to the outside world.

orthogonal. Variables that are uncorrelated (i.e., are independent of each other); when graphed, the variables can be visualized as forming a right angle in space.

P. See **Perceiving**.

panel members. The group of respondents included in a sample used to represent a particular population.

pcc. See **preference clarity category**.

pci. See **preference clarity index**.

Perceiving (P). The orientation (attitude) that indicates whether either Sensing or Intuition is an individual's

preferred way of dealing with the outside world and is likely to appear in observable behavior.

perceiving functions (processes). Sensing and Intuition.

Perceiving type. A person who has a perceiving orientation (P), that is, prefers to use a perceiving process (Sensing or Intuition) rather than a judging process (Thinking or Feeling) when dealing with the outside world.

perception. The process of taking in information.

phi coefficient. A correlation coefficient used when both variables being correlated are dichotomous.

phrase items. Items in any of the forms of the MBTI assessment that present sentences or phrases for comparison.

polar opposites. Dynamic elements that are opposite each other in both function (process) and orientation. For example, the polar opposite of dominant extraverted Thinking is inferior introverted Feeling; the opposite of dominant introverted Sensing is inferior extraverted Intuition.

polarity index. An index calculated to assess the consistency with which a person exhibits clear preferences across all the Step II facet scales.

pole. Either opposite preference category of a bipolar pair of MBTI opposites, such as the Thinking pole of the T–F pair or the Concrete pole of the Concrete–Abstract Step II facet scale.

posterior membership probability. The probability of correct classification as one or the other preference within a preference pair, for example, S or N. Posterior membership probabilities are nonlinear transformations of the **logits**.

prediction ratio. The formula used for item analysis of MBTI forms preceding Form M indicating the probability that any response is given in accordance with the total scale score (e.g., the probability that a response designed for Extraverts is given by Extraverts and not given by Introverts). The prediction ratio indicates how well any item discriminates between choices for its scored preference.

preference. One member of each pair of opposite mental functions (processes) and orientations that in type theory structure an individual's personality. Each pair of opposites is composed of two preferences: Extraversion–Introversion, Sensing–Intuition, Thinking–Feeling, Judging–Perceiving.

preference clarity. An estimate of the extent to which a respondent reports favoring one pole of a preference pair over its opposite, based on responses to the MBTI assessment. Such estimates include the preference score calculated on forms preceding Form M and the preference clarity category and preference clarity index used with Form M. *See also* **preference clarity category** and **preference clarity index**.

preference clarity category (pcc). For Form M of the MBTI Step I assessment, a term for a range of **preference clarity indexes** that designate the clarity of a reported

preference for one or the other pole of a dichotomy (preference pair).

preference clarity index (pci). For Form M of the MBTI Step I assessment, a ratio showing how consistently a person responded to the items on the scale of interest compared to the maximum possible score achievable on that scale. Analogous to what in forms prior to Form M was referred to as a preference score. The functionally similar term for the MBTI Global Step I assessment is **probability index**.

preference domain. A set of characteristics that define one of the eight bipolar preferences (E, I, S, N, T, F, J, P) of the MBTI typology.

preference pair. One of the four pairs of opposites defined by the Jung-Myers approach: Extraversion–Introversion, Sensing–Intuition, Thinking–Feeling, and Judging–Perceiving.

preference scale. One of the four pairs of opposites measured by the MBTI assessment, that is, the E–I scale, S–N scale, T–F scale, and J–P scale. Because the term *scale* tends to connote an amount of a characteristic, as is typical of continuous measures, it is typically limited to discussion of the psychometric characteristics of the underlying continuous scores of the MBTI assessment. *See also* **facet scale**.

preference score. On MBTI forms that preceded Form M, a score designed to show how consistently a respondent chose one pole of a preference pair over its opposite. The corresponding term for Form M is **preference clarity index (pci)**.

probability index. For the MBTI Global Step I assessment, an estimate of how probable or likely it is that the preference indicated is accurate for the respondent and how likely it is that he or she will exhibit the same preference if retested. The functionally similar term for Form M is **preference clarity index**.

process (function). One of the four basic mental processes (functions) of **Sensing**, **Intuition**, **Thinking**, and **Feeling**.

psychological opposite. An item option that is attractive to an individual who has the preference the option is intended to elicit but is not necessarily logically opposite the other option. Item options serve as stimuli to evoke the desired response rather than as grammatically consistent pairings.

quadrature points. The number of points along the theta continuum used to arrive at a theta score. In general, increasing the number of points increases the precision of the theta estimate.

raw points. On the European Step I self-scorable and Form M self-scorable versions of the MBTI assessment, the sum of item responses for each preference in a pair. The term was also used on forms that preceded Form M. Neither the raw points of Form M nor the points of previous forms are interpretable, being only a step in determining the **preference clarity index** or **preference score**.

reconciliation. Refers to the translation process for the global research version of the MBTI assessment, whereby the in-country reviewers' comments were provided to the linguist and editor, and they, in turn, responded and reacted to suggestions and edits.

regression. See **linear regression**.

reliability coefficient (*r*). In **classical test theory (CTT)**, the consistency with which a given sample of individuals responds to the test items. Reliability coefficients are calculated for consistency over time (test-retest reliability) as well as consistency among the items on a given scale (internal consistency reliability).

reported type. The four-letter type that is indicated from a person's responses to the MBTI assessment. *Contrast true type and best-fit*, or verified, type.

S. See **Sensing**.

scale clarity class. Distinguishes between respondents showing more or less certainty in their responses. Five scale classes were used to determine MBTI Step I scoring.

second process. See **auxiliary function (process)**.

selection ratio. The ratio of the number of individuals in a particular sample occurring in a type or group of types to the number of individuals in that type or group of types in a baseline population.

selection ratio type table. A type table on a group that has been compared to another population. The table shows the frequency, percentage, selection ratio, and probability based on a chi-square analysis for each type and type grouping.

sensation. The term used by Jung and many current Jungians that corresponds to the term **Sensing** in MBTI terminology.

Sensing (S). The perceiving process that is concerned with perceptions and experiences available to the senses.

Sensing type. A person who prefers Sensing over Intuition as a way of perceiving.

shared facets of type. Typological characteristics that are held in common by types that share one or more preferences.

slope. See **three-parameter model**.

standard error (SE). An estimate of the degree to which a score is characterized by unsystematic sources of variance. The higher the standard error, the less precise the score and the more "noise" there is in the score.

Classical test theory (CTT) traditionally uses a single standard error for the entire scale; **item response theory (IRT)** provides a measure of standard error at each value of theta.

static level of interpretation. In contrast to the dynamic or developmental level of interpretation, a way of understanding type that is limited to definitions and characteristics of the four preference pairs considered separately, that is, with no attention to dynamic interactions among the preferences or developmental issues in the expression of type.

statistical interaction. A situation in which two or more independent causative variables affect a dependent

variable (the one being acted upon) in ways that are different from what their separate (independent) effects on the dependent variables would be if they were added together.

Step I. A scoring system for MBTI items that yields a respondent's four-letter type; the letters represent the respondent's indicated preferences on each of the four MBTI preference pairs. See also **MBTI Step I assessment**.

Step II. A scoring system for MBTI items that yields scores on facets that are components of the underlying pair of opposites of each of the four main scales. See also **MBTI Step II assessment**.

Step III™. A scoring system for an enlarged pool of MBTI items that yields information about a respondent's type development.

straws in the wind. The metaphor used by Isabel Myers to describe her approach to writing items for the MBTI assessment. Myers reasoned that because people's preferences are often not consciously formulated, they probably could not report about them directly. The simple items of the assessment were therefore designed as observable "pointers," or "straws in the wind," that allow inferences to be made about the direction of the underlying preference itself.

T. See **Thinking**.

TCC. See **test characteristic curve**.

tertiary function (process). The function (process) opposite the auxiliary function (process) in the dynamics of any type. For example, if the auxiliary function (process) is Thinking, the tertiary function (process) is Feeling. In theory, the tertiary function (process) is less accessible to conscious direction than the auxiliary function (process) but more accessible than the unconscious inferior function (process). Also known as the **third process**.

test characteristic curve (TCC). The composite of the information described by **item characteristic curves (ICC)** in **item response theory (IRT)**.

test standard error (TSE). The **item response theory (IRT)** equivalent of the **standard error** of measurement in **classical test theory (CTT)**. TSE shows the expected standard error that would be found when estimating theta scores from the observed item responses across the full range of possible theta scores.

tetrachoric correlation. A correlation coefficient used when both variables being correlated are in the form of artificial dichotomies.

theta (θ). In **item response theory (IRT)** terminology, test scores on a psychological characteristic that are roughly equivalent to the "true scores" of **classical test theory (CTT)**. Theta scores correspond directly to the MBTI **continuous scores** that were calculated using the **prediction ratio** method developed by Isabel Myers.

Thinking (T). Of the two opposite judging functions (processes), the one by which decisions are made by ordering choices in terms of logical cause-effect and objective analysis of relevant information.

Thinking type. A person who prefers using Thinking over Feeling as a way of making decisions or coming to conclusions.

third process. See **tertiary function (process)**.

three-parameter model. In **item response theory (IRT)**, the use of three parameters to model the relationship between item responses and the latent characteristic that the items are designed to measure. A three-parameter model was used for Form M and Form Q, a two-parameter model for the Global Step I and Step II assessments (the *c* parameter is omitted). The three parameters are:

- a* parameter: a measure of item *discrimination*; the maximum slope of the item characteristic curve. Higher discrimination values indicate generally stronger items.
- b* parameter: a measure of item *difficulty*; the item location (theta) where the slope is maximized. This theta is where the item produces the most information.
- c* parameter: defines the baseline for item endorsement rates that would be expected from individuals who score very far to the low (left) end of the theta scale.

tie-breaking formula. Though tie scores are rare for the MBTI Global Step I assessment, the tradition established for previous forms is followed when it does occur, that is, ties are broken in favor of I, N, F, or P.

traditional analysis. Psychometric methods of evaluating items for inclusion in a test that uses approaches such as the **prediction ratio** method used by Myers in her development of Form G and earlier forms.

trait. In the field of personality, a single personality characteristic that is present in all people to varying degrees. When personality is described in terms of traits, it is assumed that everyone has the same characteristics. The differences between individuals are due to how much those individuals possess of each of the traits that describe human personality. Traits typically are measured on a continuum of *less* to *more*.

trait theory. A system for explaining personality variation that postulates traits as the underlying units of descriptions. Trait theory approaches predominate among psychologists. Different trait theories vary primarily in the number and nature of the traits that are believed to be sufficient and necessary to explain human personality. In contrast to type theory and the MBTI assessment, which rest on a specific theoretical system, most trait systems derive their sets of traits empirically and do not postulate any theoretical basis for the particular array of traits used.

translation. For assessment items, has historically been used to describe the process of conversion to a language other than the one in which they were originally constructed. **Contrast adaptation.**

true type. The personality type that is the hypothetical "true" type for the individual, in contrast to **reported type** and

best-fit type (or verified type). True type is considered to be unknowable and therefore exists only as a hypothesis.

two-parameter model. The **item response theory (IRT)** model used in developing the Global Step II assessment, which uses only the *a* and *b* parameters. See **three-parameter model**.

type. In the field of personality, and specifically for the MBTI assessment, a unique combination of orientations (E or I and J or P) and mental functions (processes) (S or N and T or F) that is more than the sum of its parts. Specifically, 1 of the 16 combinations of four preferences, each with specific characteristics postulated from the dynamics of the theory. The term *type* is not used to denote a single preference.

type code. In some publications, used to denote a type, for example, ESTJ or INFJ.

type development. In type theory, the emergence and expression of the dynamic elements that make up psychological type; the ages or developmental stages when the dominant, auxiliary, tertiary, and inferior functions (processes) may be manifested and the roles these play in personality growth.

Type Differentiation Indicator (TDI). See **Form J**.

type dynamics. The interaction of the dominant, auxiliary, tertiary, and inferior functions (processes) along with the orientations of Extraversion and Introversion in which they are typically used. The dynamics of any one of the 16 types is inherent in the four letters used to specify a type, such as ISTJ or ESFP.

type dynamics interactions. The theoretically predicted interactions between the E–I and J–P preference pairs that serve to identify the forms (dominant or auxiliary) and orientations (Introverted or Extraverted) of the functions (processes).

type table. A display of the 16 types in the format developed by Isabel Myers. The type table may have only the 16 types or may be supplemented with a column at the side or bottom showing the type groupings.

type theory. As used in this manual, the original theory of typology proposed and developed by Carl G. Jung and described in his book *Psychological Types*, published in 1921.

typology. A systematic way of describing types based on characteristics that are shared in various ways. In the field of personality, a system of classifying personalities into qualitatively distinct groups (types) whose members share some characteristics with other groups while at the same time possessing additional characteristics that are not shared with any other group.

unidimensional construct. A construct that describes a single aspect and definition of whatever is being measured, and that does not have component parts that can be separately measured. **Compare multifaceted construct.**

unique facets of type. Typological characteristics that are "more than the sum of the parts" and are associated with only a single type. These are the typological

characteristics that make each of the 16 MBTI types different from every other type.

unit weighting. A simple method of calculating scale scores that involves assigning weights to each item and summing across items.

verified type. See **best-fit type**.

whole type. The basic unit of type theory, being the four dynamic elements (preferences) that interact uniquely to form one of the 16 types.

word-pair items. Items used in the MBTI assessment in which two words or terms are presented, whereby the respondent must choose the one that is most appealing.

REFERENCES

- Abrahamian, E. J., Weinberg, J., Grady, M., & Stanton, C. M. (2004). The effect of personality-aware computer-human interfaces on learning. *Journal of Universal Computer Science*, 9(1), 27–37.
- Allread, W. G., & Marras, W. S. (2006). Does personality affect the risk of developing musculoskeletal discomfort? *Theoretical Issues in Ergonomics Science* 7(2), 149–167.
- Amato, C. H., & Amato, L. H. (2005). Enhancing student team effectiveness: Application of Myers-Briggs® personality assessment in business courses. *Journal of Marketing Education*, 27, 41–51.
- Anchors, S., & Dana, R. (1989). Type and substance abuse in college age students. *Journal of Psychological Type*, 17, 26–30.
- Anchors, S., & Hay, S. (1990). Resident assistants: Are they similar or dissimilar to students from whom they are selected? *Journal of College and University Student Housing*, 20(2), 18–21.
- Anchors, S., Robbins, M. A., & Gershman, E. S. (1989). The relationship between Jungian type and persistence to graduation among college students. *Journal of Psychological Type*, 17, 20–25.
- Anchors, S., & Robinson, D. C. (1992). Psychological type and the accomplishment of student development tasks. *NASPA Journal*, 29(2), 131–135.
- Anderson, M., & Kulas, J. (2013, September). Seeing MBTI® type in others. In R. C. Thompson (Chair), *Seeing MBTI® type: Awareness, expression, and cross-cultural evaluations*. Presented at Global MBTI® Users Conference, Shanghai, China.
- Ang, M. (2002). Advanced communication skills: Conflict management and persuasion. *Academic Medicine*, 77(11), 1166.
- Antonovsky, A. (1993). The structure and properties of the Sense of Coherence Scale. *Social Science and Medicine*, 36(6), 725–733.
- Apostal, R., & Trontvent, R. (1989). College students' academic comfort and personality. *Journal of College Student Development*, 30, 210–212.
- Arain, A. A. (1968). Relationships among counseling clients' personalities, expectations, and problems (Doctoral dissertation, Rutgers University). *Dissertation Abstracts International: Section A. Humanities and Social Sciences*, 29, 4903–4904. (University Microfilms No. 68-8640)
- Arneson, J. J., & Landowski, N. (2015). *Meta-analysis of relationships between the MBTI® and NEO®*. Unpublished internal document, CPP, Inc.
- Atman, K. S. (1993). Goal accomplishment style and psychological type: Cultural variations. In R. A. Moody (Ed.), *Proceedings of Psychological Type and Culture—East and West: A Multicultural Research Symposium, January 6–8, 1993, Honolulu, Hawaii* (pp. 207–220). Gainesville, FL: Center for Applications of Psychological Type.
- Bail, F. T. (1993). An exploration of relationships among psychological type, ethnicity and computer-mediated communication. In R. A. Moody (Ed.), *Proceedings of Psychological Type and Culture—East and West: A Multicultural Research Symposium, January 6–8, 1993, Honolulu, Hawaii* (pp. 221–227). Gainesville, FL: Center for Applications of Psychological Type.
- Ball, I. L., & Geyer, P. (2006, January). *Form K of the Myers-Briggs Type Indicator® in Australia: Its robustness, interpretation and use*. Paper presented at Fifth Psychological Type and Culture—East and West: A Multicultural Research Symposium, Honolulu, HI.
- Barger, N. J., & Kirby, L. K. (1995). *The challenge of change in organizations: Helping employees thrive in the new frontier*. Boston, MA: Nicholas Brealey.
- Barger, N. J., & Kirby, L. K. (1997a). *MBTI® type and change*. Sunnyvale, CA: The Myers-Briggs Company.
- Barger, N. J., & Kirby, L. K. (1997b). *MBTI® type and change participant's guide*. Sunnyvale, CA: The Myers-Briggs Company.
- Barrett, L. (1989). Impact of teacher personality on classroom environment. *Journal of Psychological Type*, 18, 50–56.
- Barrineau, P. (1997). The type to drink: Undergraduate alcohol policy violators and personality type. *Journal of Psychological Type*, 41, 23–27.
- Bartram, D. (2013). Scalar equivalence of OPQ32: Big Five profiles of 31 countries. *Journal of Cross-Cultural Psychology*, 44(1), 61–83.
- Bathurst, J. (1999). Best-fit type vs. reported type: Demographic and environmental effects on the accuracy of the MBTI®. In *Proceedings of the Thirteenth Biennial International Conference of the Association for Psychological Type*. Gainesville, FL: Center for Applications of Psychological Type.
- Bathurst, J. (2000). The relationship between reported type and best-fit type: Evidence for administrator bias? *Journal of Psychological Type*, 54, 5–11.
- Beckham, M. H. (2012). Building momentum: The unconventional strengths of Perceiving college students. *Journal of Psychological Type*, 72(2), 27–40.
- Bedwell, S. (2015). *Emotional Judgment Inventory manual* (Rev. ed.). Oxford, England: OPP Ltd.
- Bell, S. E. (2005). *Myers-Briggs Type Indicator® and executive coaching: Participants' self-perceptions*

- about the effectiveness of the two when used together (Unpublished doctoral dissertation). Capella University.
- Belling, R. (2009). The influence of psychological type on transfer of manager's learning. *Journal of Psychological Type*, 69(8), 101–110.
- Bess, T. L., & Harvey, R. J. (2002). Bimodal score distributions and the Myers-Briggs Type Indicator®: Fact or artifact? *Journal of Personality Assessment*, 78, 176–186.
- Bess, T. L., Harvey, R. J., & Swartz, D. (2003). *Hierarchical confirmatory factor analysis of the Myers-Briggs Type Indicator®*. Paper presented at the 2003 annual conference of the Society for Industrial and Organizational Psychology, Orlando, FL.
- Betz, N. E., Borgen, F. H., & Harmon, L. W. (2005). *Skills Confidence Inventory*. Sunnyvale, CA: The Myers-Briggs Company.
- Beuke, C. J., Freeman, D. G., & Wang, S. (2006, January). *Reliability and validity of the Myers-Briggs Type Indicator® Form M when translated into traditional and simplified Chinese characters*. Paper presented at Fifth Psychological Type and Culture—East and West: A Multicultural Research Symposium, Honolulu, HI.
- Beyler, J., & Schmeck, R. R. (1992). Assessment of individual differences in preferences for holistic-analytic strategies: Evaluation of some commonly available instruments. *Educational and Psychological Measurement*, 52(3), 709–719.
- Birkman, R. W., Elizondo, F., Lee, L. G., Wadlington, P. L., & Zamzow, M. W. (2008). *The Birkman Method® manual*. Houston, TX: Birkman International.
- Bishop-Clark, C., Dietz-Uhler, B., & Fisher, A. (2006–2007). The effects of personality type on web-based distance learning. *Journal of Educational Technology Systems*, 35(4), 491–506.
- Block, J. (1978). *The Q-Sort method in personality assessment and psychiatric research*. Palo Alto, CA: Consulting Psychologists Press.
- Bouchard, T. J., Jr., & Hur, Y. M. (1998). Genetic and environmental influences on the continuous scales of the Myers-Briggs Type Indicator®: An analysis based on twins reared apart. *Journal of Personality*, 66(2), 135–149.
- Bouchard, T. J., & McGue, M. (2003). Genetic and environmental influences on human psychological differences. *Journal of Neurobiology* 54(1), 4–45.
- Bowen, J. L. (1990). The combined predictive effect of creativity level and personality type on college students' reflective judgment (Doctoral dissertation, East Texas State University). *Dissertation Abstracts International: Section A. Humanities and Social Sciences*, 51(1), 110.
- Bowen, P. L., Ferguson, C. B., Lehmann, T. H., & Rohde, F. H. (2003). Cognitive style factors affecting database query performance. *International Journal of Accounting Information Systems*, 4, 251–273.
- Box, G. E. P. (1979). Robustness is the strategy of scientific model building. In R. L. Launer & G. N. Wilkinson (Eds.), *Robustness in statistics* (pp. 201–236). New York, NY: Academic Press.
- Boyd, N. E. (1995). An examination of interpersonal relationships between student and cooperating teachers (Doctoral dissertation, University of Memphis). *Dissertation Abstracts International: Section A. Humanities and Social Sciences*, 55(10), 3076.
- Bradley-Geist, J. C., & Landis, R. S. (2012). Homogeneity of personality in occupations and organizations: A comparison of alternative statistical tests. *Journal of Business Psychology*, 27, 149–159.
- Bradshaw, S. L. (1986). *Myers-Briggs® personality types of administrative managers in long-term care facilities and their relationship to employee turnover rates and resident occupancy rates* (Unpublished doctoral dissertation). University of Texas at Austin.
- Brennan, M. D. (2010). The effect of career assessments and follow-up counseling on career decision-making self-efficacy among active-duty coast guard personnel (Doctoral dissertation, Trident University International). *Dissertation Abstracts International: Section A. Humanities and Social Sciences*, 70(7), 2339.
- Briggs, S. P., Copeland, S., & Haynes, D. (2007). Accountants for the 21st century, where are you? A five-year study of accounting students' personality preferences. *Critical Perspectives on Accounting*, 18, 511–537.
- Brocato, F. C. (1985). *Psychological type and task accomplishment in the public school management team* (Unpublished doctoral dissertation). University of Oklahoma, Norman.
- Brown, F. W., & Reilly, M. D. (2009). The Myers-Briggs Type Indicator® and transformational leadership. *Journal of Management Development*, 28(10), 916–932.
- Brown, V. L., & DeCoster, D. A. (1991). The Myers-Briggs Type Indicator® as a developmental measure: Implications for student learners in higher education. *Journal of College Student Development*, 32, 378–379.
- Brush, L. C. (1989). Training residential advisors in peer counseling: Implications of type (MBTI®/PTPI). *Journal of Psychological Type*, 17, 73–78.
- Buboltz, W. C., Jr., Johnson, P., Nichols, C., Miller, M. A., & Thomas, A. (2000). MBTI® personality types and SII Personal Style Scales (Strong Interest Inventory®). *Journal of Career Assessment*, 8(2), 131–145.
- Buckworth, J., Granello, D. H., & Belmore, J. (2002). Incorporating personality assessment into counseling to help students adopt and maintain exercise behaviors. *Journal of College Counseling*, 5, 15–25.
- Burt, K. G. (2005). *Analysis of the factors involved in technological problem solving in a college technology education classroom* (Unpublished doctoral dissertation). North Carolina State University, Raleigh.
- Campbell, D. P., & Hansen, J. C. (1981). *Manual for the SVIB-SCII Strong-Campbell Interest Inventory* (3rd ed.). Stanford, CA: Stanford University Press.
- Canning, E. A. B. (1983). An analysis of cognitive style, hemispheric preference, personality type and academic achievement in a group of economically disadvantaged secondary students (Doctoral dissertation, University of

- Pittsburgh). *Dissertation Abstracts International: Section A. Humanities and Social Sciences*, 45(2), 457.
- Capraro, R. M., & Capraro, M. M. (2002). Myers-Briggs Type Indicator® score reliability across studies: A meta-analytic reliability generalization study (Form M). *Educational & Psychological Measurement*, 62(4), 590–602.
- Capretz, L. F. (2003, February). Personality types in software engineering. *International Journal of Human-Computer Studies*, 58(2), 207–214.
- Carey, J. C., Hamilton, D. L., & Shanklin, G. (1985). Psychological type and interpersonal compatibility: Evidence for a relationship between communication style preference and relationship satisfaction in college roommates. *Journal of Psychological Type*, 10, 36–39.
- Carey, J. C., Hamilton, D. L., & Shanklin, G. (1986). Does personality similarity affect male roommates' satisfaction? *Journal of College Student Personnel*, 27(1), 65–69.
- Carlson, R. (1980). Studies of Jungian typology: II. Representations of the personal world. *Journal of Personality and Social Psychology*, 38, 801–810.
- Carlson, R., & Levy, N. (1973). Studies of Jungian typology: I. Memory, social perception, and social action. *Journal of Personality*, 41(4), 559–576.
- Carrell, P. L., & Monroe, L. B. (1993). Learning styles and composition. *Modern Language Journal*, 77(2), 148–162.
- Carskadon, T. G. (1979a). Clinical and counseling aspects of the Myers-Briggs Type Indicator®: A research review. *Research in Psychological Type*, 2, 2–31.
- Carskadon, T. G. (1979b). Behavioral differences between Extraverts and Introverts as measured by the Myers-Briggs Type Indicator®: An experimental demonstration. *Research in Psychological Type*, 2, 78–82.
- Carskadon, T. G. (1982). Myers-Briggs Type Indicator® characterizations: A Jungian horoscope? *Research in Psychological Type*, 8, 87–88.
- Carskadon, T. G., & Cook, D. D. (1982). Validity of MBTI® type descriptions as perceived by recipients unfamiliar with type. *Research in Psychological Type*, 8, 89–94.
- Carter, J. A. (1990). A comparison of personality types as determined by the Myers-Briggs Type Indicator® with the process of separate-knowing and connected-knowing procedural processing among female undergraduate and nontraditional students (Doctoral dissertation, Michigan State University). *Dissertation Abstracts International: Section A. Humanities and Social Sciences*, 51(5), 1475.
- Carver, C. S. (1989). How should multifaceted personality constructs be tested? Issues illustrated by self-monitoring, attributional style, and hardiness. *Journal of Personality and Social Psychology*, 56, 577–585.
- Caspi, A. (1998). Personality development across the life course. In W. Damon (Ed.), *Handbook of Child Psychology: Vol. 3. Social, emotional, and personality development* (5th ed., pp. 311–388). New York, NY: Wiley.
- Catoe, C. W. (1992). Developmental profiles of college students: A comparison of moral development stage and Myers-Briggs Type Indicator® personality types (Doctoral dissertation, North Carolina State University). *Dissertation Abstracts International: Section B. Sciences and Engineering*, 53(3), 1639.
- Center for Creative Leadership. (1987). Benchmarks® competency definitions. In *Benchmarks Technical Manual*. Greensboro, NC: Author.
- Chany, A. M., Parakkat, J., Yang, G., Burr, D. L., & Marras, W. S. (2006). Changes in spine loading patterns throughout the workday as a function of experience, lift frequency, and personality. *Spine Journal*, 6(3), 296–305.
- Chartrand, J. M., Robbins, S. B., & Morrill, W. H. (1997). *Career Factors Inventory*. Mountain View, CA: CPP, Inc.
- Chen, J., Tian, J., & Miao, D. (2009). Item description in translated tests: Cultural effects on factor structure of the Chinese version of the Myers-Briggs Type Indicator®–G. *Social Behavior and Personality*, 37(1), 31–40.
- Cheng, S.-H., Wang, Y.-D., Horng, R.-Y., & Huang, Y.-C. (2007, June). Person-project fit and R&D performance: A case study of Industrial Technology Research Institute of Taiwan. *R&D Management*, 37(3), 209–220.
- Chesborough, S. R. (1994). The use of the MBTI® to improve opportunities for the academic success of scholarship athletes. In *Proceedings: Orchestrating Educational Change in the 90's—The Role of Psychological Type* (pp. 119–137). Gainesville, FL: Center for Applications of Psychological Type.
- Choi, K. S., Deek, F. P., & Im, I. (2008). Exploring the underlying aspects of pair programming: The impact of personality. *Information and Software Technology*, 50, 1114–1126.
- Choong, S., & Britton, K. (2007). Character strengths and type: Exploration of covariation. *International Coaching Psychology Review* 2(1), 9–23.
- Cianci, H. R. (2007). *The Myers-Briggs Type Indicator® and self-efficacy in substance abuse treatment planning* (Unpublished doctoral dissertation). Northcentral University.
- Cohen, J. (1988). *Statistical power analysis for the behavioral sciences* (2nd ed.). Hillsdale, NJ: Erlbaum.
- Cohen, J. (1994). The earth is round ($p < .05$). *American Psychologist*, 49(12), 997–1003.
- Comrey, A. L. (1983). An evaluation of the Myers-Briggs Type Indicator®. *Academic Psychology Bulletin*, 5, 115–129.
- Conn, S. R., & Rieke, M. L. (1994). *16PF®* (5th ed.). Champaign, IL: Institute for Personality and Ability Testing.
- Connor-Smith, J. K., Compas, B. E., Wadsworth, M. E., Thomsen, A. H., & Saltzman, H. (2000). Responses to stress in adolescence: Measurement of coping and involuntary stress responses. *Journal of Counseling and Clinical Psychology*, 68, 976–992.
- Cooper, S. E., & Miller, J. A. (1991). MBTI® learning style–teaching style incongruencies. *Educational and Psychological Measurement*, 51, 699–706.
- Craig, D. L., Craig, C. H., & Sleight, C. C. (1988). Type preferences of decision-makers: Corporate and clinical. *Journal of Psychological Type*, 16, 33–37.

- Cronbach, L. J. (1951). Coefficient alpha and the internal structure of tests. *Psychometrika*, 16(3), 297–334.
- Da Cunha, A. D., & Greathead, D. (2007, May). Does personality matter? An analysis of code-review ability. *Communications of the ACM*, 50(5), 109–112.
- Dawis, R. V. (1987). Scale construction. *Journal of Counseling Psychology*, 34, 481–489.
- Dawis, R. V. (1996). The theory of work adjustment and person-environment-correspondence counseling. In D. Brown & L. Brooks (Eds.), *Career choice and development* (3rd ed.). San Francisco, CA: Jossey-Bass.
- Dawis, R. V., & Lofquist, L. R. (1984). *A psychological theory of work adjustment*. Minneapolis, MN: University of Minnesota Press.
- Deakin, P. (2006). *German MBTI® Step I® data supplement*. Oxford, England: Oxford Psychologists Press.
- DeMoss, M., McCann, G., & Belcher, L. J. (2000). Integrating type into a first-year university experience business course. In *Proceedings of the Fourth Biennial International Conference on Education of the Center for Applications of Psychological Type* (pp. 43–51). Gainesville, FL: Center for Applications of Psychological Type.
- DeVellis, R. F. (1991). *Scale development: Theory and applications*. Applied Social Research Methods Series, 26. Newbury Park, CA: Sage.
- DiRienzo, C. E., Das, J., Synn, W., Kitts, J., & McGrath, K. (2010). The relationship between MBTI® and academic performance: A study across academic disciplines. *Journal of Psychological Type*, 70(5), 53–67.
- DiTiberio, J. K. (1977). The strength of Sensing–Intuition preference on the Myers-Briggs Type Indicator® as related to empathic discrimination of overt or covert feeling messages of others (Doctoral dissertation, Michigan State University). *Dissertation Abstracts International: Section A. Humanities and Social Sciences*, 37, 5599. (University Microfilms No. 77-5789)
- DiTiberio, J. K. (1996). Education, learning styles, and cognitive styles. In A. L. Hammer (Ed.), *MBTI® applications: A decade of research on the Myers-Briggs Type Indicator®* (pp. 123–166). Mountain View, CA: CPP, Inc.
- Doerries, L. E., & Ridley, D. R. (1998). Time sensitivity and purpose in life: Contrasting theoretical perspectives of Myers-Briggs® and Victor Frankl. *Psychological Reports*, 83(1), 67–71.
- Donnay, D. A. C., Morris, M. L., Schaubhut, N. A., & Thompson, R. C. (2005). *Strong Interest Inventory® manual: Research, development, and strategies for interpretation* (Rev. ed.). Sunnyvale, CA: The Myers-Briggs Company.
- Donovan, A. J. (1994). The interaction of personality traits in applied music teaching (Doctoral dissertation, University of Southern Mississippi). *Dissertation Abstracts International: Section A. Humanities and Social Sciences*, 55(6), 1499.
- Drasgow, F., & Hulin, C. L. (1990). Item response theory. In M. D. Dunnette & L. M. Hough (Eds.), *Handbook of industrial and organizational psychology* (pp. 577–636). Palo Alto, CA: Consulting Psychologists Press.
- Drasgow, F., Levine, M. V., Tsien, S., Williams, B. A., & Mead, A. D. (1995). Fitting polytomous IRT models to multiple choice tests. *Applied Psychological Measurement*, 19, 143–165.
- Drummond, R. J., & Stoddard, A. H. (1992). Learning style and personality type. *Perceptual and Motor Skills*, 75, 99–104.
- Du Toit, F., Coetzee, S. C., & Visser, D. (2005). The relation between personality type and sense of coherence among technical workers. *South African Business Review*, 9(1), 51–65.
- Edmunds, M. (1982). Jungian personality type and imagery ability within a holistic health context (Doctoral dissertation, Pennsylvania State University). *Dissertation Abstracts International: Section B. Sciences and Engineering*, 43(3), 868.
- Edwards, A. L. (1954). *Manual for the EPPS*. New York, NY: Psychological Corporation.
- Ehrman, M. (1993). Ego boundaries revisited: Toward a model of personality and learning. In J. E. Alatis (Ed.), *Strategic interaction and language acquisition: Theory, practice and research* (pp. 331–362). Washington, DC: Georgetown University Press.
- Ehrman, M. (1994a). The Type Differentiation Indicator and adult foreign language learning success. *Journal of Psychological Type*, 30, 10–29.
- Ehrman, M. (1994b). Weakest and strongest learners in intensive language training: A study of extremes. In C. Klee (Ed.), *Faces in a crowd: Individual learners in multisection programs*. Boston, MA: Heinle & Heinle.
- Ehrman, M. (1995). *The Modern Language Aptitude Test: What does it tell us?* Unpublished manuscript.
- Ehrman, M., & Oxford, R. (1990). Adult language learning styles and strategies in an intensive training setting. *Modern Language Journal*, 74(3), 311–327.
- Ehrman, M., & Oxford, R. (1995). Cognition plus: Correlates of language learning success. *Modern Language Journal*, 79(1), 67–89.
- Elliott, G. R., & Sapp, G. L. (1988). The relationship between the Myers-Briggs Type Indicator® and the Grasha-Riechmann Student Learning Styles Questionnaire. *Journal of Psychological Type*, 14, 46–50.
- Elliott, K. M. (2006). *The effects of personality and learning style on the achievement of adult learners in community college online education: An investigation based on the Myers-Briggs Type Indicator® and the Kolb Learning Styles Inventory* (Unpublished doctoral dissertation). Capella University.
- Embree, M. C. (March 2011). Psychological type and student views of the origins debate: Understanding the culture wars. *Journal of Psychological Type*, 71(1), 1–11.

- Erskine, C. G., Westerman, G. H., & Grandy, T. G. (1986). Personality styles of first-year dental students. *Journal of Dental Education*, 50(4), 221–224.
- Evans, L. N. (1976). A psycho-temporal theory of personality: A study of the relationship between temporal orientation, affect, and personality type (Doctoral dissertation, United States International University). *Dissertation Abstracts International: Section B. Sciences and Engineering*, 37, 1875. (University Microfilms No. 76-22, 381)
- Eysenck, H. J. (1967). *The biological basis of personality*. Springfield, IL: Charles C. Thomas.
- Faucett, J. M., Morgan, E. R., Poling, T. H., & Johnson, J. (1995). MBTI® type and Kohlberg's postconventional stages of moral reasoning. *Journal of Psychological Type*, 34, 17–23.
- Felder, R. M., Felder, G. N., & Dietz, E. J. (2002). The effects of personality type on engineering student performance and attitudes. *Journal of Engineering Education*, 91(1), 3–17.
- Filbeck, G., Hatfield, P., & Horvath, P. (2005). Risk aversion and personality type. *Journal of Behavioral Finance*, 6(4), 170–180.
- Fischetti, B. A., & Mentore-Lee, J. L. (2001). Personality type and teaching: A preliminary investigation. *Bulletin of Psychological Type*, 24(3), 6, 8.
- Fisher, N. M. (1994). MBTI® type and writing-across-the-curriculum. *Journal of Psychological Type*, 28, 43–47.
- Fitzgerald, C. (Ed.). (1994). *Proceedings of The Myers-Briggs Type Indicator® and Leadership: An International Research Conference*. College Park, MD: National Leadership Institute, University of Maryland.
- Fitzgerald, C., & Kirby, L. K. (Eds.). (1997). *Developing leaders: Research and applications in psychological type and leadership development*. Boston, MA: Nicholas Brealey.
- Fleenor, J. W. (1997). The relationship between the MBTI® and measures of personality and performance in management groups. In C. Fitzgerald & L. K. Kirby (Eds.), *Developing leaders: Research and applications in psychological type and leadership development* (pp. 115–138). Boston, MA: Nicholas Brealey.
- Foster, T. F., & Pashneh-Tala, K. (2002). *The utility of personality measures in the admissions process at the United States Naval Academy* (Unpublished master's thesis). Naval Postgraduate School, Monterey, CA.
- Fourqurean, J., Meisgeier, C., & Swank, P. (1990). The link between learning style and Jungian psychological type: A finding of two bipolar preference dimensions. *Journal of Experimental Education*, 58(3), 225–237.
- Fowler, C. F. (2009). *Preference for instructional methods and MBTI® personality types in nurses* (Unpublished doctoral dissertation). North Carolina State University, Raleigh.
- Friedman, C. P., & Slatt, L. M. (1988). New results relating the Myers-Briggs Type Indicator® and medical specialty choice. *Journal of Medical Education*, 63, 325–327.
- Fry, E. (1977). Fry's readability graph: Clarifications, validity, and extension to level 17. *Journal of Reading*, 21, 249.
- Fu, J. M. (1992). The relationship between the MBTI® personality types of international students and adjustment to American society (Doctoral dissertation, University of San Francisco). *Dissertation Abstracts International: Section A. Humanities and Social Sciences*, 53(1), 69.
- Furnham, A., & Crump, J. (2007). Relationship between the MBTI® and FIRO-B® in a large British sample. *Psychological Reports*, 101(3), 970–978.
- Furnham, A., Crump, J., Batey, M., & Chamorro-Premuzic, T. (2009). Personality and ability predictors of the "Consequences" test of divergent thinking in a large non-student sample. *Personality and Individual Differences*, 46(4), 536–540.
- Gailbreath, R. D., Wagner, S. L., Moffett, R. G., & Hein, M. B. (1997). Homogeneity in behavioral preference among U.S. Army leaders. *Group Dynamics: Theory, Research, and Practice*, 1(3), 222–230.
- Gaines, M. C. (1998). Team personality and school climate. In *Proceedings of the Third Biennial International Conference on Education of the Center for Applications of Psychological Type* (pp. 13–17). Gainesville, FL: Center for Applications of Psychological Type.
- Gallen, T. (2006). Managers and strategic decisions: Does the cognitive style matter? *Journal of Management Development*, 25(2), 118–133.
- Garden, A. M. (1989). Organizational size as a variable in type analysis and employee turnover. *Journal of Psychological Type*, 17, 3–13.
- Garden, A. M. (1997). Relationships between MBTI® profiles, motivation profiles and career paths. *Journal of Psychological Type*, 41, 3–16.
- Gautsch, S. M. (1993). The effects of perceptual style with mode of instruction in a hypermedia language learning environment. In R. A. Moody (Ed.), *Proceedings of Psychological Type and Culture—East and West: A Multicultural Research Symposium, January 6–8, 1993, Honolulu, Hawaii* (pp. 229–240). Gainesville, FL: Center for Applications of Psychological Type.
- Genco, J. T., & Mitchell, C. W. (2006, June). Relationships between MBTI® personality types and coping styles: A pilot study. *Journal of Psychological Type*, 66(6), 49–57.
- Gentry, W. A., Mondore, S. P., & Cox, B. D. (2007). A study of managerial derailment characteristics and personality preferences. *Journal of Management Development*, 26(9), 857–873.
- Gilchrist, B. J. (1991). The Myers-Briggs Type Indicator® as a tool for clinical legal education. *Saint Louis University Public Law Review*, 10, 601–613.
- Gilligan, C. (1982). *In a different voice*. Cambridge, MA: Harvard University Press.
- Glaman, J. M., Jones, A. P., & Rozelle, R. M. (1996). The effects of co-worker similarity on the emergence of affect in work teams. *Group & Organizational Management*, 21(2), 192–215.
- Goby, V. P. (2006). Personality and online/offline choices: MBTI® profiles and favored communication modes in a

- Singapore study. *CyberPsychology and Behavior*, 9(1), 5–13.
- Golden, T. C. (2009). *Myers-Briggs® personality type and adolescent coping in the college search* (Unpublished doctoral dissertation). Purdue University, West Lafayette, IN.
- Gordon, V. N., Coscarelli, W. C., & Sears, S. J. (1986). Comparative assessments of individual differences in learning and career decision making. *Journal of College Student Personnel*, 27(3), 233–242.
- Gordy, C. C., & Thorne, B. M. (1994). Proofreading ability as a function of personality type. *Journal of Psychological Type*, 28, 29–36.
- Gottfredson, G. D., & Holland, J. L. (1996). *Dictionary of Holland occupational codes* (3rd ed.). Odessa, FL: Psychological Assessment Resources.
- Gough, H. G. (1965). Conceptual analysis of psychological test scores and other diagnostic variables. *Journal of Abnormal Psychology*, 70, 294–302.
- Gough, H. G. (1976). Studying creativity by means of word association tests. *Journal of Applied Psychology*, 61, 348–353.
- Gough, H. G. (1981). *Studies of the Myers-Briggs Type Indicator® in a personality assessment research institute*. Paper presented at the 4th National Conference on the Myers-Briggs Type Indicator®, Palo Alto, CA.
- Gough, H. G., & Bradley, P. (1996). *CPI® manual* (3rd ed.). Sunnyvale, CA: The Myers-Briggs Company.
- Gough, H. G., & Bradley, P. (2005). *CPI 260® manual*. Sunnyvale, CA: The Myers-Briggs Company.
- Gough, H. G., & Heilbrun, A. B. (1983). *The Adjective Check List manual*. Mountain View, CA: CPP, Inc.
- Gram, P. C., Dunn, B. R., & Ellis, D. (2005). Relationship between EEG and psychological type. *Journal of Psychological Type*, 65(5), 33–46.
- Griffin, T. D., & Salter, D. W. (1993). Psychological type and involvement in a university residence hall judicial system. *Journal of Psychological Type*, 27, 32–38.
- Hackston, J. (2005). *French MBTI® Step I® data supplement*. Oxford, England: OPP Ltd.
- Hackston, J. (2015). *Type and the cultural orientations framework*. Oxford, England: OPP Ltd.
- Haley, U. C. V., & Stumpf, S. A. (1989). Cognitive trails in strategic decision-making: Linking theories of personalities and cognitions. *Journal of Management Studies*, 26(5), 477–497.
- Hall, W. B., & MacKinnon, D. W. (1969). Personality inventory correlates of creativity among architects. *Journal of Applied Psychology*, 53(4), 322–326.
- Hammer, A. L. (1985). Psychological type and media preferences in an adult sample. *Journal of Psychological Type*, 10, 20–26.
- Hammer, A. L. (1991). *Type and coping resources*. Presented at the Bay Area Association for Psychological Type, Stanford University, Stanford, CA.
- Hammer, A. L. (1993, July). *Coping resources and scores on the Type Differentiation Indicator*. Panel presentation at the 10th Biennial International Conference of the Association for Psychological Type, Newport Beach, CA.
- Hammer, A. L. (1996a). Career management and counseling. In A. L. Hammer (Ed.), *MBTI® applications: A decade of research on the Myers-Briggs Type Indicator®* (pp. 31–54). Mountain View, CA: CPP, Inc.
- Hammer, A. L. (Ed.). (1996b). *MBTI® applications: A decade of research on the Myers-Briggs Type Indicator®*. Mountain View, CA: CPP, Inc.
- Hammer, A. L. (2003). *Introduction to Type® and retention: Retaining talent, leveraging type*. Mountain View, CA: CPP, Inc.
- Hammer, A. L. (2005a). *MBTI® Career Report user's guide*. Sunnyvale, CA: The Myers-Briggs Company.
- Hammer, A. L. (2005b). *The relationship of type to six character strengths*. Unpublished paper.
- Hammer, A. L. (2014a). *Correlations of the MBTI® with the CPI® 434 in a sample of managers*. Unpublished paper.
- Hammer, A. L. (2014b). *Type and attitudes toward and use of technology*. Unpublished paper.
- Hammer, A. L., & Huszycz G. E. (1996). Teams. In A. L. Hammer (Ed.), *MBTI® applications: A decade of research on the Myers-Briggs Type Indicator®* (pp. 81–104). Mountain View, CA: CPP, Inc.
- Hammer, A. L., & Kummerow, J. M. (1996). *Strong and MBTI® career development guide* (Rev. ed.). Mountain View, CA: CPP, Inc.
- Hammer, A. L., & Macdaid, G. P. (1992a). *MBTI® Career Report manual*. Mountain View, CA: CPP, Inc.
- Hammer, A. L., & Macdaid, G. P. (1992b). *MBTI® Career Report: Form G*. Mountain View, CA: CPP, Inc.
- Hammer, A. L., & Marting, M. S. (1987). *Coping Resources Inventory*. Mountain View, CA: CPP, Inc.
- Hammer, A. L., & Marting, S. (1988). *Manual for the Coping Resources Inventory*. Mountain View, CA: CPP, Inc.
- Hammer, A. L., & Mitchell, W. D. (1996). The distribution of MBTI® types in the U.S. by gender and ethnic group. *Journal of Psychological Type*, 37, 2–15.
- Hammer, A. L., & Schnell, E. R. (2000). *FIRO-B® technical guide*. Sunnyvale, CA: The Myers-Briggs Company.
- Hammer, A. L., & Treeger, M. (1984). The effect of demand characteristics in the testing environment on preference scores of the Myers-Briggs Type Indicator®: Preliminary data from a field study. *Journal of Psychological Type*, 8, 48–51.
- Hammett, P. (2007). *Unbalanced influence: Recognizing and resolving the impact of myth and paradox in executive performance*. Boston, MA: Nicholas Brealey.
- Hanbury, G. L., Sapat, A., & Washington, C. W. (2004, September/October). Know yourself and take charge of your own destiny: The “fit model” of leadership. *Public Administration Review*, 64(5), 566–576.
- Hardy-Jones, J., & Watson, M. M. (1990). Type differences over time: A ten-year study at the University of Hawaii. *Journal of Psychological Type*, 19, 27–34.
- Harker, J. B., & Reynierse, J. H. (1999). Independent observer ratings and the correlates of the EAR subscales with their

- behavioral descriptors. *Journal of Psychological Type*, 49, 5–19.
- Harland, D. J. (2000). *Relationship of Myers-Briggs® personality types and learner participation in face-to-face and asynchronous classroom discussions* (Unpublished doctoral dissertation). Walden University.
- Harmon, L. W., Hansen, J. C., Borgen, F. H., & Hammer, A. L. (1994). *Strong Interest Inventory® applications and technical guide*. Mountain View, CA: CPP, Inc.
- Harrington, R., & Loffredo, D. A. (2001). The relationship between life satisfaction, self-consciousness, and the Myers-Briggs Type Inventory® dimensions. *Journal of Psychology*, 135(4), 439–450.
- Harrington, R., & Loffredo, D. A. (2010). MBTI® personality type and other factors that relate to preference for online versus face-to-face instruction. *Internet and Higher Education*, 13(1/2), 89–95.
- Harrison, D. F. (1984). *The temporal dimensions of Jung's psychological typology: Testing an instructional theory of future studies with middle school students* (Unpublished doctoral dissertation). University of Florida, Gainesville.
- Harrison, D. F., & Lawrence, G. (1985). Psychological type and time orientation: Do middle school students differ in projecting their personal futures? *Journal of Psychological Type*, 9, 10–15.
- Harrison, H. A. (1995). The relationships among Jungian psychological type preferences, problems experienced and perceived problem solving capabilities for African-Americans holding graduate degrees (Doctoral dissertation, Saint Louis University). *Dissertation Abstracts International: Section A. Humanities and Social Sciences*, 55(9), 2721.
- Harvey, R. J. (1996). Reliability and validity. In A. L. Hammer (Ed.), *MBTI® applications: A decade of research on the Myers-Briggs Type Indicator®* (pp. 5–29). Mountain View, CA: CPP, Inc.
- Harvey, R. J., Murry, W. D., & Stamoulis, D. (1995). Unresolved issues in the dimensionality of the Myers-Briggs Type Indicator®. *Educational and Psychological Measurement*, 55, 535–544.
- Hautala, T. M. (2005). The effects of subordinates' personality on appraisals of transformational leadership. *Journal of Leadership and Organizational Studies*, 11(4), 84–92.
- Hautala, T. M. (2006a). The relationship between personality and transformational leadership. *Journal of Management Development*, 25(8), 777–794.
- Hautala, T. M. (2006b, January). *How different personalities experience the discussions between leader and follower in the workplace*. Paper presented at Fifth Psychological Type and Culture—East and West: A Multicultural Research Symposium, Honolulu, HI.
- Healy, C. C. (2000). Interpreting the Myers-Briggs Type Indicator® to help clients in understanding their Strong Interest Inventory®. *Journal of Career Development*, 26(4), 295–308.
- Held, J. S., & Yokomoto, C. F. (1983). Technical report writing: Effects of personality differences in the laboratory. In *ASEE Annual Conference Proceedings* (pp. 197–201).
- Helson, R. (1965). Childhood interest clusters related to creativity in women. *Journal of Consulting Psychology*, 29(4), 352–361.
- Helson, R. (1968). Effect of sibling characteristics and parental values on creative interest and achievement. *Journal of Personality*, 36(4), 589–607.
- Helson, R. (1971). Women mathematicians and the creative personality. *Journal of Consulting and Clinical Psychology*, 36(2), 210–220.
- Helson, R. (1975, October). *Studying typological patterns in "real" contexts: Research styles in women mathematicians*. Paper presented at First National Conference on the Uses of the Myers-Briggs Type Indicator®, Gainesville, FL.
- Helson, R., & Crutchfield, R. S. (1970). Creative types in mathematics. *Journal of Personality*, 38(2), 177–197.
- Herk, N. A., Thompson, R. C., Morris, M. L., & Schaubhut, N. A. (2009). *FIRO Business® technical guide*. Sunnyvale, CA: The Myers-Briggs Company.
- Hermon, M. V. (1993). Communication strategies of psychological types. In *Conscious choices, unconscious forces: Proceedings of the 10th Biennial International Conference of the Association for Psychological Type, July 8–11, 1993, Newport Beach, CA* (pp. 87–92). Gainesville, FL: Center for Applications of Psychological Type.
- Hester, C. (1990). The differential performance of MBTI® types on learning tasks. *Journal of Psychological Type*, 19, 21–26.
- Hicks, L. E. (1984). Conceptual and empirical analysis of some assumptions of an explicitly typological theory. *Journal of Personality and Social Psychology*, 46(5), 1118–1131.
- Hicks, L. E. (1985). Dichotomies and typologies: Summary and implications. *Journal of Psychological Type*, 10, 11–13.
- Hicks, L. E. (1989). Bookishness and the null hypothesis. *Journal of Psychological Type*, 17, 14–19.
- Hill, O. W., & Clark, J. L. (1993). The personality typology of black college students: Evidence for a characteristic cognitive style? *Psychological Reports*, 72, 1091–1097.
- Hinkle, K. S. (1986). An investigation of the relationships among learning style preferences, personality types, and mathematics anxiety of college students (Doctoral dissertation, University of Maryland). *Dissertation Abstracts International: Section A. Humanities and Social Sciences*, 47(7), 2437.
- Hirsh, E., Hirsh, K. W., & Hirsh, S. K. (2009). *MBTI® teambuilding program: Leader's resource guide* (2nd ed.). Sunnyvale, CA: The Myers-Briggs Company.
- Hockersmith, P. E. (1986). A study of the relationships among personality, learning and thinking variables of student teachers at Shippensburg University (Doctoral

- dissertation, Temple University). *Dissertation Abstracts International: Section A. Humanities and Social Sciences*, 47(8), 3005.
- Hofstede, G. (2003). *Culture's consequences: International differences in work-related values*. Newbury Park, CA: Sage.
- Hofstede, G. (2004). *Cultures and organizations: Software of the mind*. London, England: McGraw-Hill.
- Holland, C. L., & VanBenshoten, S. (1991, October). *Coping styles and the TDI*. Paper presented at the Conference on Advanced Applications for MBTI® Technology, Princeton, NJ.
- Holsworth, T. E. (1985). Perceptual style correlates for the MBTI®. *Journal of Psychological Type*, 10, 32–35.
- Holtzman, F. I. (1989). An examination of the relationship between hemispheric dominance and learning styles as described by Keirseyan temperament types (Doctoral dissertation, University of Tennessee). *Dissertation Abstracts International: Section A. Humanities and Social Sciences*, 50(4), 853.
- Hopkins, L. G. (1997). Relationships between dimensions of personality and job satisfaction. In *Proceedings: APT XII Biennial International Conference* (pp. 83–86). Gainesville, FL: Center for Applications of Psychological Type.
- Horacek, T. M., & Betts, N. M. (1998). College students' dietary intake and quality according to their Myers-Briggs Type Indicator® personality preferences. *Journal of Nutrition Education*, 30(6), 387–395.
- Hough, J. R., & Ogilvie, D. T. (2005, March). An empirical test of cognitive style and strategic decision outcomes. *Journal of Management Studies*, 42(2), 417–448.
- Houtz, J. C., Selby, E., Esquivel, G. B., Okoye, R. A., Peters, K. M., & Treffinger, D. J. (2003). Creativity styles and personal type. *Creativity Research Journal*, 15(4), 321–330.
- Howes, R. J., & Carskadon, T. G. (1979). Test-retest reliabilities of the Myers-Briggs Type Indicator® as a function of mood changes. *Research in Psychological Type*, 2, 67–72.
- Hu, L., & Bentler, P. M. (1999). Cutoff criteria for fit indexes in covariance structure analysis: Conventional criteria versus new alternatives. *Structural Equation Modeling*, 6(1), 1–55.
- Huszczo, G. E. (2013, July). *What does type have to do with team, leadership and change behaviors?* Paper presented at the XX Biennial Meeting of the Association for Psychological Type International (APTi), Miami, FL.
- Huszczo, G. E., & Bush, J. E. (2015). *Influence of personality type and culture on ways to communicate plans for change*. Paper presented at the Global Advances in Business Communications (GABC) 7th Annual Tricontinental Conference, Ypsilanti, MI.
- Huszczo, G. E., & Claus, V. A. (2000, August). *Does type influence your approach to change?* Paper presented at the XVIII Biennial Meeting of the Association for Psychological Type International (APTi), Dallas, TX.
- Huszczo, G. E., & Endres, M. (2013). Joint effects of gender and personality on choice of happiness strategies. *Europe's Journal of Psychology*, 9(1), 136–149.
- Inscape Publishing, Inc. (2005). DiSC® Validation Research Report. Retrieved April 7, 2010, www.resourcesunlimited.com/images/DiSC_Validation_Research_Report.pdf
- Insko, C. A., Schopler, J., Gaertner, L., Wildschut, T., Kozar, R., Pinter, B., . . . , & Montoya, M. R. (2001). Interindividual-intergroup discontinuity reduction through the anticipation of future interaction. *Journal of Personality and Social Psychology*, 80(1), 95–111.
- International Test Commission. (2005). ITC guidelines for translating and adapting tests. Retrieved from www.intestcom.org/page/16
- Ireland, M. S., & Kernan-Schloss, L. (1983). Pattern analysis of recorded daydreams, memories, and personality type. *Perceptual and Motor Skills*, 56(1), 119–125.
- Jackson, G. S. (1985). The impact of roommates on development: A causal analysis of the effects of roommate personality congruence, satisfaction and initial developmental status on end-of-quarter developmental status and grade point average (Doctoral dissertation, Auburn University). *Dissertation Abstracts International: Section A. Humanities and Social Sciences*, 46(2), 360.
- Jacobson, C. M. (1993). Cognitive styles of creativity: Relations of scores on the Kirton Adaption and Innovation Inventory and the Myers-Briggs Type Indicator® among managers in USA. *Psychological Reports*, 72, 1131–1138.
- Jaffe, J. M. (1980). The relationship of Jungian psychological predispositions to the implementation of management by objectives: A sociotechnical perspective (Doctoral dissertation, University of Southern California). *Dissertation Abstracts International: Section A. Humanities and Social Sciences*, 41(11), 4833.
- James, J. D. (1984). *The effect of choice preferences and psychological type differences on group problem solving effectiveness* (Unpublished doctoral dissertation). University of Pittsburgh.
- James, L. R., Mulaik, S. A., & Brett, J. M. (1982). *Causal analysis*. Newbury Park, CA: Sage.
- Järnlström, M., & Tsuzuki, Y. (2008, January). *Career planning, use of career strategies, and satisfaction in work and career: Insight from type theory*. Paper presented at Sixth Psychological Type and Culture—East and West: A Multicultural Research Symposium, Honolulu, HI.
- John, O. P. (1989). Towards a taxonomy of personality descriptors. In D. M. Buss and N. Cantor (Eds.), *Personality psychology: Recent trends and emerging directions* (pp. 261–271). New York, NY: Springer-Verlag.
- John, O. P. (1990). The "Big Five" factor taxonomy: Dimensions of personality in the natural language and questionnaires. In L. A. Pervin (Ed.), *Handbook of personality: Theory and research* (pp. 66–100). New York, NY: Guilford Press.

- Johnson, A. K. (1997). Conflict-handling intentions and the MBTI®: A construct validity study. *Journal of Psychological Type*, 43, 29–39.
- Johnson, C., Zimmerman, A., & Brooker, N. (1994). Implications and applications of type theory on a two-year technical college campus. In *Proceedings: Orchestrating Educational Change in the 90's—The Role of Psychological Type* (pp. 221–234). Gainesville, FL: Center for Applications of Psychological Type.
- Johnson, C. A., & Thompson, R. C. (2013). *Relationships between MBTI® preferences and work behaviors*. Unpublished manuscript. Mountain View, CA: CPP, Inc.
- Johnson, C. A., & Thompson, R. C. (2014). *Relationships between MBTI® preferences and work behaviors using self-ratings and observer ratings*. Unpublished manuscript. Mountain View, CA: CPP, Inc.
- Johnson, D. A. (1992a). Predicting promotion to management in the wholesale grocery industry using the Type Differentiation Indicator. *Journal of Psychological Type*, 23, 51–59.
- Johnson, D. A. (1992b). Test-retest reliabilities of the Myers-Briggs Type Indicator® and the Type Differentiation Indicator over a 30-month period. *Journal of Psychological Type*, 24, 54–58.
- Johnson, D. A., & Golden, J. P. (1994). How psychological type influences effective leadership: The Survey of Management Practices and the Type Differentiation Indicator (TDI). In *Proceedings of The Myers-Briggs Type Indicator® and Leadership: An International Research Conference* (pp. 89–97). College Park, MD: National Leadership Institute, University of Maryland.
- Johnson, J. A., Sample, J. A., & Jones, W. J. (1988). Self-directed learning and personality type in adult degree students. *Psychology: A Journal of Human Behavior*, 25(1), 32–36.
- Johnson, W. M. (1990). A comparative analysis of learning styles of black and white college freshmen (Doctoral dissertation, Oklahoma State University). *Dissertation Abstracts International: Section A. Humanities and Social Sciences*, 50(12), 3863.
- Jöreskog, K. G., & Sörbom, D. (1981). *LISREL V: Analysis of linear structural relationships by maximum likelihood and least squares methods*. Chicago, IL: International Educational Services.
- Jung, C. G. (1921/1971). *Psychological types*. Princeton, NJ: Princeton University Press.
- Kagan, J., & Snidman, N. (1991). Infant predictors of inhibited and uninhibited profiles. *Psychological Science*, 2(1), 40–43.
- Kahn, J. H., Nauta, M. M., Gailbreath, R. D., Tipps, J., & Chartrand, J. M. (2002). The utility of career and personality assessment in predicting academic progress. *Journal of Career Assessment*, 10(1), 3–23.
- Kahneman, D. (2011). *Thinking fast and slow*. New York, NY: Farrar, Straus and Giroux.
- Kainz, R. I., & McCaulley, M. H. (1975). [Type differences in selection to medical school]. Unpublished raw data.
- Kalsbeek, D. (2003). Campus retention: The MBTI® in institutional self-studies. In J. A. Provost & S. Anchors (Eds.), *Using the MBTI® instrument in colleges and universities* (pp. 87–122). Gainesville, FL: Center for Applications of Psychological Type.
- Kalsbeek, D., Rodgers, R., Marshall, D., Denny, D., & Nicholls, G. (1982). Balancing challenge and support: A study of degrees of similarity in suitemate personality type and perceived difference in challenge and support in a residence hall environment. *Journal of College Student Personnel*, 23, 434–442.
- Kameda, D. M., & Nyland, J. L. (2003). Relationship between psychological type and sensitivity to anxiety. *Perceptual and Motor Skills*, 97(3, Pt. 1), 789–793.
- Kankaras, M., Moors, G., & Vermunt, J. K. (2010). Testing for measurement invariance with latent class analysis. In E. Davidov, P. Schmidt, & J. Billiet (Eds.), *Cross-cultural analysis: Methods and applications* (pp. 359–384). New York, NY: Routledge.
- Katira, N., Williams, L., Wiebe, E., Miller, C., Balik, S., & Gehringer, E. (2004). On understanding compatibility of student pair programmers. In *Proceedings of the 35th SIGCSE Technical Symposium on Computer Science Education* (pp. 7–11). New York, NY: ACM.
- Katz, L., Joyner, J. W., & Seaman, N. (1999). Effects of joint interpretation of the Strong Interest Inventory® and the Myers-Briggs Type Indicator® in career choice. *Journal of Career Assessment*, 7(3), 281–297.
- Keirsey, D., & Bates, M. (1978). *Please understand me: An essay on temperament styles*. Del Mar, CA: Promethean Books.
- Kench, B. T., Beekman, R. L., Wynn, G. A., & Niman, N. B. (2010). The influence of psychological type and interpersonal needs in social dilemmas: A public goods experiment with punishment. *Journal of Psychological Type*, 70(1), 1–17.
- Kent, H., & Fisher, D. (1997, March). *Associations between teacher personality and classroom environment*. Paper presented at the Annual Meeting of the American Educational Research Association, Chicago, IL.
- Kilmann, R. H., & Thomas, K. W. (1975). Interpersonal conflict-handling behavior as reflections of Jungian personality dimensions. *Psychological Reports*, 37, 971–980.
- Kirby, L. K. (1997). Introduction: Psychological type and the Myers-Briggs Type Indicator®. In C. Fitzgerald & L. K. Kirby (Eds.), *Developing leaders: Research and applications in psychological type and leadership development* (pp. 3–32). Boston, MA: Nicholas Brealey.
- Kirby, L. K., Kendall, E., & Barger, N. J. (2007). *Type and culture: Using the MBTI® instrument in international applications*. Sunnyvale, CA: The Myers-Briggs Company.
- Kirton, M. J. (1987). *Kirton Adaption and Innovation Inventory (KAI) manual* (2nd ed.). Harfield, England: Occupational Research Center.

- Kise, J. A. G. (2005). Coaching teachers for change: Using the concepts of psychological type to reframe teacher resistance. *Journal of Psychological Type*, 65(6), 47–58.
- Kitchens, A. N., & Marsh, E. G. (2007). Using the MBTI® instrument with elementary education majors in a learning community linking college math course and freshman seminar. *Journal of Psychological Type*, 67(1), 1–10.
- Knapp, R. H., & Lapuc, P. S. (1965). Time imagery, introversion and fantasied preoccupations in simulated isolation. *Perceptual and Motor Skills*, 20, 327–330.
- Komisin, L. K. (1992). Personality type and suicidal behaviors in college students. *Journal of Psychological Type*, 24, 24–32.
- Krause, N. A., & Thompson, R. C. (2008, August). *The efficacy of electronic administration and interpretation of personality measures*. Paper presented at the American Psychological Association 116th Annual Convention, Boston, MA.
- Krumboltz, J. D. (1991). *Career Beliefs Inventory manual* (Rev. ed.). Mountain View, CA: CPP, Inc.
- Krumboltz, J. D. (1992). Comment: The wisdom of indecision. *Journal of Vocational Behavior*, 41, 239–244.
- Kummerow, J. M. (1986). *Verifying your type preferences*. Gainesville, FL: Center for Applications of Psychological Type.
- Kummerow, J. M. (1988). A methodology for verifying type: Research results. *Journal of Psychological Type*, 15, 20–25.
- Kummerow, J. M., & Quenk, N. L. (1992a). *Interpretive guide for the MBTI® Expanded Analysis Report*. Mountain View, CA: CPP, Inc.
- Kummerow, J. M., & Quenk, N. L. (1992b). *Workbook for the MBTI® Expanded Analysis Report*. Mountain View, CA: CPP, Inc.
- Kummerow, J. M., & Quenk, N. L. (2018). *Understanding your MBTI® Step II® results* (2nd ed.). Sunnyvale, CA: The Myers-Briggs Company.
- LaBarbera, P. A., Weingard, P., & Yorkston, E. A. (1998). Matching the message to the mind: Advertising imagery and consumer processing styles. *Journal of Advertising Research*, September–October, pp. 29–43.
- Lamphere, G. I. (1985). The relationship between teacher and student personality and its effects on teacher perception of student (Doctoral dissertation, United States International University). *Dissertation Abstracts International: Section A. Humanities and Social Sciences*, 46(6), 1564.
- Lawrence, E. T. (2007). *A study of Myers-Briggs® psychological type and change style among hospital executives: Implications for leading organizational change in hospitals* (Unpublished doctoral dissertation). Colorado School of Professional Psychology, Colorado Springs.
- Lawrence, G. (1984). A synthesis of learning style research involving the MBTI®. *Journal of Psychological Type*, 8, 2–15.
- Lawrence, G. (1993). *People types and tiger stripes*. Gainesville, FL: Center for Applications of Psychological Type.
- Lawrence, G. (1998). *Descriptions of the sixteen types*. Gainesville, FL: Center for Applications of Psychological Type.
- Lawrence, G., & Martin, C. R. (2005). Profile of your MBTI® results. [Handout]. Gainesville, FL: Center for Applications of Psychological Type.
- Leary, M. M., Reilly, M. D., & Brown, F. W. (2009). A study of personality preferences and emotional intelligence. *Leadership and Organization Development Journal*, 30(5), 421–434.
- Lee, K. M., & Nass, C. (n.d.). Social-psychological origins of feelings of presence: Creating social presence with machine-generated voices. Research Note, Department of Communication. Stanford University.
- Leong, F. T. L., Hardin, E. E., & Gaylor, M. (2005). Career specialty choice: A combined research-intervention project. *Journal of Vocational Behavior*, 67, 69–86.
- Levine, D. M. (1988). Jungian personality type and recreational reading patterns (Doctoral dissertation, University of North Carolina at Chapel Hill). *Dissertation Abstracts International: Section B. Sciences and Engineering*, 58(12), 3716.
- Li, S., & Liu, C. J. (2008, July). Individual differences in a switch from risk-averse preferences for gains to risk-seeking preferences for losses: Can personality variables predict the risk preferences? *Journal of Risk Research*, 11(5), 673–686.
- Liddell, D. L., Halpin, G., & Halpin, W. G. (1992). The measure of moral orientation: Measuring the ethics of care and justice. *Journal of College Student Development*, 33(4), 325–330.
- Loffredo, D. A., & Opt, S. K. (2006). Argumentativeness and the Myers-Briggs Type Indicator® preferences. *Journal of Psychological Type*, 66(7), 59–68.
- Loffredo, D. A., Opt, S. K., & Harrington, R. (2008). Communicator style and MBTI® Extraversion–Introversion domains. *Journal of Psychological Type*, 68(4), 29–36.
- Lofquist, L. H., & Dawis, R. V. (1969). *Adjustment to work*. New York, NY: Appleton-Century-Crofts.
- Lombardo, M. M., & McCauley, C. D. (1994). *Benchmarks®: A manual and trainer's guide*. Greensboro, NC: Center for Creative Leadership.
- Lu, H.-P., & Hsiao, K.-L. (2010). The influence of extro/introversion on the intention to pay for social networking sites. *Information & Management*, 47, 150–157.
- Ludford, P. J., & Terveen, L. G. (2003, September). Does an individual's Myers-Briggs Type Indicator® preference influence task-oriented technology use? Paper presented at Interact 2003—Ninth IFIP TC13 International Conference on Human-Computer Interaction, Zurich, Switzerland.

- Lueder, D. L. (1986a). Psychological types of the 100 top executive educators in North America. *Journal of Psychological Type*, 12, 8–12.
- Lueder, D. L. (1986b). The “rising stars” in educational administration: A corollary to psychological types and leadership styles. *Journal of Psychological Type*, 12, 13–15.
- Luh, S. P. (1991). A study of learning styles, personality types and brain hemispheric preferences of teacher education majors (Doctoral dissertation, Drake University). *Dissertation Abstracts International: Section A. Humanities and Social Sciences*, 51(12), 4067.
- Lynch, A. Q., & Sellers, P. A. (1996). Preferences for different educational environments and psychological type: A comparison of adult learners and traditional age college students. *Journal of Psychological Type*, 39, 18–29.
- Macdavid, G. P., Kainz, R. I., & McCaulley, M. H. (1984). *The University of Florida longitudinal study: A ten-year follow-up*. Unpublished paper.
- MacKinnon, D. W. (1960). The highly effective individual. *Teachers College Record*, 61, 367–378.
- MacKinnon, D. W. (1962a). The nature and nurture of creative talent. *American Psychologist*, 17, 484–495.
- MacKinnon, D. W. (1962b). The personality correlates of creativity: A study of American architects. In G. S. Nielsen (Ed.), *Personality research: Proceedings of the XIV International Congress of Applied Psychology*, 2, 11–39.
- MacKinnon, D. W. (1965). Personality and the realization of creative potential. *American Psychologist*, 20, 273–281.
- MacKinnon, D. W. (1971). *Creativity and the transliminal experience*. *Journal of Creative Behavior*, 5(4), 227–241.
- Magidson, J. (1997). Type vs. trait debate rekindles. *Bulletin of Psychological Type*, 20(4).
- Magidson, J. (2014, March). Advantages of the latent class approach. In R. C. Thompson (Chair), *Multiple approaches to building a personality assessment using a multinational data set*. Symposium conducted at the ATP 2014 Innovations in Testing Conference, Scottsdale, AZ.
- Magidson, J. (2015, May). *Using a scale-adjusted latent class model to establish measurement equivalence in cross-cultural surveys: An application with the Myers-Briggs Type Indicator® (MBTI®)*. Paper presented at the Modern Modeling Methods (M3) Conference, University of Connecticut, Storrs.
- Magidson, J., & Vermunt, J. K. (2001). Latent class factor and cluster models, bi-plots, and related graphical displays. *Sociological Methodology*, 31(1), 223–264.
- Magidson, J., & Vermunt, J. K. (2004). Latent class models. In D. Kaplan (Ed.), *The Sage handbook of quantitative methodology for the social sciences* (pp. 175–198). Thousand Oaks, CA: Sage.
- Mann, H., Siegler, M., & Osmond, H. (1968). The many worlds of time. *Journal of Analytical Psychology*, 13(1), 33–56.
- Mann, H., Siegler, M., & Osmond, H. (1971). The psychotypology of time. In H. Yaker, H. Osmond, & F. Cheek (Eds.), *The future of time: Man's temporal environment* (pp. 142–178). Garden City, NY: Doubleday.
- Manske, D. H. (1988). A study of the relationship of preference scores on the Myers-Briggs Type Indicator® and the reading comprehension of adult readers (Doctoral dissertation, Memphis State University). *Dissertation Abstracts International: Section A. Humanities and Social Sciences*, 49(2), 227.
- Marcus, S. K. (1976). Jungian typology and time orientation (Doctoral dissertation, United States International University). *Dissertation Abstracts International: Section B. Sciences and Engineering*, 37(3), 1409.
- Marioles, N. S., Strickert, D. P., & Hammer, A. L. (1996). Attraction, satisfaction, and psychological types of couples. *Journal of Psychological Type*, 36, 16–27.
- Martin, B. F. (2011). *The relationship of personality preferences and type to health-promoting behaviors, alcohol use, and cigarette smoking* (Unpublished doctoral dissertation). University of Kentucky, Lexington.
- Maslach, C., Jackson, S. E., & Leiter, M. P. (1996). *Maslach Burnout Inventory manual* (3rd ed.). Mountain View, CA: CPP, Inc.
- Mathew, P., & Bhatewara, S. (2006). Personality differences and preferred styles of conflict management among managers. *Abhigyan*, 23, 38–45.
- May, A. O. (1992). The learning styles, personality and temperament types of eighth- and twelfth-grade urban African-American and white students: A comparative study (Doctoral dissertation, Seattle University). *Dissertation Abstracts International: Section A. Humanities and Social Sciences*, 52(7), 2472.
- Maybee, R. G., & Carter, J. L. (1999). *The relationship of psychological type to ten indices of perceived stress*. Paper presented at Mind, Body & Personality: The Role of Type in Mental and Physical Health. The Second Biennial Clinical Conference, Center for Applications of Psychological Type, Gainesville, FL.
- McBride, M. H. (1988). *The theory of psychological type congruence for advertisers revisited*. Paper presented at 4th Annual Conference of the Atlantic Marketing Association.
- McCary, P. W. (1970). The effects of small self-understanding groups on the self-concept and anxiety level when group composition has been varied (Doctoral dissertation, University of Michigan). *Dissertation Abstracts International: Section A. Humanities and Social Sciences*, 31, 2112. (University Microfilms No. 70-20, 491)
- McCaulley, M. H. (1977). *The Myers longitudinal medical study* (Monograph II). Gainesville, FL: Center for Applications of Psychological Type.
- McCaulley, M. H. (1985). The selection ratio type table: A research strategy for comparing type distributions. *Journal of Psychological Type*, 10, 46–56.
- McCaulley, M. H. (1990). The MBTI® and individual pathways in engineering design. *Engineering Education*, 80, 537–542.
- McCaulley, M. H., Macdavid, G. P., & Magidson, J. (1997). *Introducing a new graphical modeling technique for profiling type tables*. Paper presented at the 12th

- International Conference of the Association for Psychological Type, Boston, MA.
- McCaulley, M. H., & Natter, F. L. (1974). Psychological (Myers-Briggs®) type differences in education. In F. L. Natter & S. A. Rollins (Eds.), *The governor's task force on disruptive youth: Phase II report*. Tallahassee, FL: Office of the Governor.
- McCrae, R. R., & Costa, P. T., Jr. (1989). Reinterpreting the Myers-Briggs Type Indicator® from the perspective of the five-factor model of personality. *Journal of Personality*, 57(1), 17–40.
- McCrae, R. R., & Costa, P. T., Jr. (2010). *NEO™ inventories professional manual*. Lutz, FL: PAR.
- McGuigan, R. L. (1999). Determining leisure preference, personality or demographics? In B. Dubois, T. M. Lowrey, L. J. Shrum, & M. Vanhuele (Eds.), *European Advances in Consumer Research*, 4, 224–231.
- McPeck, R. W., & Breiner, J. (2013). Convergent validity of the MBTI® and MMTIC® assessments. *Journal of Psychological Type*, 73(2), 7–20.
- McPeck, R. W., & Breiner, J. (2014). Implicit, explicit, and best-fit assessment of psychological type: Explorations in “shoes off” measurement. *Journal of Psychological Type*, 74(2), 11–28.
- McPeck, R. W., Breiner, J., Murphy, E., Brock, C., Grossman, L., Loeb, M., & Tallevi, L. (2013). Student type, teacher type, and type training: CAPT type and education research 2008–2011 project summary. *Journal of Psychological Type*, 73(3), 21–54.
- McPeck, R. W., & Martin, C. R. (2012). Searching for expert consensus in describing function-attitudes. *Journal of Psychological Type*, 73(3), 41–64.
- McPeck, R. W., Urquhart, C., Breiner, J. F., Holland, D. F., & Cavalleri, D. (2011). The impact on student academic performance and attitudes of psychological type and its introduction to the classroom. *Journal of Psychological Type*, 71(3), 54–71.
- Melear, C. T., & Richardson, S. (1994). Learning styles of African American children which correspond to the MBTI®. In *Proceedings: Orchestrating Educational Change in the 90's—The Role of Psychological Type* (pp. 11–22). Gainesville, FL: Center for Applications of Psychological Type.
- Miles, R. E., & Snow, C. C. (1978). *Organizational strategy, structure, and process*. New York, NY: McGraw-Hill.
- Mills, J., Robey, D., & Smith, L. (1985). Conflict-handling and personality dimensions of project-management personnel. *Psychological Reports*, 57, 1135–1143.
- Mitchell, W. D. (1981). *Type and social climate in a large organization*. Unpublished manuscript.
- Mitchell, W. D. (1991). A test of type theory using the TDI. *Journal of Psychological Type*, 22, 15–26.
- Mitchell, W. D. (1993). Are types real? In *Conscious choices, unconscious forces: Proceedings of the 10th Biennial International Conference of the Association for Psychological Type, July 8–11, 1993, Newport Beach, CA*. Gainesville, FL: Center for Applications of Psychological Type.
- Mitchell, W. D. (2001). A full dynamic model of type. *Journal of Psychological Type*, 59, 12–28.
- Mitchell, W. D. (2006). Validation of the full dynamic model of type. *Journal of Psychological Type*, 66(5), 35–48.
- Mitchell, W. D., Quenk, N. L., & Kummerow, J. M. (1997). *MBTI® Step II®: A description of the subscales*. Mountain View, CA: CPP, Inc.
- Moody, R. (1988). Personality preferences and foreign language learning. *Modern Language Journal*, 72(4), 389–401.
- Moore, D., & Bayne, R. (1997). Preference for Sensing or Intuition and reading for pleasure. *Journal of Psychological Type*, 41, 38–42.
- Morris, M. L. (2012, July). The future of the MBTI® assessment: A new scoring approach based on latent class analysis. In R. T. Thompson, (Chair), *The present and future of the MBTI® assessment across the globe*. Symposium conducted at the 30th International Congress of Psychology, Cape Town, South Africa.
- Morris, M. L., & Cooley, T. C. (2015, May). Scalar equivalence of the MBTI® assessment across 18 countries. In R. Thompson (Chair), *Type and culture symposium*, conducted at the meeting of JvR People Development in Africa Conference, Kruger National Park, South Africa.
- Munro, N., Chilimanzi, Y., & O'Neill, V. (2012). Character strengths and psychological type in university peer educators. *South African Journal of Psychology*, 42(1), 15–24.
- Murphy, E., & Meisgeier, C. H. (2008). *MMTIC® manual: A guide to the development and use of the Murphy-Meisgeier Type Indicator for Children®* (2nd ed.). Gainesville, FL: Center for Applications of Psychological Type.
- Musselwhite, W. C., & Ingram, R. P. (2003). *Change Style Indicator facilitator guide*. Greensboro, NC: Discovery Learning Press.
- Myers, I. B. (1962a). *Introduction to Type®* (1st ed.). Swarthmore, PA: Author.
- Myers, I. B. (1962b). *Manual: The Myers-Briggs Type Indicator®*. Princeton, NJ: Educational Testing Service.
- Myers, I. B. (1980). *Introduction to Type®* (3rd ed.). Palo Alto, CA: Consulting Psychologists Press.
- Myers, I. B. (1987). *Introduction to Type®* (4th ed.). Palo Alto, CA: Consulting Psychologists Press.
- Myers, I. B. (2015). *Introduction to Myers-Briggs® Type* (7th ed.). Sunnyvale, CA: The Myers-Briggs Company.
- Myers, I. B., & Briggs, K. C. (1996). *MBTI® Step II® Expanded Profile*. Mountain View, CA: CPP, Inc.
- Myers, I. B., & McCaulley, M. H. (1985). *Manual: A guide to the development and use of the Myers-Briggs Type Indicator®* (2nd ed.). Mountain View, CA: CPP, Inc.
- Myers, I. B., McCaulley, M. H., Quenk, N. L., & Hammer, A. L. (1998). *MBTI® manual: A guide to the development and use of the Myers-Briggs Type Indicator® Instrument* (3rd ed.). Mountain View, CA: CPP, Inc.

- Myers, I. B., McCaulley, M. H., Quenk, N. L., Hammer, A. L., & Mitchell, W. D. (2009). *MBTI® Step III™ manual*. Mountain View, CA: CPP, Inc.
- Myers, I. B., with Kirby, L. K., & Myers, K. D. (1998). *Introduction to Type®* (6th ed.). Mountain View, CA: CPP, Inc.
- Myers, I. B., with Myers, P. B. (1980/1995). *Gifts differing: Understanding personality type*. Boston, MA: Nicholas Brealey.
- Myers, K. D., & Kirby L. K. (1994). *Introduction to Type® dynamics and development: Exploring the next level of type*. Sunnyvale, CA: The Myers-Briggs Company.
- Nelson, P. A., Thorne, A., & Shapiro, L. A. (2011). I'm outgoing, she's reserved: The reciprocal dynamics of personality in close friendships in young adulthood. *Journal of Personality*, 79(5), 1113–1147.
- Nevill, D. D., & Super, D. E. (1986). *The Values Scale*. Palo Alto, CA: Consulting Psychologists Press.
- Newman, J. (1985). Hemisphere specialization and Jungian typology: Evidence for a relationship. *Bulletin of Psychological Type*, 10(2), 13, 25–27.
- Newman, L. E. (1979). Personality types of therapist and client and their use in counseling. *Research in Psychological Type*, 2, 46–55.
- Nightingale, J. A. (1973). The relationship of Jungian type to death concern and time perspective (Doctoral dissertation, University of South Carolina). *Dissertation Abstracts International: Section B. Sciences and Engineering*, 33, 3956. (University Microfilms No. 73-3609)
- Norris, C., Meisgeier, C., & Hermond, D. (1998). Values and personality style of future educational leaders. In *Proceedings of the Third Biennial International Conference on Education of the Center for Applications of Psychological Type* (pp. 19–26). Gainesville, FL: Center for Applications of Psychological Type.
- Nyland, J. L., Ybarra, K. M., Sammut, K. L., Rienecker, E. M., & Kameda, D. M. (2000). Interaction of psychological type and anxiety sensitivity on academic achievement. *Perceptual and Motor Skills*, 90(3, Pt. 1), 731–739.
- O'Haire, T. D., & Marcia, J. E. (1980). Some personality characteristics associated with Ananda Marga meditations: A pilot study. *Perceptual and Motor Skills*, 51, 447–452.
- OPP Ltd. (2011). *MBTI® Step I® instrument: European data supplement*. Oxford, England: Author.
- Opt, S. K., & Loffredo, D. A. (2000). Rethinking communication apprehension: A Myers-Briggs® perspective. *Journal of Psychology*, 124, 556–570.
- Opt, S. K., & Loffredo, D. A. (2003). Communicator image and Myers-Briggs Type Indicator® Extraversion–Introversion. *Journal of Psychology*, 137(6), 560–568.
- Opt, S. K., & Loffredo, D. A. (2006). Receiver apprehension and MBTI® preferences. *Journal of Psychological Type*, 66(3), 15–22.
- O'Shea, T., & Mamchur, C. (1989). Using a loglinear model to analyze MBTI® distributions. *Journal of Psychological Type*, 17, 56–65.
- Otis, G. D., & Louks, J. L. (1997). Rebelliousness and psychological distress in a sample of introverted veterans. *Journal of Psychological Type*, 40, 20–30.
- Owen, M. B., & Liles, R. T. (1999). Relationship of the Myers-Briggs Type Indicator® to internet use by educators. In *Proceedings of APT-XIII, the Thirteenth Biennial International Conference of the Association for Psychological Type* (pp. 65–70). Gainesville, FL: Center for Applications of Psychological Type.
- Oxford, R., & Ehrman, M. (1988). Psychological type and adult language learning strategies: A pilot study. *Journal of Psychological Type*, 16, 22–32.
- Oxford, R., Nyikos, M., & Ehrman, M. (1988). Vive la difference? Reflections on sex differences in use of language learning strategies. *Foreign Language Annals*, 21(4), 321–329.
- Palmiere, L. (1972). Intro-Extra-version as an organizing principle in fantasy production. *Journal of Analytical Psychology*, 17(2), 116–131.
- Pearman, R. R. (1993). *Diversity denied: Type bias in manager selection*. Paper presented at the 11th International Conference of the Association for Psychological Type, Kansas City, MO.
- Pearman, R. R., & Fleenor, J. (1996). Differences in observed and self-reported qualities of psychological types. *Journal of Psychological Type*, 39, 3–17.
- Pencheva, E. S., Vassileva, S. I., & Bozhinva, R. B. (2006, January). *Psychological type and the regulation of individual behavior*. Paper presented at Fifth Psychological Type and Culture—East and West: A Multicultural Research Conference, Honolulu, HI.
- Penn, B. K. (1992). Correlations among learning styles, clinical specialties, and personality types of U.S. Army nurses (Doctoral dissertation, University of Texas at Austin). *Dissertation Abstracts International: Section A. Humanities and Social Sciences*, 53(2), 393.
- Percival, T. Q., Smitheram, V., & Kelly, M. (1992). Myers-Briggs Type Indicator® and conflict-handling intention: An interactive approach. *Journal of Psychological Type*, 23, 10–16.
- Peterson, C., & Seligman, M. E. P. (2004). *Character strengths and virtues*. Oxford, England: Oxford University Press and American Psychological Association.
- Petty, M. L. (1985). A profile of college student leaders relating personality types, learning styles, and leadership approaches (Doctoral dissertation, Florida State University). *Dissertation Abstracts International: Section A. Humanities and Social Sciences*, 46(12), 3621.
- Posey, A. M., Thorne, B. M., & Carskadon, T. G. (1999). Differential validity and comparative type distributions of blacks and whites on the Myers-Briggs Type Indicator®. *Journal of Psychological Type*, 48, 6–21.

- Power, S. J., & Lundsten, L. L. (1997). Studies that compare type theory and left-brain/right-brain theory. *Journal of Psychological Type*, 43, 22–28.
- Provost, J. A. (1985). Type watching and college attrition. *Journal of Psychological Type*, 9, 16–23.
- Provost, J. A. (1991). Tracking freshman difficulties in the class of 1993. *Journal of Psychological Type*, 21, 35–39.
- Provost, J. A., & Anchors, S. (2003). Student involvement and activities. In J. A. Provost & S. Anchors (Eds.), *Using the MBTI® instrument in colleges and universities* (pp. 261–283). Gainesville, FL: Center for Applications of Psychological Type.
- Quast, L. N., & Hansen, T. L. (1994). The relationship between MBTI® Expanded Analysis Report (EAR) scores and leaders' management behaviors. In *Proceedings of The Myers-Briggs Type Indicator® and Leadership: An International Research Conference* (pp. 151–162). College Park, MD: National Leadership Institute, University of Maryland.
- Quenk, N. L. (1966). Fantasy and personal outlook: A study of daydreaming as a function of optimism, pessimism, realism and anxiety (Doctoral dissertation, University of California, Berkeley). *Dissertation Abstracts International: Section B. Sciences and Engineering*, 27, 970. (University Microfilms No. 66-8364)
- Quenk, N. L. (1985). Directionality of the auxiliary function. *Bulletin of Psychological Type*, 8(1), 27–29.
- Quenk, N. L. (1993). *Beside ourselves: Our hidden personality in everyday life*. Boston, MA: Nicholas Brealey.
- Quenk, N. L. (2000). *In the grip: Understanding type, stress, and the inferior function* (2nd ed.). Sunnyvale, CA: The Myers-Briggs Company.
- Quenk, N. L. (2002). *Was that really me? How everyday stress brings out our hidden personality*. Boston, MA: Nicholas Brealey.
- Quenk, N. L. (2004). K. D. Myers, personal communication, 2004.
- Quenk, N. L. (2009a). *Essentials of Myers-Briggs Type Indicator® assessment* (2nd ed.). Hoboken, NJ: Wiley.
- Quenk (2009b). Unpublished data.
- Quenk, N. L. (2012). *MBTI® Stress Management Report: Form M*. Sunnyvale, CA: The Myers-Briggs Company.
- Quenk, N. L., & Albert, M. (1975). *A taxonomy of physician work settings* (Study Report #2 to Bureau of Health Resources Development, Health Resource Administration, Contract No. 1-MI-24197). Albuquerque, NM: University of New Mexico.
- Quenk, N. L., Hammer, A. L., & Majors, M. S. (2001). *MBTI® Step II® manual: Exploring the next level of type with the Myers-Briggs Type Indicator® Form Q*. Sunnyvale, CA: The Myers-Briggs Company.
- Quenk, N. L., & Kummerow, J. M. (1996). *MBTI® Step II® Expanded Interpretive Report: Form K*. Mountain View, CA: CPP, Inc.
- Quenk, N. L., & Kummerow, J. M. (2001). *MBTI® Step II® Interpretive Report: Form Q*. Sunnyvale, CA: The Myers-Briggs Company.
- Quenk, N. L., & Kummerow, J. M. (2019). *MBTI® Step II® user's guide: Practitioner's tool for making the most of global Step II® interpretations* (2nd ed.). Sunnyvale, CA: The Myers-Briggs Company.
- Quenk, N. L., & Quenk, A. T. (1996). Counseling and psychotherapy. In A. L. Hammer (Ed.), *MBTI® applications: A decade of research on the Myers-Briggs Type Indicator®* (pp. 105–122). Mountain View, CA: CPP, Inc.
- Quintero, A. J., Segal, L. S., King, T. S., & Black, K. P. (2009, October). The personal interview: Assessing the potential for personality similarity to bias the selection of orthopaedic residents. *Academic Medicine*, 84(10), 1364–1372.
- Reeder, R. M., & McPeck, R. W. (2011). Using psychological type-related teaching tools to improve reading comprehension. *Journal of Psychological Type*, 71, 46–53.
- Reise, S. P., & Waller, N. G. (1990). Fitting the two-parameter model to personality data. *Applied Psychological Measurement*, 14(1), 45–58.
- Reynierse, J. H. (1997). An MBTI® model of entrepreneurship and bureaucracy: The psychological types of business entrepreneurs compared to business managers and executives. *Journal of Psychological Type*, 40, 3–19.
- Reynierse, J. H. (2009). The case against type dynamics. *Journal of Psychological Type*, 40, 1–21.
- Reynolds, B. C. (1990). *An investigation of the relationship among psychological types and health service team members' evaluations of themselves, their teams, and teammates* (Unpublished doctoral dissertation). University of Minnesota, Minneapolis.
- Rosati, P. (1997). Psychological types of Canadian engineering students. *Journal of Psychological Type*, 41, 33–37.
- Ross, C. F. J., & Francis, L. J. (2010, November/December). The relationship of intrinsic, extrinsic, and quest religious orientations to Jungian psychological type among churchgoers in England and Wales. *Mental Health, Religion and Culture*, 13(7/8), 805–819.
- Ross, C. F. J., Francis, L. J., & Craig, C. L. (2005). Dogmatism, religion, and psychological type. *Pastoral Psychology*, 53(5), 483–497.
- Ross, S. I. (1994). The relationship between cognitive style and academic achievement in African-American high school students (Doctoral dissertation, Fordham University). *Dissertation Abstracts International: Section A. Humanities and Social Sciences*, 55(4), 914.
- Roush, P. L. E. (1989). MBTI® type and voluntary attrition at the United States Naval Academy. *Journal of Psychological Type*, 18, 72–79.
- Routamaa, V., & Heinäsuo, K. (2006, January). *Type and values over generations*. Paper presented at Fifth Psychological Type and Culture—East and West: A Multicultural Research Symposium, Honolulu, HI.
- Ruble, V. E., Mahan, M. P., & Schurr, K. T. (1987). *Multiple uses of the MBTI® within an athletic program*. Paper presented

- at the 4th Annual Conference on Counseling Athletes, Springfield, MA.
- Rytting, M., Ware, R., & Prince, R. A. (1994). The impact of family ownership and CEO type on the character of companies. *Journal of Psychological Type*, 31, 32–40.
- Salter, D. W. (2000). *SETA manual: A users' guide to the Salter Environmental Type Assessment*. Palo Alto, CA: Consulting Psychologists Press.
- Salter, D. W. (2003a). Exploring the “chilly classroom” phenomenon as interactions between psychological and environmental types. *Journal of College Student Development*, 44(1), 110–121.
- Salter, D. W. (2003b). Log-linear techniques for the analysis of categorical data: A demonstration with the Myers-Briggs Type Indicator®. *Measurement and Evaluation in Counseling and Development*, 36, 106–121.
- Salter, D. W. (2006). Testing the attraction-selection-attrition model of organizational functioning: The personality of the professoriate. *Journal of Psychological Type*, 66(10), 88–97.
- Salter, D. W., Evans, N. J., & Forney, D. S. (1997). Test-retest of the Myers-Briggs Type Indicator®: An examination of dominant functioning. *Educational and Psychological Measurement*, 57, 590–597.
- Salter, D. W., Forney, D. S., & Evans, N. J. (2005). Two approaches to examining the stability of Myers-Briggs Type Indicator® scores. *Measurement and Evaluation in Counseling and Development*, 37(4), 208–219.
- Satava, D. (1997). Extraverts or Introverts: Who supervises the most CPA staff members? *Journal of Psychological Type*, 43, 40–43.
- Saunders, D. R. (1987). *Type Differentiation Indicator manual*. Mountain View, CA: CPP, Inc.
- Saunders, D. R. (1989). *MBTI® Expanded Analysis Report manual*. Mountain View, CA: CPP, Inc.
- Saunders, F. W. (1991). *Katharine and Isabel: Mother's light, daughter's journey*. Boston, MA: Nicholas Brealey.
- Schadt, D. O. (1997). The relationship of type to developmental issues of midlife women: Implications for counseling. *Journal of Psychological Type*, 43, 12–21.
- Schaefer, G. L. (1994). Relationship of Myers-Briggs Type Indicator® personality profiles to academic self-esteem (Doctoral dissertation, University of the Pacific). *Dissertation Abstracts International: Section A. Humanities and Social Sciences*, 55(5), 1228.
- Schaubhut, N. A. (2013a, July). ROI/validation Van Buren Freshman Academy. In R. C. Thompson (Chair), *CPP research on the validity of the MBTI®: Countering persistent complaints*. Presented at the XX Biennial Meeting of the Association for Psychological Type International (APTi), Miami, FL.
- Schaubhut, N. A. (2013b, September). Self-awareness of type. In R. C. Thompson (Chair), *Seeing MBTI® type: Awareness, expression, and cross-cultural evaluations*. Presented at Global MBTI® Users Conference, Shanghai, China.
- Schaubhut, N. A., Herk, N. A., & Thompson, R. C. (2009). *MBTI® Form M manual supplement*. Sunnyvale, CA: The Myers-Briggs Company.
- Schaubhut, N. A., Long, J. S., & Thompson, R. C. (2011, August). *What is your money type?* Poster session presented at the APTi XVIII Biennial Conference, San Francisco, CA.
- Schaubhut, N. A., & Thompson, R. C. (2011a). *MBTI® Step II® manual supplement*. Sunnyvale, CA: The Myers-Briggs Company.
- Schaubhut, N. A., & Thompson, R. C. (2011b). *MBTI® type tables for college majors*. Sunnyvale, CA: The Myers-Briggs Company.
- Schaubhut, N. A., & Thompson, R. C. (2012). *MBTI® type tables for occupations* (2nd ed.). Sunnyvale, CA: The Myers-Briggs Company.
- Schaubhut, N. A., Thompson, R. C., Morris, M. L., & Arneson, J. A. (2018). *MBTI® Manual Global Supplement Series*. Sunnyvale, CA: The Myers-Briggs Company.
- Schaubhut, N., Weber, A., & Thompson, R. (2011). *Myers-Briggs® type and social media report* (Unpublished manuscript). Mountain View, CA: CPP, Inc.
- Scherdin, M. J. (1994). Vive la difference: Exploring librarian personality types using the MBTI®. In M. J. Scherdin (Ed.), *Discovering librarians: Profiles of a profession*. Chicago: Association of College and Research Libraries, American Library Association.
- Schmitt, P. M., Shupp, R., Swope, K. J., & Mayer, J. (2008, June). Pre-commitment and personality: Behavioral explanations in ultimatum games. *Journal of Economic Behavior & Organization*, 66(3–4), 597–605.
- Schneider, B. (1987). The people make the place. *Personnel Psychology*, 40(3), 437–453.
- Schneider, B., Goldstein, H. W., & Smith, D. B. (1995). The ASA framework: An update. *Personnel Psychology*, 48, 747–773.
- Schneider, B., Smith, D. B., Taylor, S., & Fleenor, J. (1998). Personality and organizations: A test of the homogeneity of personality hypothesis. *Journal of Applied Psychology*, 83(3), 462–470.
- Schroeder, C., Warner, R., & Malone, D. (1980). Effects of assignment in living units by personality types on environmental perceptions and student development. *Journal of College Student Personnel*, 21(15), 443–449.
- Schroeder, C. C., & Jackson, S. (2003). Designing residential environments. In J. A. Provost & S. Anchors (Eds.), *Using the MBTI® instrument in colleges and universities* (pp. 285–319). Gainesville, FL: Center for Applications of Psychological Type.
- Schullery, N. M., Knudstrup, P., Schullery, S. E., & Pfaff, L. A. (2009). The relationship between personality type and 360-degree evaluation of management skills. *Journal of Psychological Type*, 69(11), 141–155.
- Schullery, N. M., & Schullery, S. E. (2006). Are heterogeneous or homogeneous groups more beneficial to students? *Journal of Management Education*, 30(4), 542–556.

- Schurr, K. T., Henriksen, L. W., Alcorn, B. K., & Dillard, N. (1992). Tests and psychological types for nurses and teachers: Classroom achievement and standardized test scores measuring specific training objectives and general ability. *Journal of Psychological Type*, 23, 38–44.
- Schurr, K. T., Houlette, F., & Ellen, A. (1986). The effects of instructors and student Myers-Briggs Type Indicator® characteristics on the accuracy of grades predicted for an introductory English composition course. *Educational and Psychological Measurement*, 46(4), 989–1000.
- Schurr, K. T., Ruble, V. E., & Henriksen, L. W. (1989). Effects of different university admission practices on the MBTI® and gender composition of a student body, graduation rate, and enrollment in different departments. *Journal of Psychological Type*, 18, 24–32.
- Schurr, K. T., Ruble, V. E., Palomba, C., Pickerill, B., & Moore, D. (1997). Relationships between the MBTI® and selected aspects of Tinto's model for college attrition. *Journal of Psychological Type*, 40, 31–42.
- Schutz, W. (1978). *FIRO® awareness scales manual*. Mountain View, CA: CPP, Inc.
- Sedlock, J. R. (2005). An exploratory study of the validity of the MBTI® Team Report. *Journal of Psychological Type*, 65(1), 1–8.
- Seiden, H. M. (1970). Time perspective and styles of consciousness (Doctoral dissertation, New School for Social Research). *Dissertation Abstracts International: Section B. Sciences and Engineering*, 31, 386. (University Microfilms No. 70-11, 275)
- Severino, C. J. (1989). Helping nontraditional students achieve college literacy: A context-based study of the uses of text and teacher-student interaction (Doctoral dissertation, University of Illinois at Chicago). *Dissertation Abstracts International: Section A. Humanities and Social Sciences*, 50(6), 1585.
- Shelton, J. (1996). Health, stress, and coping. In A. L. Hammer (Ed.), *MBTI® applications: A decade of research on the Myers-Briggs Type Indicator®* (pp. 197–215). Mountain View, CA: CPP, Inc.
- Shiflett, S. C. (1989). Validity evidence for the Myers-Briggs Type Indicator® as a measure of hemispheric dominance. *Educational and Psychological Measurement*, 49(3), 741–745.
- Short, G. J., & Grasha, A. F. (1995). The relationship of MBTI® dimensions to perceptions of stress and coping strategies in managers. *Journal of Psychological Type*, 32, 13–22.
- Singer, M. (1989). Cognitive style and reading comprehension. *Journal of Psychological Type*, 17, 31–35.
- Sipps, G. J., Alexander, R. A., & Friedt, L. (1985). Item analysis of the Myers-Briggs Type Indicator®. *Educational and Psychological Measurement*, 45, 789–796.
- Sitarenios, G. (2004). *The relationship between the EQ-I®, MBTI®, and FIRO-B® in a corporate sample*. MHS Technical Report #019, Multi-Health Systems, Inc.
- Smith, A., Irely, R., & McCaulley, M. H. (1973). Self-paced instruction and college student's personality. *Engineering Education*, 63(6), 435–440.
- Smith, N. P. (1976). The influence of structural information characteristics of Jungian personality type on time horizons in decision-making (Doctoral dissertation, University of California). *Dissertation Abstracts International: Section A. Humanities and Social Sciences*, 37, 2297–2298. (University Microfilms No. 76-22, 715)
- Spokane, A. R. (1996). Holland's theory. In D. Brown & L. Brooks (Eds.), *Career choice and development* (3rd ed.). San Francisco, CA: Jossey-Bass.
- Squyres, E. M. (1980). *Time orientation and Jung's concept of psychological wholeness* (Unpublished master's thesis). Georgia State University, Atlanta.
- Steele, G. E. (1986). An investigation of the relationship between students' interests and the curricular practices of an alternative high school, through the perspective of Jung's theory of psychological types (Doctoral dissertation, Ohio State University). *Dissertation Abstracts International: Section A. Humanities and Social Sciences*, 47(10), 3616.
- Stenberg, G. (1990). *Brain and personality: Extraversion/Introversion in relation to EEG, evoked potentials and cerebral blood flow* (Unpublished doctoral dissertation). University of Lund, Sweden.
- Sundstrom, E., & Busby, P. L. (1997). Co-workers' perceptions of eight MBTI® leader types: Comparative analysis of managers' SYMLOG profiles. In C. Fitzgerald & L. K. Kirby (Eds.), *Developing leaders: Research and applications in psychological type and leadership development* (pp. 225–265). Boston, MA: Nicholas Brealey.
- Super, D. E. (1970). *The Work Values Inventory*. Boston, MA: Houghton Mifflin Harcourt.
- Taggart, W. M., Kroeck, K. G., & Escoffier, M. R. (1991). Validity evidence for the Myers-Briggs Type Indicator® as a measure of hemisphere dominance: Another view. *Educational and Psychological Measurement*, 51, 775–783.
- Thomas, A., Benne, M. R., Marr, M. J., Thomas, E. W., & Hume, R. M. (2000). The evidence remains stable: The MBTI® predicts attraction and attrition in an engineering program. *Journal of Psychological Type*, 55, 35–42.
- Thomas, K. W. (2002). *Introduction to conflict management: Improving performance using the TKI®*. Sunnyvale, CA: The Myers-Briggs Company.
- Thomas, K. W., & Kilmann, R. H. (1974). *Thomas-Kilmann Conflict Mode Instrument*. Sunnyvale, CA: The Myers-Briggs Company.
- Thomason, R. S. (1983). Relationships of chronological age, psychological type, and reading comprehension of college students (Doctoral dissertation, University of Florida). *Dissertation Abstracts International: Section A. Humanities and Social Sciences*, 45(4), 1086.
- Thompson, B., & Borrello, G. M. (1986). Construct validity of the Myers-Briggs Type Indicator®. *Educational and Psychological Measurement*, 46, 745–752.

- Thompson, B., & Borrello, G. M. (1989, January). A confirmatory factor analysis of data from the Myers-Briggs Type Indicator®. Paper presented at the annual meeting of the Southwest Educational Research Association, Houston, TX.
- Thompson, R. C. (2014, March). *Multiple approaches to building a personality assessment using a multinational data set*. Symposium conducted at the ATP 2014 Innovations in Testing Conference, Scottsdale, AZ.
- Thompson, R. C., Walling, L., Mackey, C., Anderson, M., Schaubhut, N. A., Weber, A., & Kulapaditharom, L. (2013, July). MBTI® Familiarity Survey (phase II). In R. C. Thompson (Chair), *Research on the validity of the MBTI®: Countering persistent complaints*. Presented at the XX Biennial Meeting of the Association for Psychological Type International (APTi), Miami, FL.
- Thorne, A. (1987). The press of personality: A study of conversations between Introverts and Extraverts. *Journal of Personality and Social Psychology*, 53(4), 718–726.
- Thorne, A., & Gough, H. (1991). *Portraits of type: An MBTI® research compendium*. Mountain View, CA: CPP, Inc.
- Thorne, A., Korobov, N., & Morgan, E. M. (2007, October). Channeling identity: A study of storytelling in conversations between introverted and extraverted friends. *Journal of Research in Personality*, 41(5), 1008–1031.
- Thorne, A., Shapiro, L., Cardilla, K., Korobov, N., & Nelson, P. A. (2009). Caught in the act: How extraverted and introverted friends communally cope with being recorded. *Journal of Research in Personality*, 43, 634–642.
- Tinsley, H. E. A., Tinsley, D. J., & Rushing, J. (2002). Psychological type, decision-making style, and reactions to structured career interventions. *Journal of Career Assessment*, 10, 258–280.
- Tischler, L. (1994). The MBTI® factor structure. *Journal of Psychological Type*, 31, 24–31.
- Toole, J. M. (2007). *The academic performance of introverted and extroverted students in traditional and online business courses: An analysis based on the Myers-Briggs Type Indicator®* (Unpublished doctoral dissertation). Capella University.
- Tzeng, O. C. S., Outcalt, D., Boyer, S. L., Ware, R., & Landis, D. (1984). Item validity of the Myers-Briggs Type Indicator®. *Journal of Personality Assessment*, 48, 255–256.
- Vaccaro, A. J. (1988, September). Personality clash. *Personnel Administrator*, 88–92.
- Van Rooyen, J. (1994). Creativity: An important managerial requirement. A South African perspective. In C. Fitzgerald (Ed.), *Proceedings of The Myers-Briggs Type Indicator® and Leadership: An International Research Conference* (pp. 49–59). College Park, MD: National Leadership Institute, University of Maryland.
- Van Velsor, E., & Fleenor, J. W. (1997). The MBTI® and leadership skills: Relationships between the MBTI® and four 360-degree management feedback instruments. In C. Fitzgerald & L. K. Kirby (Eds.), *Developing leaders: Research and applications in psychological type and leadership development* (pp. 139–162). Boston, MA: Nicholas Brealey.
- Van Zyl, C. J. J., & Taylor, N. (2012). Evaluating the MBTI® Form M in a South African context. *SA Journal of Industrial Psychology/SA Tydskrif vir Bedryfsielkunde*, 38(1), Art. #977, 15 pages. <http://dx.doi.org/10.4102/sajip.v38i1.977>
- Vaughan, J. A., & Knapp, R. H. (1963). A study in pessimism. *Journal of Social Psychology*, 59, 77–92.
- Veach, T. L., & Touhey, J. C. (1971). Personality correlates of accurate time perception. *Perceptual and Motor Skills*, 33(3), 765–766.
- Vermunt, J. K., & Magidson, J. (2003). Latent class models for classification. *Computational Statistics & Data Analysis*, 41(3), 531–537.
- Vermunt, J. K., & Magidson, J. (2013). *Technical guide for Latent GOLD 4.0*. Belmont, MA: Statistical Innovations, Inc.
- Vollbrecht, F. E. (1991). A study of selected personality characteristics of vocational trade and industrial education teachers (Doctoral dissertation, Texas A&M University). *Dissertation Abstracts International: Section A. Humanities and Social Sciences*, 52(6), 1990.
- von Franz, M. L. (1971). The inferior function. In M. L. von Franz & J. Hillman (Eds.), *Lectures on Jung's typology* (5th corrected printing, pp. 1–72). Dallas, TX: Spring Publications.
- Walck, C. L. (1992). The relationship between indicator type and “true type”: Slight preferences and the verification process. *Journal of Psychological Type*, 23, 17–21.
- Walck, C. L. (1996). Management and leadership. In A. L. Hammer (Ed.), *MBTI® applications: A decade of research on the Myers-Briggs Type Indicator®* (pp. 55–80). Sunnyvale, CA: CPP, Inc.
- Wallick, M. M., Cambre, K. M., & McClugage, S. G. (2000, August). Does the admissions committee select medical students in its own image? *Journal of the Louisiana State Medical Society*, 152(8), 393–397.
- Walter, K. M. (1984). Writing in three disciplines correlated with Jungian cognition styles (Doctoral dissertation, University of Texas at Austin). *Dissertation Abstracts International: Section A. Humanities and Social Sciences*, 46(2), 414.
- Wanberg, C. R. (1992). *Men and women without jobs: A study of factors predictive of psychological and physical health* (Unpublished doctoral dissertation). Iowa State University.
- Ware, R., & Yokomoto, C. (1985). Perceived accuracy of Myers-Briggs Type Indicator® descriptions using Keirsey profiles. *Journal of Psychological Type*, 10, 27–31.
- Webster, K., & Howard, P. J. (1989). MBTI®-type heterogeneity and business game results. In N. L. Quenk (Ed.), *Eighth Biennial International Conference of the Association for Psychological Type* (pp. 149–152). Boulder, CO: APT-VIII Planning Committee, University of Colorado.

- Weir, D. M. (1976). The relationship of four Jungian personality types to stated preference for high unconditional positive regard as a counseling approach (Doctoral dissertation, Southern Illinois University). *Dissertation Abstracts International: Section A. Humanities and Social Sciences*, 36, 7881. (University Microfilms No. 76-13, 298)
- Williams, R. L., Verble, J. S., Price, D. E., & Layne, B. H. (1995). Relationship between time-management practices and personality indices and types. *Journal of Psychological Type*, 34, 36–42.
- Williams, S. B., Staley, D., & Rudyk, B. (2000). The Exam Preparation Inventory: A reliability and validity study. In *Proceedings of the Fourth Biennial International Conference on Education of the Center for Applications of Psychological Type* (pp. 315–323). Gainesville, FL: Center for Applications of Psychological Type.
- Wilson, M. A., & Languis, M. L. (1989). Differences in brain electrical activity patterns between introverted and extraverted adults. *Journal of Psychological Type*, 18, 14–23.
- Wilson, M. A., & Languis, M. L. (1990). A topographic study of differences in the P300 between Introverts and Extraverts. *Brain Topography*, 2(4), 269–274.
- Wilson, M., Laposky, A. D., & Languis, M. L. (1991, July). A topographic study of electrophysiological differences between Sensing and Intuitive types. Paper presented at the 9th Meeting of the Association for Psychological Type, Richmond, VA.
- Yang, A. I. (1981). Psychological temporality: A study of temporal orientation, attitude, mood states, and personality type (Doctoral dissertation, University of Hawaii). *Dissertation Abstracts International: Section B. Sciences and Engineering*, 42(4), 1677.
- Yeakley, F. R. (1982). Communication style preferences and adjustments as an approach to studying effects of similarity in psychological type. *Research in Psychological Type*, 5, 30–48.
- Yeakley, F. R. (1983). Implications of communication style research for psychological type theory. *Research in Psychological Type*, 6, 2–13.
- Yorkston, E. A., & Labarbera, P. A. (1997). Personality-related cognitive style and advertising strategy. In *New and evolving paradigms: The emerging future of marketing*. (pp. 802–814). Chicago, IL: American Marketing Association.
- Zeidner, M., & Hammer, A. L. (1990). Life events and coping resources as predictors of stress symptoms in adolescents. *Personality and Individual Differences*, 11(7), 693–703.
- Zeidner, M., & Hammer, A. L. (1992). Coping with missile attack: Resources, strategies and outcomes. *Journal of Personality*, 60(4), 709–746.

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ABOUT THE AUTHORS

Isabel Briggs Myers (1897–1980), in collaboration with her mother, Katherine C. Briggs, created and developed the *Myers Briggs Type Indicator®* assessment based on the psychological type theory of Swiss psychiatrist Carl G. Jung. Myers spent decades carefully observing people, writing, testing, and revising items, and collecting data from high schools and medical, nursing, pharmacy, and law schools, among other places. She authored the first MBTI® manual (1962), the first edition of *Introduction to Type®* (1962), and her seminal book on type, *Gifts Differing* (1980). In the early 1970s, she cofounded the Center for Applications of Psychological Type with Dr. Mary McCaulley. She and McCaulley developed the first computer scoring program for the MBTI Step I® instrument. Although Myers was not involved in the later development of the MBTI Step II® instrument, she wrote a number of items anticipating its creation that are included in the current Step II instrument.

Mary H. McCaulley, PhD (1920–2003), was a professor of clinical psychology, a researcher, and cofounder and first president of the Center for Applications of Psychological Type (CAPT). With Isabel Myers, she developed the first MBTI training program designed to teach professionals how to use the MBTI instrument ethically and effectively. She supervised CAPT in their preparation of research data for the 1985 *MBTI® Manual*, which she coauthored with Myers. McCaulley also authored or coauthored *The Myers-Briggs Type Indicator® to Medicine and Other Health Professions*, MBTI studies of engineering students, and the *Myers-Briggs Type Indicator® Atlas*. She worked with Myers on the computer scoring program that preceded the MBTI Step II instrument.

Naomi L. Quenk, PhD, is a psychologist, author, editor, and independent consultant. She has authored or coauthored many articles and books on psychological type, including *Types of Family Practice Teachers and Residents: Characteristics of Physician Work Settings in Relation to Myers-Briggs® Preferences*; *MBTI® Manual: A Guide to the Development and Use of the Myers-Briggs Type Indicator® Instrument*; *Essentials of Myers-Briggs Type Indicator® Assessment*; *In the Grip: Understanding Type, Stress, and the Inferior Function*; *MBTI® Step II® Manual: Exploring the Next Level of Type with the Myers-Briggs Type Indicator® Form Q*; *MBTI® Step II® Interpretive Report*; *Was That Really Me? How Everyday Stress Brings Out Our Hidden Personality*; *MBTI® Step III® Manual: Exploring Personality Development Using the Myers-Briggs Type Indicator® Instrument*. Quenk won a Myers & Briggs Foundation award for her work on developing the MBTI Step III instrument.

Allen L. Hammer, PhD, is Senior Advisor and Assessment Expert with Manfreda Curtis Associates. With the MCA team he consults with organizations on team and leadership development for Fortune 100 corporations as well as nonprofits. Hammer is author or coauthor of *MBTI® Complete*, *MBTI® Comparison Report: Work Styles, Introduction to Type® and Careers*, *MBTI® Career Report*, *MBTI® Team Report*, *Introduction to Type® in College*, *Strong and MBTI® Career Report*, *Strong Interest Inventory® Manual*, *Introduction to FIRO-B® Instrument in Organizations*, *FIRO-B® Technical Guide*, and *FIRO Business® Leadership Report*. He edited *MBTI® Applications: A Decade of Research on the MBTI®*. Hammer won two Isabel Briggs Myers Research Awards for work on couples and on the MBTI Step III assessment. The Association for Psychological Type International recognized him with their Innovations Award for *MBTI® Complete*.

